

mmWave Radar Interface Control Document

Revision 3.42



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Contents

	Page
Contents	xii
List of Figures	xiii
List of Tables	xx
Revision History	xxii
1 Introduction	1
1.1 Scope	1
1.2 Intended Audience	2
2 TI mmWave Radar Sensor Communications Overview	3
2.1 Communication Link Description	3
2.2 Communication Link configuration	3
2.2.1 SPI	3
2.2.2 Mailbox	3
2.2.3 BOOT INFO registers	3
2.3 Radar Message Structure	15
2.3.1 SYNC	15
2.3.2 MSGHDR	15
2.3.3 MSGDATA	21
2.3.4 CRC	21
3 Message Processing	23
3.1 Communication protocol	23
3.2 Communication Sequence	24
3.2.1 Command/Response Sequence (Host)	24
3.2.2 Flow Diagram (Host) – Command/Response	25
3.2.3 Flow Diagram (Host) – Bootup/ Asynchronous Event	26
3.2.4 SPI Message Sequence – Command/Response	27
3.2.5 Command/Response sequence (MSS-BSS mailbox)	27
3.2.6 Flow Diagram Command/Response (MSS-BSS mailbox)	29
3.2.7 Asynchronous event sequence (BSS-MSS mailbox)	30

3.2.8	Flow Diagram Asynchronous event (BSS-MSS mailbox)	31
4	Radar Interface Messages Descriptions	33
4.1	Summary of all messages and their associated sub-blocks	33
4.2	AWR_ACK_MSG	39
4.3	AWR_NACK_MSG	40
4.4	AWR_ERROR_MSG	40
4.5	AWR_RF_STATIC_CONF_SET_MSG	41
4.6	AWR_RF_STATIC_CONF_GET_MSG	42
4.7	AWR_RF_INIT_MSG	42
4.8	AWR_RF_DYNAMIC_CONF_SET_MSG	44
4.9	AWR_RF_DYNAMIC_CONF_GET_MSG	45
4.10	AWR_RF_FRAME_TRIG_MSG	46
4.11	AWR_RF_ADVANCED_FEATURES_CONF_SET_MSG	47
4.12	AWR_RF_MONITORING_CONF_SET_MSG	47
4.13	AWR_RF_STATUS_GET_MSG	49
4.14	AWR_RF_MONITORING_REPORT_GET_MSG	50
4.15	AWR_RF_MONITORING_CONF_SET_2_MSG	50
4.16	AWR_RF_MISC_CONF_SET_MSG	51
4.17	AWR_RF_MISC_CONF_GET_MSG	52
4.18	AWR_RF_ASYNC_EVENT_MSG1	52
4.19	AWR_RF_ASYNC_EVENT_MSG2	54
4.20	AWR_DEV_RFPOWERUP_MSG	55
4.21	AWR_DEV_CONF_SET_MSG	56
4.22	AWR_DEV_CONF_GET_MSG	57
4.23	AWR_DEV_FILE_DOWNLOAD_MSG	58
4.24	AWR_DEV_FRAME_CONFIG_APPLY_MSG	58
4.25	AWR_DEV_STATUS_GET_MSG	59
4.26	AWR_DEV_ASYNC_EVENT_MSG	59
5	Radar Functional APIs	61
5.1	Sub block related to AWR_ERROR_MSG	61
5.1.1	Sub block 0x0000 – AWR_RESP_ERROR_SB	61
5.2	Sub blocks related to AWR_RF_STATIC_CONF_SET_MSG	62
5.2.1	Sub block 0x0080 – AWR_CHAN_CONF_SET_SB	62
5.2.2	Sub block 0x0082 – AWR_ADCOUT_CONF_SET_SB	66
5.2.3	Sub block 0x0083 – AWR_LOWPOWERMODE_CONF_SET_SB	67
5.2.4	Sub block 0x0084 – AWR_DYNAMICPOWERSAVE_CONF_SET_SB	68
5.2.5	Sub block 0x0085 – AWR_HIGHSPEEDINTFCLK_CONF_SET_SB	69
5.2.6	Sub block 0x0086 – AWR_RF_DEVICE_CFG_SB	70
5.2.7	Sub block 0x0087 – AWR_RF_RADAR_MISC_CTL_SB	73
5.2.8	Sub block 0x0088 – AWR_CAL_MON_FREQUENCY_LIMITS_SB	79
5.2.9	Sub block 0x0089 – AWR_RF_INIT_CALIBRATION_CONF_SB	81

5.2.10	Sub block 0x008A – AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB	83
5.2.11	Sub block 0x008B – AWR_CAL_DATA_RESTORE_SB	88
5.2.12	Sub block 0x008C – AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB	89
5.2.13	Sub block 0x008D – AWR_APLL_SYNTH_BW_CONTROL_SB	91
5.3	Sub blocks related to AWR_RF_STATIC_CONF_GET_MSG	93
5.3.1	Sub block 0x00A0 – 0x00AA – RESERVED	93
5.3.2	Sub block 0x00AB – AWR_CAL_DATA_SAVE_SB	93
5.3.3	Sub block 0x00AC – AWR_PHASE_SHIFTER_CAL_DATA_SAVE_SB	99
5.4	Sub blocks related to AWR_RF_INIT_MSG	102
5.4.1	Sub block 0x00C0 – AWR_RF_INIT_SB	102
5.5	Sub blocks related to AWR_RF_DYNAMIC_CONF_SET_MSG	103
5.5.1	Sub block 0x0100 – AWR_PROFILE_CONF_SET_SB	103
5.5.2	Sub block 0x0101 – AWR_CHIRP_CONF_SET_SB	117
5.5.3	Sub block 0x0102 – AWR_FRAME_CONF_SET_SB	119
5.5.4	Sub block 0x0103 – AWR_CONT_STREAMING_MODE_CONF_SET_SB	123
5.5.5	Sub block 0x0104 – AWR_CONT_STREAMING_MODE_EN_SB	128
5.5.6	Sub block 0x0105 – AWR_ADVANCED_FRAME_CONF_SB	129
5.5.7	Sub block 0x0106 – AWR_PERCHIRPPHASESHIFT_CONF_SB	141
5.5.8	Sub block 0x0107 – AWR_PROG_FILT_COEFF_RAM_SET_SB	143
5.5.9	Sub block 0x0108 – AWR_PROG_FILT_CONF_SET_SB	145
5.5.10	Sub block 0x0109 – AWR_CALIB_MON_TIME_UNIT_CONF_SB	146
5.5.11	Sub block 0x010A – AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB	149
5.5.12	Sub block 0x010B – AWR_DIGITAL_COMP_EST_CONTROL_SB	157
5.5.13	Sub block 0x010C – AWR_RX_GAIN_TEMPLUT_SET_SB	161
5.5.14	Sub block 0x010D – AWR_TX_GAIN_TEMPLUT_SET_SB	164
5.5.15	Sub block 0x010E – AWR_LOOPBACK_BURST_CONF_SET_SB	167
5.5.16	Sub block 0x010F – AWR_DYN_CHIRP_CONF_SET_SB	176
5.5.17	Sub block 0x0110 – AWR_DYN_PERCHIRP_PHASESHIFTER_CONF_SET_SB	179
5.5.18	Sub block 0x0111 – AWR_DYN_CHIRP_ENABLE_SB	181
5.5.19	Sub block 0x0112 – AWR_INTERCHIRP_BLOCKCONTROLS_SB	182
5.5.20	Sub block 0x0113 – AWR_SUBFRAME_START_CONF_SB	185
5.5.21	Sub block 0x0115 – AWR_ADVANCE_CHIRP_CONF_SB	186
5.5.22	Sub block 0x0116 – AWR_ADVANCE_CHIRP_GENERIC_LUT_LOAD_SB	202
5.5.23	Sub block 0x0117 – AWR_MONITOR_TYPE_TRIG_CONF_SB	211
5.5.24	Sub block 0x0118 – AWR_ADVANCE_CHIRP_DYN_LUT_ADDR_OFFSET_CFG_SB	213
5.5.25	Sub block 0x0119 – AWR_ADV_TX_GAIN_TEMPLUT_SET_SB	215
5.5.26	Sub block 0x011A – AWR_ADV_DYN_PERCHIRP_PHASESHIFTER_CONF_SET_SB	217

5.6	Sub blocks related to AWR_RF_DYNAMIC_CONF_GET_SB	219
5.6.1	Sub block 0x0120 – AWR_PROFILE_CONF_GET_SB	219
5.6.2	Sub block 0x0121 – AWR_CHIRP_CONF_GET_SB	219
5.6.3	Sub block 0x0122 – AWR_FRAME_CONF_GET_SB	220
5.6.4	Sub block 0x0123 – RESERVED	220
5.6.5	Sub block 0x0124 – RESERVED	220
5.6.6	Sub block 0x0125 – AWR_ADV_FRAME_CONF_GET_SB	220
5.6.7	Sub block 0x0126 – RESERVED	221
5.6.8	Sub block 0x0127 – RESERVED	221
5.6.9	Sub block 0x0128 – RESERVED	221
5.6.10	Sub block 0x0129 – RESERVED	221
5.6.11	Sub block 0x012A – RESERVED	221
5.6.12	Sub block 0x012B – RESERVED	221
5.6.13	Sub block 0x012C – AWR_RX_GAIN_TEMPLUT_GET_SB	221
5.6.14	Sub block 0x012D – AWR_TX_GAIN_TEMPLUT_GET_SB	221
5.6.15	Sub block 0x0139 – AWR_ADV_TX_GAIN_TEMPLUT_GET_SB	222
5.7	Sub blocks related to AWR_FRAME_TRIG_MSG	222
5.7.1	Sub block 0x0140 – AWR_FRAMESTARTSTOP_CONF_SB	222
5.8	Sub blocks related to AWR_RF_ADVANCED_FEATURES_CONF_SET_MSG	224
5.8.1	Sub block 0x0180 – AWR_BPM_COMMON_CONF_SET_SB	224
5.8.2	Sub block 0x0181 – AWR_BPM_CHIRP_CONF_SET_SB	224
5.8.3	Sub block 0x0182 – AWR_POWER_SAVE_MODE_CONF_SET_SB	225
5.9	Sub blocks related to AWR_RF_STATUS_GET_MSG	226
5.9.1	Sub block 0x0220 – AWR_RF_VERSION_GET_SB	226
5.9.2	Sub block 0x0221 – AWR_RF_CPUFAULT_STATUS_GET_SB	228
5.9.3	Sub block 0x0222 – AWR_RF_ESMFAULT_STATUS_GET_SB	230
5.9.4	Sub block 0x0223 – AWR_RF_DIEID_GET_SB	232
5.9.5	Sub block 0x0224 – AWR_RF_BOOTUPBIST_STATUS_GET_SB	233
5.9.6	Sub block 0x0226 – AWR_RF_ADV_ESMFAULT_STATUS_GET_SB	236
5.10	Sub blocks related to AWR_RF_MONITORING_REPORT_GET_MSG	240
5.10.1	Sub block 0x0260 – AWR_RF_DFE_STATISTICS_REPORT_GET_SB	240
5.10.2	Sub block 0x0261 – AWR_RF_REAL_CHAN_DFE_STATISTICS_REPORT_GET_SB	250
5.11	Sub blocks related to AWR_RF_MISC_CONF_SET_MSG	256
5.11.1	Sub block 0x02C0 – RESERVED	256
5.11.2	Sub block 0x02C1 – RESERVED	256
5.11.3	Sub block 0x02C2 – AWR_RF_TEST_SOURCE_CONFIG_SET_SB	256
5.11.4	Sub block 0x02C3 – AWR_RF_TEST_SOURCE_ENABLE_SET_SB	259
5.11.5	Sub block 0x02C4 – 0x02CB RESERVED	260
5.11.6	Sub block 0x02CC – AWR_RF_LDO_BYPASS_SB	260
5.11.7	Sub block 0x02CD – AWR_RF_PALOOPBACK_CFG_SB	262
5.11.8	Sub block 0x02CE – AWR_RF_PSLOOPBACK_CFG_SB	264
5.11.9	Sub block 0x02CF – AWR_RF_IFLOOPBACK_CFG_SB	265
5.11.10	Sub block 0x02D0 – AWR_RF_GPADC_CFG_SET_SB	266

5.11.11	Sub block 0x02D1 – RESERVED	269
5.11.12	Sub block 0x02D2 – RESERVED	269
5.11.13	Sub block 0x02D3 – RESERVED	269
5.11.14	Sub block 0x02D4 – RESERVED	269
5.11.15	Sub block 0x02D5 – AWR_RF_LOLOOPBACK_CFG_SB	269
5.12	Sub blocks related to AWR_RF_MISC_CONF_GET_MSG	271
5.12.1	Sub block 0x02E0 to 0x2E9 – RESERVED	271
5.12.2	Sub block 0x02EA – AWR_RF_TEMPERATURE_GET_SB	271
5.13	Sub blocks related to AWR_RF_ASYNC_EVENT_MSG1	272
5.13.1	Sub block 0x1000 – RESERVED	273
5.13.2	Sub block 0x1001 – RESERVED	273
5.13.3	Sub block 0x1002 – AWR_AE_RF_CPUFAULT_SB	273
5.13.4	Sub block 0x1003 – AWR_AE_RF_ESMFAULT_SB	275
5.13.5	Sub block 0x1004 – AWR_AE_RF_INITCALIBSTATUS_SB	278
5.13.6	Sub block 0x1005 – RESERVED	279
5.13.7	Sub block 0x1006 – RESERVED	279
5.13.8	Sub block 0x1007 – AWR_AE_RF_ADV_ESMFAULT_SB	279
5.13.9	Sub block 0x1008 – RESERVED	284
5.13.10	Sub block 0x1009 – RESERVED	284
5.13.11	Sub block 0x100A – AWR_AE_RF_MONITOR_TYPE_TRIGGER_DONE_SB	284
5.13.12	Sub block 0x100B – AWR_AE_RF_FRAME_TRIGGER_RDY_SB	284
5.13.13	Sub block 0x100C – AWR_AE_RF_GPADC_RESULT_DATA_SB	285
5.13.14	Sub block 0x100E – RESERVED	287
5.13.15	Sub block 0x100D – RESERVED	287
5.13.16	Sub block 0x100E – RESERVED	287
5.13.17	Sub block 0x100F – AWR_FRAME_END_AE_SB	287
5.13.18	Sub block 0x1010 – AWR_ANALOGFAULT_AE_SB	287
5.13.19	Sub block 0x1011 – AWR_CAL_MON_TIMING_FAIL_REPORT_AE_SB	288
5.13.20	Sub block 0x1012 – AWR_RUN_TIME_CALIB_SUMMARY_REPORT_AE_SB	289
5.13.21	Sub block 0x1013 – AWR_MONITOR_RF_DIG_LATENTFAULT_REPORT_AE_SB	292
5.13.22	Sub block 0x1014 – RESERVED	293
5.13.23	Sub block 0x1015 – AWR_MONITOR_REPORT_HEADER_AE_SB	293
5.13.24	Sub block 0x1016 – AWR_MONITOR_RF_DIG_PERIODIC_REPORT_AE_SB	294
5.13.25	Sub block 0x1017 – AWR_MONITOR_TEMPERATURE_REPORT_AE_SB	295
5.13.26	Sub block 0x1018 – AWR_MONITOR_RX_GAIN_PHASE_REPORT_AE_SB	297
5.13.27	Sub block 0x1019 – AWR_MONITOR_RX_NOISE_FIGURE_REPORT_AE_SB	300
5.13.28	Sub block 0x101A – AWR_MONITOR_RX_IFSTAGE_REPORT_AE_SB	302

5.13.29	Sub block 0x101B – AWR_MONITOR_TX0_POWER_REPORT_AE_SB	305
5.13.30	Sub block 0x101C – AWR_MONITOR_TX1_POWER_REPORT_AE_SB	306
5.13.31	Sub block 0x101D – AWR_MONITOR_TX2_POWER_REPORT_AE_SB	307
5.13.32	Sub block 0x101E – AWR_MONITOR_TX0_BALLBREAK_REPORT_AE_SB	308
5.13.33	Sub block 0x101F – AWR_MONITOR_TX1_BALLBREAK_REPORT_AE_SB	309
5.14	Sub blocks related to AWR_RF_ASYNC_EVENT_MSG2	310
5.14.1	Sub block 0x1020 – AWR_MONITOR_TX2_BALLBREAK_REPORT_AE_SB	310
5.14.2	Sub block 0x1021 – AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_REPORT_AE_SB	311
5.14.3	Sub block 0x1022 – AWR_MONITOR_TX0_PHASE_SHIFTER_REPORT_AE_SB	314
5.14.4	Sub block 0x1023 – AWR_MONITOR_TX1_PHASE_SHIFTER_REPORT_AE_SB	316
5.14.5	Sub block 0x1024 – AWR_MONITOR_TX2_PHASE_SHIFTER_REPORT_AE_SB	318
5.14.6	Sub block 0x1025 – AWR_MONITOR_SYNTHESIZER_FREQUENCY_REPORT_AE_SB	320
5.14.7	Sub block 0x1026 – AWR_MONITOR_EXTERNAL_ANALOG_SIGNALS_REPORT_AE_SB	321
5.14.8	Sub block 0x1027 – AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB	323
5.14.9	Sub block 0x1028 – AWR_MONITOR_TX1_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB	324
5.14.10	Sub block 0x1029 – AWR_MONITOR_TX2_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB	325
5.14.11	Sub block 0x102A – AWR_MONITOR_RX_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB	326
5.14.12	Sub block 0x102B – AWR_MONITOR_PMCLKLO_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB	327
5.14.13	Sub block 0x102C – AWR_MONITOR_GPADC_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB	329
5.14.14	Sub block 0x102D – AWR_MONITOR_PLL_CONTROL_VOLTAGE_REPORT_AE_SB	330
5.14.15	Sub block 0x102E – AWR_MONITOR_DUAL_CLOCK_COMP_REPORT_AE_SB	332
5.14.16	Sub block 0x1031 – AWR_MONITOR_RX_MIXER_IN_POWER_REPORT_AE_SB	334
5.14.17	Sub block 0x1033 – AWR_MONITOR_SYNTHESIZER_FREQUENCY_NONLIVE_REPORT_AE_SB	335
5.14.18	Sub block 0x1034 – AWR_MONITOR_TX3_POWER_REPORT_AE_SB	337

5.14.19	Sub block 0x1035 – AWR_MONITOR_TX3_BALLBREAK_REPORT_AE_SB	338
5.14.20	Sub block 0x1036 – AWR_MONITOR_TX3_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB	339
5.14.21	Sub block 0x1037 – AWR_MONITOR_TX3_PHASE_SHIFTER_REPORT_AE_SB	340
5.14.22	Sub block 0x1038 – AWR_MONITOR_ADV_TX_GAIN_PHASE_MISMATCH_REPORT_AE_SB	342
5.14.23	Sub block 0x1039 – AWR_MONITOR_TX_PHSHIFTER_DAC_REPORT_AE_SB	344
5.15	Sub blocks related to AWR_DEV_RFPOWERUP_MSG	346
5.15.1	Sub block 0x4000 – AWR_DEV_RFPOWERUP_SB	346
5.16	Sub blocks related to AWR_DEV_CONF_SET_MSG	346
5.16.1	Sub block 0x4040 – AWR_DEV_MCUCLOCK_CONF_SET_SB	346
5.16.2	Sub block 0x4041 – AWR_DEV_RX_DATA_FORMAT_CONF_SET_SB	348
5.16.3	Sub block 0x4042 – AWR_DEV_RX_DATA_PATH_CONF_SET_SB	349
5.16.4	Sub block 0x4043 – AWR_DEV_RX_DATA_PATH_LANEEN_SET_SB	352
5.16.5	Sub block 0x4044 – AWR_DEV_RX_DATA_PATH_CLK_SET_SB	353
5.16.6	Sub block 0x4045 – AWR_DEV_LVDS_CFG_SET_SB	354
5.16.7	Sub block 0x4046 – AWR_DEV_RX_CONTSTREAMING_MODE_CONF_SET_SB	356
5.16.8	Sub block 0x4047 – AWR_DEV_CSI2_CFG_SET_SB	357
5.16.9	Sub block 0x4048 – AWR_DEV_PMICCLOCK_CONF_SET_SB	359
5.16.10	Sub block 0x4049 – AWR_MSS_PERIODICTESTS_CONF_SB	364
5.16.11	Sub block 0x404A – AWR_MSS_LATENTFAULT_TEST_CONF_SB	365
5.16.12	Sub block 0x404B – AWR_DEV_TESTPATTERN_GEN_SET_SB	367
5.16.13	Sub block 0x404C – AWR_DEV_CONFIGURATION_SET_SB	370
5.16.14	Sub block 0x404D – AWR_DEV_RF_DEBUG_SIG_SET_SB	371
5.16.15	Sub block 0x404E – AWR_DEV_DEV_HSI_DELAY_DUMMY_CFG_SET_SB	372
5.17	Sub blocks related to AWR_DEV_CONF_GET_MSG	373
5.17.1	Sub block 0x4060 – AWR_DEV_MCUCLOCK_GET_SB	373
5.17.2	Sub block 0x4061 – AWR_DEV_RX_DATA_FORMAT_CONF_GET_SB	373
5.17.3	Sub block 0x4062 – AWR_DEV_RX_DATA_PATH_CONF_GET_SB	374
5.17.4	Sub block 0x4063 – AWR_DEV_RX_DATA_PATH_LANEEN_GET_SB	374
5.17.5	Sub block 0x4064 – AWR_DEV_RX_DATA_PATH_CLK_GET_SB	374
5.17.6	Sub block 0x4065 – AWR_DEV_LVDS_CFG_GET_SB	375
5.17.7	Sub block 0x4066 – AWR_DEV_RX_CONTSTREAMING_MODE_CONF_GET_SB	375
5.17.8	Sub block 0x4067 – AWR_DEV_CSI2_CFG_GET_SB	375
5.17.9	Sub block 0x4068 – AWR_DEV_PMICCLOCK_CONF_GET_SB	375
5.17.10	Sub block 0x4069 – AWR_MSS_LATENTFAULT_TEST_CONF_GET_SB	376
5.17.11	Sub block 0x406A – AWR_MSS_PERIODICTESTS_CONF_GET_SB	376
5.17.12	Sub block 0x406B – AWR_DEV_TESTPATTERN_GEN_GET_SB	376

5.18	Sub blocks related to AWR_DEV_FILE_DOWNLOAD_MSG	377
5.18.1	Sub block 0x4080 – AWR_DEV_FILE_DOWNLOAD_SB	377
5.19	Sub blocks related to AWR_DEV_FRAME_CONFIG_APPLY_MSG	377
5.19.1	Sub block 0x40C0 – AWR_DEV_FRAME_CONFIG_APPLY_SB	377
5.19.2	Sub block 0x40C1 – AWR_DEV_ADV_FRAME_CONFIG_APPLY_SB	378
5.20	Sub blocks related to AWR_DEV_STATUS_GET_MSG	380
5.20.1	Sub block 0x40E0 – AWR_MSSVERSION_GET_SB	380
5.20.2	Sub block 0x40E1 – AWR_MSSCPUFAULT_STATUS_GET_SB	382
5.20.3	Sub block 0x40E2 – AWR_MSSSESMFAULT_STATUS_GET_SB	384
5.21	Sub blocks related to AWR_DEV_ASYNC_EVENT_MSG	387
5.21.1	Sub block 0x5000 – AWR_AE_DEV_MSSPOWERUPDONE_SB	387
5.21.2	Sub block 0x5001 – AWR_AE_DEV_RFPOWERUPDONE_SB	389
5.21.3	Sub block 0x5002 – AWR_AE_MSS_CPUFAULT_SB	391
5.21.4	Sub block 0x5003 – AWR_AE_MSS_ESMFAULT_STATUS_SB	392
5.21.5	Sub block 0x5004 – RESERVED	396
5.21.6	Sub block 0x5005 – AWR_AE_MSS_BOOTERRORSTATUS_SB	396
5.21.7	Sub block 0x5006 – AWR_AE_MSS_LATENTFAULT_TESTREPORT_SB	398
5.21.8	Sub block 0x5007 – AWR_AE_MSS_PERIODICTEST_STATUS_SB	400
5.21.9	Sub block 0x5008 – AWR_AE_MSS_RFERROR_STATUS_SB	400
5.21.10	Sub block 0x5009 – RESERVED	401
5.21.11	Sub block 0x500A – RESERVED	401
5.21.12	Sub block 0x500B – RESERVED	401
6	API Programming Sequence	402
6.1	Single device mode (xWR294x, xWR254x, xWR2x4xLC, xWR2x4xECO and xWR2x4xP)	402
6.2	Single device mode (AWR1243/AWR2243)	404
6.3	Cascaded device mode	407
6.4	Continuous streaming mode (in single device case)	409
6.5	Continuous streaming (CW) mode (in cascaded device case)	411
7	API Error Handling and Error Codes	413
7.1	API Error Handling	413
7.2	API Error Codes	413
7.3	Boot on SPI Error codes	425
8	Radar Monitoring APIs	428
8.1	Common Configurations and Reports	428
8.1.1	Sub block 0x01C0 – AWR_MONITOR_RF_DIG_LATENTFAULT_CONF_SB	429
8.1.2	Sub block 0x01C1 – AWR_MONITOR_RF_DIG_PERIODIC_CONF_SB	431
8.1.3	Sub block 0x01C2 – AWR_MONITOR_ANALOG_ENABLES_CONF_SB	431
8.2	Temperature Monitor	435
8.2.1	Sub block 0x01C3 – AWR_MONITOR_TEMPERATURE_CONF_SB	436
8.3	RX Gain and Phase Monitor	438

8.3.1	Sub block 0x01C4 – AWR_MONITOR_RX_GAIN_PHASE_CONF_SB	. 438
8.4	RX Noise Monitor	. 442
8.4.1	Sub block 0x01C5 – AWR_MONITOR_RX_NOISE_FIGURE_CONF_SB	. 442
8.5	RX IF Stage Monitor	. 444
8.5.1	Sub block 0x01C6 – AWR_MONITOR_RX_IFSTAGE_CONF_SB	. 444
8.6	TX Power Monitor	. 446
8.6.1	Sub block 0x01C7 – AWR_MONITOR_TX0_POWER_CONF_SB	. 446
8.6.2	Sub block 0x01C8 – AWR_MONITOR_TX1_POWER_CONF_SB	. 448
8.6.3	Sub block 0x01C9 – AWR_MONITOR_TX2_POWER_CONF_SB	. 449
8.7	TX Ball Break Monitor	. 451
8.7.1	Sub block 0x01CA – AWR_MONITOR_TX0_BALLBREAK_CONF_SB	. 451
8.7.2	Sub block 0x01CB – AWR_MONITOR_TX1_BALLBREAK_CONF_SB	. 453
8.7.3	Sub block 0x01CC – AWR_MONITOR_TX2_BALLBREAK_CONF_SB	. 454
8.8	TX Gain and Phase Mismatch Monitoring	. 455
8.8.1	Sub block 0x01CD – AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_CONF_SB	. 455
8.9	TX Phase Shifter Monitor	. 459
8.9.1	Sub block 0x01CE – AWR_MONITOR_TX0_PHASE_SHIFTER_CONF_SB	. 460
8.9.2	Sub block 0x01CF – AWR_MONITOR_TX1_PHASE_SHIFTER_CONF_SB	. 463
8.9.3	Sub block 0x01D0 – AWR_MONITOR_TX2_PHASE_SHIFTER_CONF_SB	. 467
8.10	Synthesizer Frequency Monitoring	. 471
8.10.1	Sub block 0x01D1 – AWR_MONITOR_SYNTHESIZER_FREQUENCY_CONF_SB	. 471
8.11	External Analog Signals Monitor	. 474
8.11.1	Sub block 0x01D2 – AWR_MONITORING_EXTERNAL_ANALOG_SIGNALS_CONF_SB	. 474
8.12	Internal Analog Signals Monitor	. 477
8.12.1	Sub block 0x01D3 – AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_CONF_SB	. 478
8.12.2	Sub block 0x01D4 – AWR_MONITOR_TX1_INTERNAL_ANALOG_SIGNALS_CONF_SB	. 479
8.12.3	Sub block 0x01D5 – AWR_MONITOR_TX2_INTERNAL_ANALOG_SIGNALS_CONF_SB	. 480
8.12.4	Sub block 0x01D6 – AWR_MONITOR_RX_INTERNAL_ANALOG_SIGNALS_CONF_SB	. 481
8.12.5	Sub block 0x01D7 – AWR_MONITOR_PMCLKLO_INTERNAL_ANALOG_SIGNALS_CONF_SB	. 481
8.12.6	Sub block 0x01D8 – AWR_MONITOR_GPADC_INTERNAL_ANALOG_SIGNALS_CONF_SB	. 483
8.13	PLL Control Voltage Monitor	. 483

8.13.1	Sub block 0x01D9 – AWR_MONITOR_PLL_CONTROL_VOLTAGE_SIGNALS_CONF_SB	484
8.14	Dual Clock Comparator Based Clock Frequency Monitor	486
8.14.1	Sub block 0x01DA – AWR_MONITOR_DUAL_CLOCK_COMP_CONF_SB	487
8.15	RX Saturation Detection Monitor	488
8.15.1	Sub block 0x01DB – AWR_MONITOR_RX_SATURATION_DETECTOR_CONF_SB	488
8.15.2	Sub block 0x01DC – AWR_MONITOR_SIG_IMG_MONITOR_CONF_SB	490
8.16	RX mixer input power monitor	492
8.16.1	Sub block 0x01DD – AWR_MONITOR_RX_MIXER_IN_POWER_CONF_SB	492
8.17	Sub block 0x01DE – RESERVED	494
8.18	Analog Fault injection	494
8.18.1	Sub block 0x01DF – AWR_ANALOG_FAULT_INJECTION_CONF_SB	494
8.19	Sub blocks related to AWR_RF_MONITORING_CONF_SET_2_MSG	503
8.19.1	Sub block 0x0280 – AWR_MONITOR_TXN_POWER_CONF_SB	503
8.19.2	Sub block 0x0281 – AWR_MONITOR_TXN_BALLBREAK_CONF_SB	505
8.19.3	Sub block 0x0282 – AWR_MONITOR_TXN_INTERNAL_ANALOG_SIGNALS_CONF_SB	507
8.19.4	Sub block 0x0283 – AWR_MONITOR_TXN_PHASE_SHIFTER_CONF_SB	508
8.19.5	Sub block 0x0284 – AWR_MONITOR_ADV_TX_GAIN_PHASE_MISMATCH_CONF_SB	512
8.19.6	Sub block 0x0285 – AWR_MONITOR_VMON_CONF_SB	516
8.19.7	Sub block 0x0286 – AWR_MONITOR_OVERRIDES_AND_DITHER_CONF_SB	519
8.19.8	Sub block 0x0287 – AWR_MONITOR_TX_PHSHIFTER_DAC_CONF_SB	523
9	Unsupported Features/APIs and Debug APIs	526
9.1	Unsupported Features/APIs and Debug APIs	526
10	Chirp Parameters (CP) and Chirp Quality (CQ) data	527
10.1	Chirp Parameters data	527
10.2	Chirp Quality data	528
10.2.1	CQ1	530
10.2.2	CQ2	533
11	Chirp, Burst and Frame timings	536
11.1	Chirp Cycle Time	536
11.2	Minimum Inter Burst Time	539
11.3	Minimum Inter Sub-frame or Frame Time	542
12	Calibration and monitoring durations	543

12.1 Boot time calibration durations	543
12.2 Run time calibration durations	545
12.3 Monitoring duration	547
12.4 Software overheads	553
12.4.1 Note on idle time for clearing the watchdog (WDT)	553
12.5 Sample Application	i
Appendices	ii
A AWR2243 API changes	iii
B xWR6243 API changes	vi

List of Figures

2.1	AWR12xx, AWR22xx, AWR62xx Software Architecture	13
2.2	xWR16xx, xWR18xx, IWR68xx, xWR294x, xWR254x, xWR2x4xEco and xWR2x4xP Software Architecture	14
2.3	Radar Message Structure	15
2.4	Message Header Format	16
2.5	OPCODE Format	16
2.6	MSGLEN Format	18
2.7	FLAGS Format	18
2.8	NSBC Format	20
2.9	Message Sub block structure	21
3.1	Flow Diagram (API)	25
3.2	Flow Diagram (Asynchronous Events)	26
3.3	SPI Message Sequence	27
3.4	Command/Response Flow Diagram	29
3.5	ASYNc Event Flow Diagram	31
5.1	TX PS apply timing in a chirp	116
5.2	Frame trigger delay in case of external hardware trigger	123
5.3	Frame trigger delay in case of external hardware trigger	141
5.4	TX PS apply timing in a chirp	143
5.5	Dynamic chirp configuration use case timing diagram	182
5.6	Advance chirp parameter dither sources and program	187
5.7	Lane formats and the order of receiving the data from the lanes	356
10.1	Chirp parameter information fields	527
10.2	Chirp parameter information from DSS registers	528
10.3	CQ data start address configuration in single chirp use case	529
10.4	CQ data start address configuration in multi chirp use case	530
10.5	Time slices during RX signal and image band monitor and saturation monitor	531
10.6	CQ1 data format in memory in 16-bit mode	532
10.7	CQ1 data format in memory in 12-bit mode	533
10.8	CQ1 data format in memory in 14-bit mode	533
10.9	CQ2 data format in memory in 16-bit mode	534
10.10	CQ2 data format in memory in 12-bit mode	535
10.11	CQ2 data format in memory in 14-bit mode	535

12.1 Watchdog idle time calculation	554
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List of Tables

1.1	TI CMOS mmWave radar devices	2
2.1	RSS_CR4_BOOT_INFO_REG0	4
2.2	RSS_CR4_BOOT_INFO_REG1 (Redundant copy of REG0)	5
2.3	RSS_CR4_BOOT_INFO_REG3	6
2.4	RSS_CR4_BOOT_INFO_REG4 (Redundant copy of REG3)	8
2.5	RSS_CR4_BOOT_INFO_REG5	11
2.6	RSS_CR4_BOOT_INFO_REG6	12
2.7	Possible SYNC values and their usage	16
2.9	MSGLEN field descriptions	18
2.10	FLAGS field description	18
2.11	NSBC field description	20
2.12	Checksum computation example	20
2.13	CRC types and their polynomials	22
4.1	Summary of common Radar messages in both BSS and MSS and their associated sub blocks	33
4.2	Summary of all Radar messages in the BSS and their associated sub blocks	33
4.3	Summary of Radar messages in the MSS of front-end devices (12xx/22xx/62xx) and their associated sub blocks	38
5.1	AWR_RESP_ERROR_SB contents	61
5.2	AWR_CHAN_CONF_SET_SB contents	62
5.3	AWR_ADCOUT_CONF_SB contents	66
5.4	AWR_LOWPOWERMODE_CONF_SET_SB contents	67
5.5	AWR_DYNAMICPOWERSAVE_CONF_SET_SB contents	68
5.6	AWR_HIGHSPEEDINTFCLK_CONF_SET_SB contents	69
5.7	AWR_RF_DEVICE_CFG_SB contents	70
5.8	AWR_RF_MISC_CTL_SB contents	74
5.9	AWR_CAL_MON_FREQUENCY_LIMITS_SB contents	79
5.10	AWR_RF_INIT_CALIBRATION_CONF_SB contents	81
5.11	AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB contents	83
5.12	AWR_CAL_DATA_RESTORE_SB contents	88
5.13	AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB contents	89
5.14	Typical APLL and Synth BW settings for xWR6x43	91
5.15	Typical APLL and Synth BW settings for AWR2243	91

5.16 Typical APLL and Synth BW settings for xWR294x, xWR254x, xWR2x4xEco and xWR2x4xP devices	92
5.17 AWR_APLL_SYNTB_BW_CONTROL_SB contents	92
5.18 AWR_CAL_DATA_SAVE_SB contents	94
5.19 AWR_CAL_DATA_SAVE_SB response packet contents	94
5.20 AWR_CAL_DATA contents for AWR2243	94
5.21 AWR_CAL_DATA contents for xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP devices	96
5.22 AWR_CAL_DATA contents for xWR6x43 devices	97
5.23 AWR_PHASE_SHIFTER_CAL_DATA_SAVE_SB contents	99
5.24 AWR_PHASE_SHIFTER_CAL_DATA_SAVE_SB response packet contents	100
5.25 AWR_RF_INIT_SB contents	102
5.26 AWR_PROFILE_CONF_SB contents	104
5.27 Note on maximum sampling rate	117
5.28 AWR_CHIRP_CONF_SET_SB contents	118
5.29 AWR_FRAME_CONF_SET_SB contents	119
5.30 AWR_CONT_STREAMING_MODE_CONF_SET_SB contents	123
5.31 AWR_CONT_STREAMING_MODE_EN_SB contents	129
5.32 AWR_ADVANCED_FRAME_CONF_SB contents	129
5.33 AWR_PERCHIRPPHASESHIFT_CONF_SB contents	141
5.34 Programmable filter DFE sampling rate and max number of taps for AWR2243/xWR6243	144
5.35 Programmable filter DFE sampling rate and max number of taps for xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP	144
5.36 AWR_PROG_FILT_COEFF_RAM_SET_SB contents	144
5.37 AWR_PROG_FILT_CONF_SET_SB contents	145
5.38 AWR_CALIB_MON_TIME_UNIT_CONF_SB contents	146
5.39 AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB contents	150
5.40 AWR_DIGITAL_COMP_EST_CONTROL_SB contents	158
5.41 AWR_RX_GAIN_TEMPLUT_SET_SB contents	162
5.42 AWR_TX_GAIN_TEMPLUT_SET_SB contents	164
5.43 AWR_LOOPBACK_BURST_CONF_SET_SB contents	168
5.44 AWR_DYN_CHIRP_CONF_SET_SB contents	177
5.45 AWR_DYN_PERCHIRP_PHASESHIFTER_CONF_SB contents	180
5.46 AWR_DYN_CHIRP_ENABLE_SB contents	181
5.47 AWR_INTERCHIRP_BLOCKCONTROLS_SB contents	182
5.48 AWR_SUBFRAME_START_CONF_SB contents	185
5.49 AWR_ADVANCE_CHIRP_CONF_SB contents	189
5.50 ADV_CHIRP_FIXED_DELTA_PARAM description	199
5.51 AWR_ADVANCE_CHIRP_GENERIC_LUT_LOAD_SB contents	203
5.52 ADV_CHIRP_GENERIC_LUT_PARAM description	204
5.53 Types of Monitors	211
5.54 Monitor Categorization	212
5.55 AWR_MONITOR_TYPE_TRIG_CONF_SB contents	213
5.56 AWR_ADVANCE_CHIRP_DYN_LUT_ADDR_OFFSET_CFG_SB contents	214

5.57	AWR_ADV_TX_GAIN_TEMPLUT_SET_SB contents	216
5.58	AWR_ADV_DYN_PERCHIRP_PHASESHIFTER_CONF_SB contents	217
5.59	AWR_PROFILE_CONF_GET_SB contents	219
5.60	AWR_CHIRP_CONF_GET_SB contents	220
5.61	AWR_FRAME_CONF_GET_SB contents	220
5.62	AWR_ADV_FRAME_CONF_GET_SB contents	220
5.63	AWR_RX_GAIN_TEMPLUT_GET_SB contents	221
5.64	AWR_TX_GAIN_TEMPLUT_GET_SB contents	222
5.65	AWR_ADV_TX_GAIN_TEMPLUT_GET_SB contents	222
5.66	AWR_FRAMESTARTSTOP_CONF_SB contents	223
5.67	AWR_BPM_COMMON_CONF_SET_SB contents	224
5.68	AWR_BPM_CHIRP_CONF_SET_SB contents	225
5.69	AWR_POWER_SAVE_MODE_CONF_SET_SB contents	226
5.70	AWR_RF_VERSION_GET_SB contents	227
5.71	AWR_RF_VERSION_SB response contents	227
5.72	AWR_RF_CPUFAULT_STATUS_GET_SB contents	228
5.73	AWR_RF_CPUFAULT_STATUS_GET_SB response contents	229
5.74	AWR_RF_ESMFAULT_STATUS_GET_SB contents	230
5.75	AWR_RF_ESMFAULT_STATUS_SB response contents	231
5.76	AWR_RF_DIEID_GET_SB contents	233
5.77	AWR_RF_DIEID_STATUS_SB response contents	233
5.78	AWR_RF_BOOTUPBIST_STATUS_GET_SB contents	233
5.79	AWR_RF_BOOTUPBIST_STATUS_DATA_SB response contents	234
5.80	AWR_RF_ADV_ESMFAULT_STATUS_GET_SB contents	236
5.81	AWR_RF_ADV_ESMFAULT_STATUS_SB response contents	236
5.82	AWR_RF_DFE_STATISTICS_REPORT_GET_SB contents	240
5.83	AWR_RF_DFE_STATISTICS_REPORT_SB response contents	240
5.84	AWR_RF_REAL_CHAN_DFE_STATISTICS_REPORT_GET_SB contents	250
5.85	AWR_RF_REAL_CHAN_DFE_STATISTICS_REPORT_SB response contents	251
5.86	AWR_RF_TEST_SOURCE_CONFIG_SET_SB contents	257
5.87	AWR_RF_TEST_SOURCE_ENABLE_SET_SB contents	259
5.88	AWR_RF_LDO_BYPASS_SB contents	260
5.89	AWR_RF_PALOOPBACK_CFG_SB contents for AWR2243/xWR6243 devices	262
5.90	AWR_RF_PALOOPBACK_CFG_SB contents for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices	263
5.91	AWR_RF_PSLOOPBACK_CFG_SB contents	264
5.92	AWR_RF_IFLOOPBACK_CFG_SB contents	266
5.93	AWR_RF_GPADC_CFG_SET_SB contents	267
5.94	AWR_RF_LOLOOPBACK_CFG_SB contents	270
5.95	AWR_RF_TEMPERATURE_GET_SB contents	271
5.96	AWR_RF_TEMPERATURE_DATA_SB contents	271
5.97	AWR_AE_RF_CPUFAULT_SB response contents	273
5.98	AWR_AE_RF_ESMFAULT_STATUS_SB response contents	276
5.99	AWR_AE_RF_INITCALIBSTATUS_SB response contents	278

5.100AWR_AE_RF_ADV_ESMFAULT_STATUS_SB response contents	280
5.101AWR_AE_RF_MONITOR_TYPE_TRIGGER_DONE_SB response contents	284
5.102AWR_AE_RF_FRAME_TRIGGER_RDY_SB response contents	285
5.103AWR_AE_RF_GPADC_RESULT_DATA_SB response contents	285
5.104AWR_FRAME_END_AE_SB response contents	287
5.105AWR_ANALOGFAULT_AE_SB response contents	287
5.106AWR_CAL_MON_TIMING_FAIL_REPORT_AE_SB response contents	288
5.107AWR_RUN_TIME_CALIB_SYMMARY_REPORT_AE_SB response contents	290
5.108AWR_MONITOR_RF_DIG_LATENTFAULT_REPORT_AE_SB response contents	292
5.109AWR_MONITORING_REPORT_HEADER_AE_SB response contents	294
5.110AWR_MONITOR_RF_DIG_PERIODIC_REPORT_AE_SB contents	294
5.111AWR_MONITORING_TEMPERATURE_REPORT_AE_SB contents	295
5.112AWR_MONITOR_RX_GAIN_PHASE_REPORT_AE_SB contents	297
5.113AWR_MONITOR_RX_NOISE_FIGURE_REPORT_AE_SB contents	301
5.114AWR_MONITOR_RX_IFSTAGE_REPORT_AE_SB contents	302
5.115AWR_MONITOR_TX0_POWER_REPORT_AE_SB contents	306
5.116AWR_MONITOR_TX1_POWER_REPORT_AE_SB contents	307
5.117AWR_MONITOR_TX2_POWER_REPORT_AE_SB contents	308
5.118AWR_MONITOR_TX0_BALLBREAK_REPORT_AE_SB contents	309
5.119AWR_MONITOR_TX1_BALLBREAK_REPORT_AE_SB contents	309
5.120AWR_MONITOR_TX2_BALLBREAK_REPORT_AE_SB contents	310
5.121AWR_MONITOR_TX_GAIN_PHASE_REPORT_AE_SB contents	311
5.122AWR_MONITOR_TX0_PHASE_SHIFTER_REPORT_AE_SB contents	314
5.123AWR_MONITOR_TX1_PHASE_SHIFTER_REPORT_AE_SB contents	316
5.124AWR_MONITOR_TX2_PHASE_SHIFTER_REPORT_AE_SB contents	318
5.125AWR_MONITOR_SYNTH_FREQUENCY_REPORT_AE_SB contents	320
5.126AWR_MONITOR_EXTERNAL_ANALOG_SIGNALS_REPORT_AE_SB contents	322
5.127AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB con- tents	323
5.128AWR_MONITOR_TX1_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB con- tents	324
5.129AWR_MONITOR_TX2_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB con- tents	325
5.130AWR_MONITOR_RX_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB contents	326
5.131AWR_MONITOR_PM_CLK_LO_INTERNAL_ANALOG_SIGNALS_REPORT_AE_ SB contents	327
5.132AWR_MONITOR_GPADC_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB con- tents	329
5.133AWR_MONITOR_PLL_CONTROL_VOLTAGE_REPORT_AE_SB contents	330
5.134AWR_MONITOR_DUAL_CLOCK_COMP_REPORT_AE_SB contents	332
5.135AWR_MONITOR_RX_MIXER_IN_POWER_REPORT_AE_SB contents	334
5.136AWR_MONITOR_SYNTHESIZER_FREQUENCY_NONLIVE_REPORT_AE_SB con- tents	335
5.137AWR_MONITOR_TX3_POWER_REPORT_AE_SB contents	337

5.138AWR_MONITOR_TX3_BALLBREAK_REPORT_AE_SB contents	338
5.139AWR_MONITOR_TX3_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB con- tents	339
5.140AWR_MONITOR_TX3_PHASE_SHIFTER_REPORT_AE_SB contents	340
5.141AWR_MONITOR_ADV_TX_GAIN_PHASE_REPORT_AE_SB contents	342
5.142AWR_MONITOR_TX_PHSHIFTER_DAC_REPORT_AE_SB contents	345
5.143AWR_DEV_POWERUP_SB contents	346
5.144AWR_DEV_MCUCLOCK_CONF_SET_SB contents	347
5.145AWR_DEV_RX_DATA_FORMAT_CONF_SB contents	348
5.146AWR_DEV_RX_DATA_PATH_CONF_SB contents	349
5.147AWR_DEV_RX_DATA_PATH_LANEEN_SET_SB contents	352
5.148AWR_DEV_RX_DATA_PATH_CLK_SET_SB contents	353
5.149AWR_DEV_LVDS_CFG_SET_SB contents	354
5.150AWR_DEV_RX_CONTSTREAMING_MODE_CFG_SET_SB contents	357
5.151AWR_DEV_CSI2_CFG_SET_SB contents	357
5.152AWR_DEV_PMICCLOCK_CONF_SET_SB contents	359
5.153PMIC clock frequency across chirps in chirp-to-chirp staircase mode in an exam- ple when PMIC clock varies from 2 MHz to 2.5 MHz in 32 chirps	363
5.154AWR_MSS_PERIODICTESTS_CONF_SB contents	364
5.155AWR_MSS_LATENTFAULT_TEST_CONF_SB contents	365
5.156AWR_DEV_TESTPATTERN_GEN_SET_SB contents	368
5.157AWR_DEV_CONFIGURATION_SET_SB contents	370
5.158AWR_DEV_RF_DEBUG_SIG_SET_SB contents	371
5.159AWR_DEV_DEV_HSI_DELAY_DUMMY_CFG_SET_SB contents	372
5.160AWR_DEV_MCUCLOCK_GET_SB contents	373
5.161AWR_DEV_RX_DATA_FORMAT_CONF_GET_SB contents	374
5.162AWR_DEV_RX_DATA_PATH_CONF_GET_SB contents	374
5.163AWR_DEV_RX_DATA_PATH_LANEEN_GET_SB contents	374
5.164AWR_DEV_RX_DATA_PATH_CLK_GET_SB contents	374
5.165AWR_DEV_LVDS_CFG_GET_SB contents	375
5.166AWR_DEV_RX_CONTSTREAMING_CONF_GET_SB contents	375
5.167AWR_DEV_CSI2_CFG_GET_SB contents	375
5.168AWR_DEV_PMICCLOCK_CONF_GET_SB contents	376
5.169AWR_MSS_LATENTFAULT_CONF_GET_SB contents	376
5.170AWR_MSS_PERIODICTESTS_CONF_GET_SB contents	376
5.171AWR_DEV_TESTPATTERN_GEN_GET_SB contents	376
5.172AWR_DEV_FILE_DOWNLOAD_SB contents	377
5.173AWR_DEV_FRAME_CONFIG_APPLY_SB contents	378
5.174AWR_DEV_ADV_FRAME_CONFIG_APPLY_SB contents	378
5.175AWR_MSSVERSION_GET_SB contents	380
5.176AWR_MSSVERSION_SB contents	381
5.177AWR_MSSVERSION_SB contents	382
5.178AWR_MSSCPUFAULT_STATUS_SB contents	382
5.179AWR_MSSSESMFAULT_STATUS_GET_SB contents	384

5.180	AWR_MSESFAULT_STATUS_SB contents	384
5.181	AWR_AE_DEV_MSSPOWERUPDONE_SB contents	387
5.182	AWR_AE_DEV_RFPOWERUPDONE_SB contents	389
5.183	AWR_AE_MSS_CPUFAULT_STATUS_SB contents	391
5.184	AWR_AE_MSS_ESMFAULT_STATUS_SB contents	393
5.185	AWR_AE_MSS_BOOTERRORSTATUS_SB contents	396
5.186	AWR_AE_MSS_LATENTFAULT_TESTREPORT_SB contents	398
5.187	AWR_AE_MSS_PERIODICTEST_STATUS_SB contents	400
5.188	AWR_AE_MSS_RFERROR_STATUS_SB contents	400
6.1	Sequence of APIs to be issued to master and slave devices in cascaded mode configuration for FMCW mode measurements	408
6.2	Sequence of APIs to be issued to master and slave devices in cascaded mode configuration for FMCW mode measurements	411
7.1	BSS API error codes	414
7.2	MSS API error codes (Applicable only in AWR1243/AWR2243)	423
7.3	Bit field describing the error status during boot on SPI	426
8.1	AWR_MONITOR_RF_DIG_LATENTFAULT_CONF_SB contents	429
8.2	AWR_MONITOR_RF_DIG_PERIODIC_CONF_SB contents	431
8.3	AWR_MONITOR_ANALOG_ENABLES_CONF_SB contents	432
8.4	AWR_MONITOR_TEMPERATURE_CONF_SB contents	436
8.5	AWR_MONITOR_RX_GAIN_PHASE_CONF_SB contents	438
8.6	AWR_MONITOR_RX_NOISE_FIGURE_CONF_SB contents	443
8.7	AWR_MONITOR_RX_IFSTAGE_CONF_SB contents	444
8.8	AWR_MONITOR_TX0_POWER_CONF_SB contents	446
8.9	AWR_MONITOR_TX1_POWER_CONF_SB contents	448
8.10	AWR_MONITOR_TX2_POWER_CONF_SB contents	450
8.11	AWR_MONITOR_TX0_BALLBREAK_CONF_SB contents	452
8.12	AWR_MONITOR_TX1_BALLBREAK_CONF_SB contents	453
8.13	AWR_MONITOR_TX2_BALLBREAK_CONF_SB contents	454
8.14	AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_CONF_SB contents	456
8.15	AWR_MONITOR_TX0_PHASE_SHIFTER_CONF_SB contents	460
8.16	AWR_MONITOR_TX1_PHASE_SHIFTER_CONF_SB contents	464
8.17	AWR_MONITOR_TX2_PHASE_SHIFTER_CONF_SB contents	467
8.18	AWR_MONITOR_SYNTHESIZER_FREQUENCY_CONF_SB contents	472
8.19	AWR_MONITOR_EXTERNAL_ANALOG_SIGNALS_CONF_SB contents	475
8.20	AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_CONF_SB contents	478
8.21	AWR_MONITOR_TX1_INTERNAL_ANALOG_SIGNALS_CONF_SB contents	479
8.22	AWR_MONITOR_TX2_INTERNAL_ANALOG_SIGNALS_CONF_SB contents	480
8.23	AWR_MONITOR_RX_INTERNAL_ANALOG_SIGNALS_CONF_SB contents	481
8.24	AWR_MONITOR_PMCLKLO_INTERNAL_ANALOG_SIGNALS_CONF_SB contents	482
8.25	AWR_MONITOR_GPADC_INTERNAL_ANALOG_SIGNALS_CONF_SB contents	483

8.26	AWR_MONITOR_PLL_CONTROL_VOLTAGE_CONF_SB contents	484
8.27	DCC Clock monitor pairs	486
8.28	AWR_MONITOR_DUAL_CLOCK_COMP_CONF_SB contents	487
8.29	AWR_MONITOR_RX_SATURATION_DETECTOR_CONF_SB contents	489
8.30	AWR_MONITOR_RX_SIG_IMG_MONITOR_CONF_SB contents	491
8.31	AWR_MONITOR_MIXER_IN_POWER_CONF_SB contents	492
8.32	AWR_ANALOG_FAULT_INJECTION_CONF_SB contents	494
8.33	AWR_MONITOR_TXN_POWER_CONF_SB contents	503
8.34	AWR_MONITOR_TXN_BALLBREAK_CONF_SB contents	505
8.35	AWR_MONITOR_TXN_INTERNAL_ANALOG_SIGNALS_CONF_SB contents	507
8.36	AWR_MONITOR_TXN_PHASE_SHIFTER_CONF_SB contents	508
8.37	AWR_MONITOR_ADV_TX_GAIN_PHASE_MISMATCH_CONF_SB contents	513
8.38	AWR_MONITOR_VMON_CONF_SB contents	516
8.39	AWR_MONITOR_OVERRIDES_AND_DITHER_CONF_SB contents	519
8.40	AWR_MONITOR_TX_PHSHIFTER_DAC_CONF_SB contents	524
11.1	Minimum chirp cycle time for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices	536
11.2	Minimum inter burst time for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices	539
11.3	Minimum inter burst time for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices	540
11.4	Minimum inter sub-frame, frame time for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices	542
12.1	Duration of boot time calibrations for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP device	543
12.2	Duration of boot time calibrations for AWR2243 device	544
12.3	Duration of boot time calibrations for xWR6243 device	544
12.4	Duration of run time calibrations for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices	545
12.5	Duration of run time calibrations for AWR2243 devices	546
12.6	Duration of run time calibrations for xWR6243 devices	546
12.7	Duration of analog monitors for xWR254x, xWR2x4xECO, xWR2x4xP device	548
12.8	Duration of analog monitors for xWR294x device	549
12.9	Duration of digital monitors for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP device	550
12.10	Duration of analog monitors for AWR2243 device	551
12.11	Duration of digital monitors for AWR2243 device	551
12.12	Duration of analog monitors for xWR6243 device	552
12.13	Duration of digital monitors for xWR6243 device	552
12.14	Software overheads every FTTI that should be accounted to program CALIB_MON_TIME_UNIT and CALIBRATION_PERIODICITY	553

Revision History

Revision	Date	Description
2.0 (AWR2243)	19.08.2019	1. Base-lined from AWR1243 DFP 1.2.5 release (mmWave Radar Interface Control Document v1.7) 2. Added new parameter MISC_FUNC_CTRL to disable dither in test source configuration API at 256. 3. Added new parameter to enable VMON in analog monitor configuration API at 431. 4. Added new error code 160 in 413. 5. Added new TX Phase shifter DAC monitor settings and reports in TX internal analog signal monitors at 478 and 323. 6. updated RX gain temperature LUT RF gain code description at 161. 7. updated OSCCLKOUT_DIS description in channel config API 62.

Revision	Date	Description
2.1	10.09.2019	- Added a new device AWR2243 and its features in ICD. - Added new feature INTFRC_MASTER_EN bit in CASCADING_PINOUTCFG field in channel config API in page 62. - Added a new API AWR_APLL_SYNTH_BW_CONTROL_SB in page 91. - Added a new field LODIST_BIAS_CODE in API AWR_CAL_DATA_SAVE_SB in page 93. - Updated AWR_PROFILE_CONF_SET_SB API to include AWR2243 features in page 103. - Added a new feature MONITORING_MODE in AWR_CALIB_MON_TIME_UNIT_CONF_SB in page 146. - Added a new feature CAL_TEMP_INDEX_OVERRIDE_ENABLE in AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB in page 149. - Added a new API AWR_ADVANCE_CHIRP_CONF_SB in page 186. - Added a new API AWR_MONITOR_TYPE_TRIG_CONF_SB in page 211. - Added a new bootup and latent fault DCC monitor feature in page 233, 389, 429. - Added a new feature MONITOR_CONFIG_MODE in AWR_MONITOR_SYNTHESIZER_FREQUENCY_CONF_SB in page 471. - Added a new Async Event AWR_AE_RF_MONITOR_TYPE_TRIGGER_DONE_SB in page 284. - Added a new Async Event AWR_MONITOR_SYNTHESIZER_FREQUENCY_NONLIVE_REPORT_AE_SB in page 335. - Added a new parameter DIS_LINE_START_END in AWR_DEV_CSI2_CFG_SET_SB in page 357. - Added a new API AWR_DEV_RF_DEBUG_SIG_SET_SB in page 371. - Added a new API AWR_DEV_DEV_HSI_DELAY_DUMMY_CFG_SET_SB in page 372.

Revision	Date	Description
2.2	23.09.2019	- Added a new RX FE disable option in LOOPBACK_SEL in AWR_LOOPBACK_BURST_CONF_SET_SB API in page 167 - Updated LPF monitoring threshold and reporting values in AWR_MONITOR_RX_IFSTAGE_CONF_SB API and AWR_MONITOR_RX_IFSTAGE_REPORT_AE_SB AE in page 444 and 302 - Changed name of AWR_INTER_RX_GAIN_PHASE_CONTROL_SB to AWR_DIGITAL_COMP_EST_CONTROL_SB API and updated the fields in page 157 - Removed AWR_AE_MSS_VMON_ERRORSTATUS_SB Subblock (0x5009) - Removed VCO slope monitoring in AWR_MONITOR_PLL_CONTROL_VOLTAGE_REPORT_AE_SB for all devices 330 - Added a new enable control SYNTH_FREQ_MONITOR_NON_LIVE for non-live synth frequency monitor in AWR_MONITOR_ANALOG_ENABLES_CONF_SB in page 431 - Updated calibration and monitoring duration for AWR2243 device in page 543 - The TXOFF BPM control bits made reserved in AWR_BPM_CHIRP_CONF_SET_SB API in page 224

Revision	Date	Description
2.3	18.10.2019	- Updated Calibration structure definition for AWR2243 and IWR6843 devices in page 94 and 97 - Updated few API descriptions and added new error codes for Synth frequency monitor.

Revision	Date	Description
2.4	27.11.2019	- Updated AWR_ADVANCE_CHIRP_CONF_SB API definition to support more flexible waveform generation in page 186 - Added new AWR_ADVANCE_CHIRP_GENERIC_LUT_LOAD_SB API to load Advance chirp SW generic LUT data in page 202 - Added new Advance chirp enable flag in AWR_RF_RADAR_MISC_CTL_SB API in page 73 - Updated the definition of CHIRP_START_INDX, CHIRP_END_INDX and NUM_LOOPS in AWR_FRAME_CONF_SET_SB and AWR_ADVANCED_FRAME_CONF_SB APIs for new Advance chirp config API in page 119 and 129 - Added new noise power report fields in AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_REPORT_AE_SB AE in page 311 - Updated few API descriptions and added new error codes for new advanced chirp configuration APIs. - Added new frame stop features in AWR_FRAMESTARTSTOP_CONF_SB API in page 222 - Added new notes in AWR_CAL_MON_FREQUENCY_LIMITS_SB and AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB APIs in page 83 - Added new LODIST calibration data variable in AWR2243 Calibration structure in page 94 - Added new notes related to minimum chirp cycle time in AWR_PROFILE_CONF_SET_SB API in page 103 - Updated notes related to minimum burst time in AWR_FRAME_CONF_SET_SB and AWR_ADVANCED_FRAME_CONF_SB APIs in page 119 and 129 - Added new error codes in AWR_AE_RF_CPUFAULT_SB API in page 273 - Added dither feature in AWR_DEV_DEV_HSI_DELAY_DUMMY_CFG_SET_SB API in page 372 - Updated API programming sequence section in page 402 - Updated Table 5.15 in AWR_APLL_SYNTH_BW_CONTROL_SB API. - Updated RF gain settings and description in AWR_PROFILE_CONF_SET_SB in page 103 - Updated RF gain phase monitor report description in AWR_MONITOR_RX_GAIN_PHASE_REPORT_AE_SB in page 297 - Added new fault types in AWR_AE_MSS_CPUFAULT_SB API in page 391

Revision	Date	Description
2.5	19.12.2019	1. Added new phase shifter monitoring APIs AWR_MONITOR_TXn_PHASE_SHIFTER_CONF_SB for 3 TX in 460 which replaces legacy AWR_MONITOR_TXn_BPM_CONF_SB APIs and corresponding AE reports AWR_MONITOR_TXn_PHASE_SHIFTER_REPORT_AE_SB updated in page 314 2. Added new BSS_ANA_CTRL field in AWR_RF_DEVICE_CFG_SB API to disable inter burst power save in page 70 3. Added new field MON_CHIRP_SLOPE field in AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_CONF_SB API in page 455 4. Added new section to provide details about Chirp, Burst and Frame timings of device in page 536 5. Updated few API descriptions and added new error codes for phase shifter monitor.

Revision	Date	Description
2.6	02.01.2020	1. Updated RX_GAIN_VALUE report description in AWR_MONITOR_RX_GAIN_PHASE_REPORT_AE_SB AE in page 297 2. Added Max programmable VCO slop info in AWR_APLL_SYNTH_BW_CONTROL_SB API in page 91 3. Updated the ICD based on review comments

Revision	Date	Description
2.7	28.01.2020	1. Updated the ICD based on review comments 2. Added recommended value for MONITOR_START_TIME in AWR_MONITOR_SYNTHESIZER_FREQUENCY_CONF_SB API in page 471 3. Updated RX gain valid range in AWR_PROFILE_CONF_SET_SB in page 103 4. Removed RAMPGEN_100M clock monitoring feature from AWR_MONITOR_DUAL_CLOCK_COMP_CONF_SB API in page 487 and in AE report AWR_MONITOR_DUAL_CLOCK_COMP_REPORT_AE_SB in page 332 5. Updated few status flags in AWR_AE_DEV_MSSPOWERUPDONE_SB API in page 387 6. Updated few status flags in AWR_MSSCPUFAULT_STATUS_GET_SB API in page 382 7. Updated few status flags in AWR_AE_MSS_CPUFAULT_SB API in page 391 8. Updated few status flags in AWR_AE_MSS_BOOTERRORSTATUS_SB API in page 396 9. Updated few status flags in AWR_AE_MSS_ESMFAULT_STATUS_SB API in page 392 10. Updated few status flags in AWR_MSS_LATENTFAULT_TEST_CONF_SB API in page 365 11. Removed 900 Mbps (DDR only) data rate in AWR_DEV_RX_DATA_PATH_CLK_SET_SB API in page 353 12. Added new file type in META_IMAGE_TO_SRAM in AWR_DEV_FILE_DOWNLOAD_SB API in page 377 13. Updated Programmable filter description for AWR2243 in AWR_PROG_FILT_COEFF_RAM_SET_SB API in page 143

Revision	Date	Description
2.8	12.02.2020	- Updated MAX_TX_PHASE_SHIFTER_INTERNAL_DITHER max range in AWR_ADVANCE_CHIRP_CONF_SB API in page 186 - Updated timing of Dual clock comparator based clock monitor in Table 12.10 - Disabled DFE parity test and updated status bits in AWR_RF_BOOTUPBIST_STATUS_DATA_SB AE, AWR_MONITOR_RF_DIG_LATENTFAULT_REPORT_AE_SB AE, AWR_AE_DEV_RFPOWERUPDONE_SB AE, AWR_MONITOR_RF_DIG_LATENTFAULT_CONF_SB API, AWR_RF_ESMFAULT_STATUS_SB API and AWR_AE_RF_ESMFAULT_SB AE. - Updated TX phase shifter apply timing info note in AWR_PROFILE_CONF_SET_SB and AWR_PERCHIRPPHASESHIFT_CONF_SB API in page 103 and 141 respectively. - Added application care about notes in Communication Sequence section. - Added new error code 318

Revision	Date	Description
2.9	21.02.2020	- Added a new field ADVANCE_CHIRP_ERROR_CHK_DIS option in AWR_RF_RADAR_MISC_CTL_SB API in page 73 - Updated Synth calibration duration in page 543 - Added a new API AWR_ADVANCE_CHIRP_DYN_LUT_ADDR_OFFSET_CFG_SB in page 213

Revision	Date	Description
2.10	23.06.2020	- Updated Phase value mapping in AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB and AWR_PHASE_SHIFTER_CAL_DATA_SAVE_SB in page 99 - Addressed few review comments - Added a new field CAL_MON_TIME_UNIT_ERROR_CHK_DIS option in AWR_RF_RADAR_MISC_CTL_SB API in page 73 - Updated description in AWR_LOOPBACK_BURST_CONF_SET_SB API in page 167 - Added TX Phase shifter DAC monitor disable option in TXn internal analog signal monitors in page 478 - Updated calibration and monitor timing info as per measurement on AWR2243 ES1.1 device in page 543 - Changed the name of API from AWR_DEV_DEV_CSI2_DELAY_DUMMY_CFG_SET_SB to AWR_DEV_DEV_HSI_DELAY_DUMMY_CFG_SET_SB in page 372. - Added a new mode CQ_CP_ADC in AWR_DEV_RX_DATA_PATH_CONF_SET_SB API in page 349

Revision	Date	Description
2.11	07.07.2020	- Addressed few review comments - Updated API programming sequence section in page for cascade mode and added a new application care abouts notes 402 - Updated API field descriptions AWR_CALIB_MON_TIME_UNIT_CONF_SB in page 146 for cascade operation. - Changed field name from 'WDT_DISABLE' to 'WDT_ENABLE' in AWR_RF_DEVICE_CFG_SB API in page 70

Revision	Date	Description
2.12	28.07.2020	- Updated Max clock out frequency limit in AWR_DEV_MCUCLOCK_CONF_SET_SB and AWR_DEV_PMICCLOCK_CONF_SET_SB APIs. - Updated API field descriptions AWR_CALIB_MON_TIME_UNIT_CONF_SB in page 146 for cascade operation. - Updated max sampling rate information in Table 5.27 - Added new API Error handling section in page 413 - Added a new MSS logger disable bit field in AWR_DEV_CONFIGURATION_SET_SB API in page 370 - Updated continuous framing mode chirp and inter-burst timing -chap:chirpburstTimings

Revision	Date	Description
2.13	28.08.2020	1. Address few minor review comments. 2. Added an option to disable Digital temperature sensor disable option in AWR_MONITOR_TEMPERATURE_CONF_SB API in page 436 3. Updated max sampling rate information in Table 5.27 for low power ADC mode. 4. Updated 20G SYNC monitor conversion factors note in AWR_MONITOR_PMCLKLO_INTERNAL_ANALOG_SIGNALS_CONF_SB in page 481 5. Updated notes related to LOOPBACK_POWER and RX_GAN_VALUE in AWR_MONITOR_RX_GAIN_PHASE_REPORT_AE_SB AE in page 297

Revision	Date	Description
2.14	28.11.2020	- Updated document for xWR6243 devices - Updated Phase value mapping in OBS_PHSIFT_DATA field of AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB in page 99 - Updated Phase value mapping in OBS_PHSIFT_DATA field of AWR_PHASE_SHIFTER_CAL_DATA_SAVE_SB in page 100 - Updated AWR_APLL_SYNTH_BW_CONTROL_SB fields and description for 60GHz devices in page 91 - Added a note on boot calibrations information in non-ISM band in page 102 - Updated AWR_PROFILE_CONF_SET_SB field description for 60GHz devices in page 103 - Updated the fields CHIRP_FREQ_START_VAR, CHIRP_FREQ_SLOPE_VAR for 60GHz devices in AWR_CHIRP_CONF_SET_SB in page 117 - Updated the field descriptions of AWR_CONT_STREAMING_MODE_CONF_SET_SB for 60GHz devices in page 123 - Updated the field descriptions of AWR_LOOPBACK_BURST_CONF_SET_SB for 60GHz devices in page 167 - Updated the field descriptions of ADV_CHIRP_FIXED_DELTA_PARAM for 60GHz devices in page 199 - Updated the field descriptions of ADV_CHIRP_GENERIC_LUT_PARAM for 60GHz devices in page 204 - Added a new power saving power save API AWR_POWER_SAVE_MODE_CONF_SET_SB in page 225 - Updated DCC test description of AWR_RF_BOOTUPBIST_STATUS_DATA_SB in page 234 - Updated description of AWR_RF_GPADC_CFG_SET_SB in page 266 - Changed PROG_FILT_FATAL_PARITY_ERROR to RESERVED in AWR_AE_RF_ESMFAULT_STATUS_SB in page 276 - Updated description for AWR_MONITOR_RF_DIG_LATENTFAULT_REPORT_AE_SB in page 292 - Update the note in AWR_MONITOR_TX0_POWER_REPORT_AE_SB for 60GHz devices in page 305 - Updated MON_CHIRP_SLOPE field description of AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_CONF_SB, AWR_MONITOR_TX0_PHASE_SHIFTER_CONF_SB, AWR_MONITOR_TX1_PHASE_SHIFTER_CONF_SB, AWR_MONITOR_TX2_PHASE_SHIFTER_CONF_SB in page 455

Revision	Date	Description
2.15	01.12.2020	1. Updated PF_VCO_SELECT field of AWR_PROFILE_CONF_SB in page 103 2. Updated VCO_SELECT field of AWR_CONT_STREAMING_MODE_CONF_SET_SB in page 123 3. Updated CALIBRATION_STATUS and CALIBRATION_UPDATE fields of AWR_AE_RF_INITCALIBSTATUS_SB in page 278 4. Updated CALIBRATION_ERROR_FLAG and CALIBRATION_UPDATE_STATUS fields of AWR_RUN_TIME_CALIB_SYMMARY_REPORT_AE_SB in page 290 5. Updated STATUS_FLAGS and PLL_CONTROL_VOLTAGE_VALUES fields of AWR_MONITOR_PLL_CONTROL_VOLTAGE_REPORT_AE_SB in page 330 6. Updated SIGNAL_ENABLES field of AWR_MONITOR_PLL_CONTROL_VOLTAGE_CONF_SB in page 484 7. Updated CAL_VALIDITY_STATUS field of AWR_CAL_DATA_SAVE_SB in page 94 8. Added TX_POWER_OFFSET_VALUE field in AWR_MONITOR_TX0_POWER_CONF_SB, AWR_MONITOR_TX1_POWER_CONF_SB, AWR_MONITOR_TX2_POWER_CONF_SB in page 446 9. Added MON_START_FREQ_CONST and TX_POWER_BACKOFF field in AWR_MONITOR_TX0_BALLBREAK_CONF_SB, AWR_MONITOR_TX1_BALLBREAK_CONF_SB, AWR_MONITOR_TX2_BALLBREAK_CONF_SB in page 452

Revision	Date	Description
2.16	27.01.2021	- Updated DIGITAL_TX_FREQ_SHIFT field of AWR_DIGITAL_COMP_EST_CONTROL_SB in page 157 - Added xWR6243 API changes in page vi - Added API Error codes 319, 320 and 325 in page 413 - Updated NOTE in RF_INIT_CALIB_ENABLE_MASK field of AWR_RF_INIT_CALIBRATION_CONF_SB in page 81 - Updated field descriptions of AWR_PROFILE_CONF_SB in page 103 - Updated field descriptions of AWR_CONT_STREAMING_MODE_CONF_SET_SB in page 123 - Updated NOTE1 in AWR_FRAME_CONF_SET_SB in page 119 - Updated descriptions of AWR_DIGITAL_COMP_EST_CONTROL_SB in page 157 - Updated description of AWR_LOOPBACK_BURST_CONF_SET_SB in page 167 - Updated description of AWR_MONITOR_TYPE_TRIG_CONF_SB in page 211 - Updated description of AWR_MONITOR_RX_GAIN_PHASE_REPORT_AE_SB in page 297 - Updated description of AWR_MONITOR_PMCLKLO_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB in page 327 - Updated description of AWR_DEV_RX_DATA_PATH_LANEEN_SET_SB in page 352 - Updated LANE_POS_POL_SEL field of AWR_DEV_CSI2_CFG_SET_SB in page 357 - Updated MIN_NDIV_VAL and MAX_NDIV_VAL field description of AWR_DEV_PMICCLOCK_CONF_SET_SB in page 359 - Updated TX_GAIN_MISMATCH_THRESH field description of AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_CONF_SB in page 455

Revision	Date	Description
2.17	04.03.2021	- Updated NOTE in AWR_MONITOR_RX_GAIN_PHASE_REPORT_AE_SB in page 297 - Updated NOTE for TX_PHASE_VALUE in AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_REPORT_AE_SB AE in page 311 - Updated FREQ_LIMIT_LOW and FREQ_LIMIT_HIGH field description in AWR_CAL_MON_FREQUENCY_LIMITS_SB in 79 - Updated description for AWR_DEV_TESTPATTERN_GEN_SET_SB in 368 - Updated Calibration and Monitoring timings in 544 - Updated NOTE in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB in page 83 - Updated description for AWR_DIGITAL_COMP_EST_CONTROL_SB in page 157

Revision	Date	Description
2.18	12.03.2021	- Added RF1_RF2_FREQ_DITHER_LIMITS, RF3_FREQ_DITHER_LIMITS fields and updated RF_FREQ_BITMASK in AWR_MONITOR_RX_GAIN_PHASE_REPORT_AE_SB in page 438 - Updated APLL_RZ_TRIM_VCO field in AWR_APLL_SYNTH_BW_CONTROL_SB in page 91

Revision	Date	Description
2.19	31.03.2021	First set of API changes for xWR294x devices.

Revision	Date	Description
3.01	21.04.2021	- Rebased the ICD to the final AWR2243 DFP 2.2.3 maintenance release version. - Second set of API changes for xWR294x devices.

Revision	Date	Description
3.02	05.05.2021	- Added a bit-field to enable/disable miscellaneous profile specific features in AWR_PROFILE_CONF_SB in 103 and AWR_CONT_STREAMING_MODE_CONF_SET_SB in 123. - Updated the valid values for RX_GAIN and the new RF gain targets in AWR_PROFILE_CONF_SB, AWR_CONT_STREAMING_MODE_CONF_SET_SB and AWR_LOOPBACK_BURST_CONF_SET_SB API. - Added field for TX3 phaseshifter value in AWR_PERCHIRPPHASESHIFT_CONF_SB in 141. - Updated the MAX DFE sampling rate, real-only operating mode and the number of taps table in AWR_PROG_FILTER_COEFF_RAM_SET_SB for xWR294x devices. - Marked PROG_FILTER_FREQ_SHIFT_FACTOR field in AWR_PROG_FILTER_CONF_SET_SB as not applicable for xWR294x devices. - Added a field to enable Runtime TX Phaseshifter calibrations in AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB in 149. - Added the status field for Runtime TX phaseshifter calibration in AWR_RUN_TIME_CALIB_SUMMARY_REPORT_AE_SB at ??. - Updated loopback enable field to replace PS LB with LO LB, RX gain range and RF gain targets, added enable bit and BPM config field for TX3, LO and PA LB gain selection fields, LO and PA LB frequencies in AWR_LOOPBACK_BURST_CONF_SET_SB. - Added TX3 enable bit in AWR_DYN_CHIRP_CONF_SET_SB. - Updated the notes for the decomposition of the RX GAIN code in AWR_RX_GAIN_TEMPLUT_SET_SB into the IFA and RF gains in xWR294x. - Added a field to program the TX3 BPM value in AWR_BPM_CHIRP_CONF_SET_SB. - Updated all Q channel and IQ correction report fields in AWR_RF_DFE_STATISTICS_REPORT_GET_SB as not applicable for REAL mode and xWR294x devices. - Updated the AWR_RF_PALOOPBACK_CFG_SB API for xWR294x devices. Modified PA loopback frequency selection, QPSK and PA loopback buffer gain selection fields. - De-featured the AWR_RF_PSLOOPBACK_CFG_SB API for xWR294x devices. - Added a new API AWR_RF_LOLOOPBACK_CFG_SB API for enabling LO loopback on xWR294x devices. - De-featured the RFLDO_BYPASS_EN field in AWR_RF_LDO_BYPASS_SB for xWR294x.

Revision	Date	Description
3.03	14.05.2021	- Added TX3 fields in LUT_ADDRESS_OFFSET and ADDRESS_MASK_EN of AWR_ADVANCE_CHIRP_DYN_LUT_ADDR_OFFSET_CFG_SB in 213. - Added TX3 fields for CHIRP_TX_EN, CHIRP_BPM_VAL and TX3_PHASE_SHIFTER in ADV_CHIRP_GENERIC_LUT_PARAM table in 204. - Added TX3_PHASE_SHIFTER field in ADV_CHIRP_FIXED_DELTA_PARAM table in 199. - Added TX3_PHASE_SHIFTER field in CHIRP_PARAM_INDEX of the AWR_ADVANCE_CHIRP_CONF_SB API at 186.

Revision	Date	Description
3.04	24.05.2021	- Updated AWR_RF_INIT_CALIBRATION_CONF_SB to take the multi-TX enable for boot time TX power calibration in 81. - Updated AWR_PHASE_SHIFTER_CAL_DATA_SAVE_SB to add the TX index for TX3 and add information on the byte order of TX3 phaseshifter data in 89. - Updated AWR_PHASE_SHIFTER_CAL_DATA_SAVE_SB to add the TX index for TX3 and added information on the byte ordering of TX3 phaseshifter data in 100. - Updated the AWR_CAL_DATA_SAVE_SB and AWR_CAL_DATA_RESTORE_SB calibration data structure for xWR294x in 96. - Updated the TX_CAL_EN_CFG field in AWR_PROFILE_CONF_SB. - Moved the PF_HPF_FAST_INIT field inside PF_MISC_FEATURE_EN of AWR_PROFILE_CONF_SB API. - Added a new field RUNTIME_TX_POWER_MULTI_TX_CAL_EN inside PF_MISC_FEATURE_EN of AWR_PROFILE_CONF_SB API.

Revision	Date	Description
3.05	31.05.2021	- Added a new message command -AWR_RF_MONITORING_CONF_SET_2_MSG at 50. - Added a generic TXN power monitor configuration API - AWR_MONITOR_TXN_POWER_CONF_SB at 503. - Added a generic TXN ball-break monitor configuration API - AWR_MONITOR_TXN_BALLBREAK_CONF_SB at 505. - Added a generic TXN internal signals monitor configuration API - AWR_MONITOR_TXN_INTERNAL_ANALOG_SIGNALS_CONF_SB at 507. - Added a generic TXN phaseshifter monitor configuration API - AWR_MONITOR_TXN_PHASE_SHIFTER_CONF_SB at 508. - Added an advanced TX gain phase mismatch monitor configuration API - AWR_MONITOR_ADV_TX_GAIN_PHASE_MISMATCH_CONF_SB at 512. - Updated AWR_APLL_SYNTH_BW_CONTROL_SB to include xWR294x settings at 91. - Updated the VCO1/VCO2 frequency ranges, ADC sampling rate, Number of ADC samples for xWR294x in AWR_PROFILE_CONF_SB at 103. - Updated the DIE-ID structure to display raw hex values in AWR_RF_DIEID_STATUS_SB at 232. - Removed some notes about the reported RX gain and noise values from AWR_MONITOR_RX_GAIN_PHASE_REPORT_AE_SB at 297. - Added GPADC_REF3 status and value in AWR_MONITOR_GPADC_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB at 329. - Added TX3 power monitoring report - AWR_MONITOR_TX3_POWER_REPORT_AE_SB at 337. - Added TX3 ball-break monitoring report - AWR_MONITOR_TX3_BALLBREAK_REPORT_AE_SB at 338. - Added TX3 internal signal monitoring report - AWR_MONITOR_TX3_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB at 339. - Added TX3 phaseshifter monitoring report - AWR_MONITOR_TX3_PHASE_SHIFTER_REPORT_AE_SB at 340. - Added a new advanced TX gain phase monitoring report AWR_MONITOR_ADV_TX_GAIN_PHASE_REPORT_AE_SB at 342. - Modified AWR_MONITOR_ANALOG_ENABLES_CONF_SB to add the enables for the new monitors added at 431. - Updated AWR_MONITOR_DUAL_CLOCK_COMP_CONF_SB to display the clocks being monitored in xWR294x at 487. - Updated AWR_AE_MEAS_TX_POWER_SB to include the 4th TX measured values.

xxxviii

Revision	Date	Description
3.06	16.06.2021	1. Updated the maximum RX gain allowed to 44dB for xWR294x.

Revision	Date	Description
3.07	13.07.2021	- Added a new VMON configuration API at 516. - Marked VMON enable field and PA LDO SC enable fields in AWR_MONITOR_ANALOG_ENABLES_CONF_SB as not applicable in xWR294x at 431. - Added temperature index override fields for runtime TX phase shifter calibration at 149. - Added TX4 fields for TX_GAIN_DROP, TX_PHASE_INV, SUPPLY_LDO_FAULT and added new TX PS DAC fault injection options in 494. - Added the ECC aggregator and Bus safety monitoring status fields in AWR_RF_BOOTUPBIST_STATUS_DATA_SB at 234 and AWR_AE_DEV_RFPOWERUPDONE_SB at 389. - Added a new API to get RSS ESM status for xWR294x devices - AWR_RF_ADV_ESMFAULT_STATUS_GET_SB at 236 and its corresponding response packet details in AWR_RF_ADV_ESMFAULT_STATUS_SB at 236. - Added a new ASYNC event to report ESM errors in the xWR294x device - AWR_AE_RF_ADV_ESMFAULT_STATUS_SB at 279. - Corrected the definition of the APLL clock field in AWR_MONITOR_DUAL_CLOCK_COMP_REPORT_AE_SB at 332. - Updated AWR_ANALOGFAULT_AE_SB report by marking the supply faults and PA LDO SC faults as not applicable for xWR294x devices at 287. - Updated the expected values for GPADC_REF2_VALUE and GPADC_REF3_VALUE in AWR_MONITOR_GPADC_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB at 329

Revision	Date	Description
3.08	16.07.2021	1. Added a data dump address field for an internal debug API.

Revision	Date	Description
3.09	03.08.2021	- Added a control bit in AWR_RF_DEVICE_CFG_SB to disable the inter-burst APLL power save feature in 70. - Added an restriction on the TX_POWER_BACKOFF of the different TXs in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB when enabling multi-TX calibration (TX_POWER_CAL_CFG) in AWR_RF_INIT_CALIBRATION_CONF_SB at 81. - Updated the valid VCO1 slope values in AWR_MONITOR_PLL_CONTROL_VOLTAGE_REPORT_AE_SB for xWR294x devices at 330. - Marked RX signal image monitor and RX mixer input power monitor as reserved for xWR294x devices in AWR_MONITOR_ANALOG_ENABLES_CONF_SB at 431. - Updated the AWR_MONITOR_VMON_CONF_SB to include the reserved bytes at the end of the API at 516. - Updated AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB, AWR_MONITOR_TX1_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB, AWR_MONITOR_TX2_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB and AWR_MONITOR_TX3_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB to remove the TX PS DAC related status and report fields as inapplicable for xWR294x devices. TX PS DAC will be available as a separate configuration API and ASYNC report in xWR294x. - Marked the TX_PS_DAC_MON_THRESH field in AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_CONF_SB, AWR_MONITOR_TX1_INTERNAL_ANALOG_SIGNALS_CONF_SB, AWR_MONITOR_TX2_INTERNAL_ANALOG_SIGNALS_CONF_SB and AWR_MONITOR_TX3_INTERNAL_ANALOG_SIGNALS_CONF_SB as not applicable for xWR294x devices as the TX PS DAC monitor now has separate configuration and reports. - Added a new API for configuring the TX phase shifter DAC monitor - AWR_MONITOR_TX_PHSHIFTER_DAC_CONF_SB at 523. - Added a new monitoring ASYNC report - AWR_MONITOR_TX_PHSHIFTER_DAC_REPORT_AE_SB to report the results of the TX phase shifter DAC monitor at 344. - Added TX phase shifter DAC monitoring enable field in AWR_MONITOR_ANALOG_ENABLES_CONF_SB at 431. - Added a new API for configuring the RF parameters overrides and dither controls during monitoring - AWR_MONITOR_OVERRIDES_AND_DITHER_CONF_SB at 519.

Revision	Date	Description
3.10	10.08.2021	- Updated the limits for VCO2_SLOPE reported in AWR_MONITOR_PLL_CONTROL_VOLTAGE_REPORT_AE_SB at 330. - Corrected a few typos in AWR_MONITOR_OVERRIDES_AND_DITHER_CONF_SB at 519.

Revision	Date	Description
3.11	16.08.2021	- Added CRD_NSLOPE_MAG field to control the magnitude of the slope during controlled rampdown in AWR_RF_MISC_CTL_SB API at 73. - Added the MPU test status report bit in AWR_RF_BOOTUP_BIST_STATUS_DATA_SB at 233. - Added the MPU test status report bit in AWR_AE_DEV_RF_POWERUPDONE_SB at 389. - Defeatured AWR_RF_DFE_STATISTICS_REPORT_GET_SB for xWR294x devices and added a new API AWR_RF_REAL_CHAN_DFE_STATISTICS_REPORT_GET_SB for reading DFE statistics in xWR294x devices at 250. - Updated the RCOSC monitoring error threshold in the DCC clock pairs table at 486. - Added separate thresholds for each TX in AWR_MONITOR_TX_PHSHIFTER_DAC_CONF_SB at 523. - Increased the minimum inter-frame time from 300 to 306uS to account for the increased time taken to read DFE statistics for 7 functional profiles instead of the earlier 4 profiles at 542.

Revision	Date	Description
3.12	30.08.2021	- VMON reference values for normal and self-test operations updated in AWR_MONITOR_VMON_CONF_SB at 516. - Marked external analog signals monitor as unsupported in xWR294x devices. - Marked RX signal and image band monitor as unsupported in xWR294x devices. - Modified "Typical inter burst time", "Inter burst APLL power save time" in the Minimum Inter-Burst time table at 540. - Modified "Typical inter sub-frame/frame time" and "Test source config time" in the Min Inter-Sub-Frame/Frame time table at 542. - Added new calibrations and monitoring duration tables for xWR294x at 543. - Modified the highest RF gain target value in AWR_PROFILE_CONF_SET_SB from 35dB to 34dB.

Revision	Date	Description
3.13	07.09.2021	1. Added the recommended value for the CRD_NSLOPE_MAG in AWR_RF_MISC_CTL_SB at 73 .

Revision	Date	Description
3.14	28.09.2021	1. Added a note indicating that AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_CONF_SB has been deprecated in xWR294x devices and replaced with AWR_MONITOR_ADV_TX_GAIN_PHASE_MISMATCH_CONF_SB at 512. 2. Added a new API for dynamic per-chirp phase shifter control - AWR_ADV_DYN_PERCHIRP_PHASESHIFTER_CONF_SET_SB at page 217. The older API AWR_DYN_PERCHIRP_PHASESHIFTER_CONF_SET_SB has been marked as unsupported in xWR294x. 3. Added a new field to control the enable/disable of the RSS debug logger - LOGGER_ENABLE in AWR_RF_DEVICE_CFG_SB at page 70. 4. Added a note in FRAME_PERIODICITY field of AWR_FRAME_CONF_SET_SB on the additional dependency on monitoring periodicity. 5. Updated the maximum RF gain target in xWR294x to 34dB. 6. Additional information on the programmable filter have been added to AWR_PROG_FILT_COEFF_RAM_SET_SB at page 143.

Revision	Date	Description
3.15	19.10.2021	1. Added information on the state of the TX PAs during inter-chirp time at 68. 2. Improved the description and added recommended computation method for setting CRD_NSLOPE_MAG in AWR_RF_MISC_CTL_SB API at 74. 3. Added a new field FAST_RESET_END_TIME to control the synthesizer's fast reset end time in the idle time before a chirp in AWR_RF_MISC_CTL_SB API at 74. 4. Updated Phase value mapping in OBS_PHSIFT_DATA field of AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB in page 89. 5. Updated Phase value mapping in OBS_PHSIFT_DATA field of AWR_PHASE_SHIFTER_CAL_DATA_SAVE_SB in page 100. 6. Marked AWR_RF_GPADC_CFG_SET_SB API as unsupported in xWR294x devices at 266. 7. Updated the time taken for the application of the runtime calibration results to hardware at 545.

Revision	Date	Description
3.16	28.10.2021	1. Added TI recommended values for the FAST_RESET_END_TIME field in the AWR_RF_MISC_CTL_SB API at 74. 2. Added some more information on the firmware behavior in xWR294x devices, when the mid-frequency code of the RF band used in functional chirp profiles exceeds the cal-mon limit upper frequency - 200MHz. See NOTE 5 in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB at 83. 3. Added the 3 newly supported frequencies in the LO_LOOPBACK_FREQ field in AWR_LOOPBACK_BURST_CONF_SET_SB at 167. 4. Added the 3 newly supported frequencies in the PA_LOOPBACK_FREQ (xWR294x) field in AWR_LOOPBACK_BURST_CONF_SET_SB at 167. 5. Marked RX_NOISE_FIGURE_MONITOR as unsupported in xWR294x devices in AWR_MONITOR_ANALOG_ENABLES_CONF_SB at 431. 6. Increased the runtime calibration duration for the TX PHASE calibration in the table - Duration of run time calibrations for xWR294x devices at 545.

Revision	Date	Description
3.17	28.10.2021	1. Updated the recommended values of AWR_APLL_SYNTH_BW_CONTROL_SB API fields for xWR294x devices in the table "Typical APLL and Synth BW settings for xWR294x devices" at 92 .

Revision	Date	Description
3.18	16.11.2021	1. Modified the recommended and default values of AWR_APLL_SYNTH_BW_CONTROL_SB API fields for xWR294x devices in the table "Typical APLL and Synth BW settings for xWR294x devices" at 92. 2. Removed the field to control the enable/disable of the RSS debug logger - LOGGER_ENABLE from AWR_RF_DEVICE_CFG_SB API. 3. Added one new supported frequency (0 Hz) in the LO_LOOPBACK_FREQ field in AWR_LOOPBACK_BURST_CONF_SET_SB at 167. 4. Added one new supported frequency (0 Hz) in the PA_LOOPBACK_FREQ (xWR294x) field in AWR_LOOPBACK_BURST_CONF_SET_SB at 167.

Revision	Date	Description
3.19	08.12.2021	1. Added more information regarding the differences in API message support based on device architecture (front-end vs single-chip) in 33. 2. Changes to the AWR_APLL_SYNTH_BW_CONTROL_SB to add two additional fields to separate the SYNTH RZ and ICP trims between VCO1 and VCO2 at 91. 3. Updated the PLL clock frequency and the allowed values of the SRCLOCK_DIV field in AWR_DEV_MCUCLOCK_CONF_SET_SB at 347. 4. Updated the PLL clock frequency, the allowed values of SRC-CLOCK_DIV field in AWR_DEV_PMICCLOCK_CONF_SET_SB and the configuration examples at 359. 5. Updated the description and added notes on the usage of LDO_VMON_SC_MONITORING_EN field in the AWR_MONITOR_ANALOG_ENABLES_CONF_SB API at 431. 6. Updated the descriptions of the SYNTH_VCO_OPENLOOP and SUPPLY_LDO_FAULT fault injections to accurately describe the fault sequences used in AWR_ANALOG_FAULT_INJECTION_CONF_SB API at 494. 7. Added a note in AWR_MONITOR_VMON_CONF_SB indicating that it is applicable only in xWR294x devices and the described the reporting method of VMON faults at 516. 8. Updated the examples 1-4 that demonstrate the budgeting of calibration and monitoring time when designing the FTTI at 555.

Revision	Date	Description
3.20	16.12.2021	- Updated TX_PS_DAC_IDELTA_MIN and TX_PS_DAC_QDELTA_MIN fields in AWR_MONITOR_TX_PHSHIFTER_DAC_REPORT_AE_SB by splitting them into TX specific fields. - Marked the 2800 KHz option for HPF1/HPF2 cutoff as reserved in xWR294x devices. - Replaced AWR29xx and AWR2944 references with xWR294x. - Added a column providing the number of monitoring chirps in the monitoring durations table for xWR294x devices. - Updated the monitoring critical time for RX gain phase monitor from 260 to 290 uS. - Updated the calibration duration of periodic PD calibration from 400 to 600 uS. - Added a note to MON_DITHER_CTRL field of AWR_MONITOR_OVERRIDES_AND_DITHER_CONF_SB providing information on how enabling time dithering would impact the overall monitoring duration and critical time in an FTTI.

Revision	Date	Description
3.21	02.03.2022	- Added LSB information explicitly in the LO_LOOPBACK_FREQ and PA_LOOPBACK_FREQ (xWR294x) fields in AWR_LOOPBACK_BURST_CONF_SET_SB at 167. - Removed unsupported field HIGH_RES_TX_POWER_CTRL and its related updates to PF_TX_OUTPUT_POWER_BACK-OFF fields in both AWR_PROFILE_CONF_SB in 103 and AWR_CONT_STREAMING_CONF_SET_SB in 123. - Updated the notes in AWR_ANALOG_FAULT_INJECTION_CONF_SB API to remove the restriction that the user has to perform device reset after fault injection tests in the analog at 494. - Improved information on self-test failures during configuration of BSS VMONs through the API - AWR_MONITOR_VMON_CONF_SB at 516. - Added new API error codes at 414. - Marked the field IO_SUPPLY_INDICATOR in AWR_RF_LDO_BYPASS_SB API as RESERVED for xWR294x devices at 260. - Information regarding gain normalization of the programmable filter has been added to AWR_PROG_FILTER_COEFF_RAM_SET_SB at page 143.

Revision	Date	Description
3.22	26.04.2022	- Added a note on using different number of ADC samples in chirps within the same sub-frame in AWR_PROFILE_CONF_SET_SB at page 103. - Added a note on the possible inaccuracies seen when using the 1.4MHz HPF1/HPF2 cutoff setting in AWR_PROFILE_CONF_SET_SB at page 103. - Added a note on the possible inaccuracies seen when using the 1.4MHz HPF1/HPF2 cutoff setting in AWR_CONFIG_STREAMING_MODE_CONF_SET_SB at page 123. - Updated the bit-width of TX gain STG_CODE of xWR294x devices in AWR_TX_GAIN_TEMPLUT_SET_SB (164) and AWR_ADV_TX_GAIN_TEMPLUT_SET_SB (216) APIs. - Deprecated AWR_RF_ESMFAULT_STATUS_GET_SB in xWR294x devices and pointed to the new API - AWR_RF_ADV_ESMFAULT_STATUS_GET_SB. - Deprecated AWR_AE_RF_ESMFAULT_SB in xWR294x devices and pointed to the new ASYNC event - AWR_AE_RF_ADV_ESMFAULT_SB. - Added back the DFE parity test and DFE parity ESM G2 error indications. - Removed the note that DCC test is only applicable on AWR2243/xWR6243 devices as it is applicable in xWR294x devices as well. The below APIs/ASYNC events have been updated - AWR_RF_BOOTUPBIST_STATUS_DATA_SB, AWR_MONITOR_RF_DIG_LATENTFAULT_REPORT_AE_SB, AWR_AE_DEV_RFPOWERUPDONE_SB and AWR_MONITOR_RF_DIG_LATENTFAULT_CONF_SB. - Marked all the LPF monitoring result fields in AWR_MONITOR_RX_IFSTAGE_REPORT_AE_SB - LPF_CUTOFF_BANDEDGE_DROOP_VALUE_RX0, LPF_CUTOFF_STOPBAND_ATTEN_VALUE and LPF_CUTOFF_BANDEDGE_DROOP_VALUE_RX as DEBUG ONLY in xWR294x devices at 302. - Marked all the LPF monitoring configuration fields in AWR_MONITOR_RX_IFSTAGE_CONF_SB - LPF_CUTOFF_BANDEDGE_DROOP_THRESH, LPF_CUTOFF_STOPBAND_ATTEN_THRESH as DEBUG ONLY in xWR294x devices at 444. - Updated the API error handling section for xWR294x at 413. - Removed the note regarding the loopback path gain and phase mismatches that can show up as mismatches in the TX gain phase monitoring results. This is no longer applicable for xWR294x devices and hence this note has been removed from AWR_MONITOR_ADV_TX_GAIN_PHASE_MISMATCH_CONF_SB API at 512.

Revision	Date	Description
3.23	16.05.2022	1. Added a note to the AWR_CALIB_MON_TIME_UNIT_CONF_SB API explicitly indicating that the CALIB_MON_TIME_UNIT needs to be set within the allowed range even if no programmable monitoring or calibrations are desired by the user at 146 .

Revision	Date	Description
3.24	21.07.2022	- Updated AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB API with a new field - VCO2_RANGE_CONFIG at page 83. - Updated the valid ranges for FREQ_LIMIT_HIGH_TX0 to FREQ_LIMIT_HIGH_TX3 fields in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB API to account for the VCO2_RANGE_CONFIG which is configured by the user. - Updated valid range information for PF_INDX, PF_VCO_SELECT, PF_FREQ_START_CONST, PF_RAMP_END_TIME, fields in AWR_PROFILE_CONF_SET_SB API at page 103. - Modified the note on the xWR294x specific gain settings in PF_RX_GAIN in AWR_PROFILE_CONF_SET_SB API indicating the (RX gain, RF gain target) pairs that are not supported. - Updated additional information on PF_HPF_FAST_INIT field in AWR_PROFILE_CONF_SET_SB API. - Updated valid range information for VCO_SELECT and PF_FREQ_START_CONST fields in AWR_CONT_STREAMING_MODE_CONF_SET_SB API at page 123. - Removed PF_MISC_FEATURE_EN field from AWR_CONT_STREAMING_MODE_CONF_SET_SB API as it wasn't being used in the RSS firmware. For CW configuration, these misc feature settings are being taken from Profile 0 configuration. - Fixed the issue of contradicting recommendations for the maximum pulse width of the SYNC_IN pulse in the ICD. Updated the note in AWR_ADVANCED_FRAME_CONF_SB to recommend a max size of 4uS for the SYNC_IN at 129. - Added a note indicating the order of application for the RX signal digital compensation in AWR_DIGITAL_COMP_EST_CONTROL_SB at page 157. - Corrected an error in the IF_LOOPBACK_FREQ field table in AWR_LOOPBACK_BURST_CONF_SET_SB API at page 167 and AWR_RF_IFLOOPBACK_CFG_SB API at page 266. - Removed numerical frequency limits in the error codes description for AWR_PROFILE_CONF_SET_SB, AWR_CHIRP_CONF_SET_SB, AWR_CAL_MON_FREQUENCY_LIMITS_SB APIs at 413. - Updated the VCO extreme frequency set at which the PLL control voltage monitor is executed in AWR_MONITOR_PLL_CONTROL_VOLTAGE_SIGNALS_CONF_SB API at page 484. - Updated the Chirp Cycle Timing table at page 536.

Revision	Date	Description
3.25	25.08.2022	- Updated the MON_START_FREQ_CONST field in TX ball break monitoring configuration APIs (AWR_MONITOR_TXn_BALLBREAK_CONF_SB, AWR_MONITOR_TX0_BALLBREAK_CONF_SB, AWR_MONITOR_TX1_BALLBREAK_CONF_SB and AWR_MONITOR_TX2_BALLBREAK_CONF_SB) to document the allowed range and LSB for 77GHz devices. - Modified the interpretation and the default value of the VCO2_RANGE_CONFIG field in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB and AWR_CAL_MON_FREQUENCY_LIMITS_SB APIs. - Removed the limitation on PF_NUM_ADC_SAMPLES in the additional notes section of AWR_FRAME_CONF_SET_SB and AWR_ADVANCED_FRAME_CONF_SB APIs. - Added information on the reset state of the OSC_CLKOUT in xWR294x in the OSCCLKOUT_DIS field of AWR_RF_STATIC_CONF_SET_MSG API at 62. - Updated the valid range information for CHIRP_PROFILE_SELECT, CHIRP_TX_EN and CHIRP_BPM_VAL field in ADV_CHIRP_GENERIC_LUT_PARAM table at 204.

Revision	Date	Description
3.26	12.12.2022	- Updated the RF_GAIN_TARGET values supported in xWR294x in AWR_PROFILE_CONF_SB, AWR_CONT_STREAMING_MODE_CONF_SET_SB and AWR_LOOPBACK_BURST_CONF_SET_SB APIs. - Added a note in AWR_PROG_FILT_CONF_SET_SB API on the restriction that the programmable filter needs to be enabled for all the profiles that is configured in a frame. - TX_PS_DAC_FAULT fault injection field updated to indicate that the failure would be seen in the TX PS DAC monitoring report in xWR294x devices. - MISC_THRESH_FAULT fault injection description updated to indicate the swapping of the reported values of GPADC internal signals when fault is injected. - Added new sections detailing the command/response and ASYNC event flow in single chip devices at 27. - Added new sections detailing the BOOT INFO register configurations required in xWR294x at 3. - Added a note on PA loopback signal strength inaccuracies for higher backoff values in the PA_LOOPBACK_BUF_GAIN_SEL field in AWR_LOOPBACK_BURST_CONF_SET_SB API at 167. - De-featured the "End of Frame" mode of GLOBAL_RESET_MODE in AWR_ADVANCE_CHIRP_CONF_SB API at 186. - Added a note on the TX phase resolution supported in the hardware and how it relates to the delta increment value of AWR_ADVANCE_CHIRP_CONF_SB in 202.

Revision	Date	Description
3.27	02.03.2023	1. Added a note in BSS_DIG_CTRL field of AWR_RF_DEVICE_CFG_SB API that BSS WDT cannot be enabled if BSS dynamic frequency switching feature is enabled. 2. Updated VCO specific recommendations for the CRD_NSLOPE_MAG field in AWR_RF_RADAR_MISC_CTL_SB API at 73. 3. Updated AWR_MONITOR_GPADC_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB to maintain the previous devices report structure and highlight xWR294x changes separately. 4. Added a note to AWR_MONITOR_ANALOG_ENABLES_CONF_SB API and documented the order of execution of the RF/Analog monitors in xWR294x devices at 431. 5. Updated the description of AWR_MONITOR_SYNTHESIZER_FREQUENCY_CONF_SB API to clarify that the SYNTH_FREQ live monitor can be used to monitor functional chirps in xWR294x. On other devices, it is still de-featured and should be used only for debug. 6. Corrected multiple places in the ICD where CAL_MON_TIME_UNIT was used instead of the actual CALIB_MON_TIME_UNIT field name.

Revision	Date	Description
3.28	10.03.2023	1. Updated the valid range for the PF_FREQ_SLOPE_CONST in the API AWR_PROFILE_CONF_SET_SB in xWR294x devices at 103. 2. Added a note on RX data coherence across profiles in the API AWR_PROFILE_CONF_SET_SB at 103. 3. Added additional information on the RSS dynamic frequency switching and the RSS logger features at page 11.

Revision	Date	Description
3.29	14.03.2023	1. Updated the minimum valid value to 2.5uS for the PF_IDLE_TIME_CONST in the API AWR_PROFILE_CONF_SET_SB in xWR294x devices at 103. 2. Updated the minimum idle times used in the CRD_NSLOPE_MAG and FAST_RESET_END_TIME computation examples in AWR_RF_RADAR_MISC_CTL_SB API at page 73.

Revision	Date	Description
3.30	24.03.2023	Added a note on the TX power accuracy for backoff values greater than 15dB in the in PF_TX_OUTPUT_POWER_BACKOFF field of API AWR_PROFILE_CONF_SET_SB in xWR294x devices at 103.

Revision	Date	Description
3.31	11.07.2023	- Updated the duration of the run time TX phase shifter calibration in Table 12.4. - Updated all AWR294x references to xWR294x in the document. - Marked ROM CRC and PCR tests in RF_POWERUP_BIST_STATUS_FLAGS as reserved for xWR294x devices in Table 5.79. - Updated the mailbox communication sequences to include the recommended fix for the missing interrupt issue at 27. - Added a note about the jitter between chirp start and ADC sampling start in AWR_FRAME_CONF_SET_SB API at 119 and AWR_ADVANCED_FRAME_CONF_SB API at 129.

Revision	Date	Description
3.32	06.07.2023	First set of API changes for xWR254x. - Removed RXIFMM for xWR254x devices. - Added FRC clock source for xWR254x devices. - Removed BIST FFT self test for xWR254x devices. - Updated mailbox sequence. - Removed DFE AND parity test for xWR254x devices. - Updated synthesizer frequency monitor to support multiple profiles for xWR254x devices.

Revision	Date	Description
3.33	21.07.2023	- Added inter-chirp jitter mitigation for xWR254x devices. - Defeatured bus safety self test for xWR254x devices. - Added OSCCLKOUTETH for xWR254x devices. - Added disable option for advance chirp configuration API. - Added support for 1ns resolution of ADC start time.

Revision	Date	Description
3.34	11.10.2023	1. Updated description for CHIRP_ADC_START_TIME_VAR in Table 5.52 2. Updated description for INTER_CHIRP_JITTER_MITIGATION in 73 3. Updated NOTE in 73 4. Updated description for ESM_GROUP1_ERRORS_MSB in 236 5. Updated description for DSS_MBOX_CONFIG in 3

Revision	Date	Description
3.35	16.01.2024	1. Updated description for SYNTH_FAULT in 494 2. Updated description for LPF_CUTOFF_BANDEDGE_DROOP_THRESH and LPF_CUTOFF_STOPBAND_ATTEN_THRESH fields in AWR_MONITOR_RX_IFSTAGE_CONF_SB API in Table 8.7 3. Updated description for INTER_CHIRP_JITTER_MITIGATION of AWR_RF_RADAR_MISC_CTL_SB in 73 4. Updated timing information for xWR254x devices in Chapter 11 and Chapter 12 5. Added analog monitoring duration for xWR254x devices in Table 12.7 6. Added minimum chirp cycle time for xWR254x devices in Table 11.1 7. Updated description for PF_RX_GAIN in AWR_PROFILE_CONF_SET_SB API in 103 8. Updated description for PF_RX_GAIN in AWR_CONT_STREAMING_MODE_CONF_SET_SB API in 123 9. Updated description for PF_RX_GAIN in AWR_LOOPBACK_BURST_CONF_SET_SB API in 167 10. Updated description for LO_LOOPBACK_BUF_GAIN_SEL of AWR_LOOPBACK_BURST_CONF_SET_SB in 168

Revision	Date	Description
3.36	06.06.2024	- Updated description for Limitation 2 for AWR_ADVANCE_CHIRP_GENERIC_LUT_LOAD_SB at 210 - Updated description for 'RSS watchdog feature' of 'RSS dynamic frequency switch' section at 11 - Updated description for PF_RX_GAIN in AWR_PROFILE_CONF_SET_SB API at 103 - Updated description for PF_RX_GAIN in AWR_CONT_STREAMING_MODE_CONF_SET_SB API at 123 - Updated description for RX_GAIN in AWR_LOOPBACK_BURST_CONF_SET_SB API at 167 - Updated description for SYNTH_FAULT in AWR_ANALOG_FAULT_INJECTION_CONF_SB at 494 - Updated description for FRC_200MHZ_CLK_SRC in RSS_PROC_CTRL:RSS_CR4_BOOT_INFO_REG5 at page 11. - Updated description for DSS_MBOX_CONFIG in RSS_PROC_CTRL:RSS_CR4_BOOT_INFO_REG3 at page 7. - Updated description for DSS_MBOX_CONFIG in RSS_PROC_CTRL:RSS_CR4_BOOT_INFO_REG4 at page 9. - Updated description in 7.1 at pagerefchap:ApiErrorHandling - Updated description in 6 at pagerefchap:ApiSequence

Revision	Date	Description
3.37	14.06.2024	- Updated for xWR2x4xP devices - Updated description for MSS_MBOX_CONFIG, DSS_MBOX_CONFIG in RSS_PROC_CTRL:RSS_CR4_BOOT_INFO_REG3 at page 7. - Updated description for MSS_MBOX_CONFIG, DSS_MBOX_CONFIG in RSS_PROC_CTRL:RSS_CR4_BOOT_INFO_REG4 at page 9. - Updated description for PF_RX_GAIN in AWR_PROFILE_CONF_SET_SB API at 103 - Updated description for PF_RX_GAIN in AWR_CONT_STREAMING_MODE_CONF_SET_SB API at 123 - Updated description for RX_GAIN in AWR_LOOPBACK_BURST_CONF_SET_SB API at 167

Revision	Date	Description
3.38	17.07.2024	1. Updated description for PF_RX_GAIN in AWR_PROFILE_CONF_SET_SB API at 103 2. Updated description for PF_RX_GAIN in AWR_CONFIG_STREAMING_MODE_CONF_SET_SB API at 123 3. Updated description for RX_GAIN in AWR_LOOPBACK_BURST_CONF_SET_SB API at 167

Revision	Date	Description
3.39	06.11.2024	1. Updated for xWR2x4xP devices 2. Updated description for RF_FREQ_BITMASK in AWR_MONITOR_RX_GAIN_PHASE_CONF_SB at 438 3. Updated description for RF_FREQ_BITMASK in AWR_MONITOR_RX_NOISE_FIGURE_CONF_SB at 442 4. Updated description for RF_FREQ_BITMASK in AWR_MONITOR_TX0_POWER_CONF_SB at 446 5. Updated description for RF_FREQ_BITMASK in AWR_MONITOR_TX1_POWER_CONF_SB at 448 6. Updated description for RF_FREQ_BITMASK in AWR_MONITOR_TX2_POWER_CONF_SB at 449 7. Updated description for RF_FREQ_BITMASK in AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_CONF_SB at 512 8. Updated description for RF_FREQ_BITMASK in AWR_MONITOR_TXN_POWER_CONF_SB at 503 9. Updated description for RF_FREQ_BITMASK in AWR_MONITOR_ADV_TX_GAIN_PHASE_MISMATCH_CONF_SB at 512 10. Updated description for RX_IF_STAGES_FAULT in AWR_ANALOG_FAULT_INJECTION_CONF_SB at 494 11. Updated description for PF_TX_OUTPUT_POWER_BACK-OFF in AWR_PROFILE_CONF_SET_SB at 103 12. Updated NOTE in AWR_MONITOR_TX0_POWER_REPORT_AE_SB at 305 13. Updated NOTE in AWR_MONITOR_TX3_POWER_REPORT_AE_SB at 337 14. Updated NOTE for ENABLE_CAL_REPORT in AWR_RUNTIME_CALIBRATION_CONF_AND_TRIGGER_SB at 149 15. Updated NOTE for OSCCLKOUT_DIS in AWR_CHANNEL_CONF_SET_SB at 62

Revision	Date	Description
3.40	12.12.2024	1. Updated document for xWR2x4xECO device

Revision	Date	Description
3.41	19.05.2025	- Updated description for FRC_200MHZ_CLK_SRC in RSS_PROC_CTRL:RSS_CR4_BOOT_INFO_REG5 at page 11. - Updated description for PF_FREQ_SLOPE_CONST in AWR_PROFILE_CONF_SET_SB API at page 103. - Updated description for NUM_LOOPS in AWR_FRAME_CONF_SET_SB at page ??. - Updated description for SF1_NUM_LOOPS_PER_BURST, SF2_NUM_LOOPS_PER_BURST, SF3_NUM_LOOPS_PER_BURST, SF4_NUM_LOOPS_PER_BURST in AWR_ADVANCED_FRAME_CONF_SB at page 129. - Updated description for RF_POWERUP_BIST_STATUS_FLAGS in AWR_RF_BOOTUPBIST_STATUS_DATA_SB at page 234. - Updated description for ESM_GROUP1_ERRORS_LSB, ESM_GROUP1_ERRORS_MSB in AWR_RF_ADV_ESMFAULT_STATUS_SB at page 236. - Updated description for ESM_GROUP1_ERRORS_LSB,ESM_GROUP1_ERRORS_MSB in AWR_AE_RF_ADV_ESMFAULT_STATUS_SB at page 280. - Updated description for DIG_MON_LATENT_FAULT_STATUS in AWR_MONITOR_RF_DIG_LATENTFAULT_REPORT_AE_SB at page 292. - Updated description for BSS_POWERUP_BIST_STATUS_FLAGS in AWR_AE_DEV_RFPOWERUPDONE_SB at page 389. - Updated description for DIG_MONITORING_ENABLES in AWR_MONITOR_RF_DIG_LATENTFAULT_CONF_SB at page 429. - Added Temperature monitor timing in Monitoring duration table at page 548 and 549.

Revision	Date	Description
3.42	20.06.2025	1. Updated description for LOOPBACK_SEL (Defeatured Value 4) in AWR_LOOPBACK_BURST_CONF_SET_SB at page 168. 2. Updated description for IF_LOOPBACK_FREQ (Added 20MHz) in AWR_LOOPBACK_BURST_CONF_SET_SB at page 168. 3. Merged Chirp Cycle Timing table for xWR294x with xWR254x, xWR2x4xP devices at page 536.

NOTE1:	Please refer latest mmWave device DFP release notes for all known issues and de-featured APIs
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NOTE2:	All reserved bytes/bits in configuration API sub blocks shall be programmed with value zero. The functionality of radar device is not guaranteed if reserved bytes are not zero.
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NOTE3:	All reserved bytes/bits in API message reports (ack or AE) sub blocks shall be masked off in application.
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1 Introduction

1.1 Scope

The Scope of this document is to define interface control specifications for TI's xWR294x, xWR254x, xWR2x4xECO, xWR2x4xLC and xWR2x4xP mmWave sensor devices. These devices are enhanced version of AWR2243 device, most features/APIs of AWR2243 are supported and applicable in xWR294x (Backward Compatible), however very few of them are modified for enhancement. xWR254x can be considered as a version of xWR294x optimized for 1st dimension processing and thereby does not include a DSP processor. The interfaces have been optimized accordingly while the RF performance remains similar. xWR2x4xP can be considered as a version of xWR294x with improved RF performance, more memory for storage of radar data and support for enhanced DSP clock rate for faster processing of radar detected objects. xWR2x4xECO can be considered as a version of xWR2x4xP with RF performance similar to xWR294x and support for enhanced DSP clock rate for faster processing of radar detected objects.

The key differentiated new features in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xLC and xWR2x4xP are:

1. Max slope support upto 266MHz/us.
2. New programmable filter support
3. Improved analog RX front end noise figure and gain settings
4. New Advance flexible waveform generation API can support up-to 2048 unique chirps in a burst/frame
5. xWR294x, xWR254x, xWR2x4xP, xWR2x4xECO supports 4 TX and 4 Real-only RX chains, while AWR2243 supports 3 TX and 4 complex RX chains.
6. 4.5GHz RF bandwidth support in VCO2 for xWR294x, xWR254x, xWR2x4xP, xWR2x4xECO
7. xWR254x/xWR2x4xP/xWR2x4xECO supports inter-chirp jitter mitigation
8. Various new API features are listed in Appendix A

The wide range of TI highly integrated 77GHz and 60GHz CMOS TI mmwave sensors are tabulated in table 1.1. The mmWave device integrates all RF and Analog functionalities including VCO, PLL, PA, LNA, Mixer and ADC for multiple TX/RX channels into a single chip with integrated cortex R4 for programmability.

The device includes a Radar Sub-System (RadarSS) also called Built-in Self-Test (BSS) processor, which is responsible to configure the RF/Analog and digital front-end in real-time, as well as to periodically schedule calibration and functional safety monitoring. This enables the mmWave

front-end to be self-contained and capable of adapting itself to handle temperature and ageing effects, and to enable significant ease-of-use from an external host perspective.

This document contains the Interface Control Specification for communications on the serial interface (SPI) between the Radar device and the external host processor. The same protocol is used in all devices when the messages are sent to Radar Control subsystem (BIST subsystem) from the MCU subsystem (Master subsystem) and DSP subsystems (Not applicable for xWR254x, xWR2x4xLC).

Refer Link <http://www.ti.com/sensors/mmwave/overview.html> for more informations.

Table 1.1: TI CMOS mmWave radar devices

Frequency Type	60GHz RF Frequency	77GHz RF Frequency
TI Automotive Radar Devices	AWR6843, AWR6243	AWR1243, AWR1642, AWR1443, AWR1843, AWR2243, xWR294x, xWR254x, xWR2x4xLC, xWR2x4xP, xWR2x4xECO
TI Industrial Radar Devices	IWR6843, IWR6243	IWR1642, IWR1443, IWR1843

1.2 Intended Audience

The intended audience for this document is firmware, host software, and validation engineers needing to understand the format and contents of all communications between the Radar frontend and the host/application processor.

2 TI mmWave Radar Sensor Communications Overview

2.1 Communication Link Description

The AWR2243/xWR6243 radar device communicates with the external host processor using the SPI interface. The radar device is configured and controlled from the external host processor by sending commands to AWR2243/xWR6243 device over SPI.

The xWR1642, xWR1843, xWR6843, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xLC and xWR2x4xP radar devices are configured and controlled using the internal MCU (Master subsystem) and they communicate with an external ECU using the CAN interface.

This document details the communication protocol between radar device and external host processor using SPI in AWR2243/xWR6243. In xWR1642, xWR1843, xWR6843, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xLC and xWR2x4xP the same protocol is used to communicate between the BIST subsystem and Master subsystem over SRAM mailboxes. In xWR294x, xWR254x, xWR2x4xECO, xWR2x4xLC, xWR2x4xP in addition to the mailbox based communication interface between the BSS and the MSS, the BSS also has a few BOOT_INFO registers which have to be configured before the BSS CPU is released out of reset.

2.2 Communication Link configuration

2.2.1 SPI

This interface is synchronous. The interface includes four signals (SPICCLK, SPICS, and Data In and Data Out) and supports clock rates up to 40 MHz. The AWR2243/xWR6243 radar device is always the SPI slave and the external host processor will be the SPI master.

2.2.2 Mailbox

This interface includes a SRAM and an interrupt line from Master subsystem to BIST subsystem. A reverse channel which includes a different SRAM and a different interrupt line from the BIST subsystem to Master subsystem is used for responses which originate from BIST subsystem.

2.2.3 BOOT INFO registers

These registers are applicable only to xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP devices. These are memory mapped registers that have to be configured properly to allow the BSS firmware to boot and perform its functions. Fields indicated as CONFIG in the below registers

have to be configured by the application/secondary boot loader before the RSS CPU is unhalted.

RSS_PROC_CTRL:RSS_CR4_BOOT_INFO_REG0

This register contains boot related configuration and status fields.

Table 2.1: RSS_CR4_BOOT_INFO_REG0

Bits	Config (OR) Status	Field	Description
[15:0]	CONFIG	XTAL_FREQ	Frequency of the XTAL in MHz to be programmed by the user application.
[16:16]	STATUS	APLL_CALIB_DONE	BSS sets this to '1' if APLL calibration was performed in boot.
[17:17]	STATUS	APLL_CALIB_STATUS	PASS/FAIL status of the APLL calibration.
[18:18]	STATUS	BSS_BOOT_DONE	BSS sets this to '1' at the end of its boot up sequence to indicate that the boot has been completed.
[19:19]	STATUS	BSS_BOOT_STATUS	PASS/FAIL status of the BOOT monitors.
[22:20]	STATUS	BSS_FAULT_STATUS	Firmware fault status. 0 No fault 1 BSS firmware ASSERT 2 BSS Firmware CPU Abort 3 ESM Group 1 Error 4 ESM Group 2 Error 5 RESERVED 6 ANA LDO SC fault
[26:23]	STATUS	MAILBOX_BOOT_CONFIG_STATUS	Bits set to '1' indicate successful configuration of the mailbox system. Bits Description 23 MSS configuration success 24 DSS configuration success 26:25 Reserved
[31:27]	-	RESERVED	Set to 0.

RSS_PROC_CTRL:RSS_CR4_BOOT_INFO_REG1

This register contains the same information as in REG0 for redundancy.

Table 2.2: RSS_CR4_BOOT_INFO_REG1 (Redundant copy of REG0)

Bits	Config (OR) Status	Field	Description
[15:0]	CONFIG	XTAL_FREQ	Frequency of the XTAL in MHz to be programmed by the user application.
[16:16]	STATUS	APLL_CALIB_DONE	BSS sets this to '1' if APLL calibration was performed in boot.
[17:17]	STATUS	APLL_CALIB_STATUS	PASS/FAIL status of the APLL calibration.
[18:18]	STATUS	BSS_BOOT_DONE	BSS sets this to '1' at the end of its boot up sequence to indicate that the boot has been completed.
[19:19]	STATUS	BSS_BOOT_STATUS	PASS/FAIL status of the BOOT monitors.
[22:20]	STATUS	BSS_FAULT_STATUS	Firmware fault status. 0 No fault 1 BSS firmware ASSERT 2 BSS Firmware CPU Abort 3 ESM Group 1 Error 4 ESM Group 2 Error 5 RESERVED 6 ANA LDO SC fault
[26:23]	STATUS	MAILBOX_BOOT_CONFIG_STATUS	Bits set to '1' indicate successful configuration of the mailbox system. Bits Description 23 MSS configuration success 24 DSS configuration success 26:25 Reserved
[31:27]	-	RESERVED	Set to 0.

RSS_PROC_CTRL:RSS_CR4_BOOT_INFO_REG3

This register contains config fields for setting up the mailboxes between BSS and MSS/DSS.

NOTE:	In xWR2x4xECO, xWR2x4xP devices only one subsystem (either MSS or DSS) can communicate at a time with RadarSS .
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Table 2.3: RSS_CR4_BOOT_INFO_REG3

Bits	Config (OR) Status	Field	Description										
[7:0]	CONFIG	MSS_MBOX_CONFIG	Mailbox configuration for the MSS. <table><thead><tr><th>Bits</th><th>Description</th></tr></thead><tbody><tr><td>7</td><td>Enable communication with MSS</td></tr><tr><td>6</td><td>Sub-System selection 0 - MSS_CR5A, 1 - RESERVED</td></tr><tr><td>5:4</td><td>Buffer size in TX mailbox memory 0b00 512B 0b01 1kB 0b10 1.5kB 0b11 2kB</td></tr><tr><td>3:0</td><td>Buffer start address Offset Index of the memory for BSS Tx buffer. The offset is computed as (Buffer size * Offset Index) Note: The buffer end address, calculated as (Buffer size * (1 + Offset Index)) must be less than 8kB.</td></tr></tbody></table>	Bits	Description	7	Enable communication with MSS	6	Sub-System selection 0 - MSS_CR5A, 1 - RESERVED	5:4	Buffer size in TX mailbox memory 0b00 512B 0b01 1kB 0b10 1.5kB 0b11 2kB	3:0	Buffer start address Offset Index of the memory for BSS Tx buffer. The offset is computed as (Buffer size * Offset Index) Note: The buffer end address, calculated as (Buffer size * (1 + Offset Index)) must be less than 8kB.
Bits	Description												
7	Enable communication with MSS												
6	Sub-System selection 0 - MSS_CR5A, 1 - RESERVED												
5:4	Buffer size in TX mailbox memory 0b00 512B 0b01 1kB 0b10 1.5kB 0b11 2kB												
3:0	Buffer start address Offset Index of the memory for BSS Tx buffer. The offset is computed as (Buffer size * Offset Index) Note: The buffer end address, calculated as (Buffer size * (1 + Offset Index)) must be less than 8kB.												
[15:8]	CONFIG	DSS_MBOX_CONFIG	Mailbox configuration for the DSS. <table><thead><tr><th>Bits</th><th>Description</th></tr></thead><tbody><tr><td>7</td><td>Enable communication with DSS</td></tr><tr><td>6</td><td>Sub-System selection 0 - DSS_DSP, 1 - RESERVED</td></tr><tr><td>5:4</td><td>Buffer size in TX mailbox memory 0b00 512B 0b01 1kB 0b10 1.5kB 0b11 2kB</td></tr><tr><td>3:0</td><td>Buffer start address Offset Index of the memory for BSS Tx buffer. The offset is computed as (Buffer size * Offset Index) Note: The buffer end address, calculated as (Buffer size * (1 + Offset Index)) must be less than 8kB. NOTE: DSS_MBOX_CONFIG field is Reserved in xWR254x, xWR2x4xLC.</td></tr></tbody></table>	Bits	Description	7	Enable communication with DSS	6	Sub-System selection 0 - DSS_DSP, 1 - RESERVED	5:4	Buffer size in TX mailbox memory 0b00 512B 0b01 1kB 0b10 1.5kB 0b11 2kB	3:0	Buffer start address Offset Index of the memory for BSS Tx buffer. The offset is computed as (Buffer size * Offset Index) Note: The buffer end address, calculated as (Buffer size * (1 + Offset Index)) must be less than 8kB. NOTE: DSS_MBOX_CONFIG field is Reserved in xWR254x, xWR2x4xLC.
Bits	Description												
7	Enable communication with DSS												
6	Sub-System selection 0 - DSS_DSP, 1 - RESERVED												
5:4	Buffer size in TX mailbox memory 0b00 512B 0b01 1kB 0b10 1.5kB 0b11 2kB												
3:0	Buffer start address Offset Index of the memory for BSS Tx buffer. The offset is computed as (Buffer size * Offset Index) Note: The buffer end address, calculated as (Buffer size * (1 + Offset Index)) must be less than 8kB. NOTE: DSS_MBOX_CONFIG field is Reserved in xWR254x, xWR2x4xLC.												
[31:16]	-	RESERVED	Set to 0.										

RSS_PROC_CTRL:RSS_CR4_BOOT_INFO_REG4

This register is the redundant copy of RSS_CR4_BOOT_INFO_REG3 .

NOTE:	In xWR2x4xECO, xWR2x4xP devices only one subsystem (either MSS or DSS) can communicate at a time with RadarSS .
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Table 2.4: RSS_CR4_BOOT_INFO_REG4 (Redundant copy of REG3)

Bits	Config (OR) Status	Field	Description										
[7:0]	CONFIG	MSS_MBOX_CONFIG	Mailbox configuration for the MSS. <table><thead><tr><th>Bits</th><th>Description</th></tr></thead><tbody><tr><td>7</td><td>Enable communication with MSS</td></tr><tr><td>6</td><td>Sub-System selection 0 - MSS_CR5A, 1 - RESERVED</td></tr><tr><td>5:4</td><td>Buffer size in TX mailbox memory 0b00 512B 0b01 1kB 0b10 1.5kB 0b11 2kB</td></tr><tr><td>3:0</td><td>Buffer start address Offset Index of the memory for BSS Tx buffer. The offset is computed as (Buffer size * Offset Index) Note: The buffer end address, calculated as (Buffer size * (1 + Offset Index)) must be less than 8kB.</td></tr></tbody></table>	Bits	Description	7	Enable communication with MSS	6	Sub-System selection 0 - MSS_CR5A, 1 - RESERVED	5:4	Buffer size in TX mailbox memory 0b00 512B 0b01 1kB 0b10 1.5kB 0b11 2kB	3:0	Buffer start address Offset Index of the memory for BSS Tx buffer. The offset is computed as (Buffer size * Offset Index) Note: The buffer end address, calculated as (Buffer size * (1 + Offset Index)) must be less than 8kB.
Bits	Description												
7	Enable communication with MSS												
6	Sub-System selection 0 - MSS_CR5A, 1 - RESERVED												
5:4	Buffer size in TX mailbox memory 0b00 512B 0b01 1kB 0b10 1.5kB 0b11 2kB												
3:0	Buffer start address Offset Index of the memory for BSS Tx buffer. The offset is computed as (Buffer size * Offset Index) Note: The buffer end address, calculated as (Buffer size * (1 + Offset Index)) must be less than 8kB.												
[15:8]	CONFIG	DSS_MBOX_CONFIG	Mailbox configuration for the DSS. <table><thead><tr><th>Bits</th><th>Description</th></tr></thead><tbody><tr><td>7</td><td>Enable communication with DSS</td></tr><tr><td>6</td><td>Sub-System selection 0 - DSS_DSP, 1 - RESERVED</td></tr><tr><td>5:4</td><td>Buffer size in TX mailbox memory 0b00 512B 0b01 1kB 0b10 1.5kB 0b11 2kB</td></tr><tr><td>3:0</td><td>Buffer start address Offset Index of the memory for BSS Tx buffer. The offset is computed as (Buffer size * Offset Index) Note: The buffer end address, calculated as (Buffer size * (1 + Offset Index)) must be less than 8kB. NOTE: DSS_MBOX_CONFIG field is Reserved in xWR254x, xWR2x4xLC.</td></tr></tbody></table>	Bits	Description	7	Enable communication with DSS	6	Sub-System selection 0 - DSS_DSP, 1 - RESERVED	5:4	Buffer size in TX mailbox memory 0b00 512B 0b01 1kB 0b10 1.5kB 0b11 2kB	3:0	Buffer start address Offset Index of the memory for BSS Tx buffer. The offset is computed as (Buffer size * Offset Index) Note: The buffer end address, calculated as (Buffer size * (1 + Offset Index)) must be less than 8kB. NOTE: DSS_MBOX_CONFIG field is Reserved in xWR254x, xWR2x4xLC.
Bits	Description												
7	Enable communication with DSS												
6	Sub-System selection 0 - DSS_DSP, 1 - RESERVED												
5:4	Buffer size in TX mailbox memory 0b00 512B 0b01 1kB 0b10 1.5kB 0b11 2kB												
3:0	Buffer start address Offset Index of the memory for BSS Tx buffer. The offset is computed as (Buffer size * Offset Index) Note: The buffer end address, calculated as (Buffer size * (1 + Offset Index)) must be less than 8kB. NOTE: DSS_MBOX_CONFIG field is Reserved in xWR254x, xWR2x4xLC.												
[31:16]	-	RESERVED	Set to 0.										

RSS_PROC_CTRL:RSS_CR4_BOOT_INFO_REG5

This register contains config fields for miscellaneous features in the BSS.

Table 2.5: RSS_CR4_BOOT_INFO_REG5

Bits	Config (OR) Status	Field	Description
[2:0]	CONFIG	RSS_LOGGER_CONFIG	Configuration of the RSS debug logger. Value Description 0b000 RSS logger disabled 0b001 RSS logger enabled in UART mode 0b010 RSS logger enabled in MSS MEM DUMP mode
[3:3]	CONFIG	RSS_DYN_FREQ_SWITCH	Enable/Disable control of the BSS dynamic frequency switching feature. Value Description 0b0 RSS CPU clock always kept at 200MHz 0b1 RSS CPU clock is dynamically switched to XTAL frequency during idle and returned to 200MHz when active.
[15:4]	CONFIG	RSS_200MHZ_CLK_SRC	The 200MHz clock source selection for RSS CPU. (12-bit RSS CPU clock source selection at MSS_TOPRCM:RSS_CLK_SRC_SEL). This field is required if BSS dynamic frequency switching is enabled.
[18:16]	CONFIG	FRC_200MHZ_CLK_SRC	NOTE: Only applicable for xWR254x, xWR2x4xECP, xWR2x4xP. The 200MHz clock source selection for FRC. Table below shows the FRC clock source when dynamic frequency switching is enabled Value Clock source 0 DPLL_CORE_HSDIV0_CLKOUT3 1 DPLL_PER_HSDIV0_CLKOUT1 2-7 RESERVED FRC clock source is same as RSS_CLK when dynamic frequency switching is disabled.
[31:19]	-	RESERVED	Set to 0

RSS dynamic frequency switch : The RSS sub-system and its firmware generally executes at a fixed 200MHz clock source from the ADPLL. For reducing power consumption during inactive periods, the firmware can switch the RSS clock source from 200MHz source to XTAL. This is done only when the RSS firmware is entering into a WFI sleep mode during the inter-burst/inter-frame time periods. It doesn't enter into this mode if any calibrations or monitoring chirps are scheduled in the current inter-burst period. Anytime the RSS firmware wakes up from this state due to an event or an interrupt, the firmware first switches the clock back to a 200MHz source and then proceeds to service the interrupt. This is an optional feature that can be enabled

through RSS_PROC_CTRL:RSS_CR4_BOOT_INFO_REG5. The additional restrictions placed on the customer application if this feature is enabled have been listed below.

1. **MSS RTI C cannot be used by the application.** It can continue to use other RTIs in the MSS.
2. **RSS watchdog feature cannot be enabled when RSS dynamic frequency switching is enabled.** The customer can use application based checks (using MSS WDT to monitor RSS) to ensure that RSS is not in any stuck state. b15 WDT test in RF_POWERUP_BIST_STATUS_FLAGS of AWR_RF_BOOTUPBIST_STATUS_DATA_SB is expected to be 0 when RSS dynamic frequency switching is enabled.
3. **FRAME_TIMING_MONITORING_EN cannot be enabled in AWR_MONITOR_RF_DIG_PERIODIC_CONF_SB.** The customer application can use alternate methods to monitor the frame timing like periodic monitoring of the FTTI using the MSS PMU to timestamp the temperature monitoring ASYNC reports and computing the period between them. Another option is periodic monitoring of the FRAME_START (burst start) interrupt coming to the MSS. Measuring the time period between two FRAME_START interrupts using the MSS PMU, we should get a more accurate estimate of the frame timing. However, this means that when the customer uses complicated waveforms like advanced frame configuration, they would have to compute the expected value of the current burst period (including IDLE periods), before comparing against the measured FRAME_START -> FRAME_START time. Please refer to the TRM for more details on the FRAME_START interrupt.

RSS logger : RSS logger data generated by the RSS firmware provides information of the RSS control flow and also provides additional data like calibration/monitoring debug data as it is executed. This can be used to debug any RSS related issues. In xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP, the RSS supports enable/disable as well as 2 modes (UART / MSS memory dump) of transferring the logger data to an external system.

RSS_PROC_CTRL:RSS_CR4_BOOT_INFO_REG6

This register contains config fields for miscellaneous features in the BSS.

Table 2.6: RSS_CR4_BOOT_INFO_REG6

Bits	Config (OR) Status	Field	Description
[31:0]	CONFIG	BSS_LOGGER_MEM_BUF_ADDR	32 bit address of the MSS L2 memory dump for the BSS logger (2kB required). The BSS interprets this buffer as 2 chunks of 1kB each for PING-PONG operation. Valid range: 0xC0200000 - 0xC02EF7FC Note: This field is only applicable if the BSS logger is enabled in the MSS MEM DUMP mode. In other cases, this can be set to zero.

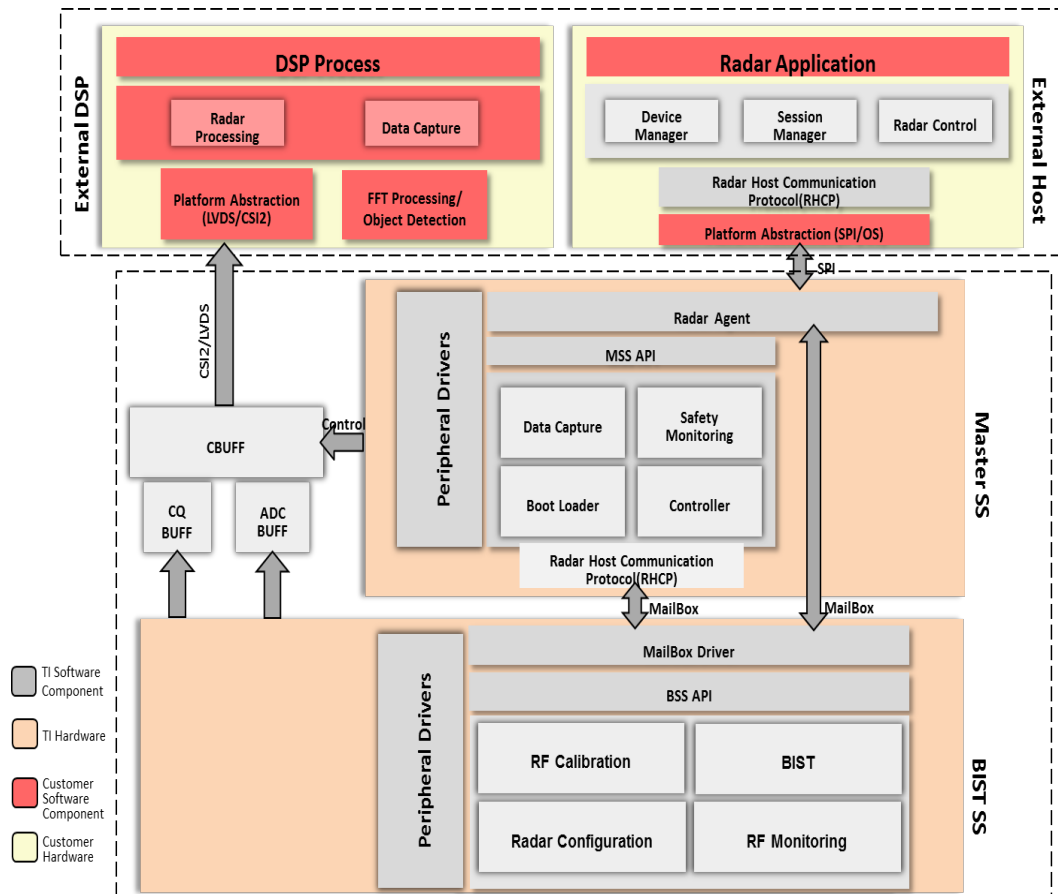


Figure 2.1: AWR12xx, AWR22xx, AWR62xx Software Architecture

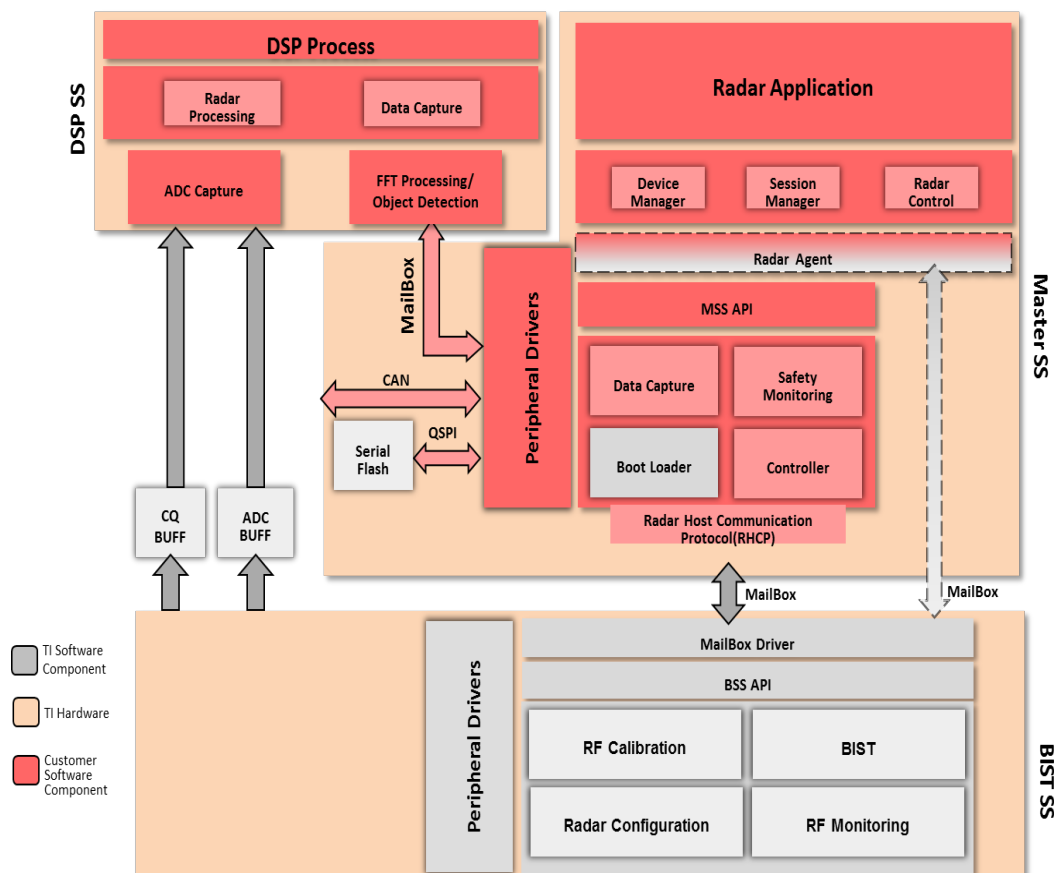


Figure 2.2: xWR16xx, xWR18xx, IWR68xx, xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP Software Architecture

2.3 Radar Message Structure

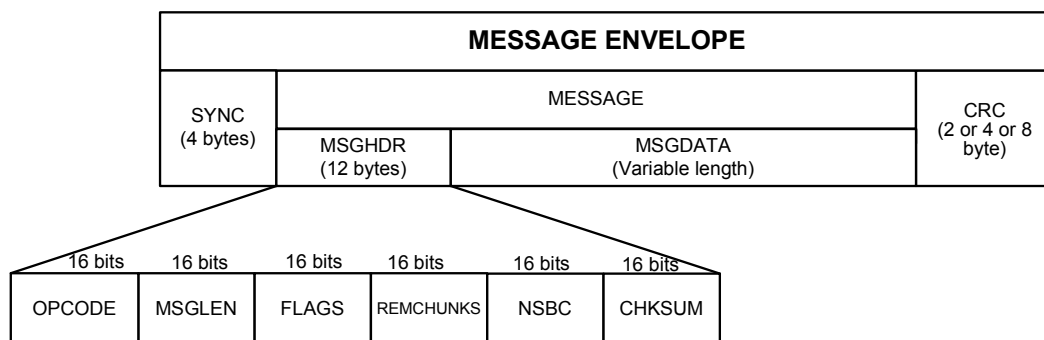


Figure 2.3: Radar Message Structure

Each message is sent in a message envelope, which starts with four special bytes called a sync pattern. Next, the message envelope contains the actual message and a CRC converted to a stream of bytes. Figure 2.3 defines the general form of radar messages. All communication messages between external host processor and the radar device will follow this message format. Each message consists of a 12-byte message header, variable length message data followed by a variable length CRC.

NOTE:

The CRC and all the fields in the message headers and message data that are larger than one byte are sent in little-endian byte order i.e. the least significant byte is sent first.

A message envelope contains only one message.

2.3.1 SYNC

SYNC is a unique 4 byte pattern which marks the start of the message. It can take one of the following 3 values, in memory all the bytes are stored in little endian format (least significant byte first).

2.3.2 MSGHDR

Figure 2.4 defines the content of the message header. Each radar message must begin with this 12 byte message header in little endian format.

OPCODE

The OPCODE is unique for a given message type. Figure 2.5 defines the OPCODE format.

Table 2.7: Possible SYNC values and their usage

SYNC word value	Description
0x43211234	Messages from master to slave indicating a new command
0x87655678	Messages from external host to device indicating the host is now ready to receive a message from the device This pattern is defined as CNYS in this document.
0xABCD DCBA	Messages from slave to master

OPCODE (16 bits)	LENGTH (16 bits)	FLAGS (16 bits)	REMCHUNKS (16 bits)	NSBC (16 bits)	CHKSUM (16 bits)
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Figure 2.4: Message Header Format

Bits	Field	Description
[3:0]	DIRECTION	Direction of command 0000 Invalid 0001 Communication between Host to BSS 0010 Communication between BSS to Host 0011 Communication between Host to DSS 0100 Communication between DSS to Host 0101 Communication between Host to Master 0110 Communication between Master to Host 0111 Communication between BSS to Master 1000 Communication between Master to BSS 1001 Communication between BSS to DSS 1010 Communication between DSS to BSS 1011 Communication between Master to DSS 1100 Communication between DSS to Master 1101 RESERVED 1110 RESERVED 1111 RESERVED
[5:4]	MSGTYPE	Message type 00 COMMAND 01 RESPONSE (ACK or ERROR) 10 NACK 11 ASYNC
[15:6]	MSGID	Message ID 0x00 AWR_ERROR_MSG 0x01 RESERVED 0x02 RESERVED 0x03 RESERVED 0x04 AWR_RF_STATIC_CONF_SET_MSG 0x05 AWR_RF_STATIC_CONF_GET_MSG 0x06 AWR_RF_INIT_MSG 0x07 RESERVED 0x08 AWR_RF_DYNAMIC_CONF_SET_MSG 0x09 AWR_RF_DYNAMIC_CONF_GET_MSG

		0x0C	AWR_RF_ADVANCED_FEATURES_CONF_SET_MSG
		0x0D	RESERVED
		0x0E	AWR_RF_MONITORING_CONF_SET_MSG
		0x0F	RESERVED
		0x10	RESERVED
		0x11	AWR_RF_STATUS_GET_MSG
		0x12	RESERVED
		0x13	AWR_RF_MONITORING_REPORT_GET_MSG
		0x14	AWR_RF_MONITORING_CONF_SET_2_MSG
		0x15	RESERVED
		0x16	AWR_RF_MISC_CONF_SET_MSG
		0x17	AWR_RF_MISC_CONF_GET_MSG
		0x18	RESERVED
		0x19	RESERVED
		0x80	AWR_RF_ASYNC_EVENT_MSG1
		0x81	AWR_RF_ASYNC_EVENT_MSG2
		0x200	AWR_DEV_RFPOWERUP_MSG
		0x201	RESERVED
		0x202	AWR_DEV_CONF_SET_MSG
		0x203	AWR_DEV_CONF_GET_MSG
		0x204	AWR_DEV_FILE_DOWNLOAD_MSG
		0x205	RESERVED
		0x206	AWR_DEV_FRAME_CONFIG_APPLY_MSG
		0x207	AWR_DEV_STATUS_GET_MSG
		0x208	RESERVED
		0x209	RESERVED
		0x20A	RESERVED
		0x20B	RESERVED
		0x20C	RESERVED
		0x20D	RESERVED
		0x280	AWR_DEV_ASYNC_EVENT_MSG

LENGTH

The length field contains the length of the message in bytes including the message header, message data and CRC. Note that length field does not include the length of the sync field. The minimum length of the message is 12 bytes and maximum is 252 bytes. The message length

minus CRC length must also be a multiple of 4 bytes.

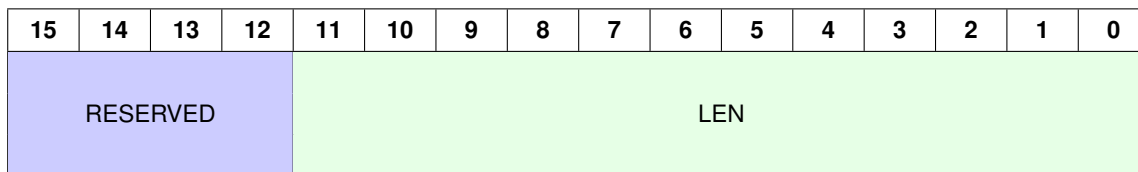


Figure 2.6: MSGLEN Format

Table 2.9: MSGLEN field descriptions

Bits	Field	Description
[11:0]	LEN	Message length in bytes (It includes message header, message data and CRC)
[15:12]	RESERVED	Keep these bits as 0s

FLAGS

The FLAGS is used to control the communication between the radar device and external host

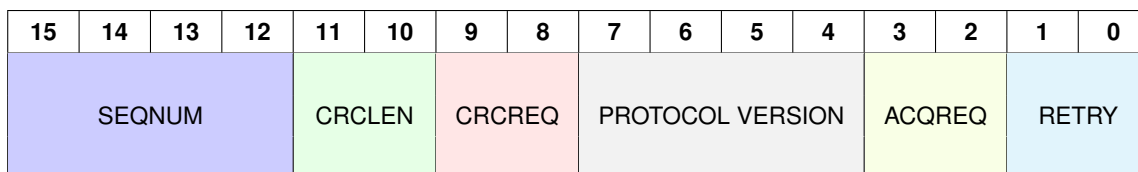


Figure 2.7: FLAGS Format

Table 2.10: FLAGS field description

Bits	Field	Description
[1:0]	RETRY	RETRY Value 00 New message 11 Retransmitted message 01 RESERVED 10 RESERVED

Continued on next page

Table 2.10 – continued from previous page

[3:2]	ACKREQ	Acknowledgement Request type 00 Acknowledgement is requested for the current message 11 Acknowledgement is not requested for the current message 01 RESERVED 10 RESERVED
[7:4]	PROTOCOL VERSION	Version number of the protocol that is used to communicate with the device (4 bits)
[9:8]	CRCREQ	CRC request type 00 CRC is appended to the message 11 CRC is not appended to the message 01 RESERVED 10 RESERVED
[11:10]	CRCLEN	Length of CRC appended to the message 00 16-bit CRC 01 32-bit CRC 10 64-bit CRC 11 RESERVED
[15:12]	SEQNUM	4 bit sequence number of the message. Sequence number is reset to 0 after a device boot and each new message has the incremented sequence number. Whenever the same message is retransmitted, the sequence number is not incremented.

NOTE: It is recommended to always append CRC to the message to prevent any message integrity issues

REMCHUNKS

If the message length is larger than 256 bytes, then it is split into multiple chunks of sizes less than 256 bytes. When this field is non-zero, this field indicates the number of remaining chunks that are to be expected.

NSBC

The message may contain several configuration sub blocks with structure as defined in Figure 2.3. The NSBC field indicates the total number sub blocks inside the message data.

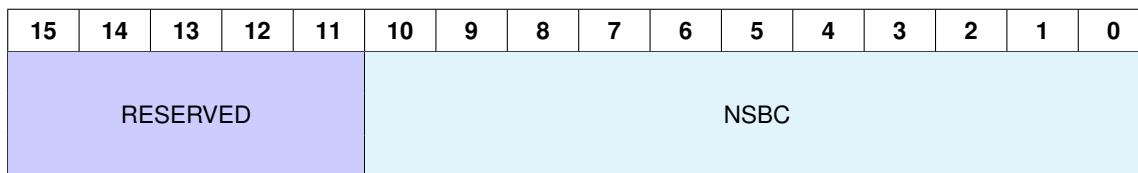


Figure 2.8: NSBC Format

Table 2.11: NSBC field description

Bits	Field	Description
[10:0]	NSBC	Number of sub blocks in the message
[15:11]	RESERVED	Keep these bits as 0s

CHKSUM

The message header is protected by a 16-bit checksum to enable the receiver to check the integrity of the message header. The checksum is computed on MSGHDR only (MSGID, MSGLEN, FLAGS, REMCHUNKS and NSBC fields). Note that SYNC field is not included in checksum calculation.

Checksum is 16-bit one's complement of the one's complement sum of all 16-bit words in the message header (Ref. <https://tools.ietf.org/html/rfc1071>).

For e.g., suppose the message header contents looks like this

Table 2.12: Checksum computation example

Field	Value
OPCODE	0x0281
MSGLEN	0x0800
FLAGS	0x040C
REMCHUNKS	0x0000
NSBC	0x0001
CHKSUM	0xF171

The receiver will compute the checksum as follows $0x0281 + 0x0800 = 0x0A81$.

Then, $0x0A81 + 0x040C = 0x0E8D$.

Then, $0x0E8D + 0x0000 = 0x0E8D$.

Then, $0x0E8D + 0x0001 = 0x0E8E$.

The carry bits generated beyond 16 bits should be added back to result

Ones complement of $0x0E8E$ is $0xF171$ which matches with the received checksum.

2.3.3 MSGDATA

The message data contains the actual message specific data for the message. The message data contains sub blocks with structure as defined in Figure 2.9. More than one sub block can be appended in the MSGDATA to reduce the overall communication latency. The total number of sub blocks in MSGDATA is indicated in the NSBC field in the MSGHDR.

All data fields are aligned so that their offset in message is a multiple of the field size in bytes. For e.g. a 32 bit field in the message will be aligned to a 4 byte boundary and a 16 bit field will be aligned to a 2 byte boundary. This makes it possible to create a structure definition for the message for easy data access in most environments.

Any reserved (currently unused) fields in the messages should be always set as 0 when sent and ignored when received. This way those fields may be taken to use in later interface versions without modifying all old software.

All data structure in sub-blocks assumed to be in little endian format. For big endian Host system byte swap is required to match with defined protocol.

MSGDATA		
SBLKID (16 bits)	SBLKLEN (16 bits)	SBLKDATA (Variable length)

Figure 2.9: Message Sub block structure

SBLKID Unique ID of the sub block

SBLKLEN Length of the sub block in bytes

SBLKDATA Data corresponding to the sub block

2.3.4 CRC

This is a CRC which is appended to the message data to protect the integrity of the message. The CRC is computed on all the bytes in the MSGHDR and MSGDATA. Note that SYNC is not included in CRC calculation.

3 different types of CRCs can be used – 16 bit, 32 bit or 64 bit. The choice of the CRC type is indicated in the FLAGS field in the MSGHDR.

The 32 bit CRC is recommended CRC to be used in SPI protocol.

The polynomials used for each type of CRC calculation are

Table 2.13: CRC types and their polynomials

CRC type	Polynomial	Remarks
16 bit	$x^{16} + x^{12} + x^5 + 1$	16-bit CRC-CCITT
32 bit	$x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$	CRC-32 (used in Ethernet)
64 bit	$x^{64} + x^4 + x^3 + x + 1$	CRC-64-ISO (HDLC)

NOTE: Device SPI protocol Limitation: The CRC length of the message or Async-event shall be multiple of 4 bytes to enable reliable retry recovery mechanism in case of any checksum failure in a message.

3 Message Processing

3.1 Communication protocol

When requested by the message transmitter, all correctly formatted radar messages are acknowledged by the receiver. This request for an acknowledgement is specified in FLAGS field of the MSGHDR (message header) field (see Section 2.3.2). A correctly formatted message is one that is formatted properly with a SYNC, MSGHDR, MSGDATA and CRC and that passes the CRC test when received. If an incorrectly formatted message is received, the radar device responds with a NACK message (MSGTYPE field in the MSGHDR set to NACK response). If a correctly formatted message is received, and after processing the message no errors are encountered, the radar device responds with an ACK response. In case of errors on a correctly formatted message, the radar device responds with an ERROR response.

The ACK response is a radar message which contains SYNC, MSGHDR, MSGDATA and CRC. In case the MSGTYPE was COMMAND_GET the MSGDATA for ACK response will contain the parameter values read by the radar device.

The NACK response is a radar message with only SYNC, MSGHDR and CRC. It does not contain MSGDATA.

For most commands the radar device prepares the acknowledgments and response packets immediately on reception. In certain cases, higher priority events in the system delay the execution of external communication function. The response time to command is a function of:

- Speed of the selected communication channel
- Although typical radar command/response occurs within a few hundreds of microseconds, it is recommended that host software wait up to 1 millisecond for response or acknowledgment before timing out on nonresponse.

The radar communication protocol is defined as follows

1. The host sends a message to the radar device requesting an acknowledgement. Host sets a timeout period of 1 ms for a response from the radar device.
2. The radar device checks the CHKSUM field for Message header validity and checks the MSGDATA field for correctness and also computes the CRC of the message and compares it with the received CRC.
 - If the computed CHKSUM does not match the received CHKSUM, the radar device does not send any response. The transmitter will timeout and eventually resend the command again with RETRY flag set

- If the CRC matches and all parameters are valid/correct, the radar device sends an ACK to the host
 - If the CRC matches, but any parameter in the message is invalid/incorrect, then the radar device sends an ERROR response to the host
 - If the CRC does not match, the radar device sends a NACK response to the host
3. On reception of the ACK, the host can send the next command to the radar device.
 4. If the host receives a NACK from the radar device within the timeout period, it sends the message again without the RETRY flag set.
 5. If the host does not receive any response from the radar device within the timeout period then it sends the same command with the RETRY flag set.

3.2 Communication Sequence

3.2.1 Command/Response Sequence (Host)

1. Host prepares the message as defined by protocol in Section [2.3](#)
2. Host writes the message to the communication channel and starts Retry Timer (~1 ms)
3. Host then waits for HOST IRQ high Interrupt
 - a. If IRQ is received, go to Step 4
 - b. If Retry Time expires, Enable Retry Flag and go to Step 2
4. Host writes CNYS (SYNC word = 0x5678 0x8765) and Dummy bytes (0xFFFF 0xFFFF 0xFFFF 0xFFFF 0xFFFF 0xFFFF) on communication channel
5. Host waits for low on Host IRQ line
 - a. If Host IRQ line is low, go to Step 6
 - b. If Retry Time expires, Flag Error
6. Host reads the header from communication channel
7. Host checks the validity of header (verify checksum)
 - a. If header is valid, parse the header and go to Step 8
 - b. If header is invalid, ignore the header and reports to error to Application
8. Host reads the payload from communication channel
9. Host checks the validity of the message (verify CRC)
 - a. If message is valid, process the message
 - b. If message is invalid, go to Step 2 with new sequence number

3.2.2 Flow Diagram (Host) – Command/Response

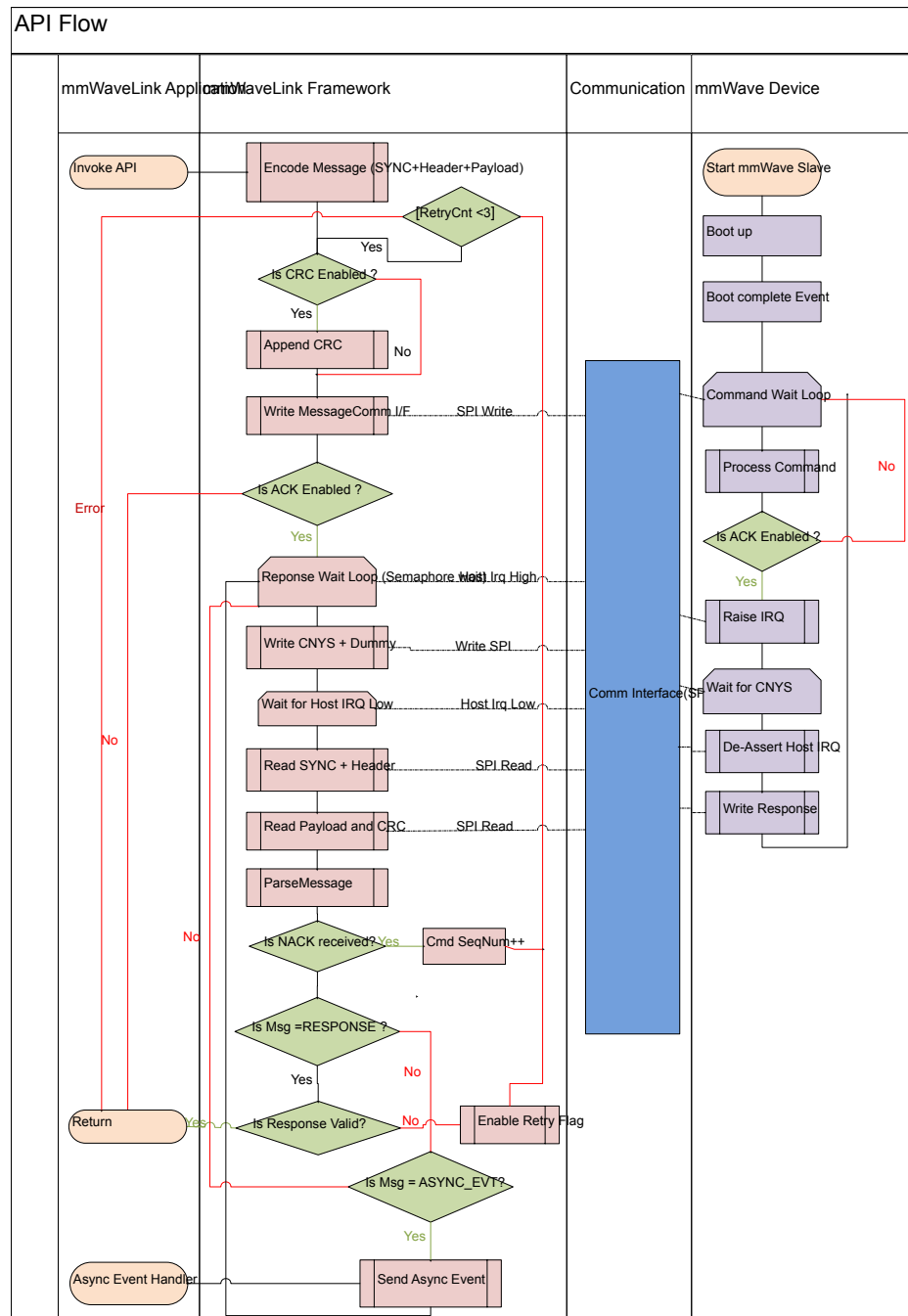


Figure 3.1: Flow Diagram (API)

3.2.3 Flow Diagram (Host) – Bootup/ Asynchronous Event

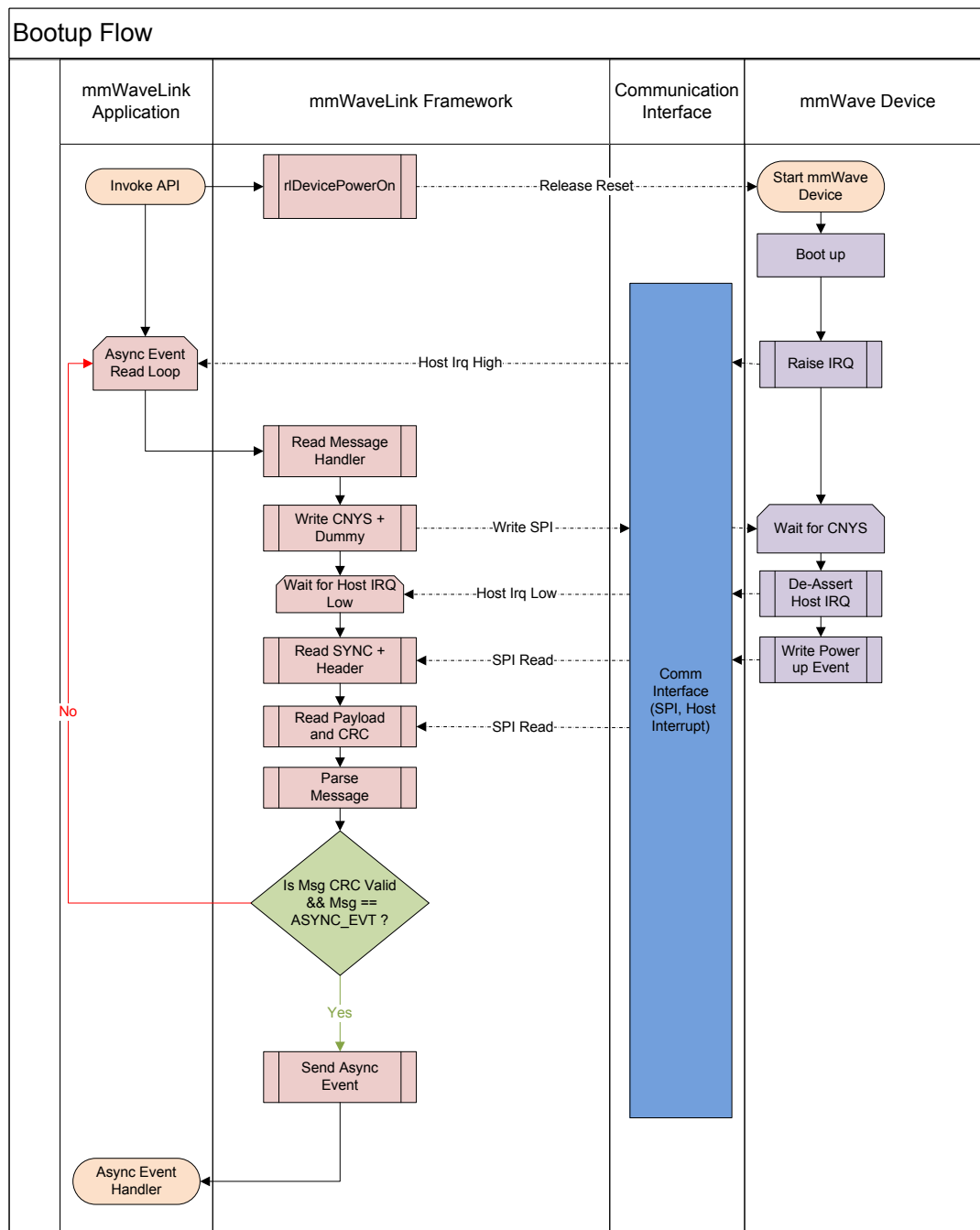


Figure 3.2: Flow Diagram (Asynchronous Events)

3.2.4 SPI Message Sequence – Command/Response

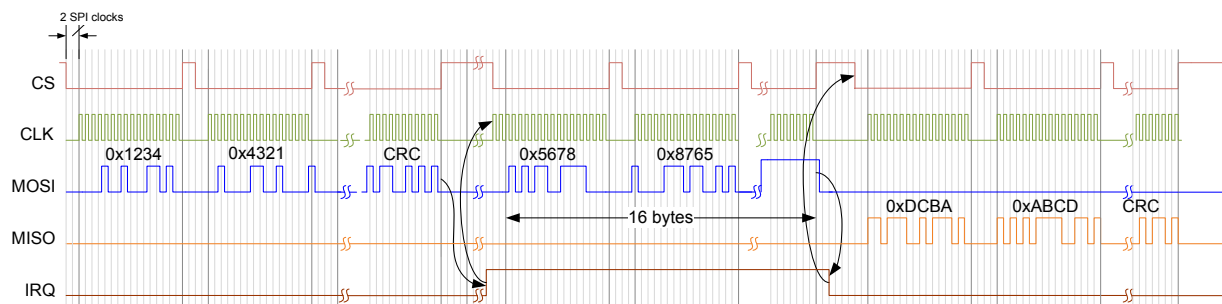


Figure 3.3: SPI Message Sequence

NOTE:

1. Host should ensure that there is a delay of at least 2 SPI clocks between CS going low and start of SPI clock
2. Host should ensure that CS is toggled for every 16 bits of transfer via SPI
3. There should be a delay of at least 2 SPI Clocks between consecutive CS
4. SPI needs to be operated at Mode 0 (Phase 1, Polarity 0)
5. SPI word length should be 16 bit (Half word)

3.2.5 Command/Response sequence (MSS-BSS mailbox)

In single chip devices like xWR16xx, xWR18xx, xWR6843, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP the host application executes from the MSS and/or DSS. Communication between the host application and the RSS/BSS takes place through mailbox SRAMs within the device. This section will describe the communication flow in these devices taking xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP as an example.

1. Host (MSS) prepares the message as defined by protocol in Sec 2.3.
2. MSS writes the message in BSS mailbox.
3. MSS polls `RSS_PROC_CTRL->RSS_CR4_MBOX_READ_REQ` and keeps writing 1 to `MSS_CTRL:MSS_CR5A_MBOX_WRITE_DONE_PROC_X` until the read-value is 1 to trigger an interrupt to BSS.

4. This generates a mailbox interrupt to the BSS.
5. BSS reads RSS_PROC_CTRL->RSS_CR4_MBOX_READ_REQ register and finds the processor that has triggered the interrupt.
6. BSS waits for 250 clock-cycles.
7. BSS clears the interrupt by writing 1 to RSS_PROC_CTRL->RSS_CR4_MBOX_READ_REQ register bit for that particular processor.
8. BSS reads and interprets the message in the mailbox.
9. BSS polls MSS_CTRL : MSS_CR5A_MBOX_READ_DONE and keeps writing 1 to RSS_PROC_CTRL : RSS_CR4_MBOX_READ_DONE_ACK register bit corresponding to that particular processor until the read-value is 1 to trigger an interrupt to MSS.
10. This generates a mailbox interrupt to the MSS.
11. MSS reads the MSS_CTRL : MSS_CR5A_MBOX_READ_DONE to check which processor has interrupted it.
12. MSS waits for 250 clock-cycles.
13. MSS clears the interrupt by writing 0x1 to MSS_CTRL : MSS_CR5A_MBOX_READ_DONE register bit.

3.2.6 Flow Diagram Command/Response (MSS-BSS mailbox)

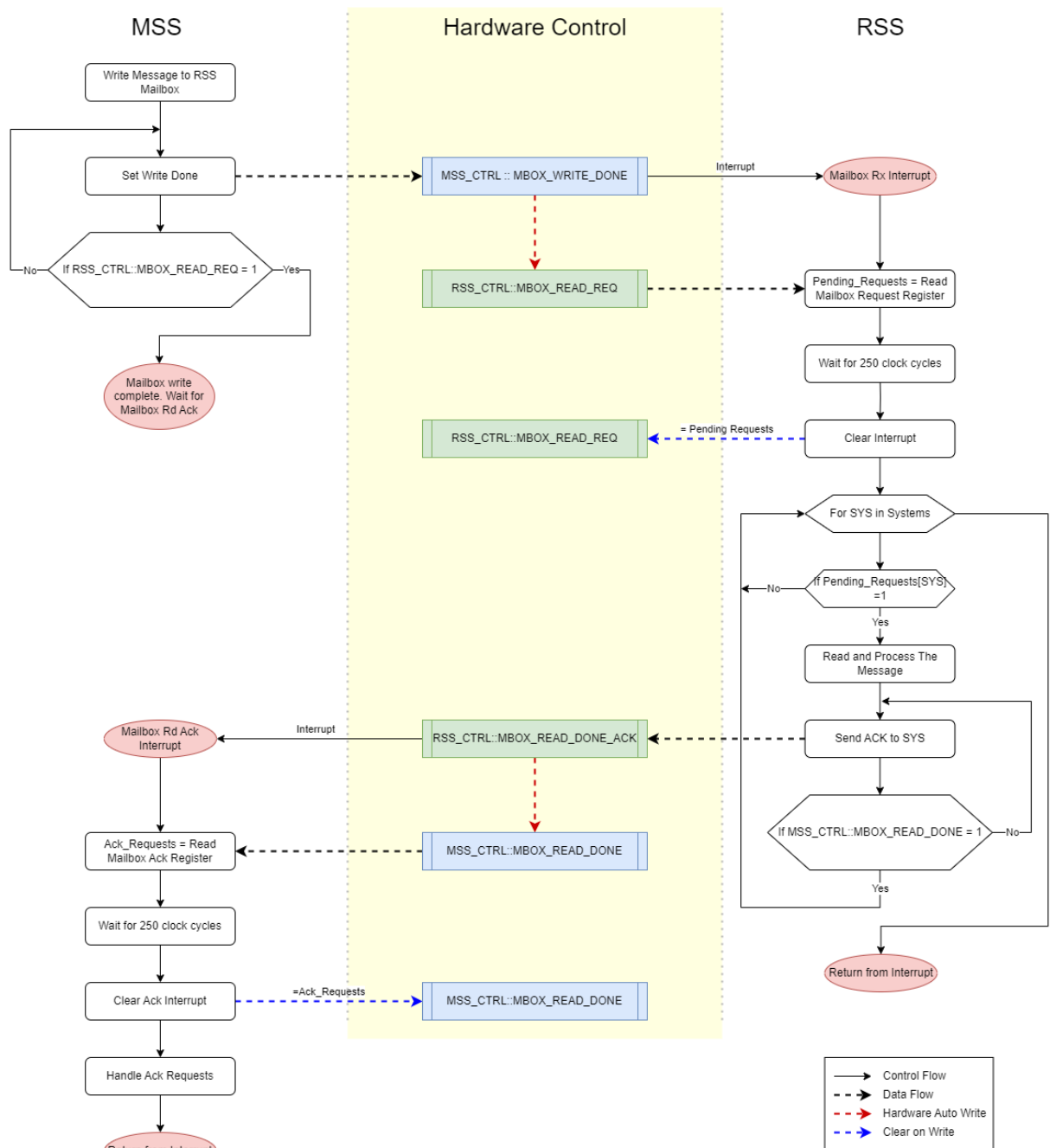


Figure 3.4: Command/Response Flow Diagram

3.2.7 Asynchronous event sequence (BSS-MSS mailbox)

This section will describe the Asynchronous Event communication flow from BSS to the MSS (Host) in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP as an example.

1. BSS prepares the ASYNC event report message as defined by protocol.
2. BSS writes the message in MSS mailbox.
3. BSS polls MSS_CTRL->MSS_CR5A_MBOX_READ_REQ and keeps on writing 1 to RSS_PROC_CTRL:RSS_CR4_MBOX_WRITE_DONE_PROC_X until MSS_CTRL->MSS_CR5A_MBOX_READ_REQ is 1 to trigger an interrupt to MSS.
4. This generates a mailbox interrupt to the MSS.
5. MSS reads MSS_CTRL->MSS_CR5A_MBOX_READ_REQ register and finds the processor that has triggered the interrupt.
6. MSS waits for 250 clock cycles.
7. MSS clears the interrupt by writing 1 to MSS_CTRL->MSS_CR5A_MBOX_READ_REQ register bit for that particular processor.
8. MSS reads and interprets the ASYNC event message in the mailbox.
9. MSS polls RSS_PROC_CTRL : RSS_CR4_MBOX_READ_DONE and keeps on writing 1 to MSS_CTRL : MSS_CR5A_MBOX_READ_DONE_ACK until RSS_PROC_CTRL : RSS_CR4_MBOX_READ_DONE is 1 to trigger an interrupt to BSS.
10. This generates a mailbox interrupt to the BSS.
11. BSS reads the RSS_PROC_CTRL : RSS_CR4_MBOX_READ_DONE to check which processor has interrupted it.
12. BSS waits for 250 clock cycles.
13. BSS clears the interrupt by writing 0x1 to RSS_PROC_CTRL : RSS_CR4_MBOX_READ_DONE register bit.

3.2.8 Flow Diagram Asynchronous event (BSS-MSS mailbox)

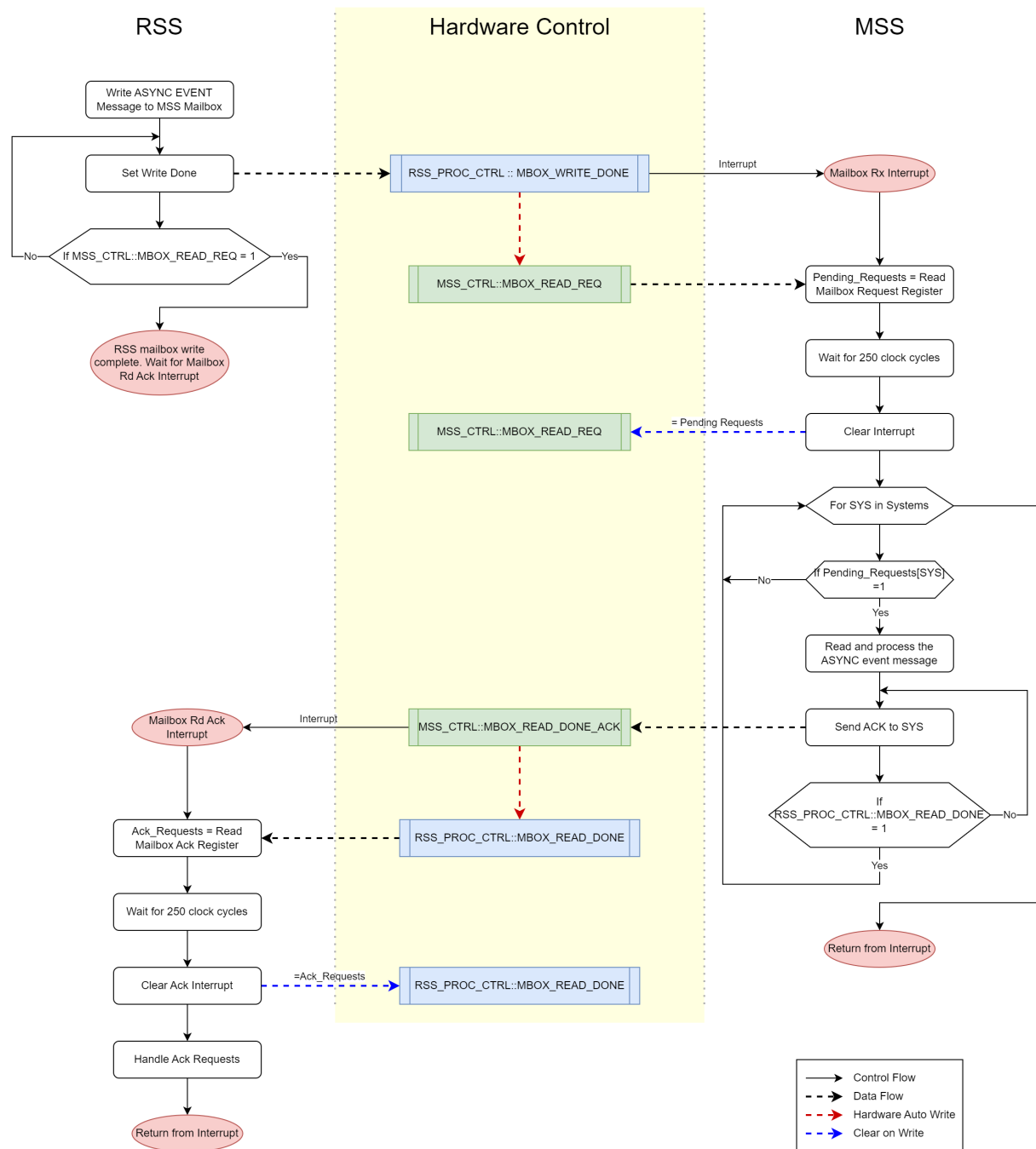


Figure 3.5: ASYNC Event Flow Diagram

**Application
Care Abouts:**

1. Retry of RF Power up message is unsupported.
2. HOST is recommended to wait for RF Power Async msg before any further APIs are issued. Lack of RF Power up Async msg should be treated as bootup failure.
3. It is recommended to wait for Async event for Latent fault injection API before the next CMD is issued.
4. HOST to ensure a delay of 30us in response to the HOST_IRQ interrupt, to allow for a SPI DMA configuration in device post HOST_IRQ set high.
5. It is recommended to use 232 as the chunk size in mmWavelink/HOST when firmware download is done through SPI.

4 Radar Interface Messages Descriptions

This section describes all the radar interface messages that are used in communication with the radar transceiver. Depending on the architecture of the device, some of the messages might not be supported. Messages towards/from the MSS firmware are only supported in front-end devices like xWR12/xWR22/xWR62xx. In single-chip devices (xWR16/xWR18/xWR68/xWR29xx), these messages are not supported and can only be used as a reference for implementation in the MSS application by the user.

These messages are categorized based on type of messages and each message consist of multiple configuration sub-blocks. Each sub-block does a unique configuration of the device, they are grouped as static and dynamic messages. The Async Event (AE) response sub-blocks generated in the device are grouped as AE messages.

4.1 Summary of all messages and their associated sub-blocks

Table 4.1: Summary of common Radar messages in both BSS and MSS and their associated sub blocks

Radar Messages	Associated sub-blocks
AWR_ACK_MSG	NA
AWR_NACK_MSG	NA
AWR_ERROR_MSG	AWR_RESP_ERROR_SB

Table 4.2: Summary of all Radar messages in the BSS and their associated sub blocks

Radar Messages	Associated sub-blocks
AWR_RF_STATIC_CONF_SET_MSG	AWR_CHAN_CONF_SET_SB AWR_ADCOUT_CONF_SET_SB AWR_LOWPOWERMODE_CONF_SET_SB AWR_DYNAMICPOWERSAVE_CONF_SET_SB AWR_HIGHSPEEDINTFCLK_CONF_SET_SB AWR_RF_DEVICE_CFG_SB AWR_RF_RADAR_MISC_CTL_SB AWR_CAL_MON_FREQUENCY_LIMITS_SB AWR_RF_INIT_CALIBRATION_CONF_SB AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB

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Table 4.2 – continued from previous page

Radar Messages	Associated sub-blocks
	AWR_CAL_DATA_RESTORE_SB AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB AWR_APLL_SYNTH_BW_CONTROL_SB
AWR_RF_STATIC_CONF_GET_MSG	AWR_CAL_DATA_SAVE_SB AWR_PHASE_SHIFTER_CAL_DATA_SAVE_SB
AWR_RF_INIT_MSG	AWR_RF_INIT_SB
AWR_RF_DYNAMIC_CONF_SET_MSG	AWR_PROFILE_CONF_SET_SB AWR_CHIRP_CONF_SET_SB AWR_FRAME_CONF_SET_SB AWR_CONT_STREAMING_MODE_CONF_SET_SB AWR_CONT_STREAMING_MODE_EN_SB AWR_ADVANCED_FRAME_CONF_SB AWR_PERCHIRPPHASESHIFT_CONF_SB AWR_PROG_FILT_COEFF_RAM_SET_SB AWR_PROG_FILT_CONF_SET_SB AWR_CALIB_MON_TIME_UNIT_CONF_SB AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB AWR_DIGITAL_COMP_EST_CONTROL_SB AWR_RX_GAIN_TEMPLUT_SET_SB AWR_TX_GAIN_TEMPLUT_SET_SB AWR_LOOPBACK_BURST_CONF_SET_SB AWR_DYN_CHIRP_CONF_SET_SB AWR_DYN_PERCHIRP_PHASESHIFTER_CONF_SB AWR_DYN_CHIRP_ENABLE_SB AWR_INTERCHIRP_BLOCKCONTROLS_SB AWR_SUBFRAME_START_CONF_SB AWR_ADVANCE_CHIRP_CONF_SB AWR_ADVANCE_CHIRP_GENERIC_LUT_LOAD_SB AWR_MONITOR_TYPE_TRIG_CONF_SB AWR_ADVANCE_CHIRP_DYN_LUT_ADDR_OFFSET_CFG_SB
AWR_RF_DYNAMIC_CONF_GET_MSG	AWR_PROFILE_CONF_GET_SB AWR_CHIRP_CONF_GET_SB

Continued on next page

Table 4.2 – continued from previous page

Radar Messages	Associated sub-blocks
	AWR_FRAME_CONF_GET_SB AWR_ADVANCED_FRAME_CONF_GET_SB AWR_RX_GAIN_TEMPLUT_GET_SB AWR_TX_GAIN_TEMPLUT_GET_SB
AWR_RF_FRAME_TRIG_MSG	AWR_FRAMESTARTSTOP_CONF_SB
AWR_RF_ADVANCED_FEATURES_CONF_SET_MSG	AWR_BPM_COMMON_CONF_SET_SB AWR_BPM_CHIRP_CONF_SET_SB
AWR_RF_MONITORING_CONF_SET_MSG	AWR_MONITOR_RF_DIG_LATENTFAULT_CONF_SB AWR_MONITOR_RF_DIG_PERIODIC_CONF_SB AWR_MONITOR_ANALOG_ENABLES_CONF_SB AWR_MONITOR_TEMPERATURE_CONF_SB AWR_MONITOR_RX_GAIN_PHASE_CONF_SB AWR_MONITOR_RX_NOISE_FIGURE_CONF_SB AWR_MONITOR_RX_IFSTAGE_CONF_SB AWR_MONITOR_TX0_POWER_CONF_SB AWR_MONITOR_TX1_POWER_CONF_SB AWR_MONITOR_TX2_POWER_CONF_SB AWR_MONITOR_TX0_BALLBREAK_CONF_SB AWR_MONITOR_TX1_BALLBREAK_CONF_SB AWR_MONITOR_TX2_BALLBREAK_CONF_SB AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_CONF_SB AWR_MONITOR_TX0_PHASE_SHIFTER_CONF_SB AWR_MONITOR_TX1_PHASE_SHIFTER_CONF_SB AWR_MONITOR_TX2_PHASE_SHIFTER_CONF_SB AWR_MONITOR_SYNTHESIZER_FREQUENCY_CONF_SB AWR_MONITOR_EXTERNAL_ANALOG_SIGNALS_CONF_SB AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_CONF_SB AWR_MONITOR_TX1_INTERNAL_ANALOG_SIGNALS_CONF_SB AWR_MONITOR_TX2_INTERNAL_ANALOG_SIGNALS_CONF_SB AWR_MONITOR_RX_INTERNAL_ANALOG_SIGNALS_CONF_SB

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Table 4.2 – continued from previous page

Radar Messages	Associated sub-blocks
	AWR_MONITOR_PMCLKLO_INTERNAL_ANALOG_SIGNALS_CONF_SB AWR_MONITOR_GPADC_INTERNAL_ANALOG_SIGNALS_CONF_SB AWR_MONITOR_PLL_CONTROL_VOLTAGE_SIGNALS_CONF_SB AWR_MONITOR_DUAL_CLOCK_COMP_CONF_SB AWR_MONITOR_RX_SATURATION_DETECTOR_CONF_SB AWR_MONITOR_SIG_IMG_MONITOR_CONF_SB AWR_MONITOR_RX_MIXER_IN_POWER_CONF_SB AWR_ANALOG_FAULT_INJECTION_CONF_SB
AWR_RF_MONITORING_CONF_SET_2_MSG	AWR_MONITOR_TXN_POWER_CONF_SB AWR_MONITOR_TXN_BALLBREAK_CONF_SB AWR_MONITOR_TXN_INTERNAL_ANALOG_SIGNALS_CONF_SB AWR_MONITOR_TXN_PHASE_SHIFTER_CONF_SB AWR_MONITOR_ADV_TX_GAIN_PHASE_MISMATCH_CONF_SB AWR_MONITOR_VMON_CONF_SB AWR_MONITOR_OVERRIDES_AND_DITHER_CONF_SB AWR_MONITOR_TX_PHSHIFTER_DAC_CONF_SB
AWR_RF_MONITORING_REPORT_GET_MSG	AWR_RF_DFE_STATISTICS_REPORT_GET_SB
AWR_RF_STATUS_GET_MSG	AWR_RF_VERSION_GET_SB AWR_RF_CPUFAULT_STATUS_GET_SB AWR_RF_ESMFAULT_STATUS_GET_SB AWR_RF_DIEID_GET_SB AWR_RF_BOOTUPBIST_STATUS_GET_SB AWR_RF_ADV_ESMFAULT_STATUS_GET_SB
AWR_RF_MISC_CONF_SET_MSG	AWR_RF_TEST_SOURCE_CONFIG_SET_SB AWR_RF_TEST_SOURCE_ENABLE_SET_SB AWR_RF_LDO_BYPASS_SB AWR_RF_PALOOPBACK_CFG_SB AWR_RF_PSLOOPBACK_CFG_SB AWR_RF_IFLOOPBACK_CFG_SB AWR_RF_GPADC_CFG_SET_SB

Continued on next page

Table 4.2 – continued from previous page

Radar Messages	Associated sub-blocks
	AWR_RF_LOLOOPBACK_CFG_SB
AWR_RF_MISC_CONF_GET_MSG	AWR_RF_TEMPERATURE_GET_SB
AWR_RF_ASYNC_EVENT_MSG1	AWR_AE_RF_CPUFAULT_SB AWR_AE_RF_ESMFAULT_SB AWR_AE_RF_INITCALIBSTATUS_SB AWR_AE_RF_MONITOR_TYPE_TRIGGER_DONE_SB AWR_AE_RF_FRAME_TRIGGER_RDY_SB AWR_AE_RF_GPADC_RESULT_DATA_SB AWR_FRAME_END_AE_SB AWR_ANALOGFAULT_AE_SB AWR_CAL_MON_TIMING_FAIL_REPORT_AE_SB AWR_RUN_TIME_CALIBRATION_SUMMARY_REPORT_AE_SB AWR_MONITOR_RF_DIG_LATENTFAULT_REPORT_AE_SB AWR_MONITOR_REPORT_HEADER_AE_SB AWR_MONITOR_RF_DIG_PERIODIC_REPORT_AE_SB AWR_MONITOR_TEMPERATURE_REPORT_AE_SB AWR_MONITOR_RX_GAIN_PHASE_REPORT_AE_SB AWR_MONITOR_RX_NOISE_FIGURE_REPORT_AE_SB AWR_MONITOR_RX_IFSTAGE_REPORT_AE_SB AWR_MONITOR_TX0_POWER_REPORT_AE_SB AWR_MONITOR_TX1_POWER_REPORT_AE_SB AWR_MONITOR_TX2_POWER_REPORT_AE_SB AWR_MONITOR_TX0_BALLBREAK_REPORT_AE_SB AWR_MONITOR_TX1_BALLBREAK_REPORT_AE_SB
AWR_RF_ASYNC_EVENT_MSG2	AWR_MONITOR_TX2_BALLBREAK_REPORT_AE_SB AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_REPORT_AE_SB AWR_MONITOR_TX0_PHASE_SHIFTER_REPORT_AE_SB AWR_MONITOR_TX1_PHASE_SHIFTER_REPORT_AE_SB AWR_MONITOR_TX2_PHASE_SHIFTER_REPORT_AE_SB AWR_MONITOR_SYNTHESIZER_FREQUENCY_REPORT_AE_SB AWR_MONITOR_EXTERNAL_ANALOG_SIGNALSREPORT_AE_SB

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Table 4.2 – continued from previous page

Radar Messages	Associated sub-blocks
	AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB AWR_MONITOR_TX1_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB AWR_MONITOR_TX2_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB AWR_MONITOR_RX_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB AWR_MONITOR_PMCLKLO_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB AWR_MONITOR_GPADC_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB AWR_MONITOR_PLL_CONTROL_VOLTAGE_REPORT_AE_SB AWR_MONITOR_DUAL_CLOCK_COMP_REPORT_AE_SB AWR_MONITOR_RX_MIXER_IN_POWER_REPORT_AE_SB AWR_MONITOR_SYNTHESIZER_FREQUENCY_NONLIVE_REPORT_AE_SB

Table 4.3: Summary of Radar messages in the MSS of front-end devices (12xx/22xx/62xx) and their associated sub blocks

Radar Messages	Associated sub-blocks
AWR_DEV_RFPOWERUP_MSG	AWR_DEV_RFPOWERUP_SB
AWR_DEV_CONF_SET_MSG	AWR_DEV_MCUCLOCK_CONF_SET_SB AWR_DEV_RX_DATA_FORMAT_CONF_SET_SB AWR_DEV_RX_DATA_PATH_CONF_SET_SB AWR_DEV_RX_DATA_PATH_LANEEN_SET_SB AWR_DEV_RX_DATA_PATH_CLK_SET_SB AWR_DEV_LVDS_CFG_SET_SB AWR_DEV_RX_CONTSTREAMING_MODE_CONF_SET_SB AWR_DEV_CSI2_CFG_SET_SB AWR_DEV_PMICCLOCK_CONF_SET_SB AWR_MSS_LATENTFAULT_TEST_CONF_SB AWR_MSS_PERIODICTESTS_CONF_SB AWR_DEV_TESTPATTERN_GEN_SET_SB AWR_DEV_CONFIGURATION_SET_SB AWR_DEV_RF_DEBUG_SIG_SET_SB

Continued on next page

Table 4.3 – continued from previous page

Radar Messages	Associated sub-blocks
	AWR_DEV_DEV_HSI_DELAY_DUMMY_CFG_SET_SB
AWR_DEV_CONF_GET_MSG	AWR_DEV_MCUCLOCK_GET_SB AWR_DEV_RX_DATA_FORMAT_CONF_GET_SB AWR_DEV_RX_DATA_PATH_CONF_GET_SB AWR_DEV_RX_DATA_PATH_LANEEN_GET_SB AWR_DEV_RX_DATA_PATH_CLK_GET_SB AWR_DEV_LVDS_CFG_GET_SB AWR_DEV_RX_CONTSTREAMING_MODE_CONF_GET_SB AWR_DEV_CSI2_CFG_GET_SB AWR_DEV_PMICCLOCK_CONF_GET_SB AWR_MSS_LATENTFAULT_TEST_CONF_GET_SB AWR_MSS_PERIODICCONF_GET_SB AWR_DEV_TESTPATTERN_GEN_GET_SB
AWR_DEV_FILE_DOWNLOAD_MSG	AWR_DEV_FILE_DOWNLOAD_SB
AWR_DEV_FRAME_CONFIG_APPLY_MSG	AWR_DEV_FRAME_CONFIG_APPLY_SB AWR_DEV_ADV_FRAME_CONFIG_APPLY_SB
AWR_DEV_STATUS_GET_MSG	AWR_MSSVERSION_GET_SB AWR_MSSCPUFAULT_STATUS_GET_SB AWR_MSESFAULT_STATUS_GET_SB
AWR_DEV_ASYNC_EVENT_MSG	AWR_AE_DEV_MSSPOWERUPDONE_SB AWR_AE_DEV_RFPOWERUPDONE_SB AWR_AE_MSS_CPUFAULT_SB AWR_AE_MSS_ESMFAULT_SB AWR_AE_MSS_BOOTERRORSTATUS_SB AWR_AE_MSS_LATENTFAULT_TESTREPORT_SB AWR_AE_MSS_PERIODICTEST_STATUS_SB AWR_AE_MSS_RFERROR_STATUS_SB

4.2 AWR_ACK_MSG

The AWR_ACK_MSG is sent by the radar transceiver on a successful reception of a command after its CRC check.

ACK messages are sent out for every command from the device. In case of any error AWR_ERROR_MSG sub-block will be sent out part of ACK message.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0xABCDDCBA												
OPCODE	2	<table border="1"> <thead> <tr> <th>Bits</th> <th>Variable name</th> <th>Value</th> </tr> </thead> <tbody> <tr> <td>b3:0</td> <td>DIRECTION</td> <td>See Table 2.8</td> </tr> <tr> <td>b5:4</td> <td>MSGTYPE</td> <td>01</td> </tr> <tr> <td>b15:6</td> <td>MSGID</td> <td>Same as MSGID in the command</td> </tr> </tbody> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	01	b15:6	MSGID	Same as MSGID in the command
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	01												
b15:6	MSGID	Same as MSGID in the command												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
CRC	Variable	Based on CRCLEN field in FLAGS												

4.3 AWR_NACK_MSG

The AWR_NACK_MSG is sent by the radar transceiver if the CRC check of the command fails.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0xABCDDCBA												
OPCODE	2	<table border="1"> <thead> <tr> <th>Bits</th> <th>Variable name</th> <th>Value</th> </tr> </thead> <tbody> <tr> <td>b3:0</td> <td>DIRECTION</td> <td>See Table 2.8</td> </tr> <tr> <td>b5:4</td> <td>MSGTYPE</td> <td>10</td> </tr> <tr> <td>b15:6</td> <td>MSGID</td> <td>Same as MSGID in the command</td> </tr> </tbody> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	10	b15:6	MSGID	Same as MSGID in the command
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	10												
b15:6	MSGID	Same as MSGID in the command												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
CRC	Variable	Based on CRCLEN field in FLAGS												

4.4 AWR_ERROR_MSG

The AWR_RF_ERROR_MSG is sent by the radar transceiver on finding errors in the command send by host.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0xABCDDCBA												
OPCODE	2	<table border="1"> <thead> <tr> <th>Bits</th> <th>Variable name</th> <th>Value</th> </tr> </thead> <tbody> <tr> <td>b3:0</td> <td>DIRECTION</td> <td>See Table 2.8</td> </tr> <tr> <td>b5:4</td> <td>MSGTYPE</td> <td>01</td> </tr> <tr> <td>b15:6</td> <td>MSGID</td> <td>0x00</td> </tr> </tbody> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	01	b15:6	MSGID	0x00
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	01												
b15:6	MSGID	0x00												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
MSGDATA	Variable	Supported sub blocks AWR_RESP_ERROR_SB												
CRC	Variable	Based on CRCLEN field in FLAGS												

4.5 AWR_RF_STATIC_CONF_SET_MSG

Static configuration sub-blocks are grouped as static messages. These messages are mostly static settings shall be configured once in radar transceiver after power cycle.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0x43211234												
OPCODE	2	<table border="1"> <thead> <tr> <th>Bits</th> <th>Variable name</th> <th>Value</th> </tr> </thead> <tbody> <tr> <td>b3:0</td> <td>DIRECTION</td> <td>See Table 2.8</td> </tr> <tr> <td>b5:4</td> <td>MSGTYPE</td> <td>00</td> </tr> <tr> <td>b15:6</td> <td>MSGID</td> <td>0x04</td> </tr> </tbody> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	00	b15:6	MSGID	0x04
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	00												
b15:6	MSGID	0x04												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
MSGDATA	Variable	Supported sub blocks AWR_CHAN_CONF_SET_SB												

		AWR_ADCOUT_CONF_SET_SB AWR_LOWPOWERMODE_CONF_SET_SB AWR_DYNAMICPOWERSAVE_CONF_SET_SB AWR_HIGHSPEEDINTFCLK_CONF_SET_SB AWR_RF_DEVICE_CFG_SB AWR_RF_RADAR_MISC_CTL_SB AWR_CAL_MON_FREQUENCY_LIMITS_SB AWR_RF_INIT_CALIBRATION_CONF_SB AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB AWR_CAL_DATA_RESTORE_SB AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB AWR_APLL_SYNTH_BW_CONTROL_SB
CRC	Variable	Based on CRCLEN field in FLAGS

4.6 AWR_RF_STATIC_CONF_GET_MSG

Static GET messages can be used to read the static configuration settings from the radar transceiver.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0x43211234												
OPCODE	2	<table border="1"> <thead> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> </thead> <tbody> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>00</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x05</td></tr> </tbody> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	00	b15:6	MSGID	0x05
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	00												
b15:6	MSGID	0x05												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
MSGDATA	Variable	Supported sub blocks AWR_CAL_DATA_SAVE_SB AWR_PHASE_SHIFTER_CAL_DATA_SAVE_SB												
CRC	Variable	Based on CRCLEN field in FLAGS												

4.7 AWR_RF_INIT_MSG

RF initialization message does the boot time calibration of the radar transceiver.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0x43211234												
OPCODE	2	<table> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>00</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x06</td></tr> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	00	b15:6	MSGID	0x06
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	00												
b15:6	MSGID	0x06												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
MSGDATA	Variable	Supported sub blocks AWR_RF_INIT_SB												
CRC	Variable	Based on CRCLEN field in FLAGS												

4.8 AWR_RF_DYNAMIC_CONF_SET_MSG

Dynamic configuration sub-blocks are grouped as dynamic messages. These messages are mostly dynamic settings configures the radar transceiver profiles, chirp and frames (waveform), these settings can be updated dynamically to achieve the dynamic waveform generation.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0x43211234												
OPCODE	2	<table> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>00</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x08</td></tr> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	00	b15:6	MSGID	0x08
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	00												
b15:6	MSGID	0x08												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
MSGDATA	Variable	Supported sub blocks												

		AWR_PROFILE_CONF_SET_SB AWR_CHIRP_CONF_SET_SB AWR_FRAME_CONF_SET_SB AWR_CONT_STREAMING_MODE_CONF_SET_SB AWR_CONT_STREAMING_MODE_EN_SB AWR_ADVANCED_FRAME_CONF_SB AWR_PERCHIRPPHASESHIFT_CONF_SB AWR_PROG_FILT_COEFF_RAM_SET_SB AWR_PROG_FILT_CONF_SET_SB AWR_CALIB_MON_TIME_UNIT_CONF_SB AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB AWR_DIGITAL_COMP_EST_CONTROL_SB AWR_RX_GAIN_TEMPLUT_SET_SB AWR_TX_GAIN_TEMPLUT_SET_SB AWR_LOOPBACK_BURST_CONF_SET_SB AWR_DYN_CHIRP_CONF_SET_SB AWR_DYN_PERCHIRP_PHASESHIFTER_CONF_SB AWR_DYN_CHIRP_ENABLE_SB AWR_INTERCHIRP_BLOCKCONTROLS_SB AWR_SUBFRAME_START_CONF_SB AWR_ADVANCE_CHIRP_CONF_SB AWR_ADVANCE_CHIRP_GENERIC_LUT_LOAD_SB AWR_MONITOR_TYPE_TRIG_CONF_SB AWR_ADVANCE_CHIRP_DYN_LUT_ADDR_OFFSET_CFG_SB
CRC	Variable	Based on CRCLEN field in FLAGS

4.9 AWR_RF_DYNAMIC_CONF_GET_MSG

Dynamic GET messages can be used to read the dynamic configuration settings from the radar transceiver.

Field Name	Number of bytes	Description
------------	-----------------	-------------

SYNC	4	Value = 0x43211234												
OPCODE	2	<table> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>00</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x09</td></tr> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	00	b15:6	MSGID	0x09
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	00												
b15:6	MSGID	0x09												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
MSGDATA	Variable	Supported sub blocks AWR_PROFILE_CONF_GET_SB AWR_CHIRP_CONF_GET_SB AWR_FRAME_CONF_GET_SB AWR_ADVANCED_FRAME_CONF_GET_SB AWR_RX_GAIN_TEMPLUT_GET_SB AWR_TX_GAIN_TEMPLUT_GET_SB												
CRC	Variable	Based on CRCLen field in FLAGS												

4.10 AWR_RF_FRAME_TRIG_MSG

Frame trigger message for the radar transceiver to start the waveform.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0x43211234												
OPCODE	2	<table> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>00</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x0A</td></tr> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	00	b15:6	MSGID	0x0A
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	00												
b15:6	MSGID	0x0A												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												

CHKSUM	2	See Section 2.3.2
MSGDATA	Variable	Supported sub blocks AWR_FRAMESTARTSTOP_CONF_SB
CRC	Variable	Based on CRCLLEN field in FLAGS

4.11 AWR_RF_ADVANCED_FEATURES_CONF_SET_MSG

Advance configuration messages for radar transceiver.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0x43211234												
OPCODE	2	<table> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>00</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x0C</td></tr> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	00	b15:6	MSGID	0x0C
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	00												
b15:6	MSGID	0x0C												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
MSGDATA	Variable	Supported sub blocks AWR_BPM_COMMON_CONF_SET_SB AWR_BPM_CHIRP_CONF_SET_SB												
CRC	Variable	Based on CRCLLEN field in FLAGS												

4.12 AWR_RF_MONITORING_CONF_SET_MSG

Monitoring configuration message sub-blocks for radar transceiver.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0x43211234												
OPCODE	2	<table> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>00</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x0E</td></tr> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	00	b15:6	MSGID	0x0E
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	00												
b15:6	MSGID	0x0E												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
MSGDATA	Variable	Supported sub blocks AWR_MONITOR_RF_DIG_LATENTFAULT_CONF_SB AWR_MONITOR_RF_DIG_PERIODIC_CONF_SB AWR_MONITOR_ANALOG_ENABLES_CONF_SB AWR_MONITOR_TEMPERATURE_SONF_SB AWR_MONITOR_RX_GAIN_PHASE_CONF_SB AWR_MONITOR_RX_NOISE_FIGURE_CONF_SB AWR_MONITOR_RX_IFSTAGE_CONF_SB AWR_MONITOR_TX0_POWER_CONF_SB AWR_MONITOR_TX1_POWER_CONF_SB AWR_MONITOR_TX2_POWER_CONF_SB AWR_MONITOR_TX0_BALLBREAK_CONF_SB AWR_MONITOR_TX1_BALLBREAK_CONF_SB AWR_MONITOR_TX2_BALLBREAK_CONF_SB AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_CONF_SB AWR_MONITOR_TX0_PHASE_SHIFTER_CONF_SB AWR_MONITOR_TX1_PHASE_SHIFTER_CONF_SB AWR_MONITOR_TX2_PHASE_SHIFTER_CONF_SB AWR_MONITOR_SYNTHESIZER_FREQUENCY_CONF_SB AWR_MONITOR_EXTERNAL_ANALOG_SIGNALS_CONF_SB AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_CONF_SB												

		AWR_MONITOR_TX1_INTERNAL_ANALOG_SIGNALS_CONF_SB AWR_MONITOR_TX2_INTERNAL_ANALOG_SIGNALS_CONF_SB AWR_MONITOR_RX_INTERNAL_ANALOG_SIGNALS_CONF_SB AWR_MONITOR_PMCLKLO_INTERNAL_ANALOG_SIGNALS_CONF_SB AWR_MONITOR_GPADC_INTERNAL_ANALOG_SIGNALS_CONF_SB AWR_MONITOR_PLL_CONTROL_VOLTAGE_SIGNALS_CONF_SB AWR_MONITOR_DUAL_CLOCK_COMP_CONF_SB AWR_MONITOR_RX_SATURATION_DETECTOR_CONF_SB AWR_MONITOR_SIG_IMG_MONITOR_CONF_SB AWR_MONITOR_RX_MIXER_IN_POWER_CONF_SB AWR_ANALOG_FAULT_INJECTION_CONF_SB
CRC	Variable	Based on CRCLEN field in FLAGS

4.13 AWR_RF_STATUS_GET_MSG

Radar transceiver status GET messages.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0x43211234												
OPCODE	2	<table> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>00</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x11</td></tr> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	00	b15:6	MSGID	0x11
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	00												
b15:6	MSGID	0x11												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
MSGDATA	Variable	Supported sub blocks												

		AWR_RF_VERSION_GET_SB AWR_RF_CPUFAULT_STATUS_GET_SB AWR_RF_ESMFAULT_STATUS_GET_SB AWR_RF_DIEID_GET_SB AWR_RF_BOOTUPBIST_STATUS_GET_SB AWR_RF_ADV_ESMFAULT_STATUS_GET_SB
CRC	Variable	Based on CRCLEN field in FLAGS

4.14 AWR_RF_MONITORING_REPORT_GET_MSG

DFE statistics report GET message from Radar transceiver.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0x43211234												
OPCODE	2	<table border="1"> <thead> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> </thead> <tbody> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>00</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x13</td></tr> </tbody> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	00	b15:6	MSGID	0x13
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	00												
b15:6	MSGID	0x13												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
MSGDATA	Variable	Supported sub blocks AWR_RF_DFE_STATISTICS_REPORT_GET_SB												
CRC	Variable	Based on CRCLEN field in FLAGS												

4.15 AWR_RF_MONITORING_CONF_SET_2_MSG

Monitoring configuration message 2 sub-blocks for radar transceiver.

Field Name	Number of bytes	Description
SYNC	4	Value = 0x43211234

OPCODE	2	Bits	Variable name	Value
		b3:0	DIRECTION	See Table 2.8
		b5:4	MSGTYPE	00
		b15:6	MSGID	0x14
MSGLEN	2	Length of the message in bytes (do not include sync length)		
FLAGS	2	See Section 2.3.2		
REMCHUNKS	2	Value = 0		
NSBC	2	Number of sub blocks contained in the message		
CHKSUM	2	See Section 2.3.2		
MSGDATA	Variable	Supported sub blocks AWR_MONITOR_TXN_POWER_CONF_SB AWR_MONITOR_TXN_BALLBREAK_CONF_SB AWR_MONITOR_TXN_INTERNAL_ANALOG_SIG- NALS_CONF_SB AWR_MONITOR_TXN_PHASE_SHIFTER_CONF_SB AWR_MONITOR_ADV_TX_GAIN_PHASE_MISMATCH_CONF_SB		
CRC	Variable	Based on CRCLEN field in FLAGS		

4.16 AWR_RF_MISC_CONF_SET_MSG

Miscellaneous configuration message sub-blocks for radar transceiver.

Field Name	Number of bytes	Description
SYNC	4	Value = 0x43211234
OPCODE	2	Bits
		Variable name
		Value
		b3:0 DIRECTION See Table 2.8
		b5:4 MSGTYPE 00
		b15:6 MSGID 0x16
MSGLEN	2	Length of the message in bytes (do not include sync length)
FLAGS	2	See Section 2.3.2
REMCHUNKS	2	Value = 0
NSBC	2	Number of sub blocks contained in the message
CHKSUM	2	See Section 2.3.2

MSGDATA	Variable	Supported sub blocks AWR_RF_TEST_SOURCE_CONFIG_SET_SB AWR_RF_TEST_SOURCE_ENABLE_SET_SB AWR_RF_LDO_BYPASS_SB AWR_RF_PALOOPBACK_CFG_SB AWR_RF_PSLOOPBACK_CFG_SB AWR_RF_IFLOOPBACK_CFG_SB AWR_RF_GPADC_CFG_SET_SB AWR_RF_LOLOOPBACK_CFG_SB
CRC	Variable	Based on CRCLEN field in FLAGS

4.17 AWR_RF_MISC_CONF_GET_MSG

Miscellaneous configuration GET messages from radar transceiver.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0x43211234												
OPCODE	2	<table border="1"> <thead> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> </thead> <tbody> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>00</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x17</td></tr> </tbody> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	00	b15:6	MSGID	0x17
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	00												
b15:6	MSGID	0x17												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
MSGDATA	Variable	Supported sub blocks AWR_RF_TEMPERATURE_GET_SB												
CRC	Variable	Based on CRCLEN field in FLAGS												

4.18 AWR_RF_ASYNC_EVENT_MSG1

The AWR_RF_ASYNC_EVENT_MSG1 message is sent by the radar transceiver to the host. This message indicates that specific events have occurred within the device.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0xABCDDCBA												
OPCODE	2	<table> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>11</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x80</td></tr> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	11	b15:6	MSGID	0x80
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	11												
b15:6	MSGID	0x80												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
MSGDATA	Variable	Supported sub blocks AWR_AE_RF_CPUFAULT_SB AWR_AE_RF_ESMFAULT_SB AWR_AE_RF_INITCALIBSTATUS_SB AWR_AE_RF_MONITOR_TYPE_TRIGGER_DONE_SB AWR_AE_RF_FRAME_TRIGGER_RDY_SB AWR_AE_RF_GPADC_RESULT_DATA_SB AWR_FRAME_END_AE_SB AWR_ANALOGFAULT_AE_SB AWR_CAL_MON_TIMING_FAIL_REPORT_AE_SB AWR_RUN_TIME_CALIBRATION_SUMMARY_REPORT_AE_SB AWR_MONITOR_RF_DIG_LATENTFAULT_REPORT_AE_SB AWR_MONITOR_REPORT_HEADER_AE_SB AWR_MONITOR_RF_DIG_PERIODIC_REPORT_AE_SB AWR_MONITOR_TEMPERATURE_REPORT_AE_SB AWR_MONITOR_RX_GAIN_PHASE_REPORT_AE_SB AWR_MONITOR_RX_NOISE_FIGURE_REPORT_AE_SB AWR_MONITOR_RX_IFSTAGE_REPORT_AE_SB AWR_MONITOR_TX0_POWER_REPORT_AE_SB												

		AWR_MONITOR_TX1_POWER_REPORT_AE_SB AWR_MONITOR_TX2_POWER_REPORT_AE_SB AWR_MONITOR_TX0_BALLBREAK_REPORT_AE_SB AWR_MONITOR_TX1_BALLBREAK_REPORT_AE_SB
CRC	Variable	Based on CRCLEN field in FLAGS

4.19 AWR_RF_ASYNC_EVENT_MSG2

The AWR_RF_ASYNC_EVENT_MSG2 message is sent by the radar transceiver to the host. This message indicates that specific events have occurred within the device.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0xABCDDCBA												
OPCODE	2	<table> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>11</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x81</td></tr> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	11	b15:6	MSGID	0x81
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	11												
b15:6	MSGID	0x81												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
MSGDATA	Variable	Supported sub blocks AWR_MONITOR_TX2_BALLBREAK_REPORT_AE_SB AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_REPORT_AE_SB AWR_MONITOR_TX0_PHASE_SHIFTER_REPORT_AE_SB AWR_MONITOR_TX1_PHASE_SHIFTER_REPORT_AE_SB AWR_MONITOR_TX2_PHASE_SHIFTER_REPORT_AE_SB AWR_MONITOR_SYNTHESIZER_FREQUENCY_REPORT_AE_SB AWR_MONITOR_EXTERNAL_ANALOG_SIGNALSREPORT_AE_SB												

		AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB AWR_MONITOR_TX1_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB AWR_MONITOR_TX2_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB AWR_MONITOR_RX_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB AWR_MONITOR_PMCLKLO_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB AWR_MONITOR_GPADC_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB AWR_MONITOR_PLL_CONTROL_VOLTAGE_REPORT_AE_SB AWR_MONITOR_DUAL_CLOCK_COMP_REPORT_AE_SB AWR_MONITOR_RX_MIXER_IN_POWER_REPORT_AE_SB AWR_MONITOR_SYNTHESIZER_FREQUENCY_NONLIVE_REPORT_AE_SB
CRC	Variable	Based on CRCLEN field in FLAGS

4.20 AWR_DEV_RFPOWERUP_MSG

The AWR_DEV_RFPOWERUP_MSG is sent by the host to the MSS. This message indicates that BSS/RadarSS can now be powered up.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0x43211234												
OPCODE	2	<table> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>00</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x200</td></tr> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	00	b15:6	MSGID	0x200
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	00												
b15:6	MSGID	0x200												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												

NSBC	2	Number of sub blocks contained in the message
CHKSUM	2	See Section 2.3.2
MSGDATA	Variable	Supported sub blocks AWR_DEV_RFPOWERUP_SB
CRC	Variable	Based on CRCLEN field in FLAGS

4.21 AWR_DEV_CONF_SET_MSG

The AWR_DEV_CONF_SET_MSG message sub-blocks are sent by the host to the radar transceiver. These sub-blocks configures MSS data path and monitoring settings.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0x43211234												
OPCODE	2	<table> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>00</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x202</td></tr> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	00	b15:6	MSGID	0x202
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	00												
b15:6	MSGID	0x202												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
MSGDATA	Variable	Supported sub blocks AWR_DEV_MCUCLOCK_CONF_SET_SB AWR_DEV_RX_DATA_FORMAT_CONF_SET_SB AWR_DEV_RX_DATA_PATH_CONF_SET_SB AWR_DEV_RX_DATA_PATH_LANEEN_SET_SB AWR_DEV_RX_DATA_PATH_CLK_SET_SB AWR_DEV_LVDS_CFG_SET_SB AWR_DEV_RX_CONTSTREAMING_MODE_CONF_SET_SB AWR_DEV_CSI2_CFG_SET_SB AWR_DEV_PMICCLOCK_CONF_SET_SB AWR_MSS_LATENTFAULT_TEST_CONF_SB AWR_MSS_PERIODICTESTS_CONF_SB												

		AWR_DEV_TESTPATTERN_GEN_SET_SB AWR_DEV_CONFIGURATION_SET_SB AWR_DEV_RF_DEBUG_SIG_SET_SB AWR_DEV_DEV_HSI_DELAY_DUMMY_CFG_SET_SB
CRC	Variable	Based on CRCLEN field in FLAGS

4.22 AWR_DEV_CONF_GET_MSG

The AWR_DEV_CONF_GET_MSG is sent by the host to the radar transceiver to read back the configuration values from MSS.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0x43211234												
OPCODE	2	<table border="1"> <thead> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> </thead> <tbody> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>00</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x203</td></tr> </tbody> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	00	b15:6	MSGID	0x203
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	00												
b15:6	MSGID	0x203												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
MSGDATA	Variable	Supported sub blocks AWR_DEV_MCUCLOCK_GET_SB AWR_DEV_RX_DATA_FORMAT_CONF_GET_SB AWR_DEV_RX_DATA_PATH_CONF_GET_SB AWR_DEV_RX_DATA_PATH_LANEEN_GET_SB AWR_DEV_RX_DATA_PATH_CLK_GET_SB AWR_DEV_LVDS_CFG_GET_SB AWR_DEV_RX_CONTSTREAMING_MODE_CONF_GET_SB AWR_DEV_CSI2_CFG_GET_SB AWR_DEV_PMICCLOCK_CONF_GET_SB AWR_MSS_LATENTFAULT_TEST_CONF_GET_SB AWR_MSS_PERIODICCONF_GET_SB												

		AWR_DEV_TESTPATTERN_GEN_GET_SB
CRC	Variable	Based on CRCLEN field in FLAGS

4.23 AWR_DEV_FILE_DOWNLOAD_MSG

The AWR_DEV_FILE_DOWNLOAD_MSG is sent by the host to MSS. This message sends a file to be written into the device.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0x43211234												
OPCODE	2	<table> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>00</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x204</td></tr> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	00	b15:6	MSGID	0x204
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	00												
b15:6	MSGID	0x204												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
MSGDATA	Variable	Supported sub blocks AWR_DEV_FILE_DOWNLOAD_SB												
CRC	Variable	Based on CRCLEN field in FLAGS												

4.24 AWR_DEV_FRAME_CONFIG_APPLY_MSG

The AWR_DEV_FRAME_CONFIG_APPLY_MSG is sent by the host to MSS. This message indicates to MSS to apply all the regular framing mode configurations related to ADC buffer and CBUFF.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0x43211234												
OPCODE	2	<table> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>00</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x206</td></tr> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	00	b15:6	MSGID	0x206
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	00												
b15:6	MSGID	0x206												

MSGLEN	2	Length of the message in bytes (do not include sync length)
FLAGS	2	See Section 2.3.2
REMCHUNKS	2	Value = 0
NSBC	2	Number of sub blocks contained in the message
CHKSUM	2	See Section 2.3.2
MSGDATA	Variable	Supported sub blocks AWR_DEV_FRAME_CONFIG_APPLY_SB AWR_DEV_ADV_FRAME_CONFIG_APPLY_SB
CRC	Variable	Based on CRCLEN field in FLAGS

4.25 AWR_DEV_STATUS_GET_MSG

The AWR_DEV_STATUS_GET_MSG is sent by the host to MSS to get some status information from the MSS device.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0x43211234												
OPCODE	2	<table> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>00</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x207</td></tr> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	00	b15:6	MSGID	0x207
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	00												
b15:6	MSGID	0x207												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
MSGDATA	Variable	Supported sub blocks AWR_MSSVERSION_GET_SB AWR_MSSCPUFAULT_STATUS_GET_SB AWR_MSSSESMFAULT_STATUS_GET_SB												
CRC	Variable	Based on CRCLEN field in FLAGS												

4.26 AWR_DEV_ASYNC_EVENT_MSG

The AWR_DEV_ASYNC_EVENT_MSG is sent by the radar transceiver MSS to the host. This message indicates that specific events have occurred within the MSS device.

Field Name	Number of bytes	Description												
SYNC	4	Value = 0xABCDDCBA												
OPCODE	2	<table> <tr> <th>Bits</th><th>Variable name</th><th>Value</th></tr> <tr> <td>b3:0</td><td>DIRECTION</td><td>See Table 2.8</td></tr> <tr> <td>b5:4</td><td>MSGTYPE</td><td>11</td></tr> <tr> <td>b15:6</td><td>MSGID</td><td>0x280</td></tr> </table>	Bits	Variable name	Value	b3:0	DIRECTION	See Table 2.8	b5:4	MSGTYPE	11	b15:6	MSGID	0x280
Bits	Variable name	Value												
b3:0	DIRECTION	See Table 2.8												
b5:4	MSGTYPE	11												
b15:6	MSGID	0x280												
MSGLEN	2	Length of the message in bytes (do not include sync length)												
FLAGS	2	See Section 2.3.2												
REMCHUNKS	2	Value = 0												
NSBC	2	Number of sub blocks contained in the message												
CHKSUM	2	See Section 2.3.2												
MSGDATA	Variable	Supported sub blocks AWR_AE_DEV_MSSPOWERUPDONE_SB AWR_AE_DEV_RFPOWERUPDONE_SB AWR_AE_MSS_CPUFAULT_SB AWR_AE_MSS_ESMFAULT_SB AWR_AE_MSS_BOOTERRORSTATUS_SB AWR_AE_MSS_LATENTFAULT_TESTREPORT_SB AWR_AE_MSS_PERIODICTEST_STATUS_SB AWR_AE_MSS_RFERROR_STATUS_SB												
CRC	Variable	Based on CRCLEN field in FLAGS												

5 Radar Functional APIs

This section describes all the radar interface sub blocks that are used in messages for communicating with the radar transceiver. Some of the sub blocks are status responses from the radar device. All the API sub-blocks defined in this document are applicable to all mmWave Radar Sensors unless it is specified in the sub-block.

5.1 Sub block related to AWR_ERROR_MSG

5.1.1 Sub block 0x0000 – AWR_RESP_ERROR_SB

This sub block contains the error response for an API command.
Table 5.1 describes the contents of this sub block.

Table 5.1: AWR_RESP_ERROR_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0000
SBLKLEN	2	Value = 8
API_RESP	2	0x0001 ERROR_CMD: Incorrect MSGID 0x0002 ERROR_CMD: No Sub block found in the MSG 0x0003 ERROR_CMD: Incorrect Sub block ID 0x0004 ERROR_CMD: Incorrect Sub block Length 0x0005 ERROR_CMD: Incorrect Sub block data 0x0006 ERROR_PROC: Error in processing the command 0x0007 ERROR_FILECRCMISMATCH: File CRC mismatched 0x0008 ERROR_FILETYPEMISMATCH: File type mismatched w.r.t. magic number 0x0009 See Section 7 for details on error codes from each API - 0xFFFF

API_RESP_ER- ROR_SBC_ID	2	0x0000 Sub-Block ID in which Error Occurred for sub – block related errors 0xFFFF
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5.2 Sub blocks related to AWR_RF_STATIC_CONF_SET_MSG

5.2.1 Sub block 0x0080 – AWR_CHAN_CONF_SET_SB

This sub block contains static device configurations (applicable for the given power cycle) - how many RX and TX channels are needed for operation. It also defines static configurations related to whether the sensor uses a single mmWave or multiple chips to realize a larger antenna array (multiple is applicable only in AWR2243/xWR6243). Table 5.2 describes the contents of this sub block.

Table 5.2: AWR_CHAN_CONF_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0080
SBLKLEN	2	Value = 12
RX_CHAN_EN	2	This field specifies which RX channels are to be enabled Bit Description b0 RX_CHAN0_EN 0 Disable RX Channel 0 1 Enable RX Channel 0 b1 RX_CHAN1_EN 0 Disable RX Channel 1 1 Enable RX Channel 1 b2 RX_CHAN2_EN 0 Disable RX Channel 2 1 Enable RX Channel 2 b3 RX_CHAN3_EN 0 Disable RX Channel 3 1 Enable RX Channel 3 b15:4 RESERVED 0b000000000000

Continued on next page

Table 5.2 – continued from previous page

TX_CHAN_EN	2	This field specifies which TX channels are to be enabled <table><tr><th>Bit</th><th>Description</th></tr><tr><td>b0</td><td>TX_CHAN0_EN</td></tr><tr><td>0</td><td>Disable TX Channel 0</td></tr><tr><td>1</td><td>Enable TX Channel 0</td></tr><tr><td>b1</td><td>TX_CHAN1_EN</td></tr><tr><td>0</td><td>Disable TX Channel 1</td></tr><tr><td>1</td><td>Enable TX Channel 1</td></tr><tr><td>b2</td><td>TX_CHAN2_EN (3rd Tx is supported only on some of the devices, Please refer device data sheet)</td></tr><tr><td>0</td><td>Disable TX Channel 2</td></tr><tr><td>1</td><td>Enable TX Channel 2</td></tr><tr><td>b3</td><td>TX_CHAN3_EN (4th Tx is supported only on some of the devices, Please refer device data sheet)</td></tr><tr><td>0</td><td>Disable TX Channel 3</td></tr><tr><td>1</td><td>Enable TX Channel 3</td></tr><tr><td>b15:4</td><td>RESERVED</td></tr></table> 0b000000000000	Bit	Description	b0	TX_CHAN0_EN	0	Disable TX Channel 0	1	Enable TX Channel 0	b1	TX_CHAN1_EN	0	Disable TX Channel 1	1	Enable TX Channel 1	b2	TX_CHAN2_EN (3rd Tx is supported only on some of the devices, Please refer device data sheet)	0	Disable TX Channel 2	1	Enable TX Channel 2	b3	TX_CHAN3_EN (4th Tx is supported only on some of the devices, Please refer device data sheet)	0	Disable TX Channel 3	1	Enable TX Channel 3	b15:4	RESERVED
Bit	Description																													
b0	TX_CHAN0_EN																													
0	Disable TX Channel 0																													
1	Enable TX Channel 0																													
b1	TX_CHAN1_EN																													
0	Disable TX Channel 1																													
1	Enable TX Channel 1																													
b2	TX_CHAN2_EN (3rd Tx is supported only on some of the devices, Please refer device data sheet)																													
0	Disable TX Channel 2																													
1	Enable TX Channel 2																													
b3	TX_CHAN3_EN (4th Tx is supported only on some of the devices, Please refer device data sheet)																													
0	Disable TX Channel 3																													
1	Enable TX Channel 3																													
b15:4	RESERVED																													
CASCADING_CFG	2	This field specifies the cascading configuration. <table><tr><th>Value</th><th>Description</th></tr><tr><td>0x0000</td><td>SINGLECHIP: Single mmWave sensor application</td></tr><tr><td>0x0001</td><td>MULTICHIP_MASTER: Multiple cascade sensor application. This device is a master chip and generates LO and conveys to other slave sensors. This is applicable only for device which supports cascading.</td></tr><tr><td>0x0002</td><td>MULTICHIP_SLAVE: Multiple cascade sensor application. This device is a slave chip and uses LO conveyed to it by the master sensor. This is applicable only for device which supports cascading.</td></tr></table> MULTICHIP_MASTER and MULTICHIP_SLAVE are in general referred to as MULTICHIP applications, where larger antenna array sizes are possible in comparison with SINGLECHIP cases. Please refer device data sheet for cascading capability and 20G SYNC pins.	Value	Description	0x0000	SINGLECHIP: Single mmWave sensor application	0x0001	MULTICHIP_MASTER: Multiple cascade sensor application. This device is a master chip and generates LO and conveys to other slave sensors. This is applicable only for device which supports cascading.	0x0002	MULTICHIP_SLAVE: Multiple cascade sensor application. This device is a slave chip and uses LO conveyed to it by the master sensor. This is applicable only for device which supports cascading.																				
Value	Description																													
0x0000	SINGLECHIP: Single mmWave sensor application																													
0x0001	MULTICHIP_MASTER: Multiple cascade sensor application. This device is a master chip and generates LO and conveys to other slave sensors. This is applicable only for device which supports cascading.																													
0x0002	MULTICHIP_SLAVE: Multiple cascade sensor application. This device is a slave chip and uses LO conveyed to it by the master sensor. This is applicable only for device which supports cascading.																													

Continued on next page

Table 5.2 – continued from previous page

		Bit	Description
CASCADING_ PINOUTCFG	2	b0	FM_CW_CLKOUT_MASTER_DIS Applicable only in MUTICHIP_MASTER. Default value is 0 0 Enable FM_CW_CLKOUT on master 1 Disable FM_CW_CLKOUT on master
		b1	FM_CW_SYNCOUT_MASTER_DIS Applicable only in MULTICHIP_MASTER. Default value is 0 0 Enable FM_CW_SYNCOUT on master 1 Disable FM_CW_SYNCOUT on master
		b2	FM_CW_CLKOUT_SLAVE_EN Applicable only in MULTICHIP_SLAVE. Default value is 0 0 Disable FM_CW_CLKOUT on slave 1 Enable FM_CW_CLKOUT on slave
		b3	FM_CW_SYNCOUT_SLAVE_EN Applicable only in MULTICHIP_SLAVE. Default value is 0 0 Disable FM_CW_SYNCOUT on slave 1 Enable FM_CW_SYNCOUT on slave
		b4	INTLO_MASTER_EN Applicable only in MULTICHIP_MASTER. Default value is 0 0 Use externally looped back LO 1 Use internal LO in master Externally looped-back LO mode is useful when length-matching the 20 GHz path between master and slave devices.
		b5	OSCCLKOUT_DIS AWR2243/xWR6243: OSCCLK is enabled at power up. xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP: OSCCLK is disabled at power up. This field can be used to either disable or enable it. 0 Enable OSCCLKOUT 1 Disable OSCCLKOUT Note: This feature is supported only on AWR2243/xWR6243 devices. It is recommended to disable the OSC clock when the device is configured in SINGLE_CHIP or SLAVE mode.

Continued on next page

Table 5.2 – continued from previous page

CASCADING_ PINOUTCFG	2	b6	INTFRC_MASTER_EN Applicable only in MULTICHIP_MASTER device. Default value is 0 0 Use externally looped back FRC SyncIn 1 Use internal FRC SyncIn in master Externally looped-back FRC SyncIn mode is useful when length-matching the SyncIn path between master and slave devices.
		b7	OSCCLKOUTETH_EN This field can be used to enable or disable OSCCLK- OUTETH. 0 Disable OSCCLKOUTETH 1 Enable OSCCLKOUTETH OSCCLKETH is disabled at powerup. Enabling of both OSCLKOUT and OSCLKETH at the same time is not supported. Only one of them should be enabled. NOTE: This feature is supported only on xWR254x, xWR2x4xECO, xWR2x4xP devices.
		b8	OSCCLKDIV This field can be used to enable or disable divider for OSCLKOUT / OSCCLKOUTETH. 0 Divide by 1 1 Divide by 2 Divider is common to both OSCLKOUT / OSCLK- OUTETH. Programmed value applicable for enabled output. NOTE: This feature is supported only on xWR254x, xWR2x4xECO, xWR2x4xP devices.
		b12:9	OSCLKOUTETH_DRV_VAL These bits configure the drive strength of the OSC_ CLKOUT_ETH output. Programming 0 produces a high-Z output state. Bit Drive strength b9 0.5X b10 1X b11 2X b12 3X NOTE: This feature is supported only on xWR254x, xWR2x4xECO, xWR2x4xP devices.
		b15:13	RESERVED

5.2.2 Sub block 0x0082 – AWR_ADCOUT_CONF_SET_SB

This sub block contains static device configurations (applicable for the given power cycle) - regarding the data format of the ADC output (including the digital filtering).

Table 5.3 describes the contents of this sub block.

Table 5.3: AWR_ADCOUT_CONF_SB contents

Field Name	Number of bytes	Description																
SBLKID	2	Value = 0x0082																
SBLKLEN	2	Value = 12																
NUM_ADC_BITS	1	<table><tr><th>Bit</th><th>Description</th></tr><tr><td>b1:0</td><td>Value Definition</td></tr><tr><td>00</td><td>12 bits</td></tr><tr><td>01</td><td>14 bits</td></tr><tr><td>10</td><td>16 bits</td></tr><tr><td>Other</td><td>RESERVED</td></tr><tr><td>b7:2</td><td>RESERVED</td></tr><tr><td>0b000000</td><td></td></tr></table>	Bit	Description	b1:0	Value Definition	00	12 bits	01	14 bits	10	16 bits	Other	RESERVED	b7:2	RESERVED	0b000000	
Bit	Description																	
b1:0	Value Definition																	
00	12 bits																	
01	14 bits																	
10	16 bits																	
Other	RESERVED																	
b7:2	RESERVED																	
0b000000																		
FULL_SCALE_REDUCTION_FACTOR	1	Number of bits to reduce ADC full scale by Valid range: 0 to (16 – Number of ADC bits) For e.g. for 12 bit ADC output, this field can take values 0, 1, 2 or 3 For 14 bit ADC output, this field can take values 0, 1 or 2 For 16 bit ADC output, this field can take only value 0 Example: If the user desires 12 bit ADC output, then the digital front end (DFE) chain drops 4 LSBs before placing the data in ADC buffer (DFE output is 16 bits wide). If the user sets FULL_SCALE_REDUCTION_FACTOR as 1, then the DFE will drop only 3 LSBs but still restricting the data in ADC buffer to be within $\pm 2^{12}$. This allows wider ADC swings in smaller signal conditions.																

Continued on next page

Table 5.3 – continued from previous page

ADC_OUT_FMT	2	For AWR2243/xWR6243 devices: Bits Description b1:0 Value Definition 00 Real 01 Complex 1x (image band filtered out) 10 Complex 2x (image band visible) 11 Pseudo Real b15:2 RESERVED 0b0000000000000000 For xWR294x, xWR254x, xWR2x4xECo, xWR2x4xP devices: Bits Description b1:0 Value Definition 00 Real 01 Reserved 10 Reserved 11 Reserved b15:2 RESERVED 0b0000000000000000
RESERVED	2	0x0000
RESERVED	2	0x0000

5.2.3 Sub block 0x0083 – AWR_LOWPOWERMODE_CONF_SET_SB

This sub block contains static device configurations (applicable for this power cycle) - Sigma Delta ADC root sampling clock rate (reducing rate to half to save power in small IF bandwidth applications). In xWR6x43 devices, this API doesn't modify the ADC root sampling rate, but reduces the power by reducing the bias currents to some of the IF analog blocks.

Table 5.4 describes the contents of this sub block.

Table 5.4: AWR_LOWPOWERMODE_CONF_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0083
SBLKLEN	2	Value = 8
RESERVED	2	0x0000

Continued on next page

Table 5.4 – continued from previous page

LP_ADC_MODE	2	Value	Definition
		0x00	Regular ADC mode
		0x01	Low power ADC mode

NOTE:

For AWR2243 devices Low power mode is not recommended when IF bandwidth > 7.5MHz. For xWR6243, low power mode is not supported.

5.2.4 Sub block 0x0084 – AWR_DYNAMICPOWERSAVE_CONF_SET_SB

This sub block defines static device configuration - whether to enable dynamic power saving during inter-chirp IDLE times by turning off various circuits e.g. TX, RX, LO Distribution blocks. If Idle time + Tx start time < 10us or Idle time < 3.5us then inter-chirp dynamic power save option will be disabled, in that case, 15us of inter-burst idle time will be utilized to configure sequencer LO, TX and RX signal timings by firmware. TX PAs are always turned off in the inter-chirp time irrespective of the Idle time / Tx start time configurations.

Table 5.4 describes the contents of this sub block.

Table 5.5: AWR_DYNAMICPOWERSAVE_CONF_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0084
SBLKLEN	2	Value = 8

Continued on next page

Table 5.5 – continued from previous page

BLOCK_CFG	2	Bits	Definition
		b0	Enable power save by switching off TX during inter-chirp IDLE period 0 Disable 1 Enable Default value: 1 (power saving is enabled)
		b1	Enable power save by switching off RX during inter-chirp IDLE period 0 Disable 1 Enable Default value: 1 (power saving is enabled)
		b2	Enable power save by switching off LO Distribution blocks during inter-chirp IDLE period 0 Disable 1 Enable Default value: 1 (power saving is enabled)
		b15:3	RESERVED 0b00000000000000
RESERVED	2	0x0000	

NOTE:

All the 3 configuration bits (TX, RX and LO) should have same value, i.e. user should program value 0x7 to enable power save or 0x0 to disable the power save in BLOCK_CFG.

5.2.5 Sub block 0x0085 – AWR_HIGHSPEEDINTFCLK_CONF_SET_SB

This sub block contains static device configurations (applicable for the given power cycle) - regarding high speed interface clock rates which are related to sending the ADC data from AWR device to the host in either LVDS or CSI2 format.

Table 5.6 describes the contents of this sub block.

Table 5.6: AWR_HIGHSPEEDINTFCLK_CONF_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0085
SBLKLEN	2	Value = 8

Continued on next page

Table 5.6 – continued from previous page

HSICLKRATE_ CODE	2	This field indicates the high speed interface input clock rate, needed by the LVDS or CSI2 module. It should be N times the final serial data rate, where $N = 2$ in DDR mode and $N = 1$ in SDR mode. Bit 15:5 = Reserved (all 0). Bit 3:0 are to be set based on desired rate as follows: <table><thead><tr><th></th><th>b1:0</th><th>b1:0</th><th>b1:0</th><th>b1:0</th></tr><tr><th></th><th>00</th><th>01</th><th>10</th><th>11</th></tr></thead><tbody><tr><td>b3:2 00</td><td>Reserved</td><td>800 MHz</td><td>400 MHz</td><td>200 MHz</td></tr><tr><td>b3:2 01</td><td>Reserved</td><td>900 MHz</td><td>450 MHz</td><td>225 MHz</td></tr><tr><td>b3:2 10</td><td>Reserved</td><td>1200 MHz</td><td>600 MHz</td><td>300 MHz</td></tr><tr><td>b3:2 11</td><td>Reserved</td><td>1800 MHz</td><td>Reserved</td><td>Reserved</td></tr></tbody></table> For example, for 900 Mbps output rate with DDR, choose Bit3:0=0b1101, and for 450 Mbps output rate with SDR, choose Bit3:0=0b0110.		b1:0	b1:0	b1:0	b1:0		00	01	10	11	b3:2 00	Reserved	800 MHz	400 MHz	200 MHz	b3:2 01	Reserved	900 MHz	450 MHz	225 MHz	b3:2 10	Reserved	1200 MHz	600 MHz	300 MHz	b3:2 11	Reserved	1800 MHz	Reserved	Reserved
	b1:0	b1:0	b1:0	b1:0																												
	00	01	10	11																												
b3:2 00	Reserved	800 MHz	400 MHz	200 MHz																												
b3:2 01	Reserved	900 MHz	450 MHz	225 MHz																												
b3:2 10	Reserved	1200 MHz	600 MHz	300 MHz																												
b3:2 11	Reserved	1800 MHz	Reserved	Reserved																												
RESERVED	2	0x0000																														

5.2.6 Sub block 0x0086 – AWR_RF_DEVICE_CFG_SB

This sub block configures the direction of async event from BSS. Typically async events are sent to MSS. With this API, the user can configure the destination of async event.

Table 5.7 describes the contents of this sub block.

Table 5.7: AWR_RF_DEVICE_CFG_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0086
SBLKLEN	2	Value = 16

Continued on next page

Table 5.7 – continued from previous page

RF_AE_DIRECTION	4	Bits Definition b1:0 ASYNC_EVENT_DIR 00 BSS to MSS 01 BSS to HOST 10 BSS to DSS 11 RESERVED The ASYNC_EVENT_DIR controls the direction for following ASYNC_EVENTS 1. CPU_FAULT 2. ESM_FAULT 3. ANALOG_FAULT All other ASYNC_EVENTS are sent to the subsystem which issues the API Default value: 0b00 b3:2 MONITORING_ASYNC_EVENT_DIR 00 BSS to MSS 01 BSS to HOST 10 BSS to DSS 11 RESERVED Default value: 0b00 b31:4 RESERVED 0x0000000
AE_CONTROL	1	Bits Definition b0 FRAME_START_ASYNC_EVENT_DIS 0 Frame Start async event enable 1 Frame Start async event disable Default value: 0 b1 FRAME_STOP_ASYNC_EVENT_DIS 0 Frame Stop async event enable 1 Frame Stop async event disable Default value: 0 b7:2 RESERVED 0b000000

Continued on next page

Table 5.7 – continued from previous page

BSS_ANA_CTRL	1	Bits	Definition
		b0	INTER_BURST_POWER_SAVE_DIS 0 Inter burst power save enable (default) 1 Inter burst power save disable (Applicable only in single chip usecase)
		b1	INTER_BURST_APLL_POWER_SAVE_DIS (applicable only in xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP) 0 Inter burst APLL power save enable (default) 1 Inter burst APLL power save disable Default value: 0 This disable feature is applicable only in single chip use case. In case of cascade slave device, synth power down is done by default. This allows to disable inter burst power save feature for individual bursts in a advance frame config API to reduce inter-burst idle time requirement. The power save is done always in inter sub-frame and frame boundaries irrespective of this control bit configuration. The inter burst power save needs extra 55us burst idle time, please refer Table 11.3 for more details on inter burst time.
		b7:2	RESERVED 0b0000000
RESERVED	1	0x0000	

Continued on next page

Table 5.7 – continued from previous page

BSS_DIG_CTRL	1	Bits Definition b0 WDT_ENABLE 0 Keep watchdog disabled 1 Enable watchdog b7:1 RESERVED 0b0000000 NOTE1: WDT cannot be enabled if BSS dynamic frequency switching feature is enabled in RSS_CR4_BOOT_INFO_REG5 (BSS_DYN_FREQ_SWITCH). NOTE2: The Windowed WDT can be enabled only in Sw triggered framing Mode or in cascade mode where frames of all the devices synchronized with same clock source, if frames are triggered from Hw trigger pulse generated from un-synchronized clock then WDT can not be enabled. WDT shall be disabled if API based monitoring trigger is enabled in MONITORING_MODE in AWR_CALIB_MON_TIME_UNIT_CONF_SB. WDT shall be disabled if SUB_FRAMETRIGGER mode is enabled in AWR_ADVANCED_FRAME_CONF_SB API. Refer section 9.4 for more details on WDT timing and programming window 553
ASYNC_EVENT_CRC_CONFIG	1	Value Description 0 16 bit CRC for BSS async events 1 32 bit CRC for BSS async events 2 64 bit CRC for BSS async events
RESERVED	3	0x000000

5.2.7 Sub block 0x0087 – AWR_RF_RADAR_MISC_CTL_SB

This sub block controls miscellaneous global RF controls for e.g. per-chirp phase shifter global control.

NOTE1: Issue this API first in the sequence if AWR_PERCHIRPPHASESHIFT_CONF_SB, AWR_DYN_PERCHIRP_PHASESHIFTER_CONF_SET_SB and AWR_ADVANCE_CHIRP_CONF_SB are issued down in the sequence.

NOTE2:

1. 1 ns ADC start time resolution is available only for advance chirp configuration AWR_ADVANCE_CHIRP_CONF_SB
2. If ADC_START_TIME_RES is set to 1 in AWR_RF_MISC_CTL_SB, when AWR_ADVANCE_CHIRP_CONF_SB API is issued for CHIRP_ADC_START_TIME_VAR, it is recommended to program the below specified chirp delta parameters to 0. Only LUT based dither can be used for CHIRP_ADC_START_TIME_VAR.
 - SF0_CHIRP_PARAM_DELTA, SF1_CHIRP_PARAM_DELTA, SF2_CHIRP_PARAM_DELTA, SF3_CHIRP_PARAM_DELTA, DELTA_PARAM_UPDATE_PERIOD in AWR_ADVANCE_CHIRP_CONF_SB API.
3. LUT_CHIRP_PARAM_SCALE should be programmed to 0 when AWR_ADVANCE_CHIRP_CONF_SB API is issued for CHIRP_ADC_START_TIME_VAR. Scaling is not supported for CHIRP_ADC_START_TIME_VAR as 2 bytes are already used.
4. Bit 2 (Digital RX delay compensation enable) of DIGITAL_COMP_EN field in AWR_DIGITAL_COMP_EST_CONTROL_SB API should not be exercised when using 1ns resolution for CHIRP_ADC_START_TIME_VAR.
5. LUT_CHIRP_PARAM_SIZE should always be set to 2 bytes in AWR_ADVANCE_CHIRP_CONF_SB API when it is issued for CHIRP_ADC_START_TIME_VAR.

Table 5.8 describes the contents of this sub block.

Table 5.8: AWR_RF_MISC_CTL_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0087
SBLKLEN	2	Value = 12

Continued on next page

Table 5.8 – continued from previous page

		Bits Definition
RF_MISC_CTL	4	b0 PERCHIRP_PHASESHIFTER_EN 0 Per chirp phase shifter is disabled 1 Per chirp phase shifter is enabled This control is applicable only in devices which support phase shifter (refer data sheet). For other devices, this is a RESERVED bit and should be set to 0. Default value: 0
		b1 ADVANCE_CHIRP_CONFIG_EN 0 Advance chirp config mode is disabled 1 Advance chirp config mode is enabled This feature enables advanced mode of configuring chirps to achieve very flexible waveform generation. Default value: 0
		b2 ADVANCE_CHIRP_ERROR_CHK_DIS 0 Advance chirp parameters error check enabled in frame config API 1 Advance chirp parameters error check disabled in frame config API By default Error check is enabled for each and every parameters of advance chirp based on waveform pattern in legacy and advance frame config API, this would take around 1.8ms to process frame config for 128 chirps. This option enables the user to disable the error check in functional mode and error check can be done only during development phase. If this error check takes more than 50ms due to large number of chirps then it is recommended to disable RadarSS WDT while executing frame config API. Default value: 0

Continued on next page

Table 5.8 – continued from previous page

		b3 CAL_MON_TIME_UNIT_ERROR_CHK_DIS 0 Calibration and Monitor time error check is enabled in AWR_FRAMESTARTSTOP_CONF_SB API 1 Calibration and Monitor time error check is disabled in AWR_FRAMESTARTSTOP_CONF_SB API By default Calibration and Monitor time Error check is enabled and performed in frame start API, this error check generates AWR_CAL_MON_TIMING_FAIL_REPORT_AE_SB AE if total available idle time in a CALIB_MON_TIME_UNIT configured in AWR_CALIB_MON_TIME_UNIT_CONF_SB API is not sufficient to run all enabled calibrations and Monitors. This option enables the user to disable the error check in advance continuous farming mode where idle time is split across multiple bursts in a CALIB_MON_TIME_UNIT window. The calibration and monitors need idle time only to run critical chirps for measurements, the setup and post processing can be done in background while functional frames are running. Refer Table 12.10 and Table 12.5 for more info. Default value: 0 b4 CRD_ENABLE (only on xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices) 0 Controlled ramp down disabled on all profiles. 1 Controlled ramp down enabled on all profiles. b5 CRD_DITHER_ENABLE (only on xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices) 0 CRD Dithering disabled. 1 CRD Dithering enabled. CRD dithering should be enabled only when the CRD feature is enabled.
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Table 5.8 – continued from previous page

		b6 ADC_START_TIME_RES (only on xWR254x, xWR2x4xECO, xWR2x4xP devices) 0 10 ns 1 1 ns NOTE: Refer the list of constraints above this table in NOTE2 on other parameters when 1ns resolution is enabled.
		b7 INTER_CHIRP_JITTER_MITIGATION (only on xWR254x, xWR2x4xECO, xWR2x4xP devices) This feature enables/disables features to mitigate jitter in synchronization of sequencer clock and synthesizer clock, resulting in improved phase stability across chirps. Mitigation feature is only valid for the chirps of the same burst, there may still be jitter across bursts/frames. 0 Disable inter-chirp jitter mitigation. 1 Enable inter-chirp jitter mitigation. NOTE: If INTER_CHIRP_JITTER_MITIGATION is enabled, it is recommended to set 1. CRD_ENABLE = 1 2. Max allowed slope for CRD_NSLOPE_MAG is 400MHz/uS 3. Chirp idle time >= 6uS 4. CRD_DITHER_ENABLE = 0 b31:8 RESERVED 0b000_0000_0000_0000_0000_0000_0000

Continued on next page

Table 5.8 – continued from previous page

CRD_NSLOPE_MAG	2	Controlled Ramp Down slope (magnitude) for transitioning during the chirp's idle time from one chirp's ending RF to next chirp's starting RF. 1LSB = 48.3 KHz/us, unsigned. To be programmed based on TI Recommendations: VCO1 recommended slope (MHz/uS) = RF_Diff / ((Idle_Time - CRD_Dither_Time) * 0.7) VCO2 recommended slope (MHz/uS) = RF_Diff / ((Idle_Time - CRD_Dither_Time) * 0.8) Max allowed slope in both the above cases = 1582 MHz/uS. where, <table><tr><td>RF_Diff</td><td>Max RF Difference chirp-to-chirp in MHz. Maximum RF difference between a chirp's ending RF and next subsequent chirp's starting RF frequency across all chirps in a frame.</td></tr><tr><td>Idle_Time</td><td>Idle time of the chirp in uS. The time between a ramp end and subsequent ramp start. (Idle time is always >= 2.5 uS).</td></tr><tr><td>CRD_Dither_Time</td><td>If CRD Dither is enabled, CRD_Dither_Time is always 0.25 us otherwise it is 0 us.</td></tr></table> Example 1: VCO2 750MHz chirp with 2.5uS idle time and CRD dither enabled. Slope = 750 MHz / ((2.5 uS - 0.25 uS) * 0.8) = 416.7MHz/uS. Example 2: VCO1 500MHz chirp with 5uS idle time and CRD dither disabled. Slope = 500 MHz / ((5 uS - 0 uS) * 0.7) = 142.8MHz/uS.	RF_Diff	Max RF Difference chirp-to-chirp in MHz. Maximum RF difference between a chirp's ending RF and next subsequent chirp's starting RF frequency across all chirps in a frame.	Idle_Time	Idle time of the chirp in uS. The time between a ramp end and subsequent ramp start. (Idle time is always >= 2.5 uS).	CRD_Dither_Time	If CRD Dither is enabled, CRD_Dither_Time is always 0.25 us otherwise it is 0 us.
RF_Diff	Max RF Difference chirp-to-chirp in MHz. Maximum RF difference between a chirp's ending RF and next subsequent chirp's starting RF frequency across all chirps in a frame.							
Idle_Time	Idle time of the chirp in uS. The time between a ramp end and subsequent ramp start. (Idle time is always >= 2.5 uS).							
CRD_Dither_Time	If CRD Dither is enabled, CRD_Dither_Time is always 0.25 us otherwise it is 0 us.							

Continued on next page

Table 5.8 – continued from previous page

FAST_RESET_END_TIME	2	If set to X, fast ramp setting used during the idle time are reverted to normal setting at $X * 10\text{ns}$ before the next chirp's frequency ramp start. 1 LSB = 10ns. For example, setting this field to 40 would end the fast reset pulse 400ns before the knee of the ramp, i.e., -400ns. The fast reset disable should be set to approximately the time from when ramp jump back is completed. Example: For the case where idle time = 2.5us, CRD time is 1.75us, CRD dither is 0.25us, $\text{FAST_RESET_END_TIME} = (\text{CRD_TIME} + \text{CRD_DITHER} - \text{IDLE_TIME}) = (1.75\text{us} + 0.25\text{us} - 2.5\text{us}) = -0.5\text{us}$. $-0.5\text{us}/10\text{ns} = -50$. Therefore 50 would be programmed to this field. TI recommends -400ns as a good tradeoff between subsequent chirp linearity and settling time if using the recommended calculation for CRD slope based on idle time and chirp bandwidth. Allowed values: 1 to 8192
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5.2.8 Sub block 0x0088 – AWR_CAL_MON_FREQUENCY_LIMITS_SB

This sub block sets the limits for RF frequency transmission. This API is deprecated as a new API AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB is added to limit frequency for each TX channels in Table 5.11

Table 5.9 describes the contents of this sub block.

Table 5.9: AWR_CAL_MON_FREQUENCY_LIMITS_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0088
SBLKLEN	2	Value = 16

Continued on next page

Table 5.9 – continued from previous page

FREQ_LIMIT_LOW	2	The sensor's lower frequency limit for calibrations and monitoring is encoded in 2 bytes (16 bit unsigned number) 1 LSB = 100 MHz For 77GHz Devices(76GHz to 81Ghz): Valid range: 760 to 810 Default value: 760 (If this API is not issued) For 60GHz Devices(56GHz/57GHz to 64Ghz): Valid range: 560/570 to 640 Default value: 560/570 (If this API is not issued) NOTE: Refer to device datasheet for supported frequency ranges
FREQ_LIMIT_HIGH	2	The sensor's higher frequency limit for calibrations and monitoring is encoded in 2 bytes (16 bit unsigned number) 1 LSB = 100 MHz For 77GHz Devices (76GHz to 81Ghz): Valid range: 760 to 810 Default value: 810 (If this API is not issued) For 60GHz Devices(57.2GHz to 64Ghz): Valid range: 572 to 640 Default value: 640 (If this API is not issued) For xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: Valid range: 762 to 805/810 (based on VCO2_RANGE_CONFIG in this API) Default value: 805 (If this API is not issued) NOTE: FREQ_LIMIT_HIGH should be strictly greater than FREQ_LIMIT_LOW Examples: For an LRR device deployed in the US, one might typically configure FREQ_LIMIT_LOW to 760 and FREQ_LIMIT_HIGH to 770.
VCO2_RANGE_CONFIG	1	VCO2 frequency range configuration.

Continued on next page

Table 5.9 – continued from previous page

		Bit	Definition
		b0	VCO2 frequency range selection. VCO2 in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP only supports a maximum bandwidth of 4.5GHz. The user can select between the two allowed ranges shown below. 1 - VCO2 range is 76GHz - 80.5GHz. 0 - VCO2 range is 76.5GHz - 81GHz. Default setting: 0 (76.5-81GHz)
		b1-7	RESERVED
RESERVED	1		RESERVED 0x00
RESERVED	2		RESERVED 0x0000
RESERVED	4		RESERVED 0x0000_0000

NOTE1:	The minimum RF bandwidth shall be set to 200MHz, this is to perform internal calibration and monitoring.
NOTE2:	The limit set in this API is not applicable for functional chirps and loop-back chirps used in advanced frame config API.
NOTE3:	The TX0 frequency limit is used by default in calibrations and monitors where TX is not relevant or enabled.
NOTE4:	The RF band used in functional chirp profiles shall be within the limit set in this API.
NOTE5:	The mid frequency code of RF band used in functional chirp profiles + 200MHz shall be within the max limit set in this API.

5.2.9 Sub block 0x0089 – AWR_RF_INIT_CALIBRATION_CONF_SB

This sub block configures device to perform boot time calibration.

Table 5.10 describes the contents of this sub block.

Table 5.10: AWR_RF_INIT_CALIBRATION_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0089

Continued on next page

Table 5.10 – continued from previous page

SBLKLEN	2	Value = 16																														
RF_INIT_CALIB_ENABLE_MASK	4	Normally, upon receiving RF INIT message, the BSS performs all relevant initial calibrations. HOST can disable each boot calibration by setting the corresponding calibration bit in this field to 0x0. If disabled, the host needs to inject calibration data using AWR_CAL_DATA_RESTORE_SB API. Each of these calibrations can be selectively disabled by issuing this message before RF INIT message. <table><tr><td>Bit</td><td>Definition</td></tr><tr><td>b0</td><td>RESERVED</td></tr><tr><td>b1</td><td>RESERVED</td></tr><tr><td>b2</td><td>RESERVED</td></tr><tr><td>b3</td><td>RESERVED</td></tr><tr><td>b4</td><td>Enable LODIST calibration</td></tr><tr><td>b5</td><td>Enable RX ADC DC offset calibration</td></tr><tr><td>b6</td><td>Enable HPF cutoff calibration</td></tr><tr><td>b7</td><td>Enable LPF cutoff calibration</td></tr><tr><td>b8</td><td>Enable Peak detector calibration</td></tr><tr><td>b9</td><td>Enable TX power calibration</td></tr><tr><td>b10</td><td>Enable RX gain calibration</td></tr><tr><td>b11</td><td>Enable TX Phase calibration (Device dependent feature, please refer data sheet)</td></tr><tr><td>b12</td><td>Enable RX IQMM calibration (Device dependent feature, please refer data sheet)</td></tr><tr><td>b31:13</td><td>RESERVED</td></tr></table> 0b000_0000_0000_0000_0000 Default value: 0x1FF0 for AWR2243 / xWR6243 devices Default value: 0x0FF0 for xWR294x / xWR254x / xWR2x4xECO / xWR2x4xP devices NOTE: If calibrations are disabled then it is mandatory to restore the same.	Bit	Definition	b0	RESERVED	b1	RESERVED	b2	RESERVED	b3	RESERVED	b4	Enable LODIST calibration	b5	Enable RX ADC DC offset calibration	b6	Enable HPF cutoff calibration	b7	Enable LPF cutoff calibration	b8	Enable Peak detector calibration	b9	Enable TX power calibration	b10	Enable RX gain calibration	b11	Enable TX Phase calibration (Device dependent feature, please refer data sheet)	b12	Enable RX IQMM calibration (Device dependent feature, please refer data sheet)	b31:13	RESERVED
Bit	Definition																															
b0	RESERVED																															
b1	RESERVED																															
b2	RESERVED																															
b3	RESERVED																															
b4	Enable LODIST calibration																															
b5	Enable RX ADC DC offset calibration																															
b6	Enable HPF cutoff calibration																															
b7	Enable LPF cutoff calibration																															
b8	Enable Peak detector calibration																															
b9	Enable TX power calibration																															
b10	Enable RX gain calibration																															
b11	Enable TX Phase calibration (Device dependent feature, please refer data sheet)																															
b12	Enable RX IQMM calibration (Device dependent feature, please refer data sheet)																															
b31:13	RESERVED																															
RESERVED	2	0x00000000																														

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Table 5.10 – continued from previous page

TX_POWER_CAL_CFG	2	Number of transmitters to turn on during boot time TX power calibration. During actual operation, if more than 1 TXs are enabled during the chirp, then enabling the same TXs during calibration will have better TX output power accuracy. NOTE1: If this 16 bit field is set to 0, only 1 TX is enabled during the boot time TX power calibration. For e.g. during TX0 calibration, only TX0 will be enabled; during TX1 calibration, only TX1 will be enabled and so on. NOTE2: When this feature is enabled, all the enabled TXs should necessarily have the same TXx_POWER_BACKOFF in the AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB API. Bit Definition b3:0 TX enabled during TX0 calibration b0 - TX0, b1 - TX1, b2 - TX2, b3 - TX3 b7:4 TX enabled during TX1 calibration b4 - TX0, b5 - TX1, b6 - TX2, b7 - TX3 b11:8 TX enabled during TX2 calibration b8 - TX0, b9 - TX1, b10 - TX2, b11 - TX3 b15:12 TX enabled during TX3 calibration b12 - TX0, b13 - TX1, b14 - TX2, b15 - TX3
RESERVED	4	0x00000000

NOTE1: The APLL, SYNTH1 and SYNTH2 calibrations are always triggered by default on RF init command

5.2.10 Sub block 0x008A – AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB

This sub block sets the limits for RF frequency transmission for each TX and also TX power limits.

Table 5.11: AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x008A
SBLKLEN	2	Value = 28
FREQ_LIMIT_LOW_TX0	2	The sensor's lower frequency limit for calibrations and monitoring for TX0 is encoded in 2 bytes (16 bit unsigned number)

Continued on next page

Table 5.11 – continued from previous page

		1 LSB = 10 MHz For 77GHz Devices(76GHz to 81Ghz): Valid range: 7600 to 8100 Default value: 7600 (If this API is not issued) For 60GHz Devices(57GHz to 64Ghz): Valid range: 5700 to 6400 Default value: 5700 (If this API is not issued)
FREQ_LIMIT_ LOW_TX1	2	The sensor's lower frequency limit for calibrations and monitoring for TX1 is encoded in 2 bytes (16 bit unsigned number) 1 LSB = 10 MHz For 77GHz Devices(76GHz to 81Ghz): Valid range: 7600 to 8100 Default value: 7600 (If this API is not issued) For 60GHz Devices(57GHz to 64Ghz): Valid range: 5700 to 6400 Default value: 5700 (If this API is not issued)
FREQ_LIMIT_ LOW_TX2	2	The sensor's lower frequency limit for calibrations and monitoring for TX2 is encoded in 2 bytes (16 bit unsigned number) 1 LSB = 10 MHz For 77GHz Devices(76GHz to 81Ghz): Valid range: 7600 to 8100 Default value: 7600 (If this API is not issued) For 60GHz Devices(57GHz to 64Ghz): Valid range: 5700 to 6400 Default value: 5700 (If this API is not issued)

Continued on next page

Table 5.11 – continued from previous page

FREQ_LIMIT_HIGH_TX0	2	The sensor's higher frequency limit for calibrations and monitoring for TX0 is encoded in 2 bytes (16 bit unsigned number) 1 LSB = 10 MHz For 77GHz Devices (76GHz to 81Ghz): Valid range: 7600 to 8100 Default value: 8100 (If this API is not issued) For 60GHz Devices (57GHz to 64Ghz): Valid range: 5700 to 6400 Default value: 6400 (If this API is not issued) For xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP devices: Valid range: 7620 to 8050/8100 (based on VCO2_RANGE_CONFIG in this API) Default value: 8050 (If this API is not issued) NOTE: FREQ_LIMIT_HIGH_TXn should be strictly greater than FREQ_LIMIT_LOW_TXn
FREQ_LIMIT_HIGH_TX1	2	The sensor's higher frequency limit for calibrations and monitoring for TX1 is encoded in 2 bytes (16 bit unsigned number) 1 LSB = 10 MHz For 77GHz Devices (76GHz to 81Ghz): Valid range: 7600 to 8100 Default value: 8100 (If this API is not issued) For 60GHz Devices (57GHz to 64Ghz): Valid range: 5700 to 6400 Default value: 6400 (If this API is not issued) For xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP devices: Valid range: 7620 to 8050/8100 (based on VCO2_RANGE_CONFIG in this API) Default value: 8050 (If this API is not issued) NOTE: FREQ_LIMIT_HIGH_TXn should be strictly greater than FREQ_LIMIT_LOW_TXn

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Table 5.11 – continued from previous page

FREQ_LIMIT_HIGH_TX2	2	The sensor's higher frequency limit for calibrations and monitoring for TX2 is encoded in 2 bytes (16 bit unsigned number) 1 LSB = 10 MHz For 77GHz Devices (76GHz to 81Ghz): Valid range: 7600 to 8100 Default value: 8100 (If this API is not issued) For 60GHz Devices (57GHz to 64Ghz): Valid range: 5700 to 6400 Default value: 6400 (If this API is not issued) For xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP devices: Valid range: 7620 to 8050/8100 (based on VCO2_RANGE_CONFIG in this API) Default value: 8050 (If this API is not issued) NOTE: FREQ_LIMIT_HIGH_TXn should be strictly greater than FREQ_LIMIT_LOW_TXn
TX0_POWER_BACKOFF	1	TX0 output power back off 1 LSB = 1 dB Valid values: 0, 3, 6, 9 Default value: 0
TX1_POWER_BACKOFF	1	TX1 output power back off 1 LSB = 1 dB Valid values: 0, 3, 6, 9 Default value: 0
TX2_POWER_BACKOFF	1	TX2 output power back off 1 LSB = 1 dB Valid values: 0, 3, 6, 9 Default value: 0
TX3_POWER_BACKOFF	1	TX3 output power back off (only for xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP devices) 1 LSB = 1 dB Valid values: 0, 3, 6, 9 Default value: 0
FREQ_LIMIT_LOW_TX3	2	The sensor's lower frequency limit for calibrations and monitoring for TX3 is encoded in 2 bytes (16 bit unsigned number) (only for xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP devices) 1 LSB = 10 MHz For 77GHz Devices(76GHz to 81Ghz): Valid range: 7600 to 8100 Default value: 7600 (If this API is not issued)

Continued on next page

Table 5.11 – continued from previous page

FREQ_LIMIT_HIGH_TX3	2	The sensor's higher frequency limit for calibrations and monitoring for TX3 is encoded in 2 bytes (16 bit unsigned number) (only for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices) 1 LSB = 10 MHz For xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: Valid range: 7620 to 8050/8100 (based on VCO2_RANGE_CONFIG in this API) Default value: 8050 (If this API is not issued) NOTE: FREQ_LIMIT_HIGH_TXn should be strictly greater than FREQ_LIMIT_LOW_TXn						
VCO2_RANGE_CONFIG	1	VCO2 frequency range configuration. <table><thead><tr><th>Bit</th><th>Definition</th></tr></thead><tbody><tr><td>b0</td><td> VCO2 frequency range selection. VCO2 in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP only supports a maximum bandwidth of 4.5GHz. The user can select between the two allowed ranges shown below. 1 - VCO2 range is 76GHz - 80.5GHz. 0 - VCO2 range is 76.5GHz - 81GHz. Default setting: 0 (76.5-81GHz) </td></tr><tr><td>b1-7</td><td>RESERVED</td></tr></tbody></table>	Bit	Definition	b0	VCO2 frequency range selection. VCO2 in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP only supports a maximum bandwidth of 4.5GHz. The user can select between the two allowed ranges shown below. 1 - VCO2 range is 76GHz - 80.5GHz. 0 - VCO2 range is 76.5GHz - 81GHz. Default setting: 0 (76.5-81GHz)	b1-7	RESERVED
Bit	Definition							
b0	VCO2 frequency range selection. VCO2 in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP only supports a maximum bandwidth of 4.5GHz. The user can select between the two allowed ranges shown below. 1 - VCO2 range is 76GHz - 80.5GHz. 0 - VCO2 range is 76.5GHz - 81GHz. Default setting: 0 (76.5-81GHz)							
b1-7	RESERVED							
RESERVED	1	0x00						
RESERVED	2	0x0000						

NOTE1:	The minimum RF bandwidth 200MHz, this is to perform internal calibration and monitoring.
NOTE2:	The limit set in this API is not applicable for functional chirps and loop-back chirps used in advanced frame config API.
NOTE3:	The TX0 frequency limit is used by default in calibrations and monitors where TX is not relevant or enabled.
NOTE4:	The RF band used in functional chirp profiles shall be within the limit set in this API.
NOTE5:	The mid frequency code of RF band used in functional chirp profiles + 200MHz shall be within the max limit set in this API. In xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices, if this is not followed, the FW internally backs off any calibration test chirps starting RF to honor the Cal Mon Freq Limit. Overall, this ensures that the calibration chirps (which have up to 200MHz RF BW) don't transmit outside the Cal Mon Freq limits.
NOTE6:	The power limits in this API apply to the calibrations that transmit (TX power calibration, phase shift calibration), and the monitors that do NOT have an associated functional profile index (like TX ball break monitor). The power limits set here do NOT apply to monitors that are clearly associated with a functional profile index (like TX gain phase mismatch monitor)

5.2.11 Sub block 0x008B – AWR_CAL_DATA_RESTORE_SB

This sub block restores the calibration data which was stored previously using the [AWR_CAL_DATA_SAVE_SB](#) command. The async event [AWR_AE_RF_INITCALIBSTATUS_SB](#) will be issued after this API, this indicates success of the calibration data restore. The calibration data contents are defined in page 94 in AWR_CAL_DATA table.

Table 5.12: AWR_CAL_DATA_RESTORE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x008B
SBLKLEN	2	Value = 232
RESERVED	2	0x0000
CHUNK_ID	2	Index of the current chunk Valid range: 0 to 2
CAL_DATA	224	Calibration data chunk which was stored in non-volatile memory

- NOTE1:** All 3 chunks of 224 bytes each shall be sent to radar device to complete the restore process and to generate AWR_AE_RF_INITCALIBSTATUS_SB AE.
- NOTE2:** Refer recommended API sequence and order in page 402 for more details on sequence of issuing this API.

5.2.12 Sub block 0x008C – AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB

This sub block restores the calibration data which was stored previously using the AWR_PHASE_SHIFTER_CAL_DATA_SAVE_SB command. This is device specific feature, please refer data sheet.

- NOTE:** In xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP devices, this can be used to inject factory calibration data generated using corner reflector based external calibration. There are some limitations when using this API in xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP.
1. The injected calibration data must necessarily correspond to 25C.
 2. If external calibration data is injected, the user has to execute runtime TX phase shifter calibration using ONE_TIME_CALIB_ENABLE_MASK in AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB API before transmitting frames. This corrects the calibration data for the current temperature.

Table 5.13: AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x008C
SBLKLEN	2	Value = 136
TX_INDx	1	Index of the transmit channel for which the following data applies Valid range: 0 to 3
CAL_APPLY	1	Set this to 0x01 after applying calibration data from all transmitters. This bit will indicate the firmware to start the correction process.

Continued on next page

Table 5.13 – continued from previous page

OBS_PHSIFT_DATA	128	Observed phase shift corresponding to each desired phase shift. Index n corresponds to desired phase shift of $n \times 5.625^\circ$. For TX0 and TX3, for phase shifter setting/index $n=0$ to 63 corresponding to functional APIs (e.g. Profile Config, Per Chirp Phase Shifter, etc), the calibration data needs to be retrieved from/restored to following byte locations of TX0 and TX3 phase calibration data save/restore API: <table> <tr> <th>n</th><th>Desired phase shift</th><th>Observed phase shift is injected in the following bytes</th></tr> <tr> <td>0</td><td>0</td><td>$x5.625^\circ$ byte[1], byte[0]</td></tr> <tr> <td>1</td><td>1</td><td>$x5.625^\circ$ byte[3], byte[2]</td></tr> <tr> <td>$\vdots$</td><td>$\vdots$</td><td></td></tr> <tr> <td>62</td><td>62</td><td>$x5.625^\circ$ byte[125], byte[124]</td></tr> <tr> <td>63</td><td>63</td><td>$x5.625^\circ$ byte[127], byte[126]</td></tr> </table> For TX1 and TX2, for phase shifter setting/index $n=0$ to 63 corresponding to functional APIs (e.g. Profile Config, Per Chirp Phase Shifter, etc), the calibration data needs to be retrieved from/restored to following byte locations of TX1 and TX2 phase calibration data save/restore API: <table> <tr> <th>n</th><th>Desired phase shift</th><th>Observed phase shift is injected in the following bytes</th></tr> <tr> <td>32</td><td>32</td><td>$x5.625^\circ$ byte[1], byte[0]</td></tr> <tr> <td>33</td><td>33</td><td>$x5.625^\circ$ byte[3], byte[2]</td></tr> <tr> <td>$\vdots$</td><td>$\vdots$</td><td></td></tr> <tr> <td>62</td><td>62</td><td>$x5.625^\circ$ byte[61], byte[60]</td></tr> <tr> <td>63</td><td>63</td><td>$x5.625^\circ$ byte[63], byte[62]</td></tr> <tr> <td>0</td><td>0</td><td>$x5.625^\circ$ byte[65], byte[64]</td></tr> <tr> <td>1</td><td>1</td><td>$x5.625^\circ$ byte[67], byte[66]</td></tr> <tr> <td>$\vdots$</td><td>$\vdots$</td><td></td></tr> <tr> <td>30</td><td>30</td><td>$x5.625^\circ$ byte[125], byte[124]</td></tr> <tr> <td>31</td><td>31</td><td>$x5.625^\circ$ byte[127], byte[126]</td></tr> </table> 1 LSB = $360^\circ / 2^{10}$	n	Desired phase shift	Observed phase shift is injected in the following bytes	0	0	$x5.625^\circ$ byte[1], byte[0]	1	1	$x5.625^\circ$ byte[3], byte[2]	$\vdots$	$\vdots$		62	62	$x5.625^\circ$ byte[125], byte[124]	63	63	$x5.625^\circ$ byte[127], byte[126]	n	Desired phase shift	Observed phase shift is injected in the following bytes	32	32	$x5.625^\circ$ byte[1], byte[0]	33	33	$x5.625^\circ$ byte[3], byte[2]	$\vdots$	$\vdots$		62	62	$x5.625^\circ$ byte[61], byte[60]	63	63	$x5.625^\circ$ byte[63], byte[62]	0	0	$x5.625^\circ$ byte[65], byte[64]	1	1	$x5.625^\circ$ byte[67], byte[66]	$\vdots$	$\vdots$		30	30	$x5.625^\circ$ byte[125], byte[124]	31	31	$x5.625^\circ$ byte[127], byte[126]
n	Desired phase shift	Observed phase shift is injected in the following bytes																																																			
0	0	$x5.625^\circ$ byte[1], byte[0]																																																			
1	1	$x5.625^\circ$ byte[3], byte[2]																																																			
$\vdots$	$\vdots$																																																				
62	62	$x5.625^\circ$ byte[125], byte[124]																																																			
63	63	$x5.625^\circ$ byte[127], byte[126]																																																			
n	Desired phase shift	Observed phase shift is injected in the following bytes																																																			
32	32	$x5.625^\circ$ byte[1], byte[0]																																																			
33	33	$x5.625^\circ$ byte[3], byte[2]																																																			
$\vdots$	$\vdots$																																																				
62	62	$x5.625^\circ$ byte[61], byte[60]																																																			
63	63	$x5.625^\circ$ byte[63], byte[62]																																																			
0	0	$x5.625^\circ$ byte[65], byte[64]																																																			
1	1	$x5.625^\circ$ byte[67], byte[66]																																																			
$\vdots$	$\vdots$																																																				
30	30	$x5.625^\circ$ byte[125], byte[124]																																																			
31	31	$x5.625^\circ$ byte[127], byte[126]																																																			
RESERVED	2	0x0000																																																			

5.2.13 Sub block 0x008D – AWR_APLL_SYNTH_BW_CONTROL_SB

This is a new feature addition in **AWR2243**, **xWR6x43**, **xWR294x**, **xWR254x**, **xWR2x4xECO** and **xWR2x4xP**. This sub block is used to control bandwidth of the APLL and Synthesizer. The typical recommended settings are as below.

Table 5.14: Typical APLL and Synth BW settings for xWR6x43

S.No	Synth ICP	Synth Rtrim	APLL ICP	APLL Rtrim LPF	APLL Rtrim VCO	VCO1/ BW	VCO2 BW	APLL BW	Emission Improv	1M PN De-grad@6l	100K PN Im-prov @60G	Max Slope (MHz/us)
1	1	8	0x26	0x9	8	1.5M	1.5M	150K	2dB	2dB	0dB	250
2	3	8	0x26	0x9	8	0.75M	0.75M	150K	2dB	2dB	0dB	125
3	1	8	0x3F	0x9	8	1.5M	1.5M	300K	2dB	4dB	5dB	250
4	1	8	0x26	0x9	5	1.5M	1.5M	150K	8dB	1.5dB	0dB	250
5	3	8	0x26	0x9	5	0.75M	0.75M	150K	8dB	1.5dB	0dB	125
6	1	8	0x3F	0x9	5	1.5M	1.5M	300K	8dB	3.5dB	5dB	250
7	1	8	0x26	0x9	6	1.5M	1.5M	150K	5dB	1dB	0dB	250
8	3	8	0x26	0x9	6	0.75M	0.75M	150K	5dB	1dB	0dB	125
9	1	8	0x3F	0x9	6	1.5M	1.5M	300K	5dB	3dB	5dB	250
10	1	8	0x26	0x9	18	1.5M	1.5M	150K	0dB	0dB	0dB	250
11	3	8	0x26	0x9	18	0.75M	0.75M	150K	0dB	0dB	0dB	125
12	1	8	0x3F	0x9	18	1.5M	1.5M	300K	0dB	2dB	5dB	250

Table 5.15: Typical APLL and Synth BW settings for AWR2243

SYNTH_ ICP_TRIM	SYNTH_ RZ_TRIM	APLL_ ICP_TRIM	APLL_ RZ_TRIM	VCO1_ BW	VCO2_ BW	APLL_ BW	Description	Max VCO Slope (MHz/us)
1	8	0x26	0x9	750K	1.5M	150K	Default settings (+/-0.2% Error at 2us ADC start)	266
3	8	0x26	0x9	375K	750K	150K	Optimum for 76-77GHz VCO1 (1M, 10M PN)	100
0	8	0x26	0x9	1.3M	2.6M	150K	Synth High BW (+/-0.2% Error at 1us ADC start)	266
3	8	0x3F	0x9	375K	1.5M	300K	Optimum 100K PN	100

Table 5.16: Typical APLL and Synth BW settings for xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP devices

SYNTH_ ICP_ TRIM	SYNTH_ RZ_ TRIM	APLL_ ICP_ TRIM	APLL_ RZ_ TRIM_ LPF	APLL_ RZ_ TRIM_ VCO	VCO1_ BW	VCO2_ BW	APLL_ BW	Description
1	4	0xB	0x9	0x12	-	1.3M	158K	Default VCO2 setting
1	5	0xB	0x9	0x12	500K	-	150K	Default VCO1 setting

Table 5.17: AWR_APLL_SYNTH_BW_CONTROL_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x008D
SBLKLEN	2	Value = 20
SYNTH_ICP_TRIM , SYNTH_ICP_TRIM_VCO1 (xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP)	1	In xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices, this field sets the SYNTH ICP trim code for VCO1. In other supported devices, this field sets the common SYNTH ICP trim code for both VCO1 and VCO2. Default value in xWR294x/ xWR254x/ xWR2x4xECO/ xWR2x4xP = 0x1.
SYNTH_RZ_TRIM , SYNTH_RZ_TRIM_VCO1 (xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP)	1	In xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices, this field sets the SYNTH RZ trim code for VCO1. In other supported devices, this field sets the common SYNTH RZ trim code for both VCO1 and VCO2. Default value in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP = 0x5.
APLL_ICP_TRIM	1	APLL ICP trim code. Default value in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP = 0xB.
APLL_RZ_TRIM_LPF	1	APLL RZ LPF trim code. Default value in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP = 0x9.

Continued on next page

Table 5.17 – continued from previous page

APLL_RZ_TRIM_VCO	1	APLL RZ VCO trim code AWR2243 device: 0 : Programs the default device setting. Other values : Reserved for future use. xWR6x43 device: Values specified in the xWR6x43 table xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP device: Values specified in the xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP table Default value in xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP = 0x12.
SYNTH_ICP_TRIM_VCO2	1	In xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP devices, this field sets the SYNTH ICP trim code for VCO2. This field is RESERVED in other devices. Default value in xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP = 0x1.
SYNTH_RZ_TRIM_VCO2	1	In xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP devices, this field sets the SYNTH RZ trim code for VCO2. This field is RESERVED in other devices. Default value in xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP = 0x4.
RESERVED	9	0x0000

NOTE: Recommended to issue this AWR_APLL_SYNTH_BW_CONTROL_SB API before AWR_RF_INIT_SB API. The RF_INIT synthesizer boot calibration shall run after changing the APLL BW.

5.3 Sub blocks related to AWR_RF_STATIC_CONF_GET_MSG

5.3.1 Sub block 0x00A0 – 0x00AA – RESERVED

5.3.2 Sub block 0x00AB – AWR_CAL_DATA_SAVE_SB

This sub block reads the calibration data from the device which can be injected later using the AWR_CAL_DATA_RESTORE_SB command.

NOTE: The total size of the calibration data is 672 bytes, this has been split into 3 chunks (NUM_CHUNKS) of 224 bytes each due to SPI limitation. The Host should receive all these 3 chunks from radar device, later host can store only relevant data in non volatile memory.

Table 5.18: AWR_CAL_DATA_SAVE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x00AB
SBLKLEN	2	Value = 8
RESERVED	2	0x0000
CHUNK_ID	2	Index of the requested chunk Valid values: 0 to NUM_CHUNKS - 1

Response to the above command will contain the calibration data which is formatted as shown below

Table 5.19: AWR_CAL_DATA_SAVE_SB response packet contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x00AB
SBLKLEN	2	Value = 232
NUM_CHUNKS	2	Total number of calibration data chunks NUM_CHUNKS
CHUNK_ID	2	Current chunk number
CAL_DATA	224	Calibration data, refer device specific CAL_DATA contents below

Table 5.20: AWR_CAL_DATA contents for AWR2243

Field Name	Number of bytes	Description
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CAL_VALIDITY_STATUS	4	This field indicates the status of each calibration (0 – FAIL, 1 – PASS). If a particular calibration was not enabled, then its corresponding field should be ignored. Bit Definition (0 – FAIL, 1 – PASS) b0 RESERVED b1 APLL tuning (Ignore while store restore) b2 SYNTH VCO1 tuning (Ignore while store restore) b3 SYNTH VCO2 tuning (Ignore while store restore) b4 LODIST calibration b5 RX ADC DC offset calibration b6 HPF cutoff calibration b7 LPF cutoff calibration b8 Peak detector calibration (optional) b9 TX Power calibration (optional) b10 RX gain calibration b11 TX Phase calibration (optional) b12 RX IQMM calibration b31:13 RESERVED The recommended Validity status bits while restoring is 0x000014E0, assuming only RX_ADC_DC_CAL_DATA, HPF_CAL_DATA, LPF_CAL_DATA, RX_RF_GAIN_CAL_DATA and IQMM_CAL_DATA are stored and restored.
CAL_VALIDITY_STATUS_COPY	4	Redundant CAL_VALIDITY_STATUS value, this value should match with CAL_VALIDITY_STATUS
RESERVED	8	RESERVED
CAL_TEMPERATURE	2	Temperature at which boot calibration is done
RESERVED	14	RESERVED
RX_ADC_DC_CAL_DATA	16	Rx chain ADC DC calibration data
HPF1_CAL_DATA	1	HPF1 calibration data
HPF2_CAL_DATA	1	HPF2 calibration data
LODIST_BIAS_CODE	1	LODIST calibration data
LODIST_FREQ_INDEX	1	LODIST calibration frequency index
RESERVED	48	RESERVED
RX_RF_GAIN_CAL_DATA	8	RX RF gain calibration data

IQMM_CAL_DATA	104	RX IQMM calibration data
TX_POWER_CAL_DATA	82	TX Power calibration data
POWER_DET_CAL_DATA	326	Power detector calibration data
RESERVED	52	RESERVED

Table 5.21: AWR_CAL_DATA contents for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices

Field Name	Number of bytes	Description
CAL_VALIDITY_STATUS	4	This field indicates the status of each calibration (0 – FAIL, 1 – PASS). If a particular calibration was not enabled, then its corresponding field should be ignored. Bit Definition (0 – FAIL, 1 – PASS) b0 RESERVED b1 APLL tuning (Ignore while store restore) b2 SYNTH VCO1 tuning (Ignore while store restore) b3 SYNTH VCO2 tuning (Ignore while store restore) b4 LODIST calibration b5 RX ADC DC offset calibration b6 HPF cutoff calibration b7 LPF cutoff calibration b8 Peak detector calibration (optional) b9 TX Power calibration (optional) b10 RX gain calibration b11 TX Phase calibration (optional) b12 RESERVED b31:13 RESERVED The recommended Validity status bits while restoring is 0x000004E0, assuming only RX_ADC_DC_CAL_DATA, HPF_CAL_DATA, LPF_CAL_DATA, RX_RF_GAIN_CAL_DATA are stored and restored.
CAL_VALIDITY_STATUS_COPY	4	Redundant CAL_VALIDITY_STATUS value, this value should match with CAL_VALIDITY_STATUS
RESERVED	8	RESERVED

CAL_TEMPERA- TURE	2	Temperature at which boot calibration is done
RESERVED	14	RESERVED
RX_ADC_DC_ CAL_DATA	16	Rx chain ADC DC calibration data
HPF1_CAL_DATA	1	HPF1 calibration data
HPF2_CAL_DATA	1	HPF2 calibration data
LODIST_BIAS_ CODE	1	LODIST calibration data
LODIST_FREQ_ INDEX	1	LODIST calibration frequency index
HPF1_CAL_ DATA_300KHZ	1	HPF1 CTRIM data for 300KHz cutoff
RESERVED	47	RESERVED
RX_RF_GAIN_ CAL_DATA	10	RX RF gain calibration data
RESERVED	102	RESERVED
TX_POWER_ CAL_DATA	148	TX Power calibration data
POWER_DET_ CAL_DATA	250	Power detector calibration data
RESERVED	62	RESERVED

Table 5.22: AWR_CAL_DATA contents for xWR6x43 devices

Field Name	Number of bytes	Description
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CAL_VALIDITY_STATUS	4	This field indicates the status of each calibration (0 – FAIL, 1 – PASS). If a particular calibration was not enabled, then its corresponding field should be ignored. Bit Definition (0 – FAIL, 1 – PASS) b0 SYNTH VCO3 tuning (Available only on selected xWR6243 device variants, RESERVED for other 60GHz devices. Ignore while store restore) b1 APLL tuning (Ignore while store restore) b2 SYNTH VCO1 tuning (Ignore while store restore) b3 SYNTH VCO2 tuning (Ignore while store restore) b4 LODIST calibration (Ignore while store restore) b5 RX ADC DC offset calibration b6 HPF cutoff calibration b7 LPF cutoff calibration b8 Peak detector calibration (optional) b9 TX Power calibration (optional) b10 RX gain calibration b11 TX Phase calibration (Ignore while store restore) b12 RX IQMM calibration b31:13 RESERVED The recommended Validity status bits while restoring is 0x000014E0, assuming only RX_ADC_DC_CAL_DATA, HPF_CAL_DATA, LPF_CAL_DATA, RX_RF_GAIN_CAL_DATA and IQMM_CAL_DATA are stored and restored.
CAL_VALIDITY_STATUS_COPY	4	Redundant CAL_VALIDITY_STATUS value, this value should match with CAL_VALIDITY_STATUS
RESERVED	8	RESERVED
CAL_TEMPERATURE	2	Temperature at which boot calibration is done
RESERVED	14	RESERVED
RX_ADC_DC_CAL_DATA	16	Rx chain ADC DC calibration data
HPF1_CAL_DATA	1	HPF1 calibration data
HPF2_CAL_DATA	1	HPF2 calibration data
LODIST_BIAS_CODE	1	LODIST calibration data
RESERVED	1	RESERVED
RX_RF_GAIN_CAL_DATA	8	RX RF gain calibration data

IQMM_CAL_DATA	104	RX IQMM calibration data
TX_POWER_CAL_DATA	122	TX Power calibration data
POWER_DET_CAL_DATA	344	Power detector calibration data
RESERVED	42	RESERVED

NOTE1:	Before storing the calibration data in non volatile memory, the host shall make sure validity status of all enabled calibrations are SET to value 1 including APLL, VCO1, VCO2 and LODIST calibration validity in RF_INIT of radar device.
NOTE2:	Host can store only relevant calibration data in non volatile memory and corresponding validity bits shall be set to 1 in AWR_CAL_DATA_RESTORE_SB and rest of the validity bits should be clear to 0 before restoring the data to radar device.
NOTE3:	Host shall ignore APLL, VCO1 and VCO2 validity bits while restoring, these calibrations will be done in each device power-up.

5.3.3 Sub block 0x00AC – AWR_PHASE_SHIFTER_CAL_DATA_SAVE_SB

This sub block reads the phase shifter calibration data from the device which can be injected later using the AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB command. This is device specific feature, please refer data sheet.

NOTE:	In xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP devices, issuing this API will always return correction values from a constant LUT corresponding to 25C. In case external calibration data was injected earlier, the same data will be returned back.
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Table 5.23: AWR_PHASE_SHIFTER_CAL_DATA_SAVE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x00AC
SBLKLEN	2	Value = 8

Continued on next page

Table 5.23 – continued from previous page

TX_INDXX	1	Index of the transmitter channel for which the phase shift is desired Valid range for AWR2243/xWR6243: 0 to 2 Valid range for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP: 0 to 3
RESERVED	3	0x000000

Response to the above command will contain the phase shifter calibration data which is formatted as shown below

Table 5.24: AWR_PHASE_SHIFTER_CAL_DATA_SAVE_SB response packet contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x00AC
SBLKLEN	2	Value = 136
TX_INDXX	1	Index of the transmitter channel for which the following phase shift values applies Valid range for AWR2243/xWR6243: 0 to 2 Valid range for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP: 0 to 3
RESERVED	1	0x00

Continued on next page

Table 5.24 – continued from previous page

OBS_PHSIFT_DATA	128	Observed phase shift corresponding to each desired phase shift. Index n corresponds to desired phase shift of $n \times 5.625^\circ$. For TX0 and TX3, for phase shifter setting/index $n=0$ to 63 corresponding to functional APIs (e.g. Profile Config, Per Chirp Phase Shifter, etc), the calibration data needs to be retrieved from/restored to following byte locations of TX0 and TX3 phase calibration data save/restore API: <table> <tr> <th>n</th><th>Desired phase shift</th><th>Observed phase shift is injected in the following bytes</th></tr> <tr> <td>0</td><td>0</td><td>$x5.625^\circ$ byte[1], byte[0]</td></tr> <tr> <td>1</td><td>1</td><td>$x5.625^\circ$ byte[3], byte[2]</td></tr> <tr> <td>$\vdots$</td><td>$\vdots$</td><td></td></tr> <tr> <td>62</td><td>62</td><td>$x5.625^\circ$ byte[125], byte[124]</td></tr> <tr> <td>63</td><td>63</td><td>$x5.625^\circ$ byte[127], byte[126]</td></tr> </table> For TX1 and TX2, for phase shifter setting/index $n=0$ to 63 corresponding to functional APIs (e.g. Profile Config, Per Chirp Phase Shifter, etc), the calibration data needs to be retrieved from/restored to following byte locations of TX1 and TX2 phase calibration data save/restore API: <table> <tr> <th>n</th><th>Desired phase shift</th><th>Observed phase shift is injected in the following bytes</th></tr> <tr> <td>32</td><td>32</td><td>$x5.625^\circ$ byte[1], byte[0]</td></tr> <tr> <td>33</td><td>33</td><td>$x5.625^\circ$ byte[3], byte[2]</td></tr> <tr> <td>$\vdots$</td><td>$\vdots$</td><td></td></tr> <tr> <td>62</td><td>62</td><td>$x5.625^\circ$ byte[61], byte[60]</td></tr> <tr> <td>63</td><td>63</td><td>$x5.625^\circ$ byte[63], byte[62]</td></tr> <tr> <td>0</td><td>0</td><td>$x5.625^\circ$ byte[65], byte[64]</td></tr> <tr> <td>1</td><td>1</td><td>$x5.625^\circ$ byte[67], byte[66]</td></tr> <tr> <td>$\vdots$</td><td>$\vdots$</td><td></td></tr> <tr> <td>30</td><td>30</td><td>$x5.625^\circ$ byte[125], byte[124]</td></tr> <tr> <td>31</td><td>31</td><td>$x5.625^\circ$ byte[127], byte[126]</td></tr> </table> 1 LSB = $360^\circ / 2^{10}$	n	Desired phase shift	Observed phase shift is injected in the following bytes	0	0	$x5.625^\circ$ byte[1], byte[0]	1	1	$x5.625^\circ$ byte[3], byte[2]	$\vdots$	$\vdots$		62	62	$x5.625^\circ$ byte[125], byte[124]	63	63	$x5.625^\circ$ byte[127], byte[126]	n	Desired phase shift	Observed phase shift is injected in the following bytes	32	32	$x5.625^\circ$ byte[1], byte[0]	33	33	$x5.625^\circ$ byte[3], byte[2]	$\vdots$	$\vdots$		62	62	$x5.625^\circ$ byte[61], byte[60]	63	63	$x5.625^\circ$ byte[63], byte[62]	0	0	$x5.625^\circ$ byte[65], byte[64]	1	1	$x5.625^\circ$ byte[67], byte[66]	$\vdots$	$\vdots$		30	30	$x5.625^\circ$ byte[125], byte[124]	31	31	$x5.625^\circ$ byte[127], byte[126]
n	Desired phase shift	Observed phase shift is injected in the following bytes																																																			
0	0	$x5.625^\circ$ byte[1], byte[0]																																																			
1	1	$x5.625^\circ$ byte[3], byte[2]																																																			
$\vdots$	$\vdots$																																																				
62	62	$x5.625^\circ$ byte[125], byte[124]																																																			
63	63	$x5.625^\circ$ byte[127], byte[126]																																																			
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33	33	$x5.625^\circ$ byte[3], byte[2]																																																			
$\vdots$	$\vdots$																																																				
62	62	$x5.625^\circ$ byte[61], byte[60]																																																			
63	63	$x5.625^\circ$ byte[63], byte[62]																																																			
0	0	$x5.625^\circ$ byte[65], byte[64]																																																			
1	1	$x5.625^\circ$ byte[67], byte[66]																																																			
$\vdots$	$\vdots$																																																				
30	30	$x5.625^\circ$ byte[125], byte[124]																																																			
31	31	$x5.625^\circ$ byte[127], byte[126]																																																			
RESERVED	2	0x0000																																																			

5.4 Sub blocks related to AWR_RF_INIT_MSG

5.4.1 Sub block 0x00C0 – AWR_RF_INIT_SB

This sub block, needed to be initially issued, triggers one time calibrations such as those related to APLL and synthesizer. The BSS processor is woken up upon receiving this sub block, the RF analog and digital baseband sections are initialized.

Table 5.25 describes the content of this sub block.

Table 5.25: AWR_RF_INIT_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x00C0
SBLKLEN	2	Value = 4

NOTE1: This sub block will be acknowledged immediately but an async event [AWR_AE_RF_INITCALIBSTATUS_SB](#) from BSS will indicate that the RF initialization is complete. No commands shall be sent to BSS till the async event is received.

NOTE2: It is not recommended to issue this API in runtime multiple times. This API shall be issued only once after power cycle with or without calibration data restore operation.

NOTE3:

The following boot-time calibrations are susceptible to corruption by interference. The calibrations may result in false configuration of the RF analog sections due to corruption by interference during the calibration measurements.

- a. RX gain calibration (susceptible to interference)
- b. RX IQMM calibration (susceptible to interference)
- c. TX Phase calibration (susceptible to interference)

In the 60G band (supported by xWR6x43 devices), it is mandated by regulatory standards that transmissions in non-ISM band are capped to -10dBm. The following calibrations could violate these standards if executed in the field.

- a. TX power calibration.
- b. TX phase shifter calibration.
- c. RX IQMM calibration.

It is recommended to perform factory calibration and store the calibration data in non volatile memory using AWR_CAL_DATA_SAVE_SB API. This data can be restored to radar device using AWR_CAL_DATA_RESTORE_SB API. More info related to save restore provided in page [93](#)

5.5 Sub blocks related to AWR_RF_DYNAMIC_CONF_SET_MSG

5.5.1 Sub block 0x0100 – AWR_PROFILE_CONF_SET_SB

This sub block contains FMCW radar chirp profiles or properties (FMCW slope, chirp duration, TX power etc.). Since the device supports multiple profiles, each profile is defined in this sub block. Internal RF and analog calibrations may be triggered upon receiving this sub block and ASYNC_EVENT response sent once completed.

NOTE 1:

This API can be issued dynamically to change profile parameters. Few parameters which cannot be changed are

1. PF_NUM_ADC_SAMPLES
2. PF_DIGITAL_OUTPUT_SAMPLING_RATE
3. Programmable filter coefficients in AWR2243, xWR6243, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP
4. The dynamic profile configuration settings are applied to HW at the end of the active frame boundary (start of the frame idle time). It is recommended to issue dynamic profile config API at the beginning of the active frame (during the chirping) to apply the profile changes for immediate next frame

NOTE 2:

The number of ADC samples per chirp is a property of the profile. A sub-frame can contain chirps with different profiles. Typically, all the chirps in a sub-frame are expected to have the same number of ADC samples per chirp. However, if the user needs to implement advanced use cases with varying number of samples per chirp across chirps within a sub-frame, the following guidance may be applicable. The BSS does allow this usage by allowing different profiles to have different PF_NUM_ADC_SAMPLES, and different chirps within the same sub-frame to use different profiles. But the user-programmable/configurable MSS/DSS needs to also appropriately receive this data. One example way for this is to follow these two steps.

1. Always read out the maximum of all profiles' PF_NUM_ADC_SAMPLES after each chirp by appropriate programming of ADCBUF parameters.
2. Discard the excess data from those chirps which have lesser PF_NUM_ADC_SAMPLES.

Table 5.26 describes the contents of this sub block.

Table 5.26: AWR_PROFILE_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0100
SBLKLEN	2	Value = 48

Continued on next page

Table 5.26 – continued from previous page

PF_INDXX	2	The profile index for which the rest of the fields are applicable for. Valid range for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: 0 to 6 Valid range for other devices: 0 to 3
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Table 5.26 – continued from previous page

PF_VCO_SELECT	1	Bit	Description
		b0	FORCE_VCO_SEL (Not supported for production in xWR6243 , debug purpose only) 0 Use internal VCO selection 1 Forced external VCO selection
		b2:1	VCO_SEL (Not supported for production in xWR6243 , debug purpose only) 0 VCO1 77G: 76–78GHz in AWR2243, 77G: 76-77GHz in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP, 60G: 57-60.75GHz 1 VCO2 77G: 77–81GHz in AWR2243, 77G: 76-80.5GHz or 76.5-81GHz in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP (based on VCO2_RANGE_CONFIG in AWR_CAL_MON_ FREQUENCY_TX_POWER_LIMITS_SB). 60G: 60–64GHz 2 VCO3 77G: RESERVED. Set it to 0b0, xWR6243: 56–58GHz NOTE: 1. xWR1xxx devices: There is an overlap region of 77-78 GHz in which any of the VCOs can be used, for other regions use only the VCO which can work in that region. For e.g. for 76-78 GHz use only VCO1 and for 77-81GHz use only VCO2, for 77-78 GHz, any VCO can be used. Also note that users can inter-mix chirps from different VCOs within the same frame. 2. xWR6x43 device: There is an overlap region of 60-60.75 GHz in which any of the VCOs can be used. 3. AWR2243 device: VCO2 range is 76 - 81GHz (5GHz RF Bandwidth). There is an overlap region of 76-78 GHz in which any of the VCOs can be used. 4. xWR6243 device: VCO3 is available only on selected xWR6243 device variants. Ther is an overlap region of 57-58 GHz in which VCO1 or VCO3 can be used. 5. xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: VCO2 range is 76-80.5GHz or 76.5-81GHz based on VCO2_RANGE_CONFIG in AWR_CAL_MON_FREQUENY_TX_POWER_LIMITS_SB (4.5GHz RF bandwidth). There is an overlap region of 76-77GHz in which any of the VCOs can be used.

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Table 5.26 – continued from previous page

		b7:3 RESERVED 0b000000
PF_CALLUT_UPDATE	1	Bit Description b0 RETAIN_TXCAL_LUT 0 Update TX calibration LUT 1 Do not update TX calibration LUT b1 RETAIN_RXCAL_LUT 0 Update RX calibration LUT and update RX IQMM correction 1 Do not update RX calibration LUT b7:2 RESERVED (set it to 0b000000) Normally, whenever Profile Config API is issued, the TX/RX analog settings are recomputed and their properties can change. If Profile Config API is issued for a second time then the user has an option to ensure the TX/RX RF properties/settings are unchanged, by setting the appropriate RETAIN_RX/TXCAL_LUT bits to 1. For example, if Profile Config API is issued with minor timing parameter changes like Idle Time and without changing the TX power or RX gain or chirp RF frequency range, then the RETAIN_RX/TXCAL_LUTs can be set to ensure that only the timing parameters change without disturbing the RF analog properties.
PF_FREQ_START_CONST	4	Start frequency for this profile For 77GHz Devices (76GHz to 81Ghz): 1 LSB = $3.6e9/2^{26}$ Hz $\approx$ 53.644 Hz Valid range: 0x5471C71C to 0x5A000000 For 60GHz Devices (56GHz/57GHz to 64Ghz): 1 LSB = $2.7e9/2^{26}$ Hz $\approx$ 40.233 Hz Valid range: Only even numbers from 0x52F684BD/0x5471C71C to 0x5ED097B4 NOTE: Refer to device datasheet for supported frequency ranges NOTE: For xWR294x, xWR254x, xWR2x4xECo and xWR2x4xP, when using VCO2, this start frequency must be within the chosen VCO2_RANGE_CONFIG in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB.
PF_IDLE_TIME_CONST	4	Idle time for each profile 1 LSB = 10 ns Valid range: 250 to 524287
PF_ADC_START_TIME_CONST	4	Time of starting of ADC capture relative to the knee of the ramp 1 LSB = 10 ns Valid range: 0 to 4095

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Table 5.26 – continued from previous page

PF_RAMP_END_TIME	4	End of ramp time relative to the knee of the ramp 1 LSB = 10 ns Valid range: 2 to 500000 Ensure that the total frequency sweep is either within these ranges: 77G: 76-78 GHz or 77-81 GHz for AWR2243 77G: 76-80.5 , 76.5-81 GHz for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP based on VCO2_RANGE_CONFIG in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB. 60G: 56-58GHz or 57-60.75GHz or 60-64GHz										
PF_TX_OUTPUT_POWER_BACKOFF	4	<table><tr><th>Bits</th><th>Description</th></tr><tr><td>b7:0</td><td>TX0 output power back off</td></tr><tr><td>b15:8</td><td>TX1 output power back off</td></tr><tr><td>b23:16</td><td>TX2 output power back off</td></tr><tr><td>b31:24</td><td>TX3 output power back off (applicable only for xWR294x,xWR254x,xWR2x4xECO and xWR2x4xP devices, set to 0 otherwise)</td></tr></table> This field defines how much the transmit power should be reduced from the maximum. AWR2243 devices: 0 to 20 in 1dB steps xWR6x43 devices: 0 to 26 in 1dB steps xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: 0 to 20 in 1dB steps. TX output power inaccuracy can be high for backoff setting greater than 15dB. 0dB back-off corresponds to typically 13dBm power level in AWR2243/xWR294x/xWR254x/xWR6x43 device. 0dB back-off corresponds to typically 13.5dBm power level in xWR2x4xECO,xWR2x4xP device.	Bits	Description	b7:0	TX0 output power back off	b15:8	TX1 output power back off	b23:16	TX2 output power back off	b31:24	TX3 output power back off (applicable only for xWR294x,xWR254x,xWR2x4xECO and xWR2x4xP devices, set to 0 otherwise)
Bits	Description											
b7:0	TX0 output power back off											
b15:8	TX1 output power back off											
b23:16	TX2 output power back off											
b31:24	TX3 output power back off (applicable only for xWR294x,xWR254x,xWR2x4xECO and xWR2x4xP devices, set to 0 otherwise)											

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Table 5.26 – continued from previous page

PF_TX_PHASE_SHIFTER	4	Bits Description b1:0 RESERVED (set it to 0b00) b7:2 TX0 phase shift value 1 LSB = $360^\circ / 2^6 \approx 5.625^\circ$ b9:8 RESERVED (set it to 0b00) b15:10 TX1 phase shift value 1 LSB = $360^\circ / 2^6 \approx 5.625^\circ$ b17:16 RESERVED (set it to 0b00) b23:18 TX2 phase shift value 1 LSB = $360^\circ / 2^6 \approx 5.625^\circ$ b25:24 RESERVED (set it to 0b00) b31:26 TX3 phase shift value (only for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices) 1 LSB = $360^\circ / 2^6 \approx 5.625^\circ$ This field defines the additional phase shift to be introduced on each transmitter output. This option is supported only on selected device variants, Please refer data sheet.
PF_FREQ_SLOPE_CONST	2	Frequency slope for each profile is encoded in 2 bytes (16 bit signed number) For 77GHz Devices (76GHz to 81Ghz): 1 LSB = $3.6e9 \times 900 / 2^{26}$ Hz ≈ 48.279 kHz/μs Valid range: -5510 to 5510 (Max 266MHz/μs) xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP specific restrictions: There are additional restrictions on this field as shown below. Only X = [0 to +256, except (17, 23, 34, 45, 46, 68, 90, 92, 93, 111, 136, 137, 180, 184, 186, 222, 239)], along with +/- (2*X), +/- (4*X), +/- (8*X), +/- (16*X), +/- (32*X), +/- (64*X) are the supported slope codes. Other slope codes are not recommended. For 60GHz Devices (57GHz to 64Ghz): 1 LSB = $2.7e9 \times 900 / 2^{26}$ Hz ≈ 36.21 kHz/μs Valid range: Only even numbers in the range -6905 to 6905 (250MHz/μs) Note: Refer AWR_APLL_SYNTH_BW_CONTROL_SB BW control API for constraints on max slope.
PF_TX_START_TIME	2	Time of start of transmitter relative to the knee of the ramp 1 LSB = 10 ns Valid range: -4096 to 4095

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Table 5.26 – continued from previous page

		Positive numbers refer to start of TX after knee of the ramp and negative numbers refer to start of TX before the knee of the ramp Refer Note 6 below for more info on timing.															
PF_NUM_ADC_SAMPLES	2	Number of ADC samples to capture in a chirp for each RX Valid range: 2 to MAX_NUM_SAMPLES, where MAX_NUM_SAMPLES is such that all the enabled RX channels' data fits into 16 kB memory in AWR1243, xWR1443, AWR2243, xWR6243, xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP or 32 kB memory in xWR1642/xWR6x43/xWR1843, with each sample consuming 2 bytes for real ADC output case and 4 bytes for complex 1x and complex 2x ADC output cases. For example in AWR1243, xWR1443, AWR2243, xWR6243, xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP when the ADC buffer size is 16 kB <table> <thead> <tr> <th>Number of RX chains</th><th>ADC format</th><th>MAX_NUM_SAMPLES</th></tr> </thead> <tbody> <tr> <td>4</td><td>Complex</td><td>1024</td></tr> <tr> <td>4</td><td>Real</td><td>2048</td></tr> <tr> <td>2</td><td>Complex</td><td>2048</td></tr> <tr> <td>2</td><td>Real</td><td>4096</td></tr> </tbody> </table>	Number of RX chains	ADC format	MAX_NUM_SAMPLES	4	Complex	1024	4	Real	2048	2	Complex	2048	2	Real	4096
Number of RX chains	ADC format	MAX_NUM_SAMPLES															
4	Complex	1024															
4	Real	2048															
2	Complex	2048															
2	Real	4096															
PF_DIGITAL_OUTPUT_SAMPLING_RATE	2	ADC Sampling rate for this profile is encoded in 2 bytes (16 bit unsigned number) 1 LSB = 1 ksps Valid range: AWR2243/xWR6243 device: 2000 to 50000 (Max 20MHz IF bandwidth) xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP device: 2000 to 45000 (Max 20MHz IF bandwidth)															
PF_HPF1_CORNER_FREQ	1	xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP devices: HPF1 corner frequency for each profile is encoded in 1 byte. Note1: Both HPF1 and HPF2 necessarily must have the same corner frequency in xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP devices. Note2: Using the 1.4MHz setting might cause high inaccuracies in the cutoff frequency in both functional path and monitoring results. Value HPF1 corner frequency definition															

Continued on next page

Table 5.26 – continued from previous page

		0x00 350 kHz 0x01 700 kHz 0x02 1.4 MHz 0x03 RESERVED 0x04 300 KHz AWR2243 / xWR6243 devices HPF1 corner frequency for each profile is encoded in 1 byte Value HPF1 corner frequency definition 0x00 175 kHz 0x01 235 kHz 0x02 350 kHz 0x03 700 kHz
PF_HP2_COR- NER_FREQ	1	xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: HPF2 corner frequency for each profile is encoded in 1 byte. Note1: Both HPF1 and HPF2 necessarily must have the same corner frequency in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices. Note2: Using the 1.4MHz setting might cause high inaccuracies in the cutoff frequency in both functional path and monitoring results. Value HPF2 corner frequency definition 0x00 350 kHz 0x01 700 kHz 0x02 1.4 MHz 0x03 RESERVED 0x04 300 KHz AWR2243 / xWR6243 devices HPF2 corner frequency for each profile is encoded in 1 byte Value HPF2 corner frequency definition 0x00 350 kHz 0x01 700 kHz 0x02 1.4 MHz 0x03 2.8 MHz

Continued on next page

Table 5.26 – continued from previous page

TX_CAL_EN_CFG	2	Number of transmitters to turn on during TX power calibration. During actual operation, if more than 1 TXs are enabled during the chirp, then enabling the same TXs during calibration will have better TX output power accuracy. For this field to take effect, RUNTIME_TX_POWER_MULTI_TX_CAL_EN field in PF_MISC_FEATURE_EN needs to be enabled. If this bit is not set, only 1 TX is enabled during the TX power calibration. For e.g. during TX0 calibration, only TX0 will be enabled; during TX1 calibration, only TX1 will be enabled and so on. <table><tr><td>Bit</td><td>Definition</td></tr><tr><td>b3:0</td><td>TX enabled during TX0 calibration b0 - TX0, b1 - TX1, b2 - TX2, b3 - TX3</td></tr><tr><td>b7:4</td><td>TX enabled during TX1 calibration b4 - TX0, b5 - TX1, b6 - TX2, b7 - TX3</td></tr><tr><td>b11:8</td><td>TX enabled during TX2 calibration b8 - TX0, b9 - TX1, b10 - TX2, b11 - TX3</td></tr><tr><td>b15:12</td><td>TX enabled during TX3 calibration b12 - TX0, b13 - TX1, b14 - TX2, b15 - TX3</td></tr></table>	Bit	Definition	b3:0	TX enabled during TX0 calibration b0 - TX0, b1 - TX1, b2 - TX2, b3 - TX3	b7:4	TX enabled during TX1 calibration b4 - TX0, b5 - TX1, b6 - TX2, b7 - TX3	b11:8	TX enabled during TX2 calibration b8 - TX0, b9 - TX1, b10 - TX2, b11 - TX3	b15:12	TX enabled during TX3 calibration b12 - TX0, b13 - TX1, b14 - TX2, b15 - TX3																		
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b3:0	TX enabled during TX0 calibration b0 - TX0, b1 - TX1, b2 - TX2, b3 - TX3																													
b7:4	TX enabled during TX1 calibration b4 - TX0, b5 - TX1, b6 - TX2, b7 - TX3																													
b11:8	TX enabled during TX2 calibration b8 - TX0, b9 - TX1, b10 - TX2, b11 - TX3																													
b15:12	TX enabled during TX3 calibration b12 - TX0, b13 - TX1, b14 - TX2, b15 - TX3																													
PF_RX_GAIN	1	<table><tr><td>Bit</td><td>Definition</td></tr><tr><td>b5:0</td><td> RX_GAIN This field defines RX gain for each channel. 1 LSB = 1 dB Valid values: AWR2243 : All even values from 32 to 52 xWR6x43 : All even values from 30 to 48 xWR294x : All even values from 28 to 44 xWR254x : All even values from 28 to 44 xWR2x4xECO, xWR2x4xP : All even values from 30 to 46 </td></tr><tr><td>b7:6</td><td> RF_GAIN_TARGET The RF gain target: <table><tr><td>Value</td><td>RF gain target</td></tr><tr><td></td><td>(AWR2243/ xWR6243)</td><td>(xWR294x/ xWR254x)</td><td>(xWR2x4xECO/ xWR2x4xP)</td></tr><tr><td>00</td><td>30 dB</td><td>32 dB</td><td>34 dB</td></tr><tr><td>01</td><td>33 dB</td><td>34 dB</td><td>36 dB</td></tr><tr><td>10</td><td>36 dB</td><td>36 dB</td><td>38 dB</td></tr><tr><td>11</td><td>RESERVED</td><td>RESERVED</td><td>RESERVED</td></tr></table></td></tr></table>	Bit	Definition	b5:0	RX_GAIN This field defines RX gain for each channel. 1 LSB = 1 dB Valid values: AWR2243 : All even values from 32 to 52 xWR6x43 : All even values from 30 to 48 xWR294x : All even values from 28 to 44 xWR254x : All even values from 28 to 44 xWR2x4xECO, xWR2x4xP : All even values from 30 to 46	b7:6	RF_GAIN_TARGET The RF gain target: <table><tr><td>Value</td><td>RF gain target</td></tr><tr><td></td><td>(AWR2243/ xWR6243)</td><td>(xWR294x/ xWR254x)</td><td>(xWR2x4xECO/ xWR2x4xP)</td></tr><tr><td>00</td><td>30 dB</td><td>32 dB</td><td>34 dB</td></tr><tr><td>01</td><td>33 dB</td><td>34 dB</td><td>36 dB</td></tr><tr><td>10</td><td>36 dB</td><td>36 dB</td><td>38 dB</td></tr><tr><td>11</td><td>RESERVED</td><td>RESERVED</td><td>RESERVED</td></tr></table>	Value	RF gain target		(AWR2243/ xWR6243)	(xWR294x/ xWR254x)	(xWR2x4xECO/ xWR2x4xP)	00	30 dB	32 dB	34 dB	01	33 dB	34 dB	36 dB	10	36 dB	36 dB	38 dB	11	RESERVED	RESERVED	RESERVED
Bit	Definition																													
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b7:6	RF_GAIN_TARGET The RF gain target: <table><tr><td>Value</td><td>RF gain target</td></tr><tr><td></td><td>(AWR2243/ xWR6243)</td><td>(xWR294x/ xWR254x)</td><td>(xWR2x4xECO/ xWR2x4xP)</td></tr><tr><td>00</td><td>30 dB</td><td>32 dB</td><td>34 dB</td></tr><tr><td>01</td><td>33 dB</td><td>34 dB</td><td>36 dB</td></tr><tr><td>10</td><td>36 dB</td><td>36 dB</td><td>38 dB</td></tr><tr><td>11</td><td>RESERVED</td><td>RESERVED</td><td>RESERVED</td></tr></table>	Value	RF gain target		(AWR2243/ xWR6243)	(xWR294x/ xWR254x)	(xWR2x4xECO/ xWR2x4xP)	00	30 dB	32 dB	34 dB	01	33 dB	34 dB	36 dB	10	36 dB	36 dB	38 dB	11	RESERVED	RESERVED	RESERVED							
Value	RF gain target																													
	(AWR2243/ xWR6243)	(xWR294x/ xWR254x)	(xWR2x4xECO/ xWR2x4xP)																											
00	30 dB	32 dB	34 dB																											
01	33 dB	34 dB	36 dB																											
10	36 dB	36 dB	38 dB																											
11	RESERVED	RESERVED	RESERVED																											

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Table 5.26 – continued from previous page

		Recommended RF_GAIN_TARGET is 36dB for AWR2243/xWR6243, 36dB for xWR294x/xWR254x and 38dB for xWR2x4xECO/xWR2x4xP. The total RX gain is achieved as a sum of RF gain and IF amplifiers gain. The RF Gain Target allows the user to control the RF gain independently from the total RX gain, thus giving flexibility to the user to trade-off linearity vs. noise figure. Out of multiple gain settings for the RF stages, the firmware calibration algorithm uses the one that makes the RF gain as close as possible to the user programmed RF Gain Target. AWR2243/xWR6243: At high temperatures, the RF Gain Targets provide trade-off of approximately 4 dB in RF P1dB point vs 2 dB in noise figure. For the lowest RF Gain Target setting 30 dB, the RF gain varies linearly from 38 dB at -40C to 30 dB at 140C for nominal process corner. Since the minimum IF gain is -6 dB, The minimum achievable RX Gain varies from 32 dB at -40C to 24 dB at 140C. The maximum RX gain setting is recommended to be limited to 48dB, which can be achieved at all temperatures and RF gain target conditions. Increasing RX gain beyond 48 dB may result in degradation of in-band P1dB without improvement in noise figure. xWR294x/xWR254x: For the lowest RF Gain Target setting 32 dB, the RF gain ranges between from 30.2 dB to 35.8 dB for nominal process corner. Since the minimum IF gain is -10 dB, the minimum achievable RX Gain lies between 20.2 dB to 25.8 dB at this RF Gain Target. We recommend using RF Gain Target of 36 dB to achieve best NF while maintaining optimal Linearity (OOB-P1dB) in the FE. The maximum RX gain setting is recommended to be limited to 44dB, which can be achieved at all temperatures at 36 dB RF gain target condition. Increasing RX gain beyond 44 dB may result in degradation of in-band P1dB without significant improvement in noise figure. In addition, since the maximum IF gain is +10dB in xWR294x/xWR254x, we recommend to avoid the following pairs of (RX gain, RF gain target) settings: (44dB, 32dB).
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Table 5.26 – continued from previous page

		xWR2x4xECON/xWR2x4xP: We recommend using RF Gain Target of 38 dB to achieve best NF while maintaining optimal Linearity (OOB-P1dB) in the FE. The maximum RX gain setting is recommended to be limited to 46dB, which can be achieved at all temperatures at 38 dB RF gain target condition. Increasing RX gain beyond 46 dB may result in degradation of in-band P1dB without significant improvement in noise figure. In addition, since the maximum IF gain is +10dB in xWR2x4xECON/xWR2x4xP, we recommend to avoid the following pairs of (RX gain, RF gain target) settings: (46dB, 34dB).
PF_MISC_FEATURE_EN	1	This bit-field controls the enable, disable of miscellaneous profile specific features. This is only applicable for xWR294x, xWR254x, xWR2x4xECON, xWR2x4xP devices. Bit Feature b0 RESERVED b1 PF_HPF_FAST_INIT HPF fast init feature helps in settling the low IF signal (e.g. bumper reflection) quickly at the beginning of the chirp. When HPF fast init feature is enabled, the profile's ADC start time should be at least 2.5uS later than the profile's TX start time. Value Status 0 HPF fast init disabled. 1 HPF fast init enabled. b2 RUNTIME_TX_POWER_MULTI_TX_CAL_EN Value Status 0 Multi-TX runtime TX power cal disabled. 1 Multi-TX runtime TX power cal enabled. b3:b7 RESERVED
RESERVED	1	0x00
RESERVED	1	0x00

NOTE1:	Please refer Table ?? for details on minimum chirp duration.
NOTE2:	The max TX output power back-off only up to 20dB is supported.
NOTE3:	The RF band used in functional chirp profiles shall be within the limit set in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB API.
NOTE4:	This API takes around 700us to execute in RadarSS sub System.
NOTE5:	Phase shifter(PS) settings are applied in advance at max -5us or at $-(Idle_time - 1.28us - DfeLagTime)$ from the knee of the ramp. If idle time is $> 6.28us$ then PS is applied always at -5us and if idle time $< 6.28us$ then PS is applied at $-(Idle_time - 1.28us - DfeLagTime)$ from knee of the ramp as shown in figure below. Where DfeLagTime is internal DFE lag time (Please refer rampgen calculator).
NOTE6:	It is recommended to configure TX start time $> -5us$ or $-(Idle_time - 1.28us - DfeLagTime)$ based on PS apply time as shown in figure below.
NOTE7:	The mid frequency code of RF band used in functional chirp profiles + 200MHz shall be within the max limit set in this API.
NOTE8:	RX data coherence across profiles - Amplitude and phase coherence may not be perfect across profiles normally. Coherence across profiles can be improved if the user sets the same chirp slope, ADC Start Time, Sampling Rate, RX GAIN and HPF cutoff settings across these profiles, AND also explicitly reads one reference profile's TX calibration settings (through AWR_ADV_TX_GAIN_TEMPLUT_GET_SB) and forces the same into the other profiles (through AWR_ADV_TX_GAIN_TEMPLUT_SET_SB).

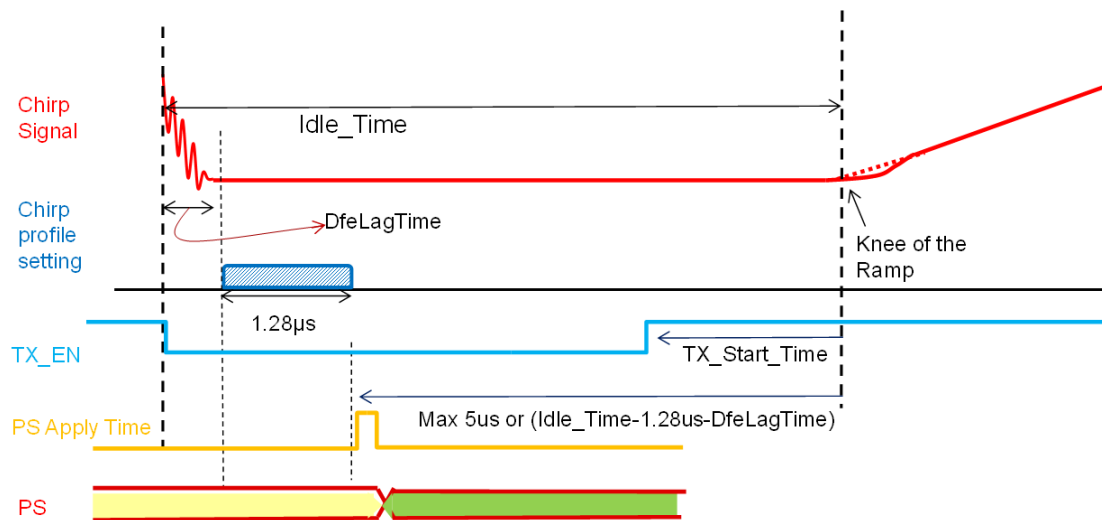


Figure 5.1: TX PS apply timing in a chirp

Table 5.27: Note on maximum sampling rate

NOTE:

The maximum sampling rate supported is limited based on the information in the table below

When device supports 20 MHz IF bandwidth (refer device data sheet)

	Real/Pseudo Real	Complex1x	Complex2x
Regular ADC mode	45 Msps	22.5 Msps	45 Msps
Low power ADC mode	18.75 Msps	9.375 Msps	18.75 Msps

When device supports 15 MHz IF bandwidth (refer device data sheet)

	Real/Pseudo Real	Complex1x	Complex2x
Regular ADC mode	37.5 Msps	18.75 Msps	37.5 Msps
Low power ADC mode	18.75 Msps	9.375 Msps	18.75 Msps

- The IF bandwidth here refers to the IF frequency of the farthest reflection desired to be detected
- Typically, the IF frequency range preserved well in the receiver baseband is $0.9 \times \text{Sampling Rate in Complex 1x}$ and $0.45 \times \text{Sampling Rate in Complex 2x and Real/Pseudo Real}$.
- The maximum sampling rates are also subject to restrictions from LVDS/CSI2 interface rate and ADC bits configurations. Typically in Complex2x mode, the maximum sampling rate would be 45 Msps.
- In Low power ADC mode, the max supported IF BW is 7.5MHz only.

5.5.2 Sub block 0x0101 – AWR_CHIRP_CONF_SET_SB

This sub block contains chirp to chirp variations on top of the chirp profiles defined in the AWR_PROFILE_CONF_SET_SB. E.g. which profile is to be used for each chirp in a frame, and small dithers in FMCW start frequency and idle time for each chirp are possible to be defined here. The

dithers used in this configuration sub block are only additive on top of programmed parameters in AWR_PROFILE_CONF_SET_SB.

Table 5.28 describes the contents of this sub block.

Table 5.28: AWR_CHIRP_CONF_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0101
SBLKLEN	2	Value = 24
CHIRP_START_INDXX	2	Valid range 0 to 511
CHIRP_END_INDXX	2	Valid range CHIRP_START_INDXX to 511
PROFILE_INDXX	2	Valid range 0 to 3
RESERVED	2	0x0000
CHIRP_FREQ_START_VAR	4	For 77GHz Devices (76GHz to 81GHz): 1 LSB = $3.6e9/2^{26} \approx 53.644$ Hz Valid range: 0 to 8388607 (450MHz) For 60GHz Devices (57GHz to 64Ghz): 1 LSB = $2.7e9/2^{26} \approx 40.23$ Hz Valid range: Only even numbers from 0 to 8388607 (337.5MHz)
CHIRP_FREQ_SLOPE_VAR	2	For 77GHz Devices (76GHz to 81GHz): 1 LSB = $3.6e9 \times 900/2^{26} \approx 48.279$ kHz Valid range: 0 to 63 (3MHz/us) For 60GHz Devices (57GHz to 64Ghz): 1 LSB = $2.7e9 \times 900/2^{26} \approx 36.21$ Hz Valid range: Only even numbers between 0 to 63 (2.3MHz/us)
CHIRP_IDLE_TIME_VAR	2	Idle time of each chirp is encoded in 2 bytes (16 bit unsigned number) 1 LSB = 10 ns Valid range: 0 to 4095
CHIRP_ADC_START_TIME_VAR	2	ADC start time of each chirp is encoded in 2 bytes (16 bit unsigned number) 1 LSB = 10 ns Valid range: 0 to 4095

Continued on next page

Table 5.28 – continued from previous page

CHIRP_TX_EN	2	TX enable selection Bit Definition b0 TX0 Enable b1 TX1 Enable b2 TX2 Enable b3 TX3 Enable (only applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices) b15:4 RESERVED 0b0_0000_0000_0000 NOTE: Maximum number of TXs that can be turned on in a chirp depends on the device data sheet specification
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5.5.3 Sub block 0x0102 – AWR_FRAME_CONF_SET_SB

This sub block defines a frame, i.e. a sequence of chirps to be transmitted subsequently, the no. of frames to be transmitted, frame periodicity and how to trigger them.

Table 5.29 describes the contents of this sub block.

Table 5.29: AWR_FRAME_CONF_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0102
SBLKLEN	2	Value = 28
RESERVED	2	May use to indicate Frame mode or Continuous chirping mode of operation.
CHIRP_START_INDXX	2	Valid range 0 to 511 NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field is not used/applicable.
CHIRP_END_INDXX	2	Valid range CHIRP_START_INDXX to 511 NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field is not used/applicable.

Continued on next page

Table 5.29 – continued from previous page

NUM_LOOPS	2	Number of times to repeat from CHIRP_START_IND_X to CHIRP_END_IND_X in each frame Valid range 1 to 255 NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field definition is modified. ADVANCE_CHIRP_CONFIG mode: This field configures the total number of chirps in a frame L. This should be programmed as per below calculation. $Y \text{ (HW loops)} = \text{ceil}(L/512)$ $X \text{ (HW RAM)} = L / Y$ where X is an integer and $1 \leq X \leq 512$ The value of L should be a multiple of 4 (assuming each chirp is min 25us duration) i.e 1, 4, 8, 12, 16, 20, ... 32768 (max). The FW needs to prepare and update HW chirp RAM dynamically in advance chirp config API, this puts some restriction on minimum number of chirps in a burst/frame. Valid range 1 to 32768
NUM_FRAMES	2	Number of frames to transmit 16 bit unsigned number Valid range: 0 to 65535 (0 for infinite frames)
RESERVED	2	0x0000
FRAME_PERIODICITY	4	$\text{PERIOD} \geq \text{Sum total time of all chirps} + \text{InterFrameBlankTime}$, where, Sum total time of all chirps = Num Loops * Num chirps * Chirp Period. <i>InterFrameBlankTime</i> is primarily for sensor calibration/-monitoring, thermal control, transferring out any safety monitoring data if requested, hardware reconfiguration for next frame, re triggering of next frame. $\text{InterFrameBlankTime} \geq 300 \mu\text{s}$ typical, refer a NOTE end of this API for more info. Add 150 μs to <i>InterFrameBlankTime</i> if data-path re-configuration needed in frame boundary due to change in profile. 1 LSB = 5 ns Valid range 300 μs to 1.342 s NOTE: If the devices' self-triggered periodic monitoring features are enabled, the user needs to set the frame period taking into consideration the monitoring periodicities (refer to CALIB_MON_TIME_UNIT in AWR_CALIB_MON_TIME_UNIT_CONF_SB)

Continued on next page

Table 5.29 – continued from previous page

TRIGGER_SELECT	2	Value	Definition
		0x0001	SWTRIGGER (Software API based triggering): Frame is triggered upon receiving AWR_FRAMESTARTSTOP_CONF_SB. There could be several tens of micro seconds uncertainty in triggering. This mode is not applicable if this device is configured as MULTICHIP_SLAVE in AWR_CHAN_CONF_SB.
		0x0002	HWTRIGGER (Hardware SYNC_IN based triggering): Each frame is triggered by rising edge of pulse in SYNC_IN pin, after receiving AWR_FRAMESTARTSTOP_CONF_SB (this is to prevent spurious transmission). W.r.t. the SYNC_IN pulse, the actual transmission has 160ns delay and 5ns uncertainty in SINGLECHIP and only a 300 ps uncertainty (due to tight inter-chip synchronization needed) in MULTICHIP sensor applications as defined in AWR_CHAN_CONF_SB. For more details please refer to device datasheet.
RESERVED	1	0x00	
RESERVED	1	0x00	
FRAME_TRIGGER_DELAY	4	Optional time delay from the SYNC_IN trigger to the occurrence of frame chirps. Applicable only in SINGLECHIP sensor applications, as defined in AWR_CHAN_CONF_SB. It is recommended only for staggering the transmission of multiple radar sensors around the car for interference avoidance, if needed. Typical range is 0 to 100 micro seconds. Units: 1 LSB = 5 ns	

NOTE1:	If hardware triggered mode is used, the SYNC_IN pulse width should be less than 4 us. Also, the minimum pulse width of SYNC_IN should be 25 ns.
NOTE2:	If frame trigger delay is used with hardware triggered mode, then external SYNC_IN pulse periodicity should take care of the configured frame trigger delay and frame periodicity. The external pulse should be issued only after the sum total of frame trigger delay and frame periodicity. See figure below
NOTE3:	If dummy chirp is used then programmer should make sure the idle time of dummy chirp $\geq 4\mu s + \text{DFE spill over time of previous chirp}$ (calculate from rampgen calculator). The first chirp of frame can not be a dummy chirp.
NOTE4:	In Hw triggered mode, the Hw pulse should be issued or periodicity of pulse is configured such that, the pulse is generated only 150us after the completion of previous frame/burst (The pulse should not be issued before end of previous frame/burst). The time delta between end of previous frame/burst and raising edge of Hw pulse recommended to be $< 300\mu s$.
NOTE5:	The PF_DIGITAL_OUTPUT_SAMPLING_RATE impacts the LVD-S/CSI2 data rate in a frame, so it is recommended to analyze timing impact if different sample rate is used across chirps in a frame.
NOTE6:	Please refer Table 11.4 for details on minimum inter-frame blank time requirements.
NOTE7:	If advance chirp configuration is enabled then this API takes around 1.8ms to execute in RadarSS sub System for 128 chirps. The error checks for each parameters of advance chirp is done in frame configuration API. This option can be disabled by using ADVANCE_CHIRP_ERROR_CHK_DIS option in AWR_RF_RADAR_MISC_CTL_SB API. If this error check takes more than 50ms due to large number of chirps then it is recommended to disable RadarSS WDT while executing this API.
NOTE9:	Due to data synchronizers in the synthesizer chirp starting path, there can be 0/1.1 ns bimodal jitter in the starting of each RF chirp wrt RX ADC sampling. This can lead to a small chirp to chirp phase jitter whose magnitude depends on IF frequency. For example, this translates to $360^\circ \times 1.1 \text{ ns} \times 1 \text{ MHz} = 0.4^\circ$ at 1-MHz IF in range dimension; 4° for 10-MHz IF, and 8° for 20-MHz IF. If the inter-chirp jitter mitigation feature is enabled in products where it is available, this jitter will be mitigated from inter-chirp perspective and be present only as an inter-burst, inter-sub-frame and inter-frame jitter, with the same magnitude.

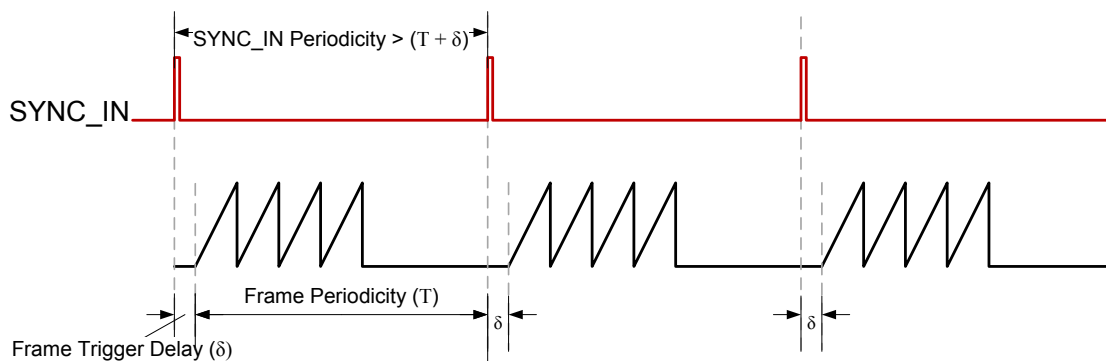


Figure 5.2: Frame trigger delay in case of external hardware trigger

5.5.4 Sub block 0x0103 – AWR_CONT_STREAMING_MODE_CONF_SET_SB

This sub block contains configuration needed to enable continuous streaming mode from the device.

NOTE: The continuous streaming mode configuration APIs are supported only for debug purpose. Please refer latest DFP release note for more info.

Table 5.30 describes the contents of this sub block.

Table 5.30: AWR_CONT_STREAMING_MODE_CONF_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0103
SBLKLEN	2	Value = 24

Continued on next page

Table 5.30 – continued from previous page

PF_FREQ_START_CONST	4	Frequency start for each profile is encoded in 4 bytes (32 bit unsigned number) For 77GHz Devices (76GHz to 81GHz): 1 LSB = $3.6e9/2^{26}$ Hz $\approx$ 53.644 Hz Valid range: 0 to 0x7FFFFFFF For 60GHz Devices (56GHz/57GHz to 64GHz): 1 LSB = $2.7e9/2^{26}$ Hz $\approx$ 40.23 Hz Valid range: Only even numbers from 0 to 0x7FFFFFFF NOTE1: Refer to device datasheet for supported frequency ranges NOTE2: For xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP, when using VCO2, this start frequency must be within the chosen VCO2_RANGE_CONFIG in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB.										
PF_TX_OUTPUT_POWER_BACKOFF	4	<table><thead><tr><th>Bits</th><th>Description</th></tr></thead><tbody><tr><td>b7:0</td><td>TX0 output power back off</td></tr><tr><td>b15:8</td><td>TX1 output power back off</td></tr><tr><td>b23:16</td><td>TX2 output power back off</td></tr><tr><td>b31:24</td><td>TX3 output power back off (applicable only for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices, set to 0 otherwise)</td></tr></tbody></table> This field defines how much the transmit power should be reduced from the maximum. 1 LSB = 1 dB	Bits	Description	b7:0	TX0 output power back off	b15:8	TX1 output power back off	b23:16	TX2 output power back off	b31:24	TX3 output power back off (applicable only for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices, set to 0 otherwise)
Bits	Description											
b7:0	TX0 output power back off											
b15:8	TX1 output power back off											
b23:16	TX2 output power back off											
b31:24	TX3 output power back off (applicable only for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices, set to 0 otherwise)											

Continued on next page

Table 5.30 – continued from previous page

PF_TX_PHASE_SHIFTER	4	Bits Description b1:0 RESERVED (set it to 0b00) b7:2 TX0 phase shift value 1 LSB = $360^\circ/2^6 \approx 5.625^\circ$ b9:8 RESERVED (set it to 0b00) b15:10 TX1 phase shift value 1 LSB = $360^\circ/2^6 \approx 5.625^\circ$ b17:16 RESERVED (set it to 0b00) b23:18 TX2 phase shift value 1 LSB = $360^\circ/2^6 \approx 5.625^\circ$ b25:24 RESERVED (set it to 0b00) b31:26 TX3 phase shift value (only for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices) 1 LSB = $360^\circ/2^6 \approx 5.625^\circ$ This field defines the additional phase shift to be introduced on each transmitter output.
PF_DIGITAL_OUTPUT_SAMPLING_RATE	2	ADC Sampling rate for each profile is encoded in 2 bytes (16 bit unsigned number) 1 LSB = 1 ksp/s Valid range 2000 to 37500
PF_HPF1_CORNER_FREQ	1	xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: HPF1 corner frequency for each profile is encoded in 1 byte. Note1: Both HPF1 and HPF2 necessarily must have the same corner frequency in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices. Note2: Using the 1.4MHz setting might cause high inaccuracies in the cutoff frequency in both functional path and monitoring results. Value HPF1 corner frequency definition 0x00 350 kHz 0x01 700 kHz 0x02 1.4 MHz 0x03 RESERVED 0x04 300 KHz AWR2243 / xWR6243 devices HPF1 corner frequency for each profile is encoded in 1 byte

Continued on next page

Table 5.30 – continued from previous page

		Value HPF1 corner frequency definition 0x00 175 kHz 0x01 235 kHz 0x02 350 kHz 0x03 700 kHz
PF_HP2_CORNER_FREQ	1	xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: HPF2 corner frequency for each profile is encoded in 1 byte. Note1: Both HPF1 and HPF2 necessarily must have the same corner frequency in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices. Note2: Using the 1.4MHz setting might cause high inaccuracies in the cutoff frequency in both functional path and monitoring results. Value HPF2 corner frequency definition 0x00 350 kHz 0x01 700 kHz 0x02 1.4 MHz 0x03 RESERVED 0x04 300 KHz AWR2243 / xWR6243 devices HPF2 corner frequency for each profile is encoded in 1 byte Value HPF2 corner frequency definition 0x00 350 kHz 0x01 700 kHz 0x02 1.4 MHz 0x03 2.8 MHz

Continued on next page

Table 5.30 – continued from previous page

PF_RX_GAIN	1	This field defines RX gain for continuous streaming mode. Bit Definition b5:0 RX_GAIN This field defines RX gain for each channel. 1 LSB = 1 dB Valid values: AWR2243 : All even values from 32 to 52 xWR6x43 : All even values from 30 to 48 xWR294x : All even values from 28 to 44 xWR254x : All even values from 28 to 44 xWR2x4xECON/xWR2x4xP : All even values from 30 to 46 b7:6 RF_GAIN_TARGET The RF gain target for AWR2243, xWR6243, xWR294x, xWR254x, xWR2x4xECON, xWR2x4xP devices: Value RF gain target <table><tr><td></td><td>(AWR2243/ xWR6243)</td><td>(xWR294x/ xWR254x)</td><td>(xWR2x4xECON/ xWR2x4xP)</td></tr><tr><td>00</td><td>30 dB</td><td>32 dB</td><td>34 dB</td></tr><tr><td>01</td><td>33 dB</td><td>34 dB</td><td>36 dB</td></tr><tr><td>10</td><td>36 dB</td><td>36 dB</td><td>38 dB</td></tr><tr><td>11</td><td>RESERVED</td><td>RESERVED</td><td>RESERVED</td></tr></table> Refer Profile configuration API for more info.		(AWR2243/ xWR6243)	(xWR294x/ xWR254x)	(xWR2x4xECON/ xWR2x4xP)	00	30 dB	32 dB	34 dB	01	33 dB	34 dB	36 dB	10	36 dB	36 dB	38 dB	11	RESERVED	RESERVED	RESERVED
	(AWR2243/ xWR6243)	(xWR294x/ xWR254x)	(xWR2x4xECON/ xWR2x4xP)																			
00	30 dB	32 dB	34 dB																			
01	33 dB	34 dB	36 dB																			
10	36 dB	36 dB	38 dB																			
11	RESERVED	RESERVED	RESERVED																			

Continued on next page

Table 5.30 – continued from previous page

VCO_SELECT	1	Bit	Description
		b0	FORCE_VCO_SEL (Not supported for production in xWR6243 , debug purpose only) 0 Use internal VCO selection 1 Forced external VCO selection
		b2:1	VCO_SEL 0 VCO1 AWR2243: 76 – 78GHz xWR294x/ xWR254x/ xWR2x4xEco/ xWR2x4xP: 76 - 77GHz 60G devices: 57 - 60.75GHz 1 VCO2 AWR2243: 77 – 81GHz xWR294x/ xWR254x/ xWR2x4xEco/ xWR2x4xP: 76 - 80.5GHz / 76.5 - 81GHz (based on VCO2_RANGE_CONFIG in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB) 60G devices: 60 – 64GHz 2 VCO3 77G devices: RESERVED Set it to 0b0 xWR6243: 56 – 58GHz NOTE: xWR6243 device: VCO3 is available only on selected xAWR6243 device variants.
		b7:3	RESERVED 0b00_0000
RESERVED	1	0x00	
RESERVED	1	0x00	

NOTE: Continuous streaming (CW) mode is useful for RF lab characterization and debug. In this mode, the device is configured to transmit a single continuous wave (CW - 0 slope) tone at a specific RF frequency continuously.

5.5.5 Sub block 0x0104 – AWR_CONT_STREAMING_MODE_EN_SB

This sub block contains configuration needed to enable continuous streaming mode from the device.

Table 5.31 describes the contents of this sub block.

Table 5.31: AWR_CONT_STREAMING_MODE_EN_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0104
SBLKLEN	2	Value = 8
CONT_STREAMING_EN	2	Value Definition 0x0000 Disable continuous streaming mode 0x0001 Enable continuous streaming mode
RESERVED	2	0x0000

5.5.6 Sub block 0x0105 – AWR_ADVANCED_FRAME_CONF_SB

This sub block contains advanced frame configuration options.

Table 5.32 describes the contents of this sub block.

Table 5.32: AWR_ADVANCED_FRAME_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0105
SBLKLEN	2	Value = 152
NUM_SUB-FRAMES	1	Number of sub frames enabled in this frame Valid range: 1 to 4
FORCE_SINGLE_PROFILE	1	Value Definition 0x00 The profile index set in Chirp Config API message governs which profile is used when that chirp is transmitted 0x01 The profile index indicated in Chirp Config message is ignored and all the chirps in each sub frame use a single profile as indicated by that sub frame's profile index set in this message. NOTE: This Field is not used/applicable for loop-back sub-frame.

Continued on next page

Table 5.32 – continued from previous page

LOOPBACK_ CFG	1	Bit	Definition
		b0	LOOPBACK_CFG_EN 0 Disable 1 Enable
		b2:1	SUB_FRAME_ID Sub frame ID for which the loop-back configura- tion applies
		b7:3	RESERVED
SUB_ FRAMETRIG- GER	1	0x00	Disabled (default mode, i.e no trigger is required in SW triggered mode and a pulse trigger is re- quired every burst start in Hw triggered mode)
		0x01	Enabled (Need to trigger each sub-frame ei- ther by SW in software triggered mode through AWR_SUBFRAME_START_CONF_SB API or HW pulse in hardware triggered mode) NOTE: Disable WDT if this mode is enabled.
SF1_PROFILE_ INDX	2	This is applicable only if FORCE_SINGLE_PROFILE is set to 0x01. Please refer to that field for description. Valid range: 0 to 3 Not applicable for loop-back sub-frame	
SF1_CHIRP_ START_INDX	2	Start index of the first chirp for the first burst in sub frame 1 Valid range: 0 to 511 This field is Not applicable for loop-back sub-frame NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field is not used/applicable.	
SF1_NUM_ UNIQUE_ CHIRPS_PER_ BURST	2	Number of unique chirps per burst Valid range: 1 to 512 This field Not applicable for loop-back sub-frame NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field is not used/applicable.	

Continued on next page

Table 5.32 – continued from previous page

SF1_NUM_LOOPS_PER_BURST	2	Number of times to loop through the unique chirps in each burst, without gaps, using HW. Valid range: 1 to 255 NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field definition is modified. ADVANCE_CHIRP_CONFIG mode: This field configures the total number of chirps in a burst L. This should be programmed as per below calculation. $Y \text{ (HW loops)} = \text{ceil}(L/512)$ $X \text{ (HW RAM)} = L / Y$ where X is an integer and $1 \leq X \leq 512$ The value of L should be a multiple of 4 (assuming each chirp is min 25us duration) i.e 1, 4, 8, 12, 16, 20, ... 32768 (max). The FW needs to prepare and update HW chirp RAM dynamically in advance chirp config API, this puts some restriction on minimum number of chirps in a burst/frame. Valid range 1 to 32768
SF1_BURST_PERIOD	4	$\text{BURST_PERIOD} \geq (\text{NUM_LOOPS_PER_BURST} \times (\text{Sum total of all unique chirp times per burst}) + \text{InterBurstBlankTime})$, where <i>InterBurstBlankTime</i> is primarily for sensor calibration / monitoring, thermal control, and some minimum time needed for triggering next burst. InterBurstBlankTime $\geq 110 \mu\text{s}$ typical, refer a NOTE end of this API for more info. NOTE: Across bursts, if the value (Sum total of all unique chirp times per burst), is not a constant, then the actual available blank time can vary and needs to be accounted for. 1 LSB = 5 ns Valid range: 55 μs to 1.342 s
SF1_CHIRP_START_INDX_OFFSET	2	The chirp start index for each burst is determined as the chirp start index of the previous burst plus SFx_START_INDX_OFFSET $\times$ BURST_INDX i.e. $\text{CHIRP_START_INDX} = \text{SFx_CHIRP_START_INDX} + (\text{SFx_CHIRP_START_INDX_OFFSET} \times \text{BURST_INDEX})$ Valid range: 0 to 511 A value of 0 can be used to repeat the same set of unique chirps across bursts. Non-zero values allow spanning a larger number of unique chirps (across bursts). This field Not applicable for loop-back sub-frame NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field is not used/applicable.

Continued on next page

Table 5.32 – continued from previous page

SF1_NUM_BURSTS	2	Number of bursts constituting this sub frame Valid range: 1 to 512 Valid range: 1 to 16 for loop-back sub-frame
SF1_NUM_OUTER_LOOPS	2	Number of times to loop over the set of above defined bursts, for this sub frame. Valid range: 1 to 64 This field is Not applicable for loop-back sub-frame
RESERVED	2	0x0000
SF1_PERIOD	4	$PERIOD \geq \text{Sum total time of all bursts} + \text{InterSubFrameBlankTime}$, where, Sum total time of all bursts = Num Outer Loops * Num Bursts * Burst Period. <i>InterSubFrameBlankTime</i> is primarily for sensor calibration/monitoring, thermal control, transferring out any safety monitoring data if requested, hardware reconfiguration for next sub frame, retriggering of next SF. $\text{InterSubFrameBlankTime} \geq 300 \mu\text{s}$ typical, refer a NOTE end of this API for more info. Add $150 \mu\text{s}$ to <i>InterSubFrameBlankTime</i> if data-path re-configuration needed in sub-frame boundary due to change in profile. 1 LSB = 5 ns Valid range $300 \mu\text{s}$ to 1.342 s
RESERVED	4	0x00000000
RESERVED	4	0x00000000
SF2_PROFILE_IDX	2	This is applicable only if FORCE_SINGLE_PROFILE is set to 0x01. Please refer to that field for description. Valid range: 0 to 3
SF2_CHIRP_START_IDX	2	Start index of the first chirp for the first burst in sub frame 2 Valid range: 0 to 511 This field Not applicable for loop-back sub-frame NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field is not used/applicable.
SF2_NUM_UNIQUE_CHIRPS_PER_BURST	2	Number of unique chirps per burst Valid range: 1 to 512 This field Not applicable for loop-back sub-frame NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field is not used/applicable.

Continued on next page

Table 5.32 – continued from previous page

SF2_NUM_LOOPS_PER_BURST	2	Number of times to loop through the unique chirps in each burst, without gaps, using HW. Valid range: 1 to 255 NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field definition is modified. ADVANCE_CHIRP_CONFIG mode: This field configures the total number of chirps in a burst L. This should be programmed as per below calculation. $Y \text{ (HW loops)} = \text{ceil}(L/512)$ $X \text{ (HW RAM)} = L / Y$ where X is an integer and $1 \leq X \leq 512$ The value of L should be a multiple of 4 (assuming each chirp is min 25us duration) i.e 1, 4, 8, 12, 16, 20, ... 32768 (max). The FW needs to prepare and update HW chirp RAM dynamically in advance chirp config API, this puts some restriction on minimum number of chirps in a burst/frame. Valid range 1 to 32768
SF2_BURST_PERIOD	4	$\text{BURST_PERIOD} \geq (\text{NUM_LOOPS_PER_BURST}) * (\text{Sum total of all unique chirp times per burst}) + \text{InterBurstBlankTime}$, where <i>InterBurstBlankTime</i> is primarily for sensor calibration / monitoring, thermal control, and some minimum time needed for triggering next burst. $\text{InterBurstBlankTime} \geq 110 \mu\text{s}$ typical, refer a NOTE end of this API for more info. NOTE: Across bursts, if the value (Sum total of all unique chirp times per burst), is not a constant, then the actual available blank time can vary and needs to be accounted for. 1 LSB = 5 ns Valid range: 55 μs to 1.342 s
SF2_CHIRP_START_INDX_OFFSET	2	The chirp start index for each burst is determined as the chirp start index of the previous burst plus SFx_START_INDX_OFFSET * BURST_INDX i.e. $\text{CHIRP_START_INDX} = \text{SFx_CHIRP_START_INDX} + (\text{SFx_CHIRP_START_INDX_OFFSET} \times \text{BURST_INDEX})$ Valid range: 0 to 511 A value of 0 can be used to repeat the same set of unique chirps across bursts. Non-zero values allow spanning a larger number of unique chirps (across bursts). This field Not applicable for loop-back sub-frame NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field is not used/applicable.

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Table 5.32 – continued from previous page

SF2_NUM_BURSTS	2	Number of bursts constituting this sub frame Valid range: 1 to 512 Valid range: 1 to 16 for loop-back sub-frame
SF2_NUM_OUTER_LOOPS	2	Number of times to loop over the set of above defined bursts, for this sub frame. Valid range: 1 to 64 This field is Not applicable for loop-back sub-frame
RESERVED	2	0x0000
SF2_PERIOD	4	$PERIOD \geq \text{Sum total time of all bursts} + \text{InterSubFrameBlankTime}$, Where, Sum total time of all bursts = Num Outer Loops * Num Bursts * Burst Period. <i>InterSubFrameBlankTime</i> is primarily for sensor calibration / monitoring, thermal control, transferring out any safety monitoring data if requested, hardware reconfiguration for next sub frame, retriggering of next SF. $\text{InterSubFrameBlankTime} \geq 300 \mu\text{s}$ typical, refer a NOTE end of this API for more info. Add $150 \mu\text{s}$ to <i>InterSubFrameBlankTime</i> if data-path re-configuration needed in sub-frame boundary due to change in profile. 1 LSB = 5 ns Valid range: $300 \mu\text{s}$ to 1.342 s
RESERVED	4	0x00000000
RESERVED	4	0x00000000
SF3_PROFILE_IDX	2	This is applicable only if FORCE_SINGLE_PROFILE is set to 0x01. Please refer to that field for description. Valid range: 0 to 3
SF3_CHIRP_START_IDX	2	Start index of the first chirp in this sub frame Valid range: 0 to 511 This field Not applicable for loop-back sub-frame NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field is not used/applicable.
SF3_NUM_UNIQUE_CHIRPS_PER_BURST	2	Number of unique chirps per burst Valid range: 1 to 512 This field Not applicable for loop-back sub-frame NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field is not used/applicable.

Continued on next page

Table 5.32 – continued from previous page

SF3_NUM_LOOPS_PER_BURST	2	Number of times to loop through the unique chirps in each burst, without gaps, using HW. Valid range: 1 to 255 NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field definition is modified. ADVANCE_CHIRP_CONFIG mode: This field configures the total number of chirps in a burst L. This should be programmed as per below calculation. $Y \text{ (HW loops)} = \text{ceil}(L/512)$ $X \text{ (HW RAM)} = L / Y$ where X is an integer and $1 \leq X \leq 512$ The value of L should be a multiple of 4 (assuming each chirp is min 25us duration) i.e 1, 4, 8, 12, 16, 20, ... 32768 (max). The FW needs to prepare and update HW chirp RAM dynamically in advance chirp config API, this puts some restriction on minimum number of chirps in a burst/frame. Valid range 1 to 32768
SF3_BURST_PERIOD	4	$\text{BURST_PERIOD} \geq (\text{NUM_LOOPS_PER_BURST}) * (\text{Sum total of all unique chirp times per burst}) + \text{InterBurstBlankTime}$, where <i>InterBurstBlankTime</i> is primarily for sensor calibration / monitoring, thermal control, and some minimum time needed for triggering next burst. $\text{InterBurstBlankTime} \geq 110 \mu\text{s}$ typical, refer a NOTE end of this API for more info. NOTE: Across bursts, if the value (Sum total of all unique chirp times per burst), is not a constant, then the actual available blank time can vary and needs to be accounted for. 1 LSB = 5 ns Valid range: 55 μs to 1.342 s
SF3_CHIRP_START_INDX_OFFSET	2	The chirp start index for each burst is determined as the chirp start index of the previous burst plus SFx_START_INDX_OFFSET * BURST_INDX i.e. $\text{CHIRP_START_INDX} = \text{SFx_CHIRP_START_INDX} + (\text{SFx_CHIRP_START_INDX_OFFSET} \times \text{BURST_INDEX})$ Valid range: 0 to 511 A value of 0 can be used to repeat the same set of unique chirps across bursts. Non-zero values allow spanning a larger number of unique chirps (across bursts). This field Not applicable for loop-back sub-frame NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field is not used/applicable.

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Table 5.32 – continued from previous page

SF3_NUM_BURSTS	2	Number of bursts constituting this sub frame Valid range: 1 to 512 Valid range: 1 to 16 for loop-back sub-frame
SF3_NUM_OUTER_LOOPS	2	Number of times to loop over the set of above defined bursts, for this sub frame. Valid range: 1 to 64 This field is Not applicable for loop-back sub-frame
RESERVED	2	0x0000
SF3_PERIOD	4	$PERIOD \geq \text{Sum total time of all bursts} + \text{InterSubFrameBlankTime}$, Where, Sum total time of all bursts = Num Outer Loops * Num Bursts * Burst Period. <i>InterSubFrameBlankTime</i> is primarily for sensor calibration / monitoring, thermal control, transferring out any safety monitoring data if requested, hardware reconfiguration for next sub frame, retriggering of next SF. $\text{InterSubFrameBlankTime} \geq 300 \mu\text{s}$ typical, refer a NOTE end of this API for more info. Add $150 \mu\text{s}$ to <i>InterSubFrameBlankTime</i> if data-path re-configuration needed in sub-frame boundary due to change in profile. 1 LSB = 5 ns Valid range: $300 \mu\text{s}$ to 1.342 s
RESERVED	4	0x00000000
RESERVED	4	0x00000000
SF4_PROFILE_IDX	2	This is applicable only if FORCE_SINGLE_PROFILE is set to 0x01. Please refer to that field for description. Valid range: 0 to 3
SF4_CHIRP_START_IDX	2	Start index of the first chirp in this sub frame Valid range: 0 to 511 This field Not applicable for loop-back sub-frame NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field is not used/applicable.
SF4_NUM_UNIQUE_CHIRPS_PER_BURST	2	Number of unique chirps per burst Valid range: 1 to 512 This field Not applicable for loop-back sub-frame NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field is not used/applicable.

Continued on next page

Table 5.32 – continued from previous page

SF4_NUM_LOOPS_PER_BURST	2	Number of times to loop through the unique chirps in each burst, without gaps, using HW. Valid range: 1 to 255 NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field definition is modified. ADVANCE_CHIRP_CONFIG mode: This field configures the total number of chirps in a burst L. This should be programmed as per below calculation. $Y \text{ (HW loops)} = \text{ceil}(L/512)$ $X \text{ (HW RAM)} = L / Y$ where X is an integer and $1 \leq X \leq 512$ The value of L should be a multiple of 4 (assuming each chirp is min 25us duration) i.e 1, 4, 8, 12, 16, 20, ... 32768 (max). The FW needs to prepare and update HW chirp RAM dynamically in advance chirp config API, this puts some restriction on minimum number of chirps in a burst/frame. Valid range 1 to 32768
SF4_BURST_PERIOD	4	$\text{BURST_PERIOD} \geq (\text{NUM_LOOPS_PER_BURST}) * (\text{Sum total of all unique chirp times per burst}) + \text{InterBurstBlankTime}$, where <i>InterBurstBlankTime</i> is primarily for sensor calibration / monitoring, thermal control, and some minimum time needed for triggering next burst. InterBurstBlankTime $\geq 110 \mu\text{s}$ typical, refer a NOTE end of this API for more info. NOTE: Across bursts, if the value (Sum total of all unique chirp times per burst), is not a constant, then the actual available blank time can vary and needs to be accounted for. 1 LSB = 5 ns Valid range: 55 μs to 1.342 s
SF4_CHIRP_START_INDX_OFFSET	2	The chirp start index for each burst is determined as the chirp start index of the previous burst plus SFx_START_INDX_OFFSET * BURST_INDX i.e. $\text{CHIRP_START_INDX} = \text{SFx_CHIRP_START_INDX} + (\text{SFx_CHIRP_START_INDX_OFFSET} \times \text{BURST_INDEX})$ Valid range: 0 to 511 A value of 0 can be used to repeat the same set of unique chirps across bursts. Non-zero values allow spanning a larger number of unique chirps (across bursts). This field Not applicable for loop-back sub-frame NOTE: If ADVANCE_CHIRP_CONFIG_EN is set then this Field is not used/applicable.

Continued on next page

Table 5.32 – continued from previous page

SF4_NUM_BURSTS	2	Number of bursts constituting this sub frame Valid range: 1 to 512 Valid range: 1 to 16 for loop-back sub-frame
SF4_NUM_OUTER_LOOPS	2	Number of times to loop over the set of above defined bursts, for this sub frame. Valid range: 1 to 64 This field is Not applicable for loop-back sub-frame
RESERVED	2	0x0000
SF4_PERIOD	4	$SF_PERIOD \geq \text{Sum total time of all bursts} + \text{InterSubFrameBlankTime}$, where, Sum total time of all bursts = Num Outer Loops * Num Bursts * Burst Period. <i>InterSubFrameBlankTime</i> is primarily for sensor calibration / monitoring, thermal control, transferring out any safety monitoring data if requested, hardware reconfiguration for next sub frame, retriggering of next SF. $\text{InterSubFrameBlankTime} \geq 300 \mu\text{s}$ typical, refer a NOTE end of this API for more info. Add $150 \mu\text{s}$ to <i>InterSubFrameBlankTime</i> if data-path re-configuration needed in sub-frame boundary due to change in profile. 1 LSB = 5 ns Valid range: $300 \mu\text{s}$ to 1.342 s
RESERVED	4	0x00000000
RESERVED	4	0x00000000
NUM_FRAMES	2	Number of frames to transmit (1 frame = all enabled sub frames). If set to 0, frames are transmitted endlessly till Frame Stop message is received. Valid range: 0 to 65535

Continued on next page

Table 5.32 – continued from previous page

TRIGGER_SELECT	2	0x0001 SWTRIGGER (Software API based triggering): Frame is triggered upon receiving AWR_FRAMESTARTSTOP_CONF_SB. There could be several tens of micro seconds uncertainty in triggering. This mode is not applicable if this device is configured as MULTICHIP_SLAVE in AWR_CHAN_CONF_SB. 0x0002 HWTRIGGER (Hardware SYNC_IN based triggering): Each frame is triggered by rising edge of pulse in SYNC_IN pin, after receiving AWR_FRAMESTARTSTOP_CONF_SB (this is to prevent spurious transmission). w.r.t. the SYNC_IN pulse, the actual transmission has 5ns uncertainty in SINGLECHIP and only a 300 ps uncertainty (due to tight inter-chip synchronization needed) in MULTICHIP sensor applications as defined in AWR_CHAN_CONF_SB.
FRAME_TRIGGER_DELAY	4	Optional time delay from the SYNC_IN trigger to the occurrence of frame chirps. Applicable only in SINGLECHIP sensor applications, as defined in AWR_CHAN_CONF_SB. It is recommended only for staggering the transmission of multiple radar sensors around the car for interference avoidance, if needed. Typical range is 0 to few tens of micro seconds. Units: 1 LSB = 5 ns
RESERVED	4	0x00000000
RESERVED	4	0x00000000

NOTE1:	If hardware trigger mode is used with SUBFRAMETRIGGER = 0, then the trigger should be issued for each burst. If SUBFRAMETRIGGER = 1, then the trigger needs to be issued for each sub-frame.
NOTE2:	If hardware triggered mode is used, the SYNC_IN pulse width should be less than 4 us. Also, the minimum pulse width of SYNC_IN should be 25 ns.
NOTE3:	If frame trigger delay is used with hardware triggered mode, then external SYNC_IN pulse periodicity should take care of the configured frame trigger delay and frame periodicity. The external pulse should be issued only after the sum total of frame trigger delay and frame periodicity. See figure below
NOTE4:	In Hw triggered mode, the Hw pulse should be issued or periodicity of pulse is configured such that, the pulse is generated only 150us after the completion of previous frame/burst (The pulse should not be issued before end of previous frame/burst). The time delta between end of previous frame/burst and raising edge of Hw pulse recommended to be < 300us.
NOTE5:	The PF_DIGITAL_OUTPUT_SAMPLING_RATE impacts the LVD-S/CSI2 data rate in a sub-frame, so it is recommended to analyze timing impact if different sample rate is used across chirps in a sub-frame.
NOTE6:	If the devices' self-triggered periodic monitoring features are enabled, the user needs to set the overall frame period taking into consideration the monitoring periodicities (refer to CALIB_MON_TIME_UNIT in AWR_CALIB_MON_TIME_UNIT_CONF_SB).
NOTE7:	Please refer Table 11.3 and Table 11.4 for details on minimum inter-burst and inter sub-frame/frame blank time requirements.
NOTE8:	If advance chirp configuration is enabled then this API takes around 1.8ms to execute in RadarSS sub System for 128 chirps. The error checks for each parameters of advance chirp is done in frame configuration API. This option can be disabled by using ADVANCE_CHIRP_ERROR_CHK_DIS option in AWR_RF_RADAR_MISC_CTL_SB API. If this error check takes more than 50ms due to large number of chirps then it is recommended to disable RadarSS WDT while executing this API.
NOTE9:	Due to data synchronizers in the synthesizer chirp starting path, there can be 0/1.1 ns bimodal jitter in the starting of each RF chirp wrt RX ADC sampling. This can lead to a small chirp to chirp phase jitter whose magnitude depends on IF frequency. For example, this translates to $360^\circ \times 1.1 \text{ ns} \times 1 \text{ MHz} = 0.4^\circ$ at 1-MHz IF in range dimension; 4° for 10-MHz IF, and 8° for 20-MHz IF. If the inter-chirp jitter mitigation feature is enabled in products where it is available, this jitter will be mitigated from inter-chirp perspective and be present only as an inter-burst, inter-sub-frame and inter-frame jitter, with the same magnitude.

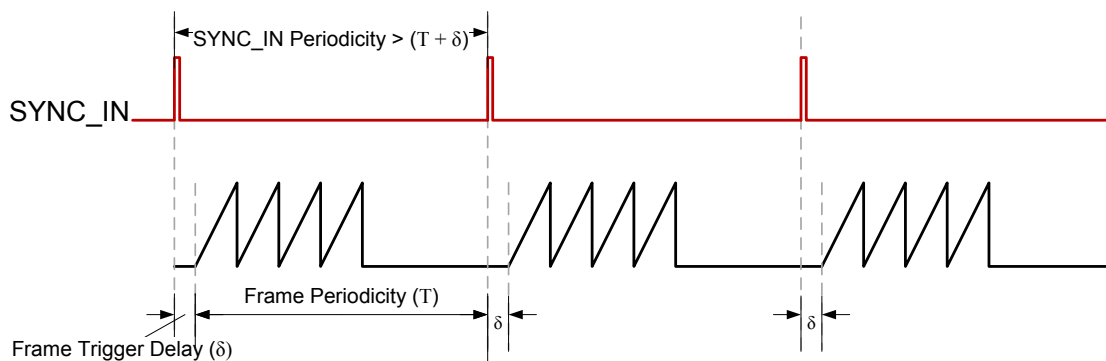


Figure 5.3: Frame trigger delay in case of external hardware trigger

5.5.7 Sub block 0x0106 – AWR_PERCHIRPPHASESHIFT_CONF_SB

This sub block defines static phase shift configurations per chirp in each of the TXs. The API is applicable only in certain devices (Please refer data sheet). This API will be honored after enabling PERCHIRP_PHASESHIFTER_EN in AWR_RF_RADAR_MISC_CTL_SB.

Table 5.33 describes the contents of this sub block.

Table 5.33: AWR_PERCHIRPPHASESHIFT_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0106
SBLKLEN	2	Value = 12
CHIRP_START_INDXX	2	Start index of the chirp for configuring the phase shifter Valid range 0 to 511
CHIRP_END_INDXX	2	End index of the chirp for configuring the phase shifter Valid range 0 to 511
TX0_PHASE_SHIFTER	1	TX0 phase shift value Bits TX0 phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 TX0 phase shift value 1 LSB = $360^\circ/2^6 = 5.625^\circ$ Valid range: 0 to 63

Continued on next page

Table 5.33 – continued from previous page

TX1_PHASE_SHIFTER	1	TX1 phase shift value Bits TX1 phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 TX1 phase shift value 1 LSB = $360^\circ/2^6 = 5.625^\circ$ Valid range: 0 to 63
TX2_PHASE_SHIFTER	1	TX2 phase shift value Bits TX2 phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 TX2 phase shift value 1 LSB = $360^\circ/2^6 = 5.625^\circ$ Valid range: 0 to 63
TX3_PHASE_SHIFTER	1	TX3 phase shift value (only applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) Bits TX3 phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 TX3 phase shift value 1 LSB = $360^\circ/2^6 = 5.625^\circ$ Valid range: 0 to 63

NOTE1: Phase shifter(PS) settings are applied in advance at max -5us or at -(Idle_time-1.28us-DfeLagTime) from the knee of the ramp. If idle time is > 6.28us then PS is applied always at -5us and if idle time < 6.28us then PS is applied at -(Idle_time-1.28us-DfeLagTime) from knee of the ramp as shown in figure below. Where DfeLagTime is internal DFE lag time (Please refer rampgen calculator).

NOTE2: It is recommended to configure TX start time > -5us or -(Idle_time-1.28us-DfeLagTime) based on PS apply time as shown in figure below.

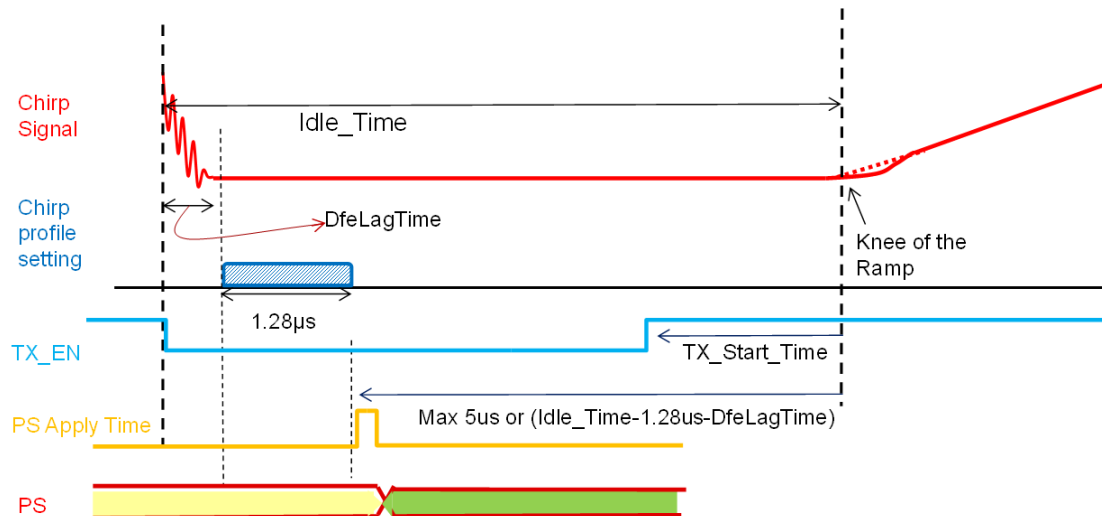


Figure 5.4: TX PS apply timing in a chirp

5.5.8 Sub block 0x0107 – AWR_PROG_FILT_COEFF_RAM_SET_SB

This sub block can be used to program the coefficients for the programmable filter. This is a new feature added in **AWR2243**, **xWR6243**, **xWR294x**, **xWR254x**, **xWR2x4xECO**, **xWR2x4xP**.

The programmable filter, when enabled, overrides the device's hard-coded digital decimation filter and allows the user to obtain any desired filter response by programming appropriate filter coefficients. The programmable filter's sampling rate is 2x Output Rate programmed for the profile. The programmable filter's output is down-sampled by 2 to form the device's ADC data output. The programmable filter allow for a trade-off between digital filter chain settling time and close-in anti-alias attenuation. In AWR2243/xWR6243, the Maximum DFE outout sampling rate in real mode is 25Msps and in complex mode is 22.5Msps. In xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP, only real ADC mode is supported. The maximum DFE output sampling rate is limited to 45 Msps.

A real-coefficient FIR with up to 63 taps (16-bit coefficients) is supported in both Complex and real output mode in AWR2243/xWR6243 devices and only in Real mode in xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP. The maximum number of filter coefficients (or taps) is subject to the limit mentioned in [5.34](#) and [5.35](#).

Filter normalization: For a Low Pass Filter example usage, in order to get 0dB gain at 0Hz (and match the device's default filter gain), the sum of all filter coefficients should be 2^{15} . A higher sum can cause filter saturation and is not recommended. A lower sum is allowed and will result in lower gain (e.g. 6dB lower if the sum is 2^{14}). Note that the maximum coefficient magnitude <

2^{15} .

Table 5.34: Programmable filter DFE sampling rate and max number of taps for AWR2243/xWR6243

DFE sampling rate F_s (Mps)	Max number of taps in Real Mode	Max number of taps in Complex Mode
≥ 25	42	NA
$\geq 12.5, < 25$	42	21
$\geq 6.25, < 12.5$	63	45
else	63	63

Table 5.35: Programmable filter DFE sampling rate and max number of taps for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP

DFE sampling rate F_s (Mps)	Max number of taps
$\geq 25, \leq 45$	18
$\geq 12.5, < 25$	42
$\geq 6.25, < 12.5$	63
else	63

NOTE: This API should be issued before AWR_PROFILE_CONF_SET_SB.

Table 5.36 describes the contents of this sub block.

Table 5.36: AWR_PROG_FILT_COEFF_RAM_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0107
SBLKLEN	2	Value = 212

Continued on next page

Table 5.36 – continued from previous page

COEFF_ARRAY	208	The array of coefficients for the programmable filter, across all profiles, to be stored in the coefficient RAMS. Each tap is a 16-bit signed <1.15, s> number. The exact set of taps to be used for a given profile can be specified through AWR_PROG_FILT_CONF_SB NOTE: All the filter taps across profiles are to be provided in one shot. There is a HW constraint that each profile's filter taps should start at four 32-bit word aligned address (i.e., the coefficients corresponding to any profile should start at array index which is a multiple of 8). Unused coefficients shall be initialized to zero.
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5.5.9 Sub block 0x0108 – AWR_PROG_FILT_CONF_SET_SB

This sub block can be used to configure the coefficients for the programmable filter and associate them to a certain profile.

The API is applicable only in xWR1642/IWR6843/xWR1843/**AWR2243**, **xWR6243**, **xWR294x**, **xWR254x**, **xWR2x4xECO**, **xWR2x4xP**. This API should be issued before AWR_PROFILE_CONF_SET_SB.

Table 5.37 describes the contents of this sub block.

Table 5.37: AWR_PROG_FILT_CONF_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0108
SBLKLEN	2	Value = 8
PROFILE_INDX	1	This field indicates the profile Index for which this configuration applies.
PROG_FILT_COEFF_START_INDEX	1	The index of the first coefficient of the programmable filter taps corresponding to this profile in the coefficient RAM programmed using AWR_PROG_FILT_COEFF_SET_SB NOTE: The profile's filter tap start index shall be 8 tap aligned (four 32-bit word aligned address).
PROG_FILT_LENGTH	1	The length (number of taps) of the filter corresponding to this profile. Together with the previous field, this determines the set of coefficients picked up from the coefficient RAM to form the filter taps for this profile. NOTE: This has to be an even number. For odd-length filters, a 0 (zero) tap needs to be appended at the end to make the length even. This is a HW constraint.

Continued on next page

Table 5.37 – continued from previous page

PROG_FILT_ FREQ_SHIFT_ FACTOR	1	Relevant only for the Complex output mode with the programmable filter. Determines the magnitude of the frequency shift do be done before filtering using the real-coefficient programmable filter. This is not applicable to xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP devices. 1 LSB = $0.01 \times F_s$ shift, where F_s is the output sampling rate, specified as PF_DIGITAL_OUTPUT_SAMPLING_RATE in AWR_PROFILE_CONF_SET_SB
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NOTE1:	PROG_FILT_COEFF_START_INDEX should be 8 tap aligned (four 32-bit word aligned address)
NOTE2:	Programmable filter should be enabled for all the profiles configured in a frame. Programmable filter getting switched (enabled or disabled) within a frame is not supported.
NOTE3:	Programmable filter APIs (AWR_PROG_FILT_COEFF_RAM_SET_SB and AWR_PROG_FILT_CONF_SET_SB) should not be issued when frames are ongoing.

5.5.10 Sub block 0x0109 – AWR_CALIB_MON_TIME_UNIT_CONF_SB

This API sub block is used to set calibration and monitoring time unit.

Table 5.38: AWR_CALIB_MON_TIME_UNIT_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0109
SBLKLEN	2	Value = 12

Continued on next page

Table 5.38 – continued from previous page

CALIB_MON_TIME_UNIT	2	Defines the basic time unit, in terms of which calibration and/or monitoring periodicities are to be defined. If any monitoring functions are desired and enabled, the monitoring infrastructure automatically inherits this time unit as the period over which the various monitors are cyclically executed; so this should be set to the desired FTTI. For calibrations, a separate CALIB_PERIODICITY can be specified, as a multiple of this time unit, in AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB NOTE: Even though calibrations many not be desired every time unit, every time unit shall be made long enough to include active chirping time, time required for all enabled calibrations and monitoring functions. 1 LSB = Duration of one frame Recommendation: See examples in Section 12 Default value in Device: 100 Valid range : 40ms to 250ms (Derive actual count value from programmed frame period) NOTE: In cascade mode this value shall be configured based on monitoring time required to monitor all cascade devices. For example in MONITORING_MODE 0 (Auto mode) typical CALIB_MON_TIME_UNIT value is 4x (4 chip cascade system) compared to single chip monitor duration. Host shall program cascade system CALIB_MON_TIME_UNIT value in all devices (4x in Auto mode), the RadarSS schedules the monitors in round robin fashion using NUM_OF_CASCADEDEV and DEVICE_ID settings. In MONITORING_MODE 1 (API based trigger), this value shall be configured based on monitoring time required to monitor all cascade devices using API based trigger.
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Table 5.38 – continued from previous page

NUM_OF_CAS- CADED_DEV	1	The number of cascaded devices in System. This configuration by default set to value 1 in single chip mode. In Cascade mode, this configuration can be set to Max num of devices in the cascade system and this needs to be set based on MONITORING_MODE setting. This control helps the device to schedule autonomous monitors in round robin fashion to avoid inter device interference. if MONITORING_MODE is 0 (Autonomous Mode - Device automatically controls the sequence of monitoring triggers), then recommended to set this configuration to Max num of devices (Example value 4 in 4-chip cascade system). if MONITORING_MODE is 1 (API based trigger Mode - Host controls the sequence of monitoring triggers), then recommended to set this configuration to value 1 irrespective of num of cascade devices. Default value: 1
DEVICE_ID	1	Device Index value for each devices in cascade System. This configuration by default set to value 0 in single chip mode. In Cascade mode, this configuration can be set to value 0, 1, 2, 3 .. depending on Max num of devices in the cascade system and this needs to be set based on MONITORING_MODE setting. This control helps the device to schedule autonomous monitors in round robin fashion to avoid inter device interference. if MONITORING_MODE is 0 (Autonomous Mode - Device automatically controls the sequence of monitoring triggers), then recommended to set this configuration to 0 (master), 1 (slave), 2(slave), 3(slave) in 4-chip cascade system. if MONITORING_MODE is 1 (API based trigger Mode - Host controls the sequence of monitoring triggers), then recommended to set this configuration to value 0 in all cascade devices. Default value: 0

Continued on next page

Table 5.38 – continued from previous page

MONITORING_ MODE	1	Monitoring mode. Mostly applicable for cascade devices (recommended) to control execution of monitoring types, refer AWR_MONITOR_TYPE_TRIG_CONF_SB for more details . 0 Autonomous monitoring trigger (default in single chip mode) 1 API based monitoring trigger (recommended in cascade mode) NOTE: This feature is supported only on AWR2243, xWR6243, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices. NOTE: Disable WDT if API based monitoring trigger is enabled.
RESERVED	3	0x00_0000

NOTE1:	Even if no explicit calibration or monitoring execution is desired by the user, CALIB_MON_TIME_UNIT field MUST be set to honor the mentioned valid range. This is because the device internally needs to perform PLL calibrations to keep them in lock, and their execution requires this parameter to be within this mentioned valid range.
NOTE2:	The Minimum total blank time in a CALIB_MON_TIME_UNIT shall be 1ms to run internal APLL and SYNTH calibrations + ~12.5% of CALIB_MON_TIME_UNIT for WDT clearing time if WDT is enabled. Refer to Table 12.10, Table 12.11 for the duration of run time monitors and Table 12.14 for software overheads.
NOTE3:	The CALIB_MON_TIME_UNIT is applicable for one frame trigger API. Once frame is stopped then FTTI will reset, CALIB_MON_TIME_UNIT is not applicable across multiple SW frame trigger API.
NOTE4:	In case of single frame configured in frame config API then set CALIB_MON_TIME_UNIT to one to run all monitors. It is recommended to use ONE_TIME_CALIB_ENABLE_MASK in AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB API to run one shot calibrations before frame trigger in single frame case.

5.5.11 Sub block 0x010A – AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB

This API is used to trigger one time calibrations instantaneously or schedule periodic run time calibrations, which will be scheduled while framing in inter-burst idle time (Min available idle time of 250 μ s is required).

Table 5.39: AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB
contents

Field Name	Number of bytes	Description																														
SBLKID	2	Value = 0x010A																														
SBLKLEN	2	Value = 24																														
ONE_TIME_CALIB_ENABLE_MASK	4	Upon receiving this trigger message, one time calibration of various RF/analog aspects are triggered if the corresponding bits in this field are set to 1. The response is in the form of an asynchronous event sent to the host. The calibrations, if enabled, are performed after the completion of any ongoing calibration cycle, and the calibration results take effect from the frame that begins after the asynchronous event response is sent from the BSS. APLL and SYNTH calibrations are done always internally irrespective of bits are enabled or not, the time required for these calibrations must be allocated. <table><tr><td>Bit</td><td>Definition</td></tr><tr><td>b0</td><td>RESERVED</td></tr><tr><td>b1</td><td>RESERVED</td></tr><tr><td>b2</td><td>RESERVED</td></tr><tr><td>b3</td><td>RESERVED</td></tr><tr><td>b4</td><td>LODIST_CALIBRATION_EN</td></tr><tr><td>b5</td><td>RESERVED</td></tr><tr><td>b6</td><td>RESERVED</td></tr><tr><td>b7</td><td>RESERVED</td></tr><tr><td>b8</td><td>PD_CALIBRATION_EN</td></tr><tr><td>b9</td><td>TX_POWER_CALIBRATION_EN</td></tr><tr><td>b10</td><td>RX_GAIN_CALIBRATION_EN</td></tr><tr><td>b11</td><td>TX_PHASESHIFTER_CALIBRATION_EN (only for xWR294x, xWR254x, xWR2x4xECP, xWR2x4xP)</td></tr><tr><td>b12</td><td>RESERVED</td></tr><tr><td>b31:13</td><td>RESERVED 0b0000_0000_0000_0000_000</td></tr></table> Default value: 0	Bit	Definition	b0	RESERVED	b1	RESERVED	b2	RESERVED	b3	RESERVED	b4	LODIST_CALIBRATION_EN	b5	RESERVED	b6	RESERVED	b7	RESERVED	b8	PD_CALIBRATION_EN	b9	TX_POWER_CALIBRATION_EN	b10	RX_GAIN_CALIBRATION_EN	b11	TX_PHASESHIFTER_CALIBRATION_EN (only for xWR294x, xWR254x, xWR2x4xECP, xWR2x4xP)	b12	RESERVED	b31:13	RESERVED 0b0000_0000_0000_0000_000
Bit	Definition																															
b0	RESERVED																															
b1	RESERVED																															
b2	RESERVED																															
b3	RESERVED																															
b4	LODIST_CALIBRATION_EN																															
b5	RESERVED																															
b6	RESERVED																															
b7	RESERVED																															
b8	PD_CALIBRATION_EN																															
b9	TX_POWER_CALIBRATION_EN																															
b10	RX_GAIN_CALIBRATION_EN																															
b11	TX_PHASESHIFTER_CALIBRATION_EN (only for xWR294x, xWR254x, xWR2x4xECP, xWR2x4xP)																															
b12	RESERVED																															
b31:13	RESERVED 0b0000_0000_0000_0000_000																															

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Table 5.39 – continued from previous page

PERIODIC_ CALIB_ENABLE_ MASK	4	Automatic periodic triggering of calibrations of various RF/analog aspects can be set up by the host issuing this message with corresponding bits in this field set to 1. <table><tr><td>Bit</td><td>Definition</td></tr><tr><td>b0</td><td>RESERVED</td></tr><tr><td>b1</td><td>RESERVED</td></tr><tr><td>b2</td><td>RESERVED</td></tr><tr><td>b3</td><td>RESERVED</td></tr><tr><td>b4</td><td>LODIST_CALIBRATION_EN</td></tr><tr><td>b5</td><td>RESERVED</td></tr><tr><td>b6</td><td>RESERVED</td></tr><tr><td>b7</td><td>RESERVED</td></tr><tr><td>b8</td><td>PD_CALIBRATION_EN</td></tr><tr><td>b9</td><td>TX_POWER_CALIBRATION_EN</td></tr><tr><td>b10</td><td>RX_GAIN_CALIBRATION_EN</td></tr><tr><td>b11</td><td>TX_PHASESHIFTER_CALIBRATION_EN (only for xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP)</td></tr><tr><td>b12</td><td>RESERVED</td></tr><tr><td>b31:13</td><td>RESERVED</td></tr></table> APLL and SYNTH calibrations are done always internally (at a periodicity of 1 second) irrespective of bits are enabled or not, the time required for these calibrations must be allocated. Refer to Table 12.5 for the duration of run time calibrations Default value: 0 NOTE: In cascade mode it is recommended to disable (set value 0 in this field) automated periodic calibration for phase synchronization.	Bit	Definition	b0	RESERVED	b1	RESERVED	b2	RESERVED	b3	RESERVED	b4	LODIST_CALIBRATION_EN	b5	RESERVED	b6	RESERVED	b7	RESERVED	b8	PD_CALIBRATION_EN	b9	TX_POWER_CALIBRATION_EN	b10	RX_GAIN_CALIBRATION_EN	b11	TX_PHASESHIFTER_CALIBRATION_EN (only for xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP)	b12	RESERVED	b31:13	RESERVED
Bit	Definition																															
b0	RESERVED																															
b1	RESERVED																															
b2	RESERVED																															
b3	RESERVED																															
b4	LODIST_CALIBRATION_EN																															
b5	RESERVED																															
b6	RESERVED																															
b7	RESERVED																															
b8	PD_CALIBRATION_EN																															
b9	TX_POWER_CALIBRATION_EN																															
b10	RX_GAIN_CALIBRATION_EN																															
b11	TX_PHASESHIFTER_CALIBRATION_EN (only for xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP)																															
b12	RESERVED																															
b31:13	RESERVED																															

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Table 5.39 – continued from previous page

CALIBRATION_ PERIODICITY	4	This field is applicable only for those calibrations which are enabled to be done periodically in the PERIODIC_CALIB_ENABLE_MASK field. This field indicates the desired periodicity of calibrations. If this field is set to N, the results of the first calibration (based on ONE_TIME_CALIB_ENABLE_MASK) are applicable for the first N CALIB_MON_TIME_UNITS. The results of the next calibration are applicable for the next N CALIB_MON_TIME_UNITS, and so on. Recommendation: Set CALIBRATION_PERIODICITY such that frequency of calibrations is greater than or equal to 1 second. 1 LSB = 1 CALIB_MON_TIME_UNIT, as specified in AWR_CALIB_MON_TIME_UNIT_CONF_SB. If the user does not wish to receive calibration reports when periodic calibrations are not enabled, then the user should set CALIBRATION_PERIODICITY to 0 Default value: 0 Valid Range: 0 (Disable), 4 to 100 (value 1 is not a valid value, this will cause internal APLL and SYNTH calibrations to stop) NOTE: In cascade mode it is recommended to disable (set value 0 in this field) automated periodic calibration for phase synchronization.												
ENABLE_CAL_ REPORT	1	<table><tr><td>Bit</td><td>Definition</td></tr><tr><td>b0</td><td>ENABLE_SUMMARY_REPORT</td></tr><tr><td>0</td><td>Summary reports are disabled</td></tr><tr><td>1</td><td>Summary reports are enabled</td></tr><tr><td></td><td>Default value: 0</td></tr><tr><td>b7:1</td><td>RESERVED</td></tr></table> NOTE1: If calibration reports are enabled, the reports will be sent every 1 second whenever internal calibrations (APLL and SYNTH) are triggered and at every CALIBRATION_PERIODICITY when the user enabled calibrations are triggered. NOTE2: If user has not enabled any one time calibrations, but if calibration report is enabled, then after issuing this API, the firmware will attempt to run the APLL and SYNTH calibrations and the calibration report will be immediately sent out. NOTE3: Recommendation is to enable the report if this API is being invoked.	Bit	Definition	b0	ENABLE_SUMMARY_REPORT	0	Summary reports are disabled	1	Summary reports are enabled		Default value: 0	b7:1	RESERVED
Bit	Definition													
b0	ENABLE_SUMMARY_REPORT													
0	Summary reports are disabled													
1	Summary reports are enabled													
	Default value: 0													
b7:1	RESERVED													

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Table 5.39 – continued from previous page

RESERVED	1	0x00
TX_POWER_CAL_MODE	1	Bit Definition b0 TX_POWER_CAL_MODE 0 Update TX gain setting from LUT and do a closed loop calibration (OLPC + CLPC) 1 Update TX gain settings from LUT only (OLPC only) OLPC: Open Loop Power Control. In this mode the TX stage codes are set based on a coarse measurement and a LUT generated for every temperature and the stage codes are picked from the LUT CLPC: Closed Loop Power Control. In this mode the TX stage codes are picked from the coarse LUT as generated in OLPC step. Later the TX power is measured and the TX stage codes are corrected to achieve the desired TX power accuracy. Default value: 0 b7:1 RESERVED
CAL_TEMP_INDEX_OVERRIDE_ENABLE	1	This field enables the Host to override the use of device's internal temperature readings for choosing front end calibration settings (e.g. bias current, Rx Gain, Tx Gain and TX phase LUT). Bit Definition b0 TX_TEMP_INDEX_OVERRIDE_EN b1 RX_TEMP_INDEX_OVERRIDE_EN b2 LODIST_TEMP_INDEX_OVERRIDE_EN b3 TXPHASE_TEMP_INDEX_OVERRIDE_EN b7:4 RESERVED Value Definition 0 Override disable 1 Override enable Default value: 0 (Override disable, use device temperature) NOTE1: This feature is supported only on AWR2243, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices. NOTE2: If any of the override bits are enabled, AWR_RF_INIT_SB should not be invoked again.

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Table 5.39 – continued from previous page

CAL_TEMP_ INDEX_TX	1	This override temperature index is used to calibrate Tx front end.	
		Index Value	Definition
		0	Index for temperature [< -30] degC
		1	Index for to temperature $[-30, -20]$ degC
		2	Index for to temperature $[-20, -10]$ degC
		3	Index for to temperature $[-10, 0]$ degC
		4	Index for to temperature $[0, 10]$ degC
		5	Index for to temperature $[10, 20]$ degC
		6	Index for to temperature $[20, 30]$ degC
		7	Index for to temperature $[30, 40]$ degC
		8	Index for to temperature $[40, 50]$ degC
		9	Index for to temperature $[50, 60]$ degC
		10	Index for to temperature $[60, 70]$ degC
		11	Index for to temperature $[70, 80]$ degC
		12	Index for to temperature $[80, 90]$ degC
		13	Index for to temperature $[90, 100]$ degC
		14	Index for to temperature $[100, 110]$ degC
		15	Index for to temperature $[110, 120]$ degC
		16	Index for to temperature $[120, 130]$ degC
		17	Index for to temperature $[130, 140]$ degC
		18	Index for to temperature $[> 140]$ degC

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Table 5.39 – continued from previous page

CAL_TEMP_ INDEX_RX	1	This override temperature index is used to calibrate Rx front end.	
		Index Value	Definition
		0	Index for temperature [< -30] degC
		1	Index for to temperature $[-30, -20]$ degC
		2	Index for to temperature $[-20, -10]$ degC
		3	Index for to temperature $[-10, 0]$ degC
		4	Index for to temperature $[0, 10]$ degC
		5	Index for to temperature $[10, 20]$ degC
		6	Index for to temperature $[20, 30]$ degC
		7	Index for to temperature $[30, 40]$ degC
		8	Index for to temperature $[40, 50]$ degC
		9	Index for to temperature $[50, 60]$ degC
		10	Index for to temperature $[60, 70]$ degC
		11	Index for to temperature $[70, 80]$ degC
		12	Index for to temperature $[80, 90]$ degC
		13	Index for to temperature $[90, 100]$ degC
		14	Index for to temperature $[100, 110]$ degC
		15	Index for to temperature $[110, 120]$ degC
		16	Index for to temperature $[120, 130]$ degC
		17	Index for to temperature $[130, 140]$ degC
		18	Index for to temperature $[> 140]$ degC

Continued on next page

Table 5.39 – continued from previous page

CAL_TEMP_INDEX_LODIST	1	This override temperature index is used to calibrate LO distribution of front end.	
		Index Value	Definition
		0	Index for temperature [< -30] degC
		1	Index for to temperature $[-30, -20]$ degC
		2	Index for to temperature $[-20, -10]$ degC
		3	Index for to temperature $[-10, 0]$ degC
		4	Index for to temperature $[0, 10]$ degC
		5	Index for to temperature $[10, 20]$ degC
		6	Index for to temperature $[20, 30]$ degC
		7	Index for to temperature $[30, 40]$ degC
		8	Index for to temperature $[40, 50]$ degC
		9	Index for to temperature $[50, 60]$ degC
		10	Index for to temperature $[60, 70]$ degC
		11	Index for to temperature $[70, 80]$ degC
		12	Index for to temperature $[80, 90]$ degC
		13	Index for to temperature $[90, 100]$ degC
		14	Index for to temperature $[100, 110]$ degC
		15	Index for to temperature $[110, 120]$ degC
		16	Index for to temperature $[120, 130]$ degC
		17	Index for to temperature $[130, 140]$ degC
		18	Index for to temperature $[> 140]$ degC
CAL_TEMP_INDEX_TXPHASE	1	This override temperature index is used to calibrate TX phase shifter of front end.	
		Index Value	Definition
		0	Index for temperature [< 5] degC
		1	Index for temperature $[5, 50]$ degC
		2	Index for temperature $[50, 95]$ degC
		3	Index for temperature $[> 95]$ degC

NOTE1:	The API AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB can be issued when the device is framing, the calibration periodicity update or one time calibrations can be done while frames are running.
NOTE2:	The CAL_TEMP_INDEX_OVERRIDE_ENABLE is supported only for one time calibrations enabled using ONE_TIME_CALIB_ENABLE_MASK, the periodic run time calibrations are recommended to be disabled using PERIODIC_CALIB_ENABLE_MASK.
NOTE3:	In case of single frame configured in frame config API then it is recommended to use ONE_TIME_CALIB_ENABLE_MASK in AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB API to run one shot calibrations before frame trigger.

5.5.12 Sub block 0x010B – AWR_DIGITAL_COMP_EST_CONTROL_SB

This API can be used to compensate various RX and gain/phase offsets and same API can be used to estimation the same using TX frequency shift.

NOTE1:	This API is supported only on AWR2243, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices. Please refer latest DFP release note for more info.
NOTE2:	Issue this API first in the sequence before AWR_PROFILE_CONF_SET_SB API.
NOTE3:	The Digital TX frequency shift enable mode in below API is for debug purpose only, the functional phase shifter will not be operational when this mode is used. It is recommended to re-issue profile config API after disabling this mode before running functional frames.
NOTE4:	An application for the digital delay compensation field may be to digitally compensate for linear IF frequency dependent phase mismatch (e.g. mismatch across devices in cascaded operation, caused by DIG_SYNC_IN path mismatches). The residual phase error can be up to +/-0.5 degree.
NOTE5:	The digital compensation is applied to the RX signal in below order: DIGTIAL_RX_DELAY_COMP, DIGITAL_RX_FREQ_SHIFT followed DIGITAL_RX_PHASE_SHIFT_COMP

Table 5.40: AWR_DIGITAL_COMP_EST_CONTROL_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x010B
SBLKLEN	2	Value = 72
PROFILE_INDX	1	This field indicates the profile Index for which this configuration applies.
DIGITAL_COMP_EN	1	This field can be used to enable or disable different digital compensation provided in this API. Bits Assignment b0 Digital RX gain compensation enable NOTE: This field is NOT applicable for xWR254x devices. b1 Digital RX phase compensation enable b2 Digital RX delay compensation enable NOTE: If ADC_START_TIME_RES is set to 1 in AWR_RF_MISC_CTL_SB, this feature is not supported. b3 Digital RX frequency shift enable b4 Digital TX frequency shift enable (For debug purpose only) b31:5 RESERVED Value 0: Disable Value 1: Enable
RESERVED	2	0x0000
DIGITAL_RX_GAIN_COMP	4	The digital gain compensation for each RX channels One byte per RX (8-bit signed number). NOTE: This field is NOT applicable for xWR254x, xWR2x4xECO, xWR2x4xP devices. Byte Assignment 0 RX0 digital gain 1 RX1 digital gain 2 RX2 digital gain 3 RX3 digital gain 1 LSB = 0.1 dB, signed Valid Range: -120 to 119

Continued on next page

Table 5.40 – continued from previous page

DIGITAL_RX_ PHASE_SHIFT_ COMP	8	The digital phase shift compensation for each RX channels Two bytes per RX Bytes Assignment 1:0 RX0 digital phase shift 3:2 RX1 digital phase shift 5:4 RX2 digital phase shift 7:6 RX3 digital phase shift 1 LSB = $360^\circ / 2^{16} \approx 0.0055^\circ$, <i>unsigned</i> Valid Range: 0 to 65535 NOTE: This field is NOT applicable when ADC_OUT_FMT is 00 (real output)
DIGITAL_RX_ DELAY_COMP	4	The digital delay compensation for each RX channels One byte per RX (8-bit unsigned number) Byte Assignment 0 RX0 digital delay 1 RX1 digital delay 2 RX2 digital delay 3 RX3 digital delay 1 LSB = $556ps / 16$, <i>unsigned</i> Valid Range: 0 to 255 The RX ADC output is delayed by this amount. The LSB becomes twice of the above if ADC low power mode is enabled.
RESERVED	16	0x00000000

Continued on next page

Table 5.40 – continued from previous page

DIGITAL_RX_ FREQ_SHIFT	8	The digital frequency shift compensation for each RX channels Two bytes per RX Bytes Assignment 1:0 RX0 digital frequency shift 3:2 RX1 digital frequency shift 5:4 RX2 digital frequency shift 7:6 RX3 digital frequency shift 1 LSB = $(ADC SamplingRateHz * floor(100MHz/ADC SamplingRateHz)) * 1/2^{16}, signed$ Valid Range: -32768 to 32767 The frequency range of interest in RX digital output is shifted by this amount. As an example, this may be used to view the spectrum beyond the conventional [0 to Output Sampling Rate] range in Complex 1X mode, say [FREQ_SHIFT to Output Sampling Rate + FREQ_SHIFT]. NOTE: This field is NOT applicable when ADC_OUT_FMT is 00 (real output)
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Table 5.40 – continued from previous page

DIGITAL_TX_FREQ_SHIFT	8	The digital frequency shift compensation for each TX channels, this is supported only for TX0 and TX1 in AWR2243. xWR6243 supports digital frequency shift for all 3 TX channels. Two bytes per TX Bytes Assignment 1:0 TX0 digital frequency shift 3:2 TX1 digital frequency shift 5:4 TX2 digital frequency shift (RESERVED for AWR2243) 7:6 TX3 digital frequency shift (applicable only for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) 1 LSB = $100MHz/2^{16}$, <i>signed</i> Valid Range: -32768 to 32767 The frequency of the TX output may be shifted wrt the RX mixer LO frequency by this amount. If such functionality is not desired, this register should be set to 0. This register cannot be used in conjunction with TX phase shifter. This may be useful in factory calibration of IF frequency dependent effects. As an example, in cascaded applications, the IF frequency at which a corner reflector's beat frequency appears at the RX mixer output can be varied using this and cascade RX IF imbalances can be measured. NOTE: The TX frequency shifting involves some sharing of digital modulation hardware, which imposes following constraints: Not more than 2 distinct non-zero values can be set in DIGITAL_TX_FREQ_SHIFT TX0, TX1, TX2, TX3 The valid values of DIGITAL_TX_FREQ_SHIFT TX2 are either 0 or same as that for TX3. Not more than 2 TX's in the device can be simultaneously enabled in the Chirp Configuration API's if using DIGITAL_TX_FREQ_SHIFT in this API
RESERVED	16	0x00000000

5.5.13 Sub block 0x010C – AWR_RX_GAIN_TEMPLUT_SET_SB

This API can be used to overwrite the RX gain temperature LUT used in firmware. This API should be issued after profile configuration API.

Table 5.41: AWR_RX_GAIN_TEMPLUT_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x010C
SBLKLEN	2	Value = 28
PROFILE_INDX	1	This field indicates the profile Index for which this configuration applies.
RESERVED	1	0x00

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Table 5.41 – continued from previous page

RX_GAIN_CODE	19	Byte0: RX gain code for temperature <-30 °C Byte1: RX gain code for temperature [-30, -20) °C Byte2: RX gain code for temperature [-20, -10) °C Byte3: RX gain code for temperature [-10, 0) °C Byte4: RX gain code for temperature [0, 10) °C Byte5: RX gain code for temperature [10, 20) °C Byte6: RX gain code for temperature [20, 30) °C Byte7: RX gain code for temperature [30, 40) °C Byte8: RX gain code for temperature [40, 50) °C Byte9: RX gain code for temperature [50, 60) °C Byte10: RX gain code for temperature [60, 70) °C Byte11: RX gain code for temperature [70, 80) °C Byte12: RX gain code for temperature [80, 90) °C Byte13: RX gain code for temperature [90, 100) °C Byte14: RX gain code for temperature [100, 110) °C Byte15: RX gain code for temperature [110, 120) °C Byte16: RX gain code for temperature [120, 130) °C Byte17: RX gain code for temperature [130, 140) °C Byte18: RX gain code for temperature ≥140 °C Each byte is encoded as follows AWR2243/xWR6243: <table><tr><td>Bits</td><td>Definition</td></tr><tr><td>b4:0</td><td> IF_GAIN_CODE IF gain is IF_GAIN_CODE ×2 – 6 dB Valid Range: Temperature < 10degC: The max IFA gain code supported is 12 (24dB). Temperature >= 10degC: The max IFA gain code supported is 15 (30dB). 1 LSB = 2 dB </td></tr><tr><td>b7:5</td><td> RF_GAIN_CODE AWR2243 device: <table><tr><td>Value</td><td>RF Gain</td></tr><tr><td>0</td><td>Maximum RF gain</td></tr><tr><td>1</td><td>Maximum RF gain – 2.5 dB</td></tr><tr><td>2</td><td>Maximum RF gain – 5 dB</td></tr></table> xWR6243 device: <table><tr><td>Value</td><td>RF Gain</td></tr><tr><td>0</td><td>Maximum RF gain</td></tr><tr><td>1</td><td>Maximum RF gain – 3.5 dB</td></tr><tr><td>2</td><td>Maximum RF gain – 6 dB</td></tr></table></td></tr></table>	Bits	Definition	b4:0	IF_GAIN_CODE IF gain is IF_GAIN_CODE ×2 – 6 dB Valid Range: Temperature < 10degC: The max IFA gain code supported is 12 (24dB). Temperature >= 10degC: The max IFA gain code supported is 15 (30dB). 1 LSB = 2 dB	b7:5	RF_GAIN_CODE AWR2243 device: <table><tr><td>Value</td><td>RF Gain</td></tr><tr><td>0</td><td>Maximum RF gain</td></tr><tr><td>1</td><td>Maximum RF gain – 2.5 dB</td></tr><tr><td>2</td><td>Maximum RF gain – 5 dB</td></tr></table> xWR6243 device: <table><tr><td>Value</td><td>RF Gain</td></tr><tr><td>0</td><td>Maximum RF gain</td></tr><tr><td>1</td><td>Maximum RF gain – 3.5 dB</td></tr><tr><td>2</td><td>Maximum RF gain – 6 dB</td></tr></table>	Value	RF Gain	0	Maximum RF gain	1	Maximum RF gain – 2.5 dB	2	Maximum RF gain – 5 dB	Value	RF Gain	0	Maximum RF gain	1	Maximum RF gain – 3.5 dB	2	Maximum RF gain – 6 dB
Bits	Definition																							
b4:0	IF_GAIN_CODE IF gain is IF_GAIN_CODE ×2 – 6 dB Valid Range: Temperature < 10degC: The max IFA gain code supported is 12 (24dB). Temperature >= 10degC: The max IFA gain code supported is 15 (30dB). 1 LSB = 2 dB																							
b7:5	RF_GAIN_CODE AWR2243 device: <table><tr><td>Value</td><td>RF Gain</td></tr><tr><td>0</td><td>Maximum RF gain</td></tr><tr><td>1</td><td>Maximum RF gain – 2.5 dB</td></tr><tr><td>2</td><td>Maximum RF gain – 5 dB</td></tr></table> xWR6243 device: <table><tr><td>Value</td><td>RF Gain</td></tr><tr><td>0</td><td>Maximum RF gain</td></tr><tr><td>1</td><td>Maximum RF gain – 3.5 dB</td></tr><tr><td>2</td><td>Maximum RF gain – 6 dB</td></tr></table>	Value	RF Gain	0	Maximum RF gain	1	Maximum RF gain – 2.5 dB	2	Maximum RF gain – 5 dB	Value	RF Gain	0	Maximum RF gain	1	Maximum RF gain – 3.5 dB	2	Maximum RF gain – 6 dB							
Value	RF Gain																							
0	Maximum RF gain																							
1	Maximum RF gain – 2.5 dB																							
2	Maximum RF gain – 5 dB																							
Value	RF Gain																							
0	Maximum RF gain																							
1	Maximum RF gain – 3.5 dB																							
2	Maximum RF gain – 6 dB																							

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Table 5.41 – continued from previous page

xWR294x, xWR254x, xWR2x4xECO, WR2x4xP device: Bits Definition b4:0 IF_GAIN_CODE IF gain is IF_GAIN_CODE $\times 2 - 10$ dB Valid Range: 0 - 10 b7:5 RF_GAIN_CODE xWR294x/ xWR254x device: Value RF Gain 0 Maximum RF gain 1 Maximum RF gain – 4 dB 2 Maximum RF gain – 7 dB 3 Maximum RF gain – 8.5 dB xWR2x4xECO, xWR2x4xP device: Value RF Gain 0 Maximum RF gain 1 Maximum RF gain – 3 dB 2 Maximum RF gain – 5.5 dB 3 Maximum RF gain – 7 dB		
RESERVED	1	0x00
RESERVED	2	0x0000

5.5.14 Sub block 0x010D – AWR_TX_GAIN_TEMPLUT_SET_SB

This API can be used to overwrite the TX gain temperature LUT used in firmware. This API should be issued after profile configuration API.

Table 5.42: AWR_TX_GAIN_TEMPLUT_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x010D
SBLKLEN	2	Value = 68
PROFILE_INDXX	1	This field indicates the profile Index for which this configuration applies
RESERVED	1	0x00

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Table 5.42 – continued from previous page

TX0_GAIN_CODE	19	Byte0: TX0 gain code for temperature <-30 °C Byte1: TX0 gain code for temperature [-30, -20) °C Byte2: TX0 gain code for temperature [-20, -10) °C Byte3: TX0 gain code for temperature [-10, 0) °C Byte4: TX0 gain code for temperature [0, 10) °C Byte5: TX0 gain code for temperature [10, 20) °C Byte6: TX0 gain code for temperature [20, 30) °C Byte7: TX0 gain code for temperature [30, 40) °C Byte8: TX0 gain code for temperature [40, 50) °C Byte9: TX0 gain code for temperature [50, 60) °C Byte10: TX0 gain code for temperature [60, 70) °C Byte11: TX0 gain code for temperature [70, 80) °C Byte12: TX0 gain code for temperature [80, 90) °C Byte13: TX0 gain code for temperature [90, 100) °C Byte14: TX0 gain code for temperature [100, 110) °C Byte15: TX0 gain code for temperature [110, 120) °C Byte16: TX0 gain code for temperature [120, 130) °C Byte17: TX0 gain code for temperature [130, 140) °C Byte18: TX0 gain code for temperature ≥140 °C Each byte is encoded as follows AW2243/xWR6243 devices: Bits Definition b5:0 STG_CODE Higher values for higher gain b7:6 RESERVED xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: Bits Definition b7:0 STG_CODE Higher values for higher gain
RESERVED	1	0x00

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Table 5.42 – continued from previous page

TX1_GAIN_CODE	19	Byte0: TX1 gain code for temperature <-30 °C Byte1: TX1 gain code for temperature [-30, -20) °C Byte2: TX1 gain code for temperature [-20, -10) °C Byte3: TX1 gain code for temperature [-10, 0) °C Byte4: TX1 gain code for temperature [0, 10) °C Byte5: TX1 gain code for temperature [10, 20) °C Byte6: TX1 gain code for temperature [20, 30) °C Byte7: TX1 gain code for temperature [30, 40) °C Byte8: TX1 gain code for temperature [40, 50) °C Byte9: TX1 gain code for temperature [50, 60) °C Byte10: TX1 gain code for temperature [60, 70) °C Byte11: TX1 gain code for temperature [70, 80) °C Byte12: TX1 gain code for temperature [80, 90) °C Byte13: TX1 gain code for temperature [90, 100) °C Byte14: TX1 gain code for temperature [100, 110) °C Byte15: TX1 gain code for temperature [110, 120) °C Byte16: TX1 gain code for temperature [120, 130) °C Byte17: TX1 gain code for temperature [130, 140) °C Byte18: TX1 gain code for temperature ≥140 °C Each byte is encoded as follows AW2243/xWR6243 devices: Bits Definition b5:0 STG_CODE Higher values for higher gain b7:6 RESERVED xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: Bits Definition b7:0 STG_CODE Higher values for higher gain
RESERVED	1	0x00

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Table 5.42 – continued from previous page

TX2_GAIN_CODE	19	Byte0: TX2 gain code for temperature <-30 °C Byte1: TX2 gain code for temperature [-30, -20) °C Byte2: TX2 gain code for temperature [-20, -10) °C Byte3: TX2 gain code for temperature [-10, 0) °C Byte4: TX2 gain code for temperature [0, 10) °C Byte5: TX2 gain code for temperature [10, 20) °C Byte6: TX2 gain code for temperature [20, 30) °C Byte7: TX2 gain code for temperature [30, 40) °C Byte8: TX2 gain code for temperature [40, 50) °C Byte9: TX2 gain code for temperature [50, 60) °C Byte10: TX2 gain code for temperature [60, 70) °C Byte11: TX2 gain code for temperature [70, 80) °C Byte12: TX2 gain code for temperature [80, 90) °C Byte13: TX2 gain code for temperature [90, 100) °C Byte14: TX2 gain code for temperature [100, 110) °C Byte15: TX2 gain code for temperature [110, 120) °C Byte16: TX2 gain code for temperature [120, 130) °C Byte17: TX2 gain code for temperature [130, 140) °C Byte18: TX2 gain code for temperature ≥140 °C Each byte is encoded as follows AW2243/xWR6243 devices: Bits Definition b5:0 STG_CODE Higher values for higher gain b7:6 RESERVED xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: Bits Definition b7:0 STG_CODE Higher values for higher gain
RESERVED	1	0x00
RESERVED	2	0x0000

5.5.15 Sub block 0x010E – AWR_LOOPBACK_BURST_CONF_SET_SB

This API can be used to introduce loopback chirps within the functional frames. This loopback chirps will be introduced only if advanced frame configuration is used where user can define which sub-frame contains loopback chirps. The following loopback configuration will apply to one burst and user can program up to 16 different loopback configurations in 16 different bursts of

a given sub-frame. User has to ensure that the corresponding sub-frame is defined in AWR_ADVANCED_FRAME_CONF_SB and sufficient time is given to allow the loopback bursts to be transmitted.

NOTE1:	If user desires to enable loopback chirps within functional frames, then this API should be issued after AWR_PROFILE_CONF_SET_SB
NOTE2:	Only profile based phase shifter is supported in loopback configuration. Per-chirp phase shifter if enabled will not be reflected in loopback chirps.
NOTE3:	For the sub-frame in which loopback is desired, user should set SFx_NUM_UNIQUE_CHIRPS_PER_BURST as 1 and can use SFx_NUM_LOOPS_PER_BURST for multiple chirps in the burst.

Table 5.43: AWR_LOOPBACK_BURST_CONF_SET_SB contents

Field Name	Number of bytes	Description														
SBLKID	2	Value = 0x010E														
SBLKLEN	2	Value = 48														
LOOPBACK_SEL	1	<table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>No loopback</td></tr><tr><td>1</td><td>IF loopback (loopback of an IF test signal into the RX IF stages)</td></tr><tr><td>2</td><td>AWR2243/xWR6243: PS loopback (loopback of an RF test signal from TX phase shifter outputs to RX LNA input) xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP: - LO loopback</td></tr><tr><td>3</td><td>AWR2243/xWR6243: PA loopback (loopback of an RF test signal from TX PA outputs to RX LNA input with On-off-keying modulation in the path) xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP: - PA loopback (loopback of an RF test signal from TX PA outputs to RX LNA input with QPSK modulation in the path)</td></tr><tr><td>4</td><td>Rx FE disabled (RX RF i.e. Mixer and LNA are disabled and no loopback is engaged) NOTE: Value 4, RX FE disabled is de-featured and not supported in xWR294x, xWR2x4xECO, xWR2x4xP devices</td></tr><tr><td>Others</td><td>RESERVED</td></tr></table>	Value	Definition	0	No loopback	1	IF loopback (loopback of an IF test signal into the RX IF stages)	2	AWR2243/xWR6243: PS loopback (loopback of an RF test signal from TX phase shifter outputs to RX LNA input) xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP: - LO loopback	3	AWR2243/xWR6243: PA loopback (loopback of an RF test signal from TX PA outputs to RX LNA input with On-off-keying modulation in the path) xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP: - PA loopback (loopback of an RF test signal from TX PA outputs to RX LNA input with QPSK modulation in the path)	4	Rx FE disabled (RX RF i.e. Mixer and LNA are disabled and no loopback is engaged) NOTE: Value 4, RX FE disabled is de-featured and not supported in xWR294x, xWR2x4xECO, xWR2x4xP devices	Others	RESERVED
Value	Definition															
0	No loopback															
1	IF loopback (loopback of an IF test signal into the RX IF stages)															
2	AWR2243/xWR6243: PS loopback (loopback of an RF test signal from TX phase shifter outputs to RX LNA input) xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP: - LO loopback															
3	AWR2243/xWR6243: PA loopback (loopback of an RF test signal from TX PA outputs to RX LNA input with On-off-keying modulation in the path) xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP: - PA loopback (loopback of an RF test signal from TX PA outputs to RX LNA input with QPSK modulation in the path)															
4	Rx FE disabled (RX RF i.e. Mixer and LNA are disabled and no loopback is engaged) NOTE: Value 4, RX FE disabled is de-featured and not supported in xWR294x, xWR2x4xECO, xWR2x4xP devices															
Others	RESERVED															

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Table 5.43 – continued from previous page

BASE_PRO- FILE_INDX	1	Base profile used for loopback chirps to configure the RF/analog/digital front end sections. But the configurations of some sections may get overwritten by the parameters configured below. Valid values 0 to 3										
BURST_INDX	1	Indicates the index of the burst in the loopback sub-frame for which this configuration applies Valid values 0 to 15										
RESERVED	1	0x00										
FREQ_CONST	4	Start frequency for loopback. The start frequency configured here should be within profile's sweep bandwidth. For 77GHz Devices (76GHz to 81GHz): 1 LSB = $3.6e9/2^{26}$ Hz $\approx$ 53.644 Hz Valid range: 0x5471C71C to 0x5A000000 For 60GHz Devices (57GHz to 64GHz): 1 LSB = $2.7e9/2^{26}$ Hz $\approx$ 40.233 Hz Valid range: Only even numbers from 0x5471C71C to 0x5ED097B4										
SLOPE_CONST	2	Frequency slope for loopback burst (16 bit signed number) For 77GHz Devices (76GHz to 81GHz): 1 LSB = $3.6e9 \times 900/2^{26} \approx$ 48.279 kHz/ μ s Valid range: -2072 to 2072 For 60GHz Devices (57GHz to 64GHz): 1 LSB = $2.7e9 \times 900/2^{26} \approx$ 36.21 kHz/ μ s Valid range: Only even numbers between -6905 and 6905										
RESERVED	2	0x0000										
TX_BACKOFF	4	<table> <tr> <th>Bits</th> <th>Definition</th> </tr> <tr> <td>b7:0</td> <td>TX0 back off 1 LSB = 1 dB</td> </tr> <tr> <td>b15:8</td> <td>TX1 back off 1 LSB = 1 dB</td> </tr> <tr> <td>b23:16</td> <td>TX2 back off 1 LSB = 1 dB</td> </tr> <tr> <td>b31:24</td> <td>TX3 back off (only for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) 1 LSB = 1 dB</td> </tr> </table> This setting is applicable only in PA loop-back mode.	Bits	Definition	b7:0	TX0 back off 1 LSB = 1 dB	b15:8	TX1 back off 1 LSB = 1 dB	b23:16	TX2 back off 1 LSB = 1 dB	b31:24	TX3 back off (only for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) 1 LSB = 1 dB
Bits	Definition											
b7:0	TX0 back off 1 LSB = 1 dB											
b15:8	TX1 back off 1 LSB = 1 dB											
b23:16	TX2 back off 1 LSB = 1 dB											
b31:24	TX3 back off (only for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) 1 LSB = 1 dB											

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Table 5.43 – continued from previous page

RX_GAIN	2	Bits	Definition
		b5:0	RX_GAIN This field defines RX gain for each profile 1 LSB = 1 dB AWR2243 devices - all even values from 32 to 52 xWR6x43 devices - all even values from 30 to 48 xWR294x devices - all even values from 28 to 44 xWR254x devices - all even values from 28 to 44 xWR2x4xECO/xWR2x4xP devices - all even values from 30 to 46 This setting is applicable in all loop-back modes.
		b7:6	RF_GAIN_TARGET RF gain for AWR2243, xWR6243, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP device: Value RF gain target (AWR2243/ (xWR294x/ (xWR2x4xECO/ xWR6243) xWR254x) xWR2x4xP) 00 30 dB 32 dB 34 dB 01 33 dB 34 dB 36 dB 10 36 dB 36 dB 38 dB 11 RESERVED RESERVED RESERVED Refer profile configuration API for more info. This setting is applicable only in PA and PS/LO loop-back modes.
		b15:8	RESERVED
TX_ENABLE	1	Bits	Definition
		b0	TX0 Enable
		b1	TX1 Enable
		b2	TX2 Enable (PS LB not supported for TX2)
TX_ENABLE	1	b3	TX3 Enable (only applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP)
		b7:4	RESERVED This setting is applicable in all loop-back modes.
RESERVED	1	0x00	

Continued on next page

Table 5.43 – continued from previous page

BPM_CONFIG	2	Bit	Definition	
		b0	RESERVED	
		b1	CONST_BPM_VAL_TX0_ON Value of Binary Phase Shift value for TX0, during chirp	
		b2	RESERVED	
		b3	CONST_BPM_VAL_TX1_ON For TX1	
		b4	RESERVED	
		b5	CONST_BPM_VAL_TX2_ON For TX2	
		b6	RESERVED	
		b7	CONST_BPM_VAL_TX3_ON (only for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) For TX3	
		b15:6	RESERVED This setting is applicable only in PA and PS loop-back modes.	
DIGITAL_CORRECTION_DISABLE	2	Bits	Digital corrections	
		b0	IQMM correction disable (Applicable only in PS and PA loopback modes, In case of IF loopback mode, IQMM is disabled by default) 0 - Enable, 1 - Disable	
		b1	Digital Inter-RX Gain and Phase correction disable 0 - Enable, 1 - Disable This setting is applicable in all loop-back modes.	
		b15:2	RESERVED	
IF_LOOPBACK_FREQ	1	Value	IF Loopback frequency	Value IF Loopback frequency
		0	180 kHz	9 5 MHz
		1	240 kHz	10 6 MHz
		2	360 kHz	11 7.5 MHz
		3	720 kHz	12 8.03 MHz
		4	1 MHz	13 9 MHz
		5	2 MHz	14 10 MHz
		6	2.5 MHz	22-15 RESERVED
		7	3 MHz	23 20 MHz
		8	4.02 MHz	255-24 RESERVED

Continued on next page

Table 5.43 – continued from previous page

IF_LOOPBACK_MAG	1	1 LSB = 10 mV Valid range: 1 to 63																																																																																																				
PS0_PGA_GAIN_INDEX (AWR2243), COMMON_BUF_GAIN_SEL (xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP)	1	AWR2243/xWR6243 devices: Programmable Gain Amplifier Setting: This configures the Phase shifter loopback path amplifier gain for TX0 based PS loopback. <table><tr><th>Value</th><th>PGA gain value</th><th>Value</th><th>PGA gain value</th></tr><tr><td>0</td><td>PGA is OFF</td><td>15</td><td>-3 dB</td></tr><tr><td>1</td><td>-22 dB</td><td>16</td><td>-2 dB</td></tr><tr><td>2</td><td>-16 dB</td><td>17</td><td>-1 dB</td></tr><tr><td>3</td><td>-15 dB</td><td>18</td><td>0 dB</td></tr><tr><td>4</td><td>-14 dB</td><td>19</td><td>1 dB</td></tr><tr><td>5</td><td>-13 dB</td><td>20</td><td>2 dB</td></tr><tr><td>6</td><td>-12 dB</td><td>21</td><td>3 dB</td></tr><tr><td>7</td><td>-11 dB</td><td>22</td><td>4 dB</td></tr><tr><td>8</td><td>-10 dB</td><td>23</td><td>5 dB</td></tr><tr><td>9</td><td>-9 dB</td><td>24</td><td>6 dB</td></tr><tr><td>10</td><td>-8 dB</td><td>25</td><td>7 dB</td></tr><tr><td>11</td><td>-7 dB</td><td>26</td><td>8 dB</td></tr><tr><td>12</td><td>-6 dB</td><td>27</td><td>9 dB</td></tr><tr><td>13</td><td>-5 dB</td><td>255-28</td><td>RESERVED</td></tr><tr><td>14</td><td>-4 dB</td><td></td><td></td></tr></table> xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: This field configures the QPSK buffer gain in both the PA and the LO loopback paths in xWR294x,xWR254x,xWR2x4xECO and xWR2x4xP. <table><tr><th>Value</th><th>Common Buffer Gain</th><th>Value</th><th>Common Buffer Gain</th></tr><tr><td>48</td><td>0dB</td><td>19</td><td>-10dB</td></tr><tr><td>29</td><td>-1dB</td><td>18</td><td>-11dB</td></tr><tr><td>25</td><td>-2dB</td><td>17</td><td>-13dB</td></tr><tr><td>24</td><td>-3dB</td><td>16</td><td>-14dB</td></tr><tr><td>23</td><td>-5dB</td><td>15</td><td>-18dB</td></tr><tr><td>22</td><td>-6dB</td><td>0</td><td>-19dB</td></tr><tr><td>21</td><td>-7dB</td><td></td><td></td></tr><tr><td>20</td><td>-8dB</td><td></td><td></td></tr></table>	Value	PGA gain value	Value	PGA gain value	0	PGA is OFF	15	-3 dB	1	-22 dB	16	-2 dB	2	-16 dB	17	-1 dB	3	-15 dB	18	0 dB	4	-14 dB	19	1 dB	5	-13 dB	20	2 dB	6	-12 dB	21	3 dB	7	-11 dB	22	4 dB	8	-10 dB	23	5 dB	9	-9 dB	24	6 dB	10	-8 dB	25	7 dB	11	-7 dB	26	8 dB	12	-6 dB	27	9 dB	13	-5 dB	255-28	RESERVED	14	-4 dB			Value	Common Buffer Gain	Value	Common Buffer Gain	48	0dB	19	-10dB	29	-1dB	18	-11dB	25	-2dB	17	-13dB	24	-3dB	16	-14dB	23	-5dB	15	-18dB	22	-6dB	0	-19dB	21	-7dB			20	-8dB		
Value	PGA gain value	Value	PGA gain value																																																																																																			
0	PGA is OFF	15	-3 dB																																																																																																			
1	-22 dB	16	-2 dB																																																																																																			
2	-16 dB	17	-1 dB																																																																																																			
3	-15 dB	18	0 dB																																																																																																			
4	-14 dB	19	1 dB																																																																																																			
5	-13 dB	20	2 dB																																																																																																			
6	-12 dB	21	3 dB																																																																																																			
7	-11 dB	22	4 dB																																																																																																			
8	-10 dB	23	5 dB																																																																																																			
9	-9 dB	24	6 dB																																																																																																			
10	-8 dB	25	7 dB																																																																																																			
11	-7 dB	26	8 dB																																																																																																			
12	-6 dB	27	9 dB																																																																																																			
13	-5 dB	255-28	RESERVED																																																																																																			
14	-4 dB																																																																																																					
Value	Common Buffer Gain	Value	Common Buffer Gain																																																																																																			
48	0dB	19	-10dB																																																																																																			
29	-1dB	18	-11dB																																																																																																			
25	-2dB	17	-13dB																																																																																																			
24	-3dB	16	-14dB																																																																																																			
23	-5dB	15	-18dB																																																																																																			
22	-6dB	0	-19dB																																																																																																			
21	-7dB																																																																																																					
20	-8dB																																																																																																					

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Table 5.43 – continued from previous page

PS1_PGA_GAIN_INDEX (AWR2243) , LO_LOOPBACK_BUF_GAIN_SEL (xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP)	1	AWR2243/xWR6243 devices:			
		Programmable Gain Amplifier Setting: This configures the Phase shifter loopback path amplifier gain for TX1 based PS loopback.			
		Value	PGA gain value	Value	PGA gain value
		0	PGA is OFF	15	-3 dB
		1	-22 dB	16	-2 dB
		2	-16 dB	17	-1 dB
		3	-15 dB	18	0 dB
		4	-14 dB	19	1 dB
		5	-13 dB	20	2 dB
		6	-12 dB	21	3 dB
		7	-11 dB	22	4 dB
		8	-10 dB	23	5 dB
		9	-9 dB	24	6 dB
		10	-8 dB	25	7 dB
		11	-7 dB	26	8 dB
		12	-6 dB	27	9 dB
		13	-5 dB	255-28	RESERVED
		14	-4 dB		
		xWR294x, xWR254x, xWR2x4xECO, WR2x4xP devices:			
		This field configures the loopback buffer in the LO loopback path in xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP.			
		NOTE: It is recommended to keep LO loopback buffer gain setting lesser than -7dB to avoid ADC saturation, especially at low temperatures.			
		Value	Loopback gain	Value	Loopback gain
		96	0 dB	28	-15 dB
		51	-1 dB	27	-16 dB
		47	-2 dB	26	-17 dB
		44	-3 dB	25	-18 dB
		42	-4 dB	24	-19 dB
		40	-5 dB	23	-21 dB
		38	-6 dB	22	-22 dB
		36	-7 dB	21	-23 dB
		35	-8 dB	20	-25 dB
		34	-9 dB	19	-26 dB
		33	-10 dB	18	-27 dB
		32	-11 dB	0	-30 dB
		31	-12 dB		
		30	-13 dB		
		29	-14 dB		
		Copyright © 2021, Texas Instruments Incorporated			

Table 5.43 – continued from previous page

PS_LOOP- BACK_FREQ (AWR2243) , LO_ LOOPBACK_ FREQ (xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP)	4	AWR2243/xWR6243 devices: Phase shifter loop back frequency in kHz: The TX phase shifter's phase shift command word is linearly varied at a rate configured by this field to achieve a frequency shift. 1 LSB = 1 kHz Bits Definition b15:0 TX0 Loopback Frequency b31:16 TX1 Loopback Frequency xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: LO loopback modulation frequency in Hz. The allowed frequencies are listed below. 1 LSB = 1 Hz. LO loopback frequency 0 Hz 180000 Hz 300000 Hz 360000 Hz 500000 Hz 1000000 Hz 1500000 Hz 2500000 Hz 3000000 Hz 4017857 Hz 4500000 Hz 5000000 Hz 7500000 Hz 8035714 Hz 9000000 Hz 20454545 Hz
RESERVED	4	RESERVED

Continued on next page

Table 5.43 – continued from previous page

PA_LOOP- BACK_FREQ (AWR2243)	2	This value is a 100 MHz divider which sets the loopback frequency: The PA output is fed to a modulator before coupling to the RX LNA input. This field govern the modulation frequency and can be used to separate the internal loopback signal from external reflections wrt IF frequency at the Receiver. For e.g. for a 1 MHz frequency, set this to 100 For a 2 MHz frequency, set this to 50 NOTE: To ensure no leakage of signal power, user has to ensure that 100MHz/LOOPBACK_FREQ is an integer multiple of bin width For e.g. if user chooses 25Msps sampling rate and 2048 samples/chirp, then LOOPBACK_FREQ of 64 (=1.5625 MHz) will ensure no leakage This field is RESERVED in xWR294x, xWR254x and xWR294xP devices, a separate field "PA_LOOPBACK_FREQ (xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP)" will be used to set the PA LB frequency.																																				
PA_LOOPBACK_ BUF_GAIN_SEL	1	This configures the PA loopback buffer gain. This is only applicable for xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP devices. <table><thead><tr><th>Value</th><th>PA LB Buffer Gain</th><th>Value</th><th>PA LB Buffer Gain</th></tr></thead><tbody><tr><td>48</td><td>0dB</td><td>19</td><td>-10dB</td></tr><tr><td>27</td><td>-1dB</td><td>18</td><td>-11dB</td></tr><tr><td>25</td><td>-2dB</td><td>17</td><td>-13dB</td></tr><tr><td>24</td><td>-3dB</td><td>16</td><td>-15dB</td></tr><tr><td>23</td><td>-4dB</td><td>15</td><td>-17dB</td></tr><tr><td>22</td><td>-5dB</td><td>14</td><td>-18dB</td></tr><tr><td>21</td><td>-6dB</td><td>0</td><td>-19dB</td></tr><tr><td>20</td><td>-7dB</td><td></td><td></td></tr></tbody></table> Note: PA loopback signal strength becomes inaccurate for gain values lower than -10dB.	Value	PA LB Buffer Gain	Value	PA LB Buffer Gain	48	0dB	19	-10dB	27	-1dB	18	-11dB	25	-2dB	17	-13dB	24	-3dB	16	-15dB	23	-4dB	15	-17dB	22	-5dB	14	-18dB	21	-6dB	0	-19dB	20	-7dB		
Value	PA LB Buffer Gain	Value	PA LB Buffer Gain																																			
48	0dB	19	-10dB																																			
27	-1dB	18	-11dB																																			
25	-2dB	17	-13dB																																			
24	-3dB	16	-15dB																																			
23	-4dB	15	-17dB																																			
22	-5dB	14	-18dB																																			
21	-6dB	0	-19dB																																			
20	-7dB																																					
RESERVED	1	0x0000																																				

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Table 5.43 – continued from previous page

PA_LOOPBACK_FREQ (xWR294x, xWR254x)	4	xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: PA loopback modulation frequency in Hz. The allowed frequencies are listed below. 1 LSB = 1 Hz. PA loopback frequency 0 Hz 180000 Hz 300000 Hz 360000 Hz 500000 Hz 1000000 Hz 1500000 Hz 2500000 Hz 3000000 Hz 4017857 Hz 4500000 Hz 5000000 Hz 7500000 Hz 8035714 Hz 9000000 Hz 20454545 Hz
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NOTE: The expected signal strength change with change in index value is only approximately indicated for PS<n>_PGA_GAIN_INDEX. Typically, the loopback path is the dominant path only in top 10 indices (highest PGA gain values). For lower indices (lower PGA gain values), parasitic paths in the RF system can start dominating the loop-back measurements, and under such conditions, inter channel imbalances measured using such LB path, and LB signal SNR etc. can show degraded performance, with the degradation attributed to the loop-back path and not the functional path/circuits/system.

5.5.16 Sub block 0x010F – AWR_DYN_CHIRP_CONF_SET_SB

This API can be used to dynamically change the chirp configuration while frames are on-going. The configuration will be stored in software and the new configuration will be applied after receiving the AWR_DYN_CHIRP_ENABLE_SB API.

Table 5.44: AWR_DYN_CHIRP_CONF_SET_SB contents

Field Name	Number of bytes	Description												
SBLKID	2	Value = 0x010F												
SBLKLEN	2	Value = 200												
CHIRP_ROW_SELECT	1	BitsDescription b3:0RESERVED b7:4If user does not wish to reconfigure all 3 chirp rows, then the following mode can be used to configure only one row per chirp which enables the user to configure 48 chirps in one API, effectively saving on the reconfiguration time. If CHIRP_ROW_SELECT[7:4] is non-zero, then the API parameters CHIRP_x_R1, CHIRP_x_R2 and CHIRP_x_R3 for 1 ≤ <i>x</i> ≤ 16 in this API would mean CHIRP(3<i>x</i> − 2)_R_{<i>y</i>}, CHIRP(3<i>x</i> − 1)_R_{<i>y</i>} and CHIRP(3<i>x</i>)_R_{<i>y</i>} where <i>y</i> is as per the below table <table><tr><th>Value</th><th>Definition</th></tr><tr><td>0b0000</td><td>Enables all 3 chirp rows to be reconfigured (default)</td></tr><tr><td>0b0001</td><td>Enables only chirp row 1 to be reconfigured</td></tr><tr><td>0b0010</td><td>Enables only chirp row 2 to be reconfigured</td></tr><tr><td>0b0011</td><td>Enables only chirp row 3 to be reconfigured</td></tr><tr><td>Others</td><td>RESERVED</td></tr></table>	Value	Definition	0b0000	Enables all 3 chirp rows to be reconfigured (default)	0b0001	Enables only chirp row 1 to be reconfigured	0b0010	Enables only chirp row 2 to be reconfigured	0b0011	Enables only chirp row 3 to be reconfigured	Others	RESERVED
Value	Definition													
0b0000	Enables all 3 chirp rows to be reconfigured (default)													
0b0001	Enables only chirp row 1 to be reconfigured													
0b0010	Enables only chirp row 2 to be reconfigured													
0b0011	Enables only chirp row 3 to be reconfigured													
Others	RESERVED													
CHIRP_SEGMENT_SELECT	1	Valid range 0 to 31. Indicates the segment of the chirp RAM that the 16 chirp definitions in this sub block map to												

Continued on next page

Table 5.44 – continued from previous page

PROGRAM_ MODE	2	Bits	Description
		b0	Value Definition
			0 Program the new configuration when AWR_DYN_CHIRP_ENABLE API is issued 1 Program the new configuration immediately NOTE: User has to ensure that the chirps which are being reconfigured are not the ones which are currently in use for chirping
		b15:1	RESERVED
CHIRP1_R1	4	Bits	Definition
		b3:0	PROFILE_IND Valid range 0 to 3
		b7:4	RESERVED
		b13:8	FREQ_SLOPE_VAR For 77GHz Devices (76GHz to 81GHz): 1 LSB = $3.6e9 \times 900/2^{26} \approx 48.279$ kHz Valid range: 0 to 63 For 60GHz Devices (57GHz to 64GHz): 1 LSB = $2.7e9 \times 900/2^{26} \approx 36.21$ kHz Valid range: Only even numbers from 0 to 63
		b15:14	RESERVED
		b19:16	TX_ENABLE
		Bit	Definition
		b0	TX0 Enable
		b1	TX1 Enable
		b2	TX2 Enable
		b2	TX3 Enable (only for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP)
		b23:20	RESERVED
		b29:24	RESERVED
		b31:30	RESERVED

Continued on next page

Table 5.44 – continued from previous page

CHIRP1_R2	4	Bits Definition b22:0 FREQ_START_VAR For 77GHz Devices (76GHz to 81GHz): 1 LSB = $3.6e9/2^{26} \approx 53.644$ Hz Valid range: 0 to 8388607 For 60GHz Devices (57GHz to 64GHz): 1 LSB = $2.7e9/2^{26} \approx 40.233$ Hz Valid range: Only even numbers from 0 to 8388607 b31:23 RESERVED
CHIRP1_R3	4	Bits Definition b11:0 IDLE_TIME_VAR 1 LSB = 10 ns Valid range: 0 to 4095 b15:12 RESERVED b27:16 ADC_START_TIME_VAR 1 LSB = 10 ns Valid range: 0 to 4095 b31:28 RESERVED
CHIRP2_R1	4	See description for CHIRP1_R1
CHIRP2_R2	4	See description for CHIRP1_R2
CHIRP2_R3	4	See description for CHIRP1_R3
...	...	...
CHIRP16_R1	4	See description for CHIRP1_R1
CHIRP16_R2	4	See description for CHIRP1_R2
CHIRP16_R3	4	See description for CHIRP1_R3

NOTE:

If user wants to update the chirp ram rows using dynamic chirp config API in runtime then it is must to use same dynamic chirp config API (instead of legacy chirp config API) to configure all chirp parameters during sensor initialization.

5.5.17 Sub block 0x0110 – AWR_DYN_PERCHIRP_PHASESHIFTER_CONF_SET_SB

This API can be used to dynamically change the per-chirp phase shifter configuration (applicable only in certain devices) while frames are on-going. The configuration will be stored in software and the new configuration will be applied after receiving the AWR_DYN_CHIRP_ENABLE_SB API.

NOTE: This API is not supported in xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP. Please use AWR_ADV_DYN_PERCHIRP_PHASESHIFTER_CONF_SET_SB for configuring all 4TX phase shifts dynamically in xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP.

Table 5.45: AWR_DYN_PERCHIRP_PHASESHIFTER_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0110
SBLKLEN	2	Value = 56
RESERVED	1	0x00
CHIRP_SEG- MENT_SELECT	1	Indicates the segment of the chirp RAM that the 16 chirp definitions in this sub block map to. Valid range 0 to 31
CHIRP1_ TX0_PHASE_ SHIFTER	1	TX0 phase shift value Bits TX0 phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 TX0 phase shift value 1 LSB = $360^\circ/2^6 = 5.625^\circ$ Valid range: 0 to 63
CHIRP1_ TX1_PHASE_ SHIFTER	1	TX1 phase shift value Bits TX1 phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 TX1 phase shift value 1 LSB = $360^\circ/2^6 = 5.625^\circ$ Valid range: 0 to 63
CHIRP1_ TX2_PHASE_ SHIFTER	1	TX2 phase shift value Bits TX1 phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 TX1 phase shift value 1 LSB = $360^\circ/2^6 = 5.625^\circ$ Valid range: 0 to 63
CHIRP2_ TX0_PHASE_ SHIFTER	1	See description for CHIRP1_TX0_PHASE_SHIFTER
CHIRP2_ TX1_PHASE_ SHIFTER	1	See description for CHIRP2_TX1_PHASE_SHIFTER

Continued on next page

Table 5.45 – continued from previous page

CHIRP2_TX2_PHASE_SHIFTER	1	See description for CHIRP3_TX2_PHASE_SHIFTER		
...	...	...		
CHIRP16_TX0_PHASE_SHIFTER	1	See description for CHIRP1_TX0_PHASE_SHIFTER		
CHIRP16_TX1_PHASE_SHIFTER	1	See description for CHIRP2_TX1_PHASE_SHIFTER		
CHIRP16_TX2_PHASE_SHIFTER	1	See description for CHIRP3_TX2_PHASE_SHIFTER		
PROGRAM_MODE	2	Bits	Description	
		b0	Value	Definition
			0	Program the new configuration when AWR_DYN_CHIRP_ENABLE API is issued
			1	Program the new configuration immediately NOTE: User has to ensure that the chirps which are being reconfigured are not the ones which are currently in use for chirping
		b15:1	RESERVED	

5.5.18 Sub block 0x0111 – AWR_DYN_CHIRP_ENABLE_SB

This API can be used to trigger the copy of chirp configuration from software to hardware. The copy will be performed at the end of the ongoing frame active window (start of the frame idle time).

Table 5.46: AWR_DYN_CHIRP_ENABLE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0111
SBLKLEN	2	Value = 8
RESERVED	4	0x00000000

NOTE: HW reconfiguration time (as shown in the figure below) is around 500 μ s. User has to ensure that AWR_DYN_CHIRP_ENABLE_SB API is issued at least 500 μ s before the end of the ongoing frame active window (start of the frame idle time) to apply configurations for next frame onwards.

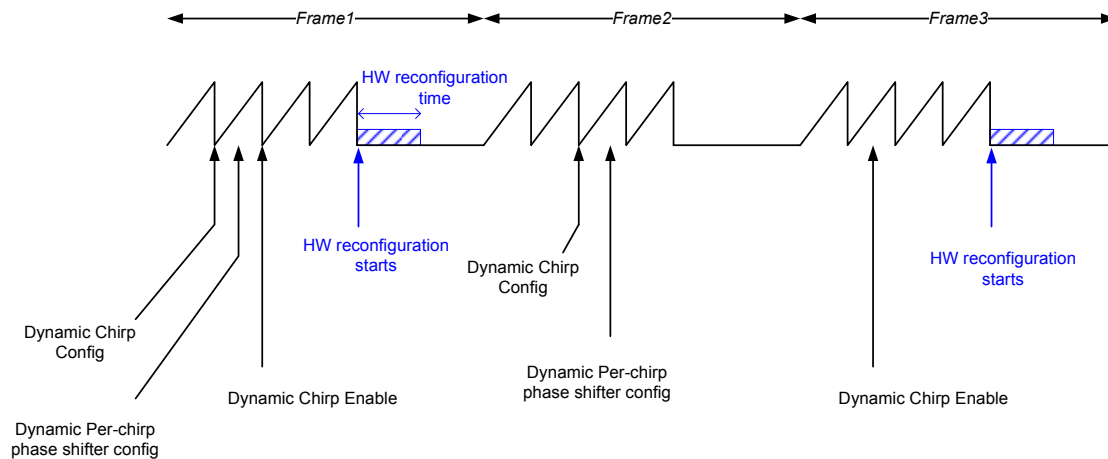


Figure 5.5: Dynamic chirp configuration use case timing diagram

5.5.19 Sub block 0x0112 – AWR_INTERCHIRP_BLOCKCONTROLS_SB

This API can be used to program the inter-chip turn on and turn off times or various RF blocks.

Table 5.47: AWR_INTERCHIRP_BLOCKCONTROLS_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0112
SBLKLEN	2	Value = 44
RX02_RF_TURN_OFF_TIME	2	Time to wait after ramp end before turning off RX0 and RX2 RF stages. 1 LSB = 10 ns Valid range: -1024 to 1023

Continued on next page

Table 5.47 – continued from previous page

RX13_RF_TURN_OFF_TIME	2	Time to wait after ramp end before turning off RX1 and RX3 RF stages. 1 LSB = 10 ns Valid range: -1024 to 1023
RX02_BB_TURN_OFF_TIME	2	Time to wait after ramp end before turning off RX0 and RX2 baseband stages. 1 LSB = 10 ns Valid range: -1024 to 1023
RX13_BB_TURN_OFF_TIME	2	Time to wait after ramp end before turning off RX1 and RX3 baseband stages. 1 LSB = 10 ns Valid range: -1024 to 1023
RX02_RF_PRE_ENABLE_TIME	2	Time before TX Start Time when RX0 and RX2 RF stages are to be put in fast-charge state. 1 LSB = 10 ns Valid range: -1024 to 1023
RX13_RF_PRE_ENABLE_TIME	2	Time before TX Start Time when RX1 and RX3 RF stages are to be put in fast-charge state. 1 LSB = 10 ns Valid range: -1024 to 1023
RX02_BB_PRE_ENABLE_TIME	2	Time before TX Start Time when RX1 and RX3 baseband stages are to be put in fast-charge state. 1 LSB = 10 ns Valid range: -1024 to 1023
RX13_BB_PRE_ENABLE_TIME	2	Time before TX Start Time when RX2 and RX4 baseband stages are to be put in fast-charge state. 1 LSB = 10 ns Valid range: -1024 to 1023
RX02_RF_TURN_ON_TIME	2	Time before TX Start Time when RX1 and RX3 RF stages are to be enabled. 1 LSB = 10 ns Valid range: -1024 to 1023
RX13_RF_TURN_ON_TIME	2	Time before TX Start Time when RX2 and RX4 RF stages are to be enabled. 1 LSB = 10 ns Valid range: -1024 to 1023
RX02_BB_TURN_ON_TIME	2	Time before TX Start Time when RX1 and RX3 baseband stages are to be enabled. 1 LSB = 10 ns Valid range: -1024 to 1023

Continued on next page

Table 5.47 – continued from previous page

RX13_BB_TURN_ON_TIME	2	Time before TX Start Time when RX2 and RX4 baseband stages are to be enabled. 1 LSB = 10 ns Valid range: -1024 to 1023
RX_LO_CHAIN_TURN_OFF_TIME	2	Time to wait after ramp end before turning off RX LO chain. 1 LSB = 10 ns Valid range: -1024 to 1023
TX_LO_CHAIN_TURN_OFF_TIME	2	Time to wait after ramp end before turning off TX LO chain. 1 LSB = 10 ns Valid range: -1024 to 1023
RX_LO_CHAIN_TURN_ON_TIME	2	Time before TX Start Time when the RX LO chain is to be enabled. 1 LSB = 10 ns Valid range: -1024 to 1023
TX_LO_CHAIN_TURN_ON_TIME	2	Time before TX Start Time when the TX LO chain is to be enabled. 1 LSB = 10 ns Valid range: -1024 to 1023
RESERVED	4	0x00000000
RESERVED	4	0x00000000

NOTE:

The minimum inter-chirp time should be greater than maximum of the following

1. $\text{abs}(\text{RX02_RF_TURN_OFF_TIME}) + \max(\text{abs}(\text{RX02_RF_PRE_ENABLE_TIME}), \text{abs}(\text{RX02_RF_TURN_ON_TIME}))$
2. $\text{abs}(\text{RX13_RF_TURN_OFF_TIME}) + \max(\text{abs}(\text{RX13_RF_PRE_ENABLE_TIME}), \text{abs}(\text{RX13_RF_TURN_ON_TIME}))$
3. $\text{abs}(\text{RX02_BB_TURN_OFF_TIME}) + \max(\text{abs}(\text{RX02_BB_PRE_ENABLE_TIME}), \text{abs}(\text{RX02_BB_TURN_ON_TIME}))$
4. $\text{abs}(\text{RX13_BB_TURN_OFF_TIME}) + \max(\text{abs}(\text{RX13_BB_PRE_ENABLE_TIME}), \text{abs}(\text{RX13_BB_TURN_ON_TIME}))$
5. $\text{abs}(\text{RX_LO_TURN_OFF_TIME}) + \text{abs}(\text{RX_LO_TURN_ON_TIME})$
6. $\text{abs}(\text{TX_LO_TURN_OFF_TIME}) + \text{abs}(\text{TX_LO_TURN_ON_TIME})$

5.5.20 Sub block 0x0113 – AWR_SUBFRAME_START_CONF_SB

This API can be used to trigger each sub-frame individually in software triggered mode. This API takes effect only when the advanced frame configuration indicates that each sub-frame needs to be individually triggered by the user.

Table 5.48: AWR_SUBFRAME_START_CONF_SB contents

Field Name	Number of bytes	Description									
SBLKID	2	Value = 0x0113									
SBLKLEN	2	Value = 8									
START_CMD	2	<table border="1"> <thead> <tr> <th>Bits</th><th>Value</th><th>Definition</th></tr> </thead> <tbody> <tr> <td>b15:0</td><td>0x0000</td><td>No effect</td></tr> <tr> <td></td><td>0x0001</td><td>Trigger next sub-frame in software triggered sub-frame mode</td></tr> </tbody> </table>	Bits	Value	Definition	b15:0	0x0000	No effect		0x0001	Trigger next sub-frame in software triggered sub-frame mode
Bits	Value	Definition									
b15:0	0x0000	No effect									
	0x0001	Trigger next sub-frame in software triggered sub-frame mode									
RESERVED	2	0x0000									

NOTE1:	If the user wishes to trigger each sub-frame independently, then after advanced frame config, the FRAME START command should be issued once using AWR_FRAMESTARTSTOP_CONF_SB. This does not start any sub-frames but it will prepare the hardware for sub-frame trigger. Next any subsequent sub-frame trigger will start the sub-frames
NOTE2:	If the user wishes to use sub-frame trigger, he has to ensure that sub-frame trigger command is issued $k \cdot N$ times where k is the number of sub-frames in each frame and N is the number of frames. If the user wishes to stop frames in between, then he has to issue the FRAME STOP command (using AWR_FRAMESTARTSTOP_CONF_SB) only after $k \cdot M$ triggers of sub-frame trigger command (where M is an integer). i.e. FRAME STOP command can be issued only at frame boundaries
NOTE3:	If software based sub-frame trigger mode is chosen by the user, watchdog feature will not be available. User has to ensure that the watchdog is disabled before enabling the software based sub-frame trigger mode.
NOTE4:	If sub-frame trigger or hardware trigger mode is used to trigger the frames/sub-frames and if frames need to be stopped before the specified number of frames, then the the FRAME_STOP command using AWR_FRAMESTARTSTOP_CONF_SB API should be issued while the frame is on-going. If the frames are stopped while the device is idle, it can lead to errors.

5.5.21 Sub block 0x0115 – AWR_ADVANCE_CHIRP_CONF_SB

This API sub-block defines the programming of advanced chirp configurations for each chirp parameters to generate a waveform pattern in a frame/burst. This API provides ability to program fixed delta increment (Delta dither) for certain chirp parameters (eg. chirp start frequency, idle time, phase shifter, etc.), on top of unique dithers selected from configurable look-up-table (LUT Dither). The configurable look-up-table is an array of values loaded into a pre-configured “Generic SW Chirp Parameter LUT” The size of the generic LUT is 12kB and user has the flexibility to program any number of unique dithers for each chirp parameters. Thus the user can achieve fixed increment, or LUT based dither, or a combination of both.

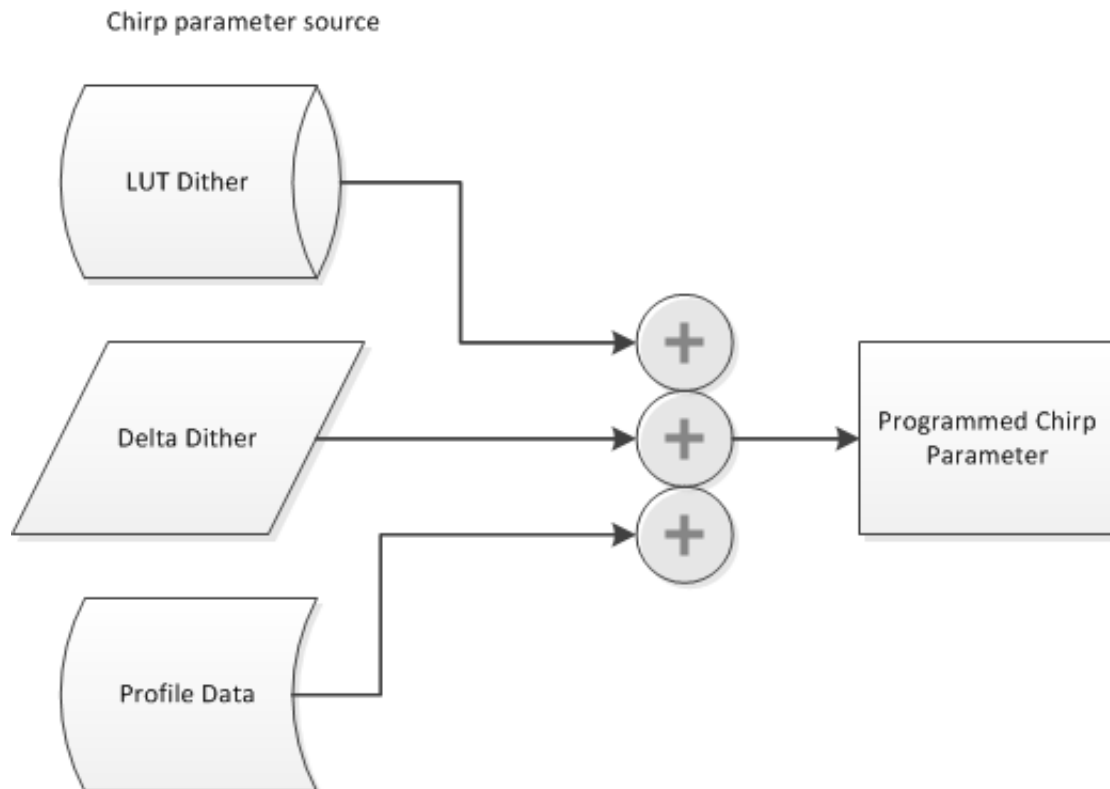


Figure 5.6: Advance chirp parameter dither sources and program

Using this API, four types of control can be achieved on each parameters of a chirp.

1. Fixed value for all chirps: To generate sequence of chirps which never changes, then only one value can be programmed in LUT (LUT Dither), i.e NUM_OF_PATTERNS (P) = 1 and LUT_PARAM_UPDATE_PERIOD (K) = 0
2. Unique chirps: Index every LUT_PARAM_UPDATE_PERIOD (K) chirps in LUT to generate unique sequence of chirps.
3. Delta increment every DELTA_PARAM_UPDATE_PERIOD (N) chirps: On top of sequence of unique chirps from LUT, the fixed delta increment (Delta dither) can be done every N chirps.
4. The set of chirp parameters across bursts and sub-frames can be different by setting offset to LUT in BURST_LUT_INDEX_OFFSET and SF_LUT_INDEX_OFFSET.

When using the Advanced Chirp Config API, there are some implications to frame config and advanced frame config APIs. Specifically, the CHIRP_START_INDX and CHIRP_END_INDX fields are no longer applicable, and the NUM_LOOPS field has a different meaning in the sense that this field now denotes the total number of chirps in the frame/burst. Please refer [AWR_FRAME_CONF_SET_SB](#) and [AWR_ADVANCED_FRAME_CONF_SB](#) APIs with the updated field descriptions as below.

The total number of chirps L in a burst should be programmed as per below calculation in frame configuration API (using the NUM_LOOPS field).

$L = X * Y$, where X is 1 to 512 (supported HW CHIRP RAM) and Y is 1 to 128 (supported HW CHIRP LOOPS) The value of L should be a multiple of 4 (assuming each chirp is min 25us duration) i.e 1, 4, 8, 12, 16, 20, ... 32768 (max). The FW needs to prepare and update HW CHIRP RAM dynamically in advance chirp config API, this puts some restriction on minimum number of chirps in a burst/frame.

NOTE1:	The Legacy AWR_CHIRP_CONF_SET_SB , AWR_DYN_CHIRP_CONF_SET_SB , AWR_PERCHIRPPHASESHIFT_CONF_SB , AWR_DYN_PERCHIRP_PHASESHIFTER_CONF_SET_SB and AWR_BPM_CHIRP_CONF_SET_SB APIs are not supported if device is configured with Advanced Chirp Config API enabled in AWR_RF_RADAR_MISC_CTL_SB or vice versa.
NOTE2:	The per chirp phase shifter and BPM configurations are part of this API.
NOTE3:	The parameters in this API are not applicable to loop-back sub-frames AWR_LOOPBACK_BURST_CONF_SET_SB . If loop-back sub-frames are needed, it is recommended to be configured in the last sub-frame (SF) of AWR_ADVANCED_FRAME_CONF_SB API.
NOTE4:	The dynamic update of this API is allowed at frame boundary along with the Generic SW Chirp Parameters, as long as the LUT addresses modified differ from the addresses used in the current ongoing frame. The dynamic chirp enable API AWR_DYN_CHIRP_ENABLE_SB shall be issued at least 500us before end of current active window of frame (500us before start of idle time of the frame) to apply the dynamic configurations in immediate next frame.
NOTE5:	The RF frequency used for measurement in monitors are derived only from profile settings (start frequency and slope) and not from the advance chirp configuration API, if fixed delta increment is used to change the start frequency every chirp, it is recommended to have a separate profile for monitors which covers full RF bandwidth of interest.

Table 5.49 describes the contents of this sub block. All the fields in this API are specific to selected CHIRP_PARAM_INDEX in this API, this API needs to be programmed ten times for each of the

chirp parameters defined in CHIRP_PARAM_INDEX field in below API.

The Delta Dither is optional and can be disabled by setting DELTA_PARAM_UPDATE_PERIOD (N) = 0 and SFn_CHIRP_PARAM_DELTA = 0.

The LUT Dither is mandatory and at least one dither parameter value (it can be value zero) shall be programmed for all chirp parameters in generic LUT, same dither value can be programmed to all chirps in a burst/frame by setting LUT_PARAM_UPDATE_PERIOD (K) = 0.

Table 5.49: AWR_ADVANCE_CHIRP_CONF_SB contents

Field Name	Number of bytes	Description																																																				
SBLKID	2	Value = 0x0115																																																				
SBLKLEN	2	Value = 60																																																				
CHIRP_PARAM_INDEX	1	This field indicates the chirp parameter that the current API configures. The mapping and availability of dither modes are as below: <table><thead><tr><th>Index</th><th>Parameter</th><th>Delta Dither</th><th>LUT Dither</th></tr></thead><tbody><tr><td>0</td><td>CHIRP_PROFILE_SELECT</td><td>No</td><td>Yes</td></tr><tr><td>1</td><td>CHIRP_FREQ_START_VAR</td><td>Yes</td><td>Yes</td></tr><tr><td>2</td><td>CHIRP_FREQ_SLOPE_VAR</td><td>Yes</td><td>Yes</td></tr><tr><td>3</td><td>CHIRP_IDLE_TIME_VAR</td><td>Yes</td><td>Yes</td></tr><tr><td>4</td><td>CHIRP_ADC_START_TIME_VAR</td><td>Yes</td><td>Yes</td></tr><tr><td>5</td><td>CHIRP_TX_EN</td><td>No</td><td>Yes</td></tr><tr><td>6</td><td>CHIRP_BPM_VAL</td><td>No</td><td>Yes</td></tr><tr><td>7</td><td>TX0_PHASE_SHIFTER</td><td>Yes</td><td>Yes</td></tr><tr><td>8</td><td>TX1_PHASE_SHIFTER</td><td>Yes</td><td>Yes</td></tr><tr><td>9</td><td>TX2_PHASE_SHIFTER</td><td>Yes</td><td>Yes</td></tr><tr><td>10</td><td>TX3_PHASE_SHIFTER</td><td>Yes</td><td>Yes</td></tr><tr><td>255-11</td><td>Reserved</td><td></td><td></td></tr></tbody></table> The parameters referred to here are the same as the ones with the name referred to in AWR_CHIRP_CONF_SET_SB and in AWR_PERCHIRPPHASESHIFT_CONF_SB.	Index	Parameter	Delta Dither	LUT Dither	0	CHIRP_PROFILE_SELECT	No	Yes	1	CHIRP_FREQ_START_VAR	Yes	Yes	2	CHIRP_FREQ_SLOPE_VAR	Yes	Yes	3	CHIRP_IDLE_TIME_VAR	Yes	Yes	4	CHIRP_ADC_START_TIME_VAR	Yes	Yes	5	CHIRP_TX_EN	No	Yes	6	CHIRP_BPM_VAL	No	Yes	7	TX0_PHASE_SHIFTER	Yes	Yes	8	TX1_PHASE_SHIFTER	Yes	Yes	9	TX2_PHASE_SHIFTER	Yes	Yes	10	TX3_PHASE_SHIFTER	Yes	Yes	255-11	Reserved		
Index	Parameter	Delta Dither	LUT Dither																																																			
0	CHIRP_PROFILE_SELECT	No	Yes																																																			
1	CHIRP_FREQ_START_VAR	Yes	Yes																																																			
2	CHIRP_FREQ_SLOPE_VAR	Yes	Yes																																																			
3	CHIRP_IDLE_TIME_VAR	Yes	Yes																																																			
4	CHIRP_ADC_START_TIME_VAR	Yes	Yes																																																			
5	CHIRP_TX_EN	No	Yes																																																			
6	CHIRP_BPM_VAL	No	Yes																																																			
7	TX0_PHASE_SHIFTER	Yes	Yes																																																			
8	TX1_PHASE_SHIFTER	Yes	Yes																																																			
9	TX2_PHASE_SHIFTER	Yes	Yes																																																			
10	TX3_PHASE_SHIFTER	Yes	Yes																																																			
255-11	Reserved																																																					

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Table 5.49 – continued from previous page

GLOBAL_RE- SET_MODE	1	This field indicates the reset mode of the programmed pattern. It indicates when the fixed delta accumulation (Delta Dither) or the programmed dither pattern from LUT (LUT Dither) resets back to its initial value. This is a global reset occurs for all the chirp parameters. This value should be same for all chirp parameter. <table><tr><td>Mode</td><td>Definition</td></tr><tr><td>0</td><td>Reset at the end of Frame Note: This reset mode is not supported when using advanced frame configuration.</td></tr><tr><td>1</td><td>Reset at the end of Sub-Frame</td></tr><tr><td>2</td><td>Reset at the end of Burst</td></tr><tr><td>255-3</td><td>Reserved</td></tr></table>		Mode	Definition	0	Reset at the end of Frame Note: This reset mode is not supported when using advanced frame configuration.	1	Reset at the end of Sub-Frame	2	Reset at the end of Burst	255-3	Reserved		
Mode	Definition														
0	Reset at the end of Frame Note: This reset mode is not supported when using advanced frame configuration.														
1	Reset at the end of Sub-Frame														
2	Reset at the end of Burst														
255-3	Reserved														
RESERVED	2	0x0000													
PARAM_CON- TROL	1	<table><tr><td>Bits</td><td>Definition</td></tr><tr><td>0</td><td>Disable the chirp parameter</td></tr><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>Chirp parameter is enabled.</td></tr><tr><td>1</td><td>Chirp parameter is disabled.</td></tr></table> NOTE: Param control should always be enabled for CHIRP_PROFILE_SELECT, CHIRP_TX_EN, CHIRP_BPM_VAL. When param control is disabled for CHIRP_FREQ_START_VAR, CHIRP_FREQ_SLOPE_VAR, CHIRP_IDLE_TIME_VAR, CHIRP_ADC_START_TIME_VAR 0 dither values are applied. When param control is disabled for TXn_PHASE_SHIFTER TX Phase shifter values are used from PF_TX_PHASE_SHIFTER field of AWR_PROFILE_CONF_SET_SB. <table><tr><td>7-1</td><td>RESERVED</td></tr></table>		Bits	Definition	0	Disable the chirp parameter	Value	Definition	0	Chirp parameter is enabled.	1	Chirp parameter is disabled.	7-1	RESERVED
Bits	Definition														
0	Disable the chirp parameter														
Value	Definition														
0	Chirp parameter is enabled.														
1	Chirp parameter is disabled.														
7-1	RESERVED														
RESERVED	3	0x0000													

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Table 5.49 – continued from previous page

DELTA_RESET_PERIOD (M)	2	Reset the delta increment (Delta Dither) sequence every M chirps Valid range: 0 – 32768 <table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>Reset only as per RESET MODE option</td></tr><tr><td>1</td><td>Delta increment is disabled</td></tr><tr><td>32768-2</td><td>Reset every M chirps in addition to RESET MODE option</td></tr></table> The reset period should be integer multiple of DELTA_PARAM_UPDATE_PERIOD (N)	Value	Definition	0	Reset only as per RESET MODE option	1	Delta increment is disabled	32768-2	Reset every M chirps in addition to RESET MODE option
Value	Definition									
0	Reset only as per RESET MODE option									
1	Delta increment is disabled									
32768-2	Reset every M chirps in addition to RESET MODE option									
DELTA_PARAM_UPDATE_PERIOD (N)	2	The chirp parameter will be incremented by SFn_CHIRP_PARAM_DELTA (Delta Dither) every N chirps. Valid range: 0 – 16384 <table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>Delta increment is disabled</td></tr><tr><td>16384-1</td><td>The fixed delta value will be incremented once after every N chirps.</td></tr></table>	Value	Definition	0	Delta increment is disabled	16384-1	The fixed delta value will be incremented once after every N chirps.		
Value	Definition									
0	Delta increment is disabled									
16384-1	The fixed delta value will be incremented once after every N chirps.									
SF0_CHIRP_PARAM_DELTA	4	This field indicates the delta increment (Delta Dither) value for sub-frame 0 (Also applicable for legacy frame config API) that should be accumulated and added to each chirp based on update period N. This accumulated fixed dither value which increments every N chirps is added to LUT dither value derived for same chirp, The sum of accumulated dither and LUT dither derived for each chirp is added to profile setting of same chirp parameter in HW. As some parameters may need signed values, all the Bytes in this field should be populated with appropriate sign extension. NOTE: If ADC_START_TIME_RES is set to 1 in AWR_RF_MISC_CTL_SB, when AWR_ADVANCE_CHIRP_CONF_SB API is issued for CHIRP_ADC_START_TIME_VAR, it is recommended to program SF0_CHIRP_PARAM_DELTA to 0. Only LUT based dither can be used for CHIRP_ADC_START_TIME_VAR. Refer to the Fixed Delta Chirp Parameter LUT description Table 5.50 for the definition of this field when each parameter is selected. This feature is enabled only for certain chirp parameter types as defined in this table.								

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Table 5.49 – continued from previous page

SF1_CHIRP_ PARAM_DELTA	4	This field indicates the delta increment (Delta Dither) value for sub-frame 1 (Not applicable for legacy frame config API) that should be accumulated and added to each chirp based on update period N. This accumulated fixed dither value which increments every N chirps is added to LUT dither value derived for same chirp, The sum of accumulated dither and LUT dither derived for each chirp is added to profile setting of same chirp parameter in HW. As some parameters may need signed values, all the Bytes in this field should be populated with appropriate sign extension. NOTE: If ADC_START_TIME_RES is set to 1 in AWR_RF_MISC_CTL_SB, when AWR_ADVANCE_CHIRP_CONF_SB API is issued for CHIRP_ADC_START_TIME_VAR, it is recommended to program SF1_CHIRP_PARAM_DELTA to 0. Only LUT based dither can be used for CHIRP_ADC_START_TIME_VAR. Refer to the Fixed Delta Chirp Parameter LUT description Table 5.50 for the definition of this field when each parameter is selected. This feature is enabled only for certain chirp parameter types as defined in this table.
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Table 5.49 – continued from previous page

SF2_CHIRP_ PARAM_DELTA	4	This field indicates the delta increment (Delta Dither) value for sub-frame 2 (Not applicable for legacy frame config API) that should be accumulated and added to each chirp based on update period N. This accumulated fixed dither value which increments every N chirps is added to LUT dither value derived for same chirp, The sum of accumulated dither and LUT dither derived for each chirp is added to profile setting of same chirp parameter in HW. As some parameters may need signed values, all the Bytes in this field should be populated with appropriate sign extension. NOTE: If ADC_START_TIME_RES is set to 1 in AWR_RF_MISC_CTL_SB, when AWR_ADVANCE_CHIRP_CONF_SB API is issued for CHIRP_ADC_START_TIME_VAR, it is recommended to program SF2_CHIRP_PARAM_DELTA to 0. Only LUT based dither can be used for CHIRP_ADC_START_TIME_VAR. Refer to the Fixed Delta Chirp Parameter LUT description table Table 5.50 for the definition of this field when each parameter is selected. This feature is enabled only for certain chirp parameter types as defined in this table.
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Table 5.49 – continued from previous page

SF3_CHIRP_PARAM_DELTA	4	This field indicates the delta increment (Delta Dither) value for sub-frame 3 (Not applicable for legacy frame config API) that should be accumulated and added to each chirp based on update period N. This accumulated fixed dither value which increments every N chirps is added to LUT dither value derived for same chirp, The sum of accumulated dither and LUT dither derived for each chirp is added to profile setting of same chirp parameter in HW. As some parameters may need signed values, all the Bytes in this field should be populated with appropriate sign extension. NOTE: If ADC_START_TIME_RES is set to 1 in AWR_RF_MISC_CTL_SB, when AWR_ADVANCE_CHIRP_CONF_SB API is issued for CHIRP_ADC_START_TIME_VAR, it is recommended to program SF3_CHIRP_PARAM_DELTA to 0. Only LUT based dither can be used for CHIRP_ADC_START_TIME_VAR. Refer to the Fixed Delta Chirp Parameter LUT description Table 5.50 for the definition of this field when each parameter is selected. This feature is enabled only for certain chirp parameter types as defined in this table.								
RESERVED	4	RESERVED								
LUT_RESET_PERIOD (J)	2	Reset the LUT sequence (LUT Dither) every J chirps Valid range: 0 – 32768 <table><thead><tr><th>Value</th><th>Definition</th></tr></thead><tbody><tr><td>0</td><td>Reset only as per RESET MODE option</td></tr><tr><td>1</td><td>Fixed 0th indexed LUT value programmed for all chirps</td></tr><tr><td>32768-2</td><td>Reset every J chirps in addition to RESET MODE option</td></tr></tbody></table> The reset period should be integer multiple of LUT_PARAM_UPDATE_PERIOD (K)	Value	Definition	0	Reset only as per RESET MODE option	1	Fixed 0th indexed LUT value programmed for all chirps	32768-2	Reset every J chirps in addition to RESET MODE option
Value	Definition									
0	Reset only as per RESET MODE option									
1	Fixed 0th indexed LUT value programmed for all chirps									
32768-2	Reset every J chirps in addition to RESET MODE option									

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Table 5.49 – continued from previous page

LUT_PARAM_UPDATE_PERIOD (K)	2	The chirp parameter (LUT Dither) will be updated with new value from LUT every K chirps. Valid range: 0 – 16384 <table><thead><tr><th>Value</th><th>Definition</th></tr></thead><tbody><tr><td>0</td><td>Fixed 0th indexed LUT value programmed for all chirps</td></tr><tr><td>16384-1</td><td>Index to LUT will be incremented once after every K chirps and corresponding LUT value is used.</td></tr></tbody></table>	Value	Definition	0	Fixed 0th indexed LUT value programmed for all chirps	16384-1	Index to LUT will be incremented once after every K chirps and corresponding LUT value is used.
Value	Definition							
0	Fixed 0th indexed LUT value programmed for all chirps							
16384-1	Index to LUT will be incremented once after every K chirps and corresponding LUT value is used.							
LUT_PATTERN_ADDRESS_OFFSET	2	This field provides the start address offset within the Generic SW Chirp Parameter LUT which holds dither parameters (LUT Dither) for this CHIRP_PARAM_INDEX. The first chirp of the burst/frame picks the dither from 0th index to LUT with this address offset and dithers for next chirps will be derived based on pattern configuration defined in this API. Address offset has to be multiple 4 bytes (word boundary) The Generic SW chirp parameters are described in Table 5.52 and it can be loaded in to LUT using AWR_ADVANCE_CHIRP_GENERIC_LUT_LOAD_SB API.						
NUM_OF_PATTERNS (P)	2	This field provides the number of unique dither parameters present in LUT (LUT Dither). Valid range: 1 to 8192 0 is not a valid number This information is used to perform array out of bound error check on index to LUT in FW.						

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Table 5.49 – continued from previous page

BURST_LUT_INDEX_OFFSET	2	Only relevant when using Advanced Frame Config API. Provides flexibility to have an offset in index to LUT (LUT Dither) from one burst to the next burst. This field provides the LUT index start offset for subsequent bursts in advanced frame config API. The chirp LUT start index for each burst is determined as the chirp LUT start index of the previous burst plus BURST_LUT_INDEX_OFFSET. This feature helps to loop set of different chirps in subsequent bursts in a sub-frame. Valid Range: 0 to P 0 – No offset (default) 1 to P – LUT index start offset for each burst. NOTE1: The first burst in second or higher sub-frame is always indexing to SF_LUT_INDEX_OFFSET parameter in LUT. NOTE2: The LUT_RESET_PERIOD can not be more than number of chirps in a burst if this feature is used.
SF_LUT_INDEX_OFFSET	2	Only relevant when using Advanced Frame Config API. Provides flexibility to have an offset in index to LUT (LUT Dither) from one subframe to the next subframe. This field provides the LUT index start offset for subsequent sub-frames in advanced frame config. The chirp LUT start index for first burst in each SF is determined as the chirp LUT start index of the previous SF plus SF_LUT_INDEX_OFFSET. This feature helps to loop set of different chirps in subsequent sub-frames. Valid Range: 0 to P 0 – No offset (default) 1 to P - LUT index start offset for each sub-frame (SF). NOTE1: The first SF in advance frame is always indexing to 0th parameter in LUT. NOTE2: The LUT_RESET_PERIOD can not be more than number of chirps in a sub-frame if this feature is used.

Continued on next page

Table 5.49 – continued from previous page

LUT_CHIRP_PARAM_SIZE	1	This field is applicable only for LUT chirp parameters (LUT Dither) of type CHIRP_FREQ_START_VAR, CHIRP_IDLE_TIME_VAR and CHIRP_ADC_START_TIME_VAR. This feature can be used to reduce the size of the parameter in LUT if dynamic range of the parameter is small. Valid Range: 0 to 2 <table><tr><td>CHIRP_PARAM_INDEX type</td><td>value 0</td><td>value 1</td><td>value 2</td></tr><tr><td>CHIRP_FREQ_START_VAR</td><td>4 bytes</td><td>2 bytes</td><td>1 byte</td></tr><tr><td>CHIRP_IDLE_TIME_VAR</td><td>2 bytes</td><td>1 byte</td><td>-</td></tr><tr><td>CHIRP_ADC_START_TIME_VAR</td><td>2 bytes</td><td>1 byte</td><td>-</td></tr></table> Default Value: 0 (default size) NOTE: If ADC_START_TIME_RES is set to 1 in AWR_RF_MISC_CTL_SB, LUT_CHIRP_PARAM_SIZE should always be set to 2 bytes in AWR_ADVANCE_CHIRP_CONF_SB API when it is issued for CHIRP_ADC_START_TIME_VAR.	CHIRP_PARAM_INDEX type	value 0	value 1	value 2	CHIRP_FREQ_START_VAR	4 bytes	2 bytes	1 byte	CHIRP_IDLE_TIME_VAR	2 bytes	1 byte	-	CHIRP_ADC_START_TIME_VAR	2 bytes	1 byte	-
CHIRP_PARAM_INDEX type	value 0	value 1	value 2															
CHIRP_FREQ_START_VAR	4 bytes	2 bytes	1 byte															
CHIRP_IDLE_TIME_VAR	2 bytes	1 byte	-															
CHIRP_ADC_START_TIME_VAR	2 bytes	1 byte	-															
LUT_CHIRP_PARAM_SCALE	1	This field is applicable only for LUT chirp parameters (LUT Dither) of type CHIRP_FREQ_START_VAR, CHIRP_IDLE_TIME_VAR and CHIRP_ADC_START_TIME_VAR. This feature can be used to reduce the size of the parameter in LUT if granularity of the resolution can be increased. Valid Range: 0 to 16 The actual parameter value for the defined chirp is given by: $2^{SCALE} * LUT_PARAM_VALUE$ Default Value: 0 (no scale) NOTE: If ADC_START_TIME_RES is set to 1 in AWR_RF_MISC_CTL_SB, LUT_CHIRP_PARAM_SCALE should be programmed to 0 when AWR_ADVANCE_CHIRP_CONF_SB API is issued for CHIRP_ADC_START_TIME_VAR. Refer to the corresponding rows in the Generic SW Chirp Parameter LUT description Table 5.52 for more info.																

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Table 5.49 – continued from previous page

MAX_TX_ PHASE_ SHIFTER_ INTERNAL_ DITHER	2	This field is applicable only if SFn_CHIRP_PARAM_DELTA increment (Delta Dither) is enabled for TXn_PHASE_SHIFTER parameter. It controls the TX phase quantization process. The device's internal TX phase shifters are 6 bit. For deriving the internal 6 bit phase, the 16 bit SFn_CHIRP_PARAM_DELTA is accumulated in the firmware every chirp. The accumulator's output is added with a random number from 0 to this field's value. The 6 MSBs of the adder's output are used as the internal 6 bit phase for that chirp. Valid Range: 0 to 4096 Default Value: 0 (no dither)
RESERVED	8	RESERVED

Fixed Delta Chirp Parameter description table:

Here is the description of SFn_CHIRP_PARAM_DELTA for each relevant parameter in Advanced Chirp Config API. This fixed delta is being incremented every N chirps as per update period defined in this API and the start value of the accumulator is 0 for first chirp. The accumulated delta is being added to LUT dither value and to the profile config setting in HW. This fixed delta increment feature helps to reduce the need of dedicated chirp RAM for each chirp if pattern can be generated in fixed increment fashion for each chirp.

Table 5.50: ADV_CHIRP_FIXED_DELTA_PARAM description

Parameter	LSB definition	Description
CHIRP_FREQ_START_VAR	For 77GHz Devices (76GHz to 81GHz): 1 LSB = $3.6e9/2^{26}$ Hz $\approx$ 53.644 Hz Signed Valid Range: -0x058E38E3 to 0x058E38E3 +/-5GHz range (Depending on max range of VCO) For 60GHz Devices (57GHz to 64GHz): 1 LSB = $2.7e9/2^{26}$ $\approx$ 40.23 Hz Signed Valid Range: -0x05ED097B to 0x05ED097B +/-4GHz range (Depending on max range of VCO)	The start frequency dither fixed delta increment value. This field is signed and has higher dynamic range compared to legacy chirp configuration API. Limitations: If accumulated delta dither + LUT dither value for a chirp is negative or $\geq$ 450MHz then Fw internally has to update the start freq of the profile of corresponding chirp dynamically in HW and this leads to limitation as mentioned below. If accumulated delta dither + LUT dither value is negative or $\geq$ +/-450MHz for 2nd chirp in a burst then due to Hw limitation the both first and 2nd chirps will be programmed with same start frequency i.e accumulated delta + LUT dither belongs to 2nd chirp. The chirps beyond 1st chirp will have proper start frequency programmed as expected, the 1st chirp start frequency would be bad, so it is recommended to discard the first chirp. NOTE: The Profile start freq + Accumulated delta dither + LUT dither for a chirp should not exceed the VCO range. NOTE: If GLOBAL_RESET_MODE is set to 0 (end of frame) or 1 (end of sub-frame) then above limitation is applicable at start of each burst of a sub-frame.

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Table 5.50 – continued from previous page

CHIRP_FREQ_ SLOPE_VAR	For 77GHz Devices (76GHz to 81GHz): $1 \text{ LSB} = 3.6e9 \times 900/2^{26} \approx 48.279 \text{ kHz Signed}$ Valid Range: -63 to 63 +/-3MHz/us range For 60GHz Devices (57GHz to 64GHz): $1 \text{ LSB} = 2.7e9 \times 900/2^{26} \approx 36.21 \text{ kHz Signed}$ Valid Range: -63 to 63 +/-2.28MHz/us range	The slope dither fixed delta increment value. This field is signed Limitations: If accumulated delta dither + LUT dither value for a chirp is negative or $\geq 3\text{MHz/us}$ then Fw internally has to update the slope of the profile of corresponding chirp dynamically in HW and this leads to limitation as mentioned below. If accumulated delta dither + LUT dither value is negative or $\geq +/-3\text{MHz}$ for 2nd chirp in a burst then due to Hw limitation the both first and 2nd chirps will be programmed with same slope i.e accumulated delta dither + LUT dither belongs to 2nd chirp. The chirps beyond 1st chirp will have proper slope programmed as expected, the 1st chirp slope would be bad, so it is recommended to discard the first chirp. NOTE: The Profile slope + Accumulated delta dither + LUT dither for a chirp should not exceed the max slope range of VCO. NOTE: If GLOBAL_RESET_MODE is set to 0 (end of frame) or 1 (end of sub-frame) then above limitation is applicable at start of each burst of a sub-frame.
CHIRP_IDLE_ TIME_VAR	1 LSB = 10 ns, unsigned Valid range: 0 to 4095 0 to 40.95us range	The idle time dither fixed delta increment value. This field is unsigned. NOTE: The Accumulated delta dither + LUT dither for a chirp should not exceed the max idle time dither value 40.95us (4095 value).

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Table 5.50 – continued from previous page

CHIRP_ADC_START_TIME_VAR	1 LSB = 10 ns, unsigned Valid range: 0 to 4095 0 to 40.95us range	The ADC start time dither fixed delta increment value. This field is unsigned. NOTE: The Accumulated delta dither + LUT dither for a chirp should not exceed the max ADC start time dither value 40.95us (4095 value). NOTE: If ADC_START_TIME_RES is set to 1 in AWR_RF_MISC_CTL_SB, only LUT based dither can be used when AWR_ADVANCE_CHIRP_CONF_SB API is issued for CHIRP_ADC_START_TIME_VAR. The chirp delta parameters specified below should be programmed to 0: - SF0_CHIRP_PARAM_DELTA, - SF1_CHIRP_PARAM_DELTA, - SF2_CHIRP_PARAM_DELTA, - SF3_CHIRP_PARAM_DELTA, - DELTA_PARAM_UPDATE_PERIOD in AWR_ADVANCE_CHIRP_CONF_SB API.
TX0_PHASE_SHIFTER	1 LSB = $360^\circ/2^{16}$ = 0.005493°, unsigned Valid range: 0 to 65535 0 to 360° range	The per chirp TX0 phase shifter fixed delta increment value. This field is unsigned and has finer resolution compared to LUT per chirp dither value. Please review the notes below for more information on the phase resolution supported in hardware and how it relates to delta increment values.
TX1_PHASE_SHIFTER	1 LSB = $360^\circ/2^{16}$ = 0.005493°, unsigned Valid range: 0 to 65535 0 to 360° range	The per chirp TX1 phase shifter fixed delta increment value. This field is unsigned and has finer resolution compared to LUT per chirp dither value. Please review the notes below for more information on the phase resolution supported in hardware and how it relates to delta increment values.

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Table 5.50 – continued from previous page

TX2_PHASE_SHIFTER	1 LSB = $360^\circ/2^{16}$ = 0.005493°, unsigned Valid range: 0 to 65535 0to360° range	The per chirp TX2 phase shifter fixed delta increment value. This field is unsigned and has finer resolution compared to LUT per chirp dither value. Please review the notes below for more information on the phase resolution supported in hardware and how it relates to delta increment values.
TX3_PHASE_SHIFTER	1 LSB = $360^\circ/2^{16}$ = 0.005493°, unsigned Valid range: 0 to 65535 0to360° range	The per chirp TX3 phase shifter fixed delta increment value. This field is unsigned and has finer resolution compared to LUT per chirp dither value. Please review the notes below for more information on the phase resolution supported in hardware and how it relates to delta increment values.

NOTE1:	If fixed delta dither is used to generate the pattern then it is recommended to program same start frequency in profile config API for each chirps in a frame. Each chirp can have different profiles associated with it except start frequency.
NOTE2:	The number of chirps programmed in a burst/frame shall be multiple of 4. Exception: a single chirp can be programmed in a burst.
NOTE3:	The device's internal TX phase shifters are 6 bit (5.625° resolution). For deriving the internal 6 bit phase, the delta increment value is accumulated in the firmware every chirp. 6 MSBs of the accumulator's output is used as the 6 bit phase for that chirp.

5.5.22 Sub block 0x0116 – AWR_ADVANCE_CHIRP_GENERIC_LUT_LOAD_SB

This API sub-block loads the unique dither values for each chirp on Generic SW Chirp Parameter LUT at offset address defined in [AWR_ADVANCE_CHIRP_CONF_SB](#) API. This LUT can be used to pre-load dither patterns for each chirp parameters and provides the flexibility to program any number of unique dithers for each chirp parameters.

NOTE1:	The Generic SW Chirp Parameter LUT can be modified by the host dynamically, as long as the LUT addresses modified differ from the addresses used in the current frame.
NOTE2:	The dynamic update of this API is effective immediately and does not depend on AWR_DYN_CHIRP_ENABLE_SB API. This might impact the ongoing chirps if timing of the update is not handled properly as if ongoing chirps use same fields/addresses in LUT. It is recommended to perform proper timing analysis before updating the LUT dynamically considering SPI communication delays.
NOTE3:	The total size of Generic SW Chirp Parameter LUT is 12kB.
NOTE4:	The start address offset of all chirp parameter in LUT shall be multiple of 4 bytes (word boundary), that means minimum 4 bytes in LUT shall be allocated to each chirp parameter.
NOTE5:	At least one dither parameter value shall be programmed for each chirp parameter type (10 types) in generic LUT, same value can be programmed to all chirps in a burst/frame using Advance chirp config API, LUT_PARAM_UPDATE_PERIOD (K) = 0 configuration.

Table [5.51](#) describes the contents of this sub block.

Table 5.51: AWR_ADVANCE_CHIRP_GENERIC_LUT_LOAD_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0116
SBLKLEN	2	Value = 232
RESERVED	12	0x0000
LUT_ADDRESS_OFFSET	2	Start address offset in LUT at which to populate the bytes of patterns. Address offset has to be multiple 4 bytes (word boundary)
NUM_OF_BYTES	2	Number of valid bytes to load in LUT Valid range: 4 to 212 bytes, must be multiple of 4 bytes.
DATA_BYTES	212	Byte array to load in to the Generic SW Chirp Parameter LUT. The description and size of the chirp parameters defined in Table 5.52 below. NOTE: The size of this sub-block is fixed to total 232 bytes, hence it is recommended to group multiple chirp parameters and send in chunks.

Generic SW Chirp Parameter LUT parameters description table:

Here is the description of chirp parameter dithers which are programmed at LUT_PATTERN_ADDRESS_OFFSET address in Generic LUT defined in Advanced Chirp Config API. The index

to this LUT is being incremented every K chirps as per update period defined in Advanced Chirp Config API and the index to LUT is 0 (at offset address) for first chirp. This LUT dither is being added to accumulated delta value and to the profile config setting in HW. This generic LUT helps to program unique dithers in device chirp RAM only for certain chirp parameters based on wave-form generation need, there is no need to program the dithers for chirp parameters which are not required to be dithered unlike legacy AWR_CHIRP_CONF_SET_SB API.

Table 5.52: ADV_CHIRP_GENERIC_LUT_PARAM description

Parameter	LSB definition and Size	Description
CHIRP_PROFILE_SELECT	Valid Range: 0 to 6 Size: 4 bits for each parameter Min Size in LUT: 4 Bytes (Up to 8 parameters)	Each byte can hold profile index parameter for 2 chirps in LUT. Index 0 and 1 refer to the first and second parameters in LUT. Bit Field b3:0 0th Profile index parameter in LUT (Mandatory field) b7:4 1st Profile index parameter in LUT (optional - in case 0th fixed profile wants to be used for all chirps in a burst/frame)

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Table 5.52 – continued from previous page

CHIRP_FREQ_ START_VAR	For 77GHz Devices (76GHz to 81GHz): 1 LSB = $3.6e9/2^{26} * 2^{SCALE}$ Hz Signed Valid Range: -0x058E38E3 to 0x058E38E3 +/-5GHz range (Depending on max range of VCO) For 60GHz Devices (57GHz to 64GHz): 1 LSB = $2.7e9/2^{26} * 2^{SCALE}$ Hz Signed Valid Range: -0x05ED097B to 0x05ED097B +/-4GHz range (Depending on max range of VCO) Size: 1 or 2 or 4 Bytes Configurable based on LUT_CHIRP_PARAM_SIZE defined in Advanced chirp config API. Min Size in LUT: 4 Bytes (Up to 4 parameters in case size is 1 byte) Scale: 0 to 16 Configurable based on LUT_CHIRP_PARAM_SCAL defined in Advanced chirp config API.	The start frequency dither value for each chirp to be added to the profile's start frequency. This value is signed and has higher dynamic range compared to legacy chirp configuration API. Limitations: If accumulated delta + LUT dither value for a chirp is -ve or $\geq 450\text{MHz}$ then Fw internally has to update the start freq of the profile of corresponding chirp dynamically in HW and this leads to limitation as mentioned below. If accumulated delta + LUT dither value is -ve or $\geq \pm 450\text{MHz}$ for 2nd chirp in a burst then due to Hw limitation the both first and 2nd chirps will be programmed with same start frequency i.e accumulated delta + LUT dither belongs to 2nd chirp. The chirps beyond 1st chirp will have proper start frequency as expected, so it is recommended to discard the first chirp in this case. NOTE: The Profile start freq + Accumulated delta + LUT dither for a chirp should not exceed the VCO range.
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Table 5.52 – continued from previous page

CHIRP_FREQ_ SLOPE_VAR	For 77GHz Devices (76GHz to 81GHz): $1 \text{ LSB} = 3.6e9 \times 900/2^{26} \approx 48.279 \text{ kHz Signed}$ Valid Range: -63 to 63 +/-3MHz/us range For 60GHz Devices (57GHz to 64GHz): $1 \text{ LSB} = 2.7e9 \times 900/2^{26} \approx 36.21 \text{ kHz Signed}$ Valid Range: -63 to 63 +/-2.28MHz/us range Size: 1 byte Min Size in LUT: 4 Bytes (Up to 4 parameters)	The slope dither value. This value is signed Limitations: If accumulated delta + LUT dither value for a chirp is -ve or $\geq 3\text{MHz/us}$ then Fw internally has to update the slope of the profile of corresponding chirp dynamically in HW and this leads to limitation as mentioned below. If accumulated delta + LUT dither value is -ve or $\geq \pm 3\text{MHz}$ for 2nd chirp in a burst then due to Hw limitation the both first and 2nd chirps will be programmed with same slope i.e accumulated delta + LUT dither belongs to 2nd chirp. The chirps beyond 1st chirp will have proper slope as expected, so it is recommended to discard the first chirp in this case. NOTE: The Profile slope + Accumulated delta + LUT dither for a chirp should not exceed the max slope range of VCO.
CHIRP_IDLE_ TIME_VAR	$1 \text{ LSB} = 10ns * 2^{SCALE}$, unsigned Valid range: 0 to 4095 0 to 40.95us range Size: 1 or 2 Bytes Configurable based on LUT_CHIRP_PARAM_SIZE defined in Advanced chirp config API. Min Size in LUT: 4 Bytes (Up to 4 parameters in case size is 1 byte) Scale: 0 to 8 Configurable based on LUT_CHIRP_PARAM_SCAL defined in Advanced chirp config API.	The idle time dither value. This value is unsigned. NOTE: The Accumulated delta dither + LUT dither for a chirp should not exceed the max idle time dither value 40.95us (4095 value).

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Table 5.52 – continued from previous page

CHIRP_ADC_START_TIME_VAR	If ADC_START_TIME_RES is set to 0 (10ns) in AWR_ADVANCE_CHIRP_CONF_SB 1 LSB = $10ns * 2^{SCALE}$, unsigned Valid range: 0 to 4095 0 to 40.95us range Size: 1 or 2 Bytes Configurable based on LUT_CHIRP_PARAM_SIZE defined in Advanced chirp config API. Min Size in LUT: 4 Bytes (Up to 4 parameters in case size is 1 byte) Scale: 0 to 8 Configurable based on LUT_CHIRP_PARAM_SCAL defined in Advanced chirp config API. If ADC_START_TIME_RES is set to 1 (1ns) in AWR_ADVANCE_CHIRP_CONF_SB 1 LSB = 1 ns for bits [15:12], unsigned 1 LSB = 10 ns for bits [11:0], unsigned Valid range: 0 to 4095 for bits [11:0] Valid range: 0 to 9 for bits [15:12] 0 to 40.959us range Size: 2 Bytes Min Size in LUT: 4 Bytes Scale: 0 Scaling is not supported with 1ns resolution	The ADC start time dither value. This value is unsigned. If ADC_START_TIME_RES is set to 0 (10ns) in AWR_ADVANCE_CHIRP_CONF_SB NOTE: The Accumulated delta dither + LUT dither for a chirp should not exceed the max ADC start time dither value 40.95us (4095 value). If ADC_START_TIME_RES is set to 1 (1ns) in AWR_ADVANCE_CHIRP_CONF_SB Adc start time is computed using values programmed in bits [11:0] and bits [15:12] Resolution for bits [11:0] is 10 ns and for bits [15:12] is 1ns ADC Start Time value in ns = CHIRP_ADC_START_TIME_VAR[11:0] * 10ns + CHIRP_ADC_START_TIME_VAR[15:12] * 1ns NOTE: The Accumulated delta dither + LUT dither for a chirp should not exceed the max ADC start time dither value 40.959us (40959 value).
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Table 5.52 – continued from previous page

CHIRP_TX_EN	Valid Range: 0 to 15	Each byte can hold TX enable mask parameter for 2 chirps in LUT. Index 0 and 1 refer to the first and second parameters in LUT.	
	Size: 4 bits for each parameter	Bit	Parameter
	Min Size in LUT: 4 Bytes (Up to 8 parameters)	Field	
		b0	TX0 enable mask for 0th parameter in LUT (Mandatory field)
		b1	TX1 enable mask for 0th parameter in LUT (Mandatory field)
		b2	TX2 enable mask for 0th parameter in LUT (Mandatory field)
		b3	TX3 enable mask for 0th parameter in LUT (Mandatory field)
		b4	TX0 enable mask for 1st parameter in LUT (optional)
		b5	TX1 enable mask for 1st parameter in LUT (optional)
		b6	TX2 enable mask for 1st parameter in LUT (optional)
	b7	TX3 enable mask for 1st parameter in LUT (optional)	

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Table 5.52 – continued from previous page

CHIRP_BPM_VAL	Valid Range: 0 to 15 Size: 4 bits for each parameter Min Size in LUT: 4 Bytes (Up to 8 parameters)	Each byte can hold TX BPM value parameter for 2 chirps in LUT. Index 0 and 1 refer to the first and second parameters in LUT. <table><tr><th>Bit Field</th><th>Parameter</th></tr><tr><td>b0</td><td>TX0 BPM value for 0th parameter in LUT (Mandatory field)</td></tr><tr><td>b1</td><td>TX1 BPM value for 0th parameter in LUT (Mandatory field)</td></tr><tr><td>b2</td><td>TX2 BPM value for 0th parameter in LUT (Mandatory field)</td></tr><tr><td>b3</td><td>TX3 BPM value for 0th parameter in LUT (Mandatory field)</td></tr><tr><td>b4</td><td>TX0 BPM value for 1st parameter in LUT (optional)</td></tr><tr><td>b5</td><td>TX1 BPM value for 1st parameter in LUT (optional)</td></tr><tr><td>b6</td><td>TX2 BPM value for 1st parameter in LUT (optional)</td></tr><tr><td>b7</td><td>TX3 BPM value for 0th parameter in LUT (Mandatory field)</td></tr></table>	Bit Field	Parameter	b0	TX0 BPM value for 0th parameter in LUT (Mandatory field)	b1	TX1 BPM value for 0th parameter in LUT (Mandatory field)	b2	TX2 BPM value for 0th parameter in LUT (Mandatory field)	b3	TX3 BPM value for 0th parameter in LUT (Mandatory field)	b4	TX0 BPM value for 1st parameter in LUT (optional)	b5	TX1 BPM value for 1st parameter in LUT (optional)	b6	TX2 BPM value for 1st parameter in LUT (optional)	b7	TX3 BPM value for 0th parameter in LUT (Mandatory field)
Bit Field	Parameter																			
b0	TX0 BPM value for 0th parameter in LUT (Mandatory field)																			
b1	TX1 BPM value for 0th parameter in LUT (Mandatory field)																			
b2	TX2 BPM value for 0th parameter in LUT (Mandatory field)																			
b3	TX3 BPM value for 0th parameter in LUT (Mandatory field)																			
b4	TX0 BPM value for 1st parameter in LUT (optional)																			
b5	TX1 BPM value for 1st parameter in LUT (optional)																			
b6	TX2 BPM value for 1st parameter in LUT (optional)																			
b7	TX3 BPM value for 0th parameter in LUT (Mandatory field)																			
TX0_PHASE_SHIFTER	1 LSB = $360^{\circ}/2^6 = 5.625^{\circ}$, unsigned Valid range: 0 to 63 0to360° range Size: 1 byte Min Size in LUT: 4 Bytes (Up to 4 parameters)	The per chirp TX0 phase shifter value. This value is unsigned <table><tr><th>Bits</th><th>TX0 phase shift definition</th></tr><tr><td>b1:0</td><td>RESERVED (set it to 0b00)</td></tr><tr><td>b7:2</td><td>TX0 phase shift value</td></tr></table>	Bits	TX0 phase shift definition	b1:0	RESERVED (set it to 0b00)	b7:2	TX0 phase shift value												
Bits	TX0 phase shift definition																			
b1:0	RESERVED (set it to 0b00)																			
b7:2	TX0 phase shift value																			
TX1_PHASE_SHIFTER	1 LSB = $360^{\circ}/2^6 = 5.625^{\circ}$, unsigned Valid range: 0 to 63 0to360° range Size: 1 byte Min Size in LUT: 4 Bytes (Up to 4 parameters)	The per chirp TX1 phase shifter value. This value is unsigned <table><tr><th>Bits</th><th>TX1 phase shift definition</th></tr><tr><td>b1:0</td><td>RESERVED (set it to 0b00)</td></tr><tr><td>b7:2</td><td>TX1 phase shift value</td></tr></table>	Bits	TX1 phase shift definition	b1:0	RESERVED (set it to 0b00)	b7:2	TX1 phase shift value												
Bits	TX1 phase shift definition																			
b1:0	RESERVED (set it to 0b00)																			
b7:2	TX1 phase shift value																			

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Table 5.52 – continued from previous page

TX2_PHASE_SHIFTER	1 LSB = $360^\circ/2^6 = 5.625^\circ$, unsigned Valid range: 0 to 63 0 to 360° range Size: 1 byte Min Size in LUT: 4 Bytes (Up to 4 parameters)	The per chirp TX2 phase shifter value. This value is unsigned Bits TX2 phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 TX2 phase shift value
TX3_PHASE_SHIFTER	1 LSB = $360^\circ/2^6 = 5.625^\circ$, unsigned Valid range: 0 to 63 0 to 360° range Size: 1 byte Min Size in LUT: 4 Bytes (Up to 4 parameters)	The per chirp TX3 phase shifter value. This value is unsigned Bits TX3 phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 TX3 phase shift value

Limitations:

Limitation 1:	The first chirp in a burst (AFC) or in a legacy frame shall be discarded due to Hw limitation in below cases: • If start frequency dither is negative for 2nd chirp of a burst/frame (either due to delta increment or due to LUT value). Workaround: program a small negative dither for 1st chirp in LUT. • If start frequency dither is $\geq \pm 450\text{MHz}$ for 2nd chirp of a burst/frame (either due to delta increment or due to LUT value). Workaround: Discard 1st chirp data. • If slope dither is negative for 2nd chirp of a burst/frame (either due to delta increment or due to LUT value). Workaround: program a small negative dither for 1st chirp in LUT. • If slope dither is $\geq \pm 3\text{MHz/us}$ for 2nd chirp of a burst/frame (either due to delta increment or due to LUT value). Workaround: Discard 1st chirp data.
Limitation 2:	The minimum chirp duration or cycle time shall be programmed as mentioned in section Chirp Cycle Time if advanced chirp feature is used.

5.5.23 Sub block 0x0117 – AWR_MONITOR_TYPE_TRIG_CONF_SB

This is only on **AWR2243, xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP** devices. This API helps to maintain monitoring timing synchronization in cascaded devices to avoid mutual interference of monitors running in different devices in the cascade sensor. The host must trigger the monitor of types below to avoid interference if MONITORING_MODE is set to '1' in [AWR_CALIB_MON_TIME_UNIT_CONF_SB](#). The monitors can be categorized into 3 types. The [AWR_AE_RF_MONITOR_TYPE_TRIGGER_DONE_SB](#) AE will be sent once monitor type is executed.

Table 5.53: Types of Monitors

Monitor Types	Description
Type 0	Non-transmitting monitor, The execution of non-transmitting monitors does not cause RF interference to monitors executing on other devices. Therefore, they can be executed in parallel across all devices in the cascade. These include monitors which receive a test signal through RX LNA and digital monitors.
Type 1	Transmitting but not receiving (test signal), The monitors that transmit but don't receive any test signal through RX LNA are not susceptible to interference. Therefore, they can be executed in parallel across all devices in the cascade, but not when monitors that receive test signals through RX LNA are executing.
Type 2	Transmitting and receiving (test signal), The monitors that transmit and also receive test signal through RX LNA are susceptible to interference. They can be executed sequentially so as to create time separation between monitoring chirps of different devices.

Table 5.54: Monitor Categorization

Monitor Type	Monitors																																				
Type 0	The run time digital monitors in AWR_MONITOR_RF_DIG_PERIODIC_CONF_SB <table> <tr> <th>Bit</th><th>Definition</th></tr> <tr> <td>b0</td><td>PERIODIC_CONFIG_REGISTER_READ_EN</td></tr> <tr> <td>b2</td><td>DFE_STC_EN</td></tr> <tr> <td>b3</td><td>FRAME_TIMING_MONITORING_EN</td></tr> </table> The analog monitors in AWR_MONITOR_ANALOG_ENABLES_CONF_SB <table> <tr> <th>Bit</th><th>Definition</th></tr> <tr> <td>b0</td><td>TEMPERATURE_MONITOR</td></tr> <tr> <td>b1</td><td>RX_GAIN_PHASE_MONITOR</td></tr> <tr> <td>b2</td><td>RX_NOISE_FIGURE_MONITOR</td></tr> <tr> <td>b3</td><td>RX_IFSTAGE_MONITOR</td></tr> <tr> <td>b14</td><td>SYNTH_FREQ_MONITOR</td></tr> <tr> <td>b15</td><td>EXTERNAL_ANALOG_SIGNALS_MONITOR</td></tr> <tr> <td>b19</td><td>INTERNAL_RX_SIGNALS_MONITOR</td></tr> <tr> <td>b20</td><td>INTERNAL_PMCLKLO_SIGNALS_MONITOR</td></tr> <tr> <td>b21</td><td>INTERNAL_GPADC_SIGNALS_MONITOR</td></tr> <tr> <td>b22</td><td>PLL_CONTROL_VOLTAGE_MONITOR</td></tr> <tr> <td>b23</td><td>DCC_CLOCK_FREQ_MONITOR</td></tr> <tr> <td>b24</td><td>RX_SATURATION_DETECTOR_MONITOR</td></tr> <tr> <td>b25</td><td>RX_SIG_IMG_BAND_MONITOR</td></tr> </table>	Bit	Definition	b0	PERIODIC_CONFIG_REGISTER_READ_EN	b2	DFE_STC_EN	b3	FRAME_TIMING_MONITORING_EN	Bit	Definition	b0	TEMPERATURE_MONITOR	b1	RX_GAIN_PHASE_MONITOR	b2	RX_NOISE_FIGURE_MONITOR	b3	RX_IFSTAGE_MONITOR	b14	SYNTH_FREQ_MONITOR	b15	EXTERNAL_ANALOG_SIGNALS_MONITOR	b19	INTERNAL_RX_SIGNALS_MONITOR	b20	INTERNAL_PMCLKLO_SIGNALS_MONITOR	b21	INTERNAL_GPADC_SIGNALS_MONITOR	b22	PLL_CONTROL_VOLTAGE_MONITOR	b23	DCC_CLOCK_FREQ_MONITOR	b24	RX_SATURATION_DETECTOR_MONITOR	b25	RX_SIG_IMG_BAND_MONITOR
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Bit	Definition																																				
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b1	RX_GAIN_PHASE_MONITOR																																				
b2	RX_NOISE_FIGURE_MONITOR																																				
b3	RX_IFSTAGE_MONITOR																																				
b14	SYNTH_FREQ_MONITOR																																				
b15	EXTERNAL_ANALOG_SIGNALS_MONITOR																																				
b19	INTERNAL_RX_SIGNALS_MONITOR																																				
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b23	DCC_CLOCK_FREQ_MONITOR																																				
b24	RX_SATURATION_DETECTOR_MONITOR																																				
b25	RX_SIG_IMG_BAND_MONITOR																																				
Type 1	The analog monitors in AWR_MONITOR_ANALOG_ENABLES_CONF_SB <table> <tr> <th>Bit</th><th>Definition</th></tr> <tr> <td>b4</td><td>TX0_POWER_MONITOR</td></tr> <tr> <td>b5</td><td>TX1_POWER_MONITOR</td></tr> <tr> <td>b6</td><td>TX2_POWER_MONITOR</td></tr> <tr> <td>b7</td><td>TX0_BALLBREAK_MONITOR</td></tr> <tr> <td>b8</td><td>TX1_BALLBREAK_MONITOR</td></tr> <tr> <td>b9</td><td>TX2_BALLBREAK_MONITOR</td></tr> <tr> <td>b16</td><td>INTERNAL_TX0_SIGNALS_MONITOR</td></tr> <tr> <td>b17</td><td>INTERNAL_TX1_SIGNALS_MONITOR</td></tr> <tr> <td>b18</td><td>INTERNAL_TX2_SIGNALS_MONITOR</td></tr> </table>	Bit	Definition	b4	TX0_POWER_MONITOR	b5	TX1_POWER_MONITOR	b6	TX2_POWER_MONITOR	b7	TX0_BALLBREAK_MONITOR	b8	TX1_BALLBREAK_MONITOR	b9	TX2_BALLBREAK_MONITOR	b16	INTERNAL_TX0_SIGNALS_MONITOR	b17	INTERNAL_TX1_SIGNALS_MONITOR	b18	INTERNAL_TX2_SIGNALS_MONITOR																
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b17	INTERNAL_TX1_SIGNALS_MONITOR																																				
b18	INTERNAL_TX2_SIGNALS_MONITOR																																				
Type 2	The analog monitors in AWR_MONITOR_ANALOG_ENABLES_CONF_SB <table> <tr> <th>Bit</th><th>Definition</th></tr> <tr> <td>b10</td><td>TX_GAIN_PHASE_MISMATCH_MONITOR</td></tr> <tr> <td>b11</td><td>TX0_PHASE_SHIFTER_MONITOR</td></tr> <tr> <td>b12</td><td>TX1_PHASE_SHIFTER_MONITOR</td></tr> <tr> <td>b13</td><td>TX2_PHASE_SHIFTER_MONITOR</td></tr> <tr> <td>b26</td><td>RX_MIXER_INPUT_POWER_MONITOR</td></tr> </table>	Bit	Definition	b10	TX_GAIN_PHASE_MISMATCH_MONITOR	b11	TX0_PHASE_SHIFTER_MONITOR	b12	TX1_PHASE_SHIFTER_MONITOR	b13	TX2_PHASE_SHIFTER_MONITOR	b26	RX_MIXER_INPUT_POWER_MONITOR																								
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b13	TX2_PHASE_SHIFTER_MONITOR																																				
b26	RX_MIXER_INPUT_POWER_MONITOR																																				

Table 5.55: AWR_MONITOR_TYPE_TRIG_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0117
SBLKLEN	2	Value = 12
MON_TRIG_TYPE_ENABLE	1	The bit mask for monitor trigger type to control sequence of execution of monitors. Bit Definition b0 Trigger Type 0 monitors b1 Trigger Type 1 monitors b2 Trigger Type 2 monitors b7:3 RESERVED 0: Disable 1: Enable
RESERVED	7	0x0

NOTE1:	The Host can trigger all 3 types of monitor at same time or can trigger each type one after other based on system requirement, in case host is triggering monitor types one after other, then it is recommended to follow order type 0, type 1 and type 2 respectively.
NOTE2:	The Host must wait for AWR_AE_RF_MONITOR_TYPE_TRIGGER_DONE_SB AE before issuing trigger for next monitor type.
NOTE3:	The Host must ensure all types of monitors are executed within defined device FTTI interval, otherwise device can not finish all the monitors within FTTI and will report failure AE AWR_CAL_MON_TIMING_FAIL_REPORT_AE_SB

5.5.24 Sub block 0x0118 – AWR_ADVANCE_CHIRP_DYN_LUT_ADDR_OFFSET_CFG_SB

This API sub-block can be used to configure LUT address offset dynamically for each chirp parameters defined in [AWR_ADVANCE_CHIRP_CONF_SB](#) API. This API helps to update only the LUT offset address for chirp parameters when LUT data is modified at frame boundary dynamically while frames are running. The dynamic chirp enable API AWR_DYN_CHIRP_ENABLE_SB shall be issued after issuing this API at least 500us before end of current active window of frame (500us before start of idle time of the frame) to apply the dynamic configurations in immediate

next frame.

Table 5.56 describes the contents of this sub block.

Table 5.56: AWR_ADVANCE_CHIRP_DYN_LUT_ADDR_OFFSET_CFG_SB
contents

Field Name	Number of bytes	Description																										
SBLKID	2	Value = 0x0118																										
SBLKLEN	2	Value = 40																										
ADDRESS_MASK_EN	2	Enable mask for LUT address offset dynamic update, the address is updated for following enabled chirp parameters. Value 1: Enable Value 0: Disable <table><tr><th>Bit</th><th>Definition</th></tr><tr><td>b0</td><td>Enable CHIRP_PROFILE_SELECT</td></tr><tr><td>b1</td><td>Enable CHIRP_FREQ_START_VAR</td></tr><tr><td>b2</td><td>Enable CHIRP_FREQ_SLOPE_VAR</td></tr><tr><td>b3</td><td>Enable CHIRP_IDLE_TIME_VAR</td></tr><tr><td>b4</td><td>Enable CHIRP_ADC_START_TIME_VAR</td></tr><tr><td>b5</td><td>Enable CHIRP_TX_EN</td></tr><tr><td>b6</td><td>Enable CHIRP_BPM_VAL</td></tr><tr><td>b7</td><td>Enable TX0_PHASE_SHIFTER</td></tr><tr><td>b8</td><td>Enable TX1_PHASE_SHIFTER</td></tr><tr><td>b9</td><td>Enable TX2_PHASE_SHIFTER</td></tr><tr><td>b10</td><td>Enable TX3_PHASE_SHIFTER</td></tr><tr><td>b15:11</td><td>RESERVED</td></tr></table>	Bit	Definition	b0	Enable CHIRP_PROFILE_SELECT	b1	Enable CHIRP_FREQ_START_VAR	b2	Enable CHIRP_FREQ_SLOPE_VAR	b3	Enable CHIRP_IDLE_TIME_VAR	b4	Enable CHIRP_ADC_START_TIME_VAR	b5	Enable CHIRP_TX_EN	b6	Enable CHIRP_BPM_VAL	b7	Enable TX0_PHASE_SHIFTER	b8	Enable TX1_PHASE_SHIFTER	b9	Enable TX2_PHASE_SHIFTER	b10	Enable TX3_PHASE_SHIFTER	b15:11	RESERVED
Bit	Definition																											
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b5	Enable CHIRP_TX_EN																											
b6	Enable CHIRP_BPM_VAL																											
b7	Enable TX0_PHASE_SHIFTER																											
b8	Enable TX1_PHASE_SHIFTER																											
b9	Enable TX2_PHASE_SHIFTER																											
b10	Enable TX3_PHASE_SHIFTER																											
b15:11	RESERVED																											
RESERVED	2	0x0000																										

Continued on next page

Table 5.56 – continued from previous page

LUT_ADDRESS_OFFSET	22	This field provides the start address offset within the Generic SW Chirp Parameter LUT which holds dither parameters (LUT Dither) for each chirp parameter, each address offset is 2 bytes. The address will be updated only if ADDRESS_MASK_EN is SET. Bytes Definition Bytes LUT_ADDRESS_OFFSET for CHIRP_PROFILE_1:0 SELECT Bytes LUT_ADDRESS_OFFSET for CHIRP_FREQ_3:2 START_VAR Bytes LUT_ADDRESS_OFFSET for CHIRP_FREQ_5:4 SLOPE_VAR Bytes LUT_ADDRESS_OFFSET for CHIRP_IDLE_7:6 TIME_VAR Bytes LUT_ADDRESS_OFFSET for CHIRP_ADC_9:8 START_TIME_VAR Bytes LUT_ADDRESS_OFFSET for CHIRP_TX_EN 11:10 Bytes LUT_ADDRESS_OFFSET for CHIRP_BPM_VAL 13:12 Bytes LUT_ADDRESS_OFFSET for TX0_PHASE_15:14 SHIFTER Bytes LUT_ADDRESS_OFFSET for TX1_PHASE_17:16 SHIFTER Bytes LUT_ADDRESS_OFFSET for TX2_PHASE_19:18 SHIFTER Bytes LUT_ADDRESS_OFFSET for TX3_PHASE_21:20 SHIFTER Address offset has to be multiple 4 bytes (word boundary)
RESERVED	10	RESERVED

5.5.25 Sub block 0x0119 – AWR_ADV_TX_GAIN_TEMPLUT_SET_SB

This is only applicable in xWR294x, xWR254x and xWR294xP devices. This API can be used to overwrite the TX gain temperature LUT for any TX channel used in firmware. This API should be issued after profile configuration API.

Table 5.57: AWR_ADV_TX_GAIN_TEMPLUT_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0119
SBLKLEN	2	Value = 44
PROFILE_INDX	1	This field indicates the profile Index for which this configuration applies
TX_INDEX	1	This field indicates the TX channel to which this configuration applies. 0 TX0 channel 1 TX1 channel 2 TX2 channel 3 TX3 channel
TX_GAIN_CODE	19	Byte0: TX gain code for temperature <-30 °C Byte1: TX gain code for temperature [-30, -20) °C Byte2: TX gain code for temperature [-20, -10) °C Byte3: TX gain code for temperature [-10, 0) °C Byte4: TX gain code for temperature [0, 10) °C Byte5: TX gain code for temperature [10, 20) °C Byte6: TX gain code for temperature [20, 30) °C Byte7: TX gain code for temperature [30, 40) °C Byte8: TX gain code for temperature [40, 50) °C Byte9: TX gain code for temperature [50, 60) °C Byte10: TX gain code for temperature [60, 70) °C Byte11: TX gain code for temperature [70, 80) °C Byte12: TX gain code for temperature [80, 90) °C Byte13: TX gain code for temperature [90, 100) °C Byte14: TX gain code for temperature [100, 110) °C Byte15: TX gain code for temperature [110, 120) °C Byte16: TX gain code for temperature [120, 130) °C Byte17: TX gain code for temperature [130, 140) °C Byte18: TX gain code for temperature ≥140 °C Each byte is encoded as follows Bits Definition b7:0 STG_CODE Higher values for higher gain
RESERVED	1	0x00

Continued on next page

Table 5.57 – continued from previous page

RESERVED	16	0x00
RESERVED	2	0x0000

5.5.26 Sub block 0x011A – AWR_ADV_DYN_PERCHIRP_PHASESHIFTER_CONF_SET_SB

This API can be used to dynamically change the per-chirp phase shifter configuration (applicable only in xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP) while frames are on-going. The configuration will be stored in software and the new configuration will be applied after receiving the AWR_DYN_CHIRP_ENABLE_SB API.

Table 5.58: AWR_ADV_DYN_PERCHIRP_PHASESHIFTER_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x011A
SBLKLEN	2	Value = 92
RESERVED	3	0x000000
CHIRP_SEGMENT_SELECT	1	Indicates the segment of the chirp RAM that the 16 chirp definitions in this sub block map to. Valid range 0 to 31
CHIRP1_TX0_PHASE_SHIFTER	1	TX0 phase shift value Bits TX0 phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 TX0 phase shift value 1 LSB = $360^\circ/2^6 = 5.625^\circ$ Valid range: 0 to 63
CHIRP1_TX1_PHASE_SHIFTER	1	TX1 phase shift value Bits TX1 phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 TX1 phase shift value 1 LSB = $360^\circ/2^6 = 5.625^\circ$ Valid range: 0 to 63

Continued on next page

Table 5.58 – continued from previous page

CHIRP1_ TX2_PHASE_ SHIFTER	1	TX2 phase shift value Bits TX2 phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 TX2 phase shift value 1 LSB = $360^\circ/2^6 = 5.625^\circ$ Valid range: 0 to 63
CHIRP1_ TX3_PHASE_ SHIFTER	1	TX3 phase shift value Bits TX3 phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 TX3 phase shift value 1 LSB = $360^\circ/2^6 = 5.625^\circ$ Valid range: 0 to 63
CHIRP2_ TX0_PHASE_ SHIFTER	1	See description for CHIRP1_TX0_PHASE_SHIFTER
CHIRP2_ TX1_PHASE_ SHIFTER	1	See description for CHIRP1_TX1_PHASE_SHIFTER
CHIRP2_ TX2_PHASE_ SHIFTER	1	See description for CHIRP1_TX2_PHASE_SHIFTER
CHIRP2_ TX3_PHASE_ SHIFTER	1	See description for CHIRP1_TX3_PHASE_SHIFTER
...	...	...
CHIRP16_ TX0_PHASE_ SHIFTER	1	See description for CHIRP1_TX0_PHASE_SHIFTER
CHIRP16_ TX1_PHASE_ SHIFTER	1	See description for CHIRP1_TX1_PHASE_SHIFTER
CHIRP16_ TX2_PHASE_ SHIFTER	1	See description for CHIRP1_TX2_PHASE_SHIFTER
CHIRP16_ TX3_PHASE_ SHIFTER	1	See description for CHIRP1_TX3_PHASE_SHIFTER
RESERVED	18	0x0

Continued on next page

Table 5.58 – continued from previous page

PROGRAM_ MODE	2	Bits	Description
		b0	Value Definition
		0	Program the new configuration when AWR_DYN_CHIRP_ENABLE API is issued
		1	Program the new configuration immediately NOTE: User has to ensure that the chirps which are being reconfigured are not the ones which are currently in use for chirping
		b15:1	RESERVED

5.6 Sub blocks related to AWR_RF_DYNAMIC_CONF_GET_SB

5.6.1 Sub block 0x0120 – AWR_PROFILE_CONF_GET_SB

This sub block reads the parameters of a given profile. The profile details are available as part of the acknowledgment. The structure is same as AWR_PROFILE_CONF_SET_SB
Table 5.59 describes the contents of this sub block.

Table 5.59: AWR_PROFILE_CONF_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0120
SBLKLEN	2	Value = 8
PROFILE_IDX	2	Valid range 0 to 3 Index of the profile which is to be read
RESERVED	2	0x0000

5.6.2 Sub block 0x0121 – AWR_CHIRP_CONF_GET_SB

This sub block reads the parameters of a given chirp. The profile details are available as part of the acknowledgement. The structure is same as AWR_CHIRP_CONF_SET_SB
Table 5.60 describes the contents of this sub block.

Table 5.60: AWR_CHIRP_CONF_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0121
SBLKLEN	2	Value = 8
CHIRP_START_INDXX	2	Valid range 0 to 511 Starting index of the chirp which is to be read
CHIRP_END_INDXX	2	Valid range 0 to 511 Ending index of the chirp which is to be read

5.6.3 Sub block 0x0122 – AWR_FRAME_CONF_GET_SB

This sub block reads the parameters of the configured frame. The profile details are available as part of the acknowledgement. The structure is same as AWR_FRAME_CONF_SET_SB
Table 5.61 describes the contents of this sub block.

Table 5.61: AWR_FRAME_CONF_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0122
SBLKLEN	2	Value = 4

5.6.4 Sub block 0x0123 – RESERVED

5.6.5 Sub block 0x0124 – RESERVED

5.6.6 Sub block 0x0125 – AWR_ADV_FRAME_CONF_GET_SB

This sub block reads the parameters of the configured frame. The profile details are available as part of the acknowledgement. The structure is same as AWR_ADVANCED_FRAME_CONF_SET_SB

Table 5.62 describes the contents of this sub block.

Table 5.62: AWR_ADV_FRAME_CONF_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0125

Continued on next page

Table 5.62 – continued from previous page

SBLKLEN	2	Value = 4
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5.6.7 Sub block 0x0126 – RESERVED
5.6.8 Sub block 0x0127 – RESERVED
5.6.9 Sub block 0x0128 – RESERVED
5.6.10 Sub block 0x0129 – RESERVED
5.6.11 Sub block 0x012A – RESERVED
5.6.12 Sub block 0x012B – RESERVED
5.6.13 Sub block 0x012C – AWR_RX_GAIN_TEMPLUT_GET_SB

This API is issued to read the temperature based RX gain LUT used by the firmware. This API should be issued after the profile configuration API. The acknowledgement packet sent in response to this API will contain the LUT. The structure is same as AWR_RX_GAIN_LUT_SET_SB.

Table 5.63: AWR_RX_GAIN_TEMPLUT_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x012C
SBLKLEN	2	Value = 8
PROFILE_INDX	1	Profile index for which the RX gain LUT is desired
RESERVED	3	0x000000

5.6.14 Sub block 0x012D – AWR_TX_GAIN_TEMPLUT_GET_SB

This API is issued to read the temperature based TX gain LUT used by the firmware. This API should be issued after the profile configuration API. The acknowledgement packet sent in response to this API will contain the LUT. The structure is same as AWR_TX_GAIN_LUT_SET_SB.

Table 5.64: AWR_TX_GAIN_TEMPLUT_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x012D
SBLKLEN	2	Value = 8
PROFILE_INDX	1	Profile index for which the TX gain LUT is desired
RESERVED	3	0x000000

5.6.15 Sub block 0x0139 – AWR_ADV_TX_GAIN_TEMPLUT_GET_SB

This API is issued to read the temperature based TX gain LUT (per TX channel) used by the firmware. This API should be issued after the profile configuration API. The acknowledgement packet sent in response to this API will contain the LUT. The structure is same as AWR_ADV_TX_GAIN_LUT_SET_SB.

Table 5.65: AWR_ADV_TX_GAIN_TEMPLUT_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0139
SBLKLEN	2	Value = 24
PROFILE_INDX	1	Profile index for which the TX gain LUT is desired
TX_INDEX	1	TX index for which the TX gain LUT is desired
RESERVED	16	0x000000
RESERVED	2	0x000000

5.7 Sub blocks related to AWR_FRAME_TRIG_MSG

5.7.1 Sub block 0x0140 – AWR_FRAMESTARTSTOP_CONF_SB

This sub block starts or stops transmission of frames.

Table 5.66 describes the contents of this sub block.

Table 5.66: AWR_FRAMESTARTSTOP_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0140
SBLKLEN	2	Value = 8
START_STOP_CMD	2	Value Definition 0x0000 Stop the transmission of frames after the current frame is over at frame boundary 0x0001 Trigger a frame in software triggered mode. In hardware SYNC_IN triggered mode, this command allows subsequent SYNC_IN trigger to be honored 0x0002 Stop the transmission of frames after the current sub-frame is over at sub-frame boundary 0x0003 Stop the transmission of frames after the current burst is over at burst boundary 0x0004 Stop the transmission of frames immediately which are waiting for HW trigger or sub-frame trigger (applicable only for HW/sub-frame triggered mode when active frames are not running)
RESERVED	2	0x0000

NOTE1:	When Frame Stop command with 'option-0' is sent to RadarSS, the frame will be stopped after completing all the chirps of a Frame/Advance frame.
NOTE2:	In non periodic Hw triggered mode or in sub-frame triggered mode, if frame needs to be stopped immediately then frame stop command with 'option-4' can be used. The 'option-4' can not be used when active frames are running.
NOTE3:	Recommended to re-issue frame configuration API if frame is not stopped at sub-frame boundary, this is to re-config CSI2 or LVDS data path configuration in MSS.

5.8 Sub blocks related to AWR_RF_ADVANCED_FEATURES_CONF_SET_MSG

5.8.1 Sub block 0x0180 – AWR_BPM_COMMON_CONF_SET_SB

This API sub block defines static configurations related to BPM (Binary Phase Modulation) feature in each of the TXs. E.g. the source of the BPM pattern (one constant value for each chirp as defined, or intra-chirp pseudo random BPM pattern as found by a programmable LFSR or a programmable sequence inside each chirp), are defined here.

Table 5.67 describes the contents of this sub block.

Table 5.67: AWR_BPM_COMMON_CONF_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0180
SBLKLEN	2	Value = 20
BPM_MODE_CFG	2	Bits Description b1:0 BPM_SRC_SEL (select source of BPM pattern) Value Definition 00 CHIRP_CONFIG_BPM (refer to AWR_BPM_CHIRP_CONF_SB) 01 RESERVED 10 RESERVED 11 RESERVED b15:2 RESERVED
RESERVED	2	0x0000
RESERVED	2	0x0000
RESERVED	2	0x0000
RESERVED	4	0x00000000
RESERVED	4	0x00000000

5.8.2 Sub block 0x0181 – AWR_BPM_CHIRP_CONF_SET_SB

This sub block defines static configurations related to BPM (Binary Phase Modulation) feature in each of the TXs.

Table 5.68 describes the contents of this sub block.

Table 5.68: AWR_BPM_CHIRP_CONF_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0181
SBLKLEN	2	Value = 12
CHIRP_START_INDXX	2	Start index of the chirp for configuring the constant BPM Valid range 0 to 511
CHIRP_END_INDXX	2	End index of the chirp for configuring the constant BPM Valid range 0 to 511
CONST_BPM_VAL	2	Bit Definition b0 RESERVED b1 CONST_BPM_VAL_TX0_TXON Value of Binary Phase Shift value for TX0, during chirp b2 RESERVED b3 CONST_BPM_VAL_TX1_TXON Value of Binary Phase Shift value for TX1, during chirp b4 RESERVED b5 CONST_BPM_VAL_TX2_TXON Value of Binary Phase Shift value for TX2, during chirp b6 RESERVED b7 CONST_BPM_VAL_TX3_TXON (only for xWR294x,xWR254x,xWR2x4xECO and xWR2x4xP) Value of Binary Phase Shift value for TX3, during chirp b15:8 RESERVED
RESERVED	2	0x0000

NOTE1: BPM values are configured using TX phase shifter and applied at TX_START_TIME.

5.8.3 Sub block 0x0182 – AWR_POWER_SAVE_MODE_CONF_SET_SB

This sub block defines the power saving modes and API configuration
Table 5.69 describes the contents of this sub block.

Table 5.69: AWR_POWER_SAVE_MODE_CONF_SET_SB contents

Field Name	Number of bytes	Description										
SBLKID	2	Value = 0x0182										
SBLKLEN	2	Value = 24										
LOWPOWER_STATE_TRANSITION_CMD	2	Low power state transition commands are defined as below <table><thead><tr><th>Mode</th><th>Definition</th></tr></thead><tbody><tr><td>0</td><td>RESERVED.</td></tr><tr><td>1</td><td>ENTER_RF_PWR_DOWN, device enters RF power down state from normal active state.</td></tr><tr><td>2</td><td>EXIT_RF_PWR_DOWN, device exits the RF power down state back to its previous state.</td></tr><tr><td>3</td><td>- RESERVED</td></tr></tbody></table> 65535 NOTE: Values other than 1 and 2 are not supported in xWR6243 device. It is not recommended to configure other values in this field, it can cause the device to be in undesirable state.	Mode	Definition	0	RESERVED.	1	ENTER_RF_PWR_DOWN, device enters RF power down state from normal active state.	2	EXIT_RF_PWR_DOWN, device exits the RF power down state back to its previous state.	3	- RESERVED
Mode	Definition											
0	RESERVED.											
1	ENTER_RF_PWR_DOWN, device enters RF power down state from normal active state.											
2	EXIT_RF_PWR_DOWN, device exits the RF power down state back to its previous state.											
3	- RESERVED											
RESERVED	2	RESERVED										
RESERVED	16	RESERVED										

NOTE 1: This API is only applicable to xWR6243 devices in this release.

NOTE 2: These low power state transitions are executed at the lowest priority in RADARSS. It is recommended to provide at least 1ms delay after issuing a AWR_POWER_SAVE_MODE_CONF_SET_SB API and before any other command is issued to the RADARSS. This will ensure that RADARSS has enough time to complete the previous transition before receiving other commands.

5.9 Sub blocks related to AWR_RF_STATUS_GET_MSG

5.9.1 Sub block 0x0220 – AWR_RF_VERSION_GET_SB

This sub block reads RF HW and FW versions. The information returned by the device will be in the format as given in AWR_RFVERSION_SB.

Table 5.70 describes the contents of the request sub block

Table 5.70: AWR_RF_VERSION_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0220
SBLKLEN	2	Value = 4

Response to AWR_RFVERSION_GET_SB

AWR_RFVERSION_SB sub block is sent by the radar device in response to AWR_RFVERSION_GET_SB. Note that SBLKID for both AWR_RFVERSION_GET_SB and AWR_RFVERSION_SB are same.

Table 5.71 describes the contents of the response sub block.

Table 5.71: AWR_RF_VERSION_SB response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0220
SBLKLEN	2	Value = 20
HW_VARIANT	1	HW variant number
HW_VERSION_MAJOR	1	HW version major number
HW_VERSION_MINOR	1	HW version minor number
BSS_FW_VERSION_MAJOR	1	BSS FW version major number
BSS_FW_VERSION_MINOR	1	BSS FW version minor number
BSS_FW_VERSION_BUILD	1	BSS FW version build number
BSS_FW_VERSION_DEBUG	1	BSS FW version debug number
BSS_FW_VERSION_YEAR	1	Year of BSS FW version release
BSS_FW_VERSION_MONTH	1	Month of BSS FW version release
BSS_FW_VERSION_DAY	1	Day of BSS FW version release

Continued on next page

Table 5.71 – continued from previous page

BSS_FW_VERSION_PATCH_MAJOR	1	BSS FW version patch major number	
BSS_FW_VERSION_PATCH_MINOR	1	BSS FW version patch minor number	
BSS_FW_VERSION_PATCH_YEAR	1	Year of BSS FW patch release	
BSS_FW_VERSION_PATCH_MONTH	1	Month of BSS FW patch release	
BSS_FW_VERSION_PATCH_DAY	1	Day of BSS FW patch release	
BSS_FW_PATCH_BUILD_DEBUG_VERSION	1	Bit	Definition
		b3:0	DEBUG version number
		b7:4	BUILD version number

5.9.2 Sub block 0x0221 – AWR_RF_CPUFAULT_STATUS_GET_SB

This sub block provides the RF BSS CPU fault information.

Table 5.72 describes the content of this sub block.

Table 5.72: AWR_RF_CPUFAULT_STATUS_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0221
SBLKLEN	2	Value = 4

AWR_RF_CPUFAULT_STATUS_SB is sent in response to AWR_RF_CPUFAULT_STATUS_GET_SB.

Table 5.73 describes the content of AWR_RF_CPUFAULT_STATUS_SB

Table 5.73: AWR_RF_CPUFAULT_STATUS_GET_SB response contents

Field Name	Number of bytes	Description																		
SBLKID	2	Value = 0x0221																		
SBLKLEN	2	Value = 36																		
FAULT_TYPE	1	<table><tr><th>Value</th><th>Definition</th></tr><tr><td>0</td><td>RF Processor Undefined Instruction Abort</td></tr><tr><td>1</td><td>RF Processor Instruction pre-fetch Abort</td></tr><tr><td>2</td><td>RF Processor Data Access Abort</td></tr><tr><td>3</td><td>RF Processor Firmware Fatal Error</td></tr><tr><td>0x4 - 0xFE</td><td>RESERVED</td></tr><tr><td>0xFF</td><td>No fault</td></tr></table>	Value	Definition	0	RF Processor Undefined Instruction Abort	1	RF Processor Instruction pre-fetch Abort	2	RF Processor Data Access Abort	3	RF Processor Firmware Fatal Error	0x4 - 0xFE	RESERVED	0xFF	No fault				
Value	Definition																			
0	RF Processor Undefined Instruction Abort																			
1	RF Processor Instruction pre-fetch Abort																			
2	RF Processor Data Access Abort																			
3	RF Processor Firmware Fatal Error																			
0x4 - 0xFE	RESERVED																			
0xFF	No fault																			
RESERVED	1	0x00																		
LINE_NUM	2	Valid only in case of FAULT type is 0x3, provides the firmware line number at which fatal error occurred.																		
FAULT_LR	4	The instruction PC address at which Fault occurred																		
FAULT_PREV_LR	4	The return address of the function from which fault function has been called (Call stack LR)																		
FAULT_SPSR	4	The CPSR register value at which fault occurred																		
FAULT_SP	4	The SP register value at which fault occurred																		
FAULT_CAUSE_ADDRESS	4	The address access at which Fault occurred (valid only for fault type 0x0 to 0x2)																		
FAULT_ERROR_STATUS	2	<table><tr><td colspan="2">The status of Error (Error Cause type – valid only for fault type 0x0 to 0x2)</td></tr><tr><td>0x000</td><td>BACKGROUND_ERR</td></tr><tr><td>0x001</td><td>ALIGNMENT_ERR</td></tr><tr><td>0x002</td><td>DEBUG_EVENT</td></tr><tr><td>0x00D</td><td>PERMISSION_ERR</td></tr><tr><td>0x008</td><td>SYNCH_EXTER_ERR</td></tr><tr><td>0x406</td><td>ASYNCH_EXTER_ERR</td></tr><tr><td>0x409</td><td>SYNCH_ECC_ERR</td></tr><tr><td>0x408</td><td>ASYNCH_ECC_ERR</td></tr></table>	The status of Error (Error Cause type – valid only for fault type 0x0 to 0x2)		0x000	BACKGROUND_ERR	0x001	ALIGNMENT_ERR	0x002	DEBUG_EVENT	0x00D	PERMISSION_ERR	0x008	SYNCH_EXTER_ERR	0x406	ASYNCH_EXTER_ERR	0x409	SYNCH_ECC_ERR	0x408	ASYNCH_ECC_ERR
The status of Error (Error Cause type – valid only for fault type 0x0 to 0x2)																				
0x000	BACKGROUND_ERR																			
0x001	ALIGNMENT_ERR																			
0x002	DEBUG_EVENT																			
0x00D	PERMISSION_ERR																			
0x008	SYNCH_EXTER_ERR																			
0x406	ASYNCH_EXTER_ERR																			
0x409	SYNCH_ECC_ERR																			
0x408	ASYNCH_ECC_ERR																			
FAULT_ERROR_SOURCE	1	<table><tr><td colspan="2">The Source of the Error (Error Source type - valid only for fault type 0x0 to 0x2)</td></tr><tr><td>0x0</td><td>ERR_SOURCE_AXI_MASTER</td></tr><tr><td>0x1</td><td>ERR_SOURCE_ATCM</td></tr><tr><td>0x2</td><td>ERR_SOURCE_BTCM</td></tr></table>	The Source of the Error (Error Source type - valid only for fault type 0x0 to 0x2)		0x0	ERR_SOURCE_AXI_MASTER	0x1	ERR_SOURCE_ATCM	0x2	ERR_SOURCE_BTCM										
The Source of the Error (Error Source type - valid only for fault type 0x0 to 0x2)																				
0x0	ERR_SOURCE_AXI_MASTER																			
0x1	ERR_SOURCE_ATCM																			
0x2	ERR_SOURCE_BTCM																			

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Table 5.73 – continued from previous page

FAULT_AXI_ERROR_TYPE	1	The AXI Error type (Error Source type - valid only for fault type 0x0 to 0x2) 0x0 AXI_DECOD_ERR 0x1 AXI_SLAVE_ERR
FAULT_ACCESS_TYPE	1	The Error Access type (Error Access type - valid only for fault type 0x0 to 0x2) 0x0 READ_ERR 0x1 WRITE_ERR
FAULT_RECOVERY_TYPE	1	The Error Recovery type (Error Recovery type - Valid only for fault type 0x0 to 0x2) 0x0 UNRECOVERY 0x1 RECOVERY
RESERVED	2	0x0000

5.9.3 Sub block 0x0222 – AWR_RF_ESMFAULT_STATUS_GET_SB

This sub block provides the information regarding additional RF sub system faults.

NOTE: This API has been deprecated on xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP devices. Please use the updated API - AWR_RF_ADV_ESMFAULT_STATUS_GET_SB.

Table 5.74 describes the content of this sub block.

Table 5.74: AWR_RF_ESMFAULT_STATUS_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0222
SBLKLEN	2	Value = 4

The response to above request is given in the AWR_RF_ESMFAULT_STATUS_SB.
Table 5.75 describes the contents of AWR_RF_ESMFAULT_STATUS_SB.

Table 5.75: AWR_RF_ESMFAULT_STATUS_SB response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0222
SBLKLEN	2	Value = 12
ESM_GROUP1_ERRORS	4	Bit Error Information 0 – No Error , 1 – ESM Error b0 RAMPGEN_SB_ERROR b1 RESERVED b2 GPADC_RAM_SB_ERROR b3 VIM_RAM_SB_ERROR b4 RESERVED b5 VIM_SELFTEST_ERROR b6 B0TCM_SB_ERROR b7 B1TCM_SB_ERROR b8 CCMR4_SELFTEST_ERROR b9 ATCM_SB_ERROR b10 RAMPGEN_SELFTEST_ERROR b11 RAMPGEN_PAR_SELFTST_ERROR b12 SEQ_EXT_SELFTEST_ERROR b13 SEQ_EXT_SB_ERROR b14 RESERVED b15 AGC_RAM_SB_ERROR b16 B1TCM_PAR_CHK_ERROR b17 B0TCM_PAR_CHK_ERROR b18 ATCM_PAR_CHK_ERROR b19 MB_MSS2BSS_SB_ERROR b20 MB_BSS2MSS_SB_ERROR b31:21 RESERVED

Continued on next page

Table 5.75 – continued from previous page

ESM_GROUP2_ ERRORS	4	Bit	Error Information
		b0	DFE_STC_ERROR
		b1	CR4_STC_ERROR
		b2	CCMR4_COMP_ERROR
		b3	B0TCM_DB_ERROR
		b4	B1TCM_DB_ERROR
		b5	ATCM_DB_ERROR
		b6	DCC_ERROR
		b7	SEQ_EXT_ERROR
		b8	SYNT_FREQ_MON_ERROR
		b9	RESERVED
		b10	RAMPGEN_DB_ERROR
		b11	BUBBLE_CORRECTION_FAIL
		b12	RAMPGEN_LOCSTEP_ERROR
		b13	RTI_RESET_ERROR
		b14	GPADC_RAM_DB_ERROR
		b15	VIM_COMP_ERROR
		b16	CR4_LIVE_LOCK_ERROR
		b17	WDT_NMI_ERROR
		b18	VIM_RAM_DB_ERROR
		b19	RAMPGEN_PAR_ERROR
		b20	SEQ_EXT_DB_ERROR
		b21	DMA_MPU_ERROR
		b22	AGC_RAM_DB_ERROR
		b23	CRC_COMP_ERROR
		b24	WAKEUP_STS_ERROR
		b25	SHORT_CIRCUIT_ERROR
		b26	B1TCM_PAR_ERROR
		b27	B0TCM_PAR_ERROR
		b28	ATCM_PAR_ERROR
		b29	MB_MSS2BSS_DB_ERROR
		b30	MB_BSS2MSS_DB_ERROR
		b31	CCC_ERROR

5.9.4 Sub block 0x0223 – AWR_RF_DIEID_GET_SB

This sub block provides the information regarding the Die ID of the device.

Table 5.76: AWR_RF_DIEID_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0223
SBLKLEN	2	Value = 4

The response to above request is given in the AWR_RF_DIEID_STATUS_SB.
Table 5.77 describes the contents of AWR_RF_DIEID_STATUS_SB.

Table 5.77: AWR_RF_DIEID_STATUS_SB response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0223
SBLKLEN	2	Value = 36
DIEID_HEXVAL0	4	Die Id Hex value 0
DIEID_HEXVAL1	4	Die Id Hex value 1
DIEID_HEXVAL2	4	Die Id Hex value 2
DIEID_HEXVAL3	4	Die Id Hex value 3
RESERVED	4	0x00000000
RESERVED	4	0x00000000
RESERVED	4	0x00000000
RESERVED	4	0x00000000

5.9.5 Sub block 0x0224 – AWR_RF_BOOTUPBIST_STATUS_GET_SB

This sub block provides the information regarding boot up self-test status.
Table 5.78 describes the content of this sub block.

Table 5.78: AWR_RF_BOOTUPBIST_STATUS_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0224
SBLKLEN	2	Value = 4

The response of this sub block will be AWR_RF_BOOTUPBIST_STATUS_DATA_SB with content as shown in Table 5.79

Table 5.79: AWR_RF_BOOTUPBIST_STATUS_DATA_SB response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0224
SBLKLEN	2	Value = 20

Continued on next page

Table 5.79 – continued from previous page

RF_POWERUP_ BIST_STATUS_ FLAGS	4	1 - PASS, 0 - FAIL	
		Bit	Status Information
		b0	ROM CRC check (RESERVED in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b1	CR4 and VIM lockstep test
		b2	RESERVED
		b3	VIM test
		b4	STC test of diagnostic
		b5	CR4 STC
		b6	CRC test
		b7	RAMPGEN memory ECC test
		b8	DFE Parity test
		b9	DFE memory ECC
		b10	RAMPGEN lockstep test
		b11	FRC lockstep test
		b12	DFE memory PBIST
		b13	RAMPGEN memory PBIST
		b14	PBIST test
		b15	WDT test
		b16	ESM test
		b17	DFE STC
		b18	RESERVED
		b19	ATCM, BTCM ECC test
		b20	ATCM, BTCM parity test
		b21	DCC test
		b22	RESERVED
		b23	RESERVED
		b24	FFT test (RESERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b25	RTI test
		b26	PCR test (RESERVED in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b27	Bus Safety test (RESERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b28	ECC aggregator test
		b29	MPU test
		b31:30	RESERVED

Continued on next page

Table 5.79 – continued from previous page

POWERUP_ TIME	4	RF BIST SS power up time 1 LSB = 5 ns
RESERVED	4	0x00000000
RESERVED	4	0x00000000

NOTE: Bootstrap digital monitoring status are not applicable for QM devices

5.9.6 Sub block 0x0226 – AWR_RF_ADV_ESMFAULT_STATUS_GET_SB

This sub block provides the information regarding additional RF sub system faults for xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP devices.

Table 5.80 describes the content of this sub block.

Table 5.80: AWR_RF_ADV_ESMFAULT_STATUS_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0226
SBLKLEN	2	Value = 4

The response to above request is given in the AWR_RF_ADV_ESMFAULT_STATUS_SB.

Table 5.81 describes the contents of AWR_RF_ADV_ESMFAULT_STATUS_SB.

Table 5.81: AWR_RF_ADV_ESMFAULT_STATUS_SB response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0226
SBLKLEN	2	Value = 28

Continued on next page

Table 5.81 – continued from previous page

ESM_GROUP1_ ERRORS_LSB	4	Bit	Error Information 0 – No Error , 1 – ESM Error
		b0	RAMPGEN_SB_ERROR
		b1	PROG_FILT_SELFTEST_ERROR
		b2	GPADC_RAM_SB_ERROR
		b3	VIM_RAM_SB_ERROR
		b4	DFE_PARITY_AND_OUT_ERROR (RESERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b5	VIM_SELFTEST_ERROR
		b6	B0TCM_SB_ERROR
		b7	B1TCM_SB_ERROR
		b8	CCMR4_SELFTEST_ERROR
		b9	ATCM_SB_ERROR
		b10	RAMPGEN_SELFTEST_ERROR
		b11	RAMPGEN_PAR_SELFTEST_ERROR
		b12	SEQ_EXT_SELFTEST_ERROR
		b13	SEQ_EXT_SB_ERROR
		b14	PROG_FILT_PARITY_ERROR
		b15	AGC_RAM_SB_ERROR
		b16	B1TCM_PAR_CHK_ERROR
		b17	B0TCM_PAR_CHK_ERROR
		b18	ATCM_PAR_CHK_ERROR
		b19	ECC_AGG_DED_ERROR
		b20	ECC_AGG_SEC_ERROR
		b21	RESERVED
		b22	RESERVED
		b23	RESERVED
		b24	RESERVED
		b25	PROG_FILT_UERR
		b26	PROG_FILT_SERR
		b27	FRC_SELFTEST_ERROR
		b28	FRC_ERROR
		b29	RESERVED
		b30	BUS_SAFETY_PCR_ERROR (RESERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b31	BUS_SAFETY_BSS_SLV_ERROR (RESERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)

Continued on next page

Table 5.81 – continued from previous page

ESM_GROUP1_ ERRORS_MSB	4	Bit	Error Information 0 – No Error , 1 – ESM Error
		b0	BUS_SAFETY_BSS_MST_ERROR (RE- SERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b1	BUS_SAFETY_STATIC_MEM_ERROR (RE- SERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b2	BUS_SAFETY_MBOX_ERROR (RESERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b3	BUS_SAFETY_SEC_AGGREGATED_ERROR (RESERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b31-b4	RESERVED
RESERVED	4	0x00000000	

Continued on next page

Table 5.81 – continued from previous page

ESM_GROUP2_ ERRORS	4	Bit	Error Information
		b0	DFE_STC_ERROR
		b1	CR4_STC_ERROR
		b2	CCMR4_COMP_ERROR
		b3	B0TCM_DB_ERROR
		b4	B1TCM_DB_ERROR
		b5	ATCM_DB_ERROR
		b6	DCC_ERROR
		b7	SEQ_EXT_ERROR
		b8	SYNT_FREQ_MON_ERROR
		b9	DFE_PARITY_OR_OUT_ERROR
		b10	RAMPGEN_DB_ERROR
		b11	BUBBLE_CORRECTION_FAIL
		b12	RAMPGEN_LOCSTEP_ERROR
		b13	RTI_RESET_ERROR
		b14	GPADC_RAM_DB_ERROR
		b15	VIM_COMP_ERROR
		b16	CR4_LIVE_LOCK_ERROR
		b17	WDT_NMI_ERROR
		b18	VIM_RAM_DB_ERROR
		b19	RAMPGEN_PAR_ERROR
		b20	SEQ_EXT_DB_ERROR
		b21	DMA_MPU_ERROR
		b22	AGC_RAM_DB_ERROR
		b23	CRC_COMP_ERROR
		b24	WAKEUP_STS_ERROR
		b25	SHORT_CIRCUIT_ERROR
		b26	B1TCM_PAR_ERROR
		b27	B0TCM_PAR_ERROR
		b28	ATCM_PAR_ERROR
		b29	RESERVED
		b30	RESERVED
		b31	CCC_ERROR
RESERVED	4	0x00000000	
RESERVED	4	0x00000000	

5.10 Sub blocks related to AWR_RF_MONITORING_REPORT_GET_MSG

5.10.1 Sub block 0x0260 – AWR_RF_DFE_STATISTICS_REPORT_GET_SB

Table 5.82 describes the content of this sub block.

NOTE: This API is not applicable for xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP devices. Please use API AWR_RF_REAL_CHAN_DFE_STATISTICS_REPORT_GET_SB for DFE statistics on xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP.

Table 5.82: AWR_RF_DFE_STATISTICS_REPORT_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0260
SBLKLEN	2	Value = 4

The response of this sub block will be AWR_RF_DFE_STATISTICS_REPORT_SB with content as shown in Table 5.83

Table 5.83: AWR_RF_DFE_STATISTICS_REPORT_SB response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0260
SBLKLEN	2	Value = 196
PF0_RX0_ICH	2	Residual DC value in I chain for profile 0, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF0_RX0_QCH	2	Residual DC value in Q chain for profile 0, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF0_RX0_ISQ	2	RMS power in I chain for profile 0, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input

Continued on next page

Table 5.83 – continued from previous page

PF0_RX0_QSQ	2	RMS power in Q chain for profile 0, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF0_RX0_IQ-CORR	4	Cross correlation between I and Q chains for profile 0, RX channel 0 (post DC and IQ mismatch correction) represented by a 32 bit signed number 1 LSB = $1V^2/2^{30}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF0_RX1_ICH	2	Residual DC value in I chain for profile 0, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF0_RX1_QCH	2	Residual DC value in Q chain for profile 0, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF0_RX1_ISQ	2	RMS power in I chain for profile 0, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF0_RX1_QSQ	2	RMS power in Q chain for profile 0, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF0_RX1_IQ-CORR	4	Cross correlation between I and Q chains for profile 0, RX channel 1 (post DC and IQ mismatch correction) represented by a 32 bit signed number 1 LSB = $1V^2/2^{30}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF0_RX2_ICH	2	Residual DC value in I chain for profile 0, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input

Continued on next page

Table 5.83 – continued from previous page

PF0_RX2_QCH	2	Residual DC value in Q chain for profile 0, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF0_RX2_ISQ	2	RMS power in I chain for profile 0, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF0_RX2_QSQ	2	RMS power in Q chain for profile 0, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF0_RX2_IQ-CORR	4	Cross correlation between I and Q chains for profile 0, RX channel 2 (post DC and IQ mismatch correction) represented by a 32 bit signed number 1 LSB = $1V^2/2^{30}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF0_RX3_ICH	2	Residual DC value in I chain for profile 0, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF0_RX3_QCH	2	Residual DC value in Q chain for profile 0, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF0_RX3_ISQ	2	RMS power in I chain for profile 0, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF0_RX3_QSQ	2	RMS power in Q chain for profile 0, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)

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Table 5.83 – continued from previous page

PF0_RX3_IQ-CORR	4	Cross correlation between I and Q chains for profile 0, RX channel 3 (post DC and IQ mismatch correction) represented by a 32 bit signed number 1 LSB = $1V^2/2^{30}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF1_RX0_ICH	2	Residual DC value in I chain for profile 1, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF1_RX0_QCH	2	Residual DC value in Q chain for profile 1, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF1_RX0_ISQ	2	RMS power in I chain for profile 1, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF1_RX0_QSQ	2	RMS power in Q chain for profile 1, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF1_RX0_IQ-CORR	4	Cross correlation between I and Q chains for profile 1, RX channel 0 (post DC and IQ mismatch correction) represented by a 32 bit signed number 1 LSB = $1V^2/2^{30}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF1_RX1_ICH	2	Residual DC value in I chain for profile 1, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF1_RX1_QCH	2	Residual DC value in Q chain for profile 1, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)

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Table 5.83 – continued from previous page

PF1_RX1_ISQ	2	RMS power in I chain for profile 1, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF1_RX1_QSQ	2	RMS power in Q chain for profile 1, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF1_RX1_IQ-CORR	4	Cross correlation between I and Q chains for profile 1, RX channel 1 (post DC and IQ mismatch correction) represented by a 32 bit signed number 1 LSB = $1V^2/2^{30}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF1_RX2_ICH	2	Residual DC value in I chain for profile 1, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF1_RX2_QCH	2	Residual DC value in Q chain for profile 1, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF1_RX2_ISQ	2	RMS power in I chain for profile 1, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF1_RX2_QSQ	2	RMS power in Q chain for profile 1, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF1_RX2_IQ-CORR	4	Cross correlation between I and Q chains for profile 1, RX channel 2 (post DC and IQ mismatch correction) represented by a 32 bit signed number 1 LSB = $1V^2/2^{30}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)

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Table 5.83 – continued from previous page

PF1_RX3_ICH	2	Residual DC value in I chain for profile 1, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF1_RX3_QCH	2	Residual DC value in Q chain for profile 1, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF1_RX3_ISQ	2	RMS power in I chain for profile 1, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF1_RX3_QSQ	2	RMS power in Q chain for profile 1, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF1_RX3_IQ-CORR	4	Cross correlation between I and Q chains for profile 1, RX channel 3 (post DC and IQ mismatch correction) represented by a 32 bit signed number 1 LSB = $1V^2/2^{30}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF2_RX0_ICH	2	Residual DC value in I chain for profile 2 RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF2_RX0_QCH	2	Residual DC value in Q chain for profile 2, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF2_RX0_ISQ	2	RMS power in I chain for profile 2, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input

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Table 5.83 – continued from previous page

PF2_RX0_QSQ	2	RMS power in Q chain for profile 2, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF2_RX0_IQ-CORR	4	Cross correlation between I and Q chains for profile 2, RX channel 0 (post DC and IQ mismatch correction) represented by a 32 bit signed number 1 LSB = $1V^2/2^{30}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF2_RX1_ICH	2	Residual DC value in I chain for profile 2, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF2_RX1_QCH	2	Residual DC value in Q chain for profile 2, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF2_RX1_ISQ	2	RMS power in I chain for profile 2, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF2_RX1_QSQ	2	RMS power in Q chain for profile 2, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF2_RX1_IQ-CORR	4	Cross correlation between I and Q chains for profile 2, RX channel 1 (post DC and IQ mismatch correction) represented by a 32 bit signed number 1 LSB = $1V^2/2^{30}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF2_RX2_ICH	2	Residual DC value in I chain for profile 2, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input

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Table 5.83 – continued from previous page

PF2_RX2_QCH	2	Residual DC value in Q chain for profile 2, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF2_RX2_ISQ	2	RMS power in I chain for profile 2, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF2_RX2_QSQ	2	RMS power in Q chain for profile 2, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF2_RX2_IQ-CORR	4	Cross correlation between I and Q chains for profile 2, RX channel 2 (post DC and IQ mismatch correction) represented by a 32 bit signed number 1 LSB = $1V^2/2^{30}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF2_RX3_ICH	2	Residual DC value in I chain for profile 2, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF2_RX3_QCH	2	Residual DC value in Q chain for profile 2, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF2_RX3_ISQ	2	RMS power in I chain for profile 2, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF2_RX3_QSQ	2	RMS power in Q chain for profile 2, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)

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Table 5.83 – continued from previous page

PF2_RX3_IQ-CORR	4	Cross correlation between I and Q chains for profile 2, RX channel 3 (post DC and IQ mismatch correction) represented by a 32 bit signed number 1 LSB = $1V^2/2^{30}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF3_RX0_ICH	2	Residual DC value in I chain for profile 3 RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF3_RX0_QCH	2	Residual DC value in Q chain for profile 3, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF3_RX0_ISQ	2	RMS power in I chain for profile 3, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF3_RX0_QSQ	2	RMS power in Q chain for profile 3, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF3_RX0_IQ-CORR	4	Cross correlation between I and Q chains for profile 3, RX channel 0 (post DC and IQ mismatch correction) represented by a 32 bit signed number 1 LSB = $1V^2/2^{30}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF3_RX1_ICH	2	Residual DC value in I chain for profile 3, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF3_RX1_QCH	2	Residual DC value in Q chain for profile 3, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)

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Table 5.83 – continued from previous page

PF3_RX1_ISQ	2	RMS power in I chain for profile 3, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF3_RX1_QSQ	2	RMS power in Q chain for profile 3, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF3_RX1_IQ-CORR	4	Cross correlation between I and Q chains for profile 3, RX channel 1 (post DC and IQ mismatch correction) represented by a 32 bit signed number 1 LSB = $1V^2/2^{30}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF3_RX2_ICH	2	Residual DC value in I chain for profile 3, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF3_RX2_QCH	2	Residual DC value in Q chain for profile 3, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF3_RX2_ISQ	2	RMS power in I chain for profile 3, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF3_RX2_QSQ	2	RMS power in Q chain for profile 3, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF3_RX2_IQ-CORR	4	Cross correlation between I and Q chains for profile 3, RX channel 2 (post DC and IQ mismatch correction) represented by a 32 bit signed number 1 LSB = $1V^2/2^{30}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)

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Table 5.83 – continued from previous page

PF3_RX3_ICH	2	Residual DC value in I chain for profile 3, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF3_RX3_QCH	2	Residual DC value in Q chain for profile 3, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF3_RX3_ISQ	2	RMS power in I chain for profile 3, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF3_RX3_QSQ	2	RMS power in Q chain for profile 3, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
PF3_RX3_IQ-CORR	4	Cross correlation between I and Q chains for profile 3, RX channel 3 (post DC and IQ mismatch correction) represented by a 32 bit signed number 1 LSB = $1V^2/2^{30}$ referred to ADC input (Not applicable for REAL mode and xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)

5.10.2 Sub block 0x0261 – AWR_RF_REAL_CHAN_DFE_STATISTICS_REPORT_GET_SB

Table 5.84 describes the content of this sub block.

Table 5.84: AWR_RF_REAL_CHAN_DFE_STATISTICS_REPORT_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0261
SBLKLEN	2	Value = 4

The response of this sub block will be AWR_RF_REAL_CHAN_DFE_STATISTICS_REPORT_SB with content as shown in Table 5.85

Table 5.85: AWR_RF_REAL_CHAN_DFE_STATISTICS_REPORT_SB
response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0261
SBLKLEN	2	Value = 116
PF0_RX0_ICH	2	Residual DC value in I chain for profile 0, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF0_RX0_ISQ	2	RMS power in I chain for profile 0, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF0_RX1_ICH	2	Residual DC value in I chain for profile 0, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF0_RX1_ISQ	2	RMS power in I chain for profile 0, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF0_RX2_ICH	2	Residual DC value in I chain for profile 0, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF0_RX2_ISQ	2	RMS power in I chain for profile 0, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF0_RX3_ICH	2	Residual DC value in I chain for profile 0, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF0_RX3_ISQ	2	RMS power in I chain for profile 0, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF1_RX0_ICH	2	Residual DC value in I chain for profile 1, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input

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Table 5.85 – continued from previous page

PF1_RX0_ISQ	2	RMS power in I chain for profile 1, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF1_RX1_ICH	2	Residual DC value in I chain for profile 1, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF1_RX1_ISQ	2	RMS power in I chain for profile 1, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF1_RX2_ICH	2	Residual DC value in I chain for profile 1, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF1_RX2_ISQ	2	RMS power in I chain for profile 1, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF1_RX3_ICH	2	Residual DC value in I chain for profile 1, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF1_RX3_ISQ	2	RMS power in I chain for profile 1, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF2_RX0_ICH	2	Residual DC value in I chain for profile 2 RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF2_RX0_ISQ	2	RMS power in I chain for profile 2, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF2_RX1_ICH	2	Residual DC value in I chain for profile 2, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input

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Table 5.85 – continued from previous page

PF2_RX1_ISQ	2	RMS power in I chain for profile 2, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF2_RX2_ICH	2	Residual DC value in I chain for profile 2, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF2_RX2_ISQ	2	RMS power in I chain for profile 2, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF2_RX3_ICH	2	Residual DC value in I chain for profile 2, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF2_RX3_ISQ	2	RMS power in I chain for profile 2, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF3_RX0_ICH	2	Residual DC value in I chain for profile 3 RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF3_RX0_ISQ	2	RMS power in I chain for profile 3, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF3_RX1_ICH	2	Residual DC value in I chain for profile 3, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF3_RX1_ISQ	2	RMS power in I chain for profile 3, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF3_RX2_ICH	2	Residual DC value in I chain for profile 3, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input

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Table 5.85 – continued from previous page

PF3_RX2_ISQ	2	RMS power in I chain for profile 3, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF3_RX3_ICH	2	Residual DC value in I chain for profile 3, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF3_RX3_ISQ	2	RMS power in I chain for profile 3, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF4_RX0_ICH	2	Residual DC value in I chain for profile 4, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF4_RX0_ISQ	2	RMS power in I chain for profile 4, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF4_RX1_ICH	2	Residual DC value in I chain for profile 4, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF4_RX1_ISQ	2	RMS power in I chain for profile 4, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF4_RX2_ICH	2	Residual DC value in I chain for profile 4, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF4_RX2_ISQ	2	RMS power in I chain for profile 4, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF4_RX3_ICH	2	Residual DC value in I chain for profile 4, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input

Continued on next page

Table 5.85 – continued from previous page

PF4_RX3_ISQ	2	RMS power in I chain for profile 4, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF5_RX0_ICH	2	Residual DC value in I chain for profile 5, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF5_RX0_ISQ	2	RMS power in I chain for profile 5, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF5_RX1_ICH	2	Residual DC value in I chain for profile 5, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF5_RX1_ISQ	2	RMS power in I chain for profile 5, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF5_RX2_ICH	2	Residual DC value in I chain for profile 5, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF5_RX2_ISQ	2	RMS power in I chain for profile 5, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF5_RX3_ICH	2	Residual DC value in I chain for profile 5, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF5_RX3_ISQ	2	RMS power in I chain for profile 5, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF6_RX0_ICH	2	Residual DC value in I chain for profile 6, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input

Continued on next page

Table 5.85 – continued from previous page

PF6_RX0_ISQ	2	RMS power in I chain for profile 6, RX channel 0 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF6_RX1_ICH	2	Residual DC value in I chain for profile 6, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF6_RX1_ISQ	2	RMS power in I chain for profile 6, RX channel 1 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF6_RX2_ICH	2	Residual DC value in I chain for profile 6, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF6_RX2_ISQ	2	RMS power in I chain for profile 6, RX channel 2 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input
PF6_RX3_ICH	2	Residual DC value in I chain for profile 6, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit signed number 1 LSB = $1V/2^{15}$ referred to ADC input
PF6_RX3_ISQ	2	RMS power in I chain for profile 6, RX channel 3 (post DC and IQ mismatch correction) represented by a 16 bit unsigned number 1 LSB = $1V^2/2^{15}$ referred to ADC input

5.11 Sub blocks related to AWR_RF_MISC_CONF_SET_MSG

5.11.1 Sub block 0x02C0 – RESERVED

5.11.2 Sub block 0x02C1 – RESERVED

5.11.3 Sub block 0x02C2 – AWR_RF_TEST_SOURCE_CONFIG_SET_SB

This sub block is used to configure the test source of BSS

NOTE1:	The test source configuration APIs are supported only for debug purpose. Please refer latest DFP release note for more info.
NOTE2:	After test source usage, it is recommend to disable the test source and issue profile configuration API again for normal functionality of radar.

Table 5.86 describes the content of this sub block.

Table 5.86: AWR_RF_TEST_SOURCE_CONFIG_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x02C2
SBLKLEN	2	Value = 72
POSITION_VEC1	[2+2+2]	Relative position in Cartesian coordinate from radar to objects, [x, y, z] (all signed, though for y, only unsigned makes sense if forward looking: our radar is on y=0 plane). Object 0 [x,y,z] 1 LSB = 1 cm Valid Range: y: 0 to 32767 cm, x & z: ± 32767 cm
VELOCITY_VEC1	[2+2+2]	Relative velocity in Cartesian coordinate, similar to position vector (all signed) Object 0 1 LSB = 1 cm/s Valid Range = ± 5000 (i.e. ± 180 kmph)
SIG_LEV_VEC1	[2]	Reflecting objects' signal level at ADC output, relative to ADC Full Scale 1 LSB = -0.1 dBFS Valid range: 0 to 950 The same field may be used to emulate enable/disable each object by programming appropriate levels.
BOUNDARY_MIN_VEC1	[2+2+2]	Boundary minimum limit for each of x, y, z. When the current position crosses this boundary, the emulator returns the corresponding coordinate to the originally programmed value. Object 1 [x,y,z] 1 LSB = 1 cm Valid Range: y: 0 to 32767 cm, x & z: ± 32767 cm

Continued on next page

Table 5.86 – continued from previous page

BOUNDARY_ MAX_VEC1	[2+2+2]	Boundary maximum limit for each of x, y, z. When the current position crosses this boundary, the emulator returns the corresponding coordinate to the originally programmed value. Object 1 [x,y,z] 1 LSB = 1 cm Valid Range: y: 0 to 32767 cm, x & z: ± 32767 cm
POSITION_VEC2	[2+2+2]	Relative position in Cartesian coordinate from radar to objects, [x, y, z] (all signed, though for y, only unsigned makes sense if forward looking: our radar is on y=0 plane). Object 1 [x,y,z] 1 LSB = 1 cm Valid Range: y: 0 to 32767 cm, x & z: ± 32767 cm
VELOCITY_VEC2	[2+2+2]	Relative velocity in Cartesian coordinate, similar to position vector (all signed) Object 1 1 LSB = 1 cm/s Valid Range = +/- 5000 (i.e. +/-180 kmph)
SIG_LEV_VEC2	[2]	Reflecting objects' signal level at ADC output, relative to ADC Full Scale 1 LSB = -0.1 dBFS Valid range: 0 to 950 The same field may be used to emulate enable/disable each object by programming appropriate levels.
BOUNDARY_ MIN_VEC2	[2+2+2]	Boundary minimum limit for each of x, y, z. When the current position crosses this boundary, the emulator returns the corresponding coordinate to the originally programmed value. Object 1 [x,y,z] 1 LSB = 1 cm Valid Range: y: 0 to 32767 cm, x & z: ± 32767 cm
BOUNDARY_ MAX_VEC2	[2+2+2]	Boundary maximum limit for each of x, y, z. When the current position crosses this boundary, the emulator returns the corresponding coordinate to the originally programmed value. Object 1 [x,y,z] 1 LSB = 1 cm Valid Range: y: 0 to 32767 cm, x & z: ± 32767 cm

Continued on next page

Table 5.86 – continued from previous page

RX_ANT_POS_XZ	8	Receiver Antenna positions to be modeled. The radar is on y=0 plane. Only x and z coordinates to be provided. 1 LSB = Wavelength/8 Valid range = ±15 wave lengths Byte 0: RX0 X coordinate (may be 0 as reference) Byte 1: RX0 Z (may be 0 as reference) Byte 2: RX1 X Byte 3: RX1 Z Byte 4: RX2 X Byte 5: RX2 Z Byte 6: RX3 X Byte 7: RX3 Z														
RESERVED	6	RESERVED														
MISC_FUNC_CTRL	1	<table><tr><th>Bits</th><th>Description</th></tr><tr><td>b0</td><td>DIS_DITHER</td></tr><tr><td></td><td>Value Definition</td></tr><tr><td>0</td><td>DITHER is enabled in test source data</td></tr><tr><td>1</td><td>DITHER is disabled in test source data</td></tr><tr><td colspan="2">Note: This feature is supported only on AWR2243 device.</td></tr><tr><td>b7:1</td><td>RESERVED</td></tr></table>	Bits	Description	b0	DIS_DITHER		Value Definition	0	DITHER is enabled in test source data	1	DITHER is disabled in test source data	Note: This feature is supported only on AWR2243 device.		b7:1	RESERVED
Bits	Description															
b0	DIS_DITHER															
	Value Definition															
0	DITHER is enabled in test source data															
1	DITHER is disabled in test source data															
Note: This feature is supported only on AWR2243 device.																
b7:1	RESERVED															
RESERVED	1	Reserved for 4 bytes alignment														

NOTE: Test source is not characterized and tuned to 60GHz in xWR6x43 devices.

5.11.4 Sub block 0x02C3 – AWR_RF_TEST_SOURCE_ENABLE_SET_SB

This sub block is used to enable test source of BSS
Table 5.87 describes the content of this sub block.

Table 5.87: AWR_RF_TEST_SOURCE_ENABLE_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x02C3
SBLKLEN	2	Value = 8

Continued on next page

Table 5.87 – continued from previous page

TS_EN	2	Bit	Definition
		b0	0 Disable (revert to normal functionality)
			1 Enable (enter test source functionality)
		b15:1	RESERVED
RESERVED	2	0x0000	

5.11.5 Sub block 0x02C4 – 0x02CB RESERVED

5.11.6 Sub block 0x02CC – AWR_RF_LDO_BYPASS_SB

This sub block enables LDO bypass option within BSS.

CAUTION: Do not enable RF LDO bypass option when the PMIC is configured to supply 1.3V to VIN_13RF1 and VIN_13RF2 analog and RF power supply inputs. This may damage the device. Typically in TI EVMs, PMIC is configured to supply 1.3V to the RF supplies.

Table 5.88 describes the content of this sub block.

Table 5.88: AWR_RF_LDO_BYPASS_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x02CC
SBLKLEN	2	Value = 8

Continued on next page

Table 5.88 – continued from previous page

RFLDO_BY-PASS_EN	2	NOTE:This field is RESERVED for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices. AWR2243/xWR6243 devices: <table><tr><td>Bit</td><td colspan="2">Description</td></tr><tr><td rowspan="3">b0</td><td>Value</td><td>Description</td></tr><tr><td>0</td><td>RF LDO not bypassed</td></tr><tr><td>1</td><td>RF LDO bypassed</td></tr><tr><td rowspan="3">b1</td><td>Value</td><td>Description</td></tr><tr><td>0</td><td>PA LDO enabled</td></tr><tr><td>1</td><td>PA LDO disabled</td></tr></table> When simultaneous 3 TX are to be used, to avoid package reliability issues, VIN_13RF2 is shorted to VOUT_PA on the board and the PA LDO should be disabled. b15:2 RESERVED The usage of these configurations is as per the table below <table><tr><th>USECASE</th><th>LDO_ BYPASS</th><th>PA_LDO_ DISABLE</th></tr><tr><td>1.3V VIN_13RF1 and VIN_13RF2 supplies</td><td>0</td><td>0</td></tr><tr><td>1.0V VIN_13RF1 and VIN_13RF2 supplies</td><td>1</td><td>0</td></tr><tr><td>1.0V VIN_13RF1 and VIN_13RF2 supplies and VIN_13RF2 shorted to VOUT_PA</td><td>1</td><td>1</td></tr></table>	Bit	Description		b0	Value	Description	0	RF LDO not bypassed	1	RF LDO bypassed	b1	Value	Description	0	PA LDO enabled	1	PA LDO disabled	USECASE	LDO_ BYPASS	PA_LDO_ DISABLE	1.3V VIN_13RF1 and VIN_13RF2 supplies	0	0	1.0V VIN_13RF1 and VIN_13RF2 supplies	1	0	1.0V VIN_13RF1 and VIN_13RF2 supplies and VIN_13RF2 shorted to VOUT_PA	1	1
Bit	Description																														
b0	Value	Description																													
	0	RF LDO not bypassed																													
	1	RF LDO bypassed																													
b1	Value	Description																													
	0	PA LDO enabled																													
	1	PA LDO disabled																													
USECASE	LDO_ BYPASS	PA_LDO_ DISABLE																													
1.3V VIN_13RF1 and VIN_13RF2 supplies	0	0																													
1.0V VIN_13RF1 and VIN_13RF2 supplies	1	0																													
1.0V VIN_13RF1 and VIN_13RF2 supplies and VIN_13RF2 shorted to VOUT_PA	1	1																													
SUPPLY_MONITOR_IRDROP	1	IR drop is the voltage drop from the PMIC output to the device pin. The user should program the voltage drop in percentage units which will be used for adjusting the thresholds for measuring the external supplies. <table><tr><td>Value</td><td>Description</td></tr><tr><td>0</td><td>IR drop of 0%</td></tr><tr><td>1</td><td>IR drop of 3%</td></tr><tr><td>2</td><td>IR drop of 6%</td></tr><tr><td>3</td><td>IR drop of 9%</td></tr></table>	Value	Description	0	IR drop of 0%	1	IR drop of 3%	2	IR drop of 6%	3	IR drop of 9%																			
Value	Description																														
0	IR drop of 0%																														
1	IR drop of 3%																														
2	IR drop of 6%																														
3	IR drop of 9%																														

Continued on next page

Table 5.88 – continued from previous page

IO_SUPPLY_INDICATOR	1	NOTE:This field is RESERVED for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices. IO supply indicator for correct monitoring of IO supply <table><tr><td>Value</td><td>Description</td></tr><tr><td>0</td><td>3.3 V IO supply</td></tr><tr><td>1</td><td>1.8 V IO supply</td></tr></table>	Value	Description	0	3.3 V IO supply	1	1.8 V IO supply
Value	Description							
0	3.3 V IO supply							
1	1.8 V IO supply							

5.11.7 Sub block 0x02CD – AWR_RF_PALOOPBACK_CFG_SB

This sub block enables/disables PA loopback for all enabled profiles. This is used to debug both the TX and RX chains are working correctly.

NOTE: The PA loop-back configuration API is supported only for debug purpose. Please refer latest DFP release note for more info.

Table 5.89 describes the content of this sub block for AWR2243/xWR6243, and Table 5.90 describes the content of this sub-block for xWR294x, xWR254x, xWR2x4xECO and xWR2x4xP devices.

NOTE: If monitoring is enabled with the loopback APIs (subblock 0x02CD, 0x02CE, 0x02CF), then loopback will not work after monitoring is complete. To use loopback with monitoring, use AWR_ADVANCED_FRAME_CONF_SB with AWR_LOOPBACK_BURST_CONF_SB.

Table 5.89: AWR_RF_PALOOPBACK_CFG_SB contents for AWR2243/xWR6243 devices

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x02CD
SBLKLEN	2	Value = 8

Continued on next page

Table 5.89 – continued from previous page

PA_LOOPBACK_FREQ	2	This value is a 100 MHz divider which sets the loopback frequency For e.g. for a 1 MHz frequency, set this to 100 For a 2 MHz frequency, set this to 50 NOTE: To ensure no leakage of signal power, user has to ensure that 100 MHz/LOOPBACK_FREQ is an integer multiple of bin width For e.g. if user choses 25 Msps sampling rate and 2048 samples/chirp, then LOOPBACK_FREQ of 64 (=1.5625 MHz) will ensure no leakage						
PA_LOOPBACK_EN	1	<table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>PA loopback is not enabled</td></tr><tr><td>1</td><td>PA loopback is enabled</td></tr></table>	Value	Description	0	PA loopback is not enabled	1	PA loopback is enabled
Value	Description							
0	PA loopback is not enabled							
1	PA loopback is enabled							
RESERVED	1	0x00						

Table 5.90: AWR_RF_PALOOPBACK_CFG_SB contents for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices

Field Name	Number of bytes	Description						
SBLKID	2	Value = 0x02CD						
SBLKLEN	2	Value = 14						
PA_LOOPBACK_FREQ	4	PA loopback modulation frequency in Hz. NOTE: Please refer to PA_LOOPBACK_FREQ field in AWR_LOOPBACK_BURST_CONF_SET_SB for the list of allowed frequencies.						
PA_LOOPBACK_EN	1	<table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>PA loopback is not enabled</td></tr><tr><td>1</td><td>PA loopback is enabled</td></tr></table>	Value	Description	0	PA loopback is not enabled	1	PA loopback is enabled
Value	Description							
0	PA loopback is not enabled							
1	PA loopback is enabled							
RESERVED	1	0x00						
COMMON_BUF_GAIN_SEL	1	This field configures the QPSK buffer gain in both the PA and the LO loopback paths in xWR294x,xWR254x,xWR2x4xECO and xWR2x4xP. NOTE: Please refer to COMMON_BUF_GAIN_SEL field in AWR_LOOPBACK_BURST_CONF_SET_SB for the list of gain codes.						
PA_LOOPBACK_BUF_GAIN_SEL	1	This configures the PA loopback buffer gain. NOTE: Please refer to PA_LOOPBACK_BUF_GAIN_SEL field in AWR_LOOPBACK_BURST_CONF_SET_SB for the list of gain codes.						

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Table 5.90 – continued from previous page

RESERVED	2	0x0000
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5.11.8 Sub block 0x02CE – AWR_RF_PSLOOPBACK_CFG_SB

This sub block enables/disables PS (phase shifter) loopback for all enabled profiles. This is used to debug the TX (before the PA) and RX chains.

NOTE: The PS loop-back configuration API is not applicable to xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices. They will have a separate LO loopback configuration API.

NOTE: The PS loop-back configuration API is supported only for debug purpose. Please refer latest DFP release note for more info.

Table 5.91 describes the content of this sub block.

NOTE: If monitoring is enabled with the loopback APIs (subblock 0x02CD, 0x02CE, 0x02CF), then loopback will not work after monitoring is complete. To use loopback with monitoring, use AWR_ADVANCED_FRAME_CONF_SB with AWR_LOOPBACK_BURST_CONF_SB.

Table 5.91: AWR_RF_PSLOOPBACK_CFG_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x02CE
SBLKLEN	2	Value = 12
PS_LOOPBACK_FREQ	2	Loop back frequency in kHz 1 LSB = 1 kHz
RESERVED	2	0x0000
PS_LOOPBACK_EN	1	Value Definition 0 PS loopback is not enabled 1 PS loopback is enabled
PS_LOOPBACK_TXID	1	Bit Definition b0 TX0 is used for loopback b1 TX1 is used for loopback b7:2 RESERVED

Continued on next page

Table 5.91 – continued from previous page

PGA_GAIN_INDEX	1	Value		PGA	gain	Value		PGA	gain
				value				value	
		0	PGA is OFF			15	-3 dB		
		1	-22 dB			16	-2 dB		
		2	-16 dB			17	-1 dB		
		3	-15 dB			18	0 dB		
		4	-14 dB			19	1 dB		
		5	-13 dB			20	2 dB		
		6	-12 dB			21	3 dB		
		7	-11 dB			22	4 dB		
		8	-10 dB			23	5 dB		
		9	-9 dB			24	6 dB		
		10	-8 dB			25	7 dB		
		11	-7 dB			26	8 dB		
		12	-6 dB			27	9 dB		
		13	-5 dB			255-28	RESERVED		
		14	-4 dB						
RESERVED	1	0x00							

NOTE:

The expected signal strength change with change in index value is only approximately indicated for PS<n>_PGA_GAIN_INDEX. Typically, the loopback path is the dominant path only in top 10 indices (highest PGA gain values). For lower indices (lower PGA gain values), parasitic paths in the RF system can start dominating the loop-back measurements, and under such conditions, inter channel imbalances measured using such LB path, and LB signal SNR etc. can show degraded performance, with the degradation attributed to the loop-back path and not the functional path/circuits/system.

5.11.9 Sub block 0x02CF – AWR_RF_IFLOOPBACK_CFG_SB

This sub block enables/disables IF loopback for all enabled profiles. This is used to debug the RX IF chain.

NOTE:

The IF loop-back configuration API is supported only for debug purpose. Please refer latest DFP release note for more info.

Table 5.92 describes the content of this sub block.

NOTE: If monitoring is enabled with the loopback APIs (subblock 0x02CD, 0x02CE, 0x02CF), then loopback will not work after monitoring is complete. To use loopback with monitoring, use AWR_ADVANCED_FRAME_CONF_SB with AWR_LOOPBACK_BURST_CONF_SB.

Table 5.92: AWR_RF_IFLOOPBACK_CFG_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x02CF
SBLKLEN	2	Value = 8
IF_LOOPBACK_FREQ	2	Value IF Loopback frequency value 0 180 kHz 1 240 kHz 2 360 kHz 3 720 kHz 4 1 MHz 5 2 MHz 6 2.5 MHz 7 3 MHz 8 4.017857 MHz 9 5 MHz 10 6 MHz 11 7.5 MHz 12 8.035714 MHz 13 9 MHz 14 10 MHz 65535- RESERVED 15
IF_LOOPBACK_EN	1	Value Definition 0 IF loopback is not enabled 1 IF loopback is enabled
RESERVED	1	0x00

5.11.10 Sub block 0x02D0 – AWR_RF_GPADC_CFG_SET_SB

This sub block enables the GPADC reads for external inputs (available only in xWR1642/xWR1843/xWR6843/AWR224)

NOTE: This API is not supported in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP. The external analog signals can be monitored using the MSS GPADC from the customer application.

Table 5.93 describes the content of this sub block.

Table 5.93: AWR_RF_GPADC_CFG_SET_SB contents

Field Name	Number of bytes	Description																
SBLKID	2	Value = 0x02D0																
SBLKLEN	2	Value = 32																
SIGNAL_INPUT_ENABLES	1	This field indicates the sets of externally fed DC signals which are to be monitored using GPADC. When each bit in this field is set, the corresponding signal is monitored. <table><tr><th>Bit</th><th>Definition</th></tr><tr><td>0</td><td>ANALOGTEST1</td></tr><tr><td>1</td><td>ANALOGTEST2</td></tr><tr><td>2</td><td>ANALOGTEST3</td></tr><tr><td>3</td><td>ANALOGTEST4</td></tr><tr><td>4</td><td>ANAMUX</td></tr><tr><td>5</td><td>VSENSE</td></tr><tr><td>Others</td><td>RESERVED</td></tr></table>	Bit	Definition	0	ANALOGTEST1	1	ANALOGTEST2	2	ANALOGTEST3	3	ANALOGTEST4	4	ANAMUX	5	VSENSE	Others	RESERVED
Bit	Definition																	
0	ANALOGTEST1																	
1	ANALOGTEST2																	
2	ANALOGTEST3																	
3	ANALOGTEST4																	
4	ANAMUX																	
5	VSENSE																	
Others	RESERVED																	
SIGNAL_BUFFER_ENABLES	1	This field indicates the sets of externally fed DC signals which are to be buffered before being fed to the GPADC. When each bit in this field is set, the corresponding signal is buffered before the GPADC. <table><tr><th>Bit</th><th>SIGNAL</th></tr><tr><td>0</td><td>ANALOGTEST1</td></tr><tr><td>1</td><td>ANALOGTEST2</td></tr><tr><td>2</td><td>ANALOGTEST3</td></tr><tr><td>3</td><td>ANALOGTEST4</td></tr><tr><td>4</td><td>ANAMUX</td></tr><tr><td>Others</td><td>RESERVED</td></tr></table>	Bit	SIGNAL	0	ANALOGTEST1	1	ANALOGTEST2	2	ANALOGTEST3	3	ANALOGTEST4	4	ANAMUX	Others	RESERVED		
Bit	SIGNAL																	
0	ANALOGTEST1																	
1	ANALOGTEST2																	
2	ANALOGTEST3																	
3	ANALOGTEST4																	
4	ANAMUX																	
Others	RESERVED																	

Continued on next page

Table 5.93 – continued from previous page

ANATEST1_CFG	2	Bit Definition b7:0 Number of samples to collect 1 sample takes 1.6 μs b15:8 Settling time 1 LSB = 0.8 μs Valid range: 0 to 12 μs Valid programming condition: all the signals that are enabled should take a total of < 100 μs including the programmed settling times and measurement time per enabled signal.
ANATEST2_CFG	2	Bit Definition b7:0 Number of samples to collect 1 sample takes 1.6 μs b15:8 Settling time 1 LSB = 0.8 μs Valid range: 0 to 12 μs Valid programming condition: all the signals that are enabled should take a total of < 100 μs including the programmed settling times and measurement time per enabled signal.
ANATEST3_CFG	2	Bit Definition b7:0 Number of samples to collect 1 sample takes 1.6 μs b15:8 Settling time 1 LSB = 0.8 μs Valid range: 0 to 12 μs Valid programming condition: all the signals that are enabled should take a total of < 100 μs including the programmed settling times and measurement time per enabled signal.
ANATEST4_CFG	2	Bit Definition b7:0 Number of samples to collect 1 sample takes 1.6 μs b15:8 Settling time 1 LSB = 0.8 μs Valid range: 0 to 12 μs Valid programming condition: all the signals that are enabled should take a total of < 100 μs including the programmed settling times and measurement time per enabled signal.

Continued on next page

Table 5.93 – continued from previous page

ANAMUX_CFG	2	Bit Definition b7:0 Number of samples to collect 1 sample takes 1.6 μ s b15:8 Settling time 1 LSB = 0.8 μ s Valid range: 0 to 12 μ s Valid programming condition: all the signals that are enabled should take a total of < 100 μ s including the programmed settling times and measurement time per enabled signal.
VSENSE_CFG	2	Bit Definition b7:0 Number of samples to collect 1 sample takes 1.6 μ s b15:8 Settling time 1 LSB = 0.8 μ s Valid range: 0 to 12 μ s Valid programming condition: all the signals that are enabled should take a total of < 100 μ s including the programmed settling times and measurement time per enabled signal.
RESERVED	2	0x0000
RESERVED	4	0x00000000
RESERVED	4	0x00000000
RESERVED	4	0x00000000

The response to the AWR_RF_GPADC_CFG_SET_SB is an async event AWR_AE_RF_GPADC_RESULT_DATA_SB which contains the measured values for each of the enabled channels.

NOTE: The actual measurement of these GPADC signal are done in inter-burst or frame idle time and the result AE sub block will be sent only after completing all the measurements.

5.11.11 Sub block 0x02D1 – RESERVED

5.11.12 Sub block 0x02D2 – RESERVED

5.11.13 Sub block 0x02D3 – RESERVED

5.11.14 Sub block 0x02D4 – RESERVED

5.11.15 Sub block 0x02D5 – AWR_RF_LOLOOPBACK_CFG_SB

This sub block enables/disables LO loopback for all enabled profiles. This is used to debug the TX (before the PA) and RX chains. This is only applicable for xWR294x, xWR254x, xWR2x4xECO,

xWR2x4xP devices.

NOTE: The LO loop-back configuration API is supported only for debug purpose. Please refer latest DFP release note for more info.

Table 5.94 describes the content of this sub block.

NOTE: If monitoring is enabled with the loopback APIs (subblock 0x02CD, 0x02CE, 0x02CF), then loopback will not work after monitoring is complete. To use loopback with monitoring, use AWR_ADVANCED_FRAME_CONF_SB with AWR_LOOPBACK_BURST_CONF_SB.

Table 5.94: AWR_RF_LOLOOPBACK_CFG_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x02D5
SBLKLEN	2	Value = 12
LO_LOOPBACK_FREQ	4	Loop back frequency in Hz NOTE: Please refer to LO_LOOPBACK_FREQ field in AWR_LOOPBACK_BURST_CONF_SET_SB for the list of allowed frequencies.
LO_LOOPBACK_EN	1	Value Definition 0 LO loopback is not enabled 1 LO loopback is enabled
COMMON_BUF_GAIN_SEL	1	This field configures the QPSK buffer gain in both the PA and the LO loopback paths in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP. NOTE: Please refer to COMMON_BUF_GAIN_SEL field in AWR_LOOPBACK_BURST_CONF_SET_SB for the list of gain codes.
LO_LOOPBACK_BUF_GAIN_SEL	1	This configures the LO loopback buffer gain. NOTE: Please refer to LO_LOOPBACK_BUF_GAIN_SEL field in AWR_LOOPBACK_BURST_CONF_SET_SB for the list of gain codes.
RESERVED	1	0x00

5.12 Sub blocks related to AWR_RF_MISC_CONF_GET_MSG

5.12.1 Sub block 0x02E0 to 0x2E9 – RESERVED

5.12.2 Sub block 0x02EA – AWR_RF_TEMPERATURE_GET_SB

This sub block provides the device temperature sensor information.

Table 5.95 describes the content of this sub block.

NOTE: Use AWR_MONITOR_TEMPERATURE_CONF_SB to read periodic temperature during functional frames instead of AWR_RF_TEMPERATURE_GET_SB

Table 5.95: AWR_RF_TEMPERATURE_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x02EA
SBLKLEN	2	Value = 4

AWR_RF_TEMPERATURE_DATA_SB sub block is sent by the radar device in response to AWR_RF_TEMPERATURE_GET_SB.

Table 5.96: AWR_RF_TEMPERATURE_DATA_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x02EA
SBLKLEN	2	Value = 28
TIME	4	BSS local Time from device power up 1 LSB = 1 ms
TEMP_RX0_SENS	2	RX0 temperature sensor reading (signed value) 1 LSB = 1°C
TEMP_RX1_SENS	2	RX1 temperature sensor reading (signed value) 1 LSB = 1°C
TEMP_RX2_SENS	2	RX2 temperature sensor reading (signed value) 1 LSB = 1°C
TEMP_RX3_SENS	2	RX3 temperature sensor reading (signed value) 1 LSB = 1°C
TEMP_TX0_SENS	2	TX0 temperature sensor reading (signed value) 1 LSB = 1°C

Continued on next page

Table 5.96 – continued from previous page

TEMP_TX1_SENS	2	TX1 temperature sensor reading (signed value) 1 LSB = 1°C
TEMP_TX2_SENS	2	TX2 temperature sensor reading (signed value) 1 LSB = 1°C
TEMP_PM_SENS	2	PM temperature sensor reading (signed value) 1 LSB = 1°C
TEMP_DIG1_SENS / TEMP_TX3_SENS	2	AWR2243/xWR6243 devices: Digital temperature sensor reading (signed value) xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: TX3 temperature sensor reading (signed value) 1 LSB = 1°C
TEMP_DIG2_SENS	2	Digital temperature sensor reading (signed value) [Applicable only in xWR1642/xWR6843/xWR1843] xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: This field is RESERVED. 1 LSB = 1°C

5.13 Sub blocks related to AWR_RF_ASYNC_EVENT_MSG1

NOTE1: All the Monitoring Async events will be sent out periodically at CALIB_MON_TIME_UNIT frame rate (FTTI). The RadarSS/BSS has a queue to hold max 8 transmit API messages (AEs or Responses), the host shall service all the AEs before start of the next FTTI epoch to avoid RadarSS Queue full CPU fault fatal error.

NOTE2: In reporting mode 1 (Quiet mode) if any failure in RadarSS analog or digital monitors, the AWR2243 device will send AWR_AE_MSS_RFERROR_STATUS_SB AE with ERROR_STATUS_FLAG 0x7 in redundant with failure monitoring report. AWR_AE_MSS_RFERROR_STATUS_SB AE is a redundant failure report for the failure in Quiet mode.

NOTE3:

The ERROR_CODE returned part of monitor AE message reports are informative purpose only, these error codes are helpful to debug the cause for monitor failure. Application can log these information and share with TI in case of any runtime errors. The information about these error codes are documented in "API Error Codes" section in page [413](#).

5.13.1 Sub block 0x1000 – RESERVED
5.13.2 Sub block 0x1001 – RESERVED
5.13.3 Sub block 0x1002 – AWR_AE_RF_CPUFAULT_SB

This sub block indicates CPU fault status of BIST SS.

Table [5.97](#) describes the content of this sub block.

Table 5.97: AWR_AE_RF_CPUFAULT_SB response contents

Field Name	Number of bytes	Description														
SBLKID	2	Value = 0x1002														
SBLKLEN	2	Value = 36														
FAULT_TYPE	1	<table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>RF Processor Undefined Instruction Abort</td></tr><tr><td>1</td><td>RF Processor Instruction pre-fetch Abort</td></tr><tr><td>2</td><td>RF Processor Data Access Abort</td></tr><tr><td>3</td><td>RF Processor Firmware Fatal Error</td></tr><tr><td>0x4 - 0xFE</td><td>RESERVED</td></tr><tr><td>0xFF</td><td>No fault</td></tr></table>	Value	Definition	0	RF Processor Undefined Instruction Abort	1	RF Processor Instruction pre-fetch Abort	2	RF Processor Data Access Abort	3	RF Processor Firmware Fatal Error	0x4 - 0xFE	RESERVED	0xFF	No fault
Value	Definition															
0	RF Processor Undefined Instruction Abort															
1	RF Processor Instruction pre-fetch Abort															
2	RF Processor Data Access Abort															
3	RF Processor Firmware Fatal Error															
0x4 - 0xFE	RESERVED															
0xFF	No fault															

Continued on next page

Table 5.97 – continued from previous page

ERROR_CODE	1	The error code for the fault occurred. The error code is defined only for few fatal errors generated either due to wrong configuration of the device or HW limitation. Error Definition Code 0 Undefined error code 1 Rampgen is not triggered from FRC or Hw pulse (FRC is running) 2 Burst start and end counts are not matching in rampgen 3 Chirp start and end counts are not matching in rampgen 4 Calibration/Monitoring chirps not finished at pre burst 5 RadarSS TX mailbox queue full 6 Sequencer extension copy error for a chirp 7 Temperature sensor data is invalid 8 Test source configuration time failure 9 - RESERVED 0xFF
LINE_NUM	2	Valid only in case of FAULT type is 0x3, provides the firmware line number at which fatal error occurred.
FAULT_LR	4	The instruction PC address at which Fault occurred
FAULT_PREV_LR	4	The return address of the function from which fault function has been called (Call stack LR)
FAULT_SPSR	4	The CPSR register value at which fault occurred
FAULT_SP	4	The SP register value at which fault occurred
FAULT_CAUSE_ADDRESS	4	The address access at which Fault occurred (valid only for fault type 0x0 to 0x2)

Continued on next page

Table 5.97 – continued from previous page

FAULT_ERROR_STATUS	2	The status of Error (Error Cause type – valid only for fault type 0x0 to 0x2) 0x000 BACKGROUND_ERR 0x001 ALIGNMENT_ERR 0x002 DEBUG_EVENT 0x00D PERMISSION_ERR 0x008 SYNCH_EXTER_ERR 0x406 ASYNCH_EXTER_ERR 0x409 SYNCH_ECC_ERR 0x408 ASYNCH_ECC_ERR
FAULT_ERROR_SOURCE	1	The Source of the Error (Error Source type - valid only for fault type 0x0 to 0x2) 0x0 ERR_SOURCE_AXI_MASTER 0x1 ERR_SOURCE_ATCM 0x2 ERR_SOURCE_BTCM
FAULT_AXI_ERROR_TYPE	1	The AXI Error type (Error Source type - valid only for fault type 0x0 to 0x2) 0x0 AXI_DECOD_ERR 0x1 AXI_SLAVE_ERR
FAULT_ACCESS_TYPE	1	The Error Access type (Error Access type - valid only for fault type 0x0 to 0x2) 0x0 READ_ERR 0x1 WRITE_ERR
FAULT_RECOVERY_TYPE	1	The Error Recovery type (Error Recovery type - Valid only for fault type 0x0 to 0x2) 0x0 UNRECOVERY 0x1 RECOVERY
RESERVED	2	0x0000

5.13.4 Sub block 0x1003 – AWR_AE_RF_ESMFAULT_SB

This sub block indicates the status of any other faults in the BIST SS.

NOTE: This ASYNC event has been deprecated on xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices. Please use the updated ASYNC event - AWR_AE_RF_ADV_ESMFAULT_SB.

Table 5.98 describes the content of this sub block.

Table 5.98: AWR_AE_RF_ESMFAULT_STATUS_SB response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1003
SBLKLEN	2	Value = 12
ESM_GROUP1_ERRORS	4	Bit Error Information 0 – No Error , 1 – ESM Error b0 RAMPGEN_SB_ERROR b1 RESERVED b2 GPADC_RAM_SB_ERROR b3 VIM_RAM_SB_ERROR b4 RESERVED b5 VIM_SELFTEST_ERROR b6 B0TCM_SB_ERROR b7 B1TCM_SB_ERROR b8 CCMR4_SELFTEST_ERROR b9 ATCM_SB_ERROR b10 RAMPGEN_SELFTEST_ERROR b11 RAMPGEN_PAR_SELFTEST_ERROR b12 SEQ_EXT_SELFTEST_ERROR b13 SEQ_EXT_SB_ERROR b14 RESERVED b15 AGC_RAM_SB_ERROR b16 B1TCM_PAR_CHK_ERROR b17 B0TCM_PAR_CHK_ERROR b18 ATCM_PAR_CHK_ERROR b19 MB_MSS2BSS_SB_ERROR b20 MB_BSS2MSS_SB_ERROR b24:21 RESERVED b25 PROG_FILT_FATAL_DB_ECC_ERROR b31:26 RESERVED

Continued on next page

Table 5.98 – continued from previous page

ESM_GROUP2_ ERRORS	4	Bit	Error Information
		b0	DFE_STC_ERROR
		b1	CR4_STC_ERROR
		b2	CCMR4_COMP_ERROR
		b3	B0TCM_DB_ERROR
		b4	B1TCM_DB_ERROR
		b5	ATCM_DB_ERROR
		b6	DCC_ERROR
		b7	SEQ_EXT_ERROR
		b8	SYNT_FREQ_MON_ERROR
		b9	RESERVED
		b10	RAMPGEN_DB_ERROR
		b11	BUBBLE_CORRECTION_FAIL
		b12	RAMPGEN_LOCSTEP_ERROR
		b13	RTI_RESET_ERROR
		b14	GPADC_RAM_DB_ERROR
		b15	VIM_COMP_ERROR
		b16	CR4_LIVE_LOCK_ERROR
		b17	WDT_NMI_ERROR
		b18	VIM_RAM_DB_ERROR
		b19	RAMPGEN_PAR_ERROR
		b20	SEQ_EXT_DB_ERROR
		b21	DMA_MPU_ERROR
		b22	AGC_RAM_DB_ERROR
		b23	CRC_COMP_ERROR
		b24	WAKEUP_STS_ERROR
		b25	SHORT_CIRCUIT_ERROR
		b26	B1TCM_PAR_ERROR
		b27	B0TCM_PAR_ERROR
		b28	ATCM_PAR_ERROR
		b29	MB_MSS2BSS_DB_ERROR
		b30	MB_BSS2MSS_DB_ERROR
		b31	CCC_ERROR

NOTE:

The Programmable filter Parity error and double bit ECC fatal errors are connected to ESM Group 1 lines, these fatal errors must be handled in Host in case of AWR2243/xWR6243 device.

5.13.5 Sub block 0x1004 – AWR_AE_RF_INITCALIBSTATUS_SB

This sub block indicates the initial calibrations of RF BIST SS are complete.
Table 5.99 describes the content of this sub block.

Table 5.99: AWR_AE_RF_INITCALIBSTATUS_SB response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1004
SBLKLEN	2	Value = 24
CALIBRATION_STATUS	4	This field indicates the status of each calibration (0 – FAIL, 1 – PASS). If a particular calibration was not enabled, then its corresponding field should be ignored. Bit Definition (0 – FAIL, 1 – PASS) b0 Synth VCO3 tuning (Available only on selected xWR6243 device variants, RESERVED for other devices) b1 APLL tuning b2 SYNTH VCO1 tuning b3 SYNTH VCO2 tuning b4 LODIST calibration b5 RX ADC DC offset calibration b6 HPF cutoff calibration b7 LPF cutoff calibration b8 Peak detector calibration b9 TX Power calibration b10 RX gain calibration b11 TX Phase calibration b12 RX IQMM calibration (not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices) b31:13 RESERVED

Continued on next page

Table 5.99 – continued from previous page

CALIBRATION_UPDATE	4	This field indicates if a particular calibration data has been updated in hardware. (0 – no update, 1 – updated) Bit Definition b0 Synth VCO3 tuning (Available only on selected xWR6243 device variants, RESERVED for other devices) b1 APLL tuning b2 SYNTH VCO1 tuning b3 SYNTH VCO2 tuning b4 LODIST calibration b5 RX ADC DC offset calibration b6 HPF cutoff calibration b7 LPF cutoff calibration b8 Peak detector calibration b9 TX Power calibration b10 RX gain calibration b11 TX Phase calibration b12 RX IQMM calibration (not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices) b31:13 RESERVED
TEMPERATURE	2	Measured temperature, based on average of temperature sensors near all enabled TX and RX channels at the time of calibration. 1 LSB = 1°C
RESERVED	2	0x0000
TIME_STAMP	4	This field indicates time stamp at the time of performing calibration updates. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)
RESERVED	4	0x00000000

5.13.6 Sub block 0x1005 – RESERVED

5.13.7 Sub block 0x1006 – RESERVED

5.13.8 Sub block 0x1007 – AWR_AE_RF_ADV_ESMFAULT_SB

This sub block indicates the status of any other faults in the BIST SS in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices.

Table 5.100 describes the content of this sub block.

Table 5.100: AWR_AE_RF_ADV_ESMFAULT_STATUS_SB response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1007
SBLKLEN	2	Value = 28

Continued on next page

Table 5.100 – continued from previous page

ESM_GROUP1_ ERRORS_LSB	4	Bit	Error Information 0 – No Error , 1 – ESM Error
		b0	RAMPGEN_SB_ERROR
		b1	PROG_FILT_SELFTEST_ERROR
		b2	GPADC_RAM_SB_ERROR
		b3	VIM_RAM_SB_ERROR
		b4	DFE_PARITY_AND_OUT_ERROR (RESERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b5	VIM_SELFTEST_ERROR
		b6	B0TCM_SB_ERROR
		b7	B1TCM_SB_ERROR
		b8	CCMR4_SELFTEST_ERROR
		b9	ATCM_SB_ERROR
		b10	RAMPGEN_SELFTEST_ERROR
		b11	RAMPGEN_PAR_SELFTEST_ERROR
		b12	SEQ_EXT_SELFTEST_ERROR
		b13	SEQ_EXT_SB_ERROR
		b14	PROG_FILT_PARITY_ERROR
		b15	AGC_RAM_SB_ERROR
		b16	B1TCM_PAR_CHK_ERROR
		b17	B0TCM_PAR_CHK_ERROR
		b18	ATCM_PAR_CHK_ERROR
		b19	ECC_AGG_DED_ERROR
		b20	ECC_AGG_SEC_ERROR
		b21	RESERVED
		b22	RESERVED
		b23	RESERVED
		b24	RESERVED
		b25	PROG_FILT_UERR
		b26	PROG_FILT_SERR
		b27	FRC_SELFTEST_ERROR
		b28	FRC_ERROR
		b29	RESERVED
		b30	BUS_SAFETY_PCR_ERROR (RESERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b31	BUS_SAFETY_BSS_SLV_ERROR (RESERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)

Continued on next page

Table 5.100 – continued from previous page

ESM_GROUP1_ ERRORS_MSB	4	Bit	Error Information 0 – No Error , 1 – ESM Error
		b0	BUS_SAFETY_BSS_MST_ERROR (RE- SERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b1	BUS_SAFETY_STATIC_MEM_ERROR (RE- SERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b2	BUS_SAFETY_MBOX_ERROR (RESERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b3	BUS_SAFETY_SEC_AGGREGATED_ERROR (RESERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b31-b4	RESERVED
RESERVED	4	0x00000000	

Continued on next page

Table 5.100 – continued from previous page

ESM_GROUP2_ ERRORS	4	Bit	Error Information
		b0	DFE_STC_ERROR
		b1	CR4_STC_ERROR
		b2	CCMR4_COMP_ERROR
		b3	B0TCM_DB_ERROR
		b4	B1TCM_DB_ERROR
		b5	ATCM_DB_ERROR
		b6	DCC_ERROR
		b7	SEQ_EXT_ERROR
		b8	SYNT_FREQ_MON_ERROR
		b9	DFE_PARITY_OR_OUT_ERROR
		b10	RAMPGEN_DB_ERROR
		b11	BUBBLE_CORRECTION_FAIL
		b12	RAMPGEN_LOCSTEP_ERROR
		b13	RTI_RESET_ERROR
		b14	GPADC_RAM_DB_ERROR
		b15	VIM_COMP_ERROR
		b16	CR4_LIVE_LOCK_ERROR
		b17	WDT_NMI_ERROR
		b18	VIM_RAM_DB_ERROR
		b19	RAMPGEN_PAR_ERROR
		b20	SEQ_EXT_DB_ERROR
		b21	DMA_MPU_ERROR
		b22	AGC_RAM_DB_ERROR
		b23	CRC_COMP_ERROR
		b24	WAKEUP_STS_ERROR
		b25	SHORT_CIRCUIT_ERROR
		b26	B1TCM_PAR_ERROR
		b27	B0TCM_PAR_ERROR
		b28	ATCM_PAR_ERROR
		b29	RESERVED
		b30	RESERVED
		b31	CCC_ERROR
RESERVED	4	0x00000000	
RESERVED	4	0x00000000	

5.13.9 Sub block 0x1008 – RESERVED

5.13.10 Sub block 0x1009 – RESERVED

5.13.11 Sub block 0x100A – AWR_AE_RF_MONITOR_TYPE_TRIGGER_DONE_SB

This sub block indicates that, the triggered monitor types are done with execution and Host can use this signal to trigger next type of monitor.

Table 5.101 describes the content of this sub block.

Table 5.101: AWR_AE_RF_MONITOR_TYPE_TRIGGER_DONE_SB response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x100A
SBLKLEN	2	Value = 16
MON_TRIG_TYPE_DONE	1	The bit mask to indicate execution status of monitor triggered type. Bit Definition b0 Done Status of Type 0 monitor trigger b1 Done Status of Type 1 monitor trigger b2 Done Status of Type 2 monitor trigger b31:3 RESERVED
RESERVED	3	0x0
TIME_STMP	4	The device time stamp at which this AE is sent out 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)
RESERVED	4	0x0

NOTE1:	The Done status for each type is cleared only once in end of FTTI interval, example the AE report for type 2 will contains done status bit set for all types.
NOTE2:	If Trigger is done with all 3 bits set (Triggering all 3 types in one go), then still this AE will be sent 3 times for 3 types irrespective of number of trigger.

5.13.12 Sub block 0x100B – AWR_AE_RF_FRAME_TRIGGER_RDY_SB

This sub block indicates that the slave device is now ready to receive the external sync in for frame triggers, this does not indicate physical trigger of frames in Hw triggered mode. In SW triggered mode, this async event indicates that frame is triggered by Sw.

Table 5.102 describes the content of this sub block.

Table 5.102: AWR_AE_RF_FRAME_TRIGGER_RDY_SB response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x100B
SBLKLEN	2	Value = 4

5.13.13 Sub block 0x100C – AWR_AE_RF_GPADC_RESULT_DATA_SB

This sub block indicates that GPADC measurement is complete and it also contains the measured data of each of the enabled channels. The data for channels which are not enabled can be ignored.

Table 5.103 describes the content of this sub block.

Table 5.103: AWR_AE_RF_GPADC_RESULT_DATA_SB response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x100C
SBLKLEN	2	Value = 76
ANATEST1_MIN_DATA	2	Minimum GPADC reading across the captured samples for ANATEST1 input 1 LSB = 1.8V/1024
ANATEST1_MAX_DATA	2	Maximum GPADC reading across the captured samples for ANATEST1 input 1 LSB = 1.8V/1024
ANATEST1_AVG_DATA	2	Average GPADC reading across the captured samples for ANATEST1 input 1 LSB = 1.8V/1024
ANATEST2_MIN_DATA	2	Minimum GPADC reading across the captured samples for ANATEST2 input 1 LSB = 1.8V/1024
ANATEST2_MAX_DATA	2	Maximum GPADC reading across the captured samples for ANATEST2 input 1 LSB = 1.8V/1024
ANATEST2_AVG_DATA	2	Average GPADC reading across the captured samples for ANATEST2 input 1 LSB = 1.8V/1024
ANATEST3_MIN_DATA	2	Minimum GPADC reading across the captured samples for ANATEST3 input 1 LSB = 1.8V/1024

Continued on next page

Table 5.103 – continued from previous page

ANATEST3_ MAX_DATA	2	Maximum GPADC reading across the captured samples for ANATEST3 input 1 LSB = 1.8V/1024
ANATEST3_ AVG_DATA	2	Average GPADC reading across the captured samples for ANATEST3 input 1 LSB = 1.8V/1024
ANATEST4_MIN_ DATA	2	Minimum GPADC reading across the captured samples for ANATEST4 input 1 LSB = 1.8V/1024
ANATEST4_ MAX_DATA	2	Maximum GPADC reading across the captured samples for ANATEST4 input 1 LSB = 1.8V/1024
ANATEST4_ AVG_DATA	2	Average GPADC reading across the captured samples for ANATEST4 input 1 LSB = 1.8V/1024
ANAMUX_MIN_ DATA	2	Minimum GPADC reading across the captured samples for ANAMUX input 1 LSB = 1.8V/1024
ANAMUX_MAX_ DATA	2	Maximum GPADC reading across the captured samples for ANAMUX input 1 LSB = 1.8V/1024
ANAMUX_AVG_ DATA	2	Average GPADC reading across the captured samples for ANAMUX input 1 LSB = 1.8V/1024
VSENSE_MIN_ DATA	2	Minimum GPADC reading across the captured samples for VSENSE input 1 LSB = 1.8V/1024
VSENSE_MAX_ DATA	2	Maximum GPADC reading across the captured samples for VSENSE input 1 LSB = 1.8V/1024
VSENSE_AVG_ DATA	2	Average GPADC reading across the captured samples for VSENSE input 1 LSB = 1.8V/1024
RESERVED	2	0x0000
RESERVED	2	0x0000
RESERVED	2	0x0000
RESERVED	2	0x0000
RESERVED	4	0x00000000
RESERVED	4	0x00000000
RESERVED	4	0x00000000
RESERVED	4	0x00000000

Continued on next page

Table 5.103 – continued from previous page

RESERVED	4	0x00000000
RESERVED	4	0x00000000
RESERVED	4	0x00000000

5.13.14 Sub block 0x100E – RESERVED
5.13.15 Sub block 0x100D – RESERVED
5.13.16 Sub block 0x100E – RESERVED
5.13.17 Sub block 0x100F – AWR_FRAME_END_AE_SB

This sub block indicates end of the frames.

Table 5.104 describes the content of this sub block.

Table 5.104: AWR_FRAME_END_AE_SB response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x100F
SBLKLEN	2	Value = 4

5.13.18 Sub block 0x1010 – AWR_ANALOGFAULT_AE_SB

This sub block indicates fault in analog supplies or LDO short circuit condition. Once a fault is detected the functionality cannot be resumed from then on and the sensor needs to be re-started.

Table 5.105: AWR_ANALOGFAULT_AE_SB response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1010
SBLKLEN	2	Value = 16
FAULT_TYPE	1	Value Definition 0 NO FAULT 1 ANALOG_SUPPLY_FAULT Others RESERVED
RESERVED	1	0x00
RESERVED	2	0x0000

Continued on next page

Table 5.105 – continued from previous page

FAULT_SIG	4	Bit	Definition
		b0	1.8V BB ANA supply fault detected (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP))
		b1	13V, 1.0V RF supply fault detected (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP))
		b2	Synth VCO LDO short circuit detected
		b3	PA LDO short circuit detected (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP)
		b31:4	RESERVED
RESERVED	4	0x00000000	

5.13.19 Sub block 0x1011 – AWR_CAL_MON_TIMING_FAIL_REPORT_AE_SB

This sub block indicates any timing failure related to calibration or monitoring.

Table 5.106 describes the content of this sub block.

Table 5.106: AWR_CAL_MON_TIMING_FAIL_REPORT_AE_SB response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1011
SBLKLEN	2	Value = 8

Continued on next page

Table 5.106 – continued from previous page

TIMING_FAILURE_CODE	2	Bit	Definition
		b0	RESERVED
		b1	0 No Failure 1 Total monitoring and calibration time do not fit in one CALIB_MON_TIME_UNIT in AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER when ONE_TIME_CALIB is enabled
		b2	0 No Failure 1 Total monitoring and calibration time do not fit in one CALIB_MON_TIME_UNIT in AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER when PERIODIC_CALIB is enabled
		b3	0 No Failure 1 Runtime timing violation: Monitoring functions or calibrations could not be completed in one CALIB_MON_TIME_UNIT
		b15:4	RESERVED
RESERVED	2	0x0000	

NOTE: In QM devices (non safety), Periodic Digital and Analog Monitoring are not supported.

5.13.20 Sub block 0x1012 – AWR_RUN_TIME_CALIB_SUMMARY_REPORT_AE_SB

This sub block indicates the calibration status (one time or run time) if the calibration reports are enabled in the AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB.

NOTE: The calibration report is sent if the calibrations are triggered due to temperature change or whenever the internal calibrations are triggered i.e. every 1 s

Table 5.107: AWR_RUN_TIME_CALIB_SYMMARY_REPORT_AE_SB
response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1012
SBLKLEN	2	Value = 24
CALIBRATION_ERROR_FLAG	4	This field indicates the status of each calibration. 1 - calibration is passed, 0 - calibration is failed or not enabled/performed at least once. Bit Definition b0 SYNTH VCO3 tuning (Available only on selected xWR6243 device variants, RESERVED for other devices) b1 APLL tuning b2 SYNTH VCO1 tuning b3 SYNTH VCO2 tuning b4 LODIST calibration b5 RESERVED b6 RESERVED b7 RESERVED b8 PD calibration b9 TX power calibration b10 RX gain calibration b11 TX phasemitter calibration (only for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) b12 RESERVED b31:13 RESERVED

Continued on next page

Table 5.107 – continued from previous page

CALIBRATION_UPDATE_STATUS	4	Each bit corresponding to a calibration indicates if each calibration resulted in a reconfiguration of RF is indicated by a value of 1 in the respective bit in this field. 0 – Analog/RF is not updated 1 – Analog/RF is updated after a respective calibration Bit Definition b0 SYNTH VCO3 tuning (Available only on selected xWR6243 device variants, RESERVED for other devices) b1 APLL tuning b2 SYNTH VCO1 tuning b3 SYNTH VCO2 tuning b4 LODIST calibration b5 RESERVED b6 RESERVED b7 RESERVED b8 PD calibration b9 TX power calibration b10 RX gain calibration b11 TX phasemitter calibration (only for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) b12 RESERVED b31:13 RESERVED
TEMPERATURE	2	Measured temperature, based on average of temperature sensors near all enabled TX and RX channels at the time of calibration. Note that this temperature will be updated only when a run-time calibration is executed due to a change in temperature by more than 10 deg C. 1 LSB = 1°C
RESERVED	2	RESERVED
TIME_STAMP	4	This field indicates time stamp at the time of performing calibration updates. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)
RESERVED	4	0x00000000

NOTE: None of the Periodic Monitoring are supported in QM devices (IWR6843 QM, xWR1443, IWR1642 and IWR1843), The Async Event sub-blocks defined below from ID 0x1015 to 0x1031 are not valid in QM devices.

5.13.21 Sub block 0x1013 – AWR_MONITOR_RF_DIG_LATENTFAULT_REPORT_AE_SB

This async event contains the status of digital monitoring for latent faults.

Table 5.108: AWR_MONITOR_RF_DIG_LATENTFAULT_REPORT_AE_SB
response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1013
SBLKLEN	2	Value = 8

Continued on next page

Table 5.108 – continued from previous page

DIG_MON_LATENT_FAULT_STATUS	4	1 – PASS, 0 – FAIL
		Bit Definition
		b0 RESERVED
		b1 CR4 and VIM lockstep test
		b2 RESERVED
		b3 VIM test
		b4 RESERVED
		b5 RESERVED
		b6 CRC test
		b7 RAMPGEN memory ECC test
		b8 DFE parity test
		b9 DFE memory ECC test
		b10 RAMPGEN lockstep test
		b11 FRC lockstep test
		b12 RESERVED
		b13 RESERVED
		b14 RESERVED
		b15 RESERVED
		b16 ESM test
		b17 DFE STC
		b18 RESERVED
		b19 ATCM, BTCM ECC test
		b20 ATCM, BTCM parity test
		b21 DCC test
		b22 RESERVED
		b23 RESERVED
		b24 FFT test (RESERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b25 RTI test
		b26 RESERVED
		b31:27 RESERVED

5.13.22 Sub block 0x1014 – RESERVED

5.13.23 Sub block 0x1015 – AWR_MONITOR_REPORT_HEADER_AE_SB

The report header includes common information across all enabled monitors like current FTTI number and current temperature.

Table 5.109: AWR_MONITORING_REPORT_HEADER_AE_SB response contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1015
SBLKLEN	2	Value = 12
FTTI_COUNT	4	FTTI free running counter value, incremented every CALIB_MON_TIME_UNIT
AVG_TEMPERATURE	2	Average temperature at which was monitoring performed
RESERVED	2	0x0000

5.13.24 Sub block 0x1016 – AWR_MONITOR_RF_DIG_PERIODIC_REPORT_AE_SB

This async event is sent periodically to indicate the status of periodic digital monitoring tests.

Table 5.110: AWR_MONITOR_RF_DIG_PERIODIC_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1016
SBLKLEN	2	Value = 12
RF_DIG_MON_PERIODIC_STATUS	4	1 – PASS, 0 – FAIL Bit Monitoring type b0 PERIODIC_CONFIG_REGISTER_READ b1 RESERVED b2 DFE_STC b3 FRAME_TIMING_MONITORING b31:4 RESERVED
TIMESTAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

NOTE1: FRAME_TIMING_MONITORING cannot be used when RSS dynamic frequency switching feature is enabled.

Note1:

5.13.25 Sub block 0x1017 – AWR_MONITOR_TEMPERATURE_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured temperature near various RF analog and digital modules. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.111: AWR_MONITORING_TEMPERATURE_REPORT_AE_SB
contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1017
SBLKLEN	2	Value = 36
STATUS_FLAGS	2	Status flags indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_ANA_TEMP_MIN b1 STATUS_ANA_TEMP_MAX b2 STATUS_DIG_TEMP_MIN (Reserved for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices) b3 STATUS_DIG_TEMP_MAX (Reserved for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices) b4 STATUS_TEMP_DIFF_THRESH b15:5 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error

Continued on next page

Table 5.111 – continued from previous page

TEMP_VALUES	20	The measured onchip temperature is reported here. Byte numbers corresponding to different temperature sensors reported in this field are here: AWR2243/xWR6243 devices: Bytes Temperature sensor 1:0 TEMP_RX0 3:2 TEMP_RX1 5:4 TEMP_RX2 7:6 TEMP_RX3 9:8 TEMP_TX0 11:10 TEMP_TX1 13:12 TEMP_TX2 15:14 TEMP_PM 17:16 TEMP_DIG1 19:18 TEMP_DIG2 (Applicable only in xWR1642/xWR6843/xWR1843) xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: Bytes Temperature sensor 1:0 TEMP_RX0 3:2 TEMP_RX1 5:4 TEMP_RX2 7:6 TEMP_RX3 9:8 TEMP_TX0 11:10 TEMP_TX1 13:12 TEMP_TX2 15:14 TEMP_PM 17:16 TEMP_TX3 19:18 RESERVED 1 LSB = 1 °C, signed number
RESERVED	4	0x00000000
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

5.13.26 Sub block 0x1018 – AWR_MONITOR_RX_GAIN_PHASE_REPORT_AE_SB

This sub block is a monitoring report which the AWR device sends to the host, containing the measured RX Gain and Phase values, Loopback Power and Noise Power. Noise Power can be used by the Host to detect the presence of interference. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.112: AWR_MONITOR_RX_GAIN_PHASE_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1018
SBLKLEN	2	Value = 72
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_RX_GAIN_ABS b1 STATUS_RX_GAIN_MISMATCH b2 STATUS_RX_GAIN_FLATNESS b3 STATUS_RX_PHASE_MISMATCH b15:4 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_INDXX	1	Profile Index for which this monitoring report applies.

Continued on next page

Table 5.112 – continued from previous page

LOOPBACK_ POWER	3	The measured average loop-back power across RX channels at each enabled RF frequency (i.e., lowest, center and highest with 60MHz dither in the profile's RF band) at LNA input is reported here. Byte numbers corresponding to different RF, in this field are here: <table><tr><td>RF1</td><td>RF2</td><td>RF3</td></tr><tr><td>(Byte 0)</td><td>(Byte 1)</td><td>(Byte 2)</td></tr><tr><td>b4:b0</td><td>b4:b0</td><td>b4:b0</td></tr></table> b7-b5 : RESERVED in each bytes 1 LSB = -2 dBm Valid Range = -62dBm to 0dBm Only the entries of enabled RF Frequencies are valid. NOTE: The Loopback power can optionally be used to improve the RX gain estimation accuracy. But time domain filtering across many successive monitoring reports is recommended to mitigate their corruption by external interference.	RF1	RF2	RF3	(Byte 0)	(Byte 1)	(Byte 2)	b4:b0	b4:b0	b4:b0											
RF1	RF2	RF3																				
(Byte 0)	(Byte 1)	(Byte 2)																				
b4:b0	b4:b0	b4:b0																				
RX_GAIN_ VALUE	24	The measured RX gain for each enabled channel, at each enabled RF frequency (i.e., lowest, center and highest in the profile's RF band) is reported here. Byte numbers corresponding to different RX and RF, in this field are here: <table><tr><td></td><td>RF1</td><td>RF2</td><td>RF3</td></tr><tr><td>RX0</td><td>1:0</td><td>9:8</td><td>17:16</td></tr><tr><td>RX1</td><td>3:2</td><td>11:10</td><td>19:18</td></tr><tr><td>RX2</td><td>5:4</td><td>13:12</td><td>21:20</td></tr><tr><td>RX3</td><td>7:6</td><td>15:14</td><td>23:22</td></tr></table> 1 LSB = 0.1 dB Only the entries of enabled RF Frequencies and enabled RX channels are valid.		RF1	RF2	RF3	RX0	1:0	9:8	17:16	RX1	3:2	11:10	19:18	RX2	5:4	13:12	21:20	RX3	7:6	15:14	23:22
	RF1	RF2	RF3																			
RX0	1:0	9:8	17:16																			
RX1	3:2	11:10	19:18																			
RX2	5:4	13:12	21:20																			
RX3	7:6	15:14	23:22																			

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Table 5.112 – continued from previous page

RX_PHASE_ VALUE	24	The measured RX phase for each enabled channel, at each enabled RF frequency is reported here. Byte numbers corresponding to different RX and RF, in this field are here: <table> <tr> <td></td> <td>RF1</td> <td>RF2</td> <td>RF3</td> </tr> <tr> <td>RX0</td> <td>1:0</td> <td>9:8</td> <td>17:16</td> </tr> <tr> <td>RX1</td> <td>3:2</td> <td>11:10</td> <td>19:18</td> </tr> <tr> <td>RX2</td> <td>5:4</td> <td>13:12</td> <td>21:20</td> </tr> <tr> <td>RX3</td> <td>7:6</td> <td>15:14</td> <td>23:22</td> </tr> </table> 1 LSB = 360°/2¹⁶ Only the entries of enabled RF Frequencies and enabled RX channels are valid. NOTE: These phases include an unknown bias common to all RX channels.		RF1	RF2	RF3	RX0	1:0	9:8	17:16	RX1	3:2	11:10	19:18	RX2	5:4	13:12	21:20	RX3	7:6	15:14	23:22
	RF1	RF2	RF3																			
RX0	1:0	9:8	17:16																			
RX1	3:2	11:10	19:18																			
RX2	5:4	13:12	21:20																			
RX3	7:6	15:14	23:22																			
RX_NOISE_ POWER1	4	The measured RX noise for each enabled channel, at each enabled RF frequency is reported here. Bit fields corresponding to different RX in RF1 and RF2 (partial in this word) are defined in this field: <table> <tr> <td></td> <td>RF1</td> <td>RF2</td> <td>RF3</td> </tr> <tr> <td>RX0</td> <td>b4:b0</td> <td>b24:b20</td> <td>-</td> </tr> <tr> <td>RX1</td> <td>b9:b5</td> <td>b29:b25</td> <td>-</td> </tr> <tr> <td>RX2</td> <td>b14:b10</td> <td>-</td> <td>-</td> </tr> <tr> <td>RX3</td> <td>b19:b15</td> <td>-</td> <td>-</td> </tr> </table> b31-b30 : RESERVED 1 LSB = -2dBm Valid Range: -62dbm to 0dBm Only the entries of enabled RF Frequencies and enabled RX channels are valid. NOTE: This field can enable the host in detecting if the corresponding gain/phase measurement was potentially corrupted by interference or not. For example, if the reported noise power exceeds significantly from typical values (e.g. based on median of the reported values in the past few 100 mili-seconds), it can indicate that the gain/phase measurement is potentially corrupted by interference. Such gain/phase measurement reports may be discarded and the results from the next monitoring interval or from other RF frequencies may be used instead.		RF1	RF2	RF3	RX0	b4:b0	b24:b20	-	RX1	b9:b5	b29:b25	-	RX2	b14:b10	-	-	RX3	b19:b15	-	-
	RF1	RF2	RF3																			
RX0	b4:b0	b24:b20	-																			
RX1	b9:b5	b29:b25	-																			
RX2	b14:b10	-	-																			
RX3	b19:b15	-	-																			

Continued on next page

Table 5.112 – continued from previous page

RX_NOISE_POWER2	4	The measured RX noise for each enabled channel, at each enabled RF frequency is reported here. Bit fields corresponding to different RX in RF2 (partial in this word) and RF3 are defined in this field: <table><thead><tr><th></th><th>RF1</th><th>RF2</th><th>RF3</th></tr></thead><tbody><tr><td>RX0</td><td>-</td><td>-</td><td>b14:b10</td></tr><tr><td>RX1</td><td>-</td><td>-</td><td>b19:b15</td></tr><tr><td>RX2</td><td>-</td><td>b4:b0</td><td>b24:b20</td></tr><tr><td>RX3</td><td>-</td><td>b9:b5</td><td>b29:b25</td></tr></tbody></table> b31-b30 : RESERVED 1 LSB = -2dBm Valid Range: -62dbm to 0dBm Only the entries of enabled RF Frequencies and enabled RX channels are valid. NOTE: This field can enable the host in detecting if the corresponding gain/phase measurement was potentially corrupted by interference or not. For example, if the reported noise power exceeds significantly from typical values (e.g. based on median of the reported values in the past few 100 milli-seconds), it can indicate that the gain/phase measurement is potentially corrupted by interference. Such gain/phase measurement reports may be discarded and the results from the next monitoring interval or from other RF frequencies may be used instead.		RF1	RF2	RF3	RX0	-	-	b14:b10	RX1	-	-	b19:b15	RX2	-	b4:b0	b24:b20	RX3	-	b9:b5	b29:b25
	RF1	RF2	RF3																			
RX0	-	-	b14:b10																			
RX1	-	-	b19:b15																			
RX2	-	b4:b0	b24:b20																			
RX3	-	b9:b5	b29:b25																			
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)																				

5.13.27 Sub block 0x1019 – AWR_MONITOR_RX_NOISE_FIGURE_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured RX noise figure values corresponding to the full IF band of a profile. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.113: AWR_MONITOR_RX_NOISE_FIGURE_REPORT_AE_SB
contents

Field Name	Number of bytes	Description																				
SBLKID	2	Value = 0x1019																				
SBLKLEN	2	Value = 52																				
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_RX_NOISE_FIGURE b15:1 RESERVED 0 – FAIL or check wasn't done 1 – PASS																				
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error																				
PROFILE_INDX	1	Profile Index for which this monitoring report applies.																				
RESERVED	3	0x000000																				
RX_NOISE_FIG- URE_VALUE	24	The measured RX input referred for each enabled channel, at each enabled RF frequency is reported here. Byte numbers corresponding to different RX and RF, in this field are here: <table><tr><td></td><td>RF1</td><td>RF2</td><td>RF3</td></tr><tr><td>RX0</td><td>1:0</td><td>9:8</td><td>17:16</td></tr><tr><td>RX1</td><td>3:2</td><td>11:10</td><td>19:18</td></tr><tr><td>RX2</td><td>5:4</td><td>13:12</td><td>21:20</td></tr><tr><td>RX3</td><td>7:6</td><td>15:14</td><td>23:22</td></tr></table> 1 LSB = 0.1 dB, signed number Only the entries of enabled RF Frequencies and enabled RX channels are valid.		RF1	RF2	RF3	RX0	1:0	9:8	17:16	RX1	3:2	11:10	19:18	RX2	5:4	13:12	21:20	RX3	7:6	15:14	23:22
	RF1	RF2	RF3																			
RX0	1:0	9:8	17:16																			
RX1	3:2	11:10	19:18																			
RX2	5:4	13:12	21:20																			
RX3	7:6	15:14	23:22																			
RESERVED	4	0x00000000																				
RESERVED	4	0x00000000																				
RESERVED	4	0x00000000																				
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)																				

NOTE: The noise monitor reports the real baseband receivers' noise figure with LNA disabled (to suppress external interference's influence). In complex receiver modes (i.e., complex 1x, complex 2x and pseudo real), the system noise figure is 3dB lower (better) than the reported number.

5.13.28 Sub block 0x101A – AWR_MONITOR_RX_IFSTAGE_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured RX IF filter attenuation values at the given IF frequencies. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.114: AWR_MONITOR_RX_IFSTAGE_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x101A
SBLKLEN	2	Value = 48
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_RX_HPF_ERROR b1 STATUS_RX_LPF_ERROR b2 STATUS_RX_IFA_GAIN_ERROR b15:3 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_INDX	1	Profile Index for which this monitoring report applies.
RESERVED	1	0x00

Continued on next page

Table 5.114 – continued from previous page

LPF_CUTOFF_ BANDEDGE_ DROOP_VALUE_ RX0	2	The RX IFA LPF cutoff band edge droop at analog LPFs intended band edge wrt in band for RX 0, I and Q channels are reported here. Byte numbers corresponding to measured band edge droop on different RX channels, in this field are here: <table> <tr> <td></td><td>I channel</td><td>Q channel</td></tr> <tr> <td>RX0</td><td>0</td><td>1</td></tr> </table> 1 LSB = 0.2dB, signed number Applicable only for the enabled channels. Note: In xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices, these values should be used for debug purposes only.		I channel	Q channel	RX0	0	1									
	I channel	Q channel															
RX0	0	1															
HPF_CUTOFF_ FREQ_ERROR_ VALUE	8	The deviations of RX IFA HPF cutoff frequency from the ideally expected values for all the enabled RX channels are reported here. $HPF_CUTOFF_FREQ_ERROR = 100 * (Measured\ Cutoff\ Frequency / Expected\ Cutoff\ Frequency) - 100$, for RX IF filter in the HPF region. Byte numbers corresponding to measured cutoff frequency error on different RX channels, in this field are here: <table> <tr> <td></td><td>I channel</td><td>Q channel</td></tr> <tr> <td>RX0</td><td>0</td><td>4</td></tr> <tr> <td>RX1</td><td>1</td><td>5</td></tr> <tr> <td>RX2</td><td>2</td><td>6</td></tr> <tr> <td>RX3</td><td>3</td><td>7</td></tr> </table> 1 LSB = 1%, signed number Applicable only for the enabled channels.		I channel	Q channel	RX0	0	4	RX1	1	5	RX2	2	6	RX3	3	7
	I channel	Q channel															
RX0	0	4															
RX1	1	5															
RX2	2	6															
RX3	3	7															

Continued on next page

Table 5.114 – continued from previous page

LPF_CUTOFF_STOPBAND_ATTEN_VALUE	8	The RX IFA LPF stop band attenuation at 2x analog LPF's band edge wrt analog LPF's band edge for all the enabled RX channels are reported here. Byte numbers corresponding to measured stop band attenuation on different RX I and Q channels, in this field are here: <table> <thead> <tr> <th></th><th>I channel</th><th>Q channel</th></tr> </thead> <tbody> <tr> <td>RX0</td><td>0</td><td>4</td></tr> <tr> <td>RX1</td><td>1</td><td>5</td></tr> <tr> <td>RX2</td><td>2</td><td>6</td></tr> <tr> <td>RX3</td><td>3</td><td>7</td></tr> </tbody> </table> 1 LSB = 0.2dB, signed number Applicable only for the enabled channels. Note: In xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices, these values should be used for debug purposes only.		I channel	Q channel	RX0	0	4	RX1	1	5	RX2	2	6	RX3	3	7
	I channel	Q channel															
RX0	0	4															
RX1	1	5															
RX2	2	6															
RX3	3	7															
RX_IFA_GAIN_ERROR_VALUE	8	The deviations of RX IFA Gain from the ideally expected values for all the enabled RX channels are reported here. Byte numbers corresponding to measured cutoff frequency error on different RX channels and HPF/LPF, in this field are here: <table> <thead> <tr> <th></th><th>I channel</th><th>Q channel</th></tr> </thead> <tbody> <tr> <td>RX0</td><td>0</td><td>4</td></tr> <tr> <td>RX1</td><td>1</td><td>5</td></tr> <tr> <td>RX2</td><td>2</td><td>6</td></tr> <tr> <td>RX3</td><td>3</td><td>7</td></tr> </tbody> </table> 1 LSB = 0.1 dB, signed number Applicable only for the enabled channels.		I channel	Q channel	RX0	0	4	RX1	1	5	RX2	2	6	RX3	3	7
	I channel	Q channel															
RX0	0	4															
RX1	1	5															
RX2	2	6															
RX3	3	7															
IFA_GAIN_EXP	1	Expected IFA gain 1 LSB = 1 dB															
RESERVED	1	0x00															

Continued on next page

Table 5.114 – continued from previous page

LPF_CUTOFF_ BANDEDGE_ DROOP_VALUE_ RX	6	The RX IFA LPF cutoff band edge droop at analog LPFs intended band edge wrt in band for RX 1 to 3, I and Q channels are reported here. Byte numbers corresponding to measured stop band edge droop on different RX channels, in this field are here: <table> <tr> <td></td><td>I channel</td><td>Q channel</td></tr> <tr> <td>RX1</td><td>0</td><td>1</td></tr> <tr> <td>RX2</td><td>2</td><td>3</td></tr> <tr> <td>RX3</td><td>4</td><td>5</td></tr> </table> 1 LSB = 0.2dB, signed number Applicable only for the enabled channels. Note: In xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices, these values should be used for debug purposes only.		I channel	Q channel	RX1	0	1	RX2	2	3	RX3	4	5
	I channel	Q channel												
RX1	0	1												
RX2	2	3												
RX3	4	5												
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)												

5.13.29 Sub block 0x101B – AWR_MONITOR_TX0_POWER_REPORT_AE_SB

NOTE1:	The TX[0:3] power monitoring accuracy degrades at high TX back-offs and is unreliable for backoffs higher than 20dB on AWR2243, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP and 26dB on xWR6x43.
NOTE2:	The 0dB back-off corresponds to typically 13dBm power level in AWR2243/xWR294x/xWR254x/xWR6x43 device.
NOTE3:	The 0dB back-off corresponds to typically 13.5dBm power level in xWR2x4xECO, xWR2x4xP device.

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured TX power values during an explicit monitoring chirp. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.115: AWR_MONITOR_TX0_POWER_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x101B
SBLKLEN	2	Value = 24
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_ABS_ERR b1 STATUS_FLATNESS_ERR b15:2 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_INDX	1	Profile Index for which this monitoring report applies
RESERVED	3	0x000000
TX_POWER_VALUE	6	The measured TX power for each enabled channel, at each enabled RF frequency is reported here. Byte numbers corresponding to different TX and RF, in this field are here: RF1 RF2 RF3 TX0 1:0 3:2 5:4 (other bytes are reserved) 1 LSB = 0.1 dBm, signed number Only the entries of enabled RF Frequencies and enabled RX channels are valid.
RESERVED	2	0x0000
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

5.13.30 Sub block 0x101C – AWR_MONITOR_TX1_POWER_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured TX power values during an explicit monitoring chirp. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.116: AWR_MONITOR_TX1_POWER_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x101C
SBLKLEN	2	Value = 24
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_ABS_ERR b1 STATUS_FLATNESS_ERR b15:2 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_INDX	1	Profile Index for which this monitoring report applies
RESERVED	3	0x000000
TX_POWER_VALUE	6	The measured TX power for each enabled channel, at each enabled RF frequency is reported here. Byte numbers corresponding to different TX and RF, in this field are here: RF1 RF2 RF3 TX1 1:0 3:2 5:4 (other bytes are reserved) 1 LSB = 0.1 dBm, signed number Only the entries of enabled RF Frequencies and enabled RX channels are valid.
RESERVED	2	0x0000
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

5.13.31 Sub block 0x101D – AWR_MONITOR_TX2_POWER_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured TX power values during an explicit monitoring chirp. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.117: AWR_MONITOR_TX2_POWER_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x101D
SBLKLEN	2	Value = 24
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_ABS_ERR b1 STATUS_FLATNESS_ERR b15:2 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_INDX	1	Profile Index for which this monitoring report applies
RESERVED	3	0x000000
TX_POWER_VALUE	6	The measured TX power for each enabled channel, at each enabled RF frequency is reported here. Byte numbers corresponding to different TX and RF, in this field are here: RF1 RF2 RF3 TX2 1:0 3:2 5:4 (other bytes are reserved) 1 LSB = 0.1 dBm, signed number Only the entries of enabled RF Frequencies and enabled RX channels are valid.
RESERVED	2	0x0000
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

5.13.32 Sub block 0x101E – AWR_MONITOR_TX0_BALLBREAK_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured TX reflection coefficient's magnitude values, meant for detecting TX ball break. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.118: AWR_MONITOR_TX0_BALLBREAK_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x101E
SBLKLEN	2	Value = 20
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_TX0_BALLBREAK b15:1 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
TX_REFL_CO-EFF_VALUE	2	The TX reflection coefficient's magnitude for this channel is reported here. 1 LSB = 0.1 dB, signed number
RESERVED	2	0x0000
RESERVED	4	0x00000000
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

5.13.33 Sub block 0x101F – AWR_MONITOR_TX1_BALLBREAK_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured TX reflection coefficient's magnitude values, meant for detecting TX ball break. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.119: AWR_MONITOR_TX1_BALLBREAK_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x101F
SBLKLEN	2	Value = 20

Continued on next page

Table 5.119 – continued from previous page

STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_TX0_BALLBREAK b15:1 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
TX_REFL_CO-EFF_VALUE	2	The TX reflection coefficient's magnitude for this channel is reported here. 1 LSB = 0.1 dB, signed number
RESERVED	2	0x0000
RESERVED	4	0x00000000
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

5.14 Sub blocks related to AWR_RF_ASYNC_EVENT_MSG2

5.14.1 Sub block 0x1020 – AWR_MONITOR_TX2_BALLBREAK_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured TX reflection coefficient's magnitude values, meant for detecting TX ball break. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.120: AWR_MONITOR_TX2_BALLBREAK_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1020
SBLKLEN	2	Value = 20

Continued on next page

Table 5.120 – continued from previous page

STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_TX0_BALLBREAK b15:1 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
TX_REFL_CO-EFF_VALUE	2	The TX reflection coefficient's magnitude for this channel is reported here. 1 LSB = 0.1 dB, signed number
RESERVED	2	0x0000
RESERVED	4	0x00000000
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

5.14.2 Sub block 0x1021 – AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured TX gain and phase mismatch values during an explicit monitoring chirp. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.121: AWR_MONITOR_TX_GAIN_PHASE_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1021
SBLKLEN	2	Value = 60

Continued on next page

Table 5.121 – continued from previous page

STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_TX_GAIN_MISMATCH b1 STATUS_TX_PHASE_MISMATCH b15:2 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_INDX	1	Profile Index for which this monitoring report applies
RF1_TX_NOISE_POWER	3	The measured wide band noise power at RF1 frequency for each enabled TX channel is reported here. Bit fields

Continued on next page

Table 5.121 – continued from previous page

TX_PHASE_ VALUE	18	The measured TX phase for each enabled channel, at each enabled RF frequency is reported here. Byte numbers corresponding to different TX and RF, in this field are here: <table><tr><td></td><td>RF1</td><td>RF2</td><td>RF3</td></tr><tr><td>TX0</td><td>1:0</td><td>7:6</td><td>13:12</td></tr><tr><td>TX1</td><td>3:2</td><td>9:8</td><td>15:14</td></tr><tr><td>TX2</td><td>5:4</td><td>11:10</td><td>17:16</td></tr></table> 1 LSB = $360^{\circ}/2^{16}$ Only the entries of enabled RF Frequencies and enabled TX channels are valid. NOTE: In the gains/phases reported here, only inter-TX mismatches carry information, and the raw values may include unknown biases (which cannot be relied on).		RF1	RF2	RF3	TX0	1:0	7:6	13:12	TX1	3:2	9:8	15:14	TX2	5:4	11:10	17:16
	RF1	RF2	RF3															
TX0	1:0	7:6	13:12															
TX1	3:2	9:8	15:14															
TX2	5:4	11:10	17:16															
RF2_TX_NOISE_ POWER	3	The measured wide band noise power at RF2 frequency for each enabled TX channel is reported here. <table><tr><th>Bit fields</th><th>Description</th></tr><tr><td>b7:0</td><td>TX0 wide band noise power at RF2 frequency</td></tr><tr><td>b15:8</td><td>TX1 wide band noise power at RF2 frequency</td></tr><tr><td>b23:16</td><td>TX2 wide band noise power at RF2 frequency</td></tr></table> 1 LSB = -1 dBm Valid Range: 0 to -63 dBm	Bit fields	Description	b7:0	TX0 wide band noise power at RF2 frequency	b15:8	TX1 wide band noise power at RF2 frequency	b23:16	TX2 wide band noise power at RF2 frequency								
Bit fields	Description																	
b7:0	TX0 wide band noise power at RF2 frequency																	
b15:8	TX1 wide band noise power at RF2 frequency																	
b23:16	TX2 wide band noise power at RF2 frequency																	
RF3_TX_NOISE_ POWER	3	The measured wide band noise power at RF3 frequency for each enabled TX channel is reported here. <table><tr><th>Bit fields</th><th>Description</th></tr><tr><td>b7:0</td><td>TX0 wide band noise power at RF3 frequency</td></tr><tr><td>b15:8</td><td>TX1 wide band noise power at RF3 frequency</td></tr><tr><td>b23:16</td><td>TX2 wide band noise power at RF3 frequency</td></tr></table> 1 LSB = -1 dBm Valid Range: 0 to -63 dBm	Bit fields	Description	b7:0	TX0 wide band noise power at RF3 frequency	b15:8	TX1 wide band noise power at RF3 frequency	b23:16	TX2 wide band noise power at RF3 frequency								
Bit fields	Description																	
b7:0	TX0 wide band noise power at RF3 frequency																	
b15:8	TX1 wide band noise power at RF3 frequency																	
b23:16	TX2 wide band noise power at RF3 frequency																	
RESERVED	2	0x0000																
TIMESTAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)																

5.14.3 Sub block 0x1022 – AWR_MONITOR_TX0_PHASE_SHIFTER_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured TX0 phase values, amplitude values and noise power. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.122: AWR_MONITOR_TX0_PHASE_SHIFTER_REPORT_AE_SB
contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1022
SBLKLEN	2	Value = 44
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_TX0_PHASE_SHIFTER_PHASE b1 STATUS_TX0_PHASE_SHIFTER_AMPLITUDE b15:2 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_INDX	1	Profile Index for which this monitoring report applies
RESERVED	3	RESERVED
PH_SHIFTER_MON_VAL1	2	The measured phase of TX0 Loop-back tone at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON1. 1 LSB = $360^\circ / 2^{16}$
PH_SHIFTER_MON_VAL2	2	The measured phase of TX0 Loop-back tone at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON2. 1 LSB = $360^\circ / 2^{16}$
PH_SHIFTER_MON_VAL3	2	The measured phase of TX0 Loop-back tone at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON3. 1 LSB = $360^\circ / 2^{16}$

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Table 5.122 – continued from previous page

PH_SHIFTER_MON_VAL4	2	The measured phase of TX0 Loop-back tone at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON4. 1 LSB = $360^\circ / 2^{16}$
TX_PS_AMPLITUDE_VAL1	2	The measured amplitude of TX0 Loopback tone power at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON1. 1 LSB = 0.1 dB, signed number
TX_PS_AMPLITUDE_VAL2	2	The measured amplitude of TX0 Loopback tone power at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON2. 1 LSB = 0.1 dB, signed number
TX_PS_AMPLITUDE_VAL3	2	The measured amplitude of TX0 Loopback tone power at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON3. 1 LSB = 0.1 dB, signed number
TX_PS_AMPLITUDE_VAL4	2	The measured amplitude of TX0 Loopback tone power at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON4. 1 LSB = 0.1 dB, signed number
TX_PS_NOISE_VAL1	1	The maximum measured wideband power across the enabled RXs of TX0 Loopback at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON1. 1 LSB = -1 dBm, Valid Range: 0 to -63 dBm
TX_PS_NOISE_VAL2	1	The maximum measured wideband power across the enabled RXs of TX0 Loopback at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON2. 1 LSB = -1 dBm, Valid Range: 0 to -63 dBm
TX_PS_NOISE_VAL3	1	The maximum measured wideband power across the enabled RXs of TX0 Loopback at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON3. 1 LSB = -1 dBm, Valid Range: 0 to -63 dBm
TX_PS_NOISE_VAL4	1	The maximum measured wideband power across the enabled RXs of TX0 Loopback at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON4. 1 LSB = -1 dBm, Valid Range: 0 to -63 dBm
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

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Table 5.122 – continued from previous page

RESERVED	8	RESERVED
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5.14.4 Sub block 0x1023 – AWR_MONITOR_TX1_PHASE_SHIFTER_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured TX1 phase values, amplitude values and noise power. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.123: AWR_MONITOR_TX1_PHASE_SHIFTER_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1023
SBLKLEN	2	Value = 44
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_TX1_PHASE_SHIFTER_PHASE b1 STATUS_TX1_PHASE_SHIFTER_AMPLITUDE b15:2 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_INDX	1	Profile Index for which this monitoring report applies
RESERVED	3	RESERVED
PH_SHIFTER_MON_VAL1	2	The measured phase of TX1 Loop-back tone at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON1. 1 LSB = $360^\circ / 2^{16}$
PH_SHIFTER_MON_VAL2	2	The measured phase of TX1 Loop-back tone at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON2. 1 LSB = $360^\circ / 2^{16}$

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Table 5.123 – continued from previous page

PH_SHIFTER_MON_VAL3	2	The measured phase of TX1 Loop-back tone at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON3. 1 LSB = $360^\circ / 2^{16}$
PH_SHIFTER_MON_VAL4	2	The measured phase of TX1 Loop-back tone at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON4. 1 LSB = $360^\circ / 2^{16}$
TX_PS_AMPLITUDE_VAL1	2	The measured amplitude of TX1 Loopback tone power at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON1. 1 LSB = 0.1 dB, signed number
TX_PS_AMPLITUDE_VAL2	2	The measured amplitude of TX1 Loopback tone power at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON2. 1 LSB = 0.1 dB, signed number
TX_PS_AMPLITUDE_VAL3	2	The measured amplitude of TX1 Loopback tone power at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON3. 1 LSB = 0.1 dB, signed number
TX_PS_AMPLITUDE_VAL4	2	The measured amplitude of TX1 Loopback tone power at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON4. 1 LSB = 0.1 dB, signed number
TX_PS_NOISE_VAL1	1	The maximum measured wideband power across the enabled RXs of TX1 Loopback at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON1. 1 LSB = -1 dBm, Valid Range: 0 to -63 dBm
TX_PS_NOISE_VAL2	1	The maximum measured wideband power across the enabled RXs of TX1 Loopback at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON2. 1 LSB = -1 dBm, Valid Range: 0 to -63 dBm
TX_PS_NOISE_VAL3	1	The maximum measured wideband power across the enabled RXs of TX1 Loopback at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON3. 1 LSB = -1 dBm, Valid Range: 0 to -63 dBm
TX_PS_NOISE_VAL4	1	The maximum measured wideband power across the enabled RXs of TX1 Loopback at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON4. 1 LSB = -1 dBm, Valid Range: 0 to -63 dBm

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Table 5.123 – continued from previous page

TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)
RESERVED	8	RESERVED

5.14.5 Sub block 0x1024 – AWR_MONITOR_TX2_PHASE_SHIFTER_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured TX2 phase values, amplitude values and noise power. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.124: AWR_MONITOR_TX2_PHASE_SHIFTER_REPORT_AE_SB
contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1024
SBLKLEN	2	Value = 44
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_TX2_PHASE_SHIFTER_PHASE b1 STATUS_TX2_PHASE_SHIFTER_AMPLITUDE b15:2 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_INDX	1	Profile Index for which this monitoring report applies
RESERVED	3	RESERVED
PH_SHIFTER_MON_VAL1	2	The measured phase of TX2 Loop-back tone at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON1. 1 LSB = $360^\circ / 2^{16}$

Continued on next page

Table 5.124 – continued from previous page

PH_SHIFTER_MON_VAL2	2	The measured phase of TX2 Loop-back tone at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON2. 1 LSB = $360^\circ / 2^{16}$
PH_SHIFTER_MON_VAL3	2	The measured phase of TX2 Loop-back tone at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON3. 1 LSB = $360^\circ / 2^{16}$
PH_SHIFTER_MON_VAL4	2	The measured phase of TX2 Loop-back tone at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON4. 1 LSB = $360^\circ / 2^{16}$
TX_PS_AMPLITUDE_VAL1	2	The measured amplitude of TX2 Loopback tone power at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON1. 1 LSB = 0.1 dB, signed number
TX_PS_AMPLITUDE_VAL2	2	The measured amplitude of TX2 Loopback tone power at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON2. 1 LSB = 0.1 dB, signed number
TX_PS_AMPLITUDE_VAL3	2	The measured amplitude of TX2 Loopback tone power at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON3. 1 LSB = 0.1 dB, signed number
TX_PS_AMPLITUDE_VAL4	2	The measured amplitude of TX2 Loopback tone power at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON4. 1 LSB = 0.1 dB, signed number
TX_PS_NOISE_VAL1	1	The maximum measured wideband power across the enabled RXs of TX2 Loopback at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON1. 1 LSB = -1 dBm, Valid Range: 0 to -63 dBm
TX_PS_NOISE_VAL2	1	The maximum measured wideband power across the enabled RXs of TX2 Loopback at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON2. 1 LSB = -1 dBm, Valid Range: 0 to -63 dBm
TX_PS_NOISE_VAL3	1	The maximum measured wideband power across the enabled RXs of TX2 Loopback at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON3. 1 LSB = -1 dBm, Valid Range: 0 to -63 dBm

Continued on next page

Table 5.124 – continued from previous page

TX_PS_NOISE_VAL4	1	The maximum measured wideband power across the enabled RXs of TX2 Loopback at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON4. 1 LSB = -1 dBm, Valid Range: 0 to -63 dBm
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)
RESERVED	8	RESERVED

5.14.6 Sub block 0x1025 – AWR_MONITOR_SYNTHESIZER_FREQUENCY_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing information related to measured frequency error during the chirp. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.125: AWR_MONITOR_SYNTH_FREQUENCY_REPORT_AE_SB
contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1025
SBLKLEN	2	Value = 32
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_SYNTH_FREQ_ERR b15:1 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_INDX	1	Profile Index for which this monitoring report applies
RESERVED	3	0x000000

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Table 5.125 – continued from previous page

MAX_FREQUENCY_ERROR_VALUE	4	This field indicates the maximum instantaneous frequency error measured during the chirps for which frequency monitoring has been enabled in the previous monitoring period. Bits Parameter b31:0 Maximum frequency error value, signed number. 1 LSB = 1 kHz.
NOTE: When multiple profiles are enabled using PROFILE_BIT_MASK_LIVE_MODE, value reported is max across all enabled profiles		
FREQUENCY_FAILURE_COUNT	4	This field indicates the number of times during chirping in the previous monitoring period in which the measured frequency error violated the allowed threshold. Frequency error threshold violation is counted every 10 ns. Bits Parameter b31:19 RESERVED b18:0 Failure count, unsigned number
RESERVED	4	0x00000000
RESERVED	4	0x00000000
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

5.14.7 Sub block 0x1026 – AWR_MONITOR_EXTERNAL_ANALOG_SIGNALS_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the external signal voltage values measured using the GPADC. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.126: AWR_MONITOR_EXTERNAL_ANALOG_SIGNALS_REPORT_
AE_SB contents

Field Name	Number of bytes	Description																
SBLKID	2	Value = 0x1026																
SBLKLEN	2	Value = 28																
STATUS_FLAGS	2	Status flags indicating pass fail results corresponding to various threshold checks under this monitor. <table><tr><td>Bit</td><td>Definition</td></tr><tr><td>b0</td><td>STATUS_ANALOGTEST1</td></tr><tr><td>b1</td><td>STATUS_ANALOGTEST2</td></tr><tr><td>b2</td><td>STATUS_ANALOGTEST3</td></tr><tr><td>b3</td><td>STATUS_ANALOGTEST4</td></tr><tr><td>b4</td><td>STATUS_ANAMUX</td></tr><tr><td>b5</td><td>STATUS_VSENSE</td></tr><tr><td>b15:6</td><td>RESERVED</td></tr></table> 0 – FAIL or check wasn't done 1 – PASS	Bit	Definition	b0	STATUS_ANALOGTEST1	b1	STATUS_ANALOGTEST2	b2	STATUS_ANALOGTEST3	b3	STATUS_ANALOGTEST4	b4	STATUS_ANAMUX	b5	STATUS_VSENSE	b15:6	RESERVED
Bit	Definition																	
b0	STATUS_ANALOGTEST1																	
b1	STATUS_ANALOGTEST2																	
b2	STATUS_ANALOGTEST3																	
b3	STATUS_ANALOGTEST4																	
b4	STATUS_ANAMUX																	
b5	STATUS_VSENSE																	
b15:6	RESERVED																	
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error																
EXTERNAL_ANALOG_SIGNAL_VALUES	12	MEASURED_VALUE <table><tr><td>Bytes</td><td>SIGNAL</td></tr><tr><td>1:0</td><td>ANALOGTEST1</td></tr><tr><td>3:2</td><td>ANALOGTEST2</td></tr><tr><td>5:4</td><td>ANALOGTEST3</td></tr><tr><td>7:6</td><td>ANALOGTEST4</td></tr><tr><td>9:8</td><td>ANAMUX</td></tr><tr><td>11:10</td><td>VSENSE</td></tr></table> 1 LSB = 1.8V/1024	Bytes	SIGNAL	1:0	ANALOGTEST1	3:2	ANALOGTEST2	5:4	ANALOGTEST3	7:6	ANALOGTEST4	9:8	ANAMUX	11:10	VSENSE		
Bytes	SIGNAL																	
1:0	ANALOGTEST1																	
3:2	ANALOGTEST2																	
5:4	ANALOGTEST3																	
7:6	ANALOGTEST4																	
9:8	ANAMUX																	
11:10	VSENSE																	
RESERVED	4	0x00000000																
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)																

5.14.8 Sub block 0x1027 – AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing information about Internal TX0 internal analog signals including Tx Phase shifter DAC monitor report. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.127: AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1027
SBLKLEN	2	Value = 16
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_SUPPLY_TX0 b1 STATUS_DCBIAS_TX0 b2 STATUS_PS_DAC_TX0 (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) b15:3 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_IDX	1	Profile Index for which this monitoring report applies
RESERVED	1	0x00
PS_DAC_IDELTA_MIN	1	Phase shifter DAC I arm delta min value across different DAC settings (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) Unit: 1 LSB = 1.8V/1024
PS_DAC_QDELTA_MIN	1	Phase shifter DAC Q arm delta min value across different DAC settings (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) Unit: 1 LSB = 1.8V/1024
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

5.14.9 Sub block 0x1028 – AWR_MONITOR_TX1_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing information about Internal TX1 internal analog signals. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.128: AWR_MONITOR_TX1_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1028
SBLKLEN	2	Value = 16
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_SUPPLY_TX1 b1 STATUS_DCBIAS_TX1 b2 STATUS_PS_DAC_TX1 (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) b15:3 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_IDX	1	Profile Index for which this monitoring report applies
RESERVED	1	0x00
PS_DAC_IDELTA_MIN	1	Phase shifter DAC I arm delta min value across different DAC settings (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) Unit: 1 LSB = 1.8V/1024
PS_DAC_QDELTA_MIN	1	Phase shifter DAC Q arm delta min value across different DAC settings (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) Unit: 1 LSB = 1.8V/1024
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

5.14.10 Sub block 0x1029 – AWR_MONITOR_TX2_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing information about Internal TX2 internal analog signals. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.129: AWR_MONITOR_TX2_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1029
SBLKLEN	2	Value = 16
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_SUPPLY_TX2 b1 STATUS_DCBIAS_TX2 b2 STATUS_PS_DAC_TX2 (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) b15:3 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_IDX	1	Profile Index for which this monitoring report applies
RESERVED	1	0x00
PS_DAC_IDELTA_MIN (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP)	1	Phase shifter DAC I arm delta min value across different DAC settings Unit: 1 LSB = 1.8V/1024

Continued on next page

Table 5.129 – continued from previous page

PS_DAC_QDELTA_MIN (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP)	1	Phase shifter DAC Q arm delta min value across different DAC settings Unit: 1 LSB = 1.8V/1024
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

5.14.11 Sub block 0x102A – AWR_MONITOR_RX_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing information about Internal RX internal analog signals. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.130: AWR_MONITOR_RX_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x102A
SBLKLEN	2	Value = 16

Continued on next page

Table 5.130 – continued from previous page

STATUS_FLAGS	2	Status flags indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_SUPPLY_RX0 b1 STATUS_SUPPLY_RX1 b2 STATUS_SUPPLY_RX2 b3 STATUS_SUPPLY_RX3 b4 STATUS_DCBIAS_RX0 b5 STATUS_DCBIAS_RX1 b6 STATUS_DCBIAS_RX2 b7 STATUS_DCBIAS_RX3 b15:8 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_INDX	1	Profile Index for which this monitoring report applies
RESERVED	3	0x000000
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

5.14.12 Sub block 0x102B – AWR_MONITOR_PMCLKLO_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing information about Internal PM, CLK and LO subsystems' internal analog signals and in cascade devices the 20GHz SYNC IN/OUT power. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.131: AWR_MONITOR_PM_CLK_LO_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x102B
SBLKLEN	2	Value = 16

Continued on next page

Table 5.131 – continued from previous page

STATUS_FLAGS	2	Status flags indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_SUPPLY_PMCLKLO b1 STATUS_DCBIAS_PMCLKLO b2 STATUS_LVDS_PMCLKLO (Use this status bit only if LVDS is used, else ignore this) b3 STATUS_SYNC_20G (Use this field only in cascade configuration) b15:4 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_IDX	1	Profile Index for which this monitoring report applies
SYNC_20G_POWER	1	(Not applicable for single-chip devices) Monitored 20GHz SYNC_IN or SYNC_OUT signal power, signed number Unit: 1 LSB = 0.5 dBm Valid Range: -63 to 63 dBm $\text{SYNC_20G_POWER_dBm} = \text{SYNC_20G_POWER} * 0.5\text{dBm}$ NOTES: SYNC_IN power (dBm): The conversion factor for SYNC_IN power at BGA pin, $\text{Power_sync_in_bga_dbm} = (0.85 * \text{SYNC_20G_POWER_dBm}) - 10$, Refer monitor app note for more info. SYNC_OUT power (dBm): The conversion factor for SYNC_OUT power at BGA pin, $\text{Power_sync_out_bga_dbm} = (\text{SYNC_20G_POWER_dBm}) + 1$, Refer monitor app note for more info.
RESERVED	2	0x000000
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

5.14.13 Sub block 0x102C – AWR_MONITOR_GPADC_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing information about the measured value of the GPADC input DC signals whose measurements were enabled. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.132: AWR_MONITOR_GPADC_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x102C
SBLKLEN	2	Value = 20
STATUS_FLAGS	2	Status flags indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_GPADC_REF1 b1 STATUS_GPADC_REF2 b2 STATUS_GPADC_REF3 (xWR294x, xWR254x, xWR2x4xEEO, xWR2x4xP only) b15:3 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
GPADC_REF1_VALUE	2	The measured GPADC outputs corresponding to internal DC signal (GPADC_REF1, expected level 0.45V) is reported here. 1 LSB = 1.8V/1024
GPADC_REF2_VALUE	2	The measured GPADC outputs corresponding to internal DC signal (GPADC_REF2, expected level is 0.6V for xWR294x, xWR254x, xWR2x4xEEO, xWR2x4xP and 1.2V for other devices) is reported here. 1 LSB = 1.8V/1024
GPADC_REF3_VALUE	2	(xWR294x, xWR254x, xWR2x4xEEO, xWR2x4xP only) The measured GPADC outputs corresponding to internal DC signal (GPADC_REF3, expected level 0.6V) is reported here. 1 LSB = 1.8V/1024
RESERVED	2	0x0000

Continued on next page

Table 5.132 – continued from previous page

TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)
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5.14.14 Sub block 0x102D – AWR_MONITOR_PLL_CONTROL_VOLTAGE_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured PLL control voltage values during explicit monitoring chirps. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.133: AWR_MONITOR_PLL_CONTROL_VOLTAGE_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x102D
SBLKLEN	2	Value = 32
STATUS_FLAGS	2	Status flags indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_APLL_VCTRL b1 STATUS_SYNTH_VCO1_VCTRL_MAX_FREQ b2 STATUS_SYNTH_VCO1_VCTRL_MIN_FREQ b3 RESERVED b4 STATUS_SYNTH_VCO2_VCTRL_MAX_FREQ b5 STATUS_SYNTH_VCO2_VCTRL_MIN_FREQ b6 RESERVED b7 STATUS_SYNTH_VCO3_VCTRL_MAX_FREQ (Reserved in AWR2243, xWR294x, xWR254x, xWR6243, xWR2x4xECO, xWR2x4xP) b8 STATUS_SYNTH_VCO3_VCTRL_MIN_FREQ (Reserved in AWR2243, xWR294x, xWR254x, xWR6243, xWR2x4xECO, xWR2x4xP) b15:9 RESERVED 0 – FAIL or check wasn't done 1 – PASS

Continued on next page

Table 5.133 – continued from previous page

ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error																											
PLL_CONTROL_VOLTAGE_VALUES	16	The measured values of PLL control voltage levels and Synthesizer VCO slopes are reported here. Byte numbers corresponding to different control voltage values reported in this field are here: <table> <thead> <tr> <th>Bytes</th><th>SIGNAL</th><th>1 LSB</th></tr> </thead> <tbody> <tr> <td>1:0</td><td>APLL_VCTRL</td><td>1 mV</td></tr> <tr> <td>3:2</td><td>SYNTH_VCO1_VCTRL_MAX_FREQ</td><td>1 mV</td></tr> <tr> <td>5:4</td><td>SYNTH_VCO1_VCTRL_MIN_FREQ</td><td>1 mV</td></tr> <tr> <td>7:6</td><td>SYNTH_VCO1_SLOPE</td><td>1 MHz/V</td></tr> <tr> <td>9:8</td><td>SYNTH_VCO2_VCTRL_MAX_FREQ</td><td>1 mV</td></tr> <tr> <td>11:10</td><td>SYNTH_VCO2_VCTRL_MIN_FREQ</td><td>1 mV</td></tr> <tr> <td>13:12</td><td>SYNTH_VCO2_SLOPE</td><td>1 MHz/V</td></tr> <tr> <td>15:14</td><td>RESERVED</td><td>RESERVED</td></tr> </tbody> </table> Only the fields corresponding to the enabled monitors are valid. The failure thresholds are based on the following: Valid VCTRL values are [140 to 1400] mV. Valid VCO1_SLOPE values are [880 to 1320] MHz/V. Valid VCO2_SLOPE values are [3520 to 6000] MHz/V. NOTE: The VCOx_SLOPE should be ignored when synth fault is injected.	Bytes	SIGNAL	1 LSB	1:0	APLL_VCTRL	1 mV	3:2	SYNTH_VCO1_VCTRL_MAX_FREQ	1 mV	5:4	SYNTH_VCO1_VCTRL_MIN_FREQ	1 mV	7:6	SYNTH_VCO1_SLOPE	1 MHz/V	9:8	SYNTH_VCO2_VCTRL_MAX_FREQ	1 mV	11:10	SYNTH_VCO2_VCTRL_MIN_FREQ	1 mV	13:12	SYNTH_VCO2_SLOPE	1 MHz/V	15:14	RESERVED	RESERVED
Bytes	SIGNAL	1 LSB																											
1:0	APLL_VCTRL	1 mV																											
3:2	SYNTH_VCO1_VCTRL_MAX_FREQ	1 mV																											
5:4	SYNTH_VCO1_VCTRL_MIN_FREQ	1 mV																											
7:6	SYNTH_VCO1_SLOPE	1 MHz/V																											
9:8	SYNTH_VCO2_VCTRL_MAX_FREQ	1 mV																											
11:10	SYNTH_VCO2_VCTRL_MIN_FREQ	1 mV																											
13:12	SYNTH_VCO2_SLOPE	1 MHz/V																											
15:14	RESERVED	RESERVED																											

Continued on next page

Table 5.133 – continued from previous page

PLL3_CONTROL_VOLTAGE_VALUES	4	The measured values of PLL control voltage levels are reported here. Byte numbers corresponding to different control voltage values reported in this field are here: <table> <tr> <th>Bytes</th><th>SIGNAL</th><th>1 LSB</th></tr> <tr> <td>1:0</td><td>SYNTH_VCO3_VCTRL_MAX_FREQ (Reserved in AWR2243, xWR294x, xWR254x, xWR6243, xWR2x4xECO, xWR2x4xP)</td><td>1 mV</td></tr> <tr> <td>3:2</td><td>SYNTH_VCO3_VCTRL_MIN_FREQ (Reserved in AWR2243, xWR294x, xWR254x, xWR6243, xWR2x4xECO, xWR2x4xP)</td><td>1 mV</td></tr> </table> Only the fields corresponding to the enabled monitors are valid. NOTE: The VCO3 control voltage monitor is for debug purposes only and not supported in production.	Bytes	SIGNAL	1 LSB	1:0	SYNTH_VCO3_VCTRL_MAX_FREQ (Reserved in AWR2243, xWR294x, xWR254x, xWR6243, xWR2x4xECO, xWR2x4xP)	1 mV	3:2	SYNTH_VCO3_VCTRL_MIN_FREQ (Reserved in AWR2243, xWR294x, xWR254x, xWR6243, xWR2x4xECO, xWR2x4xP)	1 mV
Bytes	SIGNAL	1 LSB									
1:0	SYNTH_VCO3_VCTRL_MAX_FREQ (Reserved in AWR2243, xWR294x, xWR254x, xWR6243, xWR2x4xECO, xWR2x4xP)	1 mV									
3:2	SYNTH_VCO3_VCTRL_MIN_FREQ (Reserved in AWR2243, xWR294x, xWR254x, xWR6243, xWR2x4xECO, xWR2x4xP)	1 mV									
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)									

5.14.15 Sub block 0x102E – AWR_MONITOR_DUAL_CLOCK_COMP_REPORT_AE_SB

This API is a monitoring report API which the AWR device sends to the host, containing information about the relative frequency measurements. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.134: AWR_MONITOR_DUAL_CLOCK_COMP_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x102E
SBLKLEN	2	Value = 32

Continued on next page

Table 5.134 – continued from previous page

STATUS_FLAGS	2	Status flags indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_CLK_PAIR0 b1 STATUS_CLK_PAIR1 b2 STATUS_CLK_PAIR2 b3 STATUS_CLK_PAIR3 b4 STATUS_CLK_PAIR4 b5 RESERVED b6 STATUS_CLK_PAIR6 b15:7 RESERVED 0 – FAIL or check wasn't done 1 – PASS																											
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error																											
FREQ_MEAS_VALUES	16	The measured clock frequencies from the enabled clock pair measurements are reported here. Byte numbers corresponding to different frequency measurement values reported in this field are here: <table> <tr> <th>Bytes</th><th>CLOCK PAIR</th><th>MEASURED CLOCK FREQUENCY</th></tr> <tr> <td>1:0</td><td>0</td><td>APLL_200M</td></tr> <tr> <td>3:2</td><td>1</td><td>BSS_200M</td></tr> <tr> <td>5:4</td><td>2</td><td>BSS_100M</td></tr> <tr> <td>7:6</td><td>3</td><td>GPADC_10M</td></tr> <tr> <td>9:8</td><td>4</td><td>RCOSC_10M</td></tr> <tr> <td>11:10</td><td>RESERVED</td><td>RESERVED</td></tr> <tr> <td>13:12</td><td>6</td><td>FRC_200M (xWR254x, xWR2x4xECO, xWR2x4xP only)</td></tr> <tr> <td>15:14</td><td>RESERVED</td><td>RESERVED</td></tr> </table> 1 LSB = 0.1 MHz, unsigned number	Bytes	CLOCK PAIR	MEASURED CLOCK FREQUENCY	1:0	0	APLL_200M	3:2	1	BSS_200M	5:4	2	BSS_100M	7:6	3	GPADC_10M	9:8	4	RCOSC_10M	11:10	RESERVED	RESERVED	13:12	6	FRC_200M (xWR254x, xWR2x4xECO, xWR2x4xP only)	15:14	RESERVED	RESERVED
Bytes	CLOCK PAIR	MEASURED CLOCK FREQUENCY																											
1:0	0	APLL_200M																											
3:2	1	BSS_200M																											
5:4	2	BSS_100M																											
7:6	3	GPADC_10M																											
9:8	4	RCOSC_10M																											
11:10	RESERVED	RESERVED																											
13:12	6	FRC_200M (xWR254x, xWR2x4xECO, xWR2x4xP only)																											
15:14	RESERVED	RESERVED																											
RESERVED	4	0x00000000																											
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)																											

5.14.16 Sub block 0x1031 – AWR_MONITOR_RX_MIXER_IN_POWER_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured RX mixer input voltage swing values. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.135: AWR_MONITOR_RX_MIXER_IN_POWER_REPORT_AE_SB contents

Field Name	Number of bytes	Description										
SBLKID	2	Value = 0x1031										
SBLKLEN	2	Value = 24										
STATUS_FLAGS	2	Bit STATUS_FLAG for monitor b0 STATUS_MIXER_IN_POWER_RX0 b1 STATUS_MIXER_IN_POWER_RX1 b2 STATUS_MIXER_IN_POWER_RX2 b3 STATUS_MIXER_IN_POWER_RX3 b15:4 RESERVED 0 – FAIL or check wasn't done 1 – PASS										
ERROR_CODE	2	Internal sanity check violations are reported here. Value = 0: No error										
PROFILE_INDX	1	Profile Index for which this monitoring report applies										
RESERVED	3	0x000000										
RX_MIXER_IN_VOLTAGE_VALUE	4	The measured RX mixer input voltage swing values are reported here. The byte location of the value for each receivers is tabulated here: <table><tr><th>Receiver</th><th>Byte Location</th></tr><tr><td>RX0</td><td>0</td></tr><tr><td>RX1</td><td>1</td></tr><tr><td>RX2</td><td>2</td></tr><tr><td>RX3</td><td>3</td></tr></table> 1 LSB = 1800 mV/256, unsigned number Only the entries of enabled RX channels are valid.	Receiver	Byte Location	RX0	0	RX1	1	RX2	2	RX3	3
Receiver	Byte Location											
RX0	0											
RX1	1											
RX2	2											
RX3	3											
RESERVED	4	0x00000000										
TIME_STAMP	4	When this monitoring began is indicated here. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)										

5.14.17 Sub block 0x1033 – AWR_MONITOR_SYNTHESIZER_FREQUENCY_NONLIVE_REPORT_AE_SB

This is a new feature addition in AWR2243, xWR294x, xWR254x, xWR6243, xWR2x4xECO, xWR2x4xP device. This API is a Non live Monitoring Report SB, which device sends to the host, containing information related to measured frequency error during the monitoring chirp for two profiles configurations. The device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.136: AWR_MONITOR_SYNTHESIZER_FREQUENCY_NONLIVE_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1033
SBLKLEN	2	Value = 52
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 VCO1_SYNTH_FREQ_ERR_STATUS b1 VCO2_SYNTH_FREQ_ERR_STATUS b15:2 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_INDX_0	1	VCO1 Profile index for which this monitoring report applies
RESERVED	3	0x000000
MAX_FREQUENCY_ERROR_VALUE_0	4	This field indicates the maximum instantaneous frequency error measured during the monitoring chirp for which frequency monitoring has been enabled in the previous monitoring period for VCO1 profile. Bits Parameter b31:0 Maximum frequency error value, signed number. 1 LSB = 1 kHz.

Continued on next page

Table 5.136 – continued from previous page

FREQUENCY_ FAILURE_ COUNT_0	4	This field indicates the number of times during chirping in the previous monitoring period in which the measured frequency error violated the allowed threshold for VCO1 profile. Frequency error threshold violation is counted every 10 ns. Bits Parameter b31:19 RESERVED b18:0 Failure count, unsigned number
MAX_FREQ_ FAILURE_TIME_ 0	4	This field indicates the time at which error occurred for VCO1 profile w.r.t. knee of the ramp. 1LSB = 10ns
RESERVED	4	0x00000000
PROFILE_INDX_ 1	1	VCO2 Profile index for which this monitoring report applies
RESERVED	3	0x000000
MAX_FRE- QUENCY_ER- ROR_VALUE_1	4	This field indicates the maximum instantaneous frequency error measured during the monitoring chirp for which frequency monitoring has been enabled in the previous monitoring period for VCO2 profile. Bits Parameter b31:0 Maximum frequency error value, signed number. 1 LSB = 1 kHz.
FREQUENCY_ FAILURE_ COUNT_1	4	This field indicates the number of times during chirping in the previous monitoring period in which the measured frequency error violated the allowed threshold for VCO2 profile. Frequency error threshold violation is counted every 10 ns. Bits Parameter b31:19 RESERVED b18:0 Failure count, unsigned number
MAX_FREQ_ FAILURE_TIME_ 1	4	This field indicates the time at which error occurred for VCO2 profile w.r.t. knee of the ramp. 1LSB = 10ns
RESERVED	4	0x00000000
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

5.14.18 Sub block 0x1034 – AWR_MONITOR_TX3_POWER_REPORT_AE_SB

NOTE1:	The TX[0:3] power monitoring accuracy degrades at high TX back-offs and is unreliable for backoffs higher than 20dB on AWR2243, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP and 26dB on xWR6x43.
NOTE2:	The 0dB back-off corresponds to typically 13dBm power level in AWR2243/xWR294x/xWR254x/xWR6x43 device.
NOTE3:	The 0dB back-off corresponds to typically 13.5dBm power level in xWR2x4xECO/xWR2x4xP device.

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured TX power values during an explicit monitoring chirp. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.137: AWR_MONITOR_TX3_POWER_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1034
SBLKLEN	2	Value = 24
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_ABS_ERR b1 STATUS_FLATNESS_ERR b15:2 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_IDX	1	Profile Index for which this monitoring report applies
RESERVED	3	0x000000

Continued on next page

Table 5.137 – continued from previous page

TX_POWER_ VALUE	6	The measured TX power for each enabled channel, at each enabled RF frequency is reported here. Byte numbers corresponding to different TX and RF, in this field are here: <table><tr><td></td><td>RF1</td><td>RF2</td><td>RF3</td></tr><tr><td>TX3</td><td>1:0</td><td>3:2</td><td>5:4</td></tr></table> (other bytes are reserved) 1 LSB = 0.1 dBm, signed number Only the entries of enabled RF Frequencies and enabled RX channels are valid.		RF1	RF2	RF3	TX3	1:0	3:2	5:4
	RF1	RF2	RF3							
TX3	1:0	3:2	5:4							
RESERVED	2	0x0000								
TIME_ STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)								

5.14.19 Sub block 0x1035 – AWR_MONITOR_TX3_BALLBREAK_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured TX reflection coefficient's magnitude values, meant for detecting TX ball break. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.138: AWR_MONITOR_TX3_BALLBREAK_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1035
SBLKLEN	2	Value = 20
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_TX3_BALLBREAK b15:1 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error

Continued on next page

Table 5.138 – continued from previous page

TX_REFL_CO-EFF_VALUE	2	The TX reflection coefficient's magnitude for this channel is reported here. 1 LSB = 0.1 dB, signed number
RESERVED	2	0x0000
RESERVED	4	0x00000000
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

5.14.20 Sub block 0x1036 – AWR_MONITOR_TX3_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing information about Internal TX3 internal analog signals including Tx Phase shifter DAC monitor report. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.139: AWR_MONITOR_TX3_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1036
SBLKLEN	2	Value = 16
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_SUPPLY_TX3 b1 STATUS_DCBIAS_TX3 b2 STATUS_PS_DAC_TX3 (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) b15:3 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_INDXX	1	Profile Index for which this monitoring report applies

Continued on next page

Table 5.139 – continued from previous page

RESERVED	1	0x00
PS_DAC_IDELTA_MIN	1	Phase shifter DAC I arm delta min value across different DAC settings (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) Unit: 1 LSB = 1.8V/1024
PS_DAC_QDELTA_MIN	1	Phase shifter DAC Q arm delta min value across different DAC settings (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) Unit: 1 LSB = 1.8V/1024
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

5.14.21 Sub block 0x1037 – AWR_MONITOR_TX3_PHASE_SHIFTER_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured TX3 phase values, amplitude values and noise power. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.140: AWR_MONITOR_TX3_PHASE_SHIFTER_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1037
SBLKLEN	2	Value = 44
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_TX3_PHASE_SHIFTER_PHASE b1 STATUS_TX3_PHASE_SHIFTER_AMPLITUDE b15:2 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error

Continued on next page

Table 5.140 – continued from previous page

PROFILE_INDX	1	Profile Index for which this monitoring report applies
RESERVED	3	RESERVED
PH_SHIFTER_MON_VAL1	2	The measured phase of TX3 Loop-back tone at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON1. 1 LSB = $360^\circ / 2^{16}$
PH_SHIFTER_MON_VAL2	2	The measured phase of TX3 Loop-back tone at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON2. 1 LSB = $360^\circ / 2^{16}$
PH_SHIFTER_MON_VAL3	2	The measured phase of TX3 Loop-back tone at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON3. 1 LSB = $360^\circ / 2^{16}$
PH_SHIFTER_MON_VAL4	2	The measured phase of TX3 Loop-back tone at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON4. 1 LSB = $360^\circ / 2^{16}$
TX_PS_AMPLITUDE_VAL1	2	The measured amplitude of TX3 Loopback tone power at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON1. 1 LSB = 0.1 dB, signed number
TX_PS_AMPLITUDE_VAL2	2	The measured amplitude of TX3 Loopback tone power at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON2. 1 LSB = 0.1 dB, signed number
TX_PS_AMPLITUDE_VAL3	2	The measured amplitude of TX3 Loopback tone power at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON3. 1 LSB = 0.1 dB, signed number
TX_PS_AMPLITUDE_VAL4	2	The measured amplitude of TX3 Loopback tone power at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON4. 1 LSB = 0.1 dB, signed number
TX_PS_NOISE_VAL1	1	The maximum measured wideband power across the enabled RXs of TX3 Loopback at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON1. 1 LSB = -1 dBm, Valid Range: 0 to -63 dBm

Continued on next page

Table 5.140 – continued from previous page

TX_PS_NOISE_VAL2	1	The maximum measured wideband power across the enabled RXs of TX3 Loopback at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON2. 1 LSB = -1 dBm, Valid Range: 0 to -63 dBm
TX_PS_NOISE_VAL3	1	The maximum measured wideband power across the enabled RXs of TX3 Loopback at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON3. 1 LSB = -1 dBm, Valid Range: 0 to -63 dBm
TX_PS_NOISE_VAL4	1	The maximum measured wideband power across the enabled RXs of TX3 Loopback at the RX ADC for phase shifter monitoring setting PH_SHIFTER_MON4. 1 LSB = -1 dBm, Valid Range: 0 to -63 dBm
TIME_STAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)
RESERVED	8	RESERVED

5.14.22 Sub block 0x1038 – AWR_MONITOR_ADV_TX_GAIN_PHASE_MISMATCH_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured TX gain and phase mismatch values during an explicit monitoring chirp. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.141: AWR_MONITOR_ADV_TX_GAIN_PHASE_REPORT_AE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1038
SBLKLEN	2	Value = 80

Continued on next page

Table 5.141 – continued from previous page

STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_TX_GAIN_MISMATCH b1 STATUS_TX_PHASE_MISMATCH b15:2 RESERVED 0 – FAIL or check wasn't done 1 – PASS																				
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error																				
PROFILE_INDX	1	Profile Index for which this monitoring report applies																				
RESERVED	3	0x000000																				
TX_GAIN_VALUE	24	The measured TX PA loopback tone power at the RX ADC input, for each enabled TX channel, at each enabled RF frequency is reported here. Byte numbers corresponding to different TX and RF, in this field are here: <table><tr><td></td><td>RF1</td><td>RF2</td><td>RF3</td></tr><tr><td>TX0</td><td>1:0</td><td>9:8</td><td>17:16</td></tr><tr><td>TX1</td><td>3:2</td><td>11:10</td><td>19:18</td></tr><tr><td>TX2</td><td>5:4</td><td>13:12</td><td>21:20</td></tr><tr><td>TX3</td><td>7:6</td><td>15:14</td><td>23:22</td></tr></table> 1 LSB = 0.1dBm, signed number Only the entries of enabled RF Frequencies and enabled TX channels are valid.		RF1	RF2	RF3	TX0	1:0	9:8	17:16	TX1	3:2	11:10	19:18	TX2	5:4	13:12	21:20	TX3	7:6	15:14	23:22
	RF1	RF2	RF3																			
TX0	1:0	9:8	17:16																			
TX1	3:2	11:10	19:18																			
TX2	5:4	13:12	21:20																			
TX3	7:6	15:14	23:22																			

Continued on next page

Table 5.141 – continued from previous page

TX_PHASE_ VALUE	24	The measured TX phase for each enabled channel, at each enabled RF frequency is reported here. Byte numbers corresponding to different TX and RF, in this field are here: <table><tr><td></td><td>RF1</td><td>RF2</td><td>RF3</td></tr><tr><td>TX0</td><td>1:0</td><td>9:8</td><td>17:16</td></tr><tr><td>TX1</td><td>3:2</td><td>11:10</td><td>19:18</td></tr><tr><td>TX2</td><td>5:4</td><td>13:12</td><td>21:20</td></tr><tr><td>TX3</td><td>7:6</td><td>15:14</td><td>23:22</td></tr></table> 1 LSB = $360^{\circ} / 2^{16}$ Only the entries of enabled RF Frequencies and enabled TX channels are valid. NOTE: In the gains/phases reported here, only inter-TX mismatches carry information, and the raw values may include unknown biases (which cannot be relied on).		RF1	RF2	RF3	TX0	1:0	9:8	17:16	TX1	3:2	11:10	19:18	TX2	5:4	13:12	21:20	TX3	7:6	15:14	23:22
	RF1	RF2	RF3																			
TX0	1:0	9:8	17:16																			
TX1	3:2	11:10	19:18																			
TX2	5:4	13:12	21:20																			
TX3	7:6	15:14	23:22																			
RF_TX_NOISE_ POWER	12	The measured wide band noise power for each enabled channel, at each enabled RF frequency is reported here. <table><tr><td></td><td>RF1</td><td>RF2</td><td>RF3</td></tr><tr><td>TX0</td><td>0:0</td><td>4:4</td><td>8:8</td></tr><tr><td>TX1</td><td>1:1</td><td>5:5</td><td>9:9</td></tr><tr><td>TX2</td><td>2:2</td><td>6:6</td><td>10:10</td></tr><tr><td>TX3</td><td>3:3</td><td>7:7</td><td>11:11</td></tr></table> 1 LSB = -1 dBm Valid Range: 0 to -63 dBm		RF1	RF2	RF3	TX0	0:0	4:4	8:8	TX1	1:1	5:5	9:9	TX2	2:2	6:6	10:10	TX3	3:3	7:7	11:11
	RF1	RF2	RF3																			
TX0	0:0	4:4	8:8																			
TX1	1:1	5:5	9:9																			
TX2	2:2	6:6	10:10																			
TX3	3:3	7:7	11:11																			
RESERVED	4	0x00000000																				
TIMESTAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)																				

5.14.23 Sub block 0x1039 – AWR_MONITOR_TX_PHSHIFTER_DAC_REPORT_AE_SB

This API is a Monitoring Report API which the AWR device sends to the host, containing the measured TX phase shifter DAC monitoring results during an explicit monitoring chirp. The AWR device sends this to host at the programmed periodicity or when failure occurs, as programmed by the configuration API SB.

Table 5.142: AWR_MONITOR_TX_PHSHIFTER_DAC_REPORT_AE_SB
contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x1039
SBLKLEN	2	Value = 32
STATUS_FLAGS	2	Status flag indicating pass fail results corresponding to various threshold checks under this monitor. Bit STATUS_FLAG for monitor b0 STATUS_PS_DAC_TX0 b1 STATUS_PS_DAC_TX1 b2 STATUS_PS_DAC_TX2 b3 STATUS_PS_DAC_TX3 b15:4 RESERVED 0 – FAIL or check wasn't done 1 – PASS
ERROR_CODE	2	Indicates any error reported during monitoring Value of 0 indicates no error
PROFILE_INDX	1	Profile Index for which this monitoring report applies
RESERVED	1	0x00
TX0_PS_DAC_IDELTA_MIN	1	TX0 Phase shifter DAC I arm delta min value across different DAC settings Unit: 1 LSB = 1.8V/1024
TX1_PS_DAC_IDELTA_MIN	1	TX1 Phase shifter DAC I arm delta min value across different DAC settings Unit: 1 LSB = 1.8V/1024
TX2_PS_DAC_IDELTA_MIN	1	TX2 Phase shifter DAC I arm delta min value across different DAC settings Unit: 1 LSB = 1.8V/1024
TX3_PS_DAC_IDELTA_MIN	1	TX3 Phase shifter DAC I arm delta min value across different DAC settings Unit: 1 LSB = 1.8V/1024
TX0_PS_DAC_QDELTA_MIN	1	TX0 Phase shifter DAC Q arm delta min value across different DAC settings Unit: 1 LSB = 1.8V/1024
TX1_PS_DAC_QDELTA_MIN	1	TX1 Phase shifter DAC Q arm delta min value across different DAC settings Unit: 1 LSB = 1.8V/1024

Continued on next page

Table 5.142 – continued from previous page

TX2_PS_DAC_QDELTA_MIN	1	TX2 Phase shifter DAC Q arm delta min value across different DAC settings Unit: 1 LSB = 1.8V/1024
TX3_PS_DAC_QDELTA_MIN	1	TX3 Phase shifter DAC Q arm delta min value across different DAC settings Unit: 1 LSB = 1.8V/1024
RESERVED	2	RESERVED
RESERVED	4	RESERVED
RESERVED	4	RESERVED
TIMESTAMP	4	This field indicates when the last monitoring in the enabled set was performed. 1 LSB = 1 millisecond (time stamp rolls over upon exceeding allotted bit width)

5.15 Sub blocks related to AWR_DEV_RFPOWERUP_MSG

NOTE:

All device config APIs having sub block ID $\geq 0x4000$ are applicable only for MSS in AWR2243 RF front end devices, for other xWR1443, xWR1642 and xWR1843 devices, these APIs are for reference only.

5.15.1 Sub block 0x4000 – AWR_DEV_RFPOWERUP_SB

This sub block is a command to power up the BSS

[5.143](#) describes the content of this sub block.

Table 5.143: AWR_DEV_POWERUP_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4000
SBLKLEN	2	Value = 4

5.16 Sub blocks related to AWR_DEV_CONF_SET_MSG

5.16.1 Sub block 0x4040 – AWR_DEV_MCUCLOCK_CONF_SET_SB

This sub block contains the configurations to setup the desired frequency of the MCU Clock that is output from the device.

NOTE: The Maximum supported MCU clock out is 80MHz.

Table 5.144 describes the contents of this sub block.

Table 5.144: AWR_DEV_MCUCLOCK_CONF_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4040
SBLKLEN	2	Value = 8
MCUCLOCK_CTRL	1	This field controls the enable-disable of the MCU clock. Value Description 0x0 Disable MCU clock 0x1 Enable MCU clock
MCUCLOCK_SRC	1	This field specifies the source of the MCU clock. Applicable only in case of MCU clock enable. Else ignored. Value Description 0x0 XTAL (as connected to the device) 0x2 400MHz PLL divided clock (xWR294x) 600MHz PLL divided clock (Other devices)
SRCCLOCK_DIV	1	This field specifies the division factor to be applied to source clock. Applicable only in case of MCU clock enable. Else ignored. Allowed values in xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP: Value Description 0x0 Divide by 1 0x1 Divide by 2 0xF Divide by 16 Allowed values in other devices: Value Description 0x0 Divide by 1 0x1 Divide by 2 0xFF Divide by 256 Note: The Maximum supported MCU clock out is 80MHz.
RESERVED	1	0x00

5.16.2 Sub block 0x4041 – AWR_DEV_RX_DATA_FORMAT_CONF_SET_SB

This sub block contains the configuration of the data format of the samples received over the receive chain to be transferred out to an external host over the configured data path (LVDS or CSI2).

Table 5.145 describes the content of this sub block.

Table 5.145: AWR_DEV_RX_DATA_FORMAT_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4041
SBLKLEN	2	Value = 16
RX_CHAN_EN	2	Bits Definition b0 RX_CHAN0_EN 0 Disable RX Channel 0 1 Enable RX Channel 0 b1 RX_CHAN0_EN 0 Disable RX Channel 1 1 Enable RX Channel 1 b2 RX_CHAN0_EN 0 Disable RX Channel 2 1 Enable RX Channel 2 b3 RX_CHAN0_EN 0 Disable RX Channel 3 1 Enable RX Channel 3 b15:4 RESERVED
NUM_ADC_BITS	2	Bits Definition b1:0 00 12 bits 01 14 bits 10 16 bits Other Reserved b15:2 RESERVED
ADC_OUT_FMT	2	Bits Definition b1:0 00 Real 01 Complex Other Reserved b15:2 RESERVED

Continued on next page

Table 5.145 – continued from previous page

IQ_SWAP_SEL	1	Bits	Definition
		b1:0	To swap the IQ samples (if complex format) 00 Sample interleave mode – I first 01 Sample interleave mode – Q first Other Reserved
		b7:2	RESERVED
CHAN_INTER-LEAVE	1	Bits	Definition
		b1:0	Channel interleaving of the samples stored in the ADC buffer to be transferred out on the data path. 00 Interleaved mode of storage 01 Non-interleaved mode of storage Other Reserved
		b7:2	RESERVED
RESERVED	4	0x00000000	

5.16.3 Sub block 0x4042 – AWR_DEV_RX_DATA_PATH_CONF_SET_SB

This sub block contains the configurations of the data path to transfer the captured ADC samples received over the receive chain to be transferred out to an external host.

Table 5.146 describes the content of this sub block.

Table 5.146: AWR_DEV_RX_DATA_PATH_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4042
SBLKLEN	2	Value = 12
DATA_INTF_SEL	1	This field specifies the data path selected to transfer the Radar info. Value Description 0x0 CSI2 interface select 0x1 LVDS interface select

Continued on next page

Table 5.146 – continued from previous page

DATA_TRANS_ FMT_PKT0	1	Bits	Description
		b5:0	Packet 0 content selection Value Definition 000001 ADC 000110 CP_ADC (See note at the bottom of this table) 001001 ADC_CP 110110 CP_ADC_CQ (See note at the bottom of this table) 111001 CQ_CP_ADC (See note at the bottom of this table)
		b7:6	Packet 0 virtual channel number (valid only for CSI2) Value Definition 00 Virtual channel number 0 (Default) 01 Virtual channel number 1 10 Virtual channel number 2 11 Virtual channel number 3
DATA_TRANS_ FMT_PKT1	1	Bits	Description
		b5:0	Packet 1 content selection Value Definition 000000 Suppress packet 1 transmission 001110 CP_CQ (See note at the bottom of this table) 001011 CQ_CP (See note at the bottom of this table)
		b7:6	Packet 1 virtual channel num- ber (valid only for CSI2) Value Definition 00 Virtual channel number 0 (Default) 01 Virtual channel number 1 10 Virtual channel number 2 11 Virtual channel number 3

Continued on next page

Table 5.146 – continued from previous page

CQ_CONFIG	1	This specifies the data size of CQ samples on the lanes
		Bits Description b1:0 Value Definition
		00 12 bit
		01 14 bit
		10 16 bit
CQ0_TRANS_SIZE	1	11 RESERVED
		b7:2 RESERVED
		NOTE: The CQ size can be configured only if CQ and ADC data is sent in separate packets. When ADC and CQ is sent in the same packet, then CQ size will be same as ADC data size.
		Number of samples (in 16 bit halfwords) of CQ0 data to be transferred. Valid range [32 halfwords to 128 halfwords]
		Value 0 = Disabled.
CQ1_TRANS_SIZE	1	NOTE: Ensure that the number of halfwords specified are a multiple of the number of lanes selected.
		Number of samples (in 16 bit halfwords) of CQ1 data to be transferred. Valid range [32 halfwords to 128 halfwords]
		Value 0 = Disabled.
		NOTE: Ensure that the number of halfwords specified are a multiple of the number of lanes selected.
		Number of samples (in 16 bit halfwords) of CQ2 data to be transferred. Valid range [32 halfwords to 128 halfwords]
CQ2_TRANS_SIZE	1	Value 0 = Disabled.
		NOTE: Ensure that the number of halfwords specified are a multiple of the number of lanes selected.
		Value 0 = Disabled.
		NOTE: Ensure that the number of halfwords specified are a multiple of the number of lanes selected.
		Value 0 = Disabled.
RESERVED	1	0x00

NOTE1:		CP is Chirp Parameter information which is defined for each RX as follows
Bit		Description
b11:0		Chirp number In legacy frame configuration, chirp number for starts from 1 and increments for each chirp within the frame and resets to 0 for the next frame. In advanced frame configuration chirp number starts from 1 and increments for each chirp within the burst and resets to 0 for the next burst.
b15:12		RESERVED
b17:16		Channel number The receive channel number which is encoded as
	00	RX0
	01	RX1
	10	RX2
	11	RX3
b21:18		Profile number The profile number to which the chirp belongs
b31:22		RESERVED
NOTE2:		CQ is Chirp Quality information which is defined in Section 10

5.16.4 Sub block 0x4043 – AWR_DEV_RX_DATA_PATH_LANEEN_SET_SB

This sub block contains the configurations to enables the lanes of the LVDS path to transfer Radar information to an external host.

Table 5.147 describes the content of this sub block.

Table 5.147: AWR_DEV_RX_DATA_PATH_LANEEN_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4043
SBLKLEN	2	Value = 8

Continued on next page

Table 5.147 – continued from previous page

LANE_EN	2	Bits	Description
		b0	LANE0_EN
		0	Disable lane 0
		1	Enable lane 0
		b1	LANE1_EN
		0	Disable lane 1
		1	Enable lane 1
		b2	LANE2_EN
		0	Disable lane 2
		1	Enable lane 2
		b3	LANE3_EN
		0	Disable lane 3
		1	Enable lane 3
		b15:4	RESERVED
RESERVED	2	0x0000	

5.16.5 Sub block 0x4044 – AWR_DEV_RX_DATA_PATH_CLK_SET_SB

This sub block contains the clock configurations for data transfer on the LVDS/CSI2 lanes. Table 5.148 describes the content of this sub block.

Table 5.148: AWR_DEV_RX_DATA_PATH_CLK_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4044
SBLKLEN	2	Value = 8
LANE_CLK_CFG (Selection valid only for LVDS. For CSI2, DDR is used always)	1	Bits Description
		b0 BIT_CLK_SEL
		0 SDR clock
		1 DDR clock (Only valid value for CSI2)
		b7:1 RESERVED

Continued on next page

Table 5.148 – continued from previous page

DATA_RATE	1	Data rate selection Value Description 0x01 600 Mbps (DDR only) 0x02 450 Mbps (SDR, DDR) 0x03 400 Mbps (DDR only) 0x04 300 Mbps (SDR, DDR) 0x05 225 Mbps (DDR only) 0x06 150 Mbps (DDR only) Others RESERVED
RESERVED	2	0x0000

5.16.6 Sub block 0x4045 – AWR_DEV_LVDS_CFG_SET_SB

This sub block contains the configurations of the LVDS lanes.

Table 5.149 describes the content of this sub block.

Table 5.149: AWR_DEV_LVDS_CFG_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4045
SBLKLEN	2	Value = 8
LANE_FMT_MAP	2	LANE0 Format Map. The mapping of the data on the lanes is depicted in the figure below 0x0000 Format map 0 0x0001 Format map 1

Continued on next page

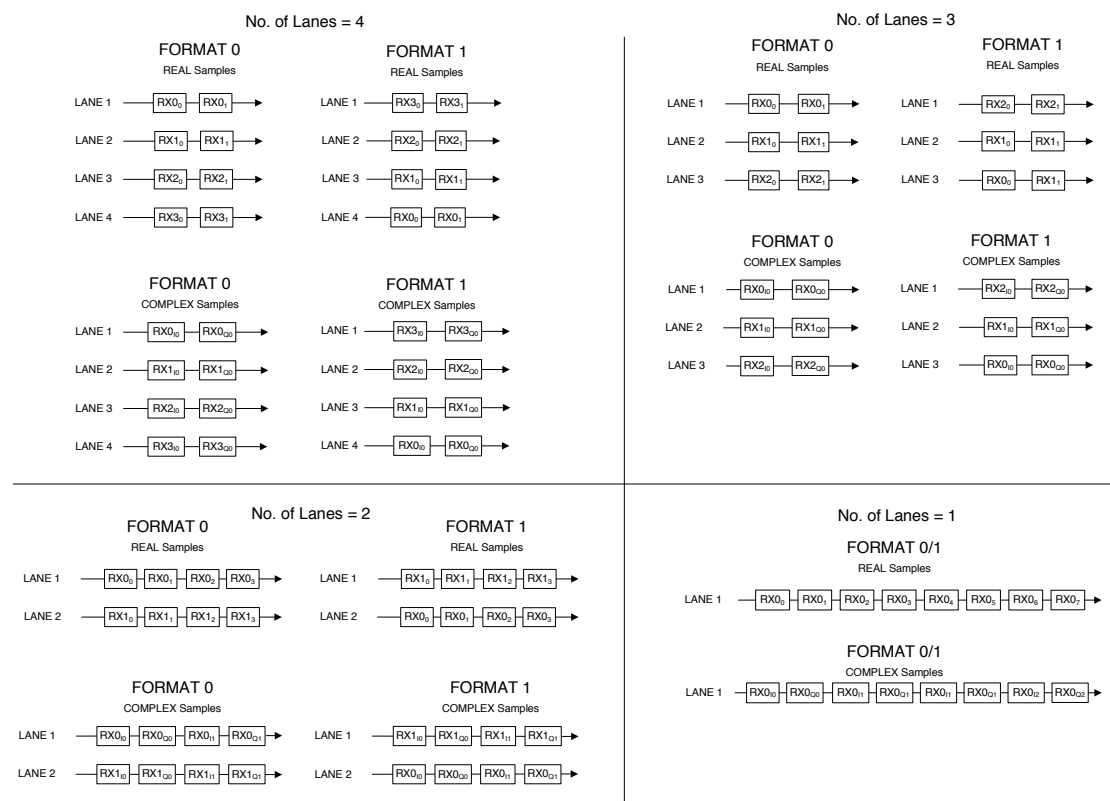
Table 5.149 – continued from previous page

LANE_PARAM_CFG	2	Bit	Description
		b0	MSB_FIRST
		0	Disable (LSB First)
		1	Enable (MSB First)
		b1	Packet End Pulse Enable
		0	Disable
		1	Enable
		b2	CRC Enable
		0	Disable
		1	Enable
		b7:3	RESERVED
		b8	Configures LSB/MSB first for CRC
		0	CRC value swapped wrt to MSB_FIRST setting
		1	CRC value follows MSB_FIRST setting
		b9	Frame clock state during idle
		0	Frame clock is held low
		1	Frame clock is held high
		b10	Frame clock period for CRC(when CRC enabled - b2)
		0	32-bit CRC is trasmitted as single sample with frame clock set to 16high, 16low configuration
		1	32-bit CRC is trasmitted as single sample with frame clock set to 8high, 8low configuration
		b11	Bit clock state during idle
		0	Bit clock toggles during idle when there are no transmission
		1	Bit clock doesn't toggle during idle when there are no transmission, the value of bit clock is held low
		b12	CRC inversion control(when CRC enabled - b2)
		0	The calcualted value of 32-bit ethernet polynomial CRC is inverted and sent out
		1	The calcualted value of 32-bit ethernet polynomial CRC is sent without inversion
		b15:13	RESERVED

Continued on next page

Table 5.149 – continued from previous page

The mapping of the 8 sample ($8 \times 16 = 128$ bit) information onto the serial interface lanes is determined by the LANE_FMT_MAP parameter. The choice of format map translating to the transfer of data on the lanes is depicted in the image below (the x axis represents time – hence the samples are as available on the lanes in time and the receiver will receive the samples in the reverse order as depicted below).


Figure 5.7: Lane formats and the order of receiving the data from the lanes

5.16.7 Sub block 0x4046 – AWR_DEV_RX_CONTSTREAMING_MODE_CONF_SET_SB

This sub block contains the configurations of the data path to transfer the captured ADC samples continuously without any break to an external host.

Table 5.150 describes the content of this sub block.

Table 5.150: AWR_DEV_RX_CONTSTREAMING_MODE_CFG_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4046
SBLKLEN	2	Value = 8
CONT_STREAMING_MODE	2	Continuous streaming mode enable Value Description 0x0 Continuous streaming mode data transfer disable 0x1 Continuous streaming mode data transfer enable
RESERVED	2	0x0000

5.16.8 Sub block 0x4047 – AWR_DEV_CSI2_CFG_SET_SB

This sub block contains the various configurations of the parameters of the CSI2 module. Table 5.151 describes the content of this sub block.

Table 5.151: AWR_DEV_CSI2_CFG_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4047
SBLKLEN	2	Value = 12

Continued on next page

Table 5.151 – continued from previous page

LANE_POS_ POL_SEL	4	Bits	Definition
		b2:0	DATA_LANE0_POS Valid values (Should be a unique position lane 0 cannot be disabled): 001b – Position 1 (default), 010b – Position 2, 011b – Position 3, 100b – Position 4, 101b – Position 5
		b3	DATA_LANE0_POL 0b – PLUSMINUS pin order, 1b – MINUSPLUS pin order
		b6:4	DATA_LANE1_POS Valid values (Should be a unique position if lane 1 is enabled, ignored if lane 1 is not enabled): 000b – Unused, 001b – Position 1, 010b – Position 2 (default), 011b – Position 3, 100b – Position 4, 101b – Position 5
		b7	DATA_LANE1_POL 0b – PLUSMINUS pin order, 1b – MINUSPLUS pin order
		b10:8	DATA_LANE2_POS Valid values (Should be a unique position if lane 2 is enabled, ignored if lane 2 is not enabled): 000b – Unused, 001b – Position 1, 010b – Position 2, 011b – Position 3, 100b – Position 4 (default), 101b – Position 5
		b11	DATA_LANE2_POL 0b – PLUSMINUS pin order, 1b – MINUSPLUS pin order
		b14:12	DATA_LANE3_POS Valid values (Should be a unique position if lane 3 is enabled, ignored if lane 3 is not enabled): 000b – Unused, 001b – Position 1, 010b – Position 2, 011b – Position 3, 100b – Position 4, 101b – Position 5 (default)
		b15	DATA_LANE3_POL 0b – PLUSMINUS pin order, 1b – MINUSPLUS pin order
		b18:16	CLOCK_POS Valid values (Should be a unique position): 0000b – Unused, 001b – Unused, 010b – Position 2, 011b – Position 3 (default), 100b – Position 4
		b19	CLOCK_POL 0b – PLUSMINUS pin order, 1b – MINUSPLUS pin order
		b31:20	RESERVED

Continued on next page

Table 5.151 – continued from previous page

DIS_LINE_START_END	1	0 – Line Start/End Enabled 1 – Line Start/End Disabled
RESERVED	3	0x00000000

5.16.9 Sub block 0x4048 – AWR_DEV_PMICCLOCK_CONF_SET_SB

This sub block contains the configurations to setup the desired frequency of the PMIC Clock that is output from the device. The configurations also allow setting up the dither values for the clock.

NOTE: The Maximum supported PMIC clock out is 20MHz.

Table 5.152 describes the contents of this sub block.

Table 5.152: AWR_DEV_PMICCLOCK_CONF_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4048
SBLKLEN	2	Value = 16
PMICCLOCK_CTRL	1	This field controls the enable-disable of the PMIC clock. Value Description 0x0 Disable PMIC clock 0x1 Enable PMIC clock
PMICCLOCK_SRC	1	This field specifies the source of the PMIC clock. Applicable only in case of PMIC clock enable. Else ignored. Value Description 0x0 XTAL (as connected to the device) 0x2 400 MHz PLL divided clock (xWR294x) 600 MHz PLL divided clock (other devices)

Continued on next page

Table 5.152 – continued from previous page

SRCCLOCK_DIV	1	This field specifies the division factor to be applied to source clock. Applicable only in case of PMIC clock enable. Else ignored. Allowed values in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP: <table><tr><th>Value</th><th>Description</th></tr><tr><td>0x0</td><td>Divide by 1</td></tr><tr><td>0x1</td><td>Divide by 2</td></tr><tr><td>...</td><td>...</td></tr><tr><td>0xF</td><td>Divide by 16</td></tr></table> Allowed values in other devices: <table><tr><th>Value</th><th>Description</th></tr><tr><td>0x0</td><td>Divide by 1</td></tr><tr><td>0x1</td><td>Divide by 2</td></tr><tr><td>...</td><td>...</td></tr><tr><td>0xFF</td><td>Divide by 256</td></tr></table> Note: The Maximum supported PMIC clock out is 20MHz.	Value	Description	0x0	Divide by 1	0x1	Divide by 2	...	...	0xF	Divide by 16	Value	Description	0x0	Divide by 1	0x1	Divide by 2	...	...	0xFF	Divide by 256
Value	Description																					
0x0	Divide by 1																					
0x1	Divide by 2																					
...	...																					
0xF	Divide by 16																					
Value	Description																					
0x0	Divide by 1																					
0x1	Divide by 2																					
...	...																					
0xFF	Divide by 256																					
MODE_SELECT	1	This field specifies the mode of operation for the PMIC clock generation. Applicable only in case of PMIC clock enable. Else ignored. <table><tr><th>Value</th><th>Description</th></tr><tr><td>0x0</td><td>Continuous mode (free running mode where the frequency change/jump is triggered based on configured number of PMIC clock ticks)</td></tr><tr><td>0x1</td><td>Chirp-to-Chirp staircase mode (frequency change/jump is triggered at every chirp boundary)</td></tr></table>	Value	Description	0x0	Continuous mode (free running mode where the frequency change/jump is triggered based on configured number of PMIC clock ticks)	0x1	Chirp-to-Chirp staircase mode (frequency change/jump is triggered at every chirp boundary)														
Value	Description																					
0x0	Continuous mode (free running mode where the frequency change/jump is triggered based on configured number of PMIC clock ticks)																					
0x1	Chirp-to-Chirp staircase mode (frequency change/jump is triggered at every chirp boundary)																					

Continued on next page

Table 5.152 – continued from previous page

FREQ_SLOPE	4	Applicable only in case of PMIC clock enable. Else ignored. Bit Description b25:0 Frequency slope value to be applied in [7.18] unsigned format 1 LSB = $1/2^{18}$ b31:26 RESERVED In continuous mode this value is accumulated every PMIC clock tick with the seed as MIN_NDIV_VAL till MAX_NDIV_VAL is reached In the stair case mode this value is accumulated every chirp with the seed as MIN_NDIV_VAL till MAX_NDIV_VAL is reached
MIN_NDIV_VAL	1	Applicable only in case of PMIC clock enable. Else ignored. Minimum allowed divider value (depends upon the highest desired clock frequency) Note: The Maximum supported PMIC clock out is 20MHz. The values can be set to get a few MHz of PMIC CLK output from the circuit, based on PMICCLOCK_SRC and SRCCLOCK_DIV. A few example configurations are given below this table.
MAX_NDIV_VAL	1	Applicable only in case of PMIC clock enable. Else ignored. Maximum allowed divider value (depends upon the lowest desired clock frequency) Note: The Maximum supported PMIC clock out is 20MHz. The values can be set to get a few MHz of PMIC CLK output from the circuit, based on PMICCLOCK_SRC and SRCCLOCK_DIV. A few example configurations are given below this table.
CLK_DITHER_EN	1	Applicable only in case of PMIC clock enable and frequency slope is non-zero. Else ignored. This field controls the enable-disable of the clock dithering. Adds a pseudo random real number (0 or 1) to the accumulated divide value. Hence it brings a random dithering of 1 LSB. Value Description 0x0 Clock dithering disabled 0x1 Clock dithering enabled
RESERVED	1	0x00

Note: The examples and their configurations below assume the use of an xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP device.

Example 1. PMIC clock with no slope in continuous mode

Objective: To configure the PMIC clock at frequency of 2 MHz with no slope.

Configurations:

1. PMICCLK_SRC = 0x2 (400 MHz PLL divided clock)
2. SRCCLOCK_DIV = 9, Reference clock = $400 \text{ MHz} / (9 + 1) = 40 \text{ MHz}$
3. MIN_NDIV_VAL = MAX_NDIV_VAL = 20 (Computed as $40 \text{ MHz} / 2.0 \text{ MHz}$)
4. FREQ_SLOPE = 0

With the above configuration, the PMIC clock frequency would be $\text{PMIC clock} = (40 \text{ MHz} / 20) = 2 \text{ MHz}$

Example 2. Dithered PMIC clock with slope in chirp-to-chirp staircase mode

Objective: To configure a dithered PMIC clock at frequencies ranging from 2 MHz to 2.5 MHz over 32 chirps.

Configurations:

1. PMICCLK_SRC = 0x2 (400 MHz PLL divided clock)
2. SRCCLOCK_DIV = 1, Reference clock = $400 \text{ MHz} / (1 + 1) = 200 \text{ MHz}$
3. MODE_SELECT = 1
4. FREQ_SLOPE = 169125 (Computed as $(\text{MAX_NDIV_VAL} - \text{MIN_NDIV_VAL}) \times 2^{18} / 31$)
5. MIN_NDIV_VAL = 80 (Computed as $200 \text{ MHz} / 2.5 \text{ MHz}$)
6. MAX_NDIV_VAL = 100 (Computed as $200 \text{ MHz} / 2.0 \text{ MHz}$)
7. CLK_DITHER_EN = 1

With the above configuration, the PMIC clock frequency would be vary between $(200 \text{ MHz} / 80)$ and $(200 \text{ MHz} / 100)$ in steps of $(200 \text{ MHz} / \lfloor (80 + (N \times \text{FREQ_SLOPE} / 2^{18} + X)) \rfloor)$ where

- N = Chirp number
- X = random fractional value in the range (0, 1) that adds the dither

The PMIC clock frequency is determined by the clock divider value which starts with a value of 100, providing a PMIC clock of 2 MHz for the 1st chirp, decrementing the divider by $\text{FREQ_SLOPE} / 2^{18} = 0.64516$ every chirp and finally reaching a value of 20 for the 32nd chirp providing a PMIC clock of 2.5 MHz.

Table 5.153: PMIC clock frequency across chirps in chirp-to-chirp staircase mode in an example when PMIC clock varies from 2 MHz to 2.5 MHz in 32 chirps

Chirp Number	PMIC Clock Frequency (MHz)	Calculation
1	2.50000	$200/(80 + 0 \times 169125/2^{18})$
2	2.48000	$200/(80 + 1 \times 169125/2^{18})$
3	2.46032	$200/(80 + 2 \times 169125/2^{18})$
4	2.44094	$200/(80 + 3 \times 169125/2^{18})$
5	2.42188	$200/(80 + 4 \times 169125/2^{18})$
6	2.40310	$200/(80 + 5 \times 169125/2^{18})$
7	2.38462	$200/(80 + 6 \times 169125/2^{18})$
8	2.36641	$200/(80 + 7 \times 169125/2^{18})$
9	2.34848	$200/(80 + 8 \times 169125/2^{18})$
10	2.33083	$200/(80 + 9 \times 169125/2^{18})$
11	2.31343	$200/(80 + 10 \times 169125/2^{18})$
12	2.29630	$200/(80 + 11 \times 169125/2^{18})$
13	2.27941	$200/(80 + 12 \times 169125/2^{18})$
14	2.26277	$200/(80 + 13 \times 169125/2^{18})$
15	2.24638	$200/(80 + 14 \times 169125/2^{18})$
16	2.23022	$200/(80 + 15 \times 169125/2^{18})$
17	2.21429	$200/(80 + 16 \times 169125/2^{18})$
18	2.19858	$200/(80 + 17 \times 169125/2^{18})$
19	2.18310	$200/(80 + 18 \times 169125/2^{18})$
20	2.16783	$200/(80 + 19 \times 169125/2^{18})$
21	2.15278	$200/(80 + 20 \times 169125/2^{18})$
22	2.13793	$200/(80 + 21 \times 169125/2^{18})$
23	2.12329	$200/(80 + 22 \times 169125/2^{18})$
24	2.10884	$200/(80 + 23 \times 169125/2^{18})$
25	2.09459	$200/(80 + 24 \times 169125/2^{18})$
26	2.08054	$200/(80 + 25 \times 169125/2^{18})$
27	2.06667	$200/(80 + 26 \times 169125/2^{18})$
28	2.05298	$200/(80 + 27 \times 169125/2^{18})$
29	2.03947	$200/(80 + 28 \times 169125/2^{18})$
30	2.02614	$200/(80 + 29 \times 169125/2^{18})$
31	2.01299	$200/(80 + 30 \times 169125/2^{18})$
32	2.00000	$200/(80 + 31 \times 169125/2^{18})$

Example 3. Dithered PMIC clock with slope in continuous mode

Objective: To configure a dithered PMIC clock at frequencies ranging from 2 MHz to 2.5 MHz over 100 μ s.

Configurations:

1. PMICCLK_SRC = 0x2 (400 MHz PLL divided clock)
2. SRCCLOCK_DIV = 1, Reference clock = 400 MHz / (1 + 1) = 200 MHz
3. MODE_SELECT = 0
4. FREQ_SLOPE = 23302 (Computed as $(\text{MAX_NDIV_VAL} - \text{MIN_NDIV_VAL}) \cdot 2^{18} / (100 \mu\text{s} \cdot (2.5 \text{ MHz} + 2 \text{ MHz})/2)$)
5. MIN_NDIV_VAL = 80 (Computed as 200 MHz/2.5 MHz)
6. MAX_NDIV_VAL = 100 (Computed as 200 MHz/2.0 MHz)
7. CLK_DITHER_EN = 1

With the above configuration, the PMIC clock frequency would be PMIC clock would vary between = (200 MHz / 80) to (200 MHz / 100) in steps of $(200 \text{ MHz} / [(80 + (N \times 23302/2^{18} + X))])$ where

- N = Iteration count that ticks every PMIC clock. The average value of PMIC clock here is $\sim 2.25 \text{ MHz}$. Hence the iteration count ticks every $(1/2.25 \text{ MHz}) \sim 0.444 \mu\text{s}$.
- X = random fractional value in the range (0, 1) that adds the dither

The PMIC clock frequency is determined by the clock divider which starts with a value of 100, on the 1st PMIC clock period, providing a PMIC clock of 2 MHz, decrementing the divider value by $23303/2^{18} = 0.08889$ every PMIC clock period of $1/2.25 \text{ MHz} \sim 0.444 \mu\text{s}$, finally reaching a value of 80 on 225th PMIC clock period, providing a PMIC clock of 2.5 MHz. Hence, the frequency varies from [2 MHz, 2.5 MHz] over 225 PMIC clock periods or $225 \times 0.444 \mu\text{s}$ or $\sim 100 \mu\text{s}$.

5.16.10 Sub block 0x4049 – AWR_MSS_PERIODICTESTS_CONF_SB

This sub block is used to trigger the periodic tests in MSS.

Table 5.154 describes the content of this sub block.

Table 5.154: AWR_MSS_PERIODICTESTS_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4049

Continued on next page

Table 5.154 – continued from previous page

SBLKLEN	2	Value = 16
PERIODICITY	4	Periodicity at which tests need to be run 1 LSB = 1 ms Minimum value is 40 ms Maximum value is 150ms NOTE: MSS Windowed WDT period is set to this periodicity and WDT can not support period more than 150ms.
TEST_EN	4	1 – Enable, 0 – Disable Bit Monitoring type b0 PERIODIC_CONFIG_REGISTER_READ_EN b1 ESM_MONITORING_EN b31:2 RESERVED
REPORTING_MODE	1	Controls when the AWR device sends the report corresponding to the periodic tests to the host. A report generically refers to both success/failure status flags. Value Definition 0 Report is sent every monitoring period 1 Report is sent only on a failure
RESERVED	3	0x000000

NOTE:

The MSS periodic monitor test run and latent tests are not recommended to run in parallel as latent tests are destructive tests which would cause periodic tests to fail.

5.16.11 Sub block 0x404A – AWR_MSS_LATENTFAULT_TEST_CONF_SB

This sub block is used to trigger the periodic latent fault tests in MSS, this API should not be issued when functional frames are running, these are destructive tests.

Table 5.155 describes the content of this sub block.

Table 5.155: AWR_MSS_LATENTFAULT_TEST_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x404A
SBLKLEN	2	Value = 16

Continued on next page

Table 5.155 – continued from previous page

TEST_EN_1	4	Bits	Definition
		b0	RESERVED
		b1	DMA self-test
		b2	RESERVED
		b3	RTI self-test
		b4	RESERVED
		b5	EDMA self-test
		b6	CRC self-test
		b7	VIM self-test
		b8	RESERVED
		b9	Mailbox self-test
		b10	RESERVED
		b11	RESERVED
		b12	Generating NERROR
		b13	MibSPI single bit error test
		b14	MibSPI double bit error test
		b15	DMA Parity error
		b16	TCMA RAM single bit errors (Not supported, refer latest release note)
		b17	TCMB RAM single bit errors (Not supported, refer latest release note)
		b18	TCMA RAM double bit errors (Not supported, refer latest release note)
		b19	TCMB RAM double bit errors (Not supported, refer latest release note)
		b20	TCMA RAM parity errors (Not supported, refer latest release note)
		b21	TCMB RAM parity errors (Not supported, refer latest release note)
		b22	RESERVED
		b23	RESERVED
		b24	DMA MPU Region tests
		b25	MSS Mailbox single bit errors
		b26	MSS Mailbox double bit errors
		b27	BSS Mailbox single bit errors
		b28	BSS Mailbox double bit errors
		b29	EDMA MPU test
		b30	EDMA parity test
		b31	CSI2 parity test

Continued on next page

Table 5.155 – continued from previous page

TEST_EN_2	4	Bits	Definition
		b0	RESERVED
		b1	RESERVED
		b2	RESERVED
		b3	VIM RAM parity test
		b4	SCI boot time test
		b31:5	RESERVED
REPORTING_MODE	1	Value	Definition
		0	Report is sent after test completion
		1	Report is send only upon a failure
TEST_MODE	1	Value	Definition
		0	Production mode. Latent faults are tested and any failures are reported
		1	Characterization mode. Faults are injected and failures are reported which allows testing of the failure reporting
RESERVED	2	0x0000	

NOTE1: The MSS latent self tests are destructive tests, which would cause corruption in ongoing SPI/mailbox transactions and may generate N-Error signals while performing ESM G2 error checks. The MIB-SPI ECC tests (b13,b14) can be destructive tests if there is an ongoing MIBSPI communication. It is recommended not to run these self tests in functional mode of operation.

NOTE2: It is recommended to wait for the latent fault test report asynchronous event after issuing this API. The MSS latent self tests cannot be issued back to back without waiting for the test report event.

5.16.12 Sub block 0x404B – AWR_DEV_TESTPATTERN_GEN_SET_SB

This sub block contains the configurations to setup the test pattern to be generated and transferred over the selected high speed interface (LVDS). This command has to be issued after the data path configurations commands are issued. This can be used to perform a sanity test of the high speed interface connectivity and correct reception.

Table 5.156 describes the contents of this sub block.

Table 5.156: AWR_DEV_TESTPATTERN_GEN_SET_SB contents

Field Name	Number of bytes	Description															
SBLKID	2	Value = 0x404B															
SBLKLEN	2	Value = 48															
TESTPATTERN_GEN_CTRL	1	This field controls the enable-disable of the generation of the test pattern. <table><tr><th>Value</th><th>Description</th></tr><tr><td>0x0</td><td>Disable test pattern generation</td></tr><tr><td>0x1</td><td>Enable test pattern generation</td></tr></table>	Value	Description	0x0	Disable test pattern generation	0x1	Enable test pattern generation									
Value	Description																
0x0	Disable test pattern generation																
0x1	Enable test pattern generation																
TESTPATTERN_GEN_TIMING	1	Number of system clocks (200 MHz) between successive samples for the test pattern gen. Applicable only in case of Test pattern enable. Else ignored.															
TESTPATTERN_PKT_SIZE	2	Number of ADC samples to capture for each RX Valid range: 64 to MAX_NUM_SAMPLES, Where MAX_NUM_SAMPLES is such that all the enabled RX channels' data fits into 16 kB memory, with each sample consuming 2 bytes for real ADC output case and 4 bytes for complex 1x and complex 2x ADC output cases. For example in AWR2243/xWR6243/AWR1243/xWR1443 when the ADC buffer size is 16 kB <table><tr><th>Number of RX chains</th><th>ADC format</th><th>MAX_NUM_SAMPLES</th></tr><tr><td>4</td><td>Complex</td><td>1024</td></tr><tr><td>4</td><td>Real</td><td>2048</td></tr><tr><td>2</td><td>Complex</td><td>2048</td></tr><tr><td>2</td><td>Real</td><td>4096</td></tr></table>	Number of RX chains	ADC format	MAX_NUM_SAMPLES	4	Complex	1024	4	Real	2048	2	Complex	2048	2	Real	4096
Number of RX chains	ADC format	MAX_NUM_SAMPLES															
4	Complex	1024															
4	Real	2048															
2	Complex	2048															
2	Real	4096															
NUM_TESTPATTERN_PKTS	4	Number of test pattern packets to send For infinite packets set it to 0															
TESTPATTERN_RX0_ICFG	4	This field specifies the values for Rx0, I channel. Applicable only in case of test pattern enable. Else ignored. <table><tr><th>Bits</th><th>Description</th></tr><tr><td>b15:0</td><td>Start offset value to be used for the first sample for the test pattern data</td></tr><tr><td>b31:16</td><td>Value to be added for each successive sample for the test pattern data</td></tr></table>	Bits	Description	b15:0	Start offset value to be used for the first sample for the test pattern data	b31:16	Value to be added for each successive sample for the test pattern data									
Bits	Description																
b15:0	Start offset value to be used for the first sample for the test pattern data																
b31:16	Value to be added for each successive sample for the test pattern data																

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Table 5.156 – continued from previous page

TESTPATTERN_ RX0_QCFG	4	This field specifies the values for Rx0, Q channel. Applicable only in case of test pattern enable. Else ignored. Bits Description b15:0 Start offset value to be used for the first sample for the test pattern data b31:16 Value to be added for each successive sample for the test pattern data
TESTPATTERN_ RX1_ICFG	4	This field specifies the values for Rx1, I channel. Applicable only in case of test pattern enable. Else ignored. Bits Description b15:0 Start offset value to be used for the first sample for the test pattern data b31:16 Value to be added for each successive sample for the test pattern data
TESTPATTERN_ RX1_QCFG	4	This field specifies the values for Rx1, Q channel. Applicable only in case of test pattern enable. Else ignored. Bits Description b15:0 Start offset value to be used for the first sample for the test pattern data b31:16 Value to be added for each successive sample for the test pattern data
TESTPATTERN_ RX2_ICFG	4	This field specifies the values for Rx2, I channel. Applicable only in case of test pattern enable. Else ignored. Bits Description b15:0 Start offset value to be used for the first sample for the test pattern data b31:16 Value to be added for each successive sample for the test pattern data
TESTPATTERN_ RX2_QCFG	4	This field specifies the values for Rx2, Q channel. Applicable only in case of test pattern enable. Else ignored. Bits Description b15:0 Start offset value to be used for the first sample for the test pattern data b31:16 Value to be added for each successive sample for the test pattern data

Continued on next page

Table 5.156 – continued from previous page

TESTPATTERN_ RX3_ICFG	4	This field specifies the values for Rx3, I channel. Applicable only in case of test pattern enable. Else ignored. Bits Description b15:0 Start offset value to be used for the first sample for the test pattern data b31:16 Value to be added for each successive sample for the test pattern data
TESTPATTERN_ RX3_QCFG	4	This field specifies the values for Rx3, Q channel. Applicable only in case of test pattern enable. Else ignored. Bits Description b15:0 Start offset value to be used for the first sample for the test pattern data b31:16 Value to be added for each successive sample for the test pattern data
RESERVED	4	0x00000000

NOTE: This test pattern can be used only in LVDS testing and bring-up

5.16.13 Sub block 0x404C – AWR_DEV_CONFIGURATION_SET_SB

This API is used to configure the CRC type for the async events from MSS. The default is 16 bit CRC if this API is not issued. The first async event after MSS powerup will have a 16 bit CRC.

Table 5.157: AWR_DEV_CONFIGURATION_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x404C
SBLKLEN	2	Value = 16
ASYNC_EVENT_ CRC_CFG	1	Value Description 0 16 bit CRC for MSS async events 1 32 bit CRC for MSS async events 2 64 bit CRC for MSS async events
MISC_DEV_CFG	1	Bit Field Description b0 Enable MSS Logger Default value : 0 (Logger Disabled) b31-1 RESERVED

Continued on next page

Table 5.157 – continued from previous page

RESERVED1	2	0x0000
RESERVED2	4	0x00000000
RESERVED3	4	0x00000000

5.16.14 Sub block 0x404D – AWR_DEV_RF_DEBUG_SIG_SET_SB

This sub-block contains the information to enable the pin-mux to bring out debug signals for the chirp cycle.

CLK_OUT signal will be output on OSC_CLKOUT pin and ADC_SIG_OUT will be output on GPIO_0 pin.

Table 5.158: AWR_DEV_RF_DEBUG_SIG_SET_SB contents

Field Name	Number of bytes	Description												
SBLKID	2	Value = 0x404D												
SBLKLEN	2	Value = 20												
CLK_OUT	2	<table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>NO_CLK_OUT, Disable clock out signal</td></tr><tr><td>1</td><td>REF_CLK_OUT, Reference clock out enable</td></tr><tr><td>2</td><td>APLL_CLK_OUT, APLL clock out enable</td></tr><tr><td>3</td><td>2P5G_SYNTH_CLK_OUT, 2.5GHz Synth clock out enable</td></tr><tr><td>4</td><td>5G_SYNTH_CLK_OUT, 5GHz Synth clock out enable</td></tr></table>	Value	Description	0	NO_CLK_OUT, Disable clock out signal	1	REF_CLK_OUT, Reference clock out enable	2	APLL_CLK_OUT, APLL clock out enable	3	2P5G_SYNTH_CLK_OUT, 2.5GHz Synth clock out enable	4	5G_SYNTH_CLK_OUT, 5GHz Synth clock out enable
Value	Description													
0	NO_CLK_OUT, Disable clock out signal													
1	REF_CLK_OUT, Reference clock out enable													
2	APLL_CLK_OUT, APLL clock out enable													
3	2P5G_SYNTH_CLK_OUT, 2.5GHz Synth clock out enable													
4	5G_SYNTH_CLK_OUT, 5GHz Synth clock out enable													
ADC_SIG_OUT	2	<table><tr><th>Bit</th><th>Description</th></tr><tr><td>b0</td><td>ADC_VALID, ADC valid signal enabled in GPIO_0</td></tr><tr><td>b31:1</td><td>RESERVED</td></tr><tr><td>0:</td><td>Disable</td></tr><tr><td>1:</td><td>Enable</td></tr></table>	Bit	Description	b0	ADC_VALID, ADC valid signal enabled in GPIO_0	b31:1	RESERVED	0:	Disable	1:	Enable		
Bit	Description													
b0	ADC_VALID, ADC valid signal enabled in GPIO_0													
b31:1	RESERVED													
0:	Disable													
1:	Enable													
RESERVED	4	0x00000000												
RESERVED	4	0x00000000												
RESERVED	4	0x00000000												

5.16.15 Sub block 0x404E – AWR_DEV_DEV_HSI_DELAY_DUMMY_CFG_SET_SB

This API can be used to increase the time between the availability of chirp data and the transfer of chirp data over HSI (CSI/LVDS) interface. It can also be used to add configurable amount of dummy data per chirp at the end of actual chirp data.

NOTE1:	The user should configure the delay (and the dummy count) such that the chirp data transmission is completed before the start of next chirp's data transmission. Excessive delay or dummy count may lead to fault which will be reported by MSS
NOTE2:	The change in Delay value configuration will reflect immediately. But any change in Dummy value configuration will reflect only after frame configuration.
NOTE3:	There is some delay T0 (even without enabling this API) between the chirp data availability to data transmission. The delay added by this API is in addition to this T0. T0 depends on ADC Sampling rate, CSI Data rate and whether the CSI Line-Start-Line-End is enabled or not. The programmer needs to account for this delay while selecting the configuration values for this API.

Table 5.159: AWR_DEV_DEV_HSI_DELAY_DUMMY_CFG_SET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x404E
SBLKLEN	2	Value = 20
ENBALE_MODE	1	This field decides if the Delay or Dummy option is enabled or disabled Mode Definition 0 No Delay or Dummy Data (Disabled) 1 HSI Data will have an additional configurable delay after ADC VALID and Configurable Dummy data after chirp data 2 Similar to configuration 0x1 but the device varies the delay in each chirp 3 - RESERVED 0xFF
RESERVED	3	0x0000

Continued on next page

Table 5.159 – continued from previous page

DELAY_VAL	2	Delay Count value Mode DELAY_VAL Definition 0 NA 1 1 LSB = 20 ns delay Delay Added = (DELAY_VAL*20ns) + 1.2us 2 Delay Added = Vary from chirp to chirp within MIN_DELAY to MAX_DELAY. MIN_DELAY = 1.1us MAX_DELAY = 1.2us + (DELAY_VAL * 20ns)
RESERVED	2	0x0000
DUMMY_VAL	2	Dummy Count value Number of dummy bytes added per chirp For 12-bit ADC data, 12 * Dummy Value For 14-bit ADC data, 14 * Dummy Value For 16-bit ADC data, 16 * Dummy Value Valid Range: 0 to 2048
RESERVED	2	0x0000
RESERVED	4	0x00000000

5.17 Sub blocks related to AWR_DEV_CONF_GET_MSG

5.17.1 Sub block 0x4060 – AWR_DEV_MCUCLOCK_GET_SB

This API is used to read the MCU clock configuration. Response packet structure will be same as AWR_DEV_MCUCLOCK_SET_SB

Table 5.160: AWR_DEV_MCUCLOCK_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4060
SBLKLEN	2	Value = 4

5.17.2 Sub block 0x4061 – AWR_DEV_RX_DATA_FORMAT_CONF_GET_SB

This API is used to read the RX data format configuration. Response packet structure will be same as AWR_DEV_RX_DATA_FORMAT_CONF_SET_SB

Table 5.161: AWR_DEV_RX_DATA_FORMAT_CONF_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4061
SBLKLEN	2	Value = 4

5.17.3 Sub block 0x4062 – AWR_DEV_RX_DATA_PATH_CONF_GET_SB

This API is used to read the RX data path configuration. Response packet structure will be same as AWR_DEV_RX_DATA_PATH_CONF_SET_SB

Table 5.162: AWR_DEV_RX_DATA_PATH_CONF_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4062
SBLKLEN	2	Value = 4

5.17.4 Sub block 0x4063 – AWR_DEV_RX_DATA_PATH_LANEEN_GET_SB

This API is used to read the RX data path lane enable configuration. Response packet structure will be same as AWR_DEV_RX_DATA_PATH_LANEEN_SET_SB

Table 5.163: AWR_DEV_RX_DATA_PATH_LANEEN_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4063
SBLKLEN	2	Value = 4

5.17.5 Sub block 0x4064 – AWR_DEV_RX_DATA_PATH_CLK_GET_SB

This API is used to read the RX data path clock configuration. Response packet structure will be same as AWR_DEV_RX_DATA_PATH_CLK_SET_SB

Table 5.164: AWR_DEV_RX_DATA_PATH_CLK_GET_SB contents

Field Name	Number of bytes	Description
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SBLKID	2	Value = 0x4064
SBLKLEN	2	Value = 4

5.17.6 Sub block 0x4065 – AWR_DEV_LVDS_CFG_GET_SB

This API is used to read the LVDS configuration. Response packet structure will be same as AWR_DEV_LVDS_CFG_SET_SB

Table 5.165: AWR_DEV_LVDS_CFG_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4065
SBLKLEN	2	Value = 4

5.17.7 Sub block 0x4066 – AWR_DEV_RX_CONTSTREAMING_MODE_CONF_GET_SB

This API is used to read the continuous streaming mode configuration. Response packet structure will be same as AWR_DEV_RX_CONTSTREAMING_MODE_CONF_SET_SB

Table 5.166: AWR_DEV_RX_CONTSTREAMING_CONF_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4066
SBLKLEN	2	Value = 4

5.17.8 Sub block 0x4067 – AWR_DEV_CSI2_CFG_GET_SB

This API is used to read the CSI2 configuration. Response packet structure will be same as AWR_DEV_CSI2_CFG_SET_SB

Table 5.167: AWR_DEV_CSI2_CFG_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4067
SBLKLEN	2	Value = 4

5.17.9 Sub block 0x4068 – AWR_DEV_PMICCLOCK_CONF_GET_SB

This API is used to read the PMIC clock configuration. Response packet structure will be same as AWR_DEV_PMICCLOCK_CONF_SET_SB

Table 5.168: AWR_DEV_PMICCLOCK_CONF_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4068
SBLKLEN	2	Value = 4

5.17.10 Sub block 0x4069 – AWR_MSS_LATENTFAULT_TEST_CONF_GET_SB

This API is used to read the MSS latent fault test configuration. Response packet structure will be same as AWR_MSS_LATENTFAULT_TEST_CONF_SET_SB

Table 5.169: AWR_MSS_LATENTFAULT_CONF_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x4069
SBLKLEN	2	Value = 4

5.17.11 Sub block 0x406A – AWR_MSS_PERIODICTESTS_CONF_GET_SB

This API is used to read the MSS periodic tests configuration. Response packet structure will be same as AWR_MSS_PERIODICTESTS_CONF_SET_SB

Table 5.170: AWR_MSS_PERIODICTESTS_CONF_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x406A
SBLKLEN	2	Value = 4

5.17.12 Sub block 0x406B – AWR_DEV_TESTPATTERN_GEN_GET_SB

This API is used to read the test pattern generation configuration. Response packet structure will be same as AWR_DEV_TESTPATTERN_GEN_SET_SB

Table 5.171: AWR_DEV_TESTPATTERN_GEN_GET_SB contents

Field Name	Number of bytes	Description
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SBLKID	2	Value = 0x406B
SBLKLEN	2	Value = 4

5.18 Sub blocks related to AWR_DEV_FILE_DOWNLOAD_MSG

5.18.1 Sub block 0x4080 – AWR_DEV_FILE_DOWNLOAD_SB

This sub block is used to send the file in chunks/parts for download into RAM.
Table 5.172 describes the content of this sub block.

Table 5.172: AWR_DEV_FILE_DOWNLOAD_SB contents

Field Name	Number of bytes	Description																		
SBLKID	2	Value = 0x4080																		
SBLKLEN	2	Value = Variable																		
FILE_TYPE	4	<table><tr><th>Value</th><th>Description</th></tr><tr><td>0x0</td><td>META_IMAGE TO SRAM</td></tr><tr><td>0x1</td><td>RESERVED</td></tr><tr><td>0x2</td><td>RESERVED</td></tr><tr><td>0x3</td><td>RESERVED</td></tr><tr><td>0x4</td><td>META_IMAGE1 TO SFLASH</td></tr><tr><td>0x5</td><td>META_IMAGE2 TO SFLASH</td></tr><tr><td>0x6</td><td>META_IMAGE3 TO SFLASH</td></tr><tr><td>0x7</td><td>META_IMAGE4 TO SFLASH</td></tr></table>	Value	Description	0x0	META_IMAGE TO SRAM	0x1	RESERVED	0x2	RESERVED	0x3	RESERVED	0x4	META_IMAGE1 TO SFLASH	0x5	META_IMAGE2 TO SFLASH	0x6	META_IMAGE3 TO SFLASH	0x7	META_IMAGE4 TO SFLASH
Value	Description																			
0x0	META_IMAGE TO SRAM																			
0x1	RESERVED																			
0x2	RESERVED																			
0x3	RESERVED																			
0x4	META_IMAGE1 TO SFLASH																			
0x5	META_IMAGE2 TO SFLASH																			
0x6	META_IMAGE3 TO SFLASH																			
0x7	META_IMAGE4 TO SFLASH																			
FILE_LENGTH	4	Length of File																		
FILE_CONTENT	Variable	Content of File, may split into multiple chunks.																		

NOTE: In the first chunk of file, FILE_TYPE and FILE_LENGTH is available and then first chunk onward these two fields will not be part of SB content

5.19 Sub blocks related to AWR_DEV_FRAME_CONFIG_APPLY_MSG

5.19.1 Sub block 0x40C0 – AWR_DEV_FRAME_CONFIG_APPLY_SB

This sub block is used to indicate to MSS to apply all the device configurations in the hardware.
This API should be used when lagacy frame config is used.
Table 5.173 describes the content of this sub block.

Table 5.173: AWR_DEV_FRAME_CONFIG_APPLY_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x40C0
SBLKLEN	2	Value = 12
NUM_CHIRPS	4	Number of chirps per frame
HALF_WORDS_PER_CHIRP	2	Number of half words in ADC buffer per chirp Example 1: In real mode, if number of ADC samples per chirp is 256 then this value will be 256 Example 2: In complex1x or complex2x modes, if number of ADC samples per chirp is 256 then this value will be 512
RESERVED	2	0x0000

5.19.2 Sub block 0x40C1 – AWR_DEV_ADV_FRAME_CONFIG_APPLY_SB

This sub block is used to indicate to MSS to apply all the advanced frame configuration settings in the hardware. This API should be used when advance frame config is used.

Table 5.174 describes the content of this sub block.

Table 5.174: AWR_DEV_ADV_FRAME_CONFIG_APPLY_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x40C1
SBLKLEN	2	Value = 40
NUM_SUB-FRAMES	1	Number of sub frames enabled in this frame Valid range: 1 to 4
RESERVED	3	0x00
SF1_TOT_NUM_CHIRPS	4	Number of chirps in sub frame 1

Continued on next page

Table 5.174 – continued from previous page

SF1_NUM_ADC_SAMPLES_PER_DATA_PKT	2	Number of half words (16 bits) of ADC samples per data packet in sub-frame 1 Example 1: In real mode, if number of ADC samples per chirp in subframe1 is 256 then this value will be 256 Example 2: In complex1x or complex2x modes, if number of ADC samples per chirp in subframe1 is 256 then this value will be 512 In AWR2243/xWR6243/AWR1243/xWR1443: Program this as the same as number of ADC samples in each chirp of this sub frame (required to be the same) Exception: Can do #chirps based ping-pong as in xWR1642 (see below), if CP/CQ are not needed. Useful for chirp stitching use case. In xWR1642/xWR1843 (For reference Only): The ADC samples corresponding to one or more chirps can be grouped and sent to the DSP as a single packet. Program this as the number of half words of ADC samples per packet. Ensure that in one sub frame, there is integer number of such packets. Maximum size of a data packet: (16384 - 1) half words.
SF1_PROC_NUM_CHIRPS_PER_DATA_PKT	1	Number of chirps per data packet to process at a time in sub-frame 1. In AWR2243/xWR6243/AWR1243/xWR1443: Program this as 1. Exception: Can be > 1 as in 1642 if CP/CQ is not needed. Useful for chirp stitching use case. In xWR1642/xWR1843 (For reference Only): The ADC samples corresponding to one or more chirps can be grouped and sent to the DSP as a single packet. Program this as the corresponding number of chirps per packet. Maximum value = 8. Note on maximum size: 8 chirps for CP and BPM.
RESERVED	1	0x00
SF2_TOT_NUM_CHIRPS	4	Number of chirps in sub-frame 2
SF2_NUM_ADC_SAMPLES_PER_DATA_PKT	2	Number of ADC Samples per data packet in sub-frame 2 Same conditions apply as in sub-frame 1.
SF2_PROC_NUM_CHIRPS_PER_DATA_PKT	1	Number of chirps per data packet to process at a time in sub-frame 2 Same conditions apply as in sub-frame 1.
RESERVED	1	0x00

Continued on next page

Table 5.174 – continued from previous page

SF3_TOT_NUM_CHIRPS	4	Number of chirps in sub-frame3
SF3_NUM_ADC_SAMPLES_PER_DATA_PKT	2	Number of ADC samples per data packet in sub-frame 3 Same conditions apply as in sub-frame 1.
SF3_PROC_NUM_CHIRPS_PER_DATA_PKT	1	Number of chirps per data packet to process at a time in sub-frame 3 Same conditions apply as in sub-frame 1.
RESERVED	1	0x00
SF4_TOT_NUM_CHIRPS	4	Number of chirps in sub-frame4
SF4_NUM_ADC_SAMPLES_PER_DATA_PKT	2	Number of ADC samples per data packet in sub-frame 4 Same conditions apply as in sub-frame 1.
SF4_PROC_NUM_CHIRPS_PER_DATA_PKT	1	Number of chirps per data packet to process at a time in sub-frame 4 Same conditions apply as in sub-frame 1.
RESERVED	1	0x00

5.20 Sub blocks related to AWR_DEV_STATUS_GET_MSG

5.20.1 Sub block 0x40E0 – AWR_MSSVERSION_GET_SB

This sub block reads MSS FW version. The information returned by the device will be in the format as given in AWR_MSSVERSION_SB.

Table 5.175 describes the contents of the request sub block

Table 5.175: AWR_MSSVERSION_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x40E0
SBLKLEN	2	Value = 4

Response to AWR_MSSVERSION_GET_SB

AWR_MSSVERSION_SB sub block is sent by the radar device in response to AWR_MSSVERSION_GET_SB. Note that SBLKID for both AWR_MSSVERSION_GET_SB and AWR_MSSVERSION_SB are same.

Table 5.176 describes the contents of the response sub block.

Table 5.176: AWR_MSSVERSION_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x40E0
SBLKLEN	2	Value = 20
HW_VARIANT	1	HW variant number
HW_VERSION_MAJOR	1	HW version major number
HW_VERSION_MINOR	1	HW version minor number
MSS_FW_VERSION_MAJOR	1	MSS FW version major number
MSS_FW_VERSION_MINOR	1	MSS FW version minor number
MSS_FW_VERSION_BUILD	1	MSS FW version build number
MSS_FW_VERSION_DEBUG	1	MSS FW version debug number
MSS_FW_VERSION_YEAR	1	Year of MSS FW version release
MSS_FW_VERSION_MONTH	1	Month of MSS FW version release
MSS_FW_VERSION_DAY	1	Day of MSS FW version release
MSS_FW_VERSION_PATCH_MAJOR	1	MSS FW version patch major number
MSS_FW_VERSION_PATCH_MINOR	1	MSS FW version patch minor number
MSS_FW_VERSION_PATCH_YEAR	1	Year of MSS FW patch release
MSS_FW_VERSION_PATCH_MONTH	1	Month of MSS FW patch release
MSS_FW_VERSION_PATCH_DAY	1	Day of MSS FW patch release

Continued on next page

Table 5.176 – continued from previous page

MSS_FW_PATCH_BUILD_DEBUG_VERSION	1	Bit	Definition
		b3:0	DEBUG version number
		b7:4	BUILD version number

5.20.2 Sub block 0x40E1 – AWR_MSSCPUFAULT_STATUS_GET_SB

This sub block provides the MSS CPU fault information.

Table 5.177 describes the content of this sub block.

Table 5.177: AWR_MSSVERSION_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x40E1
SBLKLEN	2	Value = 4

Response to AWR_MSSCPUFAULT_STATUS_GET_SB

AWR_MSSCPUFAULT_STATUS_SB is sent in response to AWR_MSSCPUFAULT_STATUS_GET_SB.

Table 5.178 describes the content of AWR_MSSCPUFAULT_STATUS_SB

Table 5.178: AWR_MSSCPUFAULT_STATUS_SB contents

Field Name	Number of bytes	Description																
SBLKID	2	Value = 0x40E1																
SBLKLEN	2	Value = 36																
FAULT_TYPE	1	<table><tr><th>Value</th><th>Definition</th></tr><tr><td>0</td><td>MSS Processor Undefined Instruction Abort</td></tr><tr><td>1</td><td>MSS Processor Instruction pre-fetch Abort</td></tr><tr><td>2</td><td>MSS Processor Data Access Abort</td></tr><tr><td>3</td><td>MSS Processor Firmware Fatal Error</td></tr><tr><td>4</td><td>MSS Processor Chirp Errors</td></tr><tr><td>5</td><td>MSS Processor Register read-back errors</td></tr><tr><td>0x6-0xFF</td><td>Reserved</td></tr></table>	Value	Definition	0	MSS Processor Undefined Instruction Abort	1	MSS Processor Instruction pre-fetch Abort	2	MSS Processor Data Access Abort	3	MSS Processor Firmware Fatal Error	4	MSS Processor Chirp Errors	5	MSS Processor Register read-back errors	0x6-0xFF	Reserved
Value	Definition																	
0	MSS Processor Undefined Instruction Abort																	
1	MSS Processor Instruction pre-fetch Abort																	
2	MSS Processor Data Access Abort																	
3	MSS Processor Firmware Fatal Error																	
4	MSS Processor Chirp Errors																	
5	MSS Processor Register read-back errors																	
0x6-0xFF	Reserved																	
RESERVED	1	0x00																

Continued on next page

Table 5.178 – continued from previous page

LINE_NUM	2	Valid only in case of FAULT type is 0x3, provides the firmware line number at which fatal error occurred.
FAULT_LR	4	The instruction PC address at which Fault occurred in case of FAULT type is 0x0 - 0x3 The register address incase of failure for Fault type 0x5
FAULT_PREV_LR	4	The return address of the function from which fault function has been called (Call stack LR) in case of FAULT type is 0x0-0x3 The register read-back value in case of FAULT type 0x5
FAULT_SPSR	4	The CPSR register value at which fault occurred in case of FAULT type is 0x0-0x3 The regisiter write value in case of FAULT type is 0x5
FAULT_SP	4	The SP register value at which fault occurred
FAULT_CAUSE_ADDRESS	4	The address access at which Fault occurred (valid only for fault type 0x0 to 0x2)
FAULT_ERROR_STATUS	2	The status of Error (Error Cause type - valid only for fault type 0x0 to 0x2) 0x000 BACKGROUND_ERR 0x001 ALIGNMENT_ERR 0x002 DEBUG_EVENT 0x00D PERMISSION_ERR 0x008 SYNCH_EXTER_ERR 0x406 ASYNCH_EXTER_ERR 0x409 SYNCH_ECC_ERR 0x408 ASYNCH_ECC_ERR
FAULT_ERROR_SOURCE	1	The Source of the Error (Error Source type - valid only for fault type 0x0 to 0x2) 0x0 ERR_SOURCE_AXI_MASTER 0x1 ERR_SOURCE_ATCM 0x2 ERR_SOURCE_BTCM
FAULT_AXI_ERROR_TYPE	1	The AXI Error type (Error Source type - valid only for fault type 0x0 to 0x2) 0x0 AXI_DECOD_ERR 0x1 AXI_SLAVE_ERR
FAULT_ACCESS_TYPE	1	The Error Access type (Error Access type - valid only for fault type 0x0 to 0x2) 0x0 READ_ERR 0x1 WRITE_ERR

Continued on next page

Table 5.178 – continued from previous page

FAULT_RECOVERY_TYPE	1	The Error Recovery type (Error Recovery type - Valid only for fault type 0x0 to 0x2) 0x0 UNRECOVERY 0x1 RECOVERY
RESERVED	2	0x0000

5.20.3 Sub block 0x40E2 – AWR_MSSESMFAULT_STATUS_GET_SB

This sub block provides the information regarding additional Master sub system faults. Table 5.179 describes the content of this sub block.

Table 5.179: AWR_MSSESMFAULT_STATUS_GET_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x40E2
SBLKLEN	2	Value = 4

The Response to above request is given in the AWR_MSSESMFAULT_STATUS_SB. Table 5.180 describes the contents of AWR_MSSESMFAULT_STATUS_SB.

Table 5.180: AWR_MSSESMFAULT_STATUS_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x40E2
SBLKLEN	2	Value = 20

Continued on next page

Table 5.180 – continued from previous page

ESM_GROUP1_ ERRORS	4	Bits	Error Information 0 – No Error , 1 – ESM Error
		b0	NERROR in sync
		b1	RESERVED
		b2	DMA MPU Region tests
		b3	DMA Parity error
		b4	RESERVED
		b5	RESERVED
		b6	DSS CSI parity Error
		b7	TPCC parity error
		b8	CBUF ECC single bit error
		b9	CBUF ECC double bit error
		b10	RESERVED
		b11	RESERVED
		b12	RESERVED
		b13	Error response from the Peripheral when a DMA transfer is done
		b14	RESERVED
		b15	VIM RAM double bit errors
		b16	RESERVED
		b17	MibSPI double bit error test
		b18	DSS TPTC0 read MPU error
		b19	RESERVED
		b20	VIM RAM single bit errors
		b21	RESERVED
		b22	FRC Lock Step Error
		b23	RESERVED
		b24	RESERVED
		b25	MibSPI single bit error test
		b26	TCMB0 RAM single bit errors
		b27	STC error
		b28	TCMB1 RAM single bit errors
		b29	DSS TPTC0 write MPU error
		b30	DCC compare error
		b31	CR4F self-test error.(test of error path by error forcing)

Continued on next page

Table 5.180 – continued from previous page

ESM_GROUP2_ ERRORS	4	Bits	Definition
		b0	TCMA RAM single bit errors
		b1	RESERVED
		b2	RESERVED
		b3	DSS TPTC1 read MPU error
		b4	DSS TPTC1 write MPU error
		b5	RESERVED
		b6	Access error interrupt from FFT ACC
		b7	VIM Self-Test Error
		b8	RESERVED
		b9	RESERVED
		b10	RESERVED
		b11	RESERVED
		b12	RESERVED
		b13	RESERVED
		b14	RESERVED
		b15	RESERVED
		b16	RESERVED
		b17	RESERVED
		b18	RESERVED
		b19	RESERVED
		b20	RESERVED
		b21	RESERVED
		b22	RESERVED
		b23	RESERVED
		b24	RESERVED
		b25	BSS to MSS ESM G2 Trigger
		b26	BSS Mailbox single bit errors
		b27	BSS Mailbox double bit errors
		b28	MSS Mailbox single bit errors
		b29	MSS Mailbox double bit errors
		b30	RESERVED
		b31	RESERVED
RESERVED	4	0x00000000	
RESERVED	4	0x00000000	

5.21 Sub blocks related to AWR_DEV_ASYNC_EVENT_MSG

5.21.1 Sub block 0x5000 – AWR_AE_DEV_MSSPOWERUPDONE_SB

This sub block indicates that Master SS power up is now complete. It also indicates the status of boot up tests done by Master SS. This async event is sent when host IRQ is enabled.

Table 5.181 describes the contents of this sub block

Table 5.181: AWR_AE_DEV_MSSPOWERUPDONE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x5000
SBLKLEN	2	Value = 24
MSS_POWERUP_TIME	4	Master SS power up time 1 LSB = 5 ns
MSS_POWERUP_STATUS	8	Refer to Table 7.3 for bit map details

Continued on next page

Table 5.181 – continued from previous page

BOOTTEST_ STATUS	8	0 – PASS, 1 – FAIL
		Bit Definition
		b0 MibSPI self-test
		b1 DMA self-test
		b2 RESERVED
		b3 RTI self-test
		b4 ESM self-test
		b5 EDMA self-test
		b6 CRC self-test
		b7 VIM self-test
		b8 MPU self-test
		b9 Mailbox self-test
		b10 RESERVED
		b11 RESERVED
		b12 RESERVED
		b13 MibSPI single bit error test
		b14 MibSPI double bit error test
		b15 DMA Parity error test
		b16 TCMA Single bit error test
		b17 TCMB Single bit error test
		b18 RESERVED
		b19 RESERVED
		b20 RESERVED
		b21 RESERVED
		b22 VIM lockstep test
		b23 CCM R4 lockstep test
		b24 DMA MPU region test
		b25 MSS Mailbox single bit error test
		b26 MSS Mailbox double bit error test
		b27 BSS Mailbox single bit error test
		b28 BSS Mailbox double bit error test
		b29 EDMA MPU test
		b30 EDMA parity test
		b31 RESERVED
		b32 RESERVED
		b33 RESERVED
		b34 PCR test
		b35 VIM RAM parity test
		b36 SCI boot time test
		b63:37 RESERVED

Table 5.181 – continued from previous page

NOTE:	The functional APIs shall be sent to radar device only after receiving AWR_AE_DEV_MSSPOWERUPDONE_SB Async-event after power cycle. In case of boot over SPI then functional APIs shall be sent to radar device only after receiving AWR_AE_MSS_BOOTERRORSTATUS_SB Async-event.
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5.21.2 Sub block 0x5001 – AWR_AE_DEV_RFPOWERUPDONE_SB

This sub block indicates that BIST SS power up is now complete.

Table 5.182 describes the contents of this sub block

Table 5.182: AWR_AE_DEV_RFPOWERUPDONE_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x5001
SBLKLEN	2	Value = 20

Continued on next page

Table 5.182 – continued from previous page

BSS_ POWERUP_ BIST_STATUS_ FLAGS	4	1 – PASS, 0 – FAIL
		Bit Status Information
		b0 ROM CRC check (RESERVED in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b1 CR4 and VIM lockstep test
		b2 RESERVED
		b3 VIM test
		b4 STC test of diagnostic
		b5 CR4 STC
		b6 CRC test
		b7 RAMPGEN memory ECC test
		b8 DFE parity test
		b9 DFE memory ECC
		b10 RAMPGEN lockstep test
		b11 FRC lockstep test
		b12 DFE memory PBIST
		b13 RAMPGEN memory PBIST
		b14 PBIST test
		b15 WDT test
		b16 ESM test
		b17 DFE STC
		b18 RESERVED
		b19 ATCM, BTCM ECC test
		b20 ATCM, BTCM parity test
		b21 DCC test
		b22 RESERVED
		b23 RESERVED
		b24 FFT test (RESERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b25 RTI test
		b26 PCR test (RESERVED in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b27 Bus Safety test (RESERVED in xWR254x, xWR2x4xECO, xWR2x4xP devices)
		b28 ECC aggregator test
		b29 MPU test
		b31:30 RESERVED

Continued on next page

Table 5.182 – continued from previous page

POWERUP_ TIME	4	RF BIST SS Power up time 1 LSB = 5 ns
RESERVED	4	0x00000000
RESERVED	4	0x00000000

5.21.3 Sub block 0x5002 – AWR_AE_MSS_CPUFAULT_SB

This sub block indicates CPU fault status of Master SS.

Table 5.183 describes the content of this sub block.

Table 5.183: AWR_AE_MSS_CPUFAULT_STATUS_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x5002
SBLKLEN	2	Value = 36
FAULT_TYPE	1	0 MSS Processor Undefined Instruction Abort 1 MSS Processor Instruction pre-fetch Abort 2 MSS Processor Data Access Abort 3 MSS Processor Firmware Fatal Error 4 MSS Processor Chirp Errors 5 MSS Processor Register read-back errors 0x6-0xFF Reserved
RESERVED	1	0x00
LINE_NUM	2	Valid only in case of FAULT type is 0x3, provides the firmware line number at which fatal error occurred.
FAULT_LR	4	The instruction PC address at which Fault occurred in case of FAULT type is 0x0 - 0x3 The register address incase of failure for Fault type 0x5
FAULT_PREV_LR	4	The return address of the function from which fault function has been called (Call stack LR) in case of FAULT type is 0x0-0x3 The register read-back value in case of FAULT type 0x5
FAULT_SPSR	4	The CPSR register value at which fault occurred in case of FAULT type is 0x0-0x3 The register write value in case of FAULT type is 0x5
FAULT_SP	4	The SP register value at which fault occurred

Continued on next page

Table 5.183 – continued from previous page

FAULT_CAUSE_ADDRESS	4	The address access at which Fault occurred (valid only for fault type 0x0 to 0x2)
FAULT_ERROR_STATUS	2	The status of Error (Error Cause type - valid only for fault type 0x0 to 0x2) 0x000 BACKGROUND_ERR 0x001 ALIGNMENT_ERR 0x002 DEBUG_EVENT 0x00D PERMISSION_ERR 0x008 SYNCH_EXTER_ERR 0x406 ASYNCH_EXTER_ERR 0x409 SYNCH_ECC_ERR 0x408 ASYNCH_ECC_ERR
FAULT_ERROR_SOURCE	1	The Source of the Error (Error Source type - valid only for fault type 0x0 to 0x2) 0x0 ERR_SOURCE_AXI_MASTER 0x1 ERR_SOURCE_ATCM 0x2 ERR_SOURCE_BTCM
FAULT_AXI_ERROR_TYPE	1	The AXI Error type (Error Source type - valid only for fault type 0x0 to 0x2) 0x0 AXI_DECOD_ERR 0x1 AXI_SLAVE_ERR
FAULT_ACCESS_TYPE	1	The Error Access type (Error Access type - valid only for fault type 0x0 to 0x2) 0x0 READ_ERR 0x1 WRITE_ERR
FAULT_RECOVERY_TYPE	1	The Error Recovery type (Error Recovery type - Valid only for fault type 0x0 to 0x2) 0x0 UNRECOVERY 0x1 RECOVERY
RESERVED	2	0x0000

5.21.4 Sub block 0x5003 – AWR_AE_MSS_ESMFAULT_STATUS_SB

This sub block indicates any other faults inside the MSS.

Table 5.184 describes the content of this sub block.

Table 5.184: AWR_AE_MSS_ESMFAULT_STATUS_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x5003
SBLKLEN	2	Value = 20

Continued on next page

Table 5.184 – continued from previous page

ESM_GROUP1_ ERRORS	4	Bits	Error Information 0 – No Error , 1 – ESM Error
		b0	NERROR in sync
		b1	RESERVED
		b2	DMA MPU Region tests
		b3	DMA Parity error
		b4	RESERVED
		b5	RESERVED
		b6	DSS CSI parity Error
		b7	TPCC parity error
		b8	CBUF ECC single bit error
		b9	CBUF ECC double bit error
		b10	RESERVED
		b11	RESERVED
		b12	RESERVED
		b13	Error response from the Peripheral when a DMA transfer is done
		b14	RESERVED
		b15	VIM RAM double bit errors
		b16	RESERVED
		b17	MibSPI double bit error test
		b18	DSS TPTC0 read MPU error
		b19	RESERVED
		b20	VIM RAM single bit errors
		b21	RESERVED
		b22	FRC Lock Step Error
		b23	RESERVED
		b24	RESERVED
		b25	MibSPI single bit error test
		b26	TCMB0 RAM single bit errors
		b27	STC error
		b28	TCMB1 RAM single bit errors
		b29	DSS TPTC0 write MPU error
		b30	DCC compare error
		b31	CR4F self-test error.(test of error path by error forcing)

Continued on next page

Table 5.184 – continued from previous page

ESM_GROUP2_ ERRORS	4	Bits	Definition
		b0	TCMA RAM single bit errors
		b1	RESERVED
		b2	RESERVED
		b3	DSS TPTC1 read MPU error
		b4	DSS TPTC1 write MPU error
		b5	RESERVED
		b6	Access error interrupt from FFT ACC
		b7	VIM Self-Test Error
		b8	RESERVED
		b9	RESERVED
		b10	RESERVED
		b11	RESERVED
		b12	RESERVED
		b13	RESERVED
		b14	RESERVED
		b15	RESERVED
		b16	RESERVED
		b17	RESERVED
		b18	RESERVED
		b19	RESERVED
		b20	RESERVED
		b21	RESERVED
		b22	RESERVED
		b23	RESERVED
		b24	RESERVED
		b25	BSS to MSS ESM G2 Trigger
		b26	BSS Mailbox single bit errors
		b27	BSS Mailbox double bit errors
		b28	MSS Mailbox single bit errors
		b29	MSS Mailbox double bit errors
		b30	RESERVED
		b31	RESERVED
RESERVED	4	0x00000000	
RESERVED	4	0x00000000	

NOTE: The FRC lockstep fatal error is connected to MSS ESM Group 1 lines, This fatal error must be handled in Host in AWR2243/xWR6243 device.

5.21.5 Sub block 0x5004 – RESERVED

5.21.6 Sub block 0x5005 – AWR_AE_MSS_BOOTERRORSTATUS_SB

This sub block indicates error status of MSS when booted over SPI. This async event is sent after the bootup over SPI is complete.

Table 5.185 describes the content of this sub block.

Table 5.185: AWR_AE_MSS_BOOTERRORSTATUS_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x5005
SBLKLEN	2	Value = 24
MSS_POWERUP_TIME	4	Master SS power up time 1 LSB = 5 ns
MSS_POWERUP_STATUS	8	Refer to Table 7.3 for bit map details

Continued on next page

Table 5.185 – continued from previous page

BOOTTEST_ STATUS	8	0 – PASS, 1 – FAIL
		Bit Definition
		b0 MibSPI self-test
		b1 DMA self-test
		b2 RESERVED
		b3 RTI self-test
		b4 ESM self-test
		b5 EDMA self-test
		b6 CRC self-test
		b7 VIM self-test
		b8 MPU self-test
		b9 Mailbox self-test
		b10 RESERVED
		b11 RESERVED
		b12 RESERVED
		b13 MibSPI single bit error test
		b14 MibSPI double bit error test
		b15 DMA Parity error test
		b16 TCMA Single bit error test
		b17 TCMB Single bit error test
		b18 RESERVED
		b19 RESERVED
		b20 RESERVED
		b21 RESERVED
		b22 VIM lockstep test
		b23 CCM R4 lockstep test
		b24 DMA MPU region test
		b25 MSS Mailbox single bit error test
		b26 MSS Mailbox double bit error test
		b27 BSS Mailbox single bit error test
		b28 BSS Mailbox double bit error test
		b29 EDMA MPU test
		b30 EDMA parity test
		b31 RESERVED
		b32 RESERVED
		b33 RESERVED
		b34 PCR test
		b35 VIM RAM parity test
		b36 SCI boot time test
		b63:37 RESERVED

Table 5.185 – continued from previous page

NOTE:	The functional APIs shall be sent to radar device only after receiving AWR_AE_MSS_BOOTERRORSTATUS_SB Async-event after boot over SPI (Flash is not connected).
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5.21.7 Sub block 0x5006 – AWR_AE_MSS_LATENTFAULT_TESTREPORT_SB

This sub block indicates the test status report of the latent fault tests.

Table 5.186 describes the content of this sub block.

Table 5.186: AWR_AE_MSS_LATENTFAULT_TESTREPORT_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x5006
SBLKLEN	2	Value = 16

Continued on next page

Table 5.186 – continued from previous page

TEST_STATUS_ FLAG1	4	1 – PASS, 0 - FAIL
		Bits Definition
		b0 RESERVED
		b1 DMA self-test
		b2 RESERVED
		b3 RTI self-test
		b4 RESERVED
		b5 EDMA self-test
		b6 CRC self-test
		b7 VIM self-test
		b8 RESERVED
		b9 Mailbox self-test
		b10 RESERVED
		b11 RESERVED
		b12 Generating NERROR
		b13 MibSPI single bit error test
		b14 MibSPI double bit error test
		b15 DMA Parity error
		b16 TCMA RAM single bit errors (Not supported, refer latest release note)
		b17 TCMB RAM single bit errors (Not supported, refer latest release note)
		b18 TCMA RAM double bit errors (Not supported, refer latest release note)
		b19 TCMB RAM double bit errors (Not supported, refer latest release note)
		b20 TCMA RAM parity errors (Not supported, refer latest release note)
		b21 TCMB RAM parity errors (Not supported, refer latest release note)
		b22 RESERVED
		b23 RESERVED
		b24 DMA MPU Region tests
		b25 MSS Mailbox single bit errors
		b26 MSS Mailbox double bit errors
		b27 BSS Mailbox single bit errors
		b28 BSS Mailbox double bit errors
		b29 EDMA MPU test
		b30 EDMA parity test
		b31 CSI2 parity test

Continued on next page

Table 5.186 – continued from previous page

TEST_STATUS_ FLAG2	4	Bits	Definition
		b0	RESERVED
		b1	RESERVED
		b2	RESERVED
		b3	VIM RAM parity test
		b4	SCI boot time test
		b31:5	RESERVED
RESERVED	4	0x00000000	

5.21.8 Sub block 0x5007 – AWR_AE_MSS_PERIODICTEST_STATUS_SB

This sub block indicates test status of the periodic tests.

Table 5.187 describes the content of this sub block.

Table 5.187: AWR_AE_MSS_PERIODICTEST_STATUS_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x5007
SBLKLEN	2	Value = 12
TEST_STATUS_ FLAG	4	1 – PASS, 0 – FAIL
		Bits Definition
		b0 Periodic read back of static registers
		b1 ESM self-test
		b31:2 RESERVED
RESERVED	4	0x00000000

5.21.9 Sub block 0x5008 – AWR_AE_MSS_RFERROR_STATUS_SB

This sub block indicates the RF error status.

Table 5.188 describes the content of this sub block.

Table 5.188: AWR_AE_MSS_RFERROR_STATUS_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x5008

Continued on next page

Table 5.188 – continued from previous page

SBLKLEN	2	Value = 12																		
ERROR_STA-TUS_FLAG	4	<table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>No fault</td></tr><tr><td>1</td><td>BSS FW assert</td></tr><tr><td>2</td><td>BSS FW abort</td></tr><tr><td>3</td><td>BSS ESM GROUP1 ERROR</td></tr><tr><td>4</td><td>BSS ESM GROUP2 ERROR</td></tr><tr><td>6-5</td><td>RESERVED</td></tr><tr><td>7</td><td>BSS monitoring failure in Mode 1(Quiet mode)</td></tr><tr><td>Others</td><td>RESERVED</td></tr></table>	Value	Definition	0	No fault	1	BSS FW assert	2	BSS FW abort	3	BSS ESM GROUP1 ERROR	4	BSS ESM GROUP2 ERROR	6-5	RESERVED	7	BSS monitoring failure in Mode 1(Quiet mode)	Others	RESERVED
Value	Definition																			
0	No fault																			
1	BSS FW assert																			
2	BSS FW abort																			
3	BSS ESM GROUP1 ERROR																			
4	BSS ESM GROUP2 ERROR																			
6-5	RESERVED																			
7	BSS monitoring failure in Mode 1(Quiet mode)																			
Others	RESERVED																			
RESERVED	4	0x00000000																		

5.21.10 Sub block 0x5009 – RESERVED

5.21.11 Sub block 0x500A – RESERVED

5.21.12 Sub block 0x500B – RESERVED

6 API Programming Sequence

6.1 Single device mode (xWR294x, xWR254x, xWR2x4xLC, xWR2x4xECO and xWR2x4xP)

This section briefly describes in which order to issue the various API SBs defined in this document for a single device.

1. Un-halt BSS (with patch loaded) either by SBL or MSS Application
2. AWR_RF_STATIC_CONF_SET_MSG
 - a. AWR_RF_DEVICE_CFG_SB
 - b. AWR_CHAN_CONF_SET_SB
 - c. AWR_ADCOUT_CONF_SET_SB
 - d. AWR_RF_LDO_BYPASS_SB
 - e. AWR_LOWPOWERMODE_CONF_SET_SB
 - f. AWR_DYNAMICPOWERSAVE_CONF_SET_SB
 - g. AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB
 - h. AWR_RF_RADAR_MISC_CTL_SB if per chirp phase shifter and Advance chirp configuration needs to be enabled.
 - i. AWR_APLL_SYNTH_BW_CONTROL_SB
3. AWR_RF_INIT_MSG
 - a. AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB (To restore factory calibration data for TX0), keep CAL_APPLY = 0
 - b. AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB (To restore factory calibration data for TX1), keep CAL_APPLY = 0
 - c. AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB (To restore factory calibration data for TX2), keep CAL_APPLY = 0
 - d. AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB (To restore factory calibration data for TX3), keep CAL_APPLY = 1
 - e. AWR_CAL_DATA_RESTORE_SB (To restore factory calibration data to avoid on field RF interference during calibration)
 - f. Wait for AWR_AE_RF_INITCALIBSTATUS_SB

- g. AWR_RF_INIT_CALIBRATION_CONF_SB (Enable only required calibration to run)
 - h. AWR_RFINIT_SB: This triggers very basic calibrations and RF initializations
 - i. Wait for AWR_AE_RF_INITCALIBSTATUS_SB
4. AWR_RF_DYNAMIC_CONF_SET_MSG
- a. AWR_PROFILE_CONF_SET_SB
 - b. Chirp configuration API
 - a. AWR_CHIRP_CONF_SET_SB or
 - b. AWR_ADVANCE_CHIRP_CONF_SB and
 - c. AWR_ADVANCE_CHIRP_GENERIC_LUT_LOAD_SB
 - c. AWR_LOOPBACK_BURST_CONF_SET_SB (if using loopback burst in advance frame config API)
 - d. AWR_FRAME_CONF_SET_SB or AWR_ADVANCED_FRAME_CONF_SB with SW mode and AWR_DEV_FRAME_CONFIG_APPLY_MSG.
 - e. AWR_CALIB_MON_TIME_UNIT_CONF_SB with CALIB_MON_TIME_UNIT value set to a value such that the total frame idle time across multiple CALIB_MON_TIME_UNITS is sufficient for all calibrations and monitoring. See Section 12 for details on calibration and monitoring durations. If any error AWR_CAL_MON_TIMING_FAIL_REPORT_AE_SB AE will be generated when frame is triggered. The calibrations and monitors will not run properly if this error is generated.
 - f. Set NUM_OF_CASCADEDEV to 1, DEVICE_ID to 0 and Set MONITORING_MODE = 0 (MONITORING_MODE 1 is recommended only in cascade mode) in AWR_CALIB_MON_TIME_UNIT_CONF_SB API
 - g. AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB (set all ONE_TIME_CALIB_ENABLE_MASK and set ENABLE_CAL_REPORT = 1)
 - h. Wait for AWR_RUN_TIME_CALIBRATION_SUMMARY_REPORT_AE_SB
 - i. AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB (set all RUN_TIME_CALIB_ENABLE_MASK and set ENABLE_CAL_REPORT = 0 to avoid receiving periodic async events)
 - j. AWR_FRAME_CONF_SET_SB or AWR_ADVANCED_FRAME_CONF_SB
5. MONITOR CONFIGURATIONS
- a. AWR_MONITOR_RF_DIG_LATENTFAULT_CONF_SB : Wait for AWR_MONITOR_RF_DIG_LATENTFAULT_REPORT_AE_SB AE. This API should not be issue when frames are running.
 - b. AWR_MONITOR_RF_DIG_PERIODIC_CONF_SB : Enable periodic digital monitors
 - c. AWR_MONITOR_ANALOG_ENABLES_CONF_SB : Enable periodic analog monitors, The corresponding monitoring configuration APIs should be issued after issuing this API. Refer latest release note for all supported monitors.

6. AWR_RF_FRAME_TRIG_MSG for frame start
 - a. AWR_FRAMESTARTSTOP_CONF_SB in Start mode (1): after this, frames get transmitted, wait for AWR_AE_RF_FRAME_TRIGGER_RDY_SB AE (optional if this async-event enabled).
7. AWR_RF_FRAME_TRIG_MSG for frame stop
 - a. AWR_FRAMESTARTSTOP_CONF_SB in Stop mode (0): after this, frames are stopped. Wait for AWR_FRAME_END_AE_SB AE. The AWR_RF_FRAME_TRIG_MSG may be issued multiple times for multiple sets of frames. Refer AWR_FRAMESTARTSTOP_CONF_SB for more frame stop options.

6.2 Single device mode (AWR1243/AWR2243)

This section briefly describes in which order to issue the various API SBs defined in this document for a single device.

1. Power up the device
2. Wait for AWR_AE_MSSPOWERUPDONE_SB
3. Wait for AWR_AE_MSS_BOOTERRORSTATUS_SB if flash is not connected (Boot over SPI)
4. AWR_DEV_CONFIGURATION_SET_SB
5. AWR_DEV_RFPOWERUP_SB
6. Wait for AWR_AE_RFPOWERUPDONE_SB
7. AWR_RF_STATIC_CONF_SET_MSG
 - a. AWR_RF_DEVICE_CFG_SB
 - b. AWR_CHAN_CONF_SET_SB
 - c. AWR_ADCOUT_CONF_SET_SB
 - d. AWR_RF_LDO_BYPASS_SB with RFLDOBYPASS_EN set to 1 if RF supply is 1.0 V
 - e. AWR_LOWPOWERMODE_CONF_SET_SB
 - f. AWR_DYNAMICPOWERSAVE_CONF_SET_SB
 - g. AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB
 - h. AWR_RF_RADAR_MISC_CTL_SB if per chirp phase shifter and Advance chirp configuration needs to be enabled.
 - i. AWR_APLL_SYNTH_BW_CONTROL_SB
8. Data path configurations

- a. AWR_DEV_RX_DATA_FORMAT_CONF_SET_SB
 - b. AWR_DEV_RX_DATA_PATH_CONF_SET_SB
 - c. AWR_DEV_RX_DATA_PATH_LANE_EN_SB
 - d. AWR_DEV_RX_DATA_PATH_CLK_SET_SB
 - e. AWR_HIGHSPEEDINTFCLK_CONF_SET_SB
 - f. AWR_DEV_LVDS_CFG_SET_SB / AWR_DEV_CSI2_CFG_SET_SB
9. AWR_RF_INIT_MSG
- a. AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB (To restore factory calibration data for TX0), keep CAL_APPLY = 0
 - b. AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB (To restore factory calibration data for TX1), keep CAL_APPLY = 0
 - c. AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB (To restore factory calibration data for TX2), keep CAL_APPLY = 0
 - d. AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB (To restore factory calibration data for TX3), keep CAL_APPLY = 1
 - e. AWR_CAL_DATA_RESTORE_SB (To restore factory calibration data to avoid on field RF interference during calibration)
 - f. Wait for AWR_AE_RF_INITCALIBSTATUS_SB
 - g. AWR_RF_INIT_CALIBRATION_CONF_SB (Enable only required calibration to run)
 - h. AWR_RFINIT_SB: This triggers very basic calibrations and RF initializations
 - i. Wait for AWR_AE_RF_INITCALIBSTATUS_SB
10. AWR_RF_DYNAMIC_CONF_SET_MSG
- a. AWR_PROG_FILT_COEFF_RAM_SET_SB
 - b. AWR_PROG_FILT_CONF_SET_SB
 - c. AWR_PROFILE_CONF_SET_SB
 - d. Chirp configuration API
 - a. AWR_CHIRP_CONF_SET_SB or
 - b. AWR_ADVANCE_CHIRP_CONF_SB and
 - c. AWR_ADVANCE_CHIRP_GENERIC_LUT_LOAD_SB
 - e. AWR_LOOPBACK_BURST_CONF_SET_SB (if using loopback burst in advance frame config API)
 - f. AWR_FRAME_CONF_SET_SB or AWR_ADVANCED_FRAME_CONF_SB with SW or HW triggered mode and AWR_DEV_FRAME_CONFIG_APPLY_MSG.

- g. AWR_CALIB_MON_TIME_UNIT_CONF_SB with CALIB_MON_TIME_UNIT value set to a value such that the total frame idle time across multiple CALIB_MON_TIME_UNITS is sufficient for all calibrations and monitoring. See Section 12 for details on calibration and monitoring durations. If any error AWR_CAL_MON_TIMING_FAIL_REPORT_AE_SB AE will be generated when frame is triggered. The calibrations and monitors will not run properly if this error is generated.
- h. Set NUM_OF_CASCADEDEV to 1, DEVICE_ID to 0 and Set MONITORING_MODE = 0 (MONITORING_MODE 1 is recommended only in cascade mode) in AWR_CALIB_MON_TIME_UNIT_CONF_SB API
- i. AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB (set all ONE_TIME_CALIB_ENABLE_MASK and set ENABLE_CAL_REPORT = 1)
- j. Wait for AWR_RUN_TIME_CALIBRATION_SUMMARY_REPORT_AE_SB
- k. AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB (set all RUN_TIME_CALIB_ENABLE_MASK and set ENABLE_CAL_REPORT = 0 to avoid receiving periodic async events)
- l. AWR_DEV_FRAME_CONFIG_APPLY_SB or AWR_DEV_ADV_FRAME_CONFIG_APPLY_SB

11. MONITOR CONFIGURATIONS

- a. AWR_MONITOR_RF_DIG_LATENTFAULT_CONF_SB : Wait for AWR_MONITOR_RF_DIG_LATENTFAULT_REPORT_AE_SB AE. This API should not be issue when frames are running.
- b. AWR_MSS_LATENTFAULT_TEST_CONF_SB : Wait for AWR_AE_MSS_LATENTFAULT_TESTREPORT_SB AE.
- c. AWR_MSS_PERIODICTESTS_CONF_SB : Enable periodic digital monitors, the monitor starts immediately after enabling this API.
- d. AWR_MONITOR_RF_DIG_PERIODIC_CONF_SB : Enable periodic digital monitors
- e. AWR_MONITOR_ANALOG_ENABLES_CONF_SB : Enable periodic analog monitors, The corresponding monitoring configuration APIs should be issued after issuing this API. Refer latest release note for all supported monitors.

12. AWR_RF_FRAME_TRIG_MSG for frame start

- a. AWR_FRAMESTARTSTOP_CONF_SB in Start mode (1): after this, frames get transmitted, wait for AWR_AE_RF_FRAME_TRIGGER_RDY_SB AE.

13. AWR_RF_FRAME_TRIG_MSG for frame stop

- a. AWR_FRAMESTARTSTOP_CONF_SB in Stop mode (0): after this, frames are stopped. Wait for AWR_FRAME_END_AE_SB AE. The AWR_RF_FRAME_TRIG_MSG may be issued multiple times for multiple sets of frames. Refer AWR_FRAMESTARTSTOP_CONF_SB for more frame stop options.

6.3 Cascaded device mode

This section briefly describes in which order to issue the various API SBs defined in this document for master and slave devices in a cascaded configuration.

When using cascaded devices, the reference clock is provided by master to slave. So unless master is powered-up and clock is available from master to slave, the slave device cannot be powered up.

Following instructions and sequence needs to be followed in cascade mode (Master and Slave mode configured in CASCADING_CFG field in AWR_CHAN_CONF_SET_SB API):

**Application
Care Abouts in
cascade mode:**

1. Disable OSC clock out (OSCCLKOUT_DIS = 1) for slaves in AWR_CHAN_CONF_SET_SB API
2. Write INTER_BURST_POWER_SAVE_DIS = 0 (Enable inter-burst power save) in AWR_R_DEVICE_CFG_SB (default setting)
3. Write WDT_ENABLE = 0 (Disable WDT) in AWR_RF_DEVICE_CFG_SB (default setting)
4. Set MONITORING_MODE = 1 (API based monitor trigger. The automated monitoring trigger is also supported in cascade mode, not used in this sequence)
5. Set NUM_OF_CASCADEDEV to 1 and DEVICE_ID to 0 in AWR_CALIB_MON_TIME_UNIT_CONF_SB (Not recommended to set other values for proper functionality in MONITORING_MODE 1)
6. Clear PERIODIC_CALIB_ENABLE_MASK and set CALIBRATION_PERIODICITY to 0 to avoid run time calibration (Automated run time calibration is not recommended in cascade mode)
7. When stopping the frames in master it is recommended to stop Slaves first and wait of frame stop AE then stop the master.
8. Follow API based monitor trigger notes in AWR_MONITOR_TYPE_TRIG_CONF_SB API section
9. Set TRIGGER_SELECT = 1 in master device and TRIGGER_SELECT = 2 in slave device in AWR_FRAME_CONF_SET_SB or AWR_ADVANCED_FRAME_CONF_SB API.

Table 6.1: Sequence of APIs to be issued to master and slave devices in cascaded mode configuration for FMCW mode measurements

Sl. No	Master device sequence	Slave device sequence
1	Power up master device	
2	Wait for AWR_AE_DEV_MSSPOWERUP-DONE_SB	
3	AWR_DEV_RFPOWERUP_SB	
4	Wait for AWR_AE_DEV_RFPOWERUP-DONE_SB	
5	AWR_CHAN_CONF_SET_SB with CAS-CADING_CFG = 0x0001. The reference clock for slave device is enabled by default	
6		Power on slave device
7		Wait for AWR_AE_DEV_MSSPOWERUP-DONE_SB
8		AWR_DEV_RFPOWERUP_SB
9		Wait for AWR_AE_DEV_RFPOWERUP-DONE_SB
10		AWR_CHAN_CONF_SET_SB with CAS-CADING_CFG = 0x0002 and disable OSC CLOCK OUT.
11	Follow all AWR_RF_STATIC_CONF_SET_MSG APIs sequence as instructed in single chip mode (point 6 in sec 6.2), please refer application Care Abouts above if any deviation from single chip	Follow all AWR_RF_STATIC_CONF_SET_MSG APIs sequence as instructed in single chip mode (point 6 in sec 6.2), please refer application Care Abouts above if any deviation from single chip
12	Follow all Data path configurations APIs sequence as instructed in single chip mode (point 7 in sec 6.2)	Follow all Data path configurations APIs sequence as instructed in single chip mode (point 7 in sec 6.2)
13	Follow all AWR_RF_INIT_MSG APIs sequence as instructed in single chip mode (point 8 in sec 6.2)	Follow all AWR_RF_INIT_MSG APIs sequence as instructed in single chip mode (point 8 in sec 6.2)
14	Follow all AWR_RF_DYNAMIC_CONF_SET_MSG APIs sequence as instructed in single chip mode (point 9 in sec 6.2), please refer application Care Abouts above if any deviation from single chip	Follow all AWR_RF_DYNAMIC_CONF_SET_MSG APIs sequence as instructed in single chip mode (point 9 in sec 6.2), please refer application Care Abouts above if any deviation from single chip
15	Follow all MONITOR CONFIGURATIONS APIs sequence as instructed in single chip mode (point 10 in sec 6.2), please refer application Care Abouts above if any deviation from single chip	Follow all MONITOR CONFIGURATIONS APIs sequence as instructed in single chip mode (point 10 in sec 6.2), please refer application Care Abouts above if any deviation from single chip

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Table 6.1 – continued from previous page

16		AWR_FRAMESTARTSOP_CONF_SB with STARTSTOP_CMD = 0x0001
17		Wait for AWR_AE_RF_FRAME_TRIGGER_RDY_SB
18	AWR_FRAMESTARTSOP_CONF_SB with STARTSTOP_CMD = 0x0001	
19	Wait for AWR_AE_RF_FRAME_TRIGGER_RDY_SB	
20	Trigger AWR_MONITOR_TYPE_TRIGGER_CONF_SB API for type 0, 1 and 2 monitors and follow instructions mentioned in this API section	Trigger AWR_MONITOR_TYPE_TRIGGER_CONF_SB API for type 0, 1 and 2 monitors and follow instructions mentioned in this API section
21		AWR_FRAMESTARTSOP_CONF_SB with STARTSTOP_CMD = 0x0000
22		Wait for AWR_FRAME_END_AE_SB AE
23	AWR_FRAMESTARTSOP_CONF_SB with STARTSTOP_CMD = 0x0000	
24	Wait for AWR_FRAME_END_AE_SB AE	

6.4 Continuous streaming mode (in single device case)

This section briefly describes in which order to issue the various API SBs defined in this document to enable continuous streaming mode on a single device

1. Power up the device
2. Wait for AWR_AE_MSSPOWERUPDONE_SB
3. AWR_DEV_CONFIGURATION_SET_SB
4. AWR_DEV_RFPOWERUP_SB
5. Wait for AWR_AE_RFPOWERUPDONE_SB
6. AWR_RF_STATIC_CONF_SET_MSG
 - a. AWR_RF_DEVICE_CFG_SB
 - b. AWR_CHAN_CONF_SET_SB
 - c. AWR_ADCOUT_CONF_SET_SB
 - d. AWR_RF_LDO_BYPASS_SB with RFLDOBYPASS_EN set to 1 if RF supply is 1.0 V
 - e. AWR_LOWPOWERMODE_CONF_SET_SB

- f. AWR_DYNAMICPOWERSAVE_CONF_SET_SB
 - g. AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB
 - h. AWR_RF_RADAR_MISC_CTL_SB if per chirp phase shifter and Advance chirp configuration needs to be enabled.
 - i. AWR_APLL_SYNTH_BW_CONTROL_SB
7. Data path configurations
- a. AWR_DEV_RX_DATA_FORMAT_CONF_SET_SB
 - b. AWR_DEV_RX_DATA_PATH_CONF_SET_SB
 - c. AWR_DEV_RX_DATA_PATH_LANE_EN_SB
 - d. AWR_DEV_RX_DATA_PATH_CLK_SET_SB
 - e. AWR_HIGHSPEEDINTFCLK_CONF_SET_SB
 - f. AWR_DEV_LVDS_CFG_SET_SB / AWR_DEV_CSI2_CFG_SET_SB
8. AWR_RF_INIT_MSG
- a. AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB (To restore factory calibration data for TX0), keep CAL_APPLY = 0
 - b. AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB (To restore factory calibration data for TX1), keep CAL_APPLY = 0
 - c. AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB (To restore factory calibration data for TX2), keep CAL_APPLY = 0
 - d. AWR_PHASE_SHIFTER_CAL_DATA_RESTORE_SB (To restore factory calibration data for TX3), keep CAL_APPLY = 1
 - e. AWR_CAL_DATA_RESTORE_SB (To restore factory calibration data to avoid on field RF interference during calibration)
 - f. Wait for AWR_AE_RF_INITCALIBSTATUS_SB
 - g. AWR_RF_INIT_CALIBRATION_CONF_SB (Enable only required calibration to run)
 - h. AWR_RFINIT_SB: This triggers very basic calibrations and RF initializations
 - i. Wait for AWR_AE_RF_INITCALIBSTATUS_SB
9. AWR_RF_DYNAMIC_CONF_SET_MSG
- a. AWR_PROG_FILT_COEFF_RAM_SET_SB
 - b. AWR_PROG_FILT_CONF_SET_SB
10. AWR_CONT_STREAMING_MODE_EN_SB to start the trigger
- a. AWR_CONT_STREAMING_MODE_CONF_SET_SB
 - b. AWR_DEV_RX_CONTSTREAMING_MODE_CONF_SET_SB

- c. AWR_CONT_STREAMING_MODE_EN_SB with CONT_STREAMING_EN = 0x0001 to start continuous streaming
11. AWR_CONT_STREAMING_MODE_EN_SB to stop the trigger
 - a. AWR_CONT_STREAMING_MODE_EN_SB with CONT_STREAMING_EN = 0x0000 to stop continuous streaming
12. Repeat steps 6-11 for a different configuration

6.5 Continuous streaming (CW) mode (in cascaded device case)

Table 6.2: Sequence of APIs to be issued to master and slave devices in cascaded mode configuration for FMCW mode measurements

Sl. No	Master device sequence	Slave device sequence
1	Power up master device	
2	Wait for AWR_AE_DEV_MSSPOWERUP-DONE_SB	
3	AWR_DEV_RFPOWERUP_SB	
4	Wait for AWR_AE_DEV_RFPOWERUP-DONE_SB	
5	AWR_CHAN_CONF_SET_SB with CAS-CADING_CFG = 0x0001. The reference clock for slave device is enabled by default	
6		Power on slave device
7		Wait for AWR_AE_DEV_MSSPOWERUP-DONE_SB
8		AWR_DEV_RFPOWERUP_SB
9		Wait for AWR_AE_DEV_RFPOWERUP-DONE_SB
10		AWR_CHAN_CONF_SET_SB with CAS-CADING_CFG = 0x0002 and disable OSC CLOCK OUT.
11	Follow all AWR_RF_STATIC_CONF_SET_MSG APIs sequence as instructed in single chip mode (point 6 in sec 6.3), please refer application Care Abouts above if any deviation from single chip	Follow all AWR_RF_STATIC_CONF_SET_MSG APIs sequence as instructed in single chip mode (point 6 in sec 6.3), please refer application Care Abouts above if any deviation from single chip
12	Follow all Data path configurations APIs sequence as instructed in single chip mode (point 7 in sec 6.3)	Follow all Data path configurations APIs sequence as instructed in single chip mode (point 7 in sec 6.3)

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Table 6.2 – continued from previous page

13	Follow all AWR_RF_INIT_MSG APIs sequence as instructed in single chip mode (point 8 in sec 6.3)	Follow all AWR_RF_INIT_MSG APIs sequence as instructed in single chip mode (point 8 in sec 6.3)
14	Follow all AWR_RF_DYNAMIC_CONF_SET_MSG APIs sequence as instructed in single chip mode (point 9 in sec 6.3), please refer application Care Abouts above if any deviation from single chip	Follow all AWR_RF_DYNAMIC_CONF_SET_MSG APIs sequence as instructed in single chip mode (point 9 in sec 6.3), please refer application Care Abouts above if any deviation from single chip
15	AWR_CONT_STREAMING_MODE_CONF_SET_SB	
16	AWR_CONT_STREAMING_MODE_EN_SB with CONT_STREAMING_EN = 0x0001 to start continuous streaming	
17		AWR_CONT_STREAMING_MODE_CONF_SET_SB with the same RF frequency configuration as in master device
18		AWR_CONT_STREAMING_MODE_EN_SB with CONT_STREAMING_EN = 0x0001 to start continuous streaming
19		AWR_CONT_STREAMING_MODE_EN_SB with CONT_STREAMING_EN = 0x0000 to stop continuous streaming
20	AWR_CONT_STREAMING_MODE_EN_SB with CONT_STREAMING_EN = 0x0000 to stop continuous streaming	
21	Repeat steps 6-20 for a different CW mode configuration	

7 API Error Handling and Error Codes

This section describes the error handling of various fault messages in device and information about error codes.

7.1 API Error Handling

The xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP device sends out error info in the form of AE messages in case of any fault in the device.

The following AE messages are Fatal errors and device shall be restarted upon receiving these messages:

1. [AWR_AE_RF_CPUFAULT_SB](#)
2. [AWR_AE_RF_ADV_ESMFAULT_SB](#) for ESM_GROUP2_ERRORS only.
3. [AWR_ANALOGFAULT_AE_SB](#)
4. [AWR_AE_RF_ADV_ESMFAULT_SB](#) for PROG_FILT_PARITY_ERROR, PROG_FILT_UERR and ECC_AGG_DED_ERROR only.

The ERROR_CODE returned part of monitor AE message reports are informative purpose only, these error codes are helpful to debug the cause for monitor failure. Application can log these information and share with TI in case of any runtime errors. The information about these error codes are documented below in "API Error Codes" section.

The Application shall handle the monitor failures reported in STATUS_FLAGS part of monitor AE message appropriately, device may not need restart in these error cases and these errors needs to be handled case by case. For example AWR_MONITOR_RX_GAIN_PHASE_REPORT_AE_SB can report failure in presence of interference, this is not a fatal error in the device. Please refer Monitoring app note for more info on monitor reports and thresholds.

The information about API_RESP Error codes returned in ACK messages part of [AWR_RESP_ERROR_SB](#) for each API commands are documented below in "API Error Codes" section, these error codes are helpful to debug the cause for API failure and errors can be corrected during development time.

7.2 API Error Codes

Table 7.1: BSS API error codes

Applicable to all API sub blocks	1	Incorrect API MSGID
	2	Sub block not found in the MSG
	3	Incorrect Sub block ID
	4	Incorrect Sub block length
	5	Incorrect Sub block data
	6	Error in processing the command
	7	Binary file CRC mismatch error
	8	Binary file type mismatch w.r.t. magic number
AWR_FRAMESTARTSTOP_CONF_SB	20	Frames are already started when the FRAME_START command was issued
	21	Frames are already stopped when the FRAME_STOP command was issued
	22	No valid frame configuration API was issued and frames are started
	23	START_STOP_CMD parameter is out of range
	129	The frame stop option-4 can not be used in Sw triggered mode
AWR_CHAN_CONF_SET_SB	24	RX_CHAN_EN parameter is out of range (Max range may vary based on device variant)
	25	TX_CHAN_EN parameter is out of range (Max range may vary based on device variant)
	26	CASCADING_CFG parameter is out of range [0, 2]
	282	Device variant does not allow cascading but API is issued to enable cascading mode
	287	Enabling OSCLKOUT and OSCLKOUTETH at the same time is not supported.
AWR_ADCOUT_CONF_SET_SB	27	NUM_ADC_BITS parameter is out of range [0, 2]
	28	ADC_OUT_FMT parameter is out of range [0, 3]
	127	FULL_SCALE_REDUCTION_FACTOR is > 0 for 16 bit ADC, or > 2 for 14 bit ADC mode or > 4 for 12 bit ADC mode
AWR_LOWPOWERMODE_CONF_SET_SB	29	LP_ADC_MODE parameter is out of range [0, 1]
	156	Regular ADC mode is used on a 5 MHz part variant device
AWR_DYNAMICPOWER_SAVE_CONF_SET_SB	30	BLOCK_CFG parameter is out of range [0, 7]
AWR_HIGHSPEEDINTFCLK_CONF_SET_SB	31	HSICLKRATECODE[1:0] is 0
	32	RESERVED

Continued on next page

Table 7.1 – continued from previous page

	33	HSICLKRATECODE[3:2] is 3 and HSICLKRATECODE[1:0] is 2
	34	HSICLKRATECODE[3:2] is 3 and HSICLKRATECODE[1:0] is 2
AWR_PROFILE_CONF_SET_SB	35	PF_INDX is ≥ 7
	36	PF_FREQ_START_CONST is not within allowed range
	37	PF_IDLE_TIME_CONST > 5.24 ms
	38	Maximum DFE spill time (refer rampgen calculator in mmWave Studio for more details) > PF_IDLE_TIME_CONST
	39	PF_ADC_START_TIME_CONST > 4095
	40	PF_RAMP_END_TIME > 524287
	41	PF_RAMP_END_TIME < PF_ADC_START_TIME_CONST + ADC_SAMPLING_TIME (ADC_SAMPLING_TIME is time taken to sample NUM_ADC_SAMPLES)
	42	PF_TX_OUTPUT_POWER_BACKOFF for TX0 > 30
	43	PF_TX_OUTPUT_POWER_BACKOFF for TX1 > 30
	44	PF_TX_OUTPUT_POWER_BACKOFF for TX2 > 30
	45	RESERVED
	46	Ramp end frequency is not within allowed range
	47	Absolute value of TX_START_TIME is > 38.45 μ s
	48	Number of ADC samples is not within [2, 8192]
	49	Output sampling rate is not within [2, MaxSamplingRate] Msps. See Table 5.27 for the MaxSamplingRate.
	50	HPF1 corner frequency is > maximum allowed
	51	HPF2 corner frequency is > maximum allowed
	52	PF_RX_GAIN is not within [24, 52] dB or PF_RX_GAIN is an odd number
	53	RESERVED
	54	RESERVED
	55	RESERVED
	56	RESERVED
	57	RESERVED
	58	RESERVED
	331	PF_TX_OUTPUT_POWER_BACKOFF for TX4 > 30
AWR_CHIRP_CONF_SET_SB	59	CHIRP_START_INDX ≥ 512
	60	CHIRP_END_INDX ≥ 512

Continued on next page

Table 7.1 – continued from previous page

	61	CHIRP_START_INDXX > CHIRP_END_INDXX
	62	PROFILE_INDXX ≥ 4
	63	If the profile corresponding to PROFILE_INDXX is not defined
	64	CHIRP_FREQ_START_VAR > 8388607
	65	CHIRP_FREQ_SLOPE_VAR > 63
	66	Chirp start frequency is outside the allowed range Chirp end frequency is outside the allowed range Chirp bandwidth is greater than maximum allowed as per device data sheet or Maximum chirp frequency is greater than maximum allowed as per device data sheet
	67	CHIRP_IDLE_TIME_VAR > 4095
	68	CHIRP_ADC_START_TIME_VAR > 4095
	69	RAMP_END_TIME < ADC_START_TIME + ADC_SAMPLING_TIME
	70	CHIRP_TX_EN > maximum simultaneous TX allowed as per device data sheet
AWR_FRAME_CONF_SET_SB	71	CHIRP_TX_EN indicates to enable a TX which is not enabled in AWR_CHAN_CONF_SET_SB
	72	CHIRP_START_INDXX ≥ 512
	73	CHIRP_END_INDXX ≥ 512
	74	CHIRP_START_INDXX > CHIRP_END_INDXX
	75	Chirp used in the frame is not configured by AWR_CHIRP_CONF_SET_SB
	76	One of the profiles used in the frame is not configured by AWR_PROF_CONF_SET_SB
	77	NUM_LOOPS is outside [1, 255]
	78	RESERVED
	79	FRAME_PERIODICITY is outside [100 μs, 1.342 s]
	80	FRAME_ON_TIME > FRAME_PERIODICITY
	81	TRIGGER_SELECT is outside [1, 2]
	82	FRAME_TRIGGER_DELAY > 100 μs
AWR_ADVANCED_FRAME_CONF_SET_SB	83	API is issued when frames are ongoing
	160	The Dummy chirps at end of frame is not supported
	84	NUM_SUBFRAMES is outside [1, 4]
	85	FORCE_SINGLE_PROFILE is outside [0, 1]
	86	FORCE_SINGLE_PROFILE ≥ 4

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Table 7.1 – continued from previous page

	87	Profile defined by FORCE_SINGLE_PROFILE is not defined
	88	SFx_CHIRP_START_INDX $\geq$ 512
	89	SFx_NUM_UNIQUE_CHIRPS_PER_BURST is outside the range [1, 512]
	90	Chirp used in the frame is not configured by AWR_CHIRP_CONF_SET_SB
	91	One of the profiles used in the frame is not configured by AWR_PROF_CONF_SET_SB
	92	SFx_NUM_LOOPS_PER_BURST is outside the range [1, 255]
	93	SFx_BURST_PERIOD is outside the range [100 μ s, 1.342 s]
	94	Burst ON time is $>$ BURST_PERIOD
	95	SFx_CHIRP_START_INDX_OFFSET $\geq$ 512
	96	SFx_CHIRP_START_INDX $\geq$ 512 or SFX_CHIRP_START_INDX + SFx_NUM_UNIQUE_CHIRPS_PER_BURST – 1 is $\geq$ 512
	97	SFx_NUM_BURSTS is outside the range [1, 512]
	98	SFx_NUM_OUTER_LOOPS is outside the range [1, 64]
	99	SFx_PERIOD is outside the range [100 μ s, 1.342 s]
	100	Subframe on time $>$ SFx_PERIOD or when TESTSOURCE is enabled, SubFrame Idle time is $<$ 150 μ s
	101	RESERVED
	102	TRIGGER_SELECT is outside the range [1, 2]
AWR_RF_TEST_SOURCE_CONFIG_SET_SB	103	FRAME_TRIGGER_DELAY is $>$ 100 μ s
	104	API is issued when frames are on going
	105	POSITION_VECx[y] $<$ 0
	106	RESERVED
	107	VELOCITY_VECx[x] $>$ 5000 or VELOCITY_VECx[y] $>$ 5000 or VELOCITY_VECx[z] $>$ 5000
	108	SIG_LEV_VECx $>$ 950
	109	RX_ANT_POS_XZ[Byte] $>$ 120
	110	RESERVED
	111	PROG_FILT_COEFF_START_INDEX is an odd number
AWR_PROG_FILT_CONF_SET_SB	112	PROFILE_INDX $\geq$ 4

Continued on next page

Table 7.1 – continued from previous page

	126	DFE mode is pseudo real
AWR_PROG_FILT_COEFF_RAM_SET_SB	113	API is issued for a non xWR1642/xWR1843 device
	126	DFE mode is pseudo real
AWR_RF_RADAR_MISC_CTL_SB	114	API is issued for an unsupported device
AWR_PERCHIRPPHASESHIFT_CONF_SB	115	CHIRP_START_INDEX $\geq$ 512
	116	CHIRP_END_INDEX $\geq$ 512
	117	CHIRP_START_INDEX > CHIRP_END_INDEX
AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB	118	Boot time calibrations are not done so cannot run runtime calibrations
	286	The forced temperature bin index is invalid
AWR_CAL_MON_FREQUENCY_LIMITS_SB	119	FREQ_LIMIT_HIGH is out side the supported range or FREQ_LIMIT_LOW > FREQ_LIMIT_HIGH
	130	The minimum RF frequency band is < 200MHz
AWR_CALIB_MON_TIME_UNIT_CONF_SB	120	CALIB_MON_TIME_UNIT $\leq$ 0
	128	NUM_OF_CASCADEDEV $\leq$ 0
AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB	121	CALIBRATION_PERIODICITY = 0
	122	API is issued when continuous streaming mode is on
	123	RX gain run time calibration was requested but boot time calibration was not performed
	124	LO distribution run time calibration was requested but boot time calibration was not performed
	125	TX power run time calibration was requested but boot time calibration was not performed
AWR_LOOPBACK_BURST_CONF_SET_SB	132	LOOPBACK_SEL is > 3
	133	BURST_INDEX $\geq$ 16
	134	Burst is not valid but loopback is enabled for this burst
AWR_DYN_CHIRP_CONF_SET_SB	135	CHIRP_SEGMENT_SELECT > 31 if CHIRP_ROW_SELECT = 0 or CHIRP_SEGMENT_SELECT > 11 if CHIRP_ROW_SELECT != 0
	159	CHIRP_ROW_SELECT > 3
AWR_DYN_PER_CHIRP_PHASESHIFTER_CONF_SB	136	CHIRP_SEGMENT_SELECT > 31
AWR_CAL_DATA_RESTORE_SB	137	CHUNK_ID $\geq$ NUM_CHUNKS

Continued on next page

Table 7.1 – continued from previous page

	138	CAL_DATA is invalid
	318	TX_IND is invalid in phase shifter restore API
AWR_INTERCHIRP_ BLOCKCONTROLS_SB	139	RX02_RF_TURN_OFF_TIME is not within the range [-1024, 1023]
	140	RX13_RF_TURN_OFF_TIME is not within the range [-1024, 1023]
	141	RX02_BB_TURN_OFF_TIME is not within the range [-1024, 1023]
	142	RX13_BB_TURN_OFF_TIME is not within the range [-1024, 1023]
	143	RX02_RF_PREENABLE_TIME is not within the range [-1024, 1023]
	144	RX13_RF_PREENABLE_TIME is not within the range [-1024, 1023]
	145	RX02_BB_PREENABLE_TIME is not within the range [-1024, 1023]
	146	RX13_BB_PREENABLE_TIME is not within the range [-1024, 1023]
	147	RX02_RF_TURN_ON_TIME is not within the range [-1024, 1023]
	148	RX13_RF_TURN_ON_TIME is not within the range [-1024, 1023]
	149	RX02_BB_TURN_ON_TIME is not within the range [-1024, 1023]
	150	RX13_BB_TURN_ON_TIME is not within the range [-1024, 1023]
	151	RX_LO_TURN_OFF_TIME is not within the range [-1024, 1023]
	152	TX_LO_TURN_OFF_TIME is not within the range [-1024, 1023]
	153	RX_LO_TURN_ON_TIME is not within the range [-1024, 1023]
	154	TX_LO_TURN_ON_TIME is not within the range [-1024, 1023]
AWR_SUBFRAME_ START_CONF_SB	155	Sub-frame start command is issued but the frame is not configured for sub frame trigger mode
AWR_ADVANCE_CHIRP_ CONF_SB	300	Invalid CHIRP_PARAM_INDEX
	301	Invalid GLOBAL_RESET_MODE
	302	Reserved

Continued on next page

Table 7.1 – continued from previous page

	303	Invalid update period DELTA_PARAM_UPDATE_PERIOD or LUT_PARAM_UPDATE_PERIOD
	304	Invalid fixed delta parameter SFn_CHIRP_PARAM_DELTA
	305	Invalid reset period DELTA_RESET_PERIOD or LUT_RESET_PERIOD
	306	Invalid LUT address LUT_PATTERN_ADDRESS_OFFSET
	307	Invalid number of patterns in LUT NUM_OF_PATTERNS
	308	Invalid LUT index offset value BURST_LUT_INDEX_OFFSET or SF_LUT_INDEX_OFFSET
	309	Invalid LUT_CHIRP_PARAM_SIZE and LUT_CHIRP_PARAM_SCALE
	310	Invalid legacy APIs are issued when advance chirp config API is enabled or vice versa
	311	All chirp parameters are not defined in advance chirp API
	312	Invalid TX phase shifter dither value MAX_TX_PHASE_SHIFTER_INTERNAL_DITHER
	313	Insufficient number of NUM_OF_PATTERNS programmed compared to actual programmed chirps (array out of bound error)
	315	Invalid num of chirps programmed in frame config API
AWR_ADVANCE_CHIRP_GENERIC_LUT_LOAD_SB	314	Invalid num of bytes NUM_OF_BYTES
Common to all monitoring configuration APIs	250	Device type is not ASILB
	251	Fault injection API or Digital latent fault API is issued when frames are ongoing
	252	Invalid reporting mode
	253	Configured profile ID is not within [0, 3]
	254	Monitoring profile ID is not configured yet
	260	Invalid RF bit mask
	281	Analog monitoring is not supported
AWR_MONITOR_RF_DIG_LATENTFAULT_CONF_SB	290	Monitoring chirp error
	251	API is issued when frames are on-going
AWR_MONITORING_EXTERNAL_ANALOG_SIGNALS_CONF_SB	255	Settling time is configured is more than 12 μ s

Continued on next page

Table 7.1 – continued from previous page

AWR_MONITOR_RX_INTERNAL_ANALOG_SIGNALS_CONF_SB	256	None of the RXs are enabled
AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_CONF_SB	257	TX0 is not enabled
AWR_MONITOR_TX1_INTERNAL_ANALOG_SIGNALS_CONF_SB	258	TX1 is not enabled
AWR_MONITOR_TX2_INTERNAL_ANALOG_SIGNALS_CONF_SB	259	TX2 is not enabled
-	261	RESERVED
-	262	RESERVED
AWR_MONITOR_TXn_BALLBREAK_CONF_SB	263	Monitored TX channel is not enabled
AWR_MONITOR_RX_GAIN_PHASE_CONF_SB AWR_MONITOR_RX_NOISE_FIGURE_CONF_SB	264	Monitored RX channel is not enabled
AWR_MONITOR_RX_GAIN_PHASE_CONF_SB	265	TX selected for RX gain phase monitor is TX2 (Only TX0 or TX1 is allowed)
	291	PD power level is less than -40dBm (Used for RX Gain Monitor)
	295	PGA Gain used for monitoring is incorrect
AWR_MONITOR_RX_SATURATION_DETECTOR_CONF_SB	266	SAT_MON_SEL is not in [0, 3]
	267	SAT_MON_PRIMARY_TIME_SLICE_DURATION is less than 0.64 μ s or greater than ADC sampling time
	268	SAT_MON_NUM_SLICES is 0 or greater than 127
	283	RX saturation monitor is not supported
AWR_MONITOR_SIG_IMG_MONITOR_CONF_SB	269	SIG_IMG_MON_NUM_SLICES is 0 or greater than 127
	270	NUM_SAMPLES_PER_PRIMARY_TIME_SLICE is odd, or less than 4 in Complex1x mode or less than 8 in non-Complex1x modes or greater than NUM_ADC_SAMPLES
	280	Signal and image band monitor is not supported
AWR_ANALOG_FAULT_INJECTION_CONF_SB	279	LDO fault inject is requested but LDOs are bypassed
AWR_MONITOR_TXn_POWER_CONF_SB	294	PD Reading incorrect (RF OFF reading higher than RF ON reading)

Continued on next page

Table 7.1 – continued from previous page

AWR_MONITOR_TXn_ BALLBREAK_CONF_SB AWR_MONITOR_RX_ INTERNAL_ANALOG_ SIGNALS_CONF_SB		
AWR_MONITOR_RX_ GAIN_PHASE_CONF_SB AWR_MONITOR_TX_ GAIN_PHASE_CONF_SB AWR_MONITOR_TXn_ PHASE_SHIFTER_ CONF_SB	292	ADC power level higher than +7 dBm or lower than -9.5 dBm
AWR_MONITOR_RX_ NOISE_FIGURE_CONF_ SB	293	Low RX noise figure (Noise Figure is less than 0 dB)
AWR_MONITOR_PM- CLKLO_INTERNAL_ ANALOG_SIGNALS_ CONF_SB	296	The 20G monitor is not supported in single chip configuration
AWR_MONITOR_ SYNTHESIZER_ FREQUENCY_CONF_SB	274	MONITOR_START_TIME is outside the specified range.
	297	MONITOR_CONFIG_MODE is invalid.
	298	The both Live and Non-live synth frequency monitors are cannot be enabled together.
AWR_MONITOR_TX_ GAIN_PHASE_CONF_SB AWR_MONITOR_TXn_ PHASE_SHIFTER_ CONF_SB	317	Invalid RX mask or the RX mask is not enabled in channel configuration API
AWR_MONITOR_TXn_ PHASE_SHIFTER_ CONF_SB	316	Invalid phase mask or at least one of the phase should be enabled for monitoring
AWR_MONITOR_TYPE_ TRIG_CONF_SB	284	RL_API_NRESP_ANA_MON_MODE_NOT_API_BASED (Monitoring trigger API is not supported in autonomous mode of operation)
	285	RL_API_NRESP_ANA_MON_TRIG_TYPE_INVALID (Monitoring trigger bit masks are all zeros in AWR_MONITOR_TYPE_TRIG_CONF_SB)
AWR_MONITOR_TXn_ BALLBREAK_CONF_SB	319	Invalid MON_START_FREQ_CONST programmed in configuration API

Continued on next page

Table 7.1 – continued from previous page

AWR_MONITOR_PLL_ CONTROL_VOLTAGE_ CONF_SB	320	VCO3 monitor not supported
AWR_POWER_SAVE_ MODE_CONF_SET_SB	325	Invalid LOWPOWER_STATE_TRANSITION_CMD in configuration API
AWR_ADV_TX_GAIN_ TEMPLUT_SET_SB	332	Invalid TX index in configuration
AWR_RF_DEVICE_CFG_ SB	335	Invalid Inter-burst power save configuration
AWR_MONITOR_ OVERRIDES_AND_ DITHER_CONF_SB	336	Invalid monitor index in configuration
	337	OVRD_TX_BACKOFF field is set to an invalid value
	338	Mon Time dither (min or max) has been set to an invalid value
AWR_MONITOR_VMON_ CONF_SB	340	VDDA VCO VMON self test failed.
	341	VDDA BB VMON self test failed.
	342	VDD RF1 VMON self test failed.
	343	VDD RF2 VMON self test failed.

Table 7.2: MSS API error codes (Applicable only in AWR1243/AWR2243)

Applicable to all API sub blocks	1	Incorrect API MSGID
	2	Sub block not found in the MSG
	3	Incorrect Sub block ID
	4	Incorrect Sub block length
	5	Incorrect Sub block data
	6	Error in processing the command
	7	Binary file CRC mismatch error
	8	Binary file type mismatch w.r.t. magic number
AWR_DEV_RX_DATA_ FORMAT_CONF_SET_SB	1001	RX_CHAN_EN > 0xF
	1002	NUM_ADC_BITS > 2
	1003	ADC_OUT_FMT > 1
	1004	IQ_SWAP_SEL > 1
	1005	CHAN_INTERLEAVE > 1
AWR_DEV_RX_DATA_ PATH_CONF_SET_SB	1006	DATA_INTF_SEL > 1
	1007	DATA_TRANS_FMT_PKT0 [5:0] not a valid value. Valid set {0x1, 0x6, 0x9, 0x36}

Continued on next page

Table 7.2 – continued from previous page

	1008	DATA_TRANS_FMT_PKT1 [5:0] not a valid value. Valid set {0x0, 0xD, 0xB}
	1050	CQ_CONFIG is out of range
AWR_DEV_RX_DATA_PATH_LANEEN_SET_SB	1009	LANE_EN > 0xF
	1010	Reserved
AWR_DEV_RX_DATA_PATH_CLK_SET_SB	1011	LANE_CLK_CFG > 1
	1012	LANE_CLK_CFG != 1 for CSI2
	1013	DATA_RATE - Invalid combination of data rate and DDR or SDR operation
AWR_DEV_LVDS_CFG_SET_SB	1014	LANE_FMT_MAP > 1
	1015	LANE_PARAM_CFG > 7
AWR_DEV_RX_CON-TSTREAMING_MODE_CONF_SET_SB	1016	CONT_STREAMING_MODE > 1
	1017	CONT_STREAMING_MODE already in requested mode
AWR_DEV_CSI2_CFG_SET_SB	1018	LANE_POS_POL_SEL [DATA_LANE0_POS] >5
	1019	LANE_POS_POL_SEL [DATA_LANE1_POS] >5
	1020	LANE_POS_POL_SEL [DATA_LANE2_POS] >5
	1021	LANE_POS_POL_SEL [DATA_LANE3_POS] >5
	1022	LANE_POS_POL_SEL [CLOCK_POS] is outside the range [2,4]
AWR_DEV_FRAME_CONFIG_APPLY_SB	1023	HALF_WORDS_PER_CHIRP is outside the range [64, 8192]
AWR_DEV_ADV_FRAME_CONFIG_APPLY_SB	1024	NUM_SUBFRAMES is outside the range [1,4]
	1025	SF1_TOT_NUM_CHIRPS is outside the range [1, 0xFFFF]
	1026	SF1_NUM_ADC_SAMPLES_PER_DATA_PKT is outside the range [64, 8192]
	1027	SF1_PROC_NUM_CHIRPS_PER_DATA_PKT != 1
	1028	SF2_TOT_NUM_CHIRPS is outside the range [1, 0xFFFF], if NUM_SUBFRAMES ≥ 2
	1029	SF2_NUM_ADC_SAMPLES_PER_DATA_PKT is outside the range [64, 8192], if NUM_SUBFRAMES ≥ 2
	1030	SF2_PROC_NUM_CHIRPS_PER_DATA_PKT != 1, if NUM_SUBFRAMES ≥ 2

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Table 7.2 – continued from previous page

	1031	SF3_TOT_NUM_CHIRPS is outside the range [1, 0xFFFF], if NUM_SUBFRAMES ≥ 3
	1032	SF3_NUM_ADC_SAMPLES_PER_DATA_PKT is outside the range [64, 8192], if NUM_SUBFRAMES ≥ 3
	1033	SF3_PROC_NUM_CHIRPS_PER_DATA_PKT $\neq 1$, if NUM_SUBFRAMES ≥ 3
	1034	SF4_TOT_NUM_CHIRPS is outside the range [1, 0xFFFF], if NUM_SUBFRAMES == 4
	1035	SF4_NUM_ADC_SAMPLES_PER_DATA_PKT is outside the range [64, 8192], if NUM_SUBFRAMES == 4
	1036	SF4_PROC_NUM_CHIRPS_PER_DATA_PKT $\neq 1$, if NUM_SUBFRAMES == 4
	1052	Invoking AWR_DEV_ADV_FRAME_CONFIG_APPLY_SB message without configuring data path
AWR_DEV_MCUCLOCK_CONF_SET_SB	1040	MCUCLOCK_CTRL is out of range
	1041	MCUCLOCK_SRC is out of range
AWR_DEV_PMICCLOCK_CONF_SET_SB	1042	PMICCLOCK_CTRL is out of range
	1043	PMICCLOCK_SRC is out of range
	1044	MODE_SELECT is out of range
	1045	FREQ_SLOPE is out of range
	1046	CLK_DITHER_EN is out of range
AWR_DEV_TESTPATTERN_GEN_SET_SB	1047	TESTPATTERN_GEN_CTRL is out of range
	1048	DATA_INTF_SEL (Data interface selected in AWR_DEV_RX_DATA_PATH_CONF_SET_SB) is SPI
AWR_MSS_LATENT_FAULT_TEST_CONF_SB	1051	RL_API_NRESP_LFAULTTEST_UNSUPPORTED_OOR (Unsupported Latent Fault test selected in AWR_MSS_LATENTFAULT_TEST_CONF_SB)

7.3 Boot on SPI Error codes

Table 7.3: Bit field describing the error status during boot on SPI

Error description	Error code	Error code bit position
CERT_AUTH_FAILURE	0x0000000000000001	BIT0
CERT_PARSER_FAILURE	0x0000000000000002	BIT1
RPRC_IMG1_AUTH_FAILURE	0x0000000000000004	BIT2
RPRC_IMG2_AUTH_FAILURE	0x0000000000000008	BIT3
RPRC_IMG3_AUTH_FAILURE	0x0000000000000010	BIT4
RPRC_HDR_NOT_FOUND	0x0000000000000020	BIT5
METAHEADER_NOT_FOUND	0x0000000000000040	BIT6
SW_ANTIROLLBACK_CHK_FAILURE	0x0000000000000080	BIT7
EFUSE_INTEGRITY_FAILURE	0x0000000000000100	BIT8
CERT_FIELD_VALIDITY_FAILURE	0x0000000000000200	BIT9
CERT_FIELD_INVALID_AUTH_KEY_INDEX	0x0000000000000400	BIT10
CERT_FIELD_INVALID_HASH_TYPE	0x0000000000000800	BIT11
CERT_FIELD_INVALID_SUBSYSTEM	0x0000000000001000	BIT12
CERT_FIELD_INVALID_DECRYPT_KEY_INDEX	0x0000000000002000	BIT13
CERT_FIELD_CEK_EFUSE_MISMATCH	0x0000000000004000	BIT14
CERT_FIELD_CEK1_EFUSE_MISMATCH	0x0000000000008000	BIT15
CERT_FIELD_CEK2_EFUSE_MISMATCH	0x0000000000010000	BIT16
CERT_FIELD_INVALID_SUBSYSTEM_BANK_ALLOCATION	0x0000000000020000	BIT17
CERT_FIELD_INVALID_TOTAL_BANKS_ALLOCATION	0x0000000000040000	BIT18
RPRC_PARSER_FILE_LENGTH_MISMATCH	0x0000000000080000	BIT19
RPRC_PARSER_MSS_FILE_OFFSET_MISMATCH	0x0000000000100000	BIT20
RPRC_PARSER_BSS_FILE_OFFSET_MISMATCH	0x0000000000200000	BIT21
RPRC_PARSER_DSS_FILE_OFFSET_MISMATCH	0x0000000000400000	BIT22
CERT_FIELD_INVALID_DECRYPT_KEY	0x0000000000800000	BIT23
CERT_FIELD_INVALID_AUTH_KEY	0x0000000001000000	BIT24
HS_DEVICE_CERT_NOT_PRESENT	0x0000000020000000	BIT25
ERROR_IN_2K_IMAGE	0x0000000040000000	BIT26

Continued on next page

Table 7.3 – continued from previous page

SHARED_MEM_ALLOC_FAILED	0x0000000080000000	BIT27
MSSIMAGE_NOT_FOUND	0x0000000100000000	BIT28
METAHEADER_NUMFILES_ERROR	0x0000000200000000	BIT29
METAHEADER_CRC_FAILURE	0x0000000400000000	BIT30
RPRC_IMG4_AUTH_FAILURE	0x0000000800000000	BIT31
RPRC_PARSER_CONFIG_FILE_OFFSET_MISMATCH	0x0000001000000000	BIT32
BOOT_EXTS_EXTRACTION_FAILURE	0x0000002000000000	BIT33
DEVICE_UID_BAD_SIZE	0x0000004000000000	BIT34
KEY_DERIVE_FUNC_BAD_SIZE	0x0000008000000000	BIT35
HMAC_BAD_SIZE	0x0000010000000000	BIT36
AES_INIT_VECTOR_BAD_SIZE	0x0000020000000000	BIT37
SECDEV_TI_KEY_ERASE_FAILED	0x0000040000000000	BIT38
SOP5_SFLASH_NOT_FOUND	0x0000080000000000	BIT39
XTAL_CLK_DETECTION_FAILED	0x0010000000000000	BIT48
CONTINUE_BOOTUP_ON_XTAL	0x0020000000000000	BIT49
DSP_POWERUP_TIMEOUT_ERR	0x0040000000000000	BIT50
MSS_LBIST_FAILED	0x0080000000000000	BIT51
DSP_LBIST_PBIST_FAILED	0x0100000000000000	BIT52
PBIST_SINGLE_PORT_MEM_FAILED	0x0200000000000000	BIT53
PBIST_TWO_PORT_MEM_FAILED	0x0400000000000000	BIT54
MEMORY_INIT_FAILED	0x0800000000000000	BIT55
MSSROM_PBIST_CRC_COMPUTATION_FAILED	0x1000000000000000	BIT56
VMON_ERROR_DETECTED	0x2000000000000000	BIT57
ESM_NERROR_DETECTED	0x8000000000000000	BIT63

8 Radar Monitoring APIs

sec:RadarMonApis

AWR monitoring can be configured through a set of API sub blocks defined in this section. Note that these APIs cover the RF/Analog related monitoring mechanisms. There are separate monitoring mechanisms for the digital logic (including the processor, memory, etc.) which are internal to the device and not explicitly enabled through these APIs.

The monitoring APIs are structured as follows. There are common configuration APIs that control the overall periodicity of monitoring, as well as, enable/disable control for each monitoring mechanism. Then, for each monitoring mechanism there is an individual API to allow the customer to set an appropriate threshold for declaring failure from that monitoring. Also, for each monitoring mechanism, there is an individual API to report soft (raw) values from that monitoring.

Refer Monitor application note for more info on exact use case and threshold configuration recommendations.

NOTE1:	Each monitor can perform monitoring on only one profile at a time. Though it is possible that different monitors can monitor different profiles simultaneously.
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NOTE2:	None of the Safety Monitoring supported in QM devices except Rx saturation and signal image monitor defined in page 488, The monitoring configurations defined below from sub-block ID 0x01C0 to 0x01DF are not valid in QM devices.
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NOTE3:	All Monitoring configurations and enable control APIs shall be issues before triggering the frames. The run time programming or configuration update for monitors are not supported while frames are running.
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NOTE4:	None of the Monitoring features are supported for VCO3 .
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8.1 Common Configurations and Reports

This section covers the APIs corresponding to the common configurations and reports.

8.1.1 Sub block 0x01C0 – AWR_MONITOR_RF_DIG_LATENTFAULT_CONF_SB

This API SB contains the consolidated configuration of all digital monitoring. This is issued by the host to the AWR device.

The enabled monitoring functions are executed when the API is issued, this API should be issued only when frames are not running, these are destructive tests. The scheduling of these monitoring should be handled in the external application. Report of these monitoring will be available in the async event [AWR_MONITOR_RF_DIG_LATENTFAULT_REPORT_AE_SB](#).

Table 8.1: AWR_MONITOR_RF_DIG_LATENTFAULT_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x01C0
SBLKLEN	2	Value = 16

Continued on next page

Table 8.1 – continued from previous page

DIG_MONITOR- ING_ENABLES	4	1 – Enable, 0 – Disabled	
		Bit	Definition
		b0	RESERVED
		b1	CR4 and VIM lockstep test
		b2	RESERVED
		b3	VIM test
		b4	RESERVED
		b5	RESERVED
		b6	CRC test
		b7	RAMPGEN memory ECC
		b8	DFE parity test
		b9	DFE memory ECC
		b10	RAMPGEN lockstep test
		b11	FRC lockstep test of diagnostic
		b12	RESERVED
		b13	RESERVED
		b14	RESERVED
		b15	RESERVED
		b16	ESM test
		b17	DFE STC
		b18	RESERVED
		b19	ATCM, BTCM ECC test
		b20	ATCM, BTCM parity test
		b21	DCC test
		b22	RESERVED
		b23	RESERVED
		b24	FFT test (RESERVED in xWR254x, xWR2x4xECC, xWR2x4xP devices)
		b25	RTI test
		b26	RESERVED
		b31:27	RESERVED
TEST_MODE	1	Value	Definition
		0	Production mode. Latent faults are tested and any failures are reported
		1	Characterization mode. Faults are injected and failures are reported which allows testing of the failure reporting path

Continued on next page

Table 8.1 – continued from previous page

RESERVED	3	0x000000
RESERVED	4	0x00000000

NOTE: The Characterization TEST_MODE is supported only for debug, in production or run time this test mode is not supported. The device reset is required after entering this mode.

8.1.2 Sub block 0x01C1 – AWR_MONITOR_RF_DIG_PERIODIC_CONF_SB

This API SB contains the consolidated configuration of all periodic digital monitoring within radar sub-system. This is issued by the host to the AWR device.

The enabled monitoring functions are executed periodically and reports are sent based on reporting mode. Report of these monitoring will be available in the async event [AWR_MONITOR_RF_DIG_PERIODIC_REPORT_AE_SB](#).

Table 8.2: AWR_MONITOR_RF_DIG_PERIODIC_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x01C1
SBLKLEN	2	Value = 16
REPORTING_MODE	1	Value Definition 0 Report is sent every monitoring period 1 Report is sent only on a failure 2 RESERVED
RESERVED	3	0x000000
PERIODIC_DIG_MON_EN	4	1 – Enable, 0 – Disable Bit Monitoring type b0 PERIODIC_CONFIG_REGISTER_READ_EN b1 RESERVED b2 DFE_STC_EN b3 FRAME_TIMING_MONITORING_EN b31:4 RESERVED
RESERVED	4	0x00000000

8.1.3 Sub block 0x01C2 – AWR_MONITOR_ANALOG_ENABLES_CONF_SB

This API SB contains the consolidated configuration of all analog monitoring. This is issued by the host to the AWR device.

The enabled monitoring functions are executed with a periodicity of CALIB_MON_TIME_UNITS number of logical frames. The host should ensure that all the enabled monitors can be completed in the available inter-frame times, based on the monitoring durations (to be provided separately).

Table 8.3: AWR_MONITOR_ANALOG_ENABLES_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x01C2
SBLKLEN	2	Value = 12

Continued on next page

Table 8.3 – continued from previous page

ANA_MON- ITORING_ ENABLES	4	If any bit in this field is set to 1, the associated monitors are enabled. The configurations and reports of each monitors are described in respective sub sections. Bit Definition b0 TEMPERATURE_MONITOR b1 RX_GAIN_PHASE_MONITOR b2 RX_NOISE_FIGURE_MONITOR (Reserved in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) b3 RX_IFSTAGE_MONITOR b4 TX0_POWER_MONITOR b5 TX1_POWER_MONITOR b6 TX2_POWER_MONITOR b7 TX0_BALLBREAK_MONITOR b8 TX1_BALLBREAK_MONITOR b9 TX2_BALLBREAK_MONITOR b10 TX_GAIN_PHASE_MISMATCH_MONITOR b11 TX0_PHASE_SHIFTER_MONITOR b12 TX1_PHASE_SHIFTER_MONITOR b13 TX2_PHASE_SHIFTER_MONITOR b14 SYNTH_FREQ_MONITOR_LIVE b15 EXTERNAL_ANALOG_SIGNALS_MONITOR (Re- served in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) b16 INTERNAL_TX0_SIGNALS_MONITOR b17 INTERNAL_TX1_SIGNALS_MONITOR b18 INTERNAL_TX2_SIGNALS_MONITOR b19 INTERNAL_RX_SIGNALS_MONITOR b20 INTERNAL_PMCLKLO_SIGNALS_MONITOR b21 INTERNAL_GPADC_SIGNALS_MONITOR b22 PLL_CONTROL_VOLTAGE_MONITOR b23 DCC_CLOCK_FREQ_MONITOR b24 RX_SATURATION_DETECTOR_MONITOR b25 RX_SIG_IMG_BAND_MONITOR (Reserved in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) b26 RX_MIXER_INPUT_POWER_MONITOR (Reserved in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) b27 RESERVED b28 SYNTH_FREQ_MONITOR_NON_LIVE b29 TX3_POWER_MONITOR (xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP only) b30 TX3_BALLBREAK_MONITOR (xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP only)	
		b31 INTERNAL_TX3_SIGNALS_MONITOR (xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP only)	433

Table 8.3 – continued from previous page

LDO_VMON_ SC_MONI- TORING_EN	2	If any bit in this field is set to 1, the associated monitors are enabled. There are no reports for these monitors. If there is any fault, the async event AWR_ANALOGFAULT_AE_SB will be sent. Bit Description b0 APLL LDO short circuit monitoring enable 0 – disable, 1 – enable b1 SYNTH VCO LDO short circuit monitoring enable 0 – disable, 1 – enable b2 PA LDO short circuit monitoring enable (<i>not applicable for xWR294x, xWR254x, xWR2x4xECON, xWR2x4xP</i>) 0 – disable, 1 – enable b3 VMON circuit monitoring enable (<i>not applicable for xWR294x, xWR254x, xWR2x4xECON, xWR2x4xP</i>) 0 – disable, 1 – enable b15:4 RESERVED Note1: The VMON circuit monitoring is not supported in AWR2243/xWR6243 device. Note2: In xWR294x, xWR254x, xWR2x4xECON, xWR2x4xP devices, there is a separate VMON monitoring configuration and enable API - AWR_MONITOR_VMON_CONF_SB. Note3: In xWR294x, xWR254x, xWR2x4xECON, xWR2x4xP devices, APLL LDO Short Circuit and VMON faults are not reported by the RSS firmware. Fault signals are mapped to the MSS ESM and should be responded to by the MSS application. Please refer to the TRM for more information.
ANA_MON- ITORING_ ENABLES_ CONT	2	If any bit in this field is set to 1, the associate monitors are enabled. The configurations and reports of each monitors are described in respective sub sections. Bit Definition b0 TX3_PHASE_SHIFTER_MONITOR (only applicable to xWR294x, xWR254x, xWR2x4xECON, xWR2x4xP) b1 TX_PHSIFTER_DAC_MONITOR (only applicable to xWR294x, xWR254x, xWR2x4xECON, xWR2x4xP) b15:2 RESERVED

NOTE:

In xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP, the enabled monitors are executed in the below order.

1. TEMPERATURE_MONITOR.
2. RX_GAIN_PHASE_MONITOR.
3. RX_IFSTAGE_MONITOR.
4. SYNTH_FREQ_MONITOR_NON_LIVE.
5. INTERNAL_RX_SIGNALS_MONITOR.
6. INTERNAL_PMCLKLO_SIGNALS_MONITOR.
7. INTERNAL_GPADC_SIGNALS_MONITOR.
8. PLL_CONTROL_VOLTAGE_MONITOR.
9. DCC_CLOCK_FREQ_MONITOR.
10. RX_SATURATION_DETECTOR_MONITOR.
11. TX_PHSHIFTER_DAC_MONITOR.
12. TX0_POWER_MONITOR.
13. TX1_POWER_MONITOR.
14. TX2_POWER_MONITOR.
15. TX3_POWER_MONITOR.
16. TX0_BALLBREAK_MONITOR.
17. TX1_BALLBREAK_MONITOR.
18. TX2_BALLBREAK_MONITOR.
19. TX3_BALLBREAK_MONITOR.
20. INTERNAL_TX0_SIGNALS_MONITOR.
21. INTERNAL_TX1_SIGNALS_MONITOR.
22. INTERNAL_TX2_SIGNALS_MONITOR.
23. INTERNAL_TX3_SIGNALS_MONITOR.
24. TX_GAIN_PHASE_MISMATCH_MONITOR.
25. TX0_PHASE_SHIFTER_MONITOR.
26. TX1_PHASE_SHIFTER_MONITOR.
27. TX2_PHASE_SHIFTER_MONITOR.
28. TX3_PHASE_SHIFTER_MONITOR.

8.2 Temperature Monitor

This section contains API SBs that configure the on chip temperature monitors and report the soft results from the monitor. The corresponding monitors are collectively named TEMPERATURE_MONITOR. These monitors observe the temperature near various RF analog and digital modules using temperature sensors and GPADC and compare them against configurable thresholds. The report is sent as an async event AWR_MONITOR_TEMPERATURE_REPORT_AE_SB.

8.2.1 Sub block 0x01C3 – AWR_MONITOR_TEMPERATURE_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to temperature monitoring. Report of this monitoring will be available in the async event [AWR_MONITOR_TEMPERATURE_REPORT_AE_SB](#).

NOTE: For AWR2243/xWR6243 devices, the digital temperature sensor monitor threshold checks (Min, Max and Delta) can be disabled by programming DIG_TEMP_THRESH_MIN and DIG_TEMP_THRESH_MAX to value Zero.

Table 8.4: AWR_MONITOR_TEMPERATURE_CONF_SB contents

Field Name	Number of bytes	Description								
SBLKID	2	Value = 0x01C3								
SBLKLEN	2	Value = 24								
REPORTING_MODE	1	<table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>Report is sent every monitoring period without threshold check</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	Report is sent every monitoring period without threshold check	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check
Value	Definition									
0	Report is sent every monitoring period without threshold check									
1	Report is send only upon a failure (after checking for thresholds)									
2	Report is sent every monitoring period with threshold check									
RESERVED	1	0x00								
ANA_TEMP_THRESH_MIN	2	The temperatures read from near the sensors near the RF analog modules are compared against a minimum threshold. The comparison result is part of the monitoring report message (Error bit is set if any measurement is outside this (minimum, maximum) range). 1 LSB = 1°C, signed number Valid range: -99°C to 199°C								
ANA_TEMP_THRESH_MAX	2	The temperatures read from near the sensors near the RF analog modules are compared against a maximum threshold. The comparison result is part of the monitoring report message (Error bit is set if any measurement is outside this (minimum, maximum) range). 1 LSB = 1°C, signed number Valid range: -99°C to 199°C								

Continued on next page

Table 8.4 – continued from previous page

DIG_TEMP_THRESH_MIN	2	xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: This field is RESERVED. AWR2243/xWR6243 devices: The temperatures read from near the sensor near the digital module are compared against a minimum threshold. The comparison result is part of the monitoring report message (Error bit is set if any measurement is outside this (minimum, maximum) range). 1 LSB = 1°C, signed number Valid range: -99°C to 199°C Value 0: Disable monitor threshold check (together with DIG_TEMP_THRESH_MAX=0)
DIG_TEMP_THRESH_MAX	2	xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: This field is RESERVED. AWR2243/xWR6243 devices: The temperatures read from near the sensor near the digital module are compared against a maximum threshold. The comparison result is part of the monitoring report message (Error bit is set if any measurement is outside this (minimum, maximum) range). 1 LSB = 1°C, signed number Valid range: -99°C to 199°C Value 0: Disable monitor threshold check (together with DIG_TEMP_THRESH_MIN=0)
TEMP_DIFF_THRESH	2	The maximum difference across temperatures read from all the enabled sensors is compared against this threshold. The comparison result is part of the monitoring report message (Error bit is set if the measured difference exceeds this field). 1 LSB = 1°C, unsigned number Valid range: 0°C to 100°C Note: For AWR2243/xWR6243 devices, digital temperature sensors can be excluded from this check by setting DIG_TEMP_THRESH_MIN and DIG_TEMP_THRESH_MAX to value 0.
RESERVED	4	0x00000000
RESERVED	4	0x00000000

8.3 RX Gain and Phase Monitor

This section contains API SBs that configure the monitors of receiver gain and phase. The corresponding monitors are collectively named RX_GAIN_PHASE_MONITOR. The report is sent as an async event [AWR_MONITOR_RX_GAIN_PHASE_REPORT_AE_SB](#).

NOTE:

It is recommended for the user to configure this monitor in verbose mode (Mode 0), so that Host can compute actual RX gain through temperature compensation and detect presence of interference using Noise Power.

In quiet mode, the user may consider programming broad thresholds for Absolute Gain Error, taking into account the temperature variation of reported RX_GAIN_VALUE.

8.3.1 Sub block 0x01C4 – AWR_MONITOR_RX_GAIN_PHASE_CONF_SB

This is a monitoring configuration API which the host sends to the AWR device, containing information related to RX gain and phase monitoring.

Table 8.5: AWR_MONITOR_RX_GAIN_PHASE_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x01C4
SBLKLEN	2	Value = 72
PROFILE_INDX	1	This field indicates the profile Index for which this monitoring configuration applies.

Continued on next page

Table 8.5 – continued from previous page

RF_FREQ_BIT-MASK	1	This field indicates the RF frequencies inside the profile's RF band at which to measure the required parameters. When each bit in this field is set, the measurement at the corresponding RF frequency is enabled w.r.t. the profile's RF band. <table><tr><td>Bit number</td><td>RF frequency</td><td>RF name</td></tr><tr><td>b0</td><td>Start RF frequency in profile's sweep bandwidth</td><td>RF1</td></tr><tr><td>b1</td><td>Center RF frequency in profile's sweep bandwidth</td><td>RF2</td></tr><tr><td>b2</td><td>End RF frequency in profile's sweep bandwidth</td><td>RF3</td></tr></table> <table><tr><td>Bit number</td><td>Value</td><td>Definition</td></tr><tr><td rowspan="2">b3</td><td></td><td>Bit for dither limits selection.</td></tr><tr><td>0</td><td>Default dither limits with 63 MHz range selected for RF1, RF2, RF3. RF1 dither limits:[0, 63] MHz RF2 dither limits:[-32,31] MHz RF3 dither limits:[-63,0] MHz</td></tr><tr><td></td><td>1</td><td>Configurable dither limits as specified in RF1_RF2_FREQ_DITHER_LIMITS and RF3_FREQ_DITHER_LIMITS</td></tr></table> The RF name column is mentioned here to set the convention for the purpose of reporting and describing many monitoring packets. The values of RF1, RF2 and RF3 are varied for each monitoring instance by adding a random dither value. The limits for uniform frequency dither can be controlled as mentioned by bit b3 in the table above.	Bit number	RF frequency	RF name	b0	Start RF frequency in profile's sweep bandwidth	RF1	b1	Center RF frequency in profile's sweep bandwidth	RF2	b2	End RF frequency in profile's sweep bandwidth	RF3	Bit number	Value	Definition	b3		Bit for dither limits selection.	0	Default dither limits with 63 MHz range selected for RF1, RF2, RF3. RF1 dither limits:[0, 63] MHz RF2 dither limits:[-32,31] MHz RF3 dither limits:[-63,0] MHz		1	Configurable dither limits as specified in RF1_RF2_FREQ_DITHER_LIMITS and RF3_FREQ_DITHER_LIMITS
Bit number	RF frequency	RF name																							
b0	Start RF frequency in profile's sweep bandwidth	RF1																							
b1	Center RF frequency in profile's sweep bandwidth	RF2																							
b2	End RF frequency in profile's sweep bandwidth	RF3																							
Bit number	Value	Definition																							
b3		Bit for dither limits selection.																							
	0	Default dither limits with 63 MHz range selected for RF1, RF2, RF3. RF1 dither limits:[0, 63] MHz RF2 dither limits:[-32,31] MHz RF3 dither limits:[-63,0] MHz																							
	1	Configurable dither limits as specified in RF1_RF2_FREQ_DITHER_LIMITS and RF3_FREQ_DITHER_LIMITS																							
REPORTING_MODE	1	<table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>Report is sent every monitoring period without threshold check</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds). It is recommended not to use quiet mode, as Host has to compute actual RX gain and need to monitor Noise power to detect presence of interference.</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	Report is sent every monitoring period without threshold check	1	Report is send only upon a failure (after checking for thresholds). It is recommended not to use quiet mode, as Host has to compute actual RX gain and need to monitor Noise power to detect presence of interference.	2	Report is sent every monitoring period with threshold check															
Value	Definition																								
0	Report is sent every monitoring period without threshold check																								
1	Report is send only upon a failure (after checking for thresholds). It is recommended not to use quiet mode, as Host has to compute actual RX gain and need to monitor Noise power to detect presence of interference.																								
2	Report is sent every monitoring period with threshold check																								

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Table 8.5 – continued from previous page

TX_SEL	1	Value	Definition
		0	TX0 is used for generating loopback signal for RX gain measurement
		1	TX1 is used for generating loopback signal for RX gain measurement
RX_GAIN_ABS_ERROR_THRESH	2	The magnitude of difference between the programmed and measured RX gain for each enabled channel at each enabled RF frequency, is compared against this threshold. The comparison result is part of the monitoring report message (Error bit is set if any measurement is above this threshold). Before the comparison, the measured gains for each RF and RX are adjusted by subtracting the offset given in the RX_GAIN_MISMATCH_OFFSET_VALUE field 1 LSB = 0.1 dB Valid range: 0 to 65535 (0 to 6553dB)	
RX_GAIN_MISMATCH_THRESH	2	The magnitude of difference between measured RX gains across the enabled channels at each enabled RF frequency is compared against this threshold. The comparison result is part of the monitoring report message (Error bit is set if the measurement is above this threshold). Before the comparison, the measured gains for each RF and RX are adjusted by subtracting the offset given in the RX_GAIN_MISMATCH_OFFSET_VALUE field. 1 LSB = 0.1 dB Valid range: 0 to 65535 (0 to 6553dB)	
RX_GAIN_FLATNESS_ERROR_THRESH	2	The magnitude of measured RX gain flatness error, for each enabled channel, is compared against this threshold. The flatness error for a channel is defined as the peak to peak variation across RF frequencies. The comparison result is part of the monitoring report message (Error bit is set if any measurement is above this threshold). Before the comparison, the measured gains for each RF and RX are adjusted by subtracting the offset given in the RX_GAIN_MISMATCH_OFFSET_VALUE field. 1 LSB = 0.1 dB Valid range: 0 to 65535 (0 to 6553dB) This flatness check is applicable only if multiple RF Frequencies are enabled, i.e., RF_FREQ_BITMASK has bit numbers 0,1,2 set.	

Continued on next page

Table 8.5 – continued from previous page

RX_PHASE_MISMATCH_THRESH	2	The magnitude of measured RX phase mismatch across the enabled channels at each enabled RF frequency is compared against this threshold. The comparison result is part of the monitoring report message (Error bit is set if any measurement is above this threshold). Before the comparison, the measured phases for each RF and RX are adjusted by subtracting the offset given in the RX_PHASE_MISMATCH_OFFSET_VALUE field. 1 LSB = $360^{\circ}/2^{16}$. Valid range: corresponding to 0° to 359.9°.																				
RX_GAIN_MISMATCH_OFFSET_VALUE	24	The offsets to be subtracted from the measured RX gain for each RX and RF before the relevant threshold comparisons are given here. Byte numbers corresponding to different RX and RF, in this field are here: <table><thead><tr><th></th><th>RF1</th><th>RF2</th><th>RF3</th></tr></thead><tbody><tr><td>RX0</td><td>1:0</td><td>9:8</td><td>17:16</td></tr><tr><td>RX1</td><td>3:2</td><td>11:10</td><td>19:18</td></tr><tr><td>RX2</td><td>5:4</td><td>13:12</td><td>21:20</td></tr><tr><td>RX3</td><td>7:6</td><td>15:14</td><td>23:22</td></tr></tbody></table> 1 LSB = 0.1 dB, signed number Only the entries of enabled RF Frequencies and enabled RX channels are considered.		RF1	RF2	RF3	RX0	1:0	9:8	17:16	RX1	3:2	11:10	19:18	RX2	5:4	13:12	21:20	RX3	7:6	15:14	23:22
	RF1	RF2	RF3																			
RX0	1:0	9:8	17:16																			
RX1	3:2	11:10	19:18																			
RX2	5:4	13:12	21:20																			
RX3	7:6	15:14	23:22																			
RX_PHASE_MISMATCH_OFFSET_VALUE	24	The offsets to be subtracted from the measured RX phase for each RX and RF before the relevant threshold comparisons are given here. <table><thead><tr><th></th><th>RF1</th><th>RF2</th><th>RF3</th></tr></thead><tbody><tr><td>RX0</td><td>1:0</td><td>9:8</td><td>17:16</td></tr><tr><td>RX1</td><td>3:2</td><td>11:10</td><td>19:18</td></tr><tr><td>RX2</td><td>5:4</td><td>13:12</td><td>21:20</td></tr><tr><td>RX3</td><td>7:6</td><td>15:14</td><td>23:22</td></tr></tbody></table> 1 LSB = $360^{\circ}/2^{16}$, unsigned number Only the entries of enabled RF Frequencies and enabled RX channels are considered.		RF1	RF2	RF3	RX0	1:0	9:8	17:16	RX1	3:2	11:10	19:18	RX2	5:4	13:12	21:20	RX3	7:6	15:14	23:22
	RF1	RF2	RF3																			
RX0	1:0	9:8	17:16																			
RX1	3:2	11:10	19:18																			
RX2	5:4	13:12	21:20																			
RX3	7:6	15:14	23:22																			

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Table 8.5 – continued from previous page

RF1_RF2_ FREQ_DITHER_ LIMITS	4	Minimum and maximum offset frequency dither limits for RF1 and RF2, when dither limit selection bit b3 of RF_FREQ_BITMASK is set to 1 Byte Description 0 RF1 offset frequency dither min limit 1 RF1 offset frequency dither max limit 2 RF2 offset frequency dither min limit 3 RF2 offset frequency dither max limit 1 LSB = 1MHz, signed number Valid Range: -128 to 127 NOTE: max limits should be greater than min limits
RF3_FREQ_ DITHER_LIMITS	2	Minimum and maximum offset frequency dither limits for RF3, when dither limit selection bit b3 of RF_FREQ_BITMASK is set to 1 Byte Description 0 RF3 offset frequency dither min limit 1 RF3 offset frequency dither max limit 1 LSB = 1MHz, signed number Valid Range: -128 to 127 NOTE: max limits should be greater than min limits
RESERVED	2	0x0000

8.4 RX Noise Monitor

This section contains API SBs that configure the monitor of receiver noise, and report the soft results from the monitor. The corresponding monitor is named RX_NOISE_FIGURE_MONITOR. The report is sent as an async event [AWR_MONITOR_RX_NOISE_FIGURE_REPORT_AE_SB](#).

8.4.1 Sub block 0x01C5 – AWR_MONITOR_RX_NOISE_FIGURE_CONF_SB

This is a monitoring configuration API which the host sends to the AWR device, containing information related to RX noise monitoring of a profile.

NOTE: The RX Noise figure monitor API is not supported in production, it can be used only for debug. Please refer latest DFP release note for more info.

Table 8.6: AWR_MONITOR_RX_NOISE_FIGURE_CONF_SB contents

Field Name	Number of bytes	Description												
SBLKID	2	Value = 0x01C5												
SBLKLEN	2	Value = 16												
PROFILE_IDX	1	This field indicates the profile Index for which this monitoring configuration applies.												
RF_FREQ_BIT-MASK	1	This field indicates the exact RF frequencies inside the profile's RF band at which to measure the required parameters. When each bit in this field is set, the measurement at the corresponding RF frequency is enabled w.r.t. the profile's RF band. <table><tr><th>Bit number</th><th>RF frequency</th><th>RF name</th></tr><tr><td>b0</td><td>Start RF frequency in profile's sweep bandwidth</td><td>RF1</td></tr><tr><td>b1</td><td>Center RF frequency in profile's sweep bandwidth</td><td>RF2</td></tr><tr><td>b2</td><td>End RF frequency in profile's sweep bandwidth</td><td>RF3</td></tr></table> The RF name column is mentioned here to set the convention for the purpose of reporting and describing many monitoring packets.	Bit number	RF frequency	RF name	b0	Start RF frequency in profile's sweep bandwidth	RF1	b1	Center RF frequency in profile's sweep bandwidth	RF2	b2	End RF frequency in profile's sweep bandwidth	RF3
Bit number	RF frequency	RF name												
b0	Start RF frequency in profile's sweep bandwidth	RF1												
b1	Center RF frequency in profile's sweep bandwidth	RF2												
b2	End RF frequency in profile's sweep bandwidth	RF3												
RESERVED	2	0x0000												
REPORTING_MODE	1	<table><tr><th>Value</th><th>Definition</th></tr><tr><td>0</td><td>Report is sent every monitoring period without threshold check</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	Report is sent every monitoring period without threshold check	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check				
Value	Definition													
0	Report is sent every monitoring period without threshold check													
1	Report is send only upon a failure (after checking for thresholds)													
2	Report is sent every monitoring period with threshold check													
RESERVED	1	0x00												
RX_NOISE_FIGURE_THRESH-OLD	2	The measured RX input referred noise figure at the enabled RF frequencies, for all channels, is compared against this threshold. The comparison result is part of the monitoring report message (Error bit is set if any measurement is above this threshold). 1 LSB = 0.1 dB Valid range: 0 to 65535 (0 to 6553dB)												
RESERVED	4	0x00000000												

NOTE: The Rx gain and phase monitoring shall be enabled when enabling Rx noise figure Monitoring. This monitors only baseband noise figure.

8.5 RX IF Stage Monitor

This section contains API SBs that configure the monitors of receiver IF filter attenuation, and report the soft results from the monitor. The corresponding monitor is named RX_IFSTAGE_MONITOR. The report is sent as an async event [AWR_MONITOR_RX_IFSTAGE_REPORT_AE_SB](#).

8.5.1 Sub block 0x01C6 – AWR_MONITOR_RX_IFSTAGE_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to RX IF filter attenuation monitoring. The report is sent as an async event [AWR_MONITOR_RX_IFSTAGE_REPORT_AE_SB](#).

Table 8.7: AWR_MONITOR_RX_IFSTAGE_CONF_SB contents

Field Name	Number of bytes	Description								
SBLKID	2	Value = 0x01C6								
SBLKLEN	2	Value = 20								
PROFILE_INDXX	1	This field indicates the Profile Index for which this monitoring configuration applies.								
REPORTING_MODE	1	<table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>Report is sent every monitoring period without threshold check</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	Report is sent every monitoring period without threshold check	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check
Value	Definition									
0	Report is sent every monitoring period without threshold check									
1	Report is send only upon a failure (after checking for thresholds)									
2	Report is sent every monitoring period with threshold check									
RESERVED	2	0x0000								
RESERVED	2	0x0000								
HPF_CUTOFF_FREQ_ERROR_THRESH	2	The absolute values of RX IF HPF cutoff percentage frequency errors are compared against the corresponding thresholds given in this field. The comparison results are part of the monitoring report message (Error bit is set if the absolute value of the errors exceeds respective thresholds). 1 LSB = 1%, unsigned number Valid range: 1% to 128%								

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Table 8.7 – continued from previous page

LPF_CUT- OFF_BAND- EDGE_DROOP_ THRESH	1	The LPF band edge droop of RX channels are compared against the corresponding thresholds given in this field (max-limit check). The comparison results are part of the monitoring report message (Error bit is set if the band edge droops exceeds respective threshold). 1 LSB = 0.2dB, unsigned number Valid range: 0 to 50dB Note1: This feature is supported only on AWR2243, xWR294x, xWR254x, xWR2x4xEco and xWR2x4xP devices. In xWR294x/xWR254x, this should be used for debug purposes only. Note2: In xWR254x/xWR2x4xEco/xWR2x4xP devices LPF monitor is executed only when LPF_CUTOFF_BANDEDGE_DROOP_THRESH and LPF_CUTOFF_STOPBAND_ATTEN_THRESH both are set to non-zero threshold values. Refer Table 12.7 for monitoring timings for LPF monitor enabled / disabled case.
LPF_CUTOFF_ STOPBAND_ ATTEN_THRESH	1	The LPF stop band attenuation at 2x analog LPF's band edge with respect to the analog LPF's band edge of RX channels are compared against the corresponding thresholds given in this field (min-limit check). The comparison results are part of the monitoring report message (Error bit is set if the stop band attenuation less than respective threshold). 1 LSB = 0.2dB, unsigned number Valid range: 0 to 50dB Note1: This feature is supported only on AWR2243, xWR294x, xWR254x, xWR2x4xEco and xWR2x4xP devices. In xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP, this should be used for debug purposes only. Note2: In xWR254x, xWR2x4xEco, xWR2x4xP devices LPF monitor is executed only when LPF_CUTOFF_BANDEDGE_DROOP_THRESH and LPF_CUTOFF_STOPBAND_ATTEN_THRESH both are set to non-zero threshold values. Refer Table 12.7 for monitoring timings for LPF monitor enabled , disabled case.
IFA_GAIN_ER- ROR_THRESH	2	The absolute deviation of RX IFA Gain from the expected gain for each enabled RX channel is compared against the thresholds given in this field. The comparison result is part of the monitoring report message (Error bit is set if the absolute value of the errors exceeds respective thresholds). 1 LSB = 0.1 dB, unsigned number Valid range: 0 to 65535 (0 to 6553dB)
RESERVED	4	0x00000000

8.6 TX Power Monitor

This section contains API SBs that configure the monitors of transmitter output power, and report the soft results from the monitor. The corresponding monitors are collectively named TXn_POWER_MONITOR where n is the TX channel number.

8.6.1 Sub block 0x01C7 – AWR_MONITOR_TX0_POWER_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to TX0 power monitoring. Absolute TX power and flatness across RF frequencies are monitored here. The report is sent as an async event [AWR_MONITOR_TX0_POWER_REPORT_AE_SB](#).

Table 8.8: AWR_MONITOR_TX0_POWER_CONF_SB contents

Field Name	Number of bytes	Description												
SBLKID	2	Value = 0x01C7												
SBLKLEN	2	Value = 20												
PROFILE_INDX	1	This field indicates the Profile Index for which this monitoring configuration applies.												
RF_FREQ_BIT-MASK	1	This field indicates the exact RF frequencies inside the profile's RF band at which to measure the required parameters. When each bit in this field is set, the measurement at the corresponding RF frequency is enabled w.r.t. the profile's RF band. <table> <tr> <th>Bit number</th><th>RF frequency</th><th>RF name</th></tr> <tr> <td>b0</td><td>Start RF frequency in profile's sweep bandwidth</td><td>RF1</td></tr> <tr> <td>b1</td><td>Center RF frequency in profile's sweep bandwidth</td><td>RF2</td></tr> <tr> <td>b2</td><td>End RF frequency in profile's sweep bandwidth</td><td>RF3</td></tr> </table> The RF Name column is mentioned here to set the convention for the purpose of reporting and describing many monitoring packets.	Bit number	RF frequency	RF name	b0	Start RF frequency in profile's sweep bandwidth	RF1	b1	Center RF frequency in profile's sweep bandwidth	RF2	b2	End RF frequency in profile's sweep bandwidth	RF3
Bit number	RF frequency	RF name												
b0	Start RF frequency in profile's sweep bandwidth	RF1												
b1	Center RF frequency in profile's sweep bandwidth	RF2												
b2	End RF frequency in profile's sweep bandwidth	RF3												
RESERVED	2	0x0000												

Continued on next page

Table 8.8 – continued from previous page

REPORTING_ MODE	1	<table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>Report is sent every monitoring period without threshold check</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	Report is sent every monitoring period without threshold check	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check
Value	Definition									
0	Report is sent every monitoring period without threshold check									
1	Report is send only upon a failure (after checking for thresholds)									
2	Report is sent every monitoring period with threshold check									
RESERVED	1	0x00								
TX_POWER_ ABSOLUTE_ER- ROR_THRESH	2	The magnitude of difference between the programmed and measured TX power for each enabled channel at each en- abled RF frequency, is compared against this threshold. The comparison result is part of the monitoring report mes- sage (Error bit is set if any measurement is above this threshold). 1 LSB = 0.1 dB Valid range: 0 to 65535 (0 to 6553dB)								
TX_POWER_ FLATNESS_ER- ROR_THRESH	2	The magnitude of measured TX power flatness error, for each enabled channel, is compared against this threshold. The flatness error for a channel is defined as the peak to peak variation across RF frequencies. The comparison re- sult is part of the monitoring report message (Error bit is set if any measurement is above this threshold). 1 LSB = 0.1 dB Valid range: 0 to 65535 (0 to 6553dB) This flatness check is applicable only if multiple RF Fre- quencies are enabled.								
RESERVED	2	0x0000								
TX_POWER_ OFFSET_VALUE	3	AWR2243, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: This field is reserved. Set it to 0x000000. xWR6243 devices: The offset values to be added with the measured TX power for each RF before the relevant threshold compar- ison. Byte numbers corresponding to different RF are <table><tr><td>RF</td><td>Byte</td></tr><tr><td>RF1</td><td>0</td></tr><tr><td>RF2</td><td>1</td></tr><tr><td>RF3</td><td>2</td></tr></table> 1 LSB = 0.1dB signed Valid range: -128 to +127 (-12.8 to 12.7dB)	RF	Byte	RF1	0	RF2	1	RF3	2
RF	Byte									
RF1	0									
RF2	1									
RF3	2									
RESERVED	1	0x00								

8.6.2 Sub block 0x01C8 – AWR_MONITOR_TX1_POWER_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to TX1 power monitoring. Absolute TX power and flatness across RF frequencies are monitored here. The report is sent as an async event [AWR_MONITOR_TX1_POWER_REPORT_AE_SB](#).

Table 8.9: AWR_MONITOR_TX1_POWER_CONF_SB contents

Field Name	Number of bytes	Description												
SBLKID	2	Value = 0x01C8												
SBLKLEN	2	Value = 20												
PROFILE_INDX	1	This field indicates the Profile Index for which this monitoring configuration applies.												
RF_FREQ_BIT-MASK	1	This field indicates the exact RF frequencies inside the profile's RF band at which to measure the required parameters. When each bit in this field is set, the measurement at the corresponding RF frequency is enabled w.r.t. the profile's RF band. <table><thead><tr><th>Bit number</th><th>RF frequency</th><th>RF name</th></tr></thead><tbody><tr><td>b0</td><td>Start RF frequency in profile's sweep bandwidth</td><td>RF1</td></tr><tr><td>b1</td><td>Center RF frequency in profile's sweep bandwidth</td><td>RF2</td></tr><tr><td>b2</td><td>End RF frequency in profile's sweep bandwidth</td><td>RF3</td></tr></tbody></table> The RF Name column is mentioned here to set the convention for the purpose of reporting and describing many monitoring packets.	Bit number	RF frequency	RF name	b0	Start RF frequency in profile's sweep bandwidth	RF1	b1	Center RF frequency in profile's sweep bandwidth	RF2	b2	End RF frequency in profile's sweep bandwidth	RF3
Bit number	RF frequency	RF name												
b0	Start RF frequency in profile's sweep bandwidth	RF1												
b1	Center RF frequency in profile's sweep bandwidth	RF2												
b2	End RF frequency in profile's sweep bandwidth	RF3												
RESERVED	2	0x0000												
REPORTING_MODE	1	<table><thead><tr><th>Value</th><th>Definition</th></tr></thead><tbody><tr><td>0</td><td>Report is sent every monitoring period without threshold check</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></tbody></table>	Value	Definition	0	Report is sent every monitoring period without threshold check	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check				
Value	Definition													
0	Report is sent every monitoring period without threshold check													
1	Report is send only upon a failure (after checking for thresholds)													
2	Report is sent every monitoring period with threshold check													
RESERVED	1	0x00												

Continued on next page

Table 8.9 – continued from previous page

TX_POWER_ABSOLUTE_ERROR_THRESH	2	The magnitude of difference between the programmed and measured TX power for each enabled channel at each enabled RF frequency, is compared against this threshold. The comparison result is part of the monitoring report message (Error bit is set if any measurement is above this threshold). 1 LSB = 0.1 dB Valid range: 0 to 65535 (0 to 6553dB)
TX_POWER_FLATNESS_ERROR_THRESH	2	The magnitude of measured TX power flatness error, for each enabled channel, is compared against this threshold. The flatness error for a channel is defined as the peak to peak variation across RF frequencies. The comparison result is part of the monitoring report message (Error bit is set if any measurement is above this threshold). 1 LSB = 0.1 dB Valid range: 0 to 65535 (0 to 6553dB) This flatness check is applicable only if multiple RF Frequencies are enabled.
RESERVED	2	0x0000
TX_POWER_OFFSET_VALUE	3	AWR2243, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: This field is reserved. Set to 0x0. xWR6243 devices: The offset values to be added with the measured TX power for each RF before the relevant threshold comparison. Byte numbers corresponding to different RF are RF Byte RF1 0 RF2 1 RF3 2 1 LSB = 0.1dB signed Valid range: -128 to +127 (-12.8 to 12.7dB)
RESERVED	1	0x00

8.6.3 Sub block 0x01C9 – AWR_MONITOR_TX2_POWER_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to TX2 power monitoring. Absolute TX power and flatness across RF frequencies are monitored here. The report is sent as an async event [AWR_MONITOR_TX2_POWER_REPORT_AE_SB](#).

Table 8.10: AWR_MONITOR_TX2_POWER_CONF_SB contents

Field Name	Number of bytes	Description												
SBLKID	2	Value = 0x01C9												
SBLKLEN	2	Value = 20												
PROFILE_IDX	1	This field indicates the Profile Index for which this monitoring configuration applies.												
RF_FREQ_BIT-MASK	1	This field indicates the exact RF frequencies inside the profile's RF band at which to measure the required parameters. When each bit in this field is set, the measurement at the corresponding RF frequency is enabled w.r.t. the profile's RF band. <table><thead><tr><th>Bit number</th><th>RF frequency</th><th>RF name</th></tr></thead><tbody><tr><td>b0</td><td>Start RF frequency in profile's sweep bandwidth</td><td>RF1</td></tr><tr><td>b1</td><td>Center RF frequency in profile's sweep bandwidth</td><td>RF2</td></tr><tr><td>b2</td><td>End RF frequency in profile's sweep bandwidth</td><td>RF3</td></tr></tbody></table> The RF Name column is mentioned here to set the convention for the purpose of reporting and describing many monitoring packets.	Bit number	RF frequency	RF name	b0	Start RF frequency in profile's sweep bandwidth	RF1	b1	Center RF frequency in profile's sweep bandwidth	RF2	b2	End RF frequency in profile's sweep bandwidth	RF3
Bit number	RF frequency	RF name												
b0	Start RF frequency in profile's sweep bandwidth	RF1												
b1	Center RF frequency in profile's sweep bandwidth	RF2												
b2	End RF frequency in profile's sweep bandwidth	RF3												
RESERVED	2	0x0000												
REPORTING_MODE	1	<table><thead><tr><th>Value</th><th>Definition</th></tr></thead><tbody><tr><td>0</td><td>Report is sent every monitoring period without threshold check</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></tbody></table>	Value	Definition	0	Report is sent every monitoring period without threshold check	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check				
Value	Definition													
0	Report is sent every monitoring period without threshold check													
1	Report is send only upon a failure (after checking for thresholds)													
2	Report is sent every monitoring period with threshold check													
RESERVED	1	0x00												
TX_POWER_ABSOLUTE_ERROR_THRESH	2	The magnitude of difference between the programmed and measured TX power for each enabled channel at each enabled RF frequency, is compared against this threshold. The comparison result is part of the monitoring report message (Error bit is set if any measurement is above this threshold). 1 LSB = 0.1 dB Valid range: 0 to 65535 (0 to 6553dB)												

Continued on next page

Table 8.10 – continued from previous page

TX_POWER_FLATNESS_ERROR_THRESH	2	The magnitude of measured TX power flatness error, for each enabled channel, is compared against this threshold. The flatness error for a channel is defined as the peak to peak variation across RF frequencies. The comparison result is part of the monitoring report message (Error bit is set if any measurement is above this threshold). 1 LSB = 0.1 dB Valid range: 0 to 65535 (0 to 6553dB) This flatness check is applicable only if multiple RF Frequencies are enabled.								
RESERVED	2	0x0000								
TX_POWER_OFFSET_VALUE	3	AWR2243, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: This field is reserved. Set to 0x0. xWR6243 devices: The offset values to be added with the measured TX power for each RF before the relevant threshold comparison. Byte numbers corresponding to different RF are <table><tr><td>RF</td><td>Byte</td></tr><tr><td>RF1</td><td>0</td></tr><tr><td>RF2</td><td>1</td></tr><tr><td>RF3</td><td>2</td></tr></table> 1 LSB = 0.1dB signed Valid range: -128 to +127 (-12.8 to 12.7dB)	RF	Byte	RF1	0	RF2	1	RF3	2
RF	Byte									
RF1	0									
RF2	1									
RF3	2									
RESERVED	1	0x00								

8.7 TX Ball Break Monitor

This section contains API SBs that configure the monitors of transmitter balls and impedance matching. The corresponding monitors are collectively named TXn_BALLBREAK_MONITOR where n is the TX channel number.

TX ball break detection is performed through measurement of TX reflection coefficient's magnitude. The breakage of a TX ball is detected by observing high reflection magnitude.

8.7.1 Sub block 0x01CA – AWR_MONITOR_TX0_BALLBREAK_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to TX ball break detection.

This API SB controls the thresholds for the reflection coefficient magnitude check and the parameters for the reflection coefficient error distance check (variation from values at the time of factory calibration). The report is sent as an async event [AWR_MONITOR_TX0_BALLBREAK_REPORT_AE_SB](#).

Table 8.11: AWR_MONITOR_TX0_BALLBREAK_CONF_SB contents

Field Name	Number of bytes	Description								
SBLKID	2	Value = 0x01CA								
SBLKLEN	2	Value = 16								
REPORTING_MODE	1	<table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>Report is sent every monitoring period without threshold check</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	Report is sent every monitoring period without threshold check	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check
Value	Definition									
0	Report is sent every monitoring period without threshold check									
1	Report is send only upon a failure (after checking for thresholds)									
2	Report is sent every monitoring period with threshold check									
RESERVED	1	0x00								
TX_REFL_COEFF_THRESH	2	The TX reflection coefficient's magnitude for each enabled channel is compared against the threshold given here. The comparison result is part of the monitoring report message (Error bit is set if the measurement is higher than this threshold, with the units of both quantities being the same). 1 LSB = 0.1 dB, signed number Valid range: -32767 to +32767 (-3276dB to +3276dB)								
MON_START_FREQ_CONST	4	AWR2243 devices: This field is reserved. Set to 0x0. xWR6x43, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: Start frequency of the monitoring chirp. For 77GHz Devices (76GHz to 80.8Ghz): 1 LSB = $3.6e9/2^{26}$ Hz $\approx$ 53.644 Hz Valid range: 0x5471C71C to 0x59C71C71 For 60GHz Devices (57GHz to 63.8Ghz): 1 LSB = $2.7e9/2^{26}$ Hz $\approx$ 40.233 Hz Valid range: Only even numbers from 0x5471C71C to 0x5E84BDA1								
TX_POWER_BACKOFF	1	AWR2243, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: This field is reserved. Set to 0x0. xWR6243 devices: TX Power Backoff settings used for ballbreak monitor 1 LSB = 1dB Valid values: 0, 3, 6, 9dB								
RESERVED	3	0x000000								

8.7.2 Sub block 0x01CB – AWR_MONITOR_TX1_BALLBREAK_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to TX ball break detection.

This API SB controls the thresholds for the reflection coefficient magnitude check and the parameters for the reflection coefficient error distance check (variation from values at the time of factory calibration). The report is sent as an async event [AWR_MONITOR_TX1_BALLBREAK_REPORT_AE_SB](#).

Table 8.12: AWR_MONITOR_TX1_BALLBREAK_CONF_SB contents

Field Name	Number of bytes	Description								
SBLKID	2	Value = 0x01CB								
SBLKLEN	2	Value = 16								
REPORTING_MODE	1	<table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>Report is sent every monitoring period without threshold check</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	Report is sent every monitoring period without threshold check	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check
Value	Definition									
0	Report is sent every monitoring period without threshold check									
1	Report is send only upon a failure (after checking for thresholds)									
2	Report is sent every monitoring period with threshold check									
RESERVED	1	0x00								
TX_REFL_COEFF_THRESH	2	The TX reflection coefficient's magnitude for each enabled channel is compared against the threshold given here. The comparison result is part of the monitoring report message (Error bit is set if the measurement is higher than this threshold, with the units of both quantities being the same). 1 LSB = 0.1 dB, signed number Valid range: -32767 to +32767 (-3276dB to +3276dB)								
MON_START_FREQ_CONST	4	AWR2243 devices: This field is reserved. Set to 0x0. xWR6x43, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: Start frequency of the monitoring chirp. For 77GHz Devices (76GHz to 80.8Ghz): 1 LSB = $3.6e9/2^{26}$ Hz $\approx$ 53.644 Hz Valid range: 0x5471C71C to 0x59C71C71 For 60GHz Devices (57GHz to 63.8Ghz): 1 LSB = $2.7e9/2^{26}$ Hz $\approx$ 40.233 Hz Valid range: Only even numbers from 0x5471C71C to 0x5E84BDA1								

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Table 8.12 – continued from previous page

TX_POWER_BACKOFF	1	AWR2243, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: This field is reserved. Set to 0x0. xWR6243 devices: TX Power Backoff settings used for ballbreak monitor 1 LSB = 1dB Valid values: 0, 3, 6, 9dB
RESERVED	3	0x000000

8.7.3 Sub block 0x01CC – AWR_MONITOR_TX2_BALLBREAK_CONF_SB

This API is a monitoring monfiguration API which the host sends to the AWR device, containing information related to TX ball break detection.

This API SB controls the thresholds for the reflection coefficient magnitude check and the parameters for the reflection coefficient error distance check (variation from values at the time of factory calibration). The report is sent as an async event [AWR_MONITOR_TX2_BALLBREAK_REPORT_AE_SB](#).

Table 8.13: AWR_MONITOR_TX2_BALLBREAK_CONF_SB contents

Field Name	Number of bytes	Description								
SBLKID	2	Value = 0x01CC								
SBLKLEN	2	Value = 16								
REPORTING_MODE	1	<table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>Report is sent every monitoring period without threshold check</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	Report is sent every monitoring period without threshold check	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check
Value	Definition									
0	Report is sent every monitoring period without threshold check									
1	Report is send only upon a failure (after checking for thresholds)									
2	Report is sent every monitoring period with threshold check									
RESERVED	1	0x00								
TX_REFL_COEFF_THRESH	2	The TX reflection coefficient's magnitude for each enabled channel is compared against the threshold given here. The comparison result is part of the monitoring report message (Error bit is set if the measurement is higher or equal to this threshold, with the units of both quantities being the same). 1 LSB = 0.1 dB, signed number Valid range: -32767 to +32767 (-3276dB to +3276dB)								

Continued on next page

Table 8.13 – continued from previous page

MON_START_ FREQ_CONST	4	AWR2243 devices: This field is reserved. Set to 0x0. xWR6x43, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: Start frequency of the monitoring chirp. For 77GHz Devices (76GHz to 80.8GHz): 1 LSB = $3.6e9/2^{26}$ Hz $\approx$ 53.644 Hz Valid range: 0x5471C71C to 0x59C71C71 For 60GHz Devices (57GHz to 63.8GHz): 1 LSB = $2.7e9/2^{26}$ Hz $\approx$ 40.233 Hz Valid range: Only even numbers from 0x5471C71C to 0x5E84BDA1
TX_POWER_ BACKOFF	1	AWR2243, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: This field is reserved. Set to 0x0. xWR6243 devices: TX Power Backoff settings used for ballbreak monitor 1 LSB = 1dB Valid values: 0, 3, 6, 9dB
RESERVED	3	0x000000

8.8 TX Gain and Phase Mismatch Monitoring

This section contains API SBs that configure the monitors of transmitter gain and phase mismatches, and report the soft results from the monitor. The corresponding monitors are collectively named TX_GAIN_PHASE_MISMATCH_MONITOR.

This monitor needs the operation of at least one RX channel. It also needs to use the RX in complex mode. Therefore, if all channels are disabled as per AWR_CHAN_CONF_SET_SB, this monitor automatically enables one RX channel. Further, this monitor automatically uses both I and Q channels of the receiver, irrespective of the ADC settings given by AWR_ADCOUT_CONF_SET_SB.

8.8.1 Sub block 0x01CD – AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to TX gain and phase mismatch monitoring. The report is sent as an async event [AWR_MONITOR_TX_GAIN_PHASE_REPORT_AE_SB](#).

NOTE: This API is not supported in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices. Please use AWR_MONITOR_ADV_TX_GAIN_PHASE_MISMATCH_CONF_SB at [512](#) to perform TX gain phase monitoring on xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices.

Table 8.14: AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_CONF_SB contents

Field Name	Number of bytes	Description												
SBLKID	2	Value = 0x01CD												
SBLKLEN	2	Value = 56												
PROFILE_INDX	1	This field indicates the Profile Index for which this monitoring configuration applies.												
RF_FREQ_BIT-MASK	1	This field indicates the exact RF frequencies inside the profile's RF band at which to measure the required parameters. When each bit in this field is set, the measurement at the corresponding RF frequency is enabled w.r.t. the profile's RF band. <table><thead><tr><th>Bit number</th><th>RF frequency</th><th>RF name</th></tr></thead><tbody><tr><td>b0</td><td>Start RF frequency in profile's sweep bandwidth</td><td>RF1</td></tr><tr><td>b1</td><td>Center RF frequency in profile's sweep bandwidth</td><td>RF2</td></tr><tr><td>b2</td><td>End RF frequency in profile's sweep bandwidth</td><td>RF3</td></tr></tbody></table> The RF Name column is mentioned here to set the convention for the purpose of reporting and describing many monitoring packets.	Bit number	RF frequency	RF name	b0	Start RF frequency in profile's sweep bandwidth	RF1	b1	Center RF frequency in profile's sweep bandwidth	RF2	b2	End RF frequency in profile's sweep bandwidth	RF3
Bit number	RF frequency	RF name												
b0	Start RF frequency in profile's sweep bandwidth	RF1												
b1	Center RF frequency in profile's sweep bandwidth	RF2												
b2	End RF frequency in profile's sweep bandwidth	RF3												
TX_EN	1	This field indicates the TX channels that should be compared for gain and phase balance. Setting the corresponding bit to 1 enables that channel for imbalance measurement. <table><thead><tr><th>Bit number</th><th>TX Channel</th></tr></thead><tbody><tr><td>b0</td><td>TX0</td></tr><tr><td>b1</td><td>TX1</td></tr><tr><td>b2</td><td>TX2</td></tr></tbody></table>	Bit number	TX Channel	b0	TX0	b1	TX1	b2	TX2				
Bit number	TX Channel													
b0	TX0													
b1	TX1													
b2	TX2													

Continued on next page

Table 8.14 – continued from previous page

RX_EN	1	This field indicates the RX channels that should be enabled for TX to RX loopback measurement. Setting the corresponding bit to 1 enables that channel for imbalance measurement. <table> <tr> <th>Bit number</th> <th>RX Channel</th> </tr> <tr> <td>b0</td> <td>RX0</td> </tr> <tr> <td>b1</td> <td>RX1</td> </tr> <tr> <td>b2</td> <td>RX2</td> </tr> <tr> <td>b3</td> <td>RX3</td> </tr> </table>	Bit number	RX Channel	b0	RX0	b1	RX1	b2	RX2	b3	RX3
Bit number	RX Channel											
b0	RX0											
b1	RX1											
b2	RX2											
b3	RX3											
REPORTING_MODE	1	<table> <tr> <th>Value</th> <th>Definition</th> </tr> <tr> <td>0</td> <td>Report is sent every monitoring period without threshold check</td> </tr> <tr> <td>1</td> <td>Report is send only upon a failure (after checking for thresholds)</td> </tr> <tr> <td>2</td> <td>Report is sent every monitoring period with threshold check</td> </tr> </table>	Value	Definition	0	Report is sent every monitoring period without threshold check	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check		
Value	Definition											
0	Report is sent every monitoring period without threshold check											
1	Report is send only upon a failure (after checking for thresholds)											
2	Report is sent every monitoring period with threshold check											
MON_CHIRP_SLOPE	1	Frequency slope for each monitoring chirp is encoded in 1 bytes (8 bit signed number) For 77GHz Devices (76GHz to 81Ghz): 1 LSB = $3.6e9 \times 900/2^{26}$ Hz ≈ 48.279 kHz/μs Valid range: -128 to +127 (Max 6.13 MHz/μs) For 60GHz Devices (57GHz to 64Ghz): 1 LSB = $2.7e9 \times 900/2^{26}$ Hz ≈ 36.21 kHz/μs Valid range: -128 to +127 (Max 4.63 MHz/μs) NOTE: Monitoring Chirp Slope can be programmed based on the emission specifications. The device transmits on air during the execution of these monitors. The host can control the monitoring emission power spectral density (dB-m/Hz) by programming this slope parameter. Each monitoring chirp is about 45us in duration. Therefore the overall RF sweep bandwidth for the monitoring chirp is given by Monitoring Chirp Slope* 45 us. Normally, low values of Monitoring Chirp Slope are recommended, as with non-zero slope in FMCW radar, any actual target reflections can potentially be interpreted as noise power and/or corrupt the loopback signal based gain/phase measurement. The user has to ensure that the RF bandwidth selected for monitoring chirp based on slope and start frequency is within FREQ_LIMIT set in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB API.										

Continued on next page

Table 8.14 – continued from previous page

TX_GAIN_MISMATCH_THRESH	2	The magnitude of difference between measured TX powers across the enabled channels at each enabled RF frequency is compared against this threshold. The comparison result is part of the monitoring report message (Error bit is set if the measurement is above this threshold). Before the comparison, the measured gains for each RF and RX are adjusted by subtracting the offset given in the TX_GAIN_MISMATCH_OFFSET_VALUE field. 1 LSB = 0.1 dB, unsigned number Valid range: 0 to 65535 (0 to 6553.5dB)																
TX_PHASE_MISMATCH_THRESH	2	The magnitude of measured TX phase mismatch across the enabled channels at each enabled RF frequency is compared against this threshold. The comparison result is part of the monitoring report message (Error bit is set if any measurement is above this threshold). Before the comparison, the measured gains for each RF and RX are adjusted by subtracting the offset given in the TX_PHASE_MISMATCH_OFFSET_VALUE field. 1 LSB = $360^{\circ}/2^{16}$, unsigned number Valid range: corresponding to 0° to 359.9°.																
TX_GAIN_MISMATCH_OFFSET_VALUE	18	The offsets to be subtracted from the measured TX gain for each TX and RF before the relevant threshold comparisons are given here. Byte numbers corresponding to different RX and RF, in this field are here: <table><thead><tr><th></th><th>RF1</th><th>RF2</th><th>RF3</th></tr></thead><tbody><tr><td>TX0</td><td>1:0</td><td>7:6</td><td>13:12</td></tr><tr><td>TX1</td><td>3:2</td><td>9:8</td><td>15:14</td></tr><tr><td>TX2</td><td>5:4</td><td>11:10</td><td>17:16</td></tr></tbody></table> 1 LSB = 0.1 dB, signed number Only the entries of enabled RF Frequencies and enabled TX channels are considered.		RF1	RF2	RF3	TX0	1:0	7:6	13:12	TX1	3:2	9:8	15:14	TX2	5:4	11:10	17:16
	RF1	RF2	RF3															
TX0	1:0	7:6	13:12															
TX1	3:2	9:8	15:14															
TX2	5:4	11:10	17:16															

Continued on next page

Table 8.14 – continued from previous page

TX_PHASE_MISMATCH_OFFSET_VALUE	18	The offsets to be subtracted from the measured TX phase for each TX and RF before the relevant threshold comparisons are given here. Byte numbers corresponding to different RX and RF, in this field are here: <table><thead><tr><th></th><th>RF1</th><th>RF2</th><th>RF3</th></tr></thead><tbody><tr><td>TX0</td><td>1:0</td><td>7:6</td><td>13:12</td></tr><tr><td>TX1</td><td>3:2</td><td>9:8</td><td>15:14</td></tr><tr><td>TX2</td><td>5:4</td><td>11:10</td><td>17:16</td></tr></tbody></table> 1 LSB = $360^{\circ}/2^{16}$. Only the entries of enabled RF Frequencies and enabled TX channels are considered.		RF1	RF2	RF3	TX0	1:0	7:6	13:12	TX1	3:2	9:8	15:14	TX2	5:4	11:10	17:16
	RF1	RF2	RF3															
TX0	1:0	7:6	13:12															
TX1	3:2	9:8	15:14															
TX2	5:4	11:10	17:16															
RESERVED	2	0x0000																
RESERVED	4	0x00000000																

NOTE:

Even when the TXs are matched, TX3 loopback path has gain and phase offsets wrt TX1 (and TX2), which get reported as mismatches in this API. These deterministic offsets can be compensated either through the OFFSET_VALUE fields (quiet mode) or through post processing by the host (verbose mode). Nominally, when the TXs are matched, TX3 - TX1 gain (i.e. loopback amplitude) is reported as -8dB. Nominally, when the TXs are matched, the reported TX3 - TX1 phase difference varies linearly with RF and it is reported as -5degree (76GHz) and 15degree (81GHz).

8.9 TX Phase Shifter Monitor

This section contains API SBs that configure the monitors of transmitter phase shifter and report the soft results from the monitor for various TX channels using TX loop-back. The corresponding monitors are collectively named TX0_PHASE_SHIFTER_MONITOR, TX1_PHASE_SHIFTER_MONITOR and TX2_PHASE_SHIFTER_MONITOR for the respective TX channels.

The phase shifter monitor will report the measured phase values in order to enable calibration of phase shifter codes at HOST. It will report tone power amplitude to provide check for amplitude stability across phase shifter codes. It will also report noise power in order to detect the chirps affected by interference.

The maximum four phases can be monitored at a time in one FTTI interval for each TX, there is an option to increment the phase by PH_SHIFTER_INC_VAL to cover all 360° phase over the time.

NOTE: The absolute gain/phase values reported by PHASE_SHIFTER_MON can exhibit smooth drifts across monitoring intervals due to slow temperature drifts. The absolute phase/gain can also exhibit abrupt jumps across temperature calibration boundaries. One way to mitigate the effects of such jumps across monitoring intervals is to rely on relative gain/phase values within the same monitoring report (e.g. assign one of the 4 phase settings in the monitoring configuration as a reference phase setting).

8.9.1 Sub block 0x01CE – AWR_MONITOR_TX0_PHASE_SHIFTER_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to TX0 TX loop back based phase shifter monitoring.

The report is sent as an async event [AWR_MONITOR_TX0_PHASE_SHIFTER_REPORT_AE_SB](#).

Table 8.15: AWR_MONITOR_TX0_PHASE_SHIFTER_CONF_SB contents

Field Name	Number of bytes	Description												
SBLKID	2	Value = 0x01CE												
SBLKLEN	2	Value = 32												
PROFILE_INDX	1	This field indicates the Profile Index for which this monitoring configuration applies.												
REPORTING_MODE	1	<table><tr><th>Value</th><th>Definition</th></tr><tr><td>0</td><td>Report is sent every monitoring period without threshold check</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	Report is sent every monitoring period without threshold check	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check				
Value	Definition													
0	Report is sent every monitoring period without threshold check													
1	Report is send only upon a failure (after checking for thresholds)													
2	Report is sent every monitoring period with threshold check													
RESERVED	2	RESERVED												
PH_SHIFTER_MON_CFG	1	<table><tr><th>Bit</th><th>Definition</th></tr><tr><td>b0</td><td>Phase shifter phase1 monitor enable bit</td></tr><tr><td>b1</td><td>Phase shifter phase2 monitor enable bit</td></tr><tr><td>b2</td><td>Phase shifter phase3 monitor enable bit</td></tr><tr><td>b3</td><td>Phase shifter phase4 monitor enable bit</td></tr><tr><td>b7:4</td><td>RESERVED</td></tr></table> Enable at least two phase settings to measure phase error and to apply threshold in reporting mode 1 and 2.	Bit	Definition	b0	Phase shifter phase1 monitor enable bit	b1	Phase shifter phase2 monitor enable bit	b2	Phase shifter phase3 monitor enable bit	b3	Phase shifter phase4 monitor enable bit	b7:4	RESERVED
Bit	Definition													
b0	Phase shifter phase1 monitor enable bit													
b1	Phase shifter phase2 monitor enable bit													
b2	Phase shifter phase3 monitor enable bit													
b3	Phase shifter phase4 monitor enable bit													
b7:4	RESERVED													

Continued on next page

Table 8.15 – continued from previous page

RX_EN	1	This field indicates the RX channels that should be enabled for TX to RX loopback measurement. Setting the corresponding bit to 1 enables that channel for measurement and the average measured value is reported out. Bit number RX Channel b0 RX0 b1 RX1 b2 RX2 b3 RX3
MON_CHIRP_SLOPE	1	Frequency slope for each monitoring chirp is encoded in 1 bytes (8 bit signed number) For 77GHz Devices (76GHz to 81Ghz): 1 LSB = $3.6e9 \times 900 / 2^{26}$ Hz ≈ 48.279 kHz/μs Valid range: -128 to +127 (Max 6.13 MHz/μs) For 60GHz Devices (57GHz to 64Ghz): 1 LSB = $2.7e9 \times 900 / 2^{26}$ Hz ≈ 36.21 kHz/μs Valid range: -128 to +127 (Max 4.63 MHz/μs) NOTE: Monitoring Chirp Slope can be programmed based on the emission specifications. The device transmits on air during the execution of these monitors. The host can control the monitoring emission power spectral density (dB-m/Hz) by programming this slope parameter. Each monitoring chirp is about 45us in duration. Therefore the overall RF sweep bandwidth for the monitoring chirp is given by Monitoring Chirp Slope* 45 us. Normally, low values of Monitoring Chirp Slope are recommended, as with non-zero slope in FMCW radar, any actual target reflections can potentially be interpreted as noise power and/or corrupt the loopback signal based gain/phase measurement. The user has to ensure that the RF bandwidth selected for monitoring chirp based on slope and start frequency is within FREQ_LIMIT set in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB API.
RESERVED	1	RESERVED

Continued on next page

Table 8.15 – continued from previous page

PH_SHIFTER_ INC_VAL1	1	Phase shifter monitoring increment value for phase1, the monitoring phase will be incremented by this value in every FTTI interval. In API based trigger MONITORING_MODE, it is important to trigger PS monitor every FTTI to make sure this logic works seamlessly. Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift increment value 1 1 LSB = 5.625°
PH_SHIFTER_ INC_VAL2	1	Phase shifter monitoring increment value for phase2, the monitoring phase will be incremented by this value in every FTTI interval. In API based trigger MONITORING_MODE, it is important to trigger PS monitor every FTTI to make sure this logic works seamlessly. Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift increment value 2 1 LSB = 5.625°
PH_SHIFTER_ INC_VAL3	1	Phase shifter monitoring increment value for phase3, the monitoring phase will be incremented by this value in every FTTI interval. In API based trigger MONITORING_MODE, it is important to trigger PS monitor every FTTI to make sure this logic works seamlessly. Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift increment value 3 1 LSB = 5.625°
PH_SHIFTER_ INC_VAL4	1	Phase shifter monitoring increment value for phase4, the monitoring phase will be incremented by this value in every FTTI interval. In API based trigger MONITORING_MODE, it is important to trigger PS monitor every FTTI to make sure this logic works seamlessly. Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift increment value 4 1 LSB = 5.625°
PH_SHIFTER_ MON1	1	TX 0 Phase shifter phase1 monitor value Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift monitor value 1 1 LSB = 5.625°

Continued on next page

Table 8.15 – continued from previous page

PH_SHIFTER_MON2	1	TX 0 Phase shifter phase2 monitor value Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift monitor value 2 1 LSB = 5.625°
PH_SHIFTER_MON3	1	TX 0 Phase shifter phase3 monitor value Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift monitor value 3 1 LSB = 5.625°
PH_SHIFTER_MON4	1	TX 0 Phase shifter phase4 monitor value Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift monitor value 4 1 LSB = 5.625°
TX_PHASE_ERROR_THRESH	2	The threshold for deviation of the TX output phase difference between the measured phase values and configured phases for each enabled phase settings. The max error is compared against the threshold given here. The measured phase is part of the monitoring report message (Error bit is set if the max deviation is higher than this threshold, with the units of both quantities being the same). 1 LSB = $360^\circ / 2^{16}$. Valid range: corresponding to 0° to 359.9°.
TX_AMPLITUDE_ERROR_THRESH	2	The threshold for deviation of the TX output amplitude difference between all enabled phase settings. The max error is compared against the threshold given here. The measured output amplitude is part of the monitoring report message (Error bit is set if the max deviation is higher than this threshold, with the units of both quantities being the same). 1 LSB = 0.1 dB Valid range: 0 to 65535 (0 to 6553dB)
RESERVED	8	0x0000

8.9.2 Sub block 0x01CF – AWR_MONITOR_TX1_PHASE_SHIFTER_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to TX1 TX loop back based phase shifter monitoring.

The report is sent as an async event [AWR_MONITOR_TX1_PHASE_SHIFTER_REPORT_AE_SB](#).

Table 8.16: AWR_MONITOR_TX1_PHASE_SHIFTER_CONF_SB contents

Field Name	Number of bytes	Description												
SBLKID	2	Value = 0x01CF												
SBLKLEN	2	Value = 32												
PROFILE_INDX	1	This field indicates the Profile Index for which this monitoring configuration applies.												
REPORTING_MODE	1	<table><tr><th>Value</th><th>Definition</th></tr><tr><td>0</td><td>Report is sent every monitoring period without threshold check</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	Report is sent every monitoring period without threshold check	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check				
Value	Definition													
0	Report is sent every monitoring period without threshold check													
1	Report is send only upon a failure (after checking for thresholds)													
2	Report is sent every monitoring period with threshold check													
RESERVED	2	RESERVED												
PH_SHIFTER_MON_CFG	1	<table><tr><th>Bit</th><th>Definition</th></tr><tr><td>b0</td><td>Phase shifter phase1 monitor enable bit</td></tr><tr><td>b1</td><td>Phase shifter phase2 monitor enable bit</td></tr><tr><td>b2</td><td>Phase shifter phase3 monitor enable bit</td></tr><tr><td>b3</td><td>Phase shifter phase4 monitor enable bit</td></tr><tr><td>b7:4</td><td>RESERVED</td></tr></table> Enable at least two phase settings to measure phase error and to apply threshold in reporting mode 1 and 2.	Bit	Definition	b0	Phase shifter phase1 monitor enable bit	b1	Phase shifter phase2 monitor enable bit	b2	Phase shifter phase3 monitor enable bit	b3	Phase shifter phase4 monitor enable bit	b7:4	RESERVED
Bit	Definition													
b0	Phase shifter phase1 monitor enable bit													
b1	Phase shifter phase2 monitor enable bit													
b2	Phase shifter phase3 monitor enable bit													
b3	Phase shifter phase4 monitor enable bit													
b7:4	RESERVED													
RX_EN	1	This field indicates the RX channels that should be enabled for TX to RX loopback measurement. Setting the corresponding bit to 1 enables that channel for measurement and the average measured value is reported out. <table><tr><th>Bit number</th><th>RX Channel</th></tr><tr><td>b0</td><td>RX0</td></tr><tr><td>b1</td><td>RX1</td></tr><tr><td>b2</td><td>RX2</td></tr><tr><td>b3</td><td>RX3</td></tr></table>	Bit number	RX Channel	b0	RX0	b1	RX1	b2	RX2	b3	RX3		
Bit number	RX Channel													
b0	RX0													
b1	RX1													
b2	RX2													
b3	RX3													

Continued on next page

Table 8.16 – continued from previous page

MON_CHIRP_ SLOPE	1	Frequency slope for each monitoring chirp is encoded in 1 bytes (8 bit signed number) For 77GHz Devices (76GHz to 81Ghz): $1 \text{ LSB} = 3.6e9 \times 900 / 2^{26} \text{ Hz} \approx 48.279 \text{ kHz}/\mu\text{s}$ Valid range: -128 to +127 (Max 6.13 MHz/μs) For 60GHz Devices (57GHz to 64Ghz): $1 \text{ LSB} = 2.7e9 \times 900 / 2^{26} \text{ Hz} \approx 36.21 \text{ kHz}/\mu\text{s}$ Valid range: -128 to +127 (Max 4.63 MHz/μs) NOTE: Monitoring Chirp Slope can be programmed based on the emission specifications. The device transmits on air during the execution of these monitors. The host can control the monitoring emission power spectral density (dB-m/Hz) by programming this slope parameter. Each monitoring chirp is about 45us in duration. Therefore the overall RF sweep bandwidth for the monitoring chirp is given by Monitoring Chirp Slope* 45 us. Normally, low values of Monitoring Chirp Slope are recommended, as with non-zero slope in FMCW radar, any actual target reflections can potentially be interpreted as noise power and/or corrupt the loopback signal based gain/phase measurement. The user has to ensure that the RF bandwidth selected for monitoring chirp based on slope and start frequency is within FREQ_LIMIT set in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB API.
RESERVED	1	RESERVED
PH_SHIFTER_ INC_VAL1	1	Phase shifter monitoring increment value for phase1, the monitoring phase will be incremented by this value in every FTTI interval. Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift increment value 1 1 LSB = 5.625°
PH_SHIFTER_ INC_VAL2	1	Phase shifter monitoring increment value for phase2, the monitoring phase will be incremented by this value in every FTTI interval. Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift increment value 2 1 LSB = 5.625°

Continued on next page

Table 8.16 – continued from previous page

PH_SHIFTER_ INC_VAL3	1	Phase shifter monitoring increment value for phase3, the monitoring phase will be incremented by this value in every FTTI interval. Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift increment value 3 1 LSB = 5.625°
PH_SHIFTER_ INC_VAL4	1	Phase shifter monitoring increment value for phase4, the monitoring phase will be incremented by this value in every FTTI interval. Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift increment value 4 1 LSB = 5.625°
PH_SHIFTER_ MON1	1	TX 1 Phase shifter phase1 monitor value Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift monitor value 1 1 LSB = 5.625°
PH_SHIFTER_ MON2	1	TX 1 Phase shifter phase2 monitor value Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift monitor value 2 1 LSB = 5.625°
PH_SHIFTER_ MON3	1	TX 1 Phase shifter phase3 monitor value Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift monitor value 3 1 LSB = 5.625°
PH_SHIFTER_ MON4	1	TX 1 Phase shifter phase4 monitor value Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift monitor value 4 1 LSB = 5.625°

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Table 8.16 – continued from previous page

TX_PHASE_ERROR_THRESH	2	The threshold for deviation of the TX output phase difference between the measured phase values and configured phases for each enabled phase settings. The max error is compared against the threshold given here. The measured phase is part of the monitoring report message (Error bit is set if the max deviation is higher than this threshold, with the units of both quantities being the same). 1 LSB = $360^\circ / 2^{16}$. Valid range: corresponding to 0° to 359.9° .
TX_AMPLITUDE_ERROR_THRESH	2	The threshold for deviation of the TX output amplitude difference between all enabled phase settings. The max error is compared against the threshold given here. The measured output amplitude is part of the monitoring report message (Error bit is set if the max deviation is higher than this threshold, with the units of both quantities being the same). 1 LSB = 0.1 dB Valid range: 0 to 65535 (0 to 6553dB)
RESERVED	8	0x0000

8.9.3 Sub block 0x01D0 – AWR_MONITOR_TX2_PHASE_SHIFTER_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to TX2 TX loop back based phase shifter monitoring.

The report is sent as an async event [AWR_MONITOR_TX2_PHASE_SHIFTER_REPORT_AE_SB](#).

Table 8.17: AWR_MONITOR_TX2_PHASE_SHIFTER_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x01D0
SBLKLEN	2	Value = 32
PROFILE_IDX	1	This field indicates the Profile Index for which this monitoring configuration applies.

Continued on next page

Table 8.17 – continued from previous page

REPORTING_ MODE	1	Value	Definition
		0	Report is sent every monitoring period without threshold check
		1	Report is send only upon a failure (after checking for thresholds)
		2	Report is sent every monitoring period with threshold check
RESERVED	2	RESERVED	
PH_SHIFTER_ MON_CFG	1	Bit	Definition
		b0	Phase shifter phase1 monitor enable bit
		b1	Phase shifter phase2 monitor enable bit
		b2	Phase shifter phase3 monitor enable bit
		b3	Phase shifter phase4 monitor enable bit
		b7:4	RESERVED
		Enable at least two phase settings to measure phase error and to apply threshold in reporting mode 1 and 2.	
RX_EN	1	This field indicates the RX channels that should be enabled for TX to RX loopback measurement. Setting the corresponding bit to 1 enables that channel for measurement and the average measured value is reported out.	
		Bit number	RX Channel
		b0	RX0
		b1	RX1
		b2	RX2
		b3	RX3

Continued on next page

Table 8.17 – continued from previous page

MON_CHIRP_ SLOPE	1	Frequency slope for each monitoring chirp is encoded in 1 bytes (8 bit signed number) For 77GHz Devices (76GHz to 81Ghz): $1 \text{ LSB} = 3.6e9 \times 900 / 2^{26} \text{ Hz} \approx 48.279 \text{ kHz}/\mu\text{s}$ Valid range: -128 to +127 (Max 6.13 MHz/μs) For 60GHz Devices (57GHz to 64Ghz): $1 \text{ LSB} = 2.7e9 \times 900 / 2^{26} \text{ Hz} \approx 36.21 \text{ kHz}/\mu\text{s}$ Valid range: -128 to +127 (Max 4.63 MHz/μs) NOTE: Monitoring Chirp Slope can be programmed based on the emission specifications. The device transmits on air during the execution of these monitors. The host can control the monitoring emission power spectral density (dB-m/Hz) by programming this slope parameter. Each monitoring chirp is about 45us in duration. Therefore the overall RF sweep bandwidth for the monitoring chirp is given by Monitoring Chirp Slope* 45 us. Normally, low values of Monitoring Chirp Slope are recommended, as with non-zero slope in FMCW radar, any actual target reflections can potentially be interpreted as noise power and/or corrupt the loopback signal based gain/phase measurement. The user has to ensure that the RF bandwidth selected for monitoring chirp based on slope and start frequency is within FREQ_LIMIT set in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB API.
RESERVED	1	RESERVED
PH_SHIFTER_ INC_VAL1	1	Phase shifter monitoring increment value for phase1, the monitoring phase will be incremented by this value in every FTTI interval. Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift increment value 1 1 LSB = 5.625°
PH_SHIFTER_ INC_VAL2	1	Phase shifter monitoring increment value for phase2, the monitoring phase will be incremented by this value in every FTTI interval. Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift increment value 2 1 LSB = 5.625°

Continued on next page

Table 8.17 – continued from previous page

PH_SHIFTER_ INC_VAL3	1	Phase shifter monitoring increment value for phase3, the monitoring phase will be incremented by this value in every FTTI interval. Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift increment value 3 1 LSB = 5.625°
PH_SHIFTER_ INC_VAL4	1	Phase shifter monitoring increment value for phase4, the monitoring phase will be incremented by this value in every FTTI interval. Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift increment value 4 1 LSB = 5.625°
PH_SHIFTER_ MON1	1	TX 2 Phase shifter phase1 monitor value Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift monitor value 1 1 LSB = 5.625°
PH_SHIFTER_ MON2	1	TX 2 Phase shifter phase2 monitor value Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift monitor value 2 1 LSB = 5.625°
PH_SHIFTER_ MON3	1	TX 2 Phase shifter phase3 monitor value Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift monitor value 3 1 LSB = 5.625°
PH_SHIFTER_ MON4	1	TX 2 Phase shifter phase4 monitor value Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift monitor value 4 1 LSB = 5.625°

Continued on next page

Table 8.17 – continued from previous page

TX_PHASE_ERROR_THRESH	2	The threshold for deviation of the TX output phase difference between the measured phase values and configured phases for each enabled phase settings. The max error is compared against the threshold given here. The measured phase is part of the monitoring report message (Error bit is set if the max deviation is higher than this threshold, with the units of both quantities being the same). 1 LSB = $360^\circ / 2^{16}$. Valid range: corresponding to 0° to 359.9° .
TX_AMPLITUDE_ERROR_THRESH	2	The threshold for deviation of the TX output amplitude difference between all enabled phase settings. The max error is compared against the threshold given here. The measured output amplitude is part of the monitoring report message (Error bit is set if the max deviation is higher than this threshold, with the units of both quantities being the same). 1 LSB = 0.1 dB Valid range: 0 to 65535 (0 to 6553dB)
RESERVED	8	0x0000

8.10 Synthesizer Frequency Monitoring

This section contains API SBs that configure the monitors of synthesizer chirp frequency, and report the soft results from the monitor. The corresponding monitor is named SYNTH_FREQ_MONITOR.

8.10.1 Sub block 0x01D1 – AWR_MONITOR_SYNTHESIZER_FREQUENCY_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to synthesizer frequency monitoring during non functional chirps (non-live). In xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP, the live monitor can be used for monitoring functional chirps. In all other devices, monitoring of live functional chirps can be used only for debug purposes. The report is sent as an async event [AWR_MONITOR_SYNTH_FREQUENCY_REPORT_AE_SB](#) for live monitor and [AWR_MONITOR_SYNTHESIZER_FREQUENCY_NONLIVE_REPORT_AE_SB](#) for nonlive monitor.

NOTE:

The synth non-live mode monitor internally generates a test chirp based on the profile associated with it. In order to limit its execution time, if the profile's ramp time exceeds 60us, the test chirp's ramp time is limited to 60us and the chirp slope is scaled to cover the profile's intended RF bandwidth.

Table 8.18: AWR_MONITOR_SYNTHESIZER_FREQUENCY_CONF_SB
contents

Field Name	Number of bytes	Description								
SBLKID	2	Value = 0x01D1								
SBLKLEN	2	Value = 16								
PROFILE_INDX	1	This field indicates the Profile Index for which this monitoring configuration applies. NOTE: Either PROFILE_INDX or PROFILE_BIT_MASK_LIVE_MODE, only one shall be enabled at a time.								
REPORTING_MODE	1	<table><tr><th>Value</th><th>Definition</th></tr><tr><td>0</td><td>Report is sent every monitoring period without threshold check</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	Report is sent every monitoring period without threshold check	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check
Value	Definition									
0	Report is sent every monitoring period without threshold check									
1	Report is send only upon a failure (after checking for thresholds)									
2	Report is sent every monitoring period with threshold check									
FREQ_ERROR_THRESH	2	During the chirp, the error of the measured instantaneous chirp frequency w.r.t. the desired value is continuously compared against this threshold. The comparison result is part of the monitoring report message (Error bit is set if the measurement is above this threshold, ever during the previous monitoring period). 1 LSB = 10 kHz Valid range: 0 to 65535 (0 to 655MHz)								
MONITOR_START_TIME	1	This field determines when the monitoring starts in each chirp relative to the start of the ramp. 1 LSB = 0.2 μ s, signed number Valid range: -25 to 25 μ s Recommended value: 6 μ s or above								

Continued on next page

Table 8.18 – continued from previous page

MONITOR_CONFIG_MODE	1	This field configures whether this monitor should be done for functional active chirps (mode 0) or non live monitor chirps. In case of non live monitor, the configuration needs to be sent twice for two VCOs (use mode 1 and 2). Value Definition 0 LIVE_CONFIG (Debug Mode), The profile configuration for live mode is picked from this API, supported only in master/single-chip mode. 1 VCO1_CONFIG, The profile configuration for Non-live mode is picked from this API for VCO1 monitor profile, supported in all modes (master, slave and single-chip). 2 VCO2_CONFIG, The profile configuration for Non-live mode is picked from this API for VCO2 monitor profile, supported in all modes (master, slave and single-chip). Note: This feature is supported only on AWR2243/xWR6243 device.
VCO_MON_EN	1	This bit mask can be used to enable/disable the monitoring of non-live VCO profiles, this helps to control monitoring of only single VCO if needed. This setting should be same in both VCO settings. bits Definition b0 Enable VCO1 non-live monitor b1 Enable VCO2 non-live monitor b31:2 RESERVED
PROFILE_BIT_MASK_LIVE_MODE	1	This bit mask can be used to enable/disable monitoring for multiple profiles in live mode. NOTE: Either PROFILE_INDX or PROFILE_BIT_MASK_LIVE_MODE, only one shall be enabled at a time.
RESERVED	4	0x00000000

NOTE1:	(Live mode) It is recommended to re-issue this configuration API each time before enabling this monitor and frame trigger. The right sequence is as below: 1. Issue Synth frequency monitor configuration API. 2. Enable Synth frequency monitor. 3. Frame start. 4. Frame stop. 5. Frame start. (Optional in case of multiple frames) 6. Frame stop. (Optional in case of multiple frames) 7. Disable Synth frequency monitor (in case disabled for some reason) 8. Issue Synth frequency monitor configuration API. 9. Enable Synth frequency monitor. 10. Frame start.
NOTE2:	In non live mode, this API can be issued twice with MONITOR_CONFIG_MODE value set to 1 and 2 respectively for two different VCOs configured in two different profiles. The consolidated report for two VCOs in non-live mode is sent in a separate AE AWR_MONITOR_SYNTHESIZER_FREQUENCY_NONLIVE_REPORT_AE_SB
NOTE3:	In non live mode, the reporting mode and VCO_MON_EN for two VCO configurations should be same.

8.11 External Analog Signals Monitor

This section contains API SBs that configure the monitors of external analog signals which are input to the device through pins ANALOGTEST1-4, ANAMUX and VSENSE (also called ADC1-6) and report the soft results from the monitor. The corresponding monitors are collectively named EXTERNAL_ANALOG_SIGNALS_MONITOR. These monitors observe various analog signals input on the pins ADC1-6 using a GPADC and compare them against internally fixed thresholds.

8.11.1 Sub block 0x01D2 – AWR_MONITORING_EXTERNAL_ANALOG_SIGNALS_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to external DC signals monitoring. The report is sent as an async event [AWR_MONITOR_EXTERNAL_ANALOG_SIGNALSREPORT_AE_SB](#).

NOTE:	This monitor is not supported in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices.
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Table [8.19](#) describes the content of this sub block.

Table 8.19: AWR_MONITOR_EXTERNAL_ANALOG_SIGNALS_CONF_SB
contents

Field Name	Number of bytes	Description																								
SBLKID	2	Value = 0x01D2																								
SBLKLEN	2	Value = 36																								
REPORTING_MODE	1	<table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>Report is sent every monitoring period without threshold check</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	Report is sent every monitoring period without threshold check	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check																
Value	Definition																									
0	Report is sent every monitoring period without threshold check																									
1	Report is send only upon a failure (after checking for thresholds)																									
2	Report is sent every monitoring period with threshold check																									
RESERVED	1	0x00																								
SIGNAL_INPUT_ENABLES	1	This field indicates the sets of externally fed DC signals which are to be monitored using GPADC. When each bit in this field is set, the corresponding signal is monitored. The monitored signals are compared against programmed limits. The comparison result is part of the monitoring report message. <table><tr><td>Bit</td><td>Location</td><td>SIGNAL</td></tr><tr><td>b0</td><td></td><td>ANALOGTEST1</td></tr><tr><td>b1</td><td></td><td>ANALOGTEST2</td></tr><tr><td>b2</td><td></td><td>ANALOGTEST3</td></tr><tr><td>b3</td><td></td><td>ANALOGTEST4</td></tr><tr><td>b4</td><td></td><td>ANAMUX</td></tr><tr><td>b5</td><td></td><td>VSENSE</td></tr><tr><td>Others</td><td></td><td>RESERVED</td></tr></table>	Bit	Location	SIGNAL	b0		ANALOGTEST1	b1		ANALOGTEST2	b2		ANALOGTEST3	b3		ANALOGTEST4	b4		ANAMUX	b5		VSENSE	Others		RESERVED
Bit	Location	SIGNAL																								
b0		ANALOGTEST1																								
b1		ANALOGTEST2																								
b2		ANALOGTEST3																								
b3		ANALOGTEST4																								
b4		ANAMUX																								
b5		VSENSE																								
Others		RESERVED																								

Continued on next page

Table 8.19 – continued from previous page

SIGNAL_BUFFER_ENABLED	1	This field indicates the sets of externally fed DC signals which are to be buffered before being fed to the GPADC. When each bit in this field is set, the corresponding signal is buffered before the GPADC. The monitored signals are compared against programmed limits. The comparison result is part of the monitoring report message. Bit SIGNAL b0 ANALOGTEST1 b1 ANALOGTEST2 b2 ANALOGTEST3 b3 ANALOGTEST4 b4 ANAMUX Others RESERVED
SIGNAL_SETTLING_TIME	6	After connecting an external signal to the GPADC, the amount of time to wait for it to settle before taking GPADC samples is programmed in this field. For each signal, after that settling time, GPADC measurements take place for 6.4 μs (averaging 4 samples of the GPADC output). The byte locations of the settling times for each signal are tabulated here: Byte SIGNAL Loca- tion 0 ANALOGTEST1 1 ANALOGTEST2 2 ANALOGTEST3 3 ANALOGTEST4 4 ANAMUX 5 VSENSE 1 LSB = 0.8 μs Valid range: 0 to 12 μs Valid programming condition: all the signals that are enabled should take a total of < 100 μs, including the programmed settling times and a fixed 6.4 μs of measurement time per enabled signal.

Continued on next page

Table 8.19 – continued from previous page

SIGNAL_ THRESH	12	The external DC signals measured on GPADC are compared against these minimum and maximum thresholds. The comparison result is part of the monitoring report message (Error bit is set if any measurement is outside this (minimum, maximum) range). <table><tr><td>Byte Location</td><td>Threshold SIGNAL</td></tr><tr><td>0</td><td>Minimum ANALOGTEST1</td></tr><tr><td>1</td><td>Minimum ANALOGTEST2</td></tr><tr><td>2</td><td>Minimum ANALOGTEST3</td></tr><tr><td>3</td><td>Minimum ANALOGTEST4</td></tr><tr><td>4</td><td>Minimum ANAMUX</td></tr><tr><td>5</td><td>Minimum VSENSE</td></tr><tr><td>6</td><td>Maximum ANALOGTEST1</td></tr><tr><td>7</td><td>Maximum ANALOGTEST2</td></tr><tr><td>8</td><td>Maximum ANALOGTEST3</td></tr><tr><td>9</td><td>Maximum ANALOGTEST4</td></tr><tr><td>10</td><td>Maximum ANAMUX</td></tr><tr><td>11</td><td>Maximum VSENSE</td></tr></table> 1 LSB = 1.8V/256 Valid range: 0 to 255	Byte Location	Threshold SIGNAL	0	Minimum ANALOGTEST1	1	Minimum ANALOGTEST2	2	Minimum ANALOGTEST3	3	Minimum ANALOGTEST4	4	Minimum ANAMUX	5	Minimum VSENSE	6	Maximum ANALOGTEST1	7	Maximum ANALOGTEST2	8	Maximum ANALOGTEST3	9	Maximum ANALOGTEST4	10	Maximum ANAMUX	11	Maximum VSENSE
Byte Location	Threshold SIGNAL																											
0	Minimum ANALOGTEST1																											
1	Minimum ANALOGTEST2																											
2	Minimum ANALOGTEST3																											
3	Minimum ANALOGTEST4																											
4	Minimum ANAMUX																											
5	Minimum VSENSE																											
6	Maximum ANALOGTEST1																											
7	Maximum ANALOGTEST2																											
8	Maximum ANALOGTEST3																											
9	Maximum ANALOGTEST4																											
10	Maximum ANAMUX																											
11	Maximum VSENSE																											
RESERVED	2	0x0000																										
RESERVED	4	0x00000000																										
RESERVED	4	0x00000000																										

8.12 Internal Analog Signals Monitor

This section contains API SBs that configure the monitors of internal analog signals in the RF analog modules and report the soft results from the monitor. The corresponding monitors are collectively named INTERNAL_ANALOG_SIGNALS_MONITOR. These monitors observe various analog nodes in the RF and analog modules using a GPADC and compare them against internally fixed thresholds.

The configuration API SBs are organized to address various analog circuits as follows:

1. TX0 Internal Analog Signals Monitoring
 - a. This monitor is called INTERNAL_TX0_SIGNALS_MONITOR.
 - b. Signal sets that are monitored: (SUPPLY_TX, PWRDET_TX)
2. TX1 Internal Analog Signals Monitoring

- a. This monitor is called INTERNAL_TX1_SIGNALS_MONITOR
 - b. Signal sets that are monitored: (SUPPLY_TX, PWRDET_TX)
3. TX2 Internal Analog Signals Monitoring
 - a. This monitor is called INTERNAL_TX2_SIGNALS_MONITOR
 - b. Signal sets that are monitored: (SUPPLY_TX, PWRDET_TX)
4. RX Internal Analog Signals Monitoring
 - a. This monitor is called INTERNAL_RX_SIGNALS_MONITOR
 - b. Signal sets that are monitored: (SUPPLY_RX, PWRDET_RX, DCBIAS_RX)
5. PM CLK LO Internal Analog Signals Monitoring
 - a. This monitor is called INTERNAL_PMCLKLO_SIGNALS_MONITOR
 - b. Signal sets that are monitored: (SUPPLY_PMCLKLO, PWRDET_PMCLKLO, DCBIAS_PMCLKLO)
6. GPADC Internal Analog Signals Monitoring
 - a. This monitor is called INTERNAL_GPADC_SIGNALS_MONITOR
 - b. Signal sets that are monitored: (GPADC_REF1, GPADC_REF2)

The results are reported in the corresponding REPORT API SBs in this section.

8.12.1 Sub block 0x01D3 – AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to TX0 Internal Analog Signals monitoring including Tx Phase shifter DAC monitor. The report is sent as an async event [AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB](#).

Table 8.20: AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x01D3
SBLKLEN	2	Value = 12
PROFILE_INDEX	1	The RF analog settings corresponding to this profile are used for monitoring the enabled signals, using test chirps (static frequency, at the center of the profile's RF frequency band).

Continued on next page

Table 8.20 – continued from previous page

REPORTING_ MODE	1	Value	Definition
		0	RESERVED
		1	Report is send only upon a failure (after checking for thresholds)
		2	Report is sent every monitoring period with threshold check
TX_PS_DAC_ MON_THRESH	2	The TX phase shifter DAC monitor delta threshold when TX_PS_DAC_MON is Enabled (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) Unit: 1 LSB = 1.8V/1024 Value 0: TX_PS_DAC_MON is disabled Valid Range: 1 to 1023	
RESERVED	4	0x00000000	

8.12.2 Sub block 0x01D4 – AWR_MONITOR_TX1_INTERNAL_ANALOG_ SIGNALS_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to TX1 Internal Analog Signals monitoring including Tx Phase shifter DAC monitor. The report is sent as an async event [AWR_MONITOR_TX1_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB](#).

**Table 8.21: AWR_MONITOR_TX1_INTERNAL_ANALOG_SIGNALS_CONF_
SB contents**

Field Name	Number of bytes	Description								
SBLKID	2	Value = 0x01D4								
SBLKLEN	2	Value = 12								
PROFILE_INDEX	1	The RF analog settings corresponding to this profile are used for monitoring the enabled signals, using test chirps (static frequency, at the center of the profile's RF frequency band).								
REPORTING_MODE	1	<table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>RESERVED</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	RESERVED	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check
Value	Definition									
0	RESERVED									
1	Report is send only upon a failure (after checking for thresholds)									
2	Report is sent every monitoring period with threshold check									

Continued on next page

Table 8.21 – continued from previous page

TX_PS_DAC_MON_THRESH	2	The TX phase shifter DAC monitor delta threshold when TX_PS_DAC_MON is Enabled (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) Unit: 1 LSB = 1.8V/1024 Value 0: TX_PS_DAC_MON is disabled Valid Range: 1 to 1023
RESERVED	4	0x00000000

8.12.3 Sub block 0x01D5 – AWR_MONITOR_TX2_INTERNAL_ANALOG_SIGNALS_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to TX2 Internal Analog Signals monitoring including Tx Phase shifter DAC monitor. The report is sent as an async event [AWR_MONITOR_TX2_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB](#).

Table 8.22: AWR_MONITOR_TX2_INTERNAL_ANALOG_SIGNALS_CONF_SB contents

Field Name	Number of bytes	Description								
SBLKID	2	Value = 0x01D5								
SBLKLEN	2	Value = 12								
PROFILE_INDEX	1	The RF analog settings corresponding to this profile are used for monitoring the enabled signals, using test chirps (static frequency, at the center of the profile's RF frequency band).								
REPORTING_MODE	1	<table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>RESERVED</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	RESERVED	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check
Value	Definition									
0	RESERVED									
1	Report is send only upon a failure (after checking for thresholds)									
2	Report is sent every monitoring period with threshold check									
TX_PS_DAC_MON_THRESH	2	The TX phase shifter DAC monitor delta threshold when TX_PS_DAC_MON is Enabled (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) Unit: 1 LSB = 1.8V/1024 Value 0: TX_PS_DAC_MON is disabled Valid Range: 1 to 1023								
RESERVED	4	0x00000000								

8.12.4 Sub block 0x01D6 – AWR_MONITOR_RX_INTERNAL_ANALOG_SIGNALS_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to RX Internal Analog Signals monitoring. The report is sent as an async event [AWR_MONITOR_RX_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB](#).

Table 8.23: AWR_MONITOR_RX_INTERNAL_ANALOG_SIGNALS_CONF_SB contents

Field Name	Number of bytes	Description								
SBLKID	2	Value = 0x01D6								
SBLKLEN	2	Value = 12								
PROFILE_INDEX	1	The RF analog settings corresponding to this profile are used for monitoring the enabled signals, using test chirps (static frequency, at the center of the profile's RF frequency band).								
REPORTING_MODE	1	<table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>RESERVED</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	RESERVED	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check
Value	Definition									
0	RESERVED									
1	Report is send only upon a failure (after checking for thresholds)									
2	Report is sent every monitoring period with threshold check									
RESERVED	2	0x0000								
RESERVED	4	0x00000000								

8.12.5 Sub block 0x01D7 – AWR_MONITOR_PMCLKLO_INTERNAL_ANALOG_SIGNALS_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to Power Management, Clock generation and LO distribution circuits' Internal Analog Signals monitoring.

The 20GHz SYNC IN/OUT monitor supported only in cascade master and slave modes. If 20G SYNC monitor is enabled in slaves, It is recommended to enable this monitor in master as well, slave devices dependent on master 20GHz SYNC settings to operate properly.

The report is sent as an async event [AWR_MONITOR_PMCLKLO_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB](#).

Table 8.24: AWR_MONITOR_PMCLKLO_INTERNAL_ANALOG_SIGNALS__CONF_SB contents

Field Name	Number of bytes	Description										
SBLKID	2	Value = 0x01D7										
SBLKLEN	2	Value = 12										
PROFILE_INDEX	1	The RF analog settings corresponding to this profile are used for monitoring the enabled signals, using test chirps (static frequency, at the center of the profile's RF frequency band). NOTE: The 20GHz SYNC monitor is always done at anywhere between ramp end frequency of last chirp in first sub-frame of a frame and 77GHz RF frequency. This measurement is done in synchronization with all cascade devices, the master device LO is ON when measurement is done on slave devices.										
REPORTING_MODE	1	<table><tr><th>Value</th><th>Definition</th></tr><tr><td>0</td><td>RESERVED</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	RESERVED	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check		
Value	Definition											
0	RESERVED											
1	Report is send only upon a failure (after checking for thresholds)											
2	Report is sent every monitoring period with threshold check											
SYNC_20G_SIG_SEL	1	This field is relevant only in cascade configuration and not applicable in single chip case <table><tr><th>Value</th><th>Definition</th></tr><tr><td>0x00</td><td>20GHz SYNC monitoring disabled</td></tr><tr><td>0x01</td><td>FMCW_SYNC_IN monitoring enabled</td></tr><tr><td>0x02</td><td>FMCW_SYNC_OUT monitoring enabled</td></tr><tr><td>0x03</td><td>FMCW_CLK_OUT monitoring enabled</td></tr></table> NOTE: The 20GHz SYNC monitor is always done at anywhere between ramp end frequency of last chirp in first sub-frame of a frame and 77GHz RF frequency. This measurement is done in synchronization with all cascade devices, the master device LO is ON when measurement is done on slave devices. If 20G SYNC monitor is enabled in slaves, It is recommended to enable this monitor in master as well, slave devices dependent on master 20GHz SYNC settings to operate properly.	Value	Definition	0x00	20GHz SYNC monitoring disabled	0x01	FMCW_SYNC_IN monitoring enabled	0x02	FMCW_SYNC_OUT monitoring enabled	0x03	FMCW_CLK_OUT monitoring enabled
Value	Definition											
0x00	20GHz SYNC monitoring disabled											
0x01	FMCW_SYNC_IN monitoring enabled											
0x02	FMCW_SYNC_OUT monitoring enabled											
0x03	FMCW_CLK_OUT monitoring enabled											
SYNC_20G_MIN_THRESH	1	The minimum threshold value of monitoring, signed number Unit: 1 LSB = 1 dBm Valid Range: -63 to +63 dBm										

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Table 8.24 – continued from previous page

SYNC_20G_ MAX_THRESH	1	The maximum threshold value of monitoring, signed number Unit: 1 LSB = 1 dBm Valid Range: -63 to +63 dBm
RESERVED	3	0x000000

8.12.6 Sub block 0x01D8 – AWR_MONITOR_GPADC_INTERNAL_ANALOG_SIGNALS_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to GPADC Internal Analog Signals monitoring. During this monitor, only the relevant circuits are ensured to be ON. The monitored signals are compared against internally chosen valid limits. The comparison result is part of the consolidated monitoring report message (Error bit for any signal set is set to 1 if any measurement in that signal set is beyond valid limits). The report is sent as an async event [AWR_MONITOR_GPADC_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB](#).

Table 8.25: AWR_MONITOR_GPADC_INTERNAL_ANALOG_SIGNALS_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x01D8
SBLKLEN	2	Value = 12
REPORTING_MODE	1	Value Definition 0 RESERVED 1 Report is send only upon a failure (after checking for thresholds) 2 Report is sent every monitoring period with threshold check
RESERVED	3	0x000000
RESERVED	4	0x00000000

8.13 PLL Control Voltage Monitor

This section contains API SBs that configure the monitors of APLL and Synthesizer VCO control voltages and report the soft results from the monitor. The corresponding monitors are collectively named PLL_CONTROL_VOLTAGE_MONITOR. These monitors observe the VCO control voltages under various conditions using the GPADC and compare them against internally fixed

thresholds. The transmitters are kept in OFF state during these measurements to avoid external emission.

8.13.1 Sub block 0x01D9 – AWR_MONITOR_PLL_CONTROL_VOLTAGE_SIGNALS_CONF_SB

This is a monitoring configuration API which the host sends to the AWR device, containing information related to APLL and Synthesizer's control voltage signals monitoring. The report is sent as an async event [AWR_MONITOR_PLL_CONTROL_VOLTAGE_REPORT_AE_SB](#).

Table 8.26: AWR_MONITOR_PLL_CONTROL_VOLTAGE_CONF_SB contents

Field Name	Number of bytes	Description								
SBLKID	2	Value = 0x01D9								
SBLKLEN	2	Value = 12								
REPORTING_MODE	1	<table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>Reserved</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	Reserved	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check
Value	Definition									
0	Reserved									
1	Report is send only upon a failure (after checking for thresholds)									
2	Report is sent every monitoring period with threshold check									
RESERVED	1	0x00								

Continued on next page

Table 8.26 – continued from previous page

SIGNAL_EN- ABLES	2	This field indicates the sets of signals which are to be monitored. When each bit in this field is set, the corresponding signal set is monitored using test chirps. Rest of the RF analog may not be ON during these test chirps. The APLL VCO control voltage can be monitored. The Synthesizer VCO control voltage for both VCO1 and VCO2 can be monitored, while operating at their respective minimum and maximum frequencies, and their respective VCO slope (Hz/V) can be monitored if both frequencies are enabled for that VCO. The monitored signals are compared against internally chosen valid limits. The comparison results are part of the monitoring report message. Bit Location SIGNAL b0 APLL_VCTRL b1 SYNTH_VCO1_VCTRL b2 SYNTH_VCO2_VCTRL b3 SYNTH_VCO3_VCTRL (Reserved in AWR2243, xWR294x, xWR254x, xWR6243, xWR2x4xEco, xWR2x4xP. Set it to 0b0) b15:4 RESERVED The synthesizer VCO extreme frequencies are: Synthesizer VCO Frequency Limits (Min, Max) For AWR2243 (76GHz to 81Ghz): VCO1 (76GHz, 78GHz) VCO2 (77GHz, 81GHz) For xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP (76GHz to 81Ghz): VCO1 (76GHz, 77GHz) VCO2 (76GHz, 80.5GHz) or (76.5GHz, 81GHz) For 60GHz Devices (57GHz to 64Ghz): VCO1 (57GHz, 60.75GHz) VCO2 (61GHz, 64GHz) Synthesizer measurements are done with TX switched off to avoid emissions. NOTE: The VCO3 control voltage monitor is for debug purposes only and not supported in production.
RESERVED	4	0x00000000

8.14 Dual Clock Comparator Based Clock Frequency Monitor

This section contains API SBs that configure the Dual Clock Comparator based monitors of clocks in the BSS digital modules and report the soft results from the monitor. The corresponding monitors are collectively named DCC_CLOCK_FREQ_MONITOR. These monitors observe the relative frequency of various clock pairs and compare the measured relative frequency errors against internally fixed thresholds.

The various clock pairs that are monitored are defined here:

CLOCK PAIR	REFERENCE CLOCK	MEASURED CLOCK	ERROR THRESH-OLD (Tentative)
0	XTAL	APLL_200M	$\pm 1.0\%$
1	APLL_200M	BSS_200M	$\pm 1.0\%$
2	APLL_200M	BSS_100M	$\pm 1.0\%$
3	APLL_200M	GPADC_10M	$\pm 2.5\%$
4	APLL_200M	RCOSC_10M	$\pm 30.0\%$
5	RESERVED	RESERVED	-
6	APLL_200M	FRC_200M (xWR254x, xWR2x4xECO, xWR2x4xP only)	$\pm 1.0\%$

The ideal frequencies of clocks involved in this monitor are given here:

Table 8.27: DCC Clock monitor pairs

CLOCK NAME	CLOCK FRE-QUENCY (MHz)	COMMENTS
XTAL	40 (xWR294x), 50 (xWR254x, xWR2x4xECO, xWR2x4xP)	Crystal clock
APLL_200M	200	200MHz clock from APLL
BSS_200M	200	BSS processor clock
BSS_100M	100	BSS internal clock
GPADC_10M	10	GPADC clock used in monitoring and calibrations
RCOSC_10M	10	RC Oscillator clock

Continued on next page

Table 8.27 – continued from previous page

FRC_200M	200	FRC clock (xWR254x, xWR2x4xECO, xWR2x4xP only)
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8.14.1 Sub block 0x01DA – AWR_MONITOR_DUAL_CLOCK_COMP_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to the DCC based clock frequency monitoring. The report is sent as an async event [AWR_MONITOR_DCC_DUAL_CLOCK_COMP_REPORT_AE_SB](#).

Table 8.28: AWR_MONITOR_DUAL_CLOCK_COMP_CONF_SB contents

Field Name	Number of bytes	Description								
SBLKID	2	Value = 0x01DA								
SBLKLEN	2	Value = 12								
REPORTING_MODE	1	<table><tr><th>Value</th><th>Definition</th></tr><tr><td>0</td><td>RESERVED check</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	RESERVED check	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check
Value	Definition									
0	RESERVED check									
1	Report is send only upon a failure (after checking for thresholds)									
2	Report is sent every monitoring period with threshold check									
RESERVED	1	0x00								

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Table 8.28 – continued from previous page

DCC_PAIR_EN- ABLES	2	This field indicates which pairs of clocks to monitor. When a bit in the field is set to 1, the firmware monitors the corresponding clock pair by deploying the hardware's Dual Clock Comparator in the corresponding DCC mode. <table><thead><tr><th>Bit</th><th>CLOCK PAIR</th></tr></thead><tbody><tr><td>b0</td><td>0 APLL_200M</td></tr><tr><td>b1</td><td>1 BSS_200M</td></tr><tr><td>b2</td><td>2 BSS_100M</td></tr><tr><td>b3</td><td>3 GPADC_10M</td></tr><tr><td>b4</td><td>4 RCOSC_10M</td></tr><tr><td>b5</td><td>5 RESERVED</td></tr><tr><td>b6</td><td>6 FRC_200M (xWR254x, xWR2x4xEco, xWR2x4xP only)</td></tr><tr><td>b15:7</td><td>RESERVED</td></tr></tbody></table> The comparison results are part of the monitoring report message. The definition of the clock pairs and their error thresholds for failure reporting are given in the table below the message definition.	Bit	CLOCK PAIR	b0	0 APLL_200M	b1	1 BSS_200M	b2	2 BSS_100M	b3	3 GPADC_10M	b4	4 RCOSC_10M	b5	5 RESERVED	b6	6 FRC_200M (xWR254x, xWR2x4xEco, xWR2x4xP only)	b15:7	RESERVED
Bit	CLOCK PAIR																			
b0	0 APLL_200M																			
b1	1 BSS_200M																			
b2	2 BSS_100M																			
b3	3 GPADC_10M																			
b4	4 RCOSC_10M																			
b5	5 RESERVED																			
b6	6 FRC_200M (xWR254x, xWR2x4xEco, xWR2x4xP only)																			
b15:7	RESERVED																			
RESERVED	4	0x00000000																		

8.15 RX Saturation Detection Monitor

This section contains API SBs that configure the monitoring of RX analog saturation detectors, and report the results from the monitor. The corresponding monitors are collectively named RX_SATURATION_DETECTOR_MONITOR and RX_SIG_IMG_BAND_MONITOR. The report is available in CQ RAM.

8.15.1 Sub block 0x01DB – AWR_MONITOR_RX_SATURATION_DETECTOR_CONF_SB

This API is a monitoring configuration API which the host sends to the mmWave device, containing information related to RX saturation detector monitoring. The report is available as CQ2 (part of CQ) in CQ RAM every chirp. The application should transfer the report from CQ RAM every chirp.

Table 8.29: AWR_MONITOR_RX_SATURATION_DETECTOR_CONF_SB
contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x01DB
SBLKLEN	2	Value = 24
PROFILE_INDX	1	This field indicates the profile index for which this monitoring configuration applies.
SAT_MON_SELECT	1	01 – Enable only the ADC saturation monitor 11 – Enable both the ADC and IFA1 saturation monitors
RESERVED	1	0x00
RESERVED	1	0x00
SAT_MON_PRIMARY_TIME_SLICE_DURATION	2	It specifies the duration of each (primary) time slice. 1 LSB = 0.16 μs. Valid range: 4 to floor(ADC sampling time us/0.16 μs) NOTES: The minimum allowed duration of each (primary) time slice is 4 LSBs = 0.64 μs. Also, the maximum number of (primary) time slices that will be monitored in a chirp is 64 so the recommendation is to set this value to correspond to (ADC sampling time / 64). If the slice is smaller, such that the ADC sampling time is longer than 64 primary slices, some regions of the valid duration of a chirp may go un-monitored.

Continued on next page

Table 8.29 – continued from previous page

SAT_MON_NUM_SLICES (N)	2	Number of (primary + secondary) time slices to monitor. Valid range: 1 to 127 NOTE1: Together with SAT_MON_PRIMARY_TIME_SLICE_DURATION, this determines the full duration of the ADC valid time that gets covered by the monitor. Primary slices = $(N+1) / 2$ Secondary slices = Primary slices - 1 NOTE2: The total monitoring duration is recommended to be programmed slightly smaller than ADC sampling time to avoid last primary slice miss in the CQ data. If this recommendation is not followed and if ADC sampling time is less than total requested monitoring duration then no error is generated but the total number of slices reported back in CQ buffer would be a different value M, which is less than user requested value of N. In such cases, there will be $(M+1)/2$ primary slices and $(M-1)/2$ secondary slices. However, if ADC sampling time is such that Secondary $(M-1)/2$ can be measured and not Primary $(M+1)/2$, then primary slice $(M+1)/2$ will not be present in the CQ buffer. In such scenario, CQ buffer will have the total number of slices reported back as M-1 instead of M.
SAT_MON_RX_CHANNEL_MASK	1	Masks RX channels used for monitoring. In every slice, saturation counts for all unmasked channels are added together, and the total is capped to 127. The 8 bits are mapped (MSB->LSB) to: [RX3Q, RX2Q, RX1Q, RX0Q, RX3I, RX2I, RX1I, RX0I] 00000000 – All channels unmasked 11111111 – All channels masked
RESERVED	1	0
RESERVED	1	0
RESERVED	1	0
RESERVED	4	0x00000000
RESERVED	4	0x00000000

8.15.2 Sub block 0x01DC – AWR_MONITOR_SIG_IMG_MONITOR_CONF_SB

This API is a monitoring configuration API which the host sends to the mmWave device, containing information related to signal and image band energy. The report is available as CQ1 (part of CQ) in CQ RAM. The application should transfer the report every chirp.

NOTE: This monitor is not supported in xWR294x, xWR254x, xWR2x4xECP, xWR2x4xP devices.

Table 8.30: AWR_MONITOR_RX_SIG_IMG_MONITOR_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x01DC
SBLKLEN	2	Value = 16
PROFILE_INDX	1	This field indicates the profile index for which this monitoring configuration applies.
SIG_IMG_MON_NUM_SLICES	1	Number of (primary + secondary) slices to monitor Valid range: 1 to 127
NUM_SAMPLES_PER_PRIMARY_TIME_SLICE	2	This field specifies the number of samples constituting each time slice. The minimum allowed value for this parameter is 4. Valid range: 4 to NUM_ADC_SAMPLES (see NOTE2 below) NOTE1: The maximum number of (primary) time slices that will be monitored in a chirp is 64, so our recommendation is that this value should at least equal (NUM_ADC_SAMPLES / 64). If the slice is smaller, such that the number of ADC samples per chirp is larger than 64 primary slices, some regions of the valid duration of a chirp may go un-monitored. NOTE2: In Complex1x mode, the minimum number of samples per slice is 4 and for other modes it is 8. Also note that number of samples should be an even number. NOTE3: The total monitoring duration is recommended to program slightly smaller than ADC sampling time
RESERVED	4	0x00000000
RESERVED	4	0x00000000

NOTE:

It is recommended to re-issue these rx saturation and/or signal image monitor configuration APIs each time before enabling these monitor and frame trigger. The right sequence is as below:

1. Issue rx saturation and/or signal image monitor configuration API.
2. Enable rx saturation and/or signal image monitor.
3. Frame start.
4. Frame stop.
5. Frame start. (Optional in case of multiple frames)
6. Frame stop. (Optional in case of multiple frames)
7. Disable rx saturation and/or signal image monitor (in case disabled for some reason)
8. Issue rx saturation and/or signal image monitor configuration API.
9. Enable rx saturation and/or signal image monitor.
10. Frame start.

8.16 RX mixer input power monitor

8.16.1 Sub block 0x01DD – AWR_MONITOR_RX_MIXER_IN_POWER_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to RX mixer input power monitoring. The report is sent as an async event [AWR_MONITOR_RX_MIXER_IN_POWER_REPORT_AE_SB](#).

NOTE:

AWR2243/6x43 devices: The RX input power monitor API is debug only API.

xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: This monitor is not supported.

Please refer latest DFP release note for more info.

Table 8.31: AWR_MONITOR_MIXER_IN_POWER_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x01DD
SBLKLEN	2	Value = 16
PROFILE_INDEX	1	The RF analog settings corresponding to this profile are used for monitoring RX mixer input power using test chirps (static frequency, at the center of the profile's RF frequency band).

Continued on next page

Table 8.31 – continued from previous page

REPORTING_ MODE	1	Value Definition 0 Report is sent every monitoring period without threshold check 1 Report is send only upon a failure (after checking for thresholds) 2 Report is sent every monitoring period with threshold check
TX_EN	1	This field indicates if and which TX channels should be enabled while measuring RX mixer input power. Setting a bit to 1 enables the corresponding TX channel. Enabling a TX channel may help find reflection power while disabling may help find interference power. Bit number TX Channel b0 TX0 b1 TX1 b2 TX2
RESERVED	1	0x00
THRESHOLDS	2	The measured RX mixer input voltage swings during this monitoring is compared against the minimum and maximum thresholds configured in this field. The comparison result is part of the monitoring report message (Status bit is cleared if any measurement is outside this (minimum, maximum) range). Byte number Threshold 0 Minimum Threshold 1 Maximum Threshold Only the RX channels enabled in the static configuration APIs are monitored. 1 LSB = 1800 mV/256, unsigned number Valid range: 0 to 255, maximum threshold $\geq$ minimum threshold
RESERVED	2	0x00000000
RESERVED	4	0x00000000

8.17 Sub block 0x01DE – RESERVED

8.18 Analog Fault injection

8.18.1 Sub block 0x01DF – AWR_ANALOG_FAULT_INJECTION_CONF_SB

This API is a fault injection API which the host sends to the AWR device. It can be used to inject faults in the analog circuits to test the corresponding monitors. After the faults are injected, the regular enabled monitors will indicate the faults in their associated reports.

NOTE1:	This API should be issued when no frames are on-going.
NOTE2:	Disable all runtime calibrations while Fault is injected.
NOTE3:	1. The fault injection should be tested by injecting one fault at a time and corresponding analog monitor should be observed, other monitors might show failure depending on type of fault, it can be discarded. 2. Once the monitoring failure has been verified, remove the fault and then proceed with other fault injection tests. 3. After all the required fault injection tests have been executed and all the injected faults have been removed, the user may begin functional frames.
NOTE4:	Some of the fault injection options are de-featured, please refer latest DFP release note for more details.

Table 8.32: AWR_ANALOG_FAULT_INJECTION_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x01DF
SBLKLEN	2	Value = 24
RESERVED	1	0x00

Continued on next page

Table 8.32 – continued from previous page

RX_GAIN_DROP	1	Primary Fault: RX Gain This field indicates which RX RF sections should have fault injected. If the fault is enabled, the RX RF gain drops significantly. The fault can be used to cause significant gain change, inter-RX gain imbalance and an uncontrolled amount of inter-RX phase imbalance. This fault can be seen in RX_GAIN_PHASE_MONITOR. <table><tr><td>Bit number</td><td>RX Channel</td></tr><tr><td>b0</td><td>RX0</td></tr><tr><td>b1</td><td>RX1</td></tr><tr><td>b2</td><td>RX2</td></tr><tr><td>b3</td><td>RX3</td></tr><tr><td>Others</td><td>RESERVED</td></tr></table> For each bit, 1 = inject fault, 0 = remove injected fault	Bit number	RX Channel	b0	RX0	b1	RX1	b2	RX2	b3	RX3	Others	RESERVED
Bit number	RX Channel													
b0	RX0													
b1	RX1													
b2	RX2													
b3	RX3													
Others	RESERVED													
RX_PHASE_INV	1	Primary Fault: RX Phase This field indicates which RX channels should have fault injected. If the fault is enabled, the RX phase gets inverted. The fault can be used to cause a controlled amount (180 deg) of inter-RX phase imbalance. This fault can be seen in RX_GAIN_PHASE_MONITOR. <table><tr><td>Bit number</td><td>RX Channel</td></tr><tr><td>b0</td><td>RX0</td></tr><tr><td>b1</td><td>RX1</td></tr><tr><td>b2</td><td>RX2</td></tr><tr><td>b3</td><td>RX3</td></tr><tr><td>Others</td><td>RESERVED</td></tr></table> For each bit, 1 = inject fault, 0 = remove injected fault	Bit number	RX Channel	b0	RX0	b1	RX1	b2	RX2	b3	RX3	Others	RESERVED
Bit number	RX Channel													
b0	RX0													
b1	RX1													
b2	RX2													
b3	RX3													
Others	RESERVED													

Continued on next page

Table 8.32 – continued from previous page

RX_HIGH_NOISE	1	Primary Fault: RX Noise This field indicates which RX channels should have fault injected. If the fault is enabled, the RX IFA square wave loopback paths are engaged to inject high noise at RX IFA input. The fault can be used to cause significant RX noise floor elevation. This fault can be seen in RX_GAIN_PHASE_MONITOR and RX_NOISE_FIGURE_MONITOR. <table><tr><td>Bit number</td><td>RX Channel</td></tr><tr><td>b0</td><td>RX0</td></tr><tr><td>b1</td><td>RX1</td></tr><tr><td>b2</td><td>RX2</td></tr><tr><td>b3</td><td>RX3</td></tr><tr><td>Others</td><td>RESERVED</td></tr></table> For each bit, 1 = inject fault, 0 = remove injected fault	Bit number	RX Channel	b0	RX0	b1	RX1	b2	RX2	b3	RX3	Others	RESERVED
Bit number	RX Channel													
b0	RX0													
b1	RX1													
b2	RX2													
b3	RX3													
Others	RESERVED													
RX_IF_STAGES_FAULT	1	Primary Fault: Cutoff frequencies of RX IFA HPF. This field indicates which RX channels should have fault injected. If the fault is enabled, the RX IFA HPF cutoff frequency becomes high (about 4 times the HPF cutoff frequency configured by AWR_PROFILE_CONF_SET_SB for monitoring profile). The fault can be used to cause the measured HPF attenuation to vary from ideal expectations. This fault can be seen in STATUS_RX_HPF_ERROR flag in report of RX_IFSTAGE_MONITOR. <table><tr><td>Bit number</td><td>RX Channel</td></tr><tr><td>b0</td><td>RX0</td></tr><tr><td>b1</td><td>RX1</td></tr><tr><td>b2</td><td>RX2</td></tr><tr><td>b3</td><td>RX3</td></tr><tr><td>Others</td><td>RESERVED</td></tr></table> For each bit, 1 = inject fault, 0 = remove injected fault NOTE: During the execution of RX_IFSTAGE_MONITOR, the RX_HIGH_NOISE faults are temporarily removed.	Bit number	RX Channel	b0	RX0	b1	RX1	b2	RX2	b3	RX3	Others	RESERVED
Bit number	RX Channel													
b0	RX0													
b1	RX1													
b2	RX2													
b3	RX3													
Others	RESERVED													

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Table 8.32 – continued from previous page

RX_LO_AMP_FAULT	1	Primary Fault: RX Mixer LO input swing reduction This field indicates which RX channels should have fault injected. If the fault is enabled, the RX mixer LO input swing is significantly reduced. The fault is primarily expected to be detected by RX_INTERNAL_ANALOG_SIGNALS_MONITOR (under PWRDET_RX category). <table><tr><td>Bit number</td><td>RX Channel</td></tr><tr><td>b0</td><td>RX0</td></tr><tr><td>b1</td><td>RX1</td></tr><tr><td>b2</td><td>RX2</td></tr><tr><td>b3</td><td>RX3</td></tr><tr><td>Others</td><td>RESERVED</td></tr></table> For each bit, 1 = inject fault, 0 = remove injected fault NOTE: This fault injection is de-featured, please refer latest release note.	Bit number	RX Channel	b0	RX0	b1	RX1	b2	RX2	b3	RX3	Others	RESERVED
Bit number	RX Channel													
b0	RX0													
b1	RX1													
b2	RX2													
b3	RX3													
Others	RESERVED													
TX_LO_AMP_FAULT	1	Primary Fault: TX PA input signal generator turning off. This field indicates which TX channels should have fault injected. If the fault is enabled, the amplifier generating TX power amplifier's LO input signal is turned off. The fault is primarily expected to be detected by TX<n>_INTERNAL_ANALOG_SIGNALS_MONITOR (under DCBIAS category). <table><tr><td>Bit number</td><td>Channel</td></tr><tr><td>b0</td><td>TX0 and TX1</td></tr><tr><td>b1</td><td>TX2 (applicable only if available in the device)</td></tr><tr><td>Others</td><td>RESERVED</td></tr></table> For each bit, 1 = inject fault, 0 = remove injected fault NOTE: This fault injection is de-featured, please refer latest release note.	Bit number	Channel	b0	TX0 and TX1	b1	TX2 (applicable only if available in the device)	Others	RESERVED				
Bit number	Channel													
b0	TX0 and TX1													
b1	TX2 (applicable only if available in the device)													
Others	RESERVED													

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Table 8.32 – continued from previous page

TX_GAIN_DROP	1	Primary Fault: TX Gain (power) This field indicates which TX RF sections should have fault injected. If the fault is enabled, the TX RF gain drops significantly. The fault can be used to cause significant TX output power change, inter-TX gain imbalance and an uncontrolled amount of inter-TX phase imbalance. This fault can be seen in TXn_POWER_MONITOR. <table><tr><td>Bit number</td><td>Channel</td></tr><tr><td>b0</td><td>TX0</td></tr><tr><td>b1</td><td>TX1</td></tr><tr><td>b2</td><td>TX2</td></tr><tr><td>b3</td><td>TX3 (xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP only)</td></tr><tr><td>Others</td><td>RESERVED</td></tr></table> For each bit, 1 = inject fault, 0 = remove injected fault	Bit number	Channel	b0	TX0	b1	TX1	b2	TX2	b3	TX3 (xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP only)	Others	RESERVED
Bit number	Channel													
b0	TX0													
b1	TX1													
b2	TX2													
b3	TX3 (xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP only)													
Others	RESERVED													

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Table 8.32 – continued from previous page

TX_PHASE_INV	1	Primary Fault: TX Phase This field indicates if TX channels should have fault injected, along with some further programmability. If the fault is enabled, the TX BPM polarity (phase) is forced to a constant value as programmed. The fault can be used to cause a controlled amount (180 degree) of inter-TX phase imbalance as well as BPM functionality failure. This fault can be seen in TX_GAIN_PHASE_MISMATCH_MONITOR and TXn_PHASE_SHIFTER_MONITOR. <table><tr><td>Bit number</td><td>TX Channel</td></tr><tr><td>b0</td><td>TX_FAULT (Common for all TX channels)</td></tr><tr><td>b1</td><td>RESERVED</td></tr><tr><td>b2</td><td>RESERVED</td></tr><tr><td>b3</td><td>TX0_BPM_VALUE</td></tr><tr><td>b4</td><td>TX1_BPM_VALUE</td></tr><tr><td>b5</td><td>TX2_BPM_VALUE</td></tr><tr><td>b6</td><td>TX3_BPM_VALUE (xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP only)</td></tr><tr><td>Others</td><td>RESERVED</td></tr></table> For each TXn_BPM_VALUE: Applicable only if TX_FAULT is enabled. Value = 0: force TX<n> BPM polarity to 0 Value = 1: force TX<n> BPM polarity to 1. NOTE: The TXn_BPM_VALUE takes effect only when TX_FAULT value is set to 1.	Bit number	TX Channel	b0	TX_FAULT (Common for all TX channels)	b1	RESERVED	b2	RESERVED	b3	TX0_BPM_VALUE	b4	TX1_BPM_VALUE	b5	TX2_BPM_VALUE	b6	TX3_BPM_VALUE (xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP only)	Others	RESERVED
Bit number	TX Channel																			
b0	TX_FAULT (Common for all TX channels)																			
b1	RESERVED																			
b2	RESERVED																			
b3	TX0_BPM_VALUE																			
b4	TX1_BPM_VALUE																			
b5	TX2_BPM_VALUE																			
b6	TX3_BPM_VALUE (xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP only)																			
Others	RESERVED																			

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Table 8.32 – continued from previous page

SYNTH_FAULT	1	Primary Fault: Synthesizer Frequency This field indicates which Synthesizer faults should be injected. SYNTH_VCO_OPENLOOP: If the fault is enabled, the synthesizer is forced in open loop mode which causes the VCO control voltage to increase to its maximum value. Additionally the monitor higher threshold is reduced by 100mV so as to ensure that the increased VCO control voltage deterministically exceeds this higher threshold and the fault can be seen. In order to avoid out of band emissions in this faulty state, this fault is injected just before the PLL_CONTROL_VOLTAGE_MONITOR is executed and released just after its completion. This fault can be seen in PLL_CONTROL_VOLTAGE_MONITOR. SYNTH_FREQ_MON_OFFSET: If the fault is enabled, the synthesizer frequency monitor's ideal frequency ramp waveform is forced to be offset from the actual ramp waveform by a constant, causing monitoring to detect failures. This fault can be seen in SYNTH_FREQ_MONITOR. <table><tr><td>Bit number</td><td>Enable Fault</td></tr><tr><td>b0</td><td>SYNTH_VCO_OPENLOOP</td></tr><tr><td>b1</td><td>SYNTH_FREQ_MON_OFFSET</td></tr><tr><td>Others</td><td>RESERVED</td></tr></table> For each bit, 1 = inject fault, 0 = remove injected fault	Bit number	Enable Fault	b0	SYNTH_VCO_OPENLOOP	b1	SYNTH_FREQ_MON_OFFSET	Others	RESERVED
Bit number	Enable Fault									
b0	SYNTH_VCO_OPENLOOP									
b1	SYNTH_FREQ_MON_OFFSET									
Others	RESERVED									

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Table 8.32 – continued from previous page

SUPPLY_LDO_FAULT	1	This field indicates whether some LDO output voltage faults should be injected or not. Bit Enable Fault b0 SUPPLY_TESTPATH_LDO_FAULT (xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) SUPPLY_LDO_RX_LODIST_FAULT (others) b7:1 RESERVED SUPPLY_LDO_FAULT: If enabled in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP device, the testpath subsystem's LDO output is disabled. In other devices, the RX LODIST LDO output voltage is modified compared to normal operation. This fault injection causes INTERNAL_PMCLKLO_SIGNALS_MONITOR to detect failure (under SUPPLY category). This fault can be seen in INTERNAL_PMCLKLO_SIGNALS_MONITOR. For each bit, 1 = inject fault, 0 = remove injected fault NOTE: This fault injection is ineffective under LDO bypass condition.
MISC_FAULT	1	This field indicates whether a few miscellaneous faults should be injected or not. Bit number Enable Fault b0 GPADC_CLK_FREQ_FAULT Others RESERVED GPADC_CLK_FREQ_FAULT: if enabled, the GPADC clock frequency is slightly increased compared to normal usage to cause BSS DCC_CLOCK_FREQ_MONITOR to detect failure. This fault can be seen in DCC_CLOCK_FREQ_MONITOR. For each bit, 1 = inject fault, 0 = remove injected fault

Continued on next page

Table 8.32 – continued from previous page

MISC_THRESH_FAULT	1	This field indicates whether faults should be forced in the threshold comparisons in the software layer of some monitors. If a fault is enabled, the logic in the min-max threshold comparisons used for failure detection is inverted, causing a fault to be reported. During these faults, no hardware fault condition is injected in the device. This fault can be seen in GPADC_INTERNAL_SIGNALS_MONITOR. Bit number Enable Fault b0 GPADC_INTERNAL_SIGNALS_MONITOR The reported values of the monitored GPADC internal signals are also swapped when this fault is enabled. Others RESERVED For each bit, 1 = inject fault, 0 = remove injected fault
TX_PS_DAC_FAULT	1	Primary Fault: TX PS DAC fault injection. This field indicates which TX PS DAC should have fault injected. If the fault is enabled, the corresponding TX's ANA DAC LDO would be turned off causing the PS DAC to fail. This fault can be seen in TX PHASE SHIFTER DAC monitoring report (AWR_MONITOR_TX_PHSIFTER_DAC_REPORT_AE_SB). This fault injection is only applicable in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices. Bit number Channel b0 TX0 b1 TX1 b2 TX2 b3 TX3 Others RESERVED For each bit, 1 = inject fault, 0 = remove injected fault
RESERVED	2	0x000000
RESERVED	4	0x00000000

8.19 Sub blocks related to AWR_RF_MONITORING_CONF_SET_2_MSG

8.19.1 Sub block 0x0280 – AWR_MONITOR_TXN_POWER_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to TXN power monitoring. Absolute TX power and flatness across RF frequencies are monitored here. The report is sent as an async event [AWR_MONITOR_TX0_POWER_REPORT_AE_SB](#) , [AWR_MONITOR_TX1_POWER_REPORT_AE_SB](#) , [AWR_MONITOR_TX2_POWER_REPORT_AE_SB](#) , [AWR_MONITOR_TX3_POWER_REPORT_AE_SB](#).

Table 8.33: AWR_MONITOR_TXN_POWER_CONF_SB contents

Field Name	Number of bytes	Description												
SBLKID	2	Value = 0x0280												
SBLKLEN	2	Value = 20												
PROFILE_INDX	1	This field indicates the Profile Index for which this monitoring configuration applies.												
RF_FREQ_BIT-MASK	1	This field indicates the exact RF frequencies inside the profile's RF band at which to measure the required parameters. When each bit in this field is set, the measurement at the corresponding RF frequency is enabled w.r.t. the profile's RF band. <table> <tr> <th>Bit number</th><th>RF frequency</th><th>RF name</th></tr> <tr> <td>b0</td><td>Start RF frequency in profile's sweep bandwidth</td><td>RF1</td></tr> <tr> <td>b1</td><td>Center RF frequency in profile's sweep bandwidth</td><td>RF2</td></tr> <tr> <td>b2</td><td>End RF frequency in profile's sweep bandwidth</td><td>RF3</td></tr> </table> The RF Name column is mentioned here to set the convention for the purpose of reporting and describing many monitoring packets.	Bit number	RF frequency	RF name	b0	Start RF frequency in profile's sweep bandwidth	RF1	b1	Center RF frequency in profile's sweep bandwidth	RF2	b2	End RF frequency in profile's sweep bandwidth	RF3
Bit number	RF frequency	RF name												
b0	Start RF frequency in profile's sweep bandwidth	RF1												
b1	Center RF frequency in profile's sweep bandwidth	RF2												
b2	End RF frequency in profile's sweep bandwidth	RF3												

Continued on next page

Table 8.33 – continued from previous page

TX_SEL_MASK	1	TX channel bit mask indicating which of the TX-N monitors have to be updated with the below configurations. Setting a bit to 1 updates the corresponding TX-N monitor to be configured. <table><tr><td>Bit number</td><td>TX index</td></tr><tr><td>b0</td><td>Configure TX0 power monitor</td></tr><tr><td>b1</td><td>Configure TX1 power monitor</td></tr><tr><td>b2</td><td>Configure TX2 power monitor</td></tr><tr><td>b3</td><td>Configure TX3 power monitor</td></tr><tr><td>b7:4</td><td>RESERVED.</td></tr></table>	Bit number	TX index	b0	Configure TX0 power monitor	b1	Configure TX1 power monitor	b2	Configure TX2 power monitor	b3	Configure TX3 power monitor	b7:4	RESERVED.
Bit number	TX index													
b0	Configure TX0 power monitor													
b1	Configure TX1 power monitor													
b2	Configure TX2 power monitor													
b3	Configure TX3 power monitor													
b7:4	RESERVED.													
RESERVED	1	0x00												
REPORTING_MODE	1	<table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>Report is sent every monitoring period without threshold check</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	Report is sent every monitoring period without threshold check	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check				
Value	Definition													
0	Report is sent every monitoring period without threshold check													
1	Report is send only upon a failure (after checking for thresholds)													
2	Report is sent every monitoring period with threshold check													
RESERVED	1	0x00												
TX_POWER_ABSOLUTE_ERROR_THRESH	2	The magnitude of difference between the programmed and measured TX power for each enabled channel at each enabled RF frequency, is compared against this threshold. The comparison result is part of the monitoring report message (Error bit is set if any measurement is above this threshold). 1 LSB = 0.1 dB Valid range: 0 to 65535 (0 to 6553dB)												
TX_POWER_FLATNESS_ERROR_THRESH	2	The magnitude of measured TX power flatness error, for each enabled channel, is compared against this threshold. The flatness error for a channel is defined as the peak to peak variation across RF frequencies. The comparison result is part of the monitoring report message (Error bit is set if any measurement is above this threshold). 1 LSB = 0.1 dB Valid range: 0 to 65535 (0 to 6553dB) This flatness check is applicable only if multiple RF Frequencies are enabled.												
RESERVED	2	0x0000												

Continued on next page

Table 8.33 – continued from previous page

TX_POWER_OFFSET_VALUE	3	AWR2243, xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices: This field is reserved. Set it to 0x000000. xWR6243 devices: The offset values to be added with the measured TX power for each RF before the relevant threshold comparison. Byte numbers corresponding to different RF are RF Byte RF1 0 RF2 1 RF3 2 1 LSB = 0.1dB signed Valid range: -128 to +127 (-12.8 to 12.7dB)
RESERVED	1	0x00

8.19.2 Sub block 0x0281 – AWR_MONITOR_TXN_BALLBREAK_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to TX ball break detection.

This API SB controls the thresholds for the reflection coefficient magnitude check and the parameters for the reflection coefficient error distance check (variation from values at the time of factory calibration). The report is sent as an async event [AWR_MONITOR_TX0_BALLBREAK_REPORT_AE_SB](#), [AWR_MONITOR_TX1_BALLBREAK_REPORT_AE_SB](#), [AWR_MONITOR_TX2_BALLBREAK_REPORT_AE_SB](#), [AWR_MONITOR_TX3_BALLBREAK_REPORT_AE_SB](#).

Table 8.34: AWR_MONITOR_TXN_BALLBREAK_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0281
SBLKLEN	2	Value = 16
REPORTING_MODE	1	Value Definition 0 Report is sent every monitoring period without threshold check 1 Report is send only upon a failure (after checking for thresholds) 2 Report is sent every monitoring period with threshold check

Continued on next page

Table 8.34 – continued from previous page

TX_SEL_MASK	1	TX channel bit mask indicating which of the TX-N monitors have to be updated with the below configurations. Setting a bit to 1 updates the corresponding TX-N monitor to be configured. <table><tr><td>Bit number</td><td>TX index</td></tr><tr><td>b0</td><td>Configure TX0 ball-break monitor</td></tr><tr><td>b1</td><td>Configure TX1 ball-break monitor</td></tr><tr><td>b2</td><td>Configure TX2 ball-break monitor</td></tr><tr><td>b3</td><td>Configure TX3 ball-break monitor</td></tr><tr><td>b7:4</td><td>RESERVED.</td></tr></table>	Bit number	TX index	b0	Configure TX0 ball-break monitor	b1	Configure TX1 ball-break monitor	b2	Configure TX2 ball-break monitor	b3	Configure TX3 ball-break monitor	b7:4	RESERVED.
Bit number	TX index													
b0	Configure TX0 ball-break monitor													
b1	Configure TX1 ball-break monitor													
b2	Configure TX2 ball-break monitor													
b3	Configure TX3 ball-break monitor													
b7:4	RESERVED.													
TX_REFL_COEFF_THRESH	2	The TX reflection coefficient's magnitude for each enabled channel is compared against the threshold given here. The comparison result is part of the monitoring report message (Error bit is set if the measurement is higher than this threshold, with the units of both quantities being the same). 1 LSB = 0.1 dB, signed number Valid range: -32767 to +32767 (-3276dB to +3276dB)												
MON_START_FREQ_CONST	4	AWR2243 devices: This field is reserved. Set to 0x0. xWR6x43, xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP devices: Start frequency of the monitoring chirp. For 77GHz Devices (76GHz to 80.8Ghz): 1 LSB = $3.6e9/2^{26}$ Hz $\approx$ 53.644 Hz Valid range: 0x5471C71C to 0x59C71C71 For 60GHz Devices (57GHz to 63.8Ghz): 1 LSB = $2.7e9/2^{26}$ Hz $\approx$ 40.233 Hz Valid range: Only even numbers from 0x5471C71C to 0x5E84BDA1												
TX_POWER_BACKOFF	1	AWR2243, xWR294x, xWR254x, xWR2x4xEco, xWR2x4xP devices: This field is reserved. Set to 0x0. xWR6243 devices: TX Power Backoff settings used for ballbreak monitor 1 LSB = 1dB Valid values: 0, 3, 6, 9dB												
RESERVED	3	0x000000												

8.19.3 Sub block 0x0282 – AWR_MONITOR_TXN_INTERNAL_ANALOG_SIGNALS_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to TX-N Internal Analog Signals monitoring including Tx Phase shifter DAC monitor. The report is sent as an async event [AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB](#), [AWR_MONITOR_TX1_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB](#), [AWR_MONITOR_TX2_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB](#), [AWR_MONITOR_TX3_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB](#).

Table 8.35: AWR_MONITOR_TXN_INTERNAL_ANALOG_SIGNALS_CONF_SB contents

Field Name	Number of bytes	Description								
SBLKID	2	Value = 0x0282								
SBLKLEN	2	Value = 12								
PROFILE_INDEX	1	The RF analog settings corresponding to this profile are used for monitoring the enabled signals, using test chirps (static frequency, at the center of the profile's RF frequency band).								
REPORTING_MODE	1	<table><tr><td>Value</td><td>Definition</td></tr><tr><td>0</td><td>RESERVED</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	RESERVED	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check
Value	Definition									
0	RESERVED									
1	Report is send only upon a failure (after checking for thresholds)									
2	Report is sent every monitoring period with threshold check									
TX_PS_DAC_MON_THRESH	2	The TX phase shifter DAC monitor delta threshold when TX_PS_DAC_MON is Enabled (Not applicable for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP) Unit: 1 LSB = 1.8V/1024 Value 0: TX_PS_DAC_MON is disabled Valid Range: 1 to 1023								

Continued on next page

Table 8.35 – continued from previous page

TX_SEL_MASK	1	TX channel bit mask indicating which of the TX-N monitors have to be updated with the below configurations. Setting a bit to 1 updates the corresponding TX-N monitor to be configured. <table><tr><td>Bit number</td><td>TX index</td></tr><tr><td>b0</td><td>Configure TX0 internal signals monitor</td></tr><tr><td>b1</td><td>Configure TX1 internal signals monitor</td></tr><tr><td>b2</td><td>Configure TX2 internal signals monitor</td></tr><tr><td>b3</td><td>Configure TX3 internal signals monitor</td></tr><tr><td>b7:4</td><td>RESERVED.</td></tr></table>	Bit number	TX index	b0	Configure TX0 internal signals monitor	b1	Configure TX1 internal signals monitor	b2	Configure TX2 internal signals monitor	b3	Configure TX3 internal signals monitor	b7:4	RESERVED.
Bit number	TX index													
b0	Configure TX0 internal signals monitor													
b1	Configure TX1 internal signals monitor													
b2	Configure TX2 internal signals monitor													
b3	Configure TX3 internal signals monitor													
b7:4	RESERVED.													
RESERVED	3	0x000000												

8.19.4 Sub block 0x0283 – AWR_MONITOR_TXN_PHASE_SHIFTER_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to TX-N TX loop back based phase shifter monitoring.

The report is sent as an async event [AWR_MONITOR_TX0_PHASE_SHIFTER_REPORT_AE_SB](#), [AWR_MONITOR_TX1_PHASE_SHIFTER_REPORT_AE_SB](#), [AWR_MONITOR_TX2_PHASE_SHIFTER_REPORT_AE_SB](#), [AWR_MONITOR_TX3_PHASE_SHIFTER_REPORT_AE_SB](#).

Table 8.36: AWR_MONITOR_TXN_PHASE_SHIFTER_CONF_SB contents

Field Name	Number of bytes	Description								
SBLKID	2	Value = 0x0283								
SBLKLEN	2	Value = 32								
PROFILE_INDX	1	This field indicates the Profile Index for which this monitoring configuration applies.								
REPORTING_MODE	1	<table><tr><th>Value</th><th>Definition</th></tr><tr><td>0</td><td>Report is sent every monitoring period without threshold check</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	Report is sent every monitoring period without threshold check	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check
Value	Definition									
0	Report is sent every monitoring period without threshold check									
1	Report is send only upon a failure (after checking for thresholds)									
2	Report is sent every monitoring period with threshold check									

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Table 8.36 – continued from previous page

RESERVED	2	RESERVED
PH_SHIFTER_MON_CFG	1	Bit Definition b0 Phase shifter phase1 monitor enable bit b1 Phase shifter phase2 monitor enable bit b2 Phase shifter phase3 monitor enable bit b3 Phase shifter phase4 monitor enable bit b7:4 RESERVED Enable at least two phase settings to measure phase error and to apply threshold in reporting mode 1 and 2.
RX_EN	1	This field indicates the RX channels that should be enabled for TX to RX loopback measurement. Setting the corresponding bit to 1 enables that channel for measurement and the average measured value is reported out. Bit number RX Channel b0 RX0 b1 RX1 b2 RX2 b3 RX3

Continued on next page

Table 8.36 – continued from previous page

MON_CHIRP_SLOPE	1	Frequency slope for each monitoring chirp is encoded in 1 bytes (8 bit signed number) For 77GHz Devices (76GHz to 81Ghz): 1 LSB = $3.6e9 \times 900/2^{26}$ Hz $\approx$ 48.279 kHz/μs Valid range: -128 to +127 (Max 6.13 MHz/μs) For 60GHz Devices (57GHz to 64Ghz): 1 LSB = $2.7e9 \times 900/2^{26}$ Hz $\approx$ 36.21 kHz/μs Valid range: -128 to +127 (Max 4.63 MHz/μs) NOTE: Monitoring Chirp Slope can be programmed based on the emission specifications. The device transmits on air during the execution of these monitors. The host can control the monitoring emission power spectral density (dB-m/Hz) by programming this slope parameter. Each monitoring chirp is about 45us in duration. Therefore the overall RF sweep bandwidth for the monitoring chirp is given by Monitoring Chirp Slope* 45 us. Normally, low values of Monitoring Chirp Slope are recommended, as with non-zero slope in FMCW radar, any actual target reflections can potentially be interpreted as noise power and/or corrupt the loopback signal based gain/phase measurement. The user has to ensure that the RF bandwidth selected for monitoring chirp based on slope and start frequency is within FREQ_LIMIT set in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB API.												
TX_SEL_MASK	1	TX channel bit mask indicating which of the TX-N monitors have to be updated with the below configurations. Setting a bit to 1 updates the corresponding TX-N monitor to be configured. <table><tr><td>Bit number</td><td>TX index</td></tr><tr><td>b0</td><td>Configure TX0 phaseshifter monitor</td></tr><tr><td>b1</td><td>Configure TX1 phaseshifter monitor</td></tr><tr><td>b2</td><td>Configure TX2 phaseshifter monitor</td></tr><tr><td>b3</td><td>Configure TX3 phaseshifter monitor</td></tr><tr><td>b7:4</td><td>RESERVED.</td></tr></table>	Bit number	TX index	b0	Configure TX0 phaseshifter monitor	b1	Configure TX1 phaseshifter monitor	b2	Configure TX2 phaseshifter monitor	b3	Configure TX3 phaseshifter monitor	b7:4	RESERVED.
Bit number	TX index													
b0	Configure TX0 phaseshifter monitor													
b1	Configure TX1 phaseshifter monitor													
b2	Configure TX2 phaseshifter monitor													
b3	Configure TX3 phaseshifter monitor													
b7:4	RESERVED.													

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Table 8.36 – continued from previous page

PH_SHIFTER_ INC_VAL1	1	Phase shifter monitoring increment value for phase1, the monitoring phase will be incremented by this value in every FTTI interval. In API based trigger MONITORING_MODE, it is important to trigger PS monitor every FTTI to make sure this logic works seamlessly. Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift increment value 1 1 LSB = 5.625°
PH_SHIFTER_ INC_VAL2	1	Phase shifter monitoring increment value for phase2, the monitoring phase will be incremented by this value in every FTTI interval. In API based trigger MONITORING_MODE, it is important to trigger PS monitor every FTTI to make sure this logic works seamlessly. Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift increment value 2 1 LSB = 5.625°
PH_SHIFTER_ INC_VAL3	1	Phase shifter monitoring increment value for phase3, the monitoring phase will be incremented by this value in every FTTI interval. In API based trigger MONITORING_MODE, it is important to trigger PS monitor every FTTI to make sure this logic works seamlessly. Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift increment value 3 1 LSB = 5.625°
PH_SHIFTER_ INC_VAL4	1	Phase shifter monitoring increment value for phase4, the monitoring phase will be incremented by this value in every FTTI interval. In API based trigger MONITORING_MODE, it is important to trigger PS monitor every FTTI to make sure this logic works seamlessly. Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift increment value 4 1 LSB = 5.625°
PH_SHIFTER_ MON1	1	TX 0 Phase shifter phase1 monitor value Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift monitor value 1 1 LSB = 5.625°

Continued on next page

Table 8.36 – continued from previous page

PH_SHIFTER_MON2	1	TX 0 Phase shifter phase2 monitor value Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift monitor value 2 1 LSB = 5.625°
PH_SHIFTER_MON3	1	TX 0 Phase shifter phase3 monitor value Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift monitor value 3 1 LSB = 5.625°
PH_SHIFTER_MON4	1	TX 0 Phase shifter phase4 monitor value Bits Phase shift definition b1:0 RESERVED (set it to 0b00) b7:2 Phase shift monitor value 4 1 LSB = 5.625°
TX_PHASE_ERROR_THRESH	2	The threshold for deviation of the TX output phase difference between the measured phase values and configured phases for each enabled phase settings. The max error is compared against the threshold given here. The measured phase is part of the monitoring report message (Error bit is set if the max deviation is higher than this threshold, with the units of both quantities being the same). 1 LSB = $360^\circ / 2^{16}$. Valid range: corresponding to 0° to 359.9°.
TX_AMPLITUDE_ERROR_THRESH	2	The threshold for deviation of the TX output amplitude difference between all enabled phase settings. The max error is compared against the threshold given here. The measured output amplitude is part of the monitoring report message (Error bit is set if the max deviation is higher than this threshold, with the units of both quantities being the same). 1 LSB = 0.1 dB Valid range: 0 to 65535 (0 to 6553dB)
RESERVED	8	0x0000

8.19.5 Sub block 0x0284 – AWR_MONITOR_ADV_TX_GAIN_PHASE_MISMATCH_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to TX gain and phase mismatch monitoring. The report is sent as an async event [AWR_MONITOR_ADV_TX_GAIN_PHASE_REPORT_AE_SB](#).

Table 8.37: AWR_MONITOR_ADV_TX_GAIN_PHASE_MISMATCH_CONF_
SB contents

Field Name	Number of bytes	Description												
SBLKID	2	Value = 0x0284												
SBLKLEN	2	Value = 68												
PROFILE_INDX	1	This field indicates the Profile Index for which this monitoring configuration applies.												
RF_FREQ_BIT-MASK	1	This field indicates the exact RF frequencies inside the profile's RF band at which to measure the required parameters. When each bit in this field is set, the measurement at the corresponding RF frequency is enabled w.r.t. the profile's RF band. <table><tr><th>Bit number</th><th>RF frequency</th><th>RF name</th></tr><tr><td>b0</td><td>Start RF frequency in profile's sweep bandwidth</td><td>RF1</td></tr><tr><td>b1</td><td>Center RF frequency in profile's sweep bandwidth</td><td>RF2</td></tr><tr><td>b2</td><td>End RF frequency in profile's sweep bandwidth</td><td>RF3</td></tr></table> The RF Name column is mentioned here to set the convention for the purpose of reporting and describing many monitoring packets.	Bit number	RF frequency	RF name	b0	Start RF frequency in profile's sweep bandwidth	RF1	b1	Center RF frequency in profile's sweep bandwidth	RF2	b2	End RF frequency in profile's sweep bandwidth	RF3
Bit number	RF frequency	RF name												
b0	Start RF frequency in profile's sweep bandwidth	RF1												
b1	Center RF frequency in profile's sweep bandwidth	RF2												
b2	End RF frequency in profile's sweep bandwidth	RF3												
TX_EN	1	This field indicates the TX channels that should be compared for gain and phase balance. Setting the corresponding bit to 1 enables that channel for imbalance measurement. <table><tr><th>Bit number</th><th>TX Channel</th></tr><tr><td>b0</td><td>TX0</td></tr><tr><td>b1</td><td>TX1</td></tr><tr><td>b2</td><td>TX2</td></tr><tr><td>b3</td><td>TX3</td></tr></table>	Bit number	TX Channel	b0	TX0	b1	TX1	b2	TX2	b3	TX3		
Bit number	TX Channel													
b0	TX0													
b1	TX1													
b2	TX2													
b3	TX3													
RX_EN	1	This field indicates the RX channels that should be enabled for TX to RX loopback measurement. Setting the corresponding bit to 1 enables that channel for imbalance measurement. <table><tr><th>Bit number</th><th>RX Channel</th></tr><tr><td>b0</td><td>RX0</td></tr><tr><td>b1</td><td>RX1</td></tr><tr><td>b2</td><td>RX2</td></tr><tr><td>b3</td><td>RX3</td></tr></table>	Bit number	RX Channel	b0	RX0	b1	RX1	b2	RX2	b3	RX3		
Bit number	RX Channel													
b0	RX0													
b1	RX1													
b2	RX2													
b3	RX3													

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Table 8.37 – continued from previous page

REPORTING_ MODE	1	Value Definition 0 Report is sent every monitoring period without threshold check 1 Report is send only upon a failure (after checking for thresholds) 2 Report is sent every monitoring period with threshold check
MON_CHIRP_ SLOPE	1	Frequency slope for each monitoring chirp is encoded in 1 bytes (8 bit signed number) For 77GHz Devices (76GHz to 81Ghz): 1 LSB = $3.6e9 \times 900 / 2^{26}$ Hz $\approx$ 48.279 kHz/μs Valid range: -128 to +127 (Max 6.13 MHz/μs) For 60GHz Devices (57GHz to 64Ghz): 1 LSB = $2.7e9 \times 900 / 2^{26}$ Hz $\approx$ 36.21 kHz/μs Valid range: -128 to +127 (Max 4.63 MHz/μs) NOTE: Monitoring Chirp Slope can be programmed based on the emission specifications. The device transmits on air during the execution of these monitors. The host can control the monitoring emission power spectral density (dB-m/Hz) by programming this slope parameter. Each monitoring chirp is about 45us in duration. Therefore the overall RF sweep bandwidth for the monitoring chirp is given by Monitoring Chirp Slope* 45 us. Normally, low values of Monitoring Chirp Slope are recommended, as with non-zero slope in FMCW radar, any actual target reflections can potentially be interpreted as noise power and/or corrupt the loopback signal based gain/phase measurement. The user has to ensure that the RF bandwidth selected for monitoring chirp based on slope and start frequency is within FREQ_LIMIT set in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB API.
TX_GAIN_ MISMATCH_ THRESH	2	The magnitude of difference between measured TX powers across the enabled channels at each enabled RF frequency is compared against this threshold. The comparison result is part of the monitoring report message (Error bit is set if the measurement is above this threshold). Before the comparison, the measured gains for each RF and RX are adjusted by subtracting the offset given in the TX_GAIN_MISMATCH_OFFSET_VALUE field. 1 LSB = 0.1 dB, unsigned number Valid range: 0 to 65535 (0 to 6553.5dB)

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Table 8.37 – continued from previous page

TX_PHASE_MISMATCH_THRESH	2	The magnitude of measured TX phase mismatch across the enabled channels at each enabled RF frequency is compared against this threshold. The comparison result is part of the monitoring report message (Error bit is set if any measurement is above this threshold). Before the comparison, the measured gains for each RF and RX are adjusted by subtracting the offset given in the TX_PHASE_MISMATCH_OFFSET_VALUE field. 1 LSB = $360^{\circ}/2^{16}$, unsigned number Valid range: corresponding to 0° to 359.9°.																				
RESERVED	2	0x0000																				
TX_GAIN_MISMATCH_OFFSET_VALUE	24	The offsets to be subtracted from the measured TX gain for each TX and RF before the relevant threshold comparisons are given here. Byte numbers corresponding to different RX and RF, in this field are here: <table><thead><tr><th></th><th>RF1</th><th>RF2</th><th>RF3</th></tr></thead><tbody><tr><td>TX0</td><td>1:0</td><td>9:8</td><td>17:16</td></tr><tr><td>TX1</td><td>3:2</td><td>11:10</td><td>19:18</td></tr><tr><td>TX2</td><td>5:4</td><td>13:12</td><td>21:20</td></tr><tr><td>TX3</td><td>7:6</td><td>15:14</td><td>23:22</td></tr></tbody></table> 1 LSB = 0.1 dB, signed number Only the entries of enabled RF Frequencies and enabled TX channels are considered.		RF1	RF2	RF3	TX0	1:0	9:8	17:16	TX1	3:2	11:10	19:18	TX2	5:4	13:12	21:20	TX3	7:6	15:14	23:22
	RF1	RF2	RF3																			
TX0	1:0	9:8	17:16																			
TX1	3:2	11:10	19:18																			
TX2	5:4	13:12	21:20																			
TX3	7:6	15:14	23:22																			
TX_PHASE_MISMATCH_OFFSET_VALUE	24	The offsets to be subtracted from the measured TX phase for each TX and RF before the relevant threshold comparisons are given here. Byte numbers corresponding to different RX and RF, in this field are here: <table><thead><tr><th></th><th>RF1</th><th>RF2</th><th>RF3</th></tr></thead><tbody><tr><td>TX0</td><td>1:0</td><td>9:8</td><td>17:16</td></tr><tr><td>TX1</td><td>3:2</td><td>11:10</td><td>19:18</td></tr><tr><td>TX2</td><td>5:4</td><td>13:12</td><td>21:20</td></tr><tr><td>TX3</td><td>7:6</td><td>15:14</td><td>23:22</td></tr></tbody></table> 1 LSB = $360^{\circ}/2^{16}$. Only the entries of enabled RF Frequencies and enabled TX channels are considered.		RF1	RF2	RF3	TX0	1:0	9:8	17:16	TX1	3:2	11:10	19:18	TX2	5:4	13:12	21:20	TX3	7:6	15:14	23:22
	RF1	RF2	RF3																			
TX0	1:0	9:8	17:16																			
TX1	3:2	11:10	19:18																			
TX2	5:4	13:12	21:20																			
TX3	7:6	15:14	23:22																			
RESERVED	4	0x00000000																				

8.19.6 Sub block 0x0285 – AWR_MONITOR_VMON_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to configuring and enabling the voltage monitors.

NOTE1:	This API is only supported in xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices. Once configured and enabled, VMON faults are not reported by the RSS firmware. Fault signals are mapped to the MSS ESM and should be responded to by the MSS application. Please refer to the TRM for more information.
NOTE2:	The BSS firmware enables/disables/self-tests the different BSS VMONs in the below given order when this API is issued. 1. VDDA_BB_1P8V 2. VDDA_VCO_1P8V 3. VDD_RF1_1P0V 4. VDD_RF2_1P0V In case self-tests are enabled and if any of the self-tests fail, the BSS would disable all the BSS VMONs and immediately return an error with the error code indicating the VMON whose self-test failed. It will not proceed ahead and attempt to configure the remaining VMONs in the API.

Table 8.38: AWR_MONITOR_VMON_CONF_SB contents

Field Name	Number of bytes	Description								
SBLKID	2	Value = 0x0285								
SBLKLEN	2	Value = 20								
VDDA_BB_1P8V_EN	1	<table><tr><th>Bit</th><th>Function</th></tr><tr><td>b0</td><td>Set to '0' to disable the VMON. Set '1' to enable the VMON.</td></tr><tr><td>b1</td><td>Set to '0' to disable self-test for the VMON. Set '1' to enable self-test for the VMON.</td></tr><tr><td>b7:b2</td><td>RESERVED</td></tr></table>	Bit	Function	b0	Set to '0' to disable the VMON. Set '1' to enable the VMON.	b1	Set to '0' to disable self-test for the VMON. Set '1' to enable self-test for the VMON.	b7:b2	RESERVED
Bit	Function									
b0	Set to '0' to disable the VMON. Set '1' to enable the VMON.									
b1	Set to '0' to disable self-test for the VMON. Set '1' to enable self-test for the VMON.									
b7:b2	RESERVED									

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Table 8.38 – continued from previous page

VDDA_BB_1P8V_REF	1	Bit range b3:b0 VDDABB UV VMON Reference Selection for normal operation. b7:b4 VDDABB UV VMON Reference Selection during self-test. VDDABB UV VMON Reference values (normal): 0x0 = 0.56V 0x1 = 0.54V 0x2 = 0.52V 0x3 = 0.5V VDDABB UV VMON Reference values (self-test): 0x0 = 0.66V 0x1 = 0.64V 0x2 = 0.62V 0x3 = 0.6V Please note that all reference values for VDDA_BB_1P8V are scaled by 3.
VDDA_VCO_1P8V_EN	1	Bit Function b0 Set to '0' to disable the VMON. Set '1' to enable the VMON. b1 Set to '0' to disable self-test for the VMON. Set '1' to enable self-test for the VMON. b7:b2 RESERVED
VDDA_VCO_1P8V_REF	1	Bit range b3:b0 VDDAVCO UV VMON Reference Selection for normal operation. b7:b4 VDDAVCO UV VMON Reference Selection during self-test. VDDAVCO UV VMON Reference values: 0x0 = 0.56V 0x1 = 0.54V 0x2 = 0.52V 0x3 = 0.5V VDDAVCO UV VMON Reference values (self-test): 0x0 = 0.66V 0x1 = 0.64V 0x2 = 0.62V 0x3 = 0.6V Please note that all reference values for VDDA_VCO_1P8V are scaled by 3.

Continued on next page

Table 8.38 – continued from previous page

VDD_RF1_1P0V_EN	1	Bit	Function
		b0	Set to '0' to disable the VMON. Set '1' to enable the VMON.
		b1	Set to '0' to disable self-test for the VMON. Set '1' to enable self-test for the VMON.
		b7:b2	RESERVED
VDD_RF1_1P0V_REF	1	Bit range	Definition
		b3:b0	VDDRF1 UV VMON Reference Selection for normal operation.
		b7:b4	VDDRF1 UV VMON Reference Selection during self-test.
			VDDRF1 UV VMON Reference values: 0x0 = 0.53V 0x1 = 0.52V 0x2 = 0.51V 0x3 = 0.5V
			VDDRF1 UV VMON Reference values (self-test): 0x0 = 0.62V 0x1 = 0.61V 0x2 = 0.6V 0x3 = 0.59V
			Please note that all reference values for VDD_RF1_1P0V are scaled by 1.8.
VDD_RF2_1P0V_EN	1	Bit	Function
		b0	Set to '0' to disable the VMON. Set '1' to enable the VMON.
		b1	Set to '0' to disable self-test for the VMON. Set '1' to enable self-test for the VMON.
		b7:b2	RESERVED

Continued on next page

Table 8.38 – continued from previous page

VDD_RF2_1P0V_REF	1	Bit range b3:b0 VDDRF2 UV VMON Reference Selection for normal operation. b7:b4 VDDRF2 UV VMON Reference Selection during self-test. VDDRF1 UV VMON Reference values: 0x0 = 0.52V 0x1 = 0.51V 0x2 = 0.49V 0x3 = 0.48V VDDRF2 UV VMON Reference values (self-test): 0x0 = 0.62V 0x1 = 0.61V 0x2 = 0.6V 0x3 = 0.59V Please note that all reference values for VDD_RF2_1P0V are scaled by 1.8.
RESERVED	4	0x00000000
RESERVED	4	0x00000000

8.19.7 Sub block 0x0286 – AWR_MONITOR_OVERRIDES_AND_DITHER_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to the monitoring overrides and dither configuration.

Table 8.39: AWR_MONITOR_OVERRIDES_AND_DITHER_CONF_SB contents

Field Name	Number of bytes	Description
SBLKID	2	Value = 0x0286
SBLKLEN	2	Value = 48

Continued on next page

Table 8.39 – continued from previous page

MON_INDEX	1	Monitoring Index to which this override API is applicable. Value Definition 0 Tx1 power monitor 1 Tx2 power monitor 2 Tx3 power monitor 3 Tx4 power monitor 4 Tx1 Ball Break monitor 5 Tx2 Ball Break monitor 6 Tx3 Ball Break monitor 7 Tx4 Ball Break monitor 8 Tx1 Phase Shifter monitor 9 Tx2 Phase Shifter monitor 10 Tx3 Phase Shifter monitor 11 Tx4 Phase Shifter monitor 12 Tx1 Internal signal monitor 13 Tx2 Internal signal monitor 14 Tx3 Internal signal monitor 15 Tx4 Internal signal monitor 16 Tx Gain Phase monitor 17 Rx Gain Phase monitor 18 Rx Internal signal monitor
MON_RF_PARAMS_OVRD_EN	1	Override enables. Bit Definition b0 Start frequency override enable b1 Slope override enable b2 TX backoff override enable b7:b3 RESERVED
RESERVED	2	RESERVED

Continued on next page

Table 8.39 – continued from previous page

OVRD_RF_ START_FREQ	12	RF start frequency. Bytes Definition Bytes 3:0 RF1 Start frequency override Bytes 7:4 RF2 Start frequency override Bytes 11:8 RF3 Start frequency override 1 LSB = $3.6e9/2^{26}$ Hz $\approx$ 53.644 Hz Valid range: 0x5471C71C to 0x5A000000 NOTE: For monitors that have RF1,2,3 (like TX Power Mon, Tx Gain phase, Rx gain phase), this will have 3 elements (RF1/RF2/RF3). For monitors that have only 1 RF (like TX Ball Break Mon, Tx Phase shifter Mon, Tx Internal signal mon and RX internal signal mon), this will have 1 element (RF2 will be used).
OVRD_RF_ SLOPE	6	RF frequency slope. Bytes Definition Bytes 1:0 RF1 Frequency slope override Bytes 3:2 RF2 Frequency slope override Bytes 5:4 RF3 Frequency slope override 1 LSB = $3.6e9 \times 900/2^{26}$ Hz $\approx$ 48.279 kHz/μs Valid range: -5510 to 5510 (Max 266MHz/us) NOTE: For monitors that have RF1,2,3 (like TX Power Mon, Tx Gain phase, Rx gain phase), this will have 3 elements (RF1/RF2/RF3). For monitors that have only 1 RF (like TX Ball Break Mon, Tx Phase shifter Mon, Tx Internal signal mon and RX internal signal mon), this will have 1 element (RF2 will be used).
OVRD_TX_ BACKOFF	1	TX backoff to be used during the monitor. 1 LSB = 1dB Applicable only for TX power monitor and TX ball break monitor.

Continued on next page

Table 8.39 – continued from previous page

MON_DITHER_CTRL	1	Control field for dither enables. Bit Definition b0 Frequency dithering enable b1 Time dithering enable b7:b2 RESERVED NOTE: If time dithering is enabled, the user has to account for the additional time incurred during monitoring chirps for computing the overall monitoring duration and critical time in an FTTI. For each monitoring chirp with dither enabled, an additional 100uS needs to be added to the critical time and the duration. The number of monitoring chirps for each analog monitor is tabulated in Table 12.8.
RF_MIN_DITHER_FREQ	6	RF1/2/3 minimum dither frequency Bytes Definition Bytes 1:0 RF1 offset frequency dither min limit Bytes 3:2 RF2 offset frequency dither min limit Bytes 5:4 RF3 offset frequency dither min limit 1 LSB = 250kHz Valid range: -128MHz to 128MHz NOTE: For monitors that have RF1,2,3 (like TX Power Mon, Tx Gain phase, Rx gain phase), this will have 3 elements (RF1/RF2/RF3). For monitors that have only 1 RF (like TX Ball Break Mon, Tx Phase shifter Mon, Tx Internal signal mon and RX internal signal mon), this will have 1 element (RF2 will be used). Note: Max limits should be greater than the min limits.

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Table 8.39 – continued from previous page

RF_MAX_ DITHER_FREQ	6	RF1/2/3 maximum dither frequency Bytes Definition Bytes 1:0 RF1 offset frequency dither max limit Bytes 3:2 RF2 offset frequency dither max limit Bytes 5:4 RF3 offset frequency dither max limit 1 LSB = 250kHz Valid range: -128MHz to 128MHz NOTE: For monitors that have RF1,2,3 (like TX Power Mon, Tx Gain phase, Rx gain phase), this will have 3 elements (RF1/RF2/RF3). For monitors that have only 1 RF (like TX Ball Break Mon, Tx Phase shifter Mon, Tx Internal signal mon and RX internal signal mon), this will have 1 element (RF2 will be used). Note: Max limits should be greater than the min limits.
MON_TIME_ DELAY_MIN_ DITHER	1	The monitoring chirp will begin after waiting an additional random delay governed by the minimum and maximum dither settings. 1 LSB = 0.5uS Valid range: 0 to 99uS. Note: Max limit should be greater than the min limit.
MON_TIME_ DELAY_MAX_ DITHER	1	The monitoring chirp will begin after waiting an additional uniform random delay governed by the minimum and maximum dither settings. 1 LSB = 0.5uS Valid range: 0 to 99uS. Note: Max limit should be greater than the min limit.
RESERVED	2	RESERVED
RESERVED	4	RESERVED

8.19.8 Sub block 0x0287 – AWR_MONITOR_TX_PHSHIFTER_DAC_CONF_SB

This API is a monitoring configuration API which the host sends to the AWR device, containing information related to the TX phaseshifter DAC monitoring. The report is sent as an async event [AWR_MONITOR_TX_PHSHIFTER_DAC_REPORT_AE_SB](#).

NOTE: This API is only applicable in xWR294x, xWR254x, xWR2x4xECo, xWR2x4xP devices. The previous generation devices have the same functionality as part of the TX1, 2, 3_INTERNAL_SIGNALS_MONITOR

Table 8.40: AWR_MONITOR_TX_PHSHIFTER_DAC_CONF_SB contents

Field Name	Number of bytes	Description												
SBLKID	2	Value = 0x0287												
SBLKLEN	2	Value = 24												
PROFILE_INDX	1	This field indicates the Profile Index for which this monitoring configuration applies.												
REPORTING_MODE	1	<table><tr><th>Value</th><th>Definition</th></tr><tr><td>0</td><td>Report is sent every monitoring period without threshold check</td></tr><tr><td>1</td><td>Report is send only upon a failure (after checking for thresholds)</td></tr><tr><td>2</td><td>Report is sent every monitoring period with threshold check</td></tr></table>	Value	Definition	0	Report is sent every monitoring period without threshold check	1	Report is send only upon a failure (after checking for thresholds)	2	Report is sent every monitoring period with threshold check				
Value	Definition													
0	Report is sent every monitoring period without threshold check													
1	Report is send only upon a failure (after checking for thresholds)													
2	Report is sent every monitoring period with threshold check													
TX_SEL_MASK	1	TX channel bit mask indicating which of the TX-N monitors have to be updated with the current configuration. Setting a bit to 1 updates the corresponding TX-N monitor to be configured. <table><tr><th>Bit</th><th>TX index</th></tr><tr><td>b0</td><td>Configure TX0 phaseshifter DAC monitor</td></tr><tr><td>b1</td><td>Configure TX1 phaseshifter DAC monitor</td></tr><tr><td>b2</td><td>Configure TX2 phaseshifter DAC monitor</td></tr><tr><td>b3</td><td>Configure TX3 phaseshifter DAC monitor</td></tr><tr><td>b7:4</td><td>RESERVED.</td></tr></table>	Bit	TX index	b0	Configure TX0 phaseshifter DAC monitor	b1	Configure TX1 phaseshifter DAC monitor	b2	Configure TX2 phaseshifter DAC monitor	b3	Configure TX3 phaseshifter DAC monitor	b7:4	RESERVED.
Bit	TX index													
b0	Configure TX0 phaseshifter DAC monitor													
b1	Configure TX1 phaseshifter DAC monitor													
b2	Configure TX2 phaseshifter DAC monitor													
b3	Configure TX3 phaseshifter DAC monitor													
b7:4	RESERVED.													
RESERVED	1	RESERVED												
TX0_PS_DAC_MON_THRESH	2	The TX0 phase shifter DAC monitor delta threshold when TX0 phaseshifter DAC monitor is Enabled Unit: 1 LSB = 1.8V/1024 Valid Range: 1 to 1023												
TX1_PS_DAC_MON_THRESH	2	The TX1 phase shifter DAC monitor delta threshold when TX1 phaseshifter DAC monitor is Enabled Unit: 1 LSB = 1.8V/1024 Valid Range: 1 to 1023												
TX2_PS_DAC_MON_THRESH	2	The TX2 phase shifter DAC monitor delta threshold when TX2 phaseshifter DAC monitor is Enabled Unit: 1 LSB = 1.8V/1024 Valid Range: 1 to 1023												
TX3_PS_DAC_MON_THRESH	2	The TX3 phase shifter DAC monitor delta threshold when TX3 phaseshifter DAC monitor is Enabled Unit: 1 LSB = 1.8V/1024 Valid Range: 1 to 1023												

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Table 8.40 – continued from previous page

RESERVED	4	RESERVED
RESERVED	4	RESERVED

9 Unsupported Features/APIs and Debug APIs

9.1 Unsupported Features/APIs and Debug APIs

The list of unsupported features, APIs and debug APIs are highlighted in latest DFP release note. Please refer latest AWR2243 DFP release note.

10 Chirp Parameters (CP) and Chirp Quality (CQ) data

10.1 Chirp Parameters data

Chirp parameter information is always updated in the CP registers DSS_REG_VBUSM__CPREG[0-3] for single chirp use case.

NOTE: Chirp Number is always reset every burst by the hardware.

		Channel 0				Channel 1				Channel 2				Channel 3			
		Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15
Bits	0	Channel Number	Reserved	Chirp Number[7:0]	Chirp Number[11:8]	Channel Number	Reserved	Chirp Number[7:0]	Chirp Number[11:8]	Channel Number	Reserved	Chirp Number[7:0]	Chirp Number[11:8]	Channel Number	Reserved	Chirp Number[7:0]	Chirp Number[11:8]
	1																
	2																
	3																
4	5	Profile Number	Reserved	Chirp Number[7:0]	Chirp Number[11:8]	Profile Number	Reserved	Chirp Number[7:0]	Chirp Number[11:8]	Profile Number	Reserved	Chirp Number[7:0]	Chirp Number[11:8]	Profile Number	Reserved	Chirp Number[7:0]	Chirp Number[11:8]
6	7																
8	9																
10	11																
12	13	Reserved	Reserved	Chirp Number[7:0]	Chirp Number[11:8]	Reserved	Reserved	Chirp Number[7:0]	Chirp Number[11:8]	Reserved	Reserved	Chirp Number[7:0]	Chirp Number[11:8]	Reserved	Reserved	Chirp Number[7:0]	Chirp Number[11:8]
14	15																
16	17																
18	19																

Figure 10.1: Chirp parameter information fields

	31	23	16	15	8	7	0
DSS_REG_VBUSM.CH0CPREG0	Byte 3	Byte 2	Byte 1	Byte 0			
DSS_REG_VBUSM.CH0CPREG1	Byte 7	Byte 6	Byte 5	Byte 4			
DSS_REG_VBUSM.CH0CPREG2	Byte 11	Byte 10	Byte 9	Byte 8			
DSS_REG_VBUSM.CH0CPREG3	Byte 15	Byte 14	Byte 13	Byte 12			

Figure 10.2: Chirp parameter information from DSS registers

For multichip use case, the CP data is available for up to 8 chirps in DSS_REG_VBUSM.CH[0-7]CPREG[0-3].

10.2 Chirp Quality data

Chirp quality information is divided into 3 parts

1. CQ0 – Wideband signal and image energy information (Reserved for future use)
2. CQ1 – RX signal and image band energy statistics
3. CQ2 – RX ADC and IF saturation information

CQ data will be available in CQ RAM which is a ping-pong memory when the CQ monitors are enabled. Currently supported CQ monitors are AWR_MONITOR_RX_SATURATION_DETECTOR_CONF_SB for CQ2 and AWR_MONITOR_SIG_IMG_MONITOR_CONF_SB for CQ1. CQ data will be refreshed every chirp by the hardware. User has to ensure that before the next chirp finishes, the current chirps' CQ data is either processed or transferred to a local memory for further processing.

NOTE:

CQ0 is not supported by firmware currently, but the CQ RAM will be updated for CQ0 data. Maximum size of CQ0 data is 256 bytes. Users should ignore the CQ RAM for CQ0.

The starting location (on 128 bit boundary) of each CQ data within the CQ memory can be configured by programming DSS_REG.CQCFG1[12:4] for CQ0, DSS_REG.CQCFG1[21:13] for CQ1 and DSS_REG.CQCFG1[30:22] for CQ2.

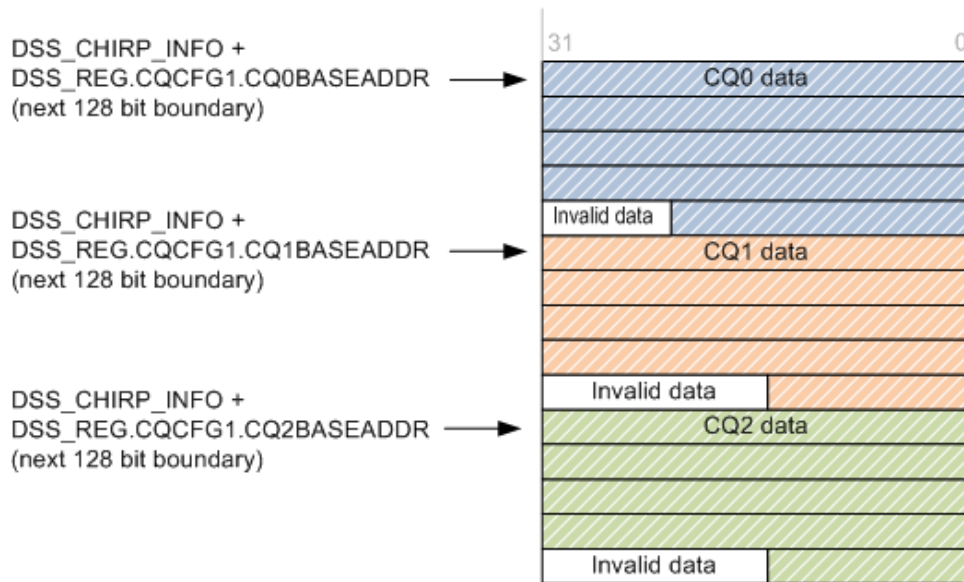


Figure 10.3: CQ data start address configuration in single chirp use case

For N-chirp use case, when user wishes to process N chirps simultaneously, then CQ0 for all N chirps will be concatenated together in memory. Similarly CQ1 and CQ2 for all N chirps will also be concatenated together.

NOTE: When CQ data is concatenated in N-chirp use case, the CQ data for new chirp starts on the next 128 bit boundary.

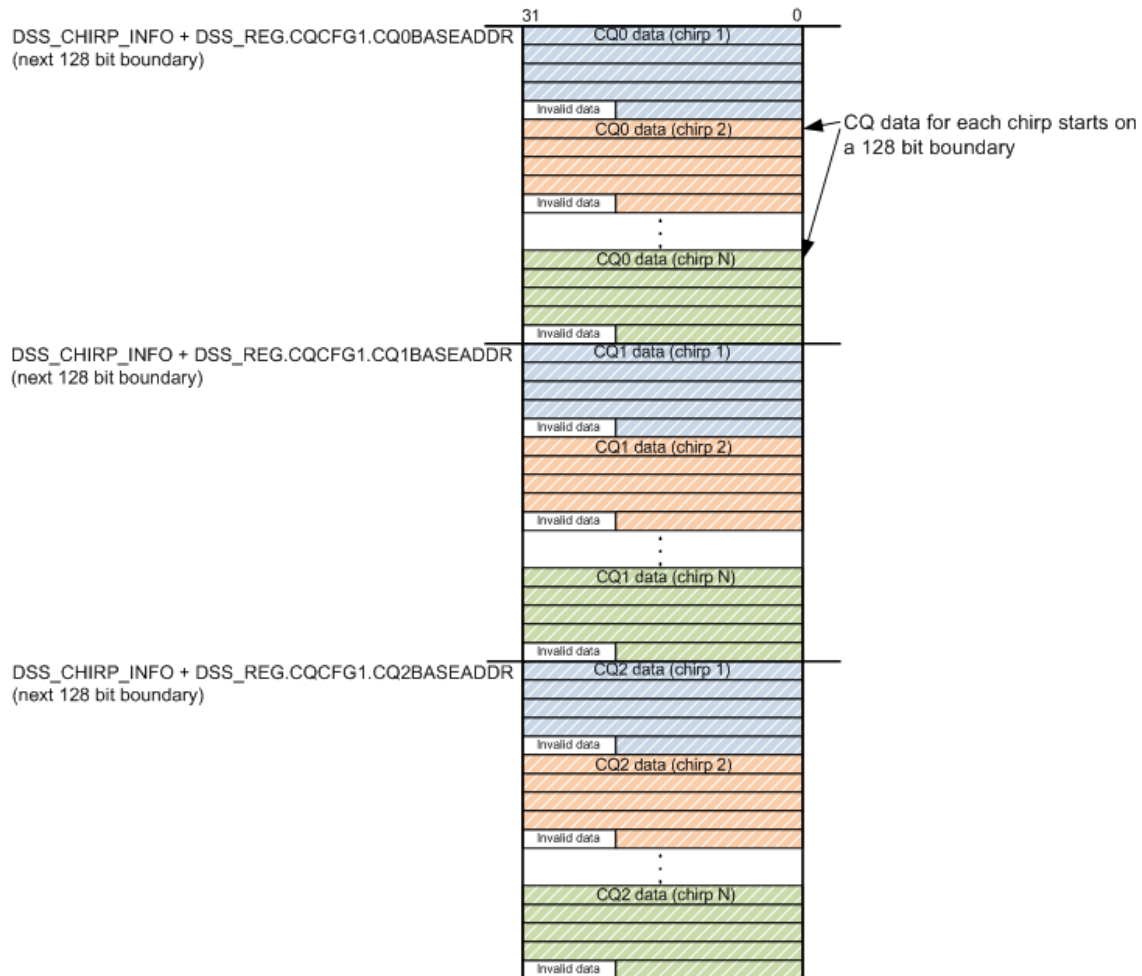


Figure 10.4: CQ data start address configuration in multi chirp use case

The CQDATAWIDTH parameter in DSS_REG.CQCFG1 defines the packing of the CQ data in the CQ memory in either 16-bit mode, 12-bit mode or in 14-bit mode.

10.2.1 CQ1

The signal band and image band are separated using a two-channel filter bank and the ADC sampling time duration is monitored in terms of primary and secondary time slices, as shown below.

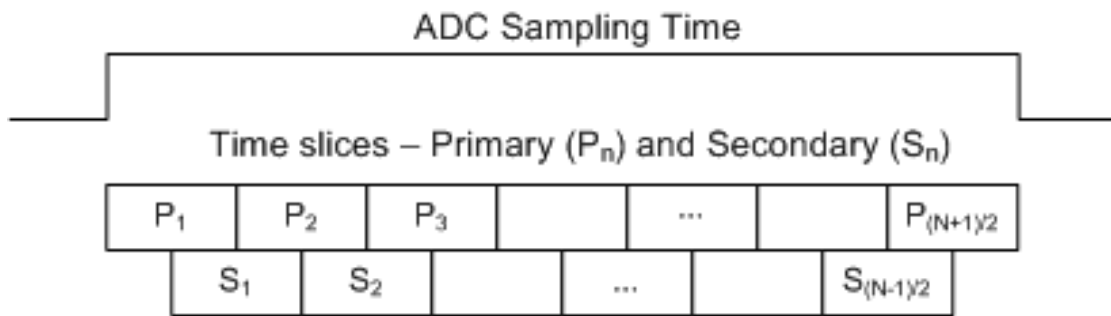


Figure 10.5: Time slices during RX signal and image band monitor and saturation monitor

For each of the two bands (signal and image), for each time slice, the input-referred average power in the slice in negative dBm is recorded as an 8-bit unsigned number, with 1 LSB = -0.5 dBm

CQ1 data is stored in memory as shown below (in 16-bit mode)

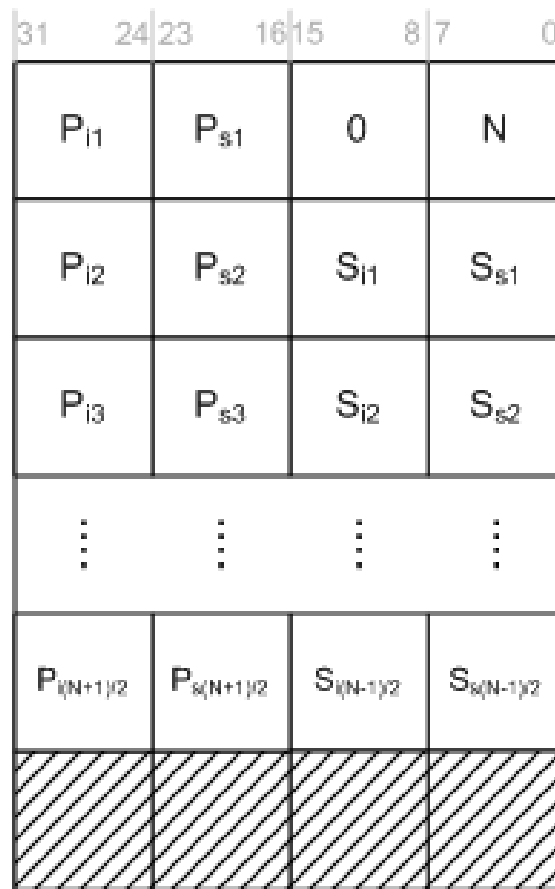


Figure 10.6: CQ1 data format in memory in 16-bit mode

N indicates the total number of primary and secondary slices which are monitored (maximum value of N is 127). P_{s,i_n} indicates the power of primary slice n for {signal, image} band and S_{s,i_n} indicates the power of secondary slice n for {signal, image} band. Each power is encoded in 8 bit unsigned number with each LSB representing -0.5 dBm.

Since maximum value of N is 127, the maximum size of CQ1 data in 16-bit mode is 256 bytes

NOTE: In real output mode, since there is no image band visibility, only the signal band statistics will be meaningful.

Similarly, in 12-bit and 14-bit modes, the CQ1 data in CQ memory will be packed as shown below. Only the relevant bits in each 16 bits of memory (either 12 bits or 14 bits) are useful and other bits are not written by hardware.

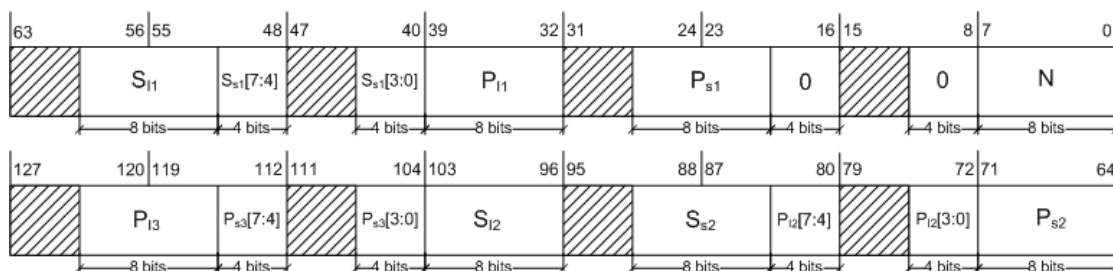


Figure 10.7: CQ1 data format in memory in 12-bit mode

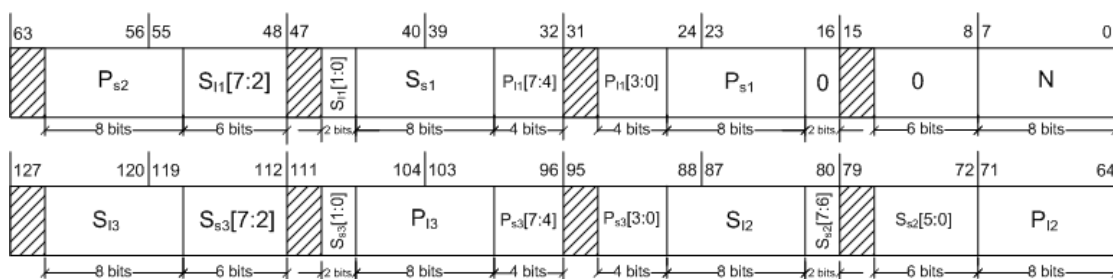


Figure 10.8: CQ1 data format in memory in 14-bit mode

10.2.2 CQ2

The analog to digital interface includes a 100 MHz bit stream indicating saturation events in the ADC/IF sections, for each channel. This one-bit indicator for each channel is monitored during the ADC sampling time duration in a time-sliced manner, as shown in Figure 10.5.

For each time slice, a saturation event count is recorded. This count is the sum of saturation event counts across all RX channels selected for monitoring, capped to a maximum count of 255 (8 bits). The saturation counts are stored in memory as shown below

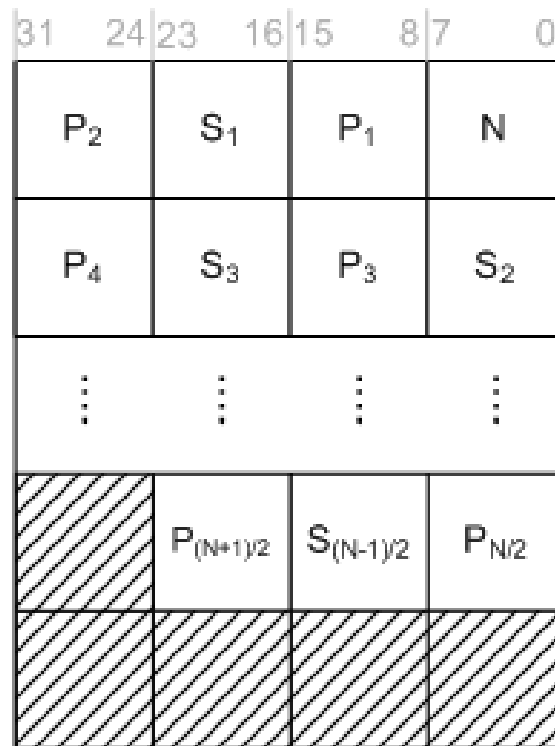


Figure 10.9: CQ2 data format in memory in 16-bit mode

N indicates the total number of primary and secondary slices which are monitored (maximum value of N is 127). P_n indicates the accumulated saturation count for all enabled RX channels in primary slice n , S_n indicates the accumulated saturation count for all enabled RX channels in secondary slice n .

Since maximum value of N is 127, the maximum size of CQ2 data in 16-bit mode is 128 bytes. Similarly, in 12-bit and 14-bit modes, the CQ2 data in CQ memory will be packed as shown below. Only the relevant bits in each 16 bits of memory (either 12 bits or 14 bits) are useful and other bits are not written by hardware.

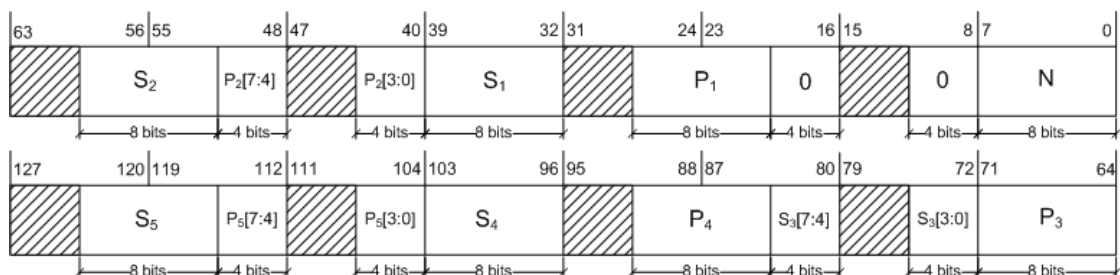


Figure 10.10: CQ2 data format in memory in 12-bit mode

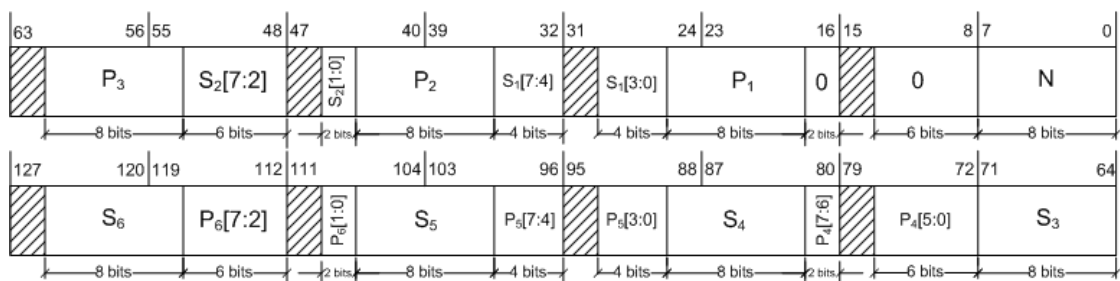


Figure 10.11: CQ2 data format in memory in 14-bit mode

11 Chirp, Burst and Frame timings

xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP device minimum chirp cycle time, inter-burst time, inter sub-frame, frame time requirements are documented in this section.

11.1 Chirp Cycle Time

NOTE1: ADC_START_TIME_RES and INTER_CHIRP_JITTER_MITIGATION is not available on xWR294x devices

Table 11.1: Minimum chirp cycle time for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices

Use case	Min Chirp cycle time (μ s)	Description
Typical chirps	13.5	The normal chirps used in a burst or a frame using legacy chirp configuration API
Advance chirps	18	Minimum advance chirp time used in a burst or a frame using advanced chirp configuration API with minimum mandatory parameters enabled (CHIRP_PROFILE_SELECT, CHIRP_TX_EN, CHIRP_BPM_VAL). When additional parameters are enabled in the advanced chirp API add the timings of enabled parameters to the minimum advance chirp time to compute the total time

Continued on next page

Table 11.1 – continued from previous page

		API	Parameter	Time in us
		AWR_ADVANCE_CHIRP_ CONF_SB	CHIRP_FREQ_START_ VAR	1.3
		AWR_ADVANCE_CHIRP_ CONF_SB	CHIRP_FREQ_SLOPE_ VAR	1
		AWR_ADVANCE_CHIRP_ CONF_SB	CHIRP_IDLE_TIME_VAR	0.9
		AWR_ADVANCE_CHIRP_ CONF_SB	CHIRP_ADC_START_ TIME_VAR	0.9
		AWR_ADVANCE_CHIRP_ CONF_SB	TXn_PHASE_SHIFTER	3.8
		AWR_RF_MISC_CTL_SB	ADC_START_TIME_RES	1.2
		AWR_RF_MISC_CTL_SB	INTER_CHIRP_JITTER_ MITIGATION	0.4
chirps in Con- tinuous framing mode	21	A single advance chirp used in a burst using either normal legacy chirp or advanced chirp configuration API. Continuous framing mode is a mode in which a single chirp is programmed in a burst using advanced frame configuration API. In this mode it is recommended to set idle time of chirp minimum 10us to save Inter chirp power save override time (Refer below table)		

11.2 Minimum Inter Burst Time

Table 11.2: Minimum inter burst time for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices

Min inter burst time	Time (μ s)	Description
Typical inter burst time	57	The minimum inter burst idle time required in normal bursts with legacy chirps configured in a advanced frame configuration API with inter burst power save disabled.
Inter burst SYNTH power save time	55	Add inter burst SYNTH power save time to minimum inter burst time if it is enabled. By default inter-burst SYNTH power save is enabled, it can be disabled (Only in single chip mode) using AWR_RF_DEVICE_CFG_SB API
Inter burst APLL power save time	37	Add inter burst APLL power save time to minimum inter burst time if it is enabled. By default inter-burst APLL power save is enabled, it can be disabled (Only in single chip mode) using AWR_RF_DEVICE_CFG_SB API
Inter chirp power save override time (power save disable)	15	Add inter chirp power save override time to minimum inter burst time if chirp idle time < 10us in a burst or can be controlled using AWR_DYNAMICPOWERSAVE_CONF_SET_SB API
Advance chirp configuration time	45	Add advance chirp configuration time to minimum inter burst time if advance chirp configuration is enabled in AWR_RF_RADAR_MISC_CTL_SB API. Not applicable for Continuous framing mode (Refer below)
Advance chirp configuration time (Continuous framing mode)	30	Add Continuous framing mode advance chirp configuration time to minimum inter burst time if advance chirp configuration is enabled in AWR_RF_RADAR_MISC_CTL_SB API. Continuous framing mode is a mode in which a single chirp is programmed in a burst using advanced frame configuration API.
Normal chirps (Continuous framing mode)	10	Add Continuous framing mode normal chirp configuration time to minimum inter burst time. Continuous framing mode is a mode in which a single chirp is programmed in a burst using advanced frame configuration API.

Table 11.3: Minimum inter burst time for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices

Min inter burst time	Time (μ s)	Description
Calibration or Monitoring chirp time	145	Add calibration or Monitoring chirp time to minimum inter burst time if calibration or monitors intended to be run in inter burst idle time. The calibration and monitoring chirps can run only in inter sub-frame or inter-frame interval if this time is not allocated in inter-burst time. Add calibration or Monitoring duration to minimum inter burst or sub-frame/frame time based on Table 12.4 and Table 12.9

11.3 Minimum Inter Sub-frame or Frame Time

Table 11.4: Minimum inter sub-frame, frame time for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices

Min inter sub-frame/frame time	Time (μ s)	Description
Typical inter sub-frame/frame time	313	The minimum inter sub-frame/frame idle time required in normal sub-frames with legacy chirps configured in a advanced frame configuration API or in a legacy frame config API. This time includes time required for minimum inter-burst idle time, inter burst power save, inter chirp power save override and single calibration/monitoring chirp time.
Advance chirp configuration time	45	Add advance chirp configuration time to minimum inter sub-frame/frame time if advance chirp configuration is enabled in AWR_RF_RADAR_MISC_CTL_SB API
Calibration or Monitoring duration	Table 12.4 and Table 12.9	Add calibration or Monitoring duration to minimum inter sub-frame/frame time based on Table 12.4 and Table 12.9
Loop-back burst configuration time	300	Add Loop-back burst configuration time to minimum inter sub-frame time for loop back sub-frames if it is enabled in advance frame config API.
Dynamic legacy chirp configuration time (for 16 chirps)	20 for 16 chirps + 500	Add dynamic legacy chirp configuration time to minimum inter frame time if dynamic chirp/phase-shifter APIs are issued in runtime.
Dynamic advance chirp configuration time (without LUT)	500	Add dynamic advance chirp configuration time to minimum inter frame time if dynamic advance chirp API is issued in runtime. The dynamic update of advance chirp generic LUT is done immediately when the API is received at BSS and there is no impact to inter frame time, however user has to take care of timing of the LUT update as it should not corrupt the ongoing chirp configuration.
Dynamic profile configuration time (for 1 profile)	1200	Add dynamic profile configuration time to minimum inter frame time if dynamic profile API is issued in runtime.
Test source config time	190	Add test source configuration time to minimum inter sub-frame time if test source API is issued.

12 Calibration and monitoring durations

12.1 Boot time calibration durations

xWR294x, xWR254x, AWR2243, xWR6243, xWR2x4xECO, xWR2x4xP device boot time calibration durations are mentioned in the tables below:

Table 12.1: Duration of boot time calibrations for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP device

Sl. No.	Calibration	Duration (μ s)
1	APLL	500
2	Synth VCO	2600
3	LO DIST	1500
4	ADC DC	600
5	HPF cutoff	3000
6	LPF cut off	200
7	Peak detector	7000
8	TX power (for each TX)	2000
9	RX gain	1500
10	TX phase (for each TX)	12 000

Table 12.2: Duration of boot time calibrations for AWR2243 device

Sl. No.	Calibration	Duration (μ s)
1	APLL	500
2	Synth VCO	2500
3	LO DIST	1500
4	ADC DC	600
5	HPF cutoff	3500
6	LPF cut off	200
7	Peak detector	7000
8	TX power (for each TX)	2000
9	RX gain	1500
10	TX phase (for each TX)	12 000
11	RX IQMM	42 000

Table 12.3: Duration of boot time calibrations for xWR6243 device

Sl. No.	Calibration	Duration (μ s)
1	APLL	500
2	Synth VCO (VCO1,VCO2)	2500
	Synth VCO (VCO1,VCO2,VCO3)	3750
3	LO DIST	12
4	ADC DC	600
5	HPF cutoff	3500
6	LPF cut off	200
7	Peak detector	8000
8	TX power (for each TX)	3000
9	RX gain	1500
10	TX phase (for each TX)	12 000
11	RX IQMM	42 000

12.2 Run time calibration durations

xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP runtime calibration durations are captured in Table 12.4, AWR2243 run time calibration durations captured in Table 12.5 and xWR6243 run time calibration durations are captured in Table 12.6. The Calibration Total duration is consist of two components,

1. Critical calibration chirp time, this is fixed to 145 μ s. Each calibration can have multiple critical chirps to complete the task, the total critical time captured in below table. The critical tasks are done in chunks of 145 μ s, refer Table 11.3 for more info.

2. Non critical setup and processing time, this is a variable component specific to each calibration, the total calibration duration captured in below table = total critical time + non critical time.

Note that the User has to ensure the total idle time in one CALIB_MON_TIME_UNIT is sufficient to fit the enabled calibrations.

Table 12.4: Duration of run time calibrations for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP devices

Sl. No.	Calibration	Total Duration (μ s)	Total Critical Time (μ s)
1	APLL	150	145
2	Synth VCO	300	290
3	LO DIST	30	0
4	Peak detector	600	580
5	TX power CLPC (for each TX and for 1 profile)	720	580
6	TX power OLPC (In case CLPC is disabled)	30	0
7	RX gain	30	0
8	TX phase	700	0
9	Application of calibration to hardware (This needs to be included always)	60	60

Table 12.5: Duration of run time calibrations for AWR2243 devices

Sl. No.	Calibration	Total Duration (μ s)	Total Critical Time (μ s)
1	APLL	150	145
2	Synth VCO	300	290
3	LO DIST	30	0
4	Peak detector	600	580
5	TX power CLPC (for each TX and for 1 profile)	800	580
6	TX power OLPC (In case CLPC is disabled)	30	0
7	RX gain	30	0
8	Application of calibration to hardware (This needs to be included always)	50	50

Table 12.6: Duration of run time calibrations for xWR6243 devices

Sl. No.	Calibration	Total Duration (μ s)
1	APLL	150
2	Synth VCO (VCO1,VCO2)	350
	Synth VCO (VCO1,VCO2,VCO3)	525
3	LO DIST	30
4	Peak detector	600
5	TX power CLPC (for each TX and for 1 profile)	800
6	TX power OLPC (In case CLPC is disabled)	30
7	RX gain	30
8	Application of calibration to hardware (This needs to be included always)	50

To configure CALIB_MON_TIME_UNIT, user has to calculate the total available IDLE time in the frames and please refer Table 11.3 for the same. The duration for all the enabled calibrations should be included and following Table 12.14 software overheads should be added to that number.

12.3 Monitoring duration

Table 12.7 lists the duration of all analog monitors for xWR254x, xWR2x4xECO, xWR2x4xP and Table 12.8 lists the duration of all analog monitors for xWR294x and Table 12.9 lists the duration of all digital monitors for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP. Table 12.10 lists the duration of all analog monitors for AWR2243 and Table 12.11 lists the duration of all digital monitors for AWR2243. Table 12.12 lists the duration of all analog monitors for xWR6243 and Table 12.13 lists the duration of all digital monitors for xWR6243.

The Monitor Total duration is consist of two components,

1. Critical Monitoring chirp time, this is fixed to 145 μ s. Each Monitor can have multiple critical chirps to complete the task, the total critical time captured in below table. The critical tasks are done in chunks of 145 μ s, refer Table 11.3 for more info.
2. Non critical setup and processing time, this is a variable component specific to each Monitoring, the total Monitor duration captured in below table = total critical time + non critical time.

Note that the User has to ensure the total idle time in one CALIB_MON_TIME_UNIT is sufficient to fit the enabled monitoring.

Table 12.7: Duration of analog monitors for xWR254x, xWR2x4xECO, xWR2x4xP device

Sl. No.	Monitors	Total Duration (μ s)	Total Critical Time (μ s)	Num of mon chirps
1	RX gain phase (assumes 1 RF frequency)	480	290	2
2	RX noise figure (assumes 1 RF frequency)	270	145	1
3a	RX IF stage (LPF monitor disabled)	750	580	4
3b	RX IF stage (LPF monitor enabled. Debug only)	2490	2320	16
4	TX power (assumes 1 TX, 1 RF frequency)	240	145	1
5	TX ballbreak (assumes 1 TX)	250	145	1
6	TX gain phase mismatch (assumes 1 TX, 1 RF frequency)	310	130	1
7	TX phase shifter (assumes 1 TX and 1 phase)	300	130	1
8	Synthesizer frequency Live	50	0	0
9	TX Internal analog signals (assumes 1 TX)	230	130	1
10	TX Phase shifter DAC monitor	2650	2610	18
11	RX internal analog signals	670	520	4
12	PMCLKLO internal analog signals	250	145	1
13	GPADC internal signals	80	70	0
14	PLL control voltage (APLL and SYNTH)	230	200	0
15	Dual clock comparator	190	190	0
16	RX saturation detector	50	0	0
17	Synthesizer frequency Non-live monitor (Single VCO)	235	145	1
18	Temperature (Critical time is accounted in Software overheads)	150	100	0

Table 12.8: Duration of analog monitors for xWR294x device

Sl. No.	Monitors	Total Duration (μ s)	Total Critical Time (μ s)	Num of mon chirps
1	RX gain phase (assumes 1 RF frequency)	480	290	2
2	RX noise figure (assumes 1 RF frequency)	270	145	1
3	RX IF stage	1110	725	5
4	TX power (assumes 1 TX, 1 RF frequency)	240	145	1
5	TX ballbreak (assumes 1 TX)	250	145	1
6	TX gain phase mismatch (assumes 1 TX, 1 RF frequency)	310	130	1
7	TX phase shifter (assumes 1 TX and 1 phase)	300	130	1
8	Synthesizer frequency Live	50	0	0
9	TX Internal analog signals (assumes 1 TX)	230	130	1
10	TX Phase shifter DAC monitor	2650	2610	18
11	RX internal analog signals	670	520	4
12	PMCLKLO internal analog signals	250	145	1
13	GPADC internal signals	80	70	0
14	PLL control voltage (APLL and SYNTH)	230	200	0
15	Dual clock comparator	140	140	0
16	RX saturation detector	50	0	0
17	Synthesizer frequency Non-live monitor (Single VCO)	235	145	1
18	Temperature (Critical time is accounted in Software overheads)	150	100	0

Table 12.9: Duration of digital monitors for xWR294x, xWR254x, xWR2x4xECO, xWR2x4xP device

Sl. No.	Monitors	Total Duration (μ s)	Total Critical Time (μ s)
1	Periodic configuration register readback	50	50
2	DFE LBIST monitoring	580	580
3	Frame timing monitoring	10	0

Table 12.10: Duration of analog monitors for AWR2243 device

Sl. No.	Monitors	Total Duration (μ s)	Total Critical Time (μ s)
1	RX gain phase (assumes 1 RF frequency)	1100	725
2	RX noise figure (assumes 1 RF frequency)	250	145
3	RX IF stage (assumes 1 RF frequency)	1300	725
4	TX power (assumes 1 TX, 1 RF frequency)	220	145
5	TX ballbreak (assumes 1 TX)	250	145
6	TX gain phase mismatch (assumes 1 TX, 1 RF frequency)	350	145
7	TX phase shifter (assumes 1 TX and 1 phase)	250	145
8	Synthesizer frequency Live	50	0
9	External analog signals (all 6 GPADC channels enabled)	150	100
10	TX Internal analog signals (assumes 1 TX)	250	145
	TX Phase shifter DAC monitor (assumes 1 TX)	2250	2175
11	RX internal analog signals	1950	1740
12	PMCLKLO internal analog signals and 20G Sync	550	435
13	GPADC internal signals	50	30
14	PLL control voltage (APLL and SYNTH)	300	260
15	Dual clock comparator (assumes 6 clock comparators)	280	270
16	RX saturation detector	50	0
17	RX signal and image band monitor	50	0
18	RX mixer input power	650	580
19	Synthesizer frequency Non-live monitor (Single VCO)	235	145

Table 12.11: Duration of digital monitors for AWR2243 device

Sl. No.	Monitors	Total Duration (μ s)	Total Critical Time (μ s)
1	Periodic configuration register readback	50	30
2	DFE LBIST monitoring	300	240
3	Frame timing monitoring	10	0

Table 12.12: Duration of analog monitors for xWR6243 device

Sl. No.	Monitors	Total Duration (μ s)
1	RX gain phase (assumes 1 RF frequency)	1250
2	RX noise figure (assumes 1 RF frequency)	250
3	RX IF stage (assumes 1 RF frequency)	1400
4	TX power (assumes 1 TX, 1 RF frequency)	250
5	TX ballbreak (assumes 1 TX)	300
6	TX gain phase mismatch (assumes 1 TX, 1 RF frequency)	400
7	TX phase shifter (assumes 1 TX and 1 phase)	250
8	Synthesizer frequency Live	50
9	External analog signals (all 6 GPADC channels enabled)	150
10	TX Internal analog signals (assumes 1 TX)	300
	TX Phase shifter DAC monitor (assumes 1 TX)	2300
11	RX internal analog signals	1950
12	PMCLKLO internal analog signals and 20G Sync	550
13	GPADC internal signals	50
14	PLL control voltage (APLL and SYNTH)	300
15	Dual clock comparator (assumes 6 clock comparators)	300
16	RX saturation detector	100
17	RX signal and image band monitor	100
18	RX mixer input power	700
19	Synthesizer frequency Non-live monitor (Single VCO)	250

Table 12.13: Duration of digital monitors for xWR6243 device

Sl. No.	Monitors	Total Duration (μ s)
1	Periodic configuration register readback	70
2	DFE LBIST monitoring	300
3	Frame timing monitoring	10

To configure CALIB_MON_TIME_UNIT, user has to calculate the total available IDLE time in the frames and please refer Table 11.3 for the same. The duration for all the enabled monitors should be included and following Table 12.14 software overheads should be added to that number.

12.4 Software overheads

When the calibrations or monitorings are enabled, the software needs certain time for reading the temperature sensors, reading the DFE statistics, preparing the calibration or monitoring reports and to clear the watchdog (WDT). All these time durations should also be accounted when computing the CALIB_MON_TIME_UNIT. The details of the software overheads are given in the Table 12.14

Table 12.14: Software overheads every FTTI that should be accounted to program CALIB_MON_TIME_UNIT and CALIBRATION_PERIODICITY

Sl. No.	Software overhead	Total Duration (μ s)	Total Critical Time (μ s)
1	Periodic monitoring of stack usage	20	20
2	Minimum monitoring duration (report formation, digital energy monitor at the end of CALIB_MON_TIME_UNIT, temperature read every CALIB_MON_TIME_UNIT)	600	100
3	Minimum run time calibration duration (report formation, temperature read every CALIB_MON_TIME_UNIT)	300	100
4	Idle time needed per FTTI for windowed watchdog (WDT)	Frame period $\times$ CALIB_MON_TIME_UNIT/8 i.e. $\sim 12.5\%$ of Frame period $\times$ CALIB_MON_TIME_UNIT is reserved for watchdog clearing time	0

12.4.1 Note on idle time for clearing the watchdog (WDT)

The clearing window of the watchdog is 12.5% of total FTTI as shown in the figure below. One FTTI can have multiple frames in legacy frame configuration or in advanced frame configuration - each frame can have multiple sub-frames and each sub-frame can have multiple bursts.

The required idle time for clearing watchdog is absolute 12.5% of the overall FTTI interval, this 12.5% clearing window can have multiple frames or subframes or bursts. The granularity of the required watchdog idle time calculation is limited to sub-frame period.

Example

A user has enabled advanced frame configuration where each frame consists of 3 sub-frames and each sub-frame is of 5 ms duration. FTTI is configured as 25 frames. Each sub-frame contains 100 chirps, each chirp consisting of 4 μ s idle time and 21 μ s ramp time. i.e. duty cycle is 50%. The watchdog clearing window and time for calibration and monitoring is calculated as follows

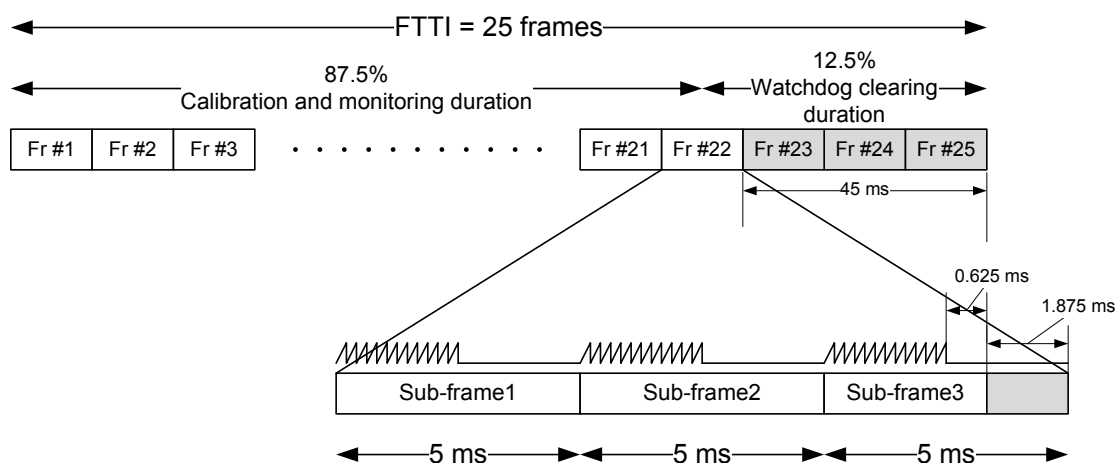


Figure 12.1: Watchdog idle time calculation

Frame duty cycle	=	50	%
Idle time per frame (50% of 15 ms)	=	7.5	ms
FTTI (15 ms $\times$ 25 frames)	=	375	ms
Available idle time per FTTI (50% of 375 ms)	=	187.5	ms
Ideal watchdog clearing window (12.5% of 375 ms)	=	46.875	ms
The calculated watchdog clearing window in firmware is as follows			
Duration of complete frames which can be fit in watchdog clearing window ($\lceil 46.875/15 \rceil \times 15$)	=	45	ms
Fractional watchdog clearing time (which will be fit in the sub-frame idle time) ($46.875 - (15 \times 3)$)	=	1.875	ms
Time available for calibration/monitoring per FTTI (21 frames $\times$ 7.5 ms) + (2 sub-frames $\times$ 2.5 ms) + 0.625 ms)	=	163.125	ms

The following examples show how the user can budget for calibration and monitoring time and configure the FTTI correctly.

Example 1

A user has enabled 4 TX, uses only 1 profile, frame configuration consists of 64 chirps, each chirp is of duration is 66 μ s (56 μ s ramp time and 10 μ s chirp idle time) and frame periodicity is 10 ms. CALIB_MON_TIME_UNIT has been set to 4. User has enabled all run time calibrations. None of the analog monitoring is enabled.

$$\begin{array}{rcl}
 \text{Frame duty cycle} & = & 42.24\% \\
 \text{Idle time per frame (57.76\% of 10 ms)} & = & 5.776 \text{ ms} \\
 \left. \begin{array}{l} \text{Idle time available for calibration/monitoring per frame} \\ \text{(100 } \mu\text{s is for frame preparation)} \end{array} \right\} & = & 5.676 \text{ ms} \\
 \left. \begin{array}{l} \text{Time needed for all run time calibrations.....} \\ 150 + 300 + 30 + 400 + (720 \times 4) + 30 + 2800 + 60 \end{array} \right\} & = & 6650 \mu\text{s} \\
 \left. \begin{array}{l} \text{Minimum time for software overheads.....} \\ 20 + 600 + 300 + (10000 \times 4/8) \end{array} \right\} & = & 5920 \mu\text{s} \\
 \left. \begin{array}{l} \text{Total time needed per FTTI for calibration} \\ 6650 \mu\text{s} + 5920 \mu\text{s} \end{array} \right\} & = & 12570 \mu\text{s}
 \end{array}$$

Total time needed per FTTI for calibration is 12.570 ms which is less than the total frame idle time in the FTTI ($5.676 \times 4 = 22.704$ ms) and hence this configuration will be honored by the MMIC. CALIB_MON_TIME_UNIT in actual time units is also within the allowed range of 40-250ms. User can set CALIBRATION_PERIODICITY as 25. With this setting calibrations are triggered once every 100 frames (i.e. once every 1 s)

Example 2

Consider another example where the frame configuration remains the same as in example 1, but frame periodicity is reduced to 6 ms.

$$\begin{array}{rcl}
 \text{Frame duty cycle} & = & 70.40\% \\
 \text{Idle time per frame (29.60\% of 6 ms)} & = & 1.776 \text{ ms} \\
 \left. \begin{array}{l} \text{Idle time available for calibration/monitoring per frame} \\ \text{(100 } \mu\text{s is for frame preparation)} \end{array} \right\} & = & 1.676 \text{ ms} \\
 \left. \begin{array}{l} \text{Time needed for all run time calibrations.....} \\ 150 + 300 + 30 + 400 + (720 \times 4) + 30 + 2800 + 60 \end{array} \right\} & = & 6650 \mu\text{s} \\
 \left. \begin{array}{l} \text{Minimum time for software overheads.....} \\ 20 + 600 + 300 + (6000 \times 4/8) \end{array} \right\} & = & 3920 \mu\text{s} \\
 \left. \begin{array}{l} \text{Total time needed per FTTI for calibration} \\ 6650 \mu\text{s} + 3920 \mu\text{s} \end{array} \right\} & = & 10570 \mu\text{s}
 \end{array}$$

Total time needed per FTTI for calibration is 10.57 ms which is more than the total frame idle time in an FTTI ($1.676 \times 4 = 6.704$ ms) and hence this configuration will **not** be honored by the MMIC. Also, CALIB_MON_TIME_UNIT in actual time units is $4 \times 6 \text{ ms} = 24 \text{ ms}$, which is **not** within the allowed range of 40-250ms.

User can set CALIB_MON_TIME_UNIT to 10 and CALIBRATION_PERIODICITY as 17. With this setting the Total time needed per FTTI for calibration is 15.07 ms which is less than the total frame idle time in an FTTI ($1.676 \times 10 = 16.760$ ms). Calibrations are triggered once every 170 frames (i.e. once every 1.02 s)

Example 3

A user has enabled 2 TX, uses 2 profiles, frame configuration consists of 32 chirps, each chirp is of duration is 90 μ s (80 μ s ramp time and 10 μ s chirp idle time) and frame periodicity is 6 ms. CALIB_MON_TIME_UNIT is set to 1. User has enabled all run time calibrations. None of the analog monitoring is enabled.

$$\begin{array}{rcl}
 \text{Frame duty cycle} & = & 48.00\% \\
 \text{Idle time per frame (52.00\% of 6 ms)} & = & 3.120 \text{ ms} \\
 \left. \begin{array}{l} \text{Idle time available for calibration/monitoring per frame} \\ \text{(100 } \mu\text{s is for frame preparation)} \end{array} \right\} & = & 3.020 \text{ ms} \\
 \left. \begin{array}{l} \text{Time needed for all run time calibrations.....} \\ 150 + 300 + 30 + 400 + (720 \times 2 \times 2) + 30 + 2800 + 60 \end{array} \right\} & = & 6650 \mu\text{s} \\
 \left. \begin{array}{l} \text{Minimum time for software overheads.....} \\ 20 + 600 + 300 + (6000 \times 1/8) \end{array} \right\} & = & 1670 \mu\text{s} \\
 \left. \begin{array}{l} \text{Total time needed per frame for calibration} \\ 6650 \mu\text{s} + 1670 \mu\text{s} \end{array} \right\} & = & 8320 \mu\text{s}
 \end{array}$$

Total time needed per FTTI for calibration is 8.320 ms which is more than the overall frame idle time in the FTTI (3.020 ms) and hence this configuration will **not** be honored by the MMIC device. Also, the CALIB_MON_TIME_UNIT in actual time units is 6 ms which is **not** within the allowed range of 40-250ms.

User can set CALIB_MON_TIME_UNIT to 7 and CALIBRATION_PERIODICITY as 24. With this setting, minimum required time is 12.82 ms and available idle time for calibration/monitoring is 3.020 ms * 7 = 21.14 ms and calibrations are triggered once every 168 frames (i.e. once every 1.008 s)

Example 4

A user has enabled 2 TX, uses 2 profiles, frame configuration consists of 32 chirps, each chirp is of duration is 90 μ s (80 μ s ramp time and 10 μ s chirp idle time) and frame periodicity is 6 ms. CALIB_MON_TIME_UNIT is set to 1. User has enabled all run time calibrations. Analog monitorings which are enabled are (a) TX output power monitor for TX0 and TX1 (b) TX Ballbreak monitor for TX0 and TX1 and (c) RX gain phase monitor. Each of the monitors are configured to be run for 1 profile and 3 RF frequencies (low, mid and high) as defined by the profile.

Frame duty cycle	=	48.00%
Idle time per frame (52.00% of 6 ms)	=	3.120 ms
Idle time available for calibration/monitoring per frame } (100 μ s is for frame preparation)	=	3.020 ms
Time needed for all run time calibrations..... }	=	6650 μ s
150 + 300 + 30 + 400 + (720 $\times$ 2 $\times$ 2) + 30 + 2800 + 60 }	=	
Time needed for all monitoring..... }	=	3380 μ s
(240 $\times$ 2 $\times$ 3) + (250 $\times$ 2) + (480 $\times$ 3) }	=	
Minimum time for software overheads..... }	=	1670 μ s
20 + 600 + 300 + (6000 $\times$ 1/8) }	=	
Total time needed per frame for calibration and monitoring }	=	11700 μ s
6650 μ s + 3380 μ s + 1670 μ s }	=	

Total time needed per FTTI for calibration is 11.700 ms which is more than the overall frame idle time in the FTTI (3.020 ms) and hence this configuration will **not** be honored by the MMIC.

User can set CALIB_MON_TIME_UNIT to 7 and CALIBRATION_PERIODICITY as 24. With this setting, minimum required time for calibration and monitoring is 16.20 ms and available idle time for calibration/monitoring is 21.14 ms. Monitoring is triggered once in 7 frames and calibration is triggered once in 168 frames (i.e. once every 1.008 s)

12.5 Sample Application

For sample application please refer DFP (device firmware package) and DFP user guide document.

Appendices

A AWR2243 API changes

The Scope of this section is to highlight the new APIs and changes to existing APIs w.r.t. AWR1243 device APIs, all the features/APIs of AWR1243 are supported and applicable to AWR2243 (Backward Compatible) in addition to new APIs unless otherwise it is explicitly captured below. Please refer mmWaveLink driver migration guide document for more info for API migration. Refer revision history for more details in page [xxii](#).

The following are the absolute new APIs added in AWR2243 device DFP.

1. [AWR_APLL_SYNTH_BW_CONTROL_SB](#)
2. [AWR_MONITOR_TYPE_TRIG_CONF_SB](#)
3. [AWR_AE_RF_MONITOR_TYPE_TRIGGER_DONE_SB](#)
4. [AWR_MONITOR_SYNTHESIZER_FREQUENCY_NONLIVE_REPORT_AE_SB](#)
5. [AWR_DEV_RF_DEBUG_SIG_SET_SB](#)
6. [AWR_DEV_DEV_HSI_DELAY_DUMMY_CFG_SET_SB](#)
7. [AWR_ADVANCE_CHIRP_CONF_SB](#)
8. [AWR_ADVANCE_CHIRP_GENERIC_LUT_LOAD_SB](#)
9. New timing information added in page [536](#)
10. [AWR_ADVANCE_CHIRP_DYN_LUT_ADDR_OFFSET_CFG_SB](#)

The following are the changes to existing APIs, backward compatibility is impacted due to these changes; however these APIs are unsupported in AWR1243 device.

1. [AWR_MONITOR_RX_IFSTAGE_CONF_SB](#)
2. [AWR_MONITOR_RX_IFSTAGE_REPORT_AE_SB](#)
3. [AWR_DIGITAL_COMP_EST_CONTROL_SB](#) (previously called [AWR_INTER_RX_GAIN_PHASE_CONTROL_SB](#))
4. [AWR_MONITOR_TX0_PHASE_SHIFTER_CONF_SB](#) (previously called [AWR_MONITOR_TX0_BPM_CONF_SB](#))
5. [AWR_MONITOR_TX1_PHASE_SHIFTER_CONF_SB](#) (previously called [AWR_MONITOR_TX1_BPM_CONF_SB](#))
6. [AWR_MONITOR_TX2_PHASE_SHIFTER_CONF_SB](#) (previously called [AWR_MONITOR_TX2_BPM_CONF_SB](#))
7. [AWR_MONITOR_TX0_PHASE_SHIFTER_REPORT_AE_SB](#) (previously called [AWR_MONITOR_TX0_BPM_REPORT_AE_SB](#))
8. [AWR_MONITOR_TX1_PHASE_SHIFTER_REPORT_AE_SB](#) (previously called [AWR_MONITOR_TX1_BPM_REPORT_AE_SB](#))
9. [AWR_MONITOR_TX2_PHASE_SHIFTER_REPORT_AE_SB](#) (previously called [AWR_MONITOR_TX2_BPM_REPORT_AE_SB](#))

TX2_BPM_REPORT_AE_SB)

The following are the updates to existing APIs in 'reserved' fields, backward compatibility is not impacted due to these changes.

1. Updated MISC_FUNC_CTRL in [AWR_RF_TEST_SOURCE_CONFIG_SET_SB](#)
2. Updated ANA_MONITORING_ENABLES and LDO_VMON_SC_MONITORING_EN in [AWR_MONITOR_ANALOG_ENABLES_CONF_SB](#)
3. Added new fields in [AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_CONF_SB](#)
4. Added new fields in [AWR_MONITOR_TX1_INTERNAL_ANALOG_SIGNALS_CONF_SB](#)
5. Added new fields in [AWR_MONITOR_TX2_INTERNAL_ANALOG_SIGNALS_CONF_SB](#)
6. Added new fields in [AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB](#)
7. Added new fields in [AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB](#)
8. Added new fields in [AWR_MONITOR_TX0_INTERNAL_ANALOG_SIGNALS_REPORT_AE_SB](#)
9. Updates to [AWR_CHAN_CONF_SET_SB](#)
10. Updates to [API AWR_CAL_DATA_SAVE_SB](#)
11. Updates to [AWR_PROFILE_CONF_SET_SB](#)
12. Added new fields in [AWR_CALIB_MON_TIME_UNIT_CONF_SB](#)
13. Added new fields in [AWR_RUN_TIME_CALIBRATION_CONF_AND_TRIGGER_SB](#)
14. Updates to [AWR_RF_BOOTUPBIST_STATUS_GET_SB](#)
15. Updates to [AWR_AE_DEV_RFPOWERUPDONE](#)
16. Updates to [AWR_MONITOR_RF_DIG_LATENTFAULT_REPORT_AE_SB](#)
17. Added new fields in [AWR_MONITOR_SYNTHESIZER_FREQUENCY_CONF_SB](#)
18. Added new fields in [AWR_DEV_CSI2_CFG_SET_SB](#)
19. Updates to [AWR_LOOPBACK_BURST_CONF_SET_SB](#)
20. Updates to [AWR_MONITOR_PLL_CONTROL_VOLTAGE_REPORT_AE_SB](#)
21. Updates to [AWR_BPM_CHIRP_CONF_SET_SB](#)
22. Updates to [AWR_RF_RADAR_MISC_CTL_SB](#)
23. Updates to [AWR_ADVANCED_FRAME_CONF_SB](#)
24. Updates to [AWR_FRAME_CONF_SET_SB](#)
25. Added new fields in [AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_REPORT_AE_SB](#)
26. Added new fields in [AWR_FRAMESTARTSTOP_CONF_SB](#)
27. Added new fields in [AWR_AE_RF_CPUFAULT_SB](#)
28. Updates to [AWR_MONITOR_RX_GAIN_PHASE_REPORT_AE_SB](#)
29. Updates to [AWR_AE_MSS_CPUFAULT_SB](#)
30. Updates to [AWR_RF_DEVICE_CFG_SB](#)
31. Updates to [AWR_MONITOR_TX_GAIN_PHASE_MISMATCH_CONF_SB](#)
32. Updates to [AWR_MONITOR_DUAL_CLOCK_COMP_CONF_SB](#)
33. Updates to [AWR_MONITOR_DUAL_CLOCK_COMP_REPORT_AE_SB](#)
34. Updates to [AWR_MSS_LATENTFAULT_TEST_CONF_SB](#)

- 35. Updates to [AWR_AE_MSS_LATENTFAULT_TESTREPORT_SB](#)
- 36. Updates to [AWR_AE_DEV_MSSPOWERUPDONE_SB](#)
- 37. Updates to [AWR_MSSCPUFAULT_STATUS_GET_SB](#)
- 38. Updates to [AWR_AE_MSS_BOOTERRORSTATUS_SB](#)
- 39. Updates to [AWR_AE_MSS_ESMFAULT_STATUS_SB](#)
- 40. Updates to [AWR_DEV_RX_DATA_PATH_CLK_SET_SB](#)
- 41. Updates to [AWR_DEV_FILE_DOWNLOAD_SB](#)
- 42. Updates to [AWR_PROG_FILT_COEFF_RAM_SET_SB](#)
- 43. Updates to [AWR_PROG_FILT_CONF_SET_SB](#)
- 44. Updates to timings in page 543
- 45. Added new fields in [AWR_APLL_SYNTH_BW_CONTROL_SB](#)
- 46. Added new fields in [AWR_MONITOR_RX_GAIN_PHASE_CONF_SB](#)

B xWR6243 API changes

The Scope of this section is to highlight the new APIs and changes to existing APIs w.r.t. AWR2243 device APIs.

The following are the absolute new APIs added in AWR6243 device DFP.

1. [AWR_POWER_SAVE_MODE_CONF_SET_SB](#)

The following are the updates to existing APIs in 'reserved' fields, backward compatibility is not impacted due to these changes.

1. Updated fields in [AWR_PROFILE_CONF_SET_SB](#) API
2. Updates to [AWR_AE_RF_INITCALIBSTATUS_SB](#)
3. Updates to [AWR_RUN_TIME_CALIB_SUMMARY_REPORT_AE_SB](#)
4. Added new fields in [AWR_MONITOR_TX0_BALLBREAK_CONF_SB](#)
5. Added new fields in [AWR_MONITOR_TX1_BALLBREAK_CONF_SB](#)
6. Added new fields in [AWR_MONITOR_TX2_BALLBREAK_CONF_SB](#)
7. Added new fields in [AWR_MONITOR_TX0_POWER_CONF_SB](#)
8. Added new fields in [AWR_MONITOR_TX1_POWER_CONF_SB](#)
9. Added new fields in [AWR_MONITOR_TX2_POWER_CONF_SB](#)
10. Added new fields in [AWR_MONITOR_PLL_CONTROL_VOLTAGE_SIGNALS_CONF_SB](#)
11. Updates to [AWR_MONITOR_PLL_CONTROL_VOLTAGE_REPORT_AE_SB](#)
12. Added new fields in [AWR_APLL_SYNTH_BW_CONTROL_SB](#)
13. Updated fields in [AWR_CONT_STREAMING_MODE_CONF_SET_SB](#)
14. Updated fields in [AWR_CAL_DATA_SAVE_SB](#)
15. Updated fields in [AWR_DIGITAL_COMP_EST_CONTROL_SB](#)

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